



Toxic Effects of Micro and Nanoplastics

Environment, Food and Human Health

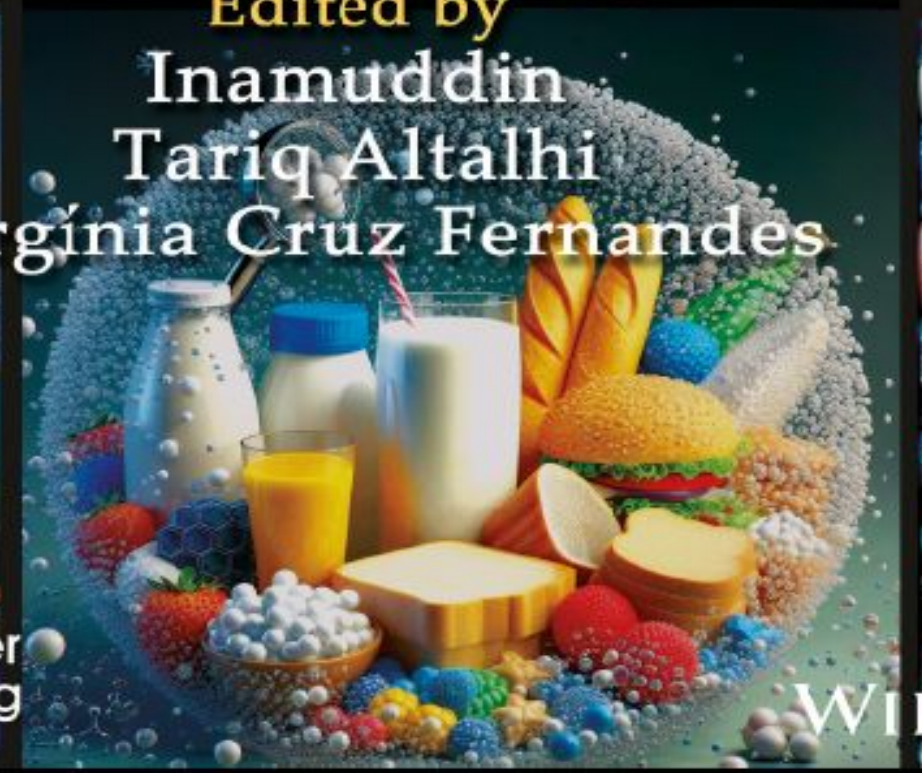






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Preface

Micro- and nanoplastics are the degradation products of large plastic compounds. These degraded polymers enter into the natural environment including air, water, and food, which leads to various significant threats to human health. The nature of these micro- and nanoplastics is persistent and consequently accumulates in the exposed person's body. Research into microplastics has shown that these particles accumulate in various human organs and impart detrimental effects on humans. To safeguard human health, analysis and remediation strategies are necessary.

This book provides a comprehensive overview on the source, distribution, life cycle assessment strategies, physico-chemical interactions, methods of analysis, toxicological investigation, and remediation strategies of micro- and nanoplastics. It is an invaluable resource for academics, researchers, post-doctoral and Ph.D. students, the polymer industry, environment agencies, food and beverage professionals, etc.

Chapter 1 explains the effect of natural processes that microplastics undergo in the environment (e.g., radiation, physical abrasion, chemical reactions, and biodegradation), which causes an increase in their ability to adsorb other pollutants and transport them. The chapter also outlines the analytical techniques used to evaluate the chemical and physical changes.

Chapter 2 presents life cycle analysis and its stages as applied to new materials called "biobased," which have emerged as an alternative to replace the use of plastics. The main focus of this chapter is to assess the

environmental impacts of bioplastics versus petrochemical plastics and their sustainability.

Chapter 3 discusses micro- and nanoplastics as an invisible threat to human health. It reviews the various routes of exposure, the phenomenon of microplastics in nourishment and nutrients, and the impact of microplastics and nanoplastics on mammalian health and their effect on marine life.

Chapter 4 explains how the small plastic particles known as micro- and nanoplastics have become a significant environmental concern due to their widespread presence in various ecosystems. The toxic effects of these particles on the environment, food, and human health are a growing concern that requires more attention and action from governments, industries, and individuals. Reducing plastic waste and promoting the use of more sustainable alternatives can help mitigate this issue and protect our planet and health.

Chapter 5 discusses the probable sources of micro- and nanoplastics, and details their hazardous effects on different environments, including terrestrial, aqueous, atmosphere, wastewater treatment plants, and their resident organisms. It also explains the journey of these particles from their production source to their final destination.

Chapter 6 covers the routes through which micro- and nanoplastics can become part of our food and their possible toxic effects on human bodies and the food chain. Two primary ways that these plastic particles enter food products is through plastic food packaging or by being ingested by animals and absorbed into plants. This chapter explains how the side effects of MPs and NPs on human lives depend on numerous factors, such as plastic chemical

functionality and biocompatibility, size, and amount of plastic ingested.

Chapter 7 discusses the microplastic, properties, types, and their impact on the environment in detail. Great attention is paid to various methods of eliminating microplastics. The chapter also includes goals and initiatives taken by the United Nations Sustainable Development Goal (SDG 14).

Chapter 8 analyzes the presence of micro- and nanoplastics in different types of beverages. It presents the classification of the analyzed beverages; the methods used for the quantification of micro- and nanoplastics; the characteristics of the particles; and their origin. Human exposure from the consumption of these products is also discussed.

Chapter 9 focuses on the effect of micro- and nanoplastics that end up in the terrestrial environment. It looks at their interaction with soil and plants while outlining their migration and accumulation inside the plant, and calculating their potential effect. The chapter also discusses the impact of micro- and nano plastics on terrestrial communities, including microbes and humans.

Chapter 10 addresses the presence of microplastics in personal care products (PCPs). The information is organized by three topics: the characterization of PM extracted from PCPs, their interactions with other substances, and toxicity. The chapter explains how the use of these products is alarming due to their wide use and risks to the environment.

Chapter 11 reveals how the various chemical compositions including plastics and microplastics are mixed into desired concentrations when manufacturing cosmetics. It discusses, too, the main sources of plastics and microplastics and their growth in India. The effect of

cosmetics on human health is explained. Finally, alternative products to plastics and microplastics for use in cosmetics are listed.

Chapter 12 delves into the detrimental impact of micro- and nanoplastics on the human genome. The introduction of such particles into the ecosystem, and ultimately to the human body, is explained. This chapter presents an thorough toxicological analysis of these particles, shedding light on the urgent need for proactive measures to safeguard our ecosystem.

Chapter 13 discusses the generation, as well as the techniques for the measurement and identification, of micro- and nanoplastics. Various degradation methods are also discussed, as are the harmful effects of plastics, nanoplastics, and microplastics. Measures to avoid the production of plastics, nanoplastics, and microplastics are emphasized.

Chapter 14 details the source and hazardous effects of micro- and nanoplastics in marine environments. Additionally, it elaborates on the damages caused by the plastic pollution on air, water, and soil. Methods for decreasing microplastics in the environment are also discussed, along with the severance of microplastics from water, sediments, and marine microbial strains associated with degrading microplastics.

Chapter 15 reviews the advances and challenges in assessing the toxicity of micro- and nanoplastics (MPs and NPs) in human beings. An analysis of 85 research articles is also presented. Results show that in most cases there is a negative effect associated with MPs and NPs, but this chapter explains how methodology differences don’t allow the establishment of cause-effect relationships.

Chapter 16 delves into the extensive impact of plasticizers and flame retardants on ecosystems and human health. The presence of plasticizers and flame retardants in various environments raises concerns about potential ecotoxicological effects. This chapter explains how bridging knowledge gaps and promoting safer alternatives are crucial to address the risks posed by these additives.

Chapter 17 details the invisible threat of micro- and nanoplastic materials on mankind and the environment. It discusses the harmful effects of inorganic and organic contaminants that are present in MPs and NPs. Inorganic contaminants primarily include heavy metals and pesticides. However, organic contaminants are persistent organic pollutants, and the impact of persistent organic pollutants and inorganic contaminants on the environment are presented in detail.

Chapter 18 compares the toxicity of microplastics, nanoplastics, and nanoparticles in the ecosystems. Smaller particles can penetrate organisms and tissues leading to more severe impacts. Their unique properties increase reactivity and oxidative stress, raising concerns about bioaccumulation and higher trophic levels. This chapter explains why urgent mitigation strategies are needed to protect ecosystems from these pervasive pollutants.

Chapter 19 discusses the occurrence and sources of micro- and nanoplastics and the pretreatments performed in the samples. Additionally, it thoroughly discusses several techniques that can be used to characterize, identify, and quantify them. Furthermore, this chapter presents a general overview of the advantages, disadvantages, and limitations of those techniques.

Chapter 20 presents new analytical approaches for the analysis of micro- and nanoplastics in the environment. Microscopic, spectroscopic, thermal, and electroanalytical

techniques are commonly used for the analysis of MPs and NPs. The chapter describes the development of analytical techniques for monitoring plastic pollution based on single and combined methods.

Chapter 21 details various enzymes that are applied for the biodegradation of micro- and nanoplastics. In this chapter, the advantages of enzymatic approaches compared to conventional methods are presented. The mechanism of enzyme-catalyzed degradation of plastics is also discussed, as are some examples from biodegradation of synthetic polymers that use various enzymes.

Chapter 22 explains how the most common physical, chemical, and biological techniques work to remove micro- and nanoplastics. Some of the most relevant findings found in the literature for each technique are presented, as are the advantages and disadvantages of each type of removal technique.

Chapter 23 details the different materials used to remove the micro- and nanoplastics from the water. This chapter explains how to use sponge/aerogel materials, materials with metals, and biochar to remove MPs and NPs. Also, remediation methods that employ powder and granulated activated carbons are presented.

Chapter 24 details the toxicity and aftereffects of micro- and nanoplastics on the marine environment and its flora and fauna. The providence of MPs and NPS and their migration to the aquatic environment, along with an analysis of various micro/nanoplastics toxicity and the propensity towards environmental implications, is also presented.

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The Editors
March 2024

1
Aging Process of Microplastics in the Environment

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Abstract

The presence of plastics in various ecosystems is an emerging worldwide environmental concern. Researchers have studied the interaction of microplastics (MPs) with other pollutants that are also present in the environment and have concluded that they act as vectors for pollution dispersion, transporting pollutants to different ecosystems, and being taken up by living organisms. The effects of natural processes that MPs undergo in the environment (UV radiation, physical abrasion, chemical reactions, and biodegradation) cause changes in their external surface, morphology, and chemical alterations that increase their ability to interact with other pollutants and transport them. Researchers have developed laboratory techniques to simulate the aging process of polymers and predict the behavior of MPs in real ecosystems. These reports highlight permanent physical and chemical changes in different properties of MPs, such as color, morphology, particle size, specific surface area, hydrophobicity, crystallinity, melting and glass transition temperature, surface groups, carbonyl index, and oxygen/carbon ratio. These properties have been measured using standard techniques (e.g., optical, fluorescence, and scanning electron microscopy, Fourier-transform infrared spectroscopy, Raman spectroscopy);

however, emerging techniques are being explored (two-dimensional correlation spectroscopy and excitation–emission matrix-parallel factor analysis), where it is possible to detect the release products of the aging process.

Keywords: Advanced oxidation process, aged microplastic, biodegradation, emerging contaminants, mechanical stress, photooxidation, pristine microplastic

1.1 Introduction

Plastics are widely used to meet various societal needs. Advantages such as light weight, low cost, and long durability have led to their expanded use and application in fields such as healthcare, engineering, construction, agriculture, and high-performance apparel [1–4]. The distribution of plastic use in industries is estimated to be 4% is used in the electrical and electronics industry and construction, 6% for transportation, 12% for consumer and institutional products, 14% for textiles, 13% for other industries, and 43% for packing [5]. Apart from their applicability, their production can also have a great impact on the environment, as they can be manufactured with fossil fuels, which have a great impact on the carbon footprint, or with natural materials, such as cellulose [2, 3]. Biobased plastics have been researched in recent years and have proven to be an alternative to fossil plastics, as they are largely derived from biomass. However, they currently account for only 1%–2% of the annual production of plastics. Nevertheless, a study of the cradle-to-grave life cycle of biobased plastic should be conducted to balance the use of fossil versus biobased plastic [6, 7]. Despite these new alternatives, low-density polyethylene (LDPE), high-density polyethylene (HDPE), polyethylene terephthalate (PET), polyvinyl chloride (PVC), polypropylene (PP), and

polystyrene (PS) are the most commonly used polymers that are produced from fossil hydrocarbons [3, 8–10].

Plastics are adaptable and inexpensive, and plastic production between 2005 and 2017 was as high as in the last 50 years, showing a tendency to increase exponentially, posing a problem in the treatment and disposal of plastic waste [1]. In 2016, the entrance of 19 to 23 million t of plastics into the aquatic environment was estimated and it is predicted that 20–53 Mt/year will be released into aquatic systems by 2030 [11]. Researchers expect that double plastic production would be achieved by 2050, meaning that approximately 8 million tons of plastic waste will escape into the oceans [12]. Single-use plastics, such as bags and straws, represent approximately half of the plastic waste generated. During the COVID-19 pandemic, protective gear containing plastics (e.g., gloves and face masks) is often used by the population, increasing the amount of plastic waste [1, 12, 13]. Zhao *et al*. reported that 6,300 million tons of plastic waste was generated in 2015, of which only 9% was recycled, 12% was incinerated, and 79% was disposed of in landfills [2]. According to Williams *et al*., of the plastic waste generated since 1950, 14% is incinerated, 40% is sent to landfills, and 14% is recycled. However, of this amount of recycled plastic only, 2% is optimally recycled, and the remaining 12% produces material with lower quality and functionality than the original, which is referred to as “downcycle” [12].

Based on this information and knowledge of the long life of plastics, humanity may face a worrying environmental problem in the coming years [14]. Governments around the world have introduced regulations to reduce the use of single-use plastics, namely imposing taxes on plastic bags and food packaging [1]. The incorrect disposal of plastics is a major problem of pollution, which may cause their entrance into the oceans [12, 15]. In recent years,

researchers have explored plastic pollution from manufacture to final disposal, and the damage caused by plastic to ecosystems, but there is still a long way to go to understand how harmful their chronic presence may be to the environment.

MPs may have two types of sources: the primary source, which is considered a direct contributor, focused on the manufacturing of MPs in various industries, such as exfoliating cleansers, cosmetics, and toothpaste [14, 16]; and the secondary source, which is considered an indirect contributor to MPs, caused by the fragmentation of large plastic pieces into small ones, which can be promoted by photodegradation, mechanical or chemical action, or other weathering processes to which plastics are subjected to improper disposal in the environment [9, 16, 17].

Plastics are subjected to physical, chemical, and biological reactions over time, resulting in the desorption of smaller particles [8, 18]. Plastics can be classified based on their size as megaplastics (>50 cm), macroplastics (>5 to 50 cm), mesoplastics (≥5 mm to 5 cm), microplastics (≥1 µm to <5 mm), and nanoplastics (<1 µm) [14, 18, 19]. MPs have become the focus of research in recent years owing to their widespread presence in ecosystems, which is considered a threat to the environment, and consequently to human health. Microplastics (MPs) have been identified in freshwater [8, 10, 14, 16, 17, 20, 21], groundwater [8, 17, 20], snow [17], ice [17], sediments [8, 16, 17, 21, 22], soils [10, 17, 22–24], terrestrial and aquatic biota [10, 14, 24], air [17, 22, 25], foodstuffs (e.g., honey and salt), tap/bottled water [17, 22, 26], and biological samples (e.g., blood, human placenta, lung tissue) [27–29]. The ubiquity of MPs in the environment and their presence in consumer products, such as food or freshwater, leads to unrestrained consumption of MPs by humans [17]. Ingestion through direct consumption of contaminated products, inhalation of

airborne contaminants with inhalable sizes of MPs and dermal contact between nanoplastics and the skin barrier are considered human exposure to MPs [22].

The biggest problem arises when researchers discover the ability of MPs to adsorb and transport various types of pollutants [9, 14, 23, 30–32]. Reports have shown that pollutants such as polycyclic aromatic hydrocarbons (PAHs), polychlorinated biphenyls (PCBs), heavy metals, pesticides and antibiotics can be transported by MPs [10, 20, 30, 33–35]. Moreover, ecotoxicity studies show the hazardous effects of MPs taken up by earthworms, mussels, gobies, seabirds, turtles, seals, mammals, fish, reptiles, and plants [8, 14, 23, 36]. Depending on their exposure and susceptibility to MPs, they are considered potentially harmful to humans because of their high surface area, which can provoke cytotoxicity, oxidative stress, and translocation to other tissues [17, 22, 37]. Their persistence may also cause chronic inflammation with possible carcinogenic effects and immune or neurodegenerative diseases [22]. Recent studies have reported that MPs have been detected in human placenta, blood, and lung tissue [27–29], increasing the urgency to understand the behavior of MPs, including their interactions, and find solutions to minimize their impact on ecosystems, organisms, and human health.

Reducing plastic use is crucial to minimize this problem. Resolutions are being made regarding single-use plastics, and some countries have defined laws banning certain types of plastics. Resolutions such as taxing the production of single-use plastic (SUP) bags, charging for the use of SUP bags, assigning recycling targets to the manufacturer, and more radically, countries such as Jakarta have already banned the use of SUP bags completely, requiring the use of environmentally friendly bags [38]. The use of biodegradable plastics contributes to solving the plastic waste problem; however, suitable microorganisms must be

present to ensure their degradation; otherwise, we will only address the problem [4]. It is crucial to harmonize policies, invest in alternative materials, and improve research on MPs, particularly in terms of ecotoxicity, namely the chronic effects on human life, which are still unknown. In addition, the focus should be on the pathways and interactions to find future solutions to minimize the impact of this problem.

1.2 Impact of MPs on the Environment

Recent studies have focused on the effects of MPs on the ecosystems where they are spread (air, terrestrial, and aquatic). As physical–chemical properties change due to the natural aging process, researchers have driven their studies on the behavior and ability of MPs to transport pollutants (e.g., PAHs, PCBs, pesticides, etc.) [10, 20, 30, 33, 34]. Adsorption and absorption constitute the sorption process and consist of the transfer of chemicals from liquids or gases to a solid, in this case MPs [39, 40]. The transfer of contaminants occurs due to interactions (van der Waals, ionic and steric forces, л–л interactions, and covalent bonds) with the surface of MPs, which is considered an adsorption process [39, 40]. Strong interactions between a low pollutant concentration and the surface of the adsorbent lead to adsorption. However, if the contaminant concentration is high, absorption occurs as soon as a large volume of pores is available to settle the pollutant molecules [39, 40]. Characteristics such as composition, size, shape, density, and chemical composition of MPs are key factors that promote the adsorption of MPs [40, 41], which may be changed by the aging processes that occur in MPs. However, environmental conditions (e.g., pH, temperature, salinity, and ionic strength) can also influence the adsorption process [42].

Studies have been carried out to predict the impact of the presence of MPs in different organisms, from aquatic (e.g., crabs and fish) to terrestrial (e.g., earthworms, nematodes, and mites) organisms. Although the responses differ according to the test organism, researchers have shown that MPs can cause hazardous effects (e.g., oxidative stress, cytotoxicity, and translocation to other tissues). Long-term contact and constant ingestion of MPs by organisms such as fish may cause effects along the food chain, leading to chronic inflammation (e.g., of the lungs) and increased risk of cancer, immune or neurodegenerative diseases, and metabolic disorders in humans [8, 22, 43, 44]. Furthermore, the harmful effects of MPs on other living organisms (plants and soil invertebrates) are of great concern [36].

Lei *et al*. investigated the effects of PS (diameter between 100 and 500 nm) on the survival rate, lifespan, motor behavior, movement-related neurons, and oxidative stress in *Caenorhabditis elegans*. After 3 days of contact, a decrease in the rate of survival, a large decrease in the organism length, and a decrease in the average life span of the nematodes were observed. In this study, it was also found that MPs can cause oxidative damage in nematodes, and the size of the particles affects their toxicity, which has far-reaching effects [43].

HDPE particles can transport chlorpyrifos (CPF), a commonly used pesticide. When this combination is in contact with mussels for 21 days, changes in the biological responses can be observed, which are greater than those induced by any stressor individually, according to the study by Fernándes *et al*. [31].

In a study performed by Bessa *et al*., 157 particles of MPs were detected, corresponding to 38 % of all fish, with 1.67 ± 0.27 (SD) MPs per fish, in three commercial fish species: sea bass (*Dicentrarchus labrax*), sea bream (*Diplodus vulgaris*), and flounder (*Platichthys flesus*). In addition,

ecotoxicological studies need to be conducted to understand the risks to fish health and the consequences of consuming these fish in humans [20]. According to a review by Rakib *et al*., approximately 25 studies have reported various effects of MPs on different marine organisms, including ingestion, translocation, and respective impacts, reduction of the feed, oxidative stress or retention in the digestive tract, or even mortality of species [45].

1.3 Pristine and Aged Microplastics

MPs are widespread in the atmosphere and in terrestrial and aquatic systems. After their release, they are subjected to natural phenomena that lead to aging [46]. Ultraviolet (UV) radiation, physical abrasion, chemical oxidation, and biodegradation cause physical and chemical changes to the MPs [46, 47]. Exposure to UV radiation, known as photooxidation, leads to rapid degradation of the polymer in the environment, resulting in color changes and the appearance of cracks [48]. Although UV radiation is considered the main cause of aging of MPs, it is important to consider other phenomena that may play a role, such as mechanical stress or physical abrasion. Waves, tides, gravel, sand, stone, water flow, and other particles surrounding MPs can affect their physical properties and render them brittle. This aging process can result in changes in crystallinity, thermal hydrophilicity, and degree of polymerization [46, 49]. Chemical reactions may also occur, promoted by reactive oxygen species (ROS) [50, 51]. Photooxidation may occur because of a chemical reaction between MPs and ROS produced from natural organic matter (NOM), NO^{-3}, and CO_2^{-3}. In addition, after exposure to sunlight, pigments can produce ROS through a series of reactions that can oxidize the polymer [50–52]. Weathering phenomena, such as UV light, mechanical erosion, or

chemical reactions, can play an important role in the life cycle of MP, as they cause changes in polymer properties.

The use of their aged in a natural environment was the better choice to study their interactions, but uncontrollable factors were associated with it. Therefore, researchers have focused on simulating the aging process in the natural environment of the laboratory, which although being more controlled may be less realistic. With constant changes in the environment and the emergence of new pollutants and possible new interactions, researchers have also focused on simulating environmental phenomena at the laboratory scale, such as photooxidation, mechanical stress, and chemical oxidation [53]. The development of these laboratory-scale aging tests is important for predicting and determining the behavior of aged MPs compared to pristine MPs, as the process of aging is very slow in the real environment [53]. According to Liu *et al*., researchers are focusing on selecting the best laboratory technologies to increase the speed of the MP aging process. The most commonly used technology is light irradiation (66.7%), followed by chemical oxidation (16.7%), heat treatment, and microbial degradation (less than 26.6%) [52]. Despite the difficulty of the developed aging techniques, researchers have focused on understanding the significant changes in MPs after the aging process: physical changes, such as color changes, cracks on the surface of MP, or chemical changes, such as the differences between spectra from the Fourier-transform infrared spectroscopy (FTIR) and Raman analysis, or even changes in crystallinity, such as an increase in melting temperature. Table 1.1 summarizes the significant changes found in MPs after the aging process in the laboratory, with different changes showing the effects of aging processes. Understanding the behavior of MPs and their interactions can be complex, considering that mechanical agents, chemical, and biological reactions constantly occur in the environment. The change in

properties has a significant impact on the adsorption behavior of MPs [54–57].

Adsorption studies were conducted with a focus on comparing the behavior of MPs in the pristine and aged states to understand the influence of altered properties. Zhang *et al*. studied the adsorption process of oxytetracycline in PS using purchased PS foams and aged PS foams made from plastic waste collected from coastal beaches. The influence of pH (between 2.0 and 10.0) and ionic strength (using sodium chloride, calcium chloride, and sodium sulfate) was tested to understand the effect on the adsorption process. The maximum adsorption capacity was observed at pH 5 for the beached PS samples, and stronger sorption of oxytetracycline in the MPs was observed in the presence of $CaCl_2$. Based on the equilibrium isotherms, it was found that aged PS had a higher adsorption capacity than pristine PS (1,520 µg g^{-1} and 27,500 µg g^{-1}, respectively) [58]. Fan *et al*. simulated the aging behavior of PS and polylactic acid (PLA) in a natural environment, exposing the MPs in a UV radiation experiment, to compare the behavior of the pristine MPs in the adsorption process of tetracycline (TC). Aged PS and PLA presented a higher adsorption capacity for these antibiotics (e.g., 5.21 mg g^{-1} and 5.49 mg g^{-1}, respectively), showing that the change of surface structure observed for the aged MPs favors the adsorption process. Compared to those of pristine PS and PLA, the adsorption capacity obtained was 2.75 mg g^{-1} and 2.51 mg g^{-1}, respectively [54]. Studies on the adsorption capacity of heavy metals in MPs showed that aged MPs presented a higher capacity than pristine MPs, as shown in the study conducted by Mao *et al*. PS was subjected to three types of aging processes in the laboratory and showed an increase in adsorption capacity in air, seawater, and pure water for five heavy metal ions (Pb^{2+}, Cu^{2+}, Cd^{2+}, Ni^{2+}, and Zn^{2+}) [56]. These studies demonstrated that pollutants could

be adsorbed by different types of MPs, and, an increase in the adsorption capacity was observed after aging.

Table 1.1 Physical and chemical changes of MPs after suffering an aging process.

Property	Agent	Changes	References
Color	Photooxidation	Yellow, opacity	[49, 64, 66, 95–98]
Morphology	Photooxidation AOP Mechanical stress Biodegradation	Cracks, flakes, roughness, biofilm colonization	[56, 66, 95–97, 99–104]
Particle size	Photooxidation AOP Mechanical stress Biodegradation	Decrease	[49, 59, 86, 88, 89, 98]
SSA	Photooxidation AOP	Increase	[33, 53–55, 58, 67, 84, 105]
Contact angle	Photooxidation AOP	Decrease	[77, 99, 100, 105, 106]
Crystallinity	Photooxidation AOP	Increase Decrease	[33, 56, 74] [75, 100, 107]
Melting temperature	Photooxidation	Increase/Decrease	[59, 100, 103, 108]
Glass transition temperature	Photooxidation	Changeable	[59, 108–111]
Surface groups	Photooxidation AOP Mechanical	New peak/band formation	[67, 78, 79, 95, 103, 112, 113]

Property	Agent	Changes	References
	stress Biodegradation		
CI	Photooxidation AOP Mechanical stress Biodegradation	Increase	[50, 92, 100, 101, 106, 107]
O/C	Photooxidation AOP Mechanical stress Biodegradation	Increase	[53, 56, 84, 90, 98, 99, 106]
Adsorption capacity	Photooxidation AOP Mechanical stress Biodegradation	Increase	[47, 96, 101, 102, 106, 114, 115]

Environmental concerns arise because all MPs spread in ecosystems constantly change under UV radiation, mechanical stress, chemical reactions, and contact with multiple contaminants. It is important to understand the pathways of MPs in the environment, their infinite interactions, and their awareness of environmental conditions.

1.4 Influence of Aging Processes in the Properties of MPs

Physical, chemical, and biological processes lead to aging, which is due to changes in size, physical resistance, surface area, thermal stability, and chemical composition. Focusing on the changes in these properties will provide a closer look

at the real environment and allow researchers to understand the different stages of aging that MPs undergo.

1.4.1 Physical Properties

MPs are exposed to physical abrasion through contact with sand and rocks, and when taken up by waves and wind [14, 48]. Their particle size constantly changes due to either fragmentation promoted by mechanical stress or photooxidation, which breaks some bonds in the MPs structure [49, 59]. Researchers are exploring the changes in physical properties, such as density, shape, color, specific surface area (SSA), and particle diameter, which decrease owing to the fragmentation/degradation of plastics [60].

The density of MPs changes over time and may play an important role in their behavior. Density gradient solutions can be used to determine the density of MPs by observing the floating or sinking state in density gradient solutions [61]. MPs have a density closer to water (0.8 to 1.4 g cm^{-3}), so the use of salt solutions, such as sodium chloride, sodium iodine, and zinc chloride, separates the MPs from other compounds [62, 63]. When the density is higher than that of water, MPs sink and settle with the sediments, while at a low density, they float [9].

Another characteristic whose changes can be identified is the color of the MPs, which usually changes after aging and can be easily identified visually or microscopically [64]. The color change is caused by the formation of chromogenic groups on the surface of the particles, as Bandow *et al*. observed on PVC particles when they were subjected to an aging process (8 weeks with UV radiation exposure of 190 MJ m^{-2}) and the particles turned yellow. This color change was the result of the formation of polyenes, and when the polyenes had more than eight C=C−C bonds, the absorption in the visible range decreased, resulting in a yellow color [64]. This change can increase the attention of organisms,

as some of them are visual predators, such as goby fish, thereby increasing the probability of ingestion [53].

The SSA is also an important physical property that is directly affected by the aging process, as it is related to the structural change of the particle [52, 65]. This property can be determined from the Brunauer–Emmett–Teller (BET) equilibrium isotherm using a nitrogen physisorption analyzer. Studies have examined the correlation between the aging process and SSA and concluded that the smaller the particle size, the larger the BET SSA [52, 66, 67].

Optical microscopy, fluorescence microscopy, and scanning electron microscopy (SEM) can be used to record the classification and abundance of large MPs with characteristic colors and morphologies [62]. The transport of MPs can be affected by the size and shape (e.g., fibers, pellets, filaments, fragments, films, foams, microbeads, and granules) of MPs [18, 37, 68]. These two characteristics may also affect adsorption because particles with a high area-to-volume ratio (smaller sizes and irregular shapes) have a high adsorption capacity for other pollutants, thereby enhancing their hazardous potential [69]. Different studies have used SEM to observe the impact of the aging process on the surface of particles. Gao *et al*. tested different MPs (PA, PET, PS, and PVC) in four environmental media (air, seawater, sand, and soil) under UV irradiation for three months. When the particles were observed with SEM, the entire surface of the MPs changed, presenting cracks, bulges, wrinkles, oxidized particles, and pores, making the MP particles susceptible to embrittlement and disintegration [46, 47, 49, 66].

Another study reported that PE exposed to UV irradiation for 6 weeks, at controlled temperature, humidity, and light irradiation, showed a surface with cracks and a significant enlargement of the surface in the SEM analysis [66]. Despite the advantages of identifying changes on the

surface of MPs, it is important to be aware of the limitations of visual identification methods because they can be influenced by the observer [62, 70, 71].

1.4.2 Chemical Properties

In addition to physical changes, MPs are subject to chemical changes that modify their crystallinity, thermal stability, and surface functional groups [72]. Polymers can be characterized according to their degree of crystallinity, which depends on the complexity of the polymer, chain configuration, and isomerism. Aging studies have shown that the crystallinity of MPs can increase owing to the degradation of amorphous polymers and chain scission reactions, suggesting that this parameter can be useful as an indicator of chain scission [59, 73]. Wang *et al*. focused their study on the adsorption capacity of PS and PE of atrazine, distinguishing between pristine and aged MPs in a UV-irradiated chamber in the laboratory for 96 h. The two MPs suffered an increase in crystallinity from 39% and 41% to 46% and 44% for the aged PS and PE, respectively, which promoted an increase in the adsorption capacity [33]. An increase in the crystallinity of PP and PS of approximately 50% was also observed after 4 weeks of UV irradiation at 60 °C [74]. However, an increase in crystallinity was not observed in any of the aging processes. Wu *et al*. observed a decrease in the crystallinity of aged PP (oxidation process with heat-activated potassium persulfate) from 56% to 49% [75]. Crystallinity changes with the aging process, depending on the composition and method of aging of the MP.

The hydrophobicity of a polymer is related to the water contact angle, which means that a larger contact angle will lead to stronger hydrophobicity [76]. Wang *et al*. studied the photo-aging process of PVC and observed that the water

angle decreases with different aging treatments that PVC was submitted, increasing its hydrophobicity [77].

To evaluate the structure of MPs, researchers have used differential scanning calorimetry (DSC), X-ray diffraction (XRD), IR spectroscopy, and inverse gas chromatography (IGC) [59].

Thermal properties, such as the glass transition temperature (Tg) and melting point (Tm), which are measured by DSC and thermogravimetric analysis, respectively, are also affected by aging. These temperature changes depend on the aging stage of the MP, and it is expected that aged MPs undergo a reduction in molecular weight caused by degradation, leading to less hindered rotation of the backbone bonds, resulting in a lower T_g [59, 73]. T_m is usually directly related to the crystallinity of MPs and is used as a reference, it is expected that as crystallinity increases, T_m also increases [73]. The aging process of MPs is usually associated with chemical reactions, such as the appearance of new functional groups on the surface of the polymers, like −HO, C=O, COOH, C=C, and C−Cl [53, 59]. Andrade *et al*. focused their research on the study of a low-cost system to simulate environmental weathering, submitting PP, PS, and PA to an incident radiation range representative of the solar irradiation that penetrates in water, surrounding temperature, and mechanical stress of the wave agitation for 3 months. FTIR analysis was used to observe the formation of new functional groups, carbonyls, and hydroperoxides formed. New bands at 3,360–3,240 cm^{-1}, corresponding to the hydroxyl group, 1,640 cm^{-1} from the double bonds or C=O groups and 1,100 cm^{-1} representing the C−O bonds, were observed in the aged MPs [78]. Tang *et al*. conducted a study on PVC under heat exposure, with a temperature of 50 °C for 12 months and 100 °C for 4 months, where the formation of novel functional groups was observed. According to the FTIR

spectra, ester carbonyl (−COO−) was observed at 1,735 cm^{-1} and between 2,500 and 3,500 cm^{-1}, the forming carboxylic acid (−COOH) as new peaks on the PVC samples aged by UV radiation [67]. The study of photo- and thermal oxidation of PE was performed by Gardette *et al*. with different contents of vinyl and t-vinyl groups in a convection oven at 100 °C for 400 h. After FTIR analysis, the appearance of characteristic peaks related to thermal oxidation was observed, with a ketone peak at 1,713 cm^{-1} [79]. FTIR is a common technique used by researchers to study the different functional groups available on the surface of MPs and is usually used with attenuated total reflection infrared spectroscopy (ATR-IR) to facilitate sample preparation [62, 63]. To determine the surface groups by FTIR, the samples were prepared with potassium bromide, whereas this sample preparation is not necessary with ATR, which makes the use of this technique advantageous [62, 63].

FTIR spectra analysis provides information about typical IR bands, such as the ratio of carbonyl groups (around 1,726 cm^{-1}) and methyl groups (around 2,851 cm^{-1}) and allows the determination of the carboxyl index (CI), which is the ratio between the carbonyl peak (ketones, carboxylic acids, esters) and the methylene reference peak. Another parameter that can be helpful in characterizing of the aging process of MPs is the O/C ratio (oxygen/carbon), which can be determined by X-ray photoelectron spectroscopy (XPS). It evaluates the integration of oxygen in MPs [80]. In general, these parameters increase with age [73]. Liu *et al*. compared the CI index of PS before and after aging and found that the C/I was 0.054 initially, and 0.201 and 0.493 after UV/Cl_2 and UV/H_2O_2 aging, respectively [81].

Although FTIR is widely used in the study of MP, emerging techniques are being developed that focus on studying the aging process of MPs without interference. Recent studies

have reported the use of two-dimensional correlation spectroscopy (2D-COS) combined with FTIR spectroscopy to analyze the routes and resulting products by univariate analysis, where the spectral change is related to aging time. The FTIR spectra provide information about specific groups, whereas this combination can provide information about the order of change in these groups [73]. In addition to this technique, excitation–emission matrix-parallel factor analysis (EEM-PARAFAC) has been used to identify the products released by MPs during the aging process using three-dimensional fluorescence spectra. Few studies have been conducted on these two techniques, but the information that can be obtained may be important for studying different aging processes [73].

1.5 Simulation in the Laboratory of the Different Aging Effects

1.5.1 Radiation

As mentioned above, UV radiation impinging on the surface of MPs is one of the most important phenomena leading to the aging of MP, promoting changes in color and surface. This photooxidation process promotes MPs aging in three phases: chain initiation, transfer, and termination. In the chain initiation phase, free radicals are formed when occurs the absorption of UV light is absorbed on the backbone of MP, breaking the C−C and C−H bonds [59, 82]. The free radicals react with oxygen and peroxyl radicals to form hydroperoxides as well as more complex and persistent free radicals, which can be responsible for autoxidation processes, leading to a chain transfer process. If these free radicals do not bond to form an inert product, the chain termination phase becomes incomplete [59, 82]. Generally, in this phase, oxygen-containing functional groups are incorporated into the chemical structure of MPs [59, 82]. To

conduct the irradiation experiments, researchers used a photochemical reactor or aging chamber, and they were able to simulate the aging of MPs over several years in a natural environment in just a few days or months, depending on the experimental conditions [52]. The types of lamps selected to simulate the aging process were metal halide, xenon, mercury, fluorescent, and UV lamps [52]. The selection of the lamps is based on the wavelength range in which we want to perform the work, with the focus on having the closest representation of sunlight. Solar radiation comprises 50% in the infrared range (IR), 44% in the visible range (VIS), and 3%–4% in the UV range [78]. The UV range can be divided into UV type A (UV-A), which accounts for approximately 95% of the UV radiation, and UV type B (UV-B), which accounts for 5% of the ground radiation. Once the goal is to age the MPs in a simulation of the real environment, UV-A is better suited to delineate a range between 360 nm and 380 nm and to simulate the degradation of MPs under sunlight, for which UV lamps are chosen for the proposal [52, 78]. However, to simulate sunlight it is also important to know the irradiance of the lamps once this parameter is defined as the amount of light incident on a unit of area (units in W m^{-2}). On a sunny day, the MPs are exposed to an irradiance of approximately 1,000 W m^{-2} and the UV lamps are not able to reach the conditions of the real environment, so researchers have proposed the use of xenon lamps [52, 78]. A study conducted by Lin *et al*. reports that bPP was exposed to UV-A irradiation with a xenon lamp for 28 days, and it was possible to observe changes in the color of MPs. The adsorption capacity of lead (Pb) of aged PP was 274 mg kg^{-1} (at pH 8) compared to 170 mg kg^{-1} of the pristine PP, once again confirming the effects of aging processes under UV irradiation on MPs [83]. To promote oxidation by UV radiation, UV and xenon lamps are the best options; however, xenon lamps should be used to simulate sunlight.

1.5.2 Chemical Oxidation and Advanced Oxidation Process (AOP)

MPs constantly move in the environment and can coexist with reactive oxygen species (ROS) produced by natural processes; however, further information is needed regarding the effects of ROS on the aging process of MPs [84]. In recent years, advanced oxidation processes (AOPs) have been increasingly used to accelerate the aging process of MP, with the free radicals responsible for initiating the oxidation reaction. Despite the differences between the natural aging process promoted by UV light (or heat) and AOP, they show similar oxidation pathways and products [73, 84]. Studies have reported the use of AOPs for oxidation with heat-activated persulfate, Fenton, photo-Fenton, O_3/H_2O_2, UV/H_2O_2, and UV/Cl_2 [53, 73, 84]. The use of free radicals leads to a faster oxidation reaction than in a real environment, which suggests that AOPs may be a possible treatment method [53, 73].

PVC was tested by measuring the release of K, Na, Ca, Mg, and potentially toxic metals into the solution after oxidation with hydrogen peroxide (H_2O_2) at 30%. A significant increase in the ability of AOP-aged PVC to release metals into the solution was observed compared to that of natural-aged PVC [85]. In addition, chemical treatment can be used to simulate MP aging. Kelkar *et al*. tested the use of chloride to produce disinfected water using different concentrations and reaction times to destroy existing bonds in the structure of MPs and introduce new bonds, such as C−Cl [59].

Li *et al*. tested six kinds of MP, such as PA, PE, PET, PP, PS, and polymethyl methacrylate (PMMA) under five different oxidation conditions, with alkali, high and low acid concentrations, H_2O_2 and Fenton. In different experiments, MPs showed changes in various properties such as size,

surface area, and adsorption potential, especially in alkaline treatment [86].

Other authors have artificially exposed LDPE, PET, and UPVC to different aging agents, such as ozone, UV-C, and solar radiation, for 120 h. SEM analysis revealed significant changes after the aging process in all MPs tested. However, the changes in LDPE with several cracks and signs of deformation were the most evident [87].

Chemical and advanced oxidation processes quickly lead to the aging of MPs. It is important to explore and test different chemical agents and conditions at a laboratory scale that can lead MPs to an aged phase, which represents the changes that occur in real ecosystems.

1.5.3 Mechanical Stress

As mentioned previously, MPs constantly change their particle size owing to the mechanical action of tides, gravel, sand, stone, water flow, etc., and when MPs are subjected to sufficient stress, the chemical bonds break [49].

Sorasan *et al*. tested the influence of mechanical degradation of PE, PP, and PS particles, and placed them in a simulated seawater tank with constant agitation for 720 h. PE, PP, and PS had a particle size distribution between 1 and 100 µm, and after the aging process, 98% of the particles (10^5–10^6 particles/mg of plastic) had a size smaller than 10 µm [88].

Fragmentation of PE was also explored by Julienne *et al*. by introducing an LDPE film in beakers filled with water under irradiation exposure (60 W/m^2) and controlled temperature (between 32 and 44 °C) for 25 weeks. After exposure, PE appeared to be fragmented, presenting a size distribution of 0.4 to 7 mm, where 62% of the average fragments were less than 5 mm in size, which corresponds to the MP definition.

In parallel with the particle size analysis, optical microscopy analysis showed cracks on the surface of PE particles [89].

Sierra *et al*. exposed PA6 to simulated weathering conditions for 11 weeks by adjusting the continuous agitation, sand erosion, sunlight irradiation and oxidative conditions. After exposure to PA6, SEM measurements showed that the surface was significantly eroded, with pores, cracks, scratches, depressions and salt deposits; consequently, MP fragmentation occurred [70].

The alteration of the particle size, morphology, density, and color, as well as chemical changes, may be caused by mechanical action.

1.5.4 Biodegradation

Besides the physical and chemical agents involved in aging, microorganisms can also promote an important degradation process of MPs [90]. Microorganisms have evolved enzymes capable of degrading and assimilating carbon sources [91]. The presence of these microorganisms interferes with the prediction of lifespan and coexistence in the environment of plastics because these microorganisms can adapt to different environments and may provide hope for a naturally enhanced removal of MPs in ecosystems [91]. The colonization of MPs by microorganisms is the beginning of biodegradation and biotransformation of MP, leading to the formation of films on their surfaces [80, 90]. Microorganisms must be able to depolymerize the matrix of MPs and mineralize monomeric compounds using enzymes in a metabolic pathway. Environmental conditions such as temperature, pH, humidity, and salinity also play an important role as they directly affect the biodegradation process. Finally, the MPs structure must provide a surface for biofilm formation [80, 91].

Sepulveda *et al*. studied the interaction of thermally aged LDPE (heated at 80 °C for 15 days, dried, and cooled to 0 °C by freezing) with *Aspergillus niger* and *Penicillium pinophilum* for 31 months. Cultures were prepared with and without ethanol as a co-substrate. Although both fungi promoted the biodegradation of LDPE, *P. pinophilum* showed a higher capacity (when ethanol was used), demonstrating that fungi can have a significant impact on the biodegradation process of MPs [92].

Chatterjee *et al*. exposed LDPE to the BP/SU1 strain of *Staphylococcus epidermis* for 3 months and significant degradation of LDPE was observed. SEM images revealed that the organism was still present after 3 months of contact, with evidence of a gradual breakdown of the polymers [93].

Tourova *et al*. investigated the biofilms of bacteria (*Erythrobacter*, *Maribacter*, *Mycobacterium*, *Pseudomonas*, and *Arenimonas*) formed on PS samples, incubated in seawater and industrial wasters for 60 days at a temperature range between 26.0 and 27.8 °C. Despite the different rates of degradation and types of bacteria present in marine and industrial waters, a significant degree of degradation has been observed through SEM analysis [93].

Over the years, researchers have explored the use of enzymes to promote the degradation of MP, and according to the review presented by Othman *et al*.: PE can be degraded by laccase, alkane hydrolase, manganese peroxidase and soybean peroxidase; PET degradation is associated to PETase and MHET; PS is degraded by a series of enzymes, such as monooxygenase, styrene oxide isomerase, phenylacet-aldehyde dehydrogenase and phenylacetyl coenzyme A ligase; for PVC degradation, there are only reports indicating the use of fungi [94].

1.6 Conclusion

The study of MPs has proven to be an area of great research interest, with multidisciplinary teams attempting to solve complex problems. In recent years, because of their classification as emerging contaminants, researchers have focused on studying their identification, quantification, behavior of MPs in different ecosystems, and potential interactions with other contaminants. Although the researchers are trying to understand the behavior of MPs, they face the problem that there are no standard procedures for sample preparation and analysis of MPs.

In the environment, MPs undergo constant changes due to weathering effects such as radiation, mechanical stress, and biodegradation, resulting in aged polymers. This aged state leads to significant physical and chemical changes in MPs, significantly increasing their toxicity in living organisms.

In this chapter, the difference between pristine and aged MPs due to natural effects or laboratory simulations is investigated, highlighting the chemical and physical changes such as color, morphology, particle size, SSA, hydrophobicity, crystallinity, melting and glass transition temperature, surface groups, CI index, O/C ratio, and adsorption capacity, showing the different pathways that can lead to an aged state. Aged MPs are usually opaque and yellow because of photooxidation. Their surfaces exhibited significant morphological changes, such as cracks and flakes, an increase in roughness, and a decrease in particle size due to fragmentation. Consequently, this fragmentation led to an increase in the SSA. Crystallinity changed as a consequence of aging. The appearance of new surface groups, such as −HO, C=O, COOH, C=C, and C−Cl, promoted by the aging process increased the CI index and C/O ratio. In general, these changes result in an increase in

adsorption capacity and toxicity when they interact with other pollutants in the environment.

Currently, researchers are still focused on the characterization of MPs using standard optical techniques, such as fluorescence and SEM, to observe the color and morphologies of MP, and FTIR and Raman spectroscopy to study the chemical groups present. However, emerging techniques should be explored, such as the 2D-COS and EEM-PARAFAC, which allow for the detection of products released during the aging process.

A commitment to decrease the use of plastics is key to reducing the presence of MPs in the environment. Governments should be aware of this problem and must keep trying to implement policies to reduce the use of plastics and prevent their dissemination in ecosystems. In addition, solutions for their removal must be explored; for example, the use of AOPs or biological treatments has shown promising results.

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2
Life Cycle Assessment (LCA) of Bioplastics

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Abstract

The use of plastics is intrinsically linked to daily activities; however, their indiscriminate use has negative consequences on the environment, such as depletion of fossil resources and accumulation of waste. In this context, it is necessary to create more sustainable alternatives to replace them. Therefore, new biobased materials have been developed, which can be bio-based, biodegradable, or both. However, it is necessary to evaluate whether these replacements decrease the environmental impacts compared to plastics of fossil origin. Life cycle analysis is a useful tool for assessing this behavior. Therefore, this chapter focuses on the systematic study of 53 scientific articles in which the environmental impact of bioplastics against petrochemical plastics is compared, generally addressing the product's entire life cycle. These studies differ in scope, functional units, and environmental impacts; however, the vast majority converge that new bio-based systems have environmental advantages. In addition to its impact on the environment, it is necessary to evaluate the technical, social, and economic aspects that make it possible to establish the global sustainability of bioplastics.

Keywords: LCA, environmental impacts, plastics, biobased, biodegradable

2.1 Introduction

In this chapter, we discuss about bioplastics, which can be biobased, biodegradable, or both. It is essential to remember that there are differences between both terms; the raw material of the biobased materials is renewable, such as the biomass of sugarcane plants, starch, corn, cassava, wheat, etc., or their respective residues. However, they may or may not be biodegradable depending on the process and other materials used to transform the product. In the other hand, the term biodegradable refers to the chemical transformation or decomposition of material into water, carbon dioxide and compost due to the action of microorganisms present in the environment; it also depends on environmental conditions such as temperature or geographical area, substrates and application. Therefore, biodegradation depends on the chemical structure of the material and can be carried out in biobased or plastics of fossil origin.

Plastics are among the main materials used in daily life. However, their usefulness comes with an environmental cost due to the use of fossil fuels for their production and their effects on the environment when they become waste [1]. The presence of plastic waste in waste management systems and natural environments has increased in recent decades because of the increased production of single-use products and the lack of end-of-life alternatives [2]. Mechanical recycling of thermoplastics and novel options such as chemolysis, pyrolysis, fluid catalytic cracking, hydrogen techniques, and gasification have been proposed for the valorization of plastic waste [3]. However, their efficiency is limited by lower collection rates and proper segregation. It is predicted that in a commercial

“business as usual” scenario, the production of plastics and the discharge of plastic waste into the environment will increase continuously [4].

Bioplastics have emerged as alternatives to decrease the environmental impact of plastics. Two pathways have been explored: the production of plastics from renewable resources (i.e., biobased materials) and the production of plastics that can be biodegraded under controlled conditions (mainly composting) when they become waste. Hence, bioplastics include biobased and biodegradable materials, as well as materials that are both biobased and biodegradable [5].

Although these new materials have gained attention and have been promoted as valid alternatives for the control of plastic pollution [6], it is necessary to assess their environmental impact through solid and reliable methodologies, which allow identification, on a case-by-case basis, if they are a better option than conventional alternatives. This can be achieved using life cycle analysis (LCA), which has been widely applied to assess bioplastics [7–11].

It is not possible to claim that all biodegradable materials are biobased or that all biobased materials are biodegradable. For example, there are biobased and biodegradable products such as polylactic acid (PLA), polyhydroxyalkanoates (PHA), polybutylene succinate (PBS), and starch mixture, biobased non-biodegradable: biopolyethylene (Bio-PE), bio polyethylene terephthalate (Bio-PET), bio polytrimethylene terephthalate (Bio-PTT), and Bio polyamide (PA) [5].

2.1.1 Life Cycle Assessment

The Life Cycle Analysis (LCA) is a sustainability tool used to quantify the aspects and potential impacts of a product, service, or activity throughout its life cycle, that is, from the origin or extraction of the raw material, the manufacturing, transportation and distribution, use, reuse and maintenance, recycling (if applicable) and until the end of use, final disposal, or waste management [12]. LCA methodology was established according to following international standards:

> ISO 14040: 2006 Life cycle analysis—Principles and framework: This international standard describes the principles and general framework for conducting a life cycle analysis (LCA) study, explaining the four phases and their relationships [13].
>
> ISO 14044: 2006 Life cycle analysis—Requirements and guidelines: This international standard describes the requirements and procedures for conducting a life cycle analysis study and life cycle inventory analysis, establishing critical review guidelines and study limits [13].

The four stages of LCA (Figure 2.1) are described below.

I. Definition of Objectives and Scope. In this stage, the objectives, applications, and reasons that motivate the study are defined, as well as the functional unit that indicates the quantifiable and measurable performance of a product, service, or process [12]. The scope limit of the system was also determined according to the established objective.

II. Life Cycle Inventory Analysis (LCI). In this stage, information gathering and the quantification of inputs and outputs for a product, service, or activity are carried out throughout its life cycle [12, 14]. There are two types of data sources: primary, which can be a direct reading of measurement equipment, invoices, samples, and logs, and secondary, which can be taken from bibliographies, databases, or expert consultations. Once the data were obtained, the validation and calculations necessary for the relationship with the functional unit were performed.

III. Life Cycle Impact Assessment (LCIA). At this stage, the magnitude and significance of the potential environmental impacts of a product, service, or activity throughout its life cycle were quantitatively assessed by associating inventory data with impact categories. The impact categories are environmental problems, such as global warming, acidification, eutrophication, water scarcity, ozone layer depletion, and toxicity. This requires a quantitative representation called a category indicator, such as the environmental impact category. For example, the indicator for global warming is equivalent to kilograms of carbon dioxide (kg CO2eq). The indicators are obtained using characterization factors that emerge from the characterization model [15] and are applied to convert the LCI analysis result multiplied by these factors [16].

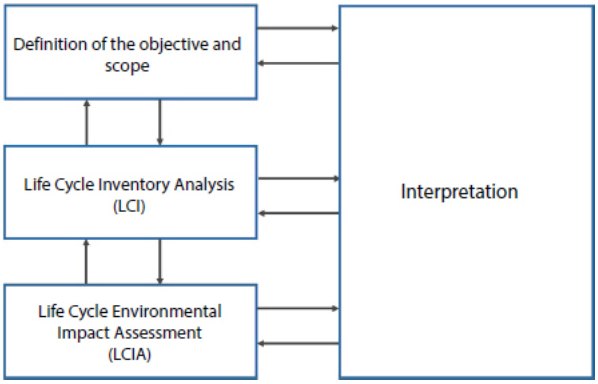


Figure 2.1 Stages of the life cycle analysis.

Source: self-made.

Different impact assessment methods have been proposed. Some focus on analyzing the final effect of environmental impact called “endpoint” and those that examine intermediate effects “midpoint.” The “endpoint” categories that are related to the final impact on the natural environment, human health, or the resources that an environmental issue of interest identifies, while the “midpoint” category evaluates the environmental impact at intermediate points of the environmental processes, based on the emission of a specific substance and its contribution to one particular environmental problem [15]. Acidification, global warming (such as depletion of the ozone layer), eutrophication, acidification, ecotoxicity, photochemical smog formation, water shortage, and a lack of non-renewable resources are examples of midpoint impacts.

There are a wide variety of mathematical models that can be used to evaluate environmental impacts. Some, such as ReCiPe (Pré Consultants), assess medium and final impacts, while CML2001 (Center of Environmental Science), IPCC (Intergovernmental Panel on Climate Change), TRACI (EPA US), IMPACT 2002+ (Swiss Federal Institute of Technology in Lausanne), and EDIP (Environmental Design of Industrial Products), among others, assess medium impacts [17].

IV. Interpretation of the Results. At this stage, the findings of inventory analysis, impact assessment, or both, are evaluated with respect to the defined objective and scope to reach conclusions and recommendations [12]. This helps decision making to determine which phase of the life cycle has the most significant environmental impacts.

2.1.2 Specialized LCA Software

There is a wide variety of software based on the life cycle analysis methodology to facilitate the calculation of environmental impacts. These programs include calculation models and databases, which vary in length and data quality. This software has interfaces

that allow the introduction of data for the preparation of the inventory and the subsequent calculation of the indicators using characterization factors to obtain the different categories of impact chosen and include the analysis of sensitivity and uncertainty. Examples of software are SimaPro (Pré Consultants), GaBi (IKO), Umberto (Ifeu-Institut), Open LCA (GreenDelta GmbH), CMLCA 4.2 (Leiden University, Institute of Environmental Sciences) [18, 19].

2.2 Purpose and Approach of this Chapter

This chapter reviews scientific publications on environmental assessments of the life cycles of bioplastic products. This review contributes to the understanding of the current state of bioplastics in comparison with plastics of fossil origin, as well as establishing points of advantages and disadvantages of the different materials through the analysis of the life cycle evaluations found in the literature.

This study adopted a Systematic Literature Review (SLR) as a method to understand the convergence of relevant topics in an investigation. SLR allows the mapping and evaluation of a specific intellectual structure to develop knowledge content [20, 21]. It is a methodological procedure that uses literature as its primary data source. SRL has a more qualitative bias, which allows it to be differentiated from bibliometric studies by applying procedures for content analysis in which the research corpus is categorized based on reflective and critical reading. The realization of this SRL followed the procedure prescribed by Pollock and Berge [20], which follows six phases: (i) clarify the goals and objectives of the research, (ii) seek relevant research, (iii) collect data, (iv) evaluate the quality of the studies, (v) synthesize evidence, and (vi) interpret the findings. The phases and activities were intended to ensure rigor and soundness in this type of investigation.

Once the research goal was defined, the Web of Science, Science Direct Freedom Collection 2020, and Scopus databases were used to search for relevant research and collect data. The bases adopted are justified because they incorporate leading journals, authors, and content from engineering, environment, and life cycle analysis, among others relevant to the field of study of this research. The main chain used to carry out the searches was ((LCA OR Life analysis cycle OR Life Analysis) AND (bioplastics)), as secondary chains were used ((LCA OR Life analysis cycle OR Life Analysis) AND (Biobased OR non-biodegradable)); ((LCA OR Life analysis cycle OR Life Analysis) AND (Biobased OR biodegradable)) and ((LCA OR Life analysis cycle OR Life Analysis) AND (biocomposites)), for each of the chains, tertiary chains were applied as shown in Figure 2.2. Boolean operators will be applied, considering the synergy of the study research areas. It is noteworthy that no temporal filter was applied to map all the productions on the topic studied. Applying the above search string returned results that were subject to analysis and selection.

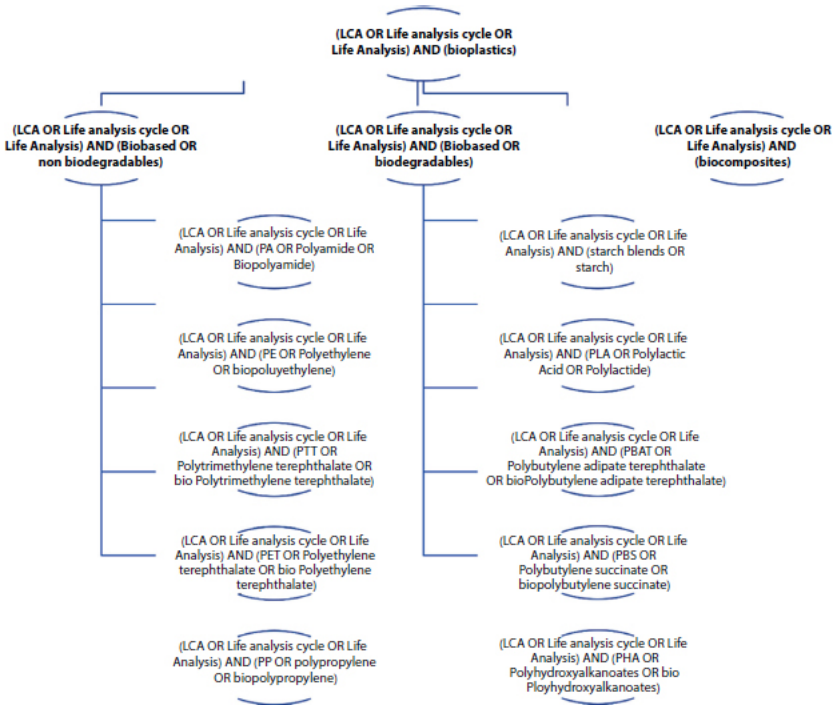


Figure 2.2 Hierarchy of information search.

The eligibility criteria were applied to evaluate the quality of the information, and inclusion and exclusion criteria were applied for evaluation of the database. The filter applied to the databases included only articles published in academic journals, excluding database articles presented at conferences, books, and others. The inclusion criteria applied were: (i) address the LCA of degradable bioplastics (starch, polylactic acid, PLA; polybutylene adipate terephthalate, PBAT; polybutylene succinate, PBS; polyhydroxyalkanoates, PHA); (ii) LCA of non-degradable bioplastics (polyamide, PA; polyethylene, PE; polytrimethylene terephthalate, PTT; polyethylene terephthalate, PET; polypropylene, PP); and (iii) LCA of biocomposites. This choice of bioplastics corresponded to the production projection estimated for 2020 according to the Bioplastics Market Development Update 2020 report [22]. With the application of these criteria, some articles referring to production processes or the synthesis of some bioplastics were removed from the database.

Finally, the corpus of analysis was constituted to compose the base sample of articles. Once the database was created with the help of electronic spreadsheets in Microsoft Excel software, the selected articles were processed. This software allowed data analysis to present results from the combination of qualitative information through categorization of the content of the articles and quantitative information through frequency analysis. The research phase also presented the study performed a relevant descriptive analysis.

An in-depth reading analysis of the corpus of articles was the final phase of this research. What made it possible to categorize the content was the careful reading of the articles, which made it possible to group, find, and compare the categories. It is possible to find online the prescriptions of the last phase of this research, Pollock and Berge [20], the same ones that were applied in this last phase, in phases (v) they synthesized the evidence, and (vi) interpreted the findings. In this study, the qualitative analysis of the articles was prioritized to constitute a matrix that represented the findings of this study as well as the application of some quantitative treatments.

2.3 Development of Life Cycle Assessments for Bioplastics

A total of 53 research articles published from 2011 to 2021 were reviewed and analyzed in detail. Their functional units, scope of the studies, impact categories, types of materials studied, and primary uses were analyzed.

2.3.1 Functional Unit and Scope Definition

The functional unit (FU) is the first step in life cycle analysis. Each study establishes it according to the product or service being evaluated so that different units can be observed (Table 2.1). The study can select it based on the mass of a material, volume of a container, number of pieces, area, or product density. Therefore, this is not universal but follows the objectives of each study.

Table 2.1 Functional unit and scope.

Reference	Functional unit	Scope				
		Cradle to grave	Cradle to gate	Gate to gate	Gate to grave	Post-consumer waste management
[29]	8.14 kg of PLA and Mater-Bi, respectively		•			
[30]	1 kg of PEF	•				
[31]	Established according to the application	•				
[32]	1 kg of LDPE polymer	•				
[33]	1 kg bioplastic resin 1 kg bioplastic product	•	•			
[27]	1 kg of polymer granulates amorphous grade 1 kg of staple fiber 1 kg of bottles		•			•
[34]	1 kg of PHB produced		•			
[35]	1 m^2 of packaging film	•				
[9]	1 kg plastic resin	•				
[36]	1 kg of PE polymer	•				
[28]	1 m2 of packaging film		•			•
[37]	10,000 boxes with dimensions of 8.0 × 10.0 × 2.5 cm of PLA, PS and PLA/starch		•			
[25]	1 cup used to contain 180 mL of coffee to be used for 1 time/day for 2 years.	•				
[38]	1 m^2 of packaging film	•				
[39]	1 kg of output		•			
[40]	1,000 units of 250-ml drinking water bottles.	•	•			
[41]	1 disposable cup of 180 mL for hot drinks from vending machines	•				
[42]	500 mL clam shell	•				
[24]	1 disposable cup of 180 mL for hot drinks from vending machines	•				
[43]	100 port-hole spacers	•				
[23]	1 kg of dry packaging material	•				
[44]	1 kg of HDPE or PET polymer		•			
[45]	1 kg In-geo biopolymer		•			

Reference	Functional unit	Scope				
		Cradle to grave	Cradle to gate	Gate to gate	Gate to grave	Post-consumer waste management
[46]	1 t of HDPE	•				
[47]	1 kg of PET bottles		•			
[26]	1 car door panel (long lifespan) 1 unit of packaging (short lifespan)	•				
[48]	1,000 boxes with capacity of 100 g per box and the total volume is 2.00 x $10^{-4}m^3$	•				
[49]	NA		•			
[50]	1.0 ton of cassava starch-based biofilm		•			
[51]	1 kg of polymer	•				
[52]	1 kg expanded-PLA tray					
[53]	Providing customers with 100,000 kg of fresh fruits for one year	•				
[54]	1 bag of 10 L capacity	•	•			
[55]	All medical devices that contained petroleum-based plastics suitable for biopolymer substitution foreach of the four types of hysterectomies	•				
[56]	Box and lid characteristics according to the material	•				
[57]	1 film with characteristics of 0.06 mm thickness and dimensions of 300 x 250 mm, to manufacture packaging bags	•				
[7]	1 kg of packaged product at the house				•	
[58]	The supply of 1,000 meals using 1,000 single use tableware	•				
[59]	100 g of plastic basin per single use	•				
[60]	100,000 kg and different sizes of pieces; 0.01, 0.1, and 1 kg	•				
[61]	12 bottles	•				
[62]	1 kg of product	•				

Reference	Functional unit	Scope				
		Cradle to grave	Cradle to gate	Gate to gate	Gate to grave	Post-consumer waste management
[63]	1 kg of PET	•				
[64]	125 g (9.5 × 14.5 × 2.5 cm) flow pack	•				
[65]	To facilitate transportation to an average household for a year, of 8.56 kg/19.01 L	•				
[66]	1 kg of PLC produced at refinery gate		•			
[67]	Single-use containers to hold 10 mL of olive oil	•				
[68]	Tray to preserve 500 g of fresh meat in an approximate volume of 1 L		•			
[69]	1 kg of waste plastics that ends up in the landfill or composting		•			
[70]	1 packaging unit intended to contain a 130 g serving of freshly cut lettuce	•				
[71]	1 kg of alternative resins in 100 years of service life according to geometry, 61 cm (24-inch) diameter and 600 cm (20-foot) length	•		•		
	Sum	31	12	1	1	2
	Percentage	60.8	23.5	2.0	2.0	3.9

HDPE, high density polyethylene; LDPE, low density polyethylene; PEF, polyethylene furandicarboxylate; PET, polyethylene terephthalate; PHB, polyhydroxybutyrate; PLA, polylactic acid; PS, polystyrene. The yellow rows indicate the studies that have two different scopes.

However, approximately 55% of the reviewed studies (Table 2.1) established their UF mass, either in kg, g, or tons. The second most used was the number of pieces (27.5%); for example, Rossi *et al*. [23] established their UF as 100 port-hole spacers. It is possible that the measurement units are combined, as in the case of Potting and van der Harst [24], who perform their evaluation with the UF of a 180 mL disposable glass to serve hot drinks in vending machines, such as what Taengwathananukool and collaborators [25], who indicate that its UF is a cup containing 180 mL of coffee to be used once a day for two years. In the last case, as it is a reusable product, the number of times it is used in a period is indicated. For this study, a few pieces and volumes were used as functional units. It is possible not to limit oneself to just a functional unit or a product, as done by Hermansson *et al*. [26], who selected two in their evaluation: one car door panel to evaluate a long-life product and one packaging unit of short useful life.

LCA has different scopes. Two of these studies did not report system limits. That is, not all of them must start and end in the same stage of the process; as can be seen in Table 2.1, the one most used (60.8%) is cradle-to-grave. Cradle-to-grave means that the evaluation is carried out from the extraction of the raw material (s) to waste management at the end of

the product's useful life. The second most commonly used scope is cradle-to-gate, and approximately 23.5% of the studies used it, where the study on the extraction of raw materials began and concluded at the door of the production site, which may or may not contemplate the transport to the user who will make use of it; however, when this happens, it must be indicated. It is possible to select only one stage of the process or service, as reported by Shen *et al*. [27] and Leceta *et al*. [28], who carried out their evaluation exclusively in the waste management post-consumer stage. Similar to the functional unit, the definition of scope is selected according to the needs of the study or data that can be carried out are available. Four studies reported two different reaches (7.8%), gate to gate and gate to grave represented 2% each, and post-consumer waste management represented 3.9%.

2.3.2 Conventional Plastics vs. Bioplastics Analyses

With LCA, it is possible to evaluate a product or service and compare two or more options. One recurrent goal is to use this tool to assess the same product with different raw materials, as shown in Table 2.2. It was shown that 62.7% of the study cases compared bioplastics based on polylactic acid (PLA) with conventional plastics or those of fossil origin, such as polyethylene (PE), polystyrene (PS), and polyethylene terephthalate (PET). Examples of this are Tsiropoulos *et al*. (2015b), who compared four materials (two bioplastics and two plastics of fossil origin), and Lorite *et al*. [53], who compared a bioplastic (PLA) against a petrochemical plastic (PET).

2.3.3 Primary Applications for Which LCA was Performed

Bioplastics are generally used to replace conventional plastics in the same application. Thus, there is no industry limit in which a possible substitution can be seen. Still, as observed in Table 2.3, flexible materials are used in a higher proportion for food packaging (35.3%), while only 15.7% of rigid materials are also applied in the same sector. Therefore, there are other target industries such as agriculture, packaging for freight transport, pipes for sewage, and medical applications.

Table 2.2 Plastics and bioplastics used in each case study.

Reference	Biobased plastics							Fossil-based pl				
	PLA	PHA PHB	BioPE	BioPET	Paper (cellulose)	Starch	Other	PE	PS	PET	PP	
[29]	•					•		•		•		
[30]							PEF	•		•		
[31]						•		•	•			
[32]		•						•				
[33]	•						PBS	•				
[27]	•		•							•		•
[34]		•										
[35]	•						Wheat	•				
[9]	•	•				•		•	•	•		
[36]			•					•				
[28]							Chitosan				•	
[37]	•					•			•			
[25]	•											
[38]							Chitosan Soja Agar				•	
[39]	•					•					•	
[40]	•									•		
[41]	•											
[42]	•					•						
[24]	•				•				•			
[43]						•			•			
[23]	•								•			
[44]			•	•				•		•		
[45]	•											
[46]			•					•				
[47]				•						•		
[26]	•											
[48]	•								•	•		
[49]	•	•					Bioethylene					
[50]						•						
[51]	•		•	•				•	•	•		
[52]	•								•			
[53]	•									•		
[54]			•				PBAT-starch blends	•				
[55]	•						Guayule derived latex	•	•		•	

Reference	Biobased plastics							Fossil-based pl				
	PLA	PHA PHB	BioPE	BioPET	Paper (cellulose)	Starch	Other	PE	PS	PET	PP	
[56]	•	•					PBS				•	
[57]	•							•				
[7]		•									•	
[58]	•								•			
[59]							Sugar cane	•				
[60]	•						Flax hemp Rice husk Wood					
[61]	•						Recycled PET Environmental Solution			•		
[62]							PBST PBSA PBAT	•		•	•	
[63]	•			•						•		
[64]	•							•		•		
[65]							Recycled paper Mater-Bi®	•			•	
[66]							PLC		•			
[67]	•		•					•		•		
[68]	•								•	•	•	
[69]	•		•					•				
[70]	•						Bactericidal ZnO Nanoparticles				•	
[71]			•				Pristine nanoclays	•				

Cell, cellulose; PBAT, Poly butylene adipate-co-terephthalate; PBSA, poly butylene succinate-co-butylene adipate; PBS, polybutylene succinate; PE, polyethylene; PEF, polyethylene furandicarboxylate; PET, polyethylene terephthalate; PHA, polyhydroxyalkanoates; PHB, polyhydroxybutyrate; PLA, polylactic acid; PLC, poly limonene carbonate; PP, polypropylene; PS, polystyrene; PSBT, polybutylene succinate terephthalate.

Table 2.3 Applications of plastics and bioplastics assessed in life cycle analyses.

Reference	Application					
	Food packaging (flexible)	**Food packaging (rigid)**	**Agricultural and livestock**	**Packaging**	**Another**	**Not described**
[29]		•				
[30]			•			
[31]					Publicity Transport Ovenproof	
[32]						•
[33]					Garbage bag	
[27]		•	•		Injection moulding Textiles and nonwovens	
[34]					Biogas	
[35]	•					
[9]			•			
[36]				•		
[28]	•					
[37]		•				
[25]		•				
[38]				•		
[39]				•		
[40]	•					
[41]	•					
[42]						
[24]	•	•				
[43]				•		
[23]	•					
[44]						•
[45]						•
[46]						•
[47]		•				
[26]				•	Packaging Motoring	
[48]				•		
[49]						•
[50]	•					
[51]	•	•				
[52]	•					

Reference	Application					
	Food packaging (flexible)	**Food packaging (rigid)**	**Agricultural and livestock**	**Packaging**	**Another**	**Not described**
[53]	•					
[54]					Garbage bag	
[55]					Medical products	
[56]	•					
[57]	•					
[7]	•					
[58]	•					
[59]					Medical products	
[60]					Filling material	
[61]		•				
[62]	•		•			
[63]						
[64]	•					
[65]					Carrier bags Container of garbage	
[66]						•
[67]				•		
[68]	•					
[69]						•
[70]	•					
[71]					Road drainage pipe.	
Sum	18	8	4	6	12	7
Percentage	35.3	15.7	7.8	11.8	23.5	13.7

2.3.4 Evaluation Methods and Impact Categories Analyzed

One of the objectives of LCA is to process the data and transform it into information that allows analysis and decision-making decisions to achieve environmental sustainability and to determine the potential environmental impacts of a process, service, or product. In a study system, there is variability in methodologies that relate to the environmental aspects associated with the procedures. The studies analyzed in this review used various impact methods (Table 2.4). The most commonly used methods are the IPCC, to calculate the impacts of climate change, its own calculations, and CML. It is curious that ReCiPe, the method with the most impact categories evaluated, was moderately used in the studies presented. It is important to mention that the different evaluation methods may

show differences between the characterization factors used to represent the results in equivalent units of each substance, and the type of substances included in each impact category, and the quantity. Therefore, when comparing the tudy results, it is important to clearly mention the methodologies used for each environmental impact evaluated. However, the evaluation method selected was not mentioned in five of the studies [36, 56, 65, 68, 72].

The reduction in greenhouse gas emissions is one of the categories expected to be favored by replacing materials of fossil origin with bioplastics. The review shows that all studies considered this category in their impact evaluations. Of these studies, at least 12 analyzed climate change as a potential impact.

The international standards LCA ISO 14040 and ISO 14044 provide general guidelines. Instead of defining the methodology to be adopted, the studies differed in a variety of standard and natural impact categories, ranging from one or two to all. The ReCiPe Midpoint methodology evaluates 18 impact categories that contain factors with perspectives that represent a set of options on issues such as time or expectations that proper management or future technological development can prevent future damage, for which it was used as a basis to identify the categories most considered in the studies (Table 2.4). In the impact categories considered, there was a large bias towards global warming potential (GWP, 100% of studies), highlighting an uneven distribution and use of non-renewable energy (NREU) (although the latter was not shown here as not part of the ReCiPe categories), followed by terrestrial acidification potential (TA), freshwater eutrophication (FE), marine eutrophication (ME), and terrestrial ecotoxicity (TE) of 8% each. Stratospheric ozone depletion (SOD) was evaluated in 7% of the studies, while land use (LU) was assessed in 6% and the rest were lower than this percentage (Figure 2.3).

2.3.5 End of Life (EoL) Scenarios

Choosing the end-of-life treatment of a plastic or bioplastic material and making the decisions that surround the modeling of the treatment processes varies widely among different studies. Among these are the possible obtainments of co-products during the treatment (energy recovery). This assignment may affect the environmental load assigned to the evaluated products [66, 73, 74].

Simplifying the end-of-life process in a life cycle analysis can lead to erroneous or biased results because the system will not be fully represented. In the same way, contemplating perfectly closed waste management can lead to bias in the results (example, send 100% of waste to recycling or landfill) [73]. Treatment options for bioplastics and fossil-based plastics are generally different. Some bioplastics can enter the conventional waste streams. However, in the case of recycling, they will not be able to enter the same stream as petrochemicals. However, petrochemical plastics cannot easily enter the processes that are often chosen for bioplastics, such as anaerobic digestion or composting [29, 35, 38, 41]. To ensure an adequate comparison at the end of life, realistic and perfectly differentiated scenarios were considered. Table 2.5 presents a summary of the end-of-life scenarios for each of the analyzed studies, including contemplating anaerobic digestion, composting, incineration, landfill, and a column of combinations in which the analysis indicates different proportions of materials going to each of the treatments (for example, 50% incineration and 50% recycling).

Table 2.4 Impact methods and impact categories selected per study.

Reference	Impact method	GWP	SOD	IR	OFHH	FPMF	OFTE	TA	FE	ME	TE	EW	EM	HCT	I
[29]	IPCC	•													
[30]	Calculations of GHG	•													
[31]	CML	•	•	•	•			•	•	•	•	•	•	•	•
[32]	Calculations	•	•					•			•				
[33]	Eco-Indicator 95 and CML 2	•													
[27]	NA	•													
[34]	TRACI	•	•					•	•	•	•			•	•
[35]	ReCiPe	•	•	•	•	•	•	•	•	•	•	•	•	•	•
[9]	TRACI	•													
[36]	NA	•													
[28]	Eco-Indicator 99	•	•					•	•	•	•			•	•
[37]	IPCC and other calculations	•						•	•						
[25]	Calculations	•	•						•	•	•				
[38]	Eco-Indicator 99	•	•					•	•	•	•			•	•
[39]	TRACI	•	•					•	•	•	•			•	•
[40]	CML	•						•			•			•	
[41]	CML and CED	•	•	•				•	•	•	•	•	•		
[42]	Impact 2000+ and ReCiPe	•	•	•	•	•	•	•	•	•	•	•	•	•	•
[24]	CML and CED	•													
•[43]	Various	•	•	•				•							
[23]	Impact 2000+	•	•	•	•		•	•	•	•	•				
[44]	IPCC and Impact 200×	•									•				
[45]	CML	•													
[46]	ILCD and ReCiPe	•	•	•	•	•			•	•	•				
[47]	TRACI and ReCiPe	•	•	•	•				•	•					
[26]	CML	•					•	•	•	•					

Reference	Impact method	GWP	SOD	IR	OFHH	FPMF	OFTE	TA	FE	ME	TE	EW	EM	HCT	
[48]	IPCC	•													
[49]	IPCC	•													
[50]	ReCiPe and CED	•	•	•					•	•	•	•			
[51]	TRACI v2.1	•	•	•			•	•	•	•				•	•
[52]	Impact 2000+	•		•							•				
[53]	Impact 2000+	•		•	•		•		•	•	•	•	•	•	•
[54]	Eco-Indicator 99	•				•		•					•		
[55]	TRACI	•	•	•				•			•			•	•
[56]	NA	•	•	•	•	•	•	•	•	•	•	•	•	•	•
[57]	IPCC	•													
[7]	IPCC	•													
[58]	Various	•	•	•	•	•			•	•	•	•	•	•	•
[59]	ReCiPe	•	•		•			•		•					
[60]	Calculations	•						•	•	•	•	•		•	•
[61]	Impact 2000+	•	•	•		•		•	•	•				•	•
[72]	CML	•						•	•	•	•				
[62]	IPCC	•													
[63]	GREET model	•													
[64]	IPCC	•													
[65]	NA	•	•					•	•	•					
[66]	ReCiPe	•	•	•	•	•	•	•	•	•	•	•	•	•	•
[67]	ILCD	•		•					•					•	•
[68]	NA	•	•	•	•	•		•	•	•	•				
[69]	GREET model	•													
[70]	ReCiPe	•	•	•	•	•	•	•	•	•	•	•	•	•	•
[71]	IPCC	•													
	SUM	52	25	20	13	10	9	26	27	26	25	11	10	18	1
	Percentage	100	48	38	25	19	17	50	52	50	48	21	19	35	3

GWP, global warming potential; SOD, stratospheric ozone depletion; IR, ionizing radiation; OFHH: ozone formation human health; FPMF, fine particulate matter formation; OFTE, ozone formation Terrestrial ecosystems; TA, terrestrial acidification; FE, freshwater eutrophication; ME, marine eutrophication; TE, terrestrial ecotoxicity; EW, freshwater ecotoxicity; EM, marine ecotoxicity; HCT, human carcinogenic toxicity; HNCT, human non-carcinogenic toxicity; LU, land use; MRS, mineral resource scarcity; FRS, fossil resource scarcity; WC, water consumption; IPCC, Intergovernmental Panel on Climate Change; TRACI, Tool for the Reduction and Assessment of Chemical and Other Environmental Impacts; GREET, Greenhouse Gases, Regulated Emissions, and Energy Use in Technologies; NA, not available; ILCD, International Reference Life Cycle Data System; CED, cumulated energy demand; GHG, greenhouse gas.

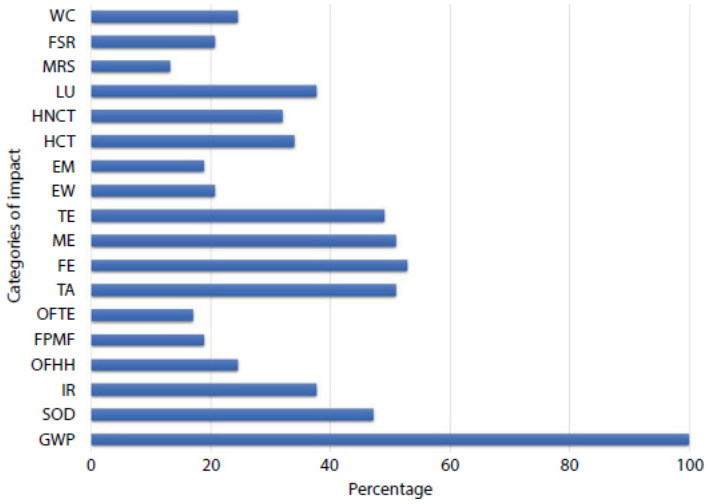


Figure 2.3 Percentage of studies that assessed each impact category. GWP, global warming potential; SOD, stratospheric ozone depletion; IR, ionizing radiation; OFHH, ozone formation human health; FPMF, fine particulate matter formation; OFTE, ozone formation terrestrial ecosystems; TA, terrestrial acidification; FE, freshwater eutrophication; ME, marine eutrophication; TE, terrestrial ecotoxicity; EW, freshwater ecotoxicity; EM, marine ecotoxicity; HCT, human carcinogenic toxicity; HNCT, human non-carcinogenic toxicity; LU, land use; MRS, mineral resource scarcity; FRS, fossil resource scarcity; WC, water consumption.

It can be observed that some studies take several treatment options to make the process more realistic and comparable. Thirty of the studies considered incineration, 21 landfills, 16 composting, 13 included recycling, and 12 anaerobic digestions. In addition, eight of the studies raised scenarios with different percentages for various processes.

The variability of different studies at the end of life makes comparisons difficult. A single waste treatment was explored [32, 36, 61, 66], while others considered hybrid scenarios that better reflect the current flow of plastics, combining approaches, or proposing variable contribution percentages in multiple methods [23, 25, 27, 28, 33, 38, 40–42, 48]. Other studies indicate or do not consider the final phase of the LCA, which may make comparisons even more difficult as Table 2.5 can be seen by gray rows [9, 30, 37, 39, 44, 45, 47, 50, 52].

Table 2.5 End-of-life scenarios.

Reference	Anaerobic digestion	Composting	Incineration	Recycling	Landfill	Combinations
[29]	•[1]	•[1]	•	•		
[30]						
[31]	•	•[2]			•	
[32]			•[3]			
[33]						•[4]
[27]						•[4]
[34]	•	•	•[5]		•	
[35]		•[1]	•[3]			
[9]						
[36]			•			
[28]	•[1]					•[4]
[37]						
[25]			•		•	
[38]	•[1]					•[4]
[39]						
[40]						•[4]
[41]	•[1]	•[1]	•	•[1]		
[42]						•[4]
[24]	•	•	•	•		
[43]						•[4]
[23]	•	•	•	•	•	
[44]						
[45]						
[46]			•	•		
[47]						
[26]	•[1]		•			
[48]						•[4]
[49]			•		•	
[50]						
[51]		•	•	•	•[5]	
[52]						
[53]		•	•		•	
[54]		•	•		•	
[55]					•	
[56]		•	•	•	•	
[57]			•	•	•[5]	
[7]	•	•			•	

Reference	Anaerobic digestion	Composting	Incineration	Recycling	Landfill	Combinations
[58]		•	x[5]		•[5]	
[59]			•			
[60]	•		•	•	•	
[61]					•	
[72]			•			
[62]		•	•			
[63]			•			
[64]			•		•	
[65]			•	•[5]	•	
[66]	•					
[67]			•			
[68]			•	•	•	
[69]		•			•	
[70]		•	•	•	•	
[71]			•	•	•	
Sum	12	16	30	13	21	8

[1]For bioplastic only; [2]Considering two composting systems: home and industrial; [3]Considering incineration with energy recovery; [4]Considering the different scenarios with percentage; and [5]For the petrochemical plastic only. Lines in yellow indicate studies that do not specify end-of-life.

If 100% of a single waste destination does not provide realistic results, it offers the technical potential to compare waste management options. If the objectives of LCA are to compare the environmental impacts of bioplastics with the implications of petrochemical plastics, it is essential to model realistic and appropriate options, in addition to considering the sensitivity of the results to these options [73].

Benefits and drawbacks of the biobased identified through LCA. Many LCA studies of fossil plastics and biobased plastics have focused on the comparison of these materials to determine the convenience of the new materials (biobased), through the assessment of their general environmental behavior, to determine the result of one or more specific environmental impacts, to see the economic viability, or the combination of all. Table 2.6 summarizes the results of each study, mainly associated with the base raw material of the biobased. Some of the identified benefits are as follows:

- In general, biobased products have a better environmental performance than conventional or fossil-based plastics in terms of GHG emissions, global warming, and post-consumer treatment, according to Kikuchi *et al*. [36], Mahalle *et al*. [39] and Leceta *et al*. [28] report, respectively.
- Biobased plastics may help minimize the impact of fossil fuel depletion [42, 47, 51].
- Some biobased products such as PLA, PBS, and starch blends, among others, in waste treatment scenarios, such as composting, recycling, and incineration, have less environmental impact on GHG emissions, global warming, and consumption of raw materials than plastics of fossil origin [35, 58, 67].
- If the raw material is organic waste, the environmental impacts on land use may be lower than plastics of fossil origin or biobased ones, where the raw material is grown exclusively for this purpose [45].

However, there are disadvantages to the production or use of biobased materials, as they cannot become viable when the use or change of soil to cultivate the raw material is considered in the analysis because of the demand for resources and the application of fertilizers.

2.4 Discussion

This chapter reviews the life cycle assessment of the biobased materials. For this purpose, an SRL was carried out to obtain the most relevant articles that could glimpse the corpus of this chapter on biobased LCA.

2.4.1 Evaluation Methods and Impact Categories

If we focus on replacing plastics of fossil origin versus biobased, the different studies have shown that some concerns may arise in phases such as extraction, production, and final disposal. Therefore, holistic evaluations are proposed that consider the widespread impacts and not only those expected, such as the case of global warming or climate change, in which possible mitigations in the value chain can be glimpsed. Most studies have concluded that biobases have a lower impact on global warming. However, in categories such as stratospheric ozone formation, acidification, and eutrophication, biobased plastics present a more significant contribution than plastics of fossil origin because of the harvesting of materials and the use of fertilizers and pesticides. For example, Deng *et al*. [35] determined an increase in four, three, and two times for acidification, photochemical oxidation, and eutrophication, respectively, of the impacts of PLA concerning the PE of fossil origin. Hermansson *et al*. [26] found that PS and PP of fossil origin contributed more significantly to the remaining categories evaluated. The above discussion considers the production of PLA using renewable energy.

Table 2.6 Main results of each case study.

Reference	Main results
[29]	Incineration, composting and anaerobic digestion processes have underperforming environmental than mechanical recycling process. Even so, the results cannot be taken as a commercial reference standard to replace PET or PE in place of PLA or Mater-Bi or back.
[30]	If PET were replaced by PEF, the use of non-renewable energy would decrease by approximately 12 to 16 % and greenhouse gas emissions by 11% to 20%. The evaluation was carried out with data modeling, it is necessary to do it physically. A land use change assessment should be carried out.
[31]	According to these case studies, biobased have a better environmental performance than petrochemical polymers, mainly GHG emissions with lower in the waste screening phase. Biobased materials are in the process of development, so they will probably require improvement.
[32]	With modelling it is possible to evaluate a product that is not on the market against a conventional alternative. The evaluation was carried out with data modeling, it is necessary to do it physically. Wide range of sensitivity can convert positive to negative environmental performance, based on assumptions and methodology.
[33]	Considering the reach from the cradle to the grave and the global warming potential, the environmental performance of biobased is better than that of fossil-based plastics. If the reach from cradle to gate is considered, polylactic acid biobased requires optimization of its process to improve its environmental performance compared to a plastic of fossil origin.
[27]	BioPET has better environmental performance than PET of fossil origin. It is necessary to improve the properties of biobased as well as integrate recycling at the end of their useful life, to improve their environmental performance. The separation for the recycling of biobased is more complex than that of polymers of fossil origin. It is necessary to improve the properties of biobased as well as integrate recycling at the end of their useful life, to improve their environmental performance. The separation for the recycling of biobased is more complex than that of polymers of fossil origin.
[34]	The right mix of waste-to-energy technologies could enable the production of imported energy and carbon-free PHBs. The recovery of PHB for biobased production is hampered by high cost on a large scale due to solvent demand and solvent recovery and environmental concerns by the large amounts of chlorinated solvents that may be present and that they are not suitable for the mass production of biobased.
[35]	Biobased have better environmental performance than fossil-based polymers, as well as incineration with energy recovery at the end of their useful life. During the cultivation and drying of the raw material there is a poor environmental performance compared to PLA and PE.
[9]	In general, biobased may have better environmental performance than polymers of fossil origin.

Reference	Main results
	Inventories need to be updated as production methods change.
[36]	The greenhouse gas emissions during the life cycle of bio-PET were lower than those of fossil PET.
[28]	In general, the environmental behavior of chitosan-based material is better than fossil PP. Chitosan-based plastic requires optimization during its manufacture to improve its environmental performance.
[37]	You can see the areas for improvement. PLA was not proven to have better environmental performance than PS.
[25]	The use of biobased coffee cups can have better environmental performance than melanin cups. Biobased cups require adequate disposal at the end of their useful life.
[38]	Biobased films minimize environmental burdens associated with fossil fuels. To reduce the impacts of chitosan and agar films during their manufacture and in the extraction of raw material in the soy base, it is necessary to redesign the process in these stages.
[39]	The biobased material of PLA has better environmental performance than PP, mainly in the category of global warming impact. The PLA biobased material is a laboratory scale prototype.
[40]	In general, PLA biobased material has better environmental performance than conventional PET bottle. PLA bottles may perform better environmentally if you have proper waste management, including composting, incineration, and recycling.
[41]	A single data set can lead to biased results.
[42]	NA
[24]	Information was generated for further investigations. The data did not allow to indicate which glass is preferable to use.
[43]	The biobased is highly recyclable by composting. It allows to reduce this type of waste in landfills. The environmental behavior of the biobased is better than the PS of fossil origin. It is only a prototype of biobased; it is required to prove it in the industry.
[23]	Results may vary if packaging is heavily contaminated with food residue.
[44]	Biobased have less negative environmental impact, on mental and human health, than materials of petrochemical origin. To produce raw material from biobased, the use of fertilizers and pesticides increase their environmental impacts.
[45]	There is technical and economic feasibility to use agricultural or industrial wastes, wood or specific crops for fermentation in the production of PLA. Further research is required for this biobased.
[46]	By using beetroot and wheat to make the biological PE material, there was a reduction in GHG emissions and fuel consumption. Nine out of 11 environmental impacts of biobased has worse environmental behavior than material of fossil origin.
[47]	Bio-PET had 21% less global warming potential and required 22% less fossil fuels than fossil-based PET.

Reference	Main results
	In bio-PET the extraction stages and the pre-processing of the raw material are more intensive in emissions than the petrochemical material.
[26]	It is possible to carry out the assignment in volume or mass, according to the information available. More research is required regarding the recycling of biobased at the end of their useful life to determine the most favorable impact reduction conditions.
[48]	PLA boxes are more eco-efficient if 100 % of them are composted as they present very low GWP impact value, but if no change in land use is made for growing the biobased. On the balance sheet, if the composting of PLA is not considered at the end of its useful life, the eco-efficiency results would be less favorable than that of materials of fossil origin.
[49]	It is possible that replacing fossil materials with biobased is more favorable to reduce GHG emissions. Including uncertainty in the calculations can be beneficial to have more precision in the results. In biobased materials, the results can be affected in the general balance of GHG emissions if a raw material with a specific crop for it is used than if a residue is used.
[50]	It is possible to redesign the development of the biobased by identifying the processes that have the greatest environmental impacts. The cultivation of the raw material, as well as the manufacture of films and the production of additives, are the processes that presented the greatest impacts.
[51]	Biobased had less impact on global warming and fossil fuel depletion than petrochemical materials. There are no studies that show consequences for the formation and use of biobased, research is lacking.
[52]	In general, the PLA material had better environmental performance than that of fossil origin. The stage of greatest environmental impact of PLA was during the cultivation of the raw material and transport to the factory. Due to the transportation of raw materials, PLA has a greater environmental impact than PS.
[53]	According to the general balance, the PLA container had less impact on the environment and human health than the PET material. PLA containers can extend the shelf life of food. Incineration is more feasible for PET than for this biobased.
[54]	The use of bioPE is highly recommended to produce garbage bags. Further research is required for the life cycle of this biobased and its application.
[55]	Biobased -based materials had less impact on health and energy demand than fossil-based materials. Agricultural inputs related to biobased present greater environmental impacts.
[56]	PBS had the best environmental and economic indicators than PLA, PHA and PP. PLA with 100% mechanical recycling had the highest environmentally and economically indicator.

Reference	Main results
	Composting is the least viable option for this group of materials at the end of their useful life.
[57]	It is possible to mix PLA with other materials. PLA mixed with other organic materials can have a higher global warming potential value than fossil-based materials. More research is required regarding the materials and proportions with which it is feasible to mix PLA.
[7]	Bio-based packaging, lower GHG emissions than fossil-based packaging in a landfill. The LCA does not address all the significant environmental impacts associated with plastics, particularly accumulation in the oceans.
[58]	The use of biodegradable material and recycling at the end of life has the best environmental performance in catering in quick service restaurants. In this case, the nature and use of the different materials have limitations to recover energy at the end of their useful life.
[59]	The biobased had better environmental performance than PE. The use of land to produce the raw material of the biobased and the transport have a less positive environmental behavior than the fossil material.
[60]	To produce biological material fill can have better environmental and economic performance. Further investigation of other plastic compounds and their filler combinations is required.
[61]	Biobased bottles have better environmental behavior than conventional PET ones. Optimization of the use of transport is required, that is, consider more sustainable transport to reduce potential impacts on the environment.
[62]	2nd generation biobased showed better environmental performance than plastics of fossil origin. The management of land use to produce 2nd generation biobased shows a special environmental impact. 1st generation biobased require further optimization research, since they do not show much difference in environmental behavior than plastics of fossil origin.
[63]	The use of raw materials for the manufacture of biobased can reduce GHG emissions. The use of transport and land to produce the biobased presents the highest GHG emissions. More data will be required to generate more information and to be more accurate.
[64]	Biobased have less environmental impact than materials of fossil origin. Consumers tend to choose more fossil fuel containers than biobased.
[65]	The number of times the bag can be reused primarily as a waste container affects the LCA, therefore, the one with the best environmental performance is PE, which is a disadvantage for single-use bags such as paper ones. recycling.
[66]	PLC production performed better in mitigating climate change than PS by capturing carbon through copolymerization and fixing biogenic carbon through photosynthesis. The PS outperformed the PLC in the categories of eutrophication impact,

Reference	Main results
	water consumption, and ozone depletion, so life cycle improvements need to be made.
[67]	Biobased packaging has better environmental performance if incineration is considered at the end of the product's life. The agricultural activities generate that the biobased has a worse environmental performance than that of fossil origin in the category of ecosystem quality impact.
[68]	Innovative or biobased designs could help reduce the environmental footprint of meat trays. PLA trays have worse environmental performance than extruded polystyrene.
[69]	Waste management strategies require further research to select the best end-of-life option for biobased. In general, biobased had a better environmental performance than those of fossil origin.
[70]	PLA coated with bactericidal ZnO nanoparticles had better environmental performance and extended the shelf life of lettuces than PP. Increasing expiration periods due to the combination of different changes in the life cycle of freshly cut products is a field that has not yet been investigated much.
[71]	Bio-PE has lower GHG emissions than other materials. The economic cost in the production of 1 kg of bio-PE is higher than virgin fossil PE, mainly attributed to the conversion of biomass.

Some authors have found that biobased plastics are more environmentally favorable in terms of climate change and the depletion of fossil resources; however, plastics of fossil origin present a lower contribution in categories such as acidification, which can even be double that of bioplastics [35, 54].

Many studies have evaluated the impact of bio-PE [26, 32, 36, 44, 49, 54]. Kikuchi *et al*. [36] conclude that PE of fossil origin has up to four times more significant impacts than bio-PE in the emission of greenhouse gases. In contrast, [32] found that PET of fossil origin has a lower impact on acidification and eutrophication than bio-PET. It has also been shown that the place of production can be an essential factor: the production of bio-PET based on sugar cane in Brazil and its subsequent transport to the United States (US) has fewer effects on GHG emissions compared to producing bio-PE from corn starch and fossil-based PE produced in the US, but is higher than the bio-PE based on switch-grass produced in the US [49].

Studies that only included global warming as an impact category may erroneously assume that one material is environmentally preferable over another without considering that this impact can be offset in another impact category.

Another category of impact that can be severely harmed by promoting biobased production is land use. The production of biobased materials requires large amounts of soil for harvesting and evaluating the impacts of this process. The change in land use, for example, from the production of land use for food production concerning the production of materials for biobased is notable, as less than 50 % of the studies analyzed to consider this category, which in some localities could be a decisive factor for the change from fossil plastics to biobased. Considering this criterion, environmental impact analysis can drastically change the conclusions of the LCA [29].

It is essential to establish specific guidelines that can provide a more significant notion of the impact associated with the replacement of materials. Consensus has been reached on

the use of specific categories and methodologies to elaborate LCAs for bioplastics.

2.4.2 End of Life (EoL)

Evaluating the end of the useful life of materials can have many variations, and it is not easy to make a specific decision based on this parameter. Treatment and management options for different materials differ depending on their nature. Assuming a specific scenario in which the flow of plastics and biobased materials is derived will often not provide a realistic vision of the impacts of one type of material. For comparison, it is necessary to address real scenarios for the management and treatment of fossil plastics and biobased materials.

PLA has been found to have a higher impact in more impact categories when incineration is considered [35]. In contrast, [48, 54] concluded that the best EoL alternative for PLA is composting.

It has also been shown that incineration with energy recovery may have lower impacts than anaerobic digestion and composting, making the question of biodegradability of bioplastics debatable [23].

There is no rigor in the end-of-life scenario adopted for the different studies, which favors a lack of realistic comparisons and increases the bias in the comparison [41].

2.5 Concluding Remarks

Overall, the life cycle analyses consulted throughout the chapter helped investigate new options for replacing one or more fossil-based plastics with biobased plastics. The main steps involved in carrying out the LCA methodology were analyzed. The functional unit, scope of the studies, impact categories, types of materials studied, and their primary uses stood out. Although a variety of functional units were used, most studies applied a cradle-to-grave scope.

It was observed how different analysis scenarios were used to have points of view in which some considered various functional units, scopes, environmental impacts, methodologies, among others, according to the needs of the study. However, it is essential to standardize LCAs, so that they can be comparable, reproducible, and repeatable to facilitate conclusions on the same basis.

In terms of applications, a bias was observed in the study of plastics and biobased materials used for food packaging. Given the versatility and actual uses of these materials, it is necessary to implement LCAs focused on different industries such as agriculture, packaging for the transport of goods, pipes for wastewater, medical applications, and all sectors that have extensive use of plastics.

The analysis methods need to be standardized to allow us to see a holistic panorama of the impacts, as taking global warming as the only parameter of comparison will not provide the real conditions of other environmental aspects. The choice of treatment at the end of the useful life of fossil plastic or biobased and the decision-making surrounding the modeling of the treatment processes present a wide variation between the different studies. Appropriate real options must be modeled, and the sensitivity of the results to these options must be considered.

Further investigation of bioplastics is required to obtain more information regarding their feasibility for use. In general, these materials need to be industrially tested because some studies have only evaluated prototypes at the laboratory level.

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3
Micro- and Nanoplastics—An Invisible Threat to Human Health

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Abstract

Microplastics are microscopic plastic particles produced by the decomposition of plastics, which are prevalent in the surroundings and damage both nature and humanity. They were recognized in a wide array of aquatic animals, drinking water, and culinary items, such as salt and pre-packaged food. Microplastics and nanoplastics have grown rapidly owing to their occurrence in various ecosystem components. Humans are exposed through an assortment of pathways, including ingestion, inhalation, and skin interactions. Following consumption, the monomers of plastic can impede perilous biological functions in humans, causing fractures of the immunological and endocrine systems, as well as having a severe influence on reproduction, mobility, and development. The COVID-19 epidemic has influenced human health and the financial prudence of various nations. It also has an effect on the surroundings because of the enormous amount of waste in the form of abandoned personal shielding gear, with possibly calamitous long-term repercussions if speedy action is not performed. This chapter discusses

microplastics and their means of exposure, contamination of food items, and effects on human and marine health.

Keywords: Nanoplastics, microplastics, routes of exposure, toxicity, marine health, environmental effect

3.1 Introduction

Synthetic polymers originally appeared in the late 1800s, but their acceptance hit the roof after World War II. Since its inception as a phenol-formaldehyde resin, plastic has matured to become one of the most widely used materials (i.e., Bakelite). Plastic after its discovery was projected to improve the living circumstances of humans; on the other hand, it turned out to be a grave menace to our ecosystem and the entire world [1]. Degrading plastic waste yields microplastics with a diameter of less than 5 mm or nanoplastic particles that are 1–100 or 1,000 nm in diameter. The prevailing research provides additional information and divisions of microplastics concerning the diameter of plastic particles. When originally described in 2004, the name “microplastic” refers to plastic shards approximately 20 µm in diameter. While these early publications mentioned minute particles, they did not define microplastics [1]. Microplastics are further classified according to their source: primary if they are purposefully liberated into the surroundings, such as cosmetics and cleaning products containing abrasives and microbeads, and secondary if they are produced indirectly through degradation processes, including domestic waste, improper plastic waste disposal, and automobile emissions. [2]. Almost 360 million metric tons of plastic are manufactured annually, 40% of which is utilized for one-time use purposes, and the bulk of it is dumped into nature [3].

The United States National Oceanographic and Atmospheric Agency (NOAA) sponsored a worldwide conference on microplastics in Washington, 2008, which was the first of its kind and created a comprehensive way to define all particles less than 5 mm to be considered microplastics. Microplastic samples are often classified into distinct categories, depending on their observed architectures [1]. Relative to the initial form of plastics, the breakdown procedure takes place on the surface of plastics, and their continuance in the atmosphere; microplastics can have a variety of forms (films, spheres, fragments, foam, beads, fibers, and pellets). While most soil microorganisms and plants cannot absorb large microplastics, monomers (e.g., nanoplastics) can easily enter cells and pose an ecological threat [2].The omnipresence of plastic in the surroundings of food packaging for commodities, such as milk and related products, poultry, seafood, and drinks, is a serious environmental concern. Interactions between food and its plastic wrapping invariably alter the quality of foodstuff, which leads to contamination in food, induced by interactions with substances in packaging components, which can affect the nutritional value and alter the safety of ingestion [1]. Plastics are ubiquitous as they are extensively used for various purposes ranging from disposable diagnostic kits to food packaging and technological equipment [2].The abundance of microplastics in the surrounding areas is creating widespread contamination. Microplastics are efficiently integrated into the surroundings and persist for an extended period owing to their properties such as water insolubility, non-degradability, synthetic high polymer composition, and size of <5 mm. As a part of diverse aquatic ecosystems, it heavily affects the food chain. The detrimental effects of these contaminants on reproduction, eating behavior, and growth of various marine species have

been studied [1]. Microplastics and nanoplastics are readily absorbed by marine life, where they biomagnify and bioaccumulate, ending in higher orders in the food chains [3]. Because microplastics are ubiquitous in a variety of aquatic environments, marine creatures, including benthos, are either directly or indirectly exposed to them. As a result, humans are exposed to these toxins by consuming various seafoods [1]. According to scientists, these micro- and nanoplastic particles can be absorbed by eating, aspiration, and cutaneous contact. Microplastics have also been found in the placenta, feces, and colectomy samples of humans [3]. This book chapter deals with the severe problems caused by the high polymer composition of plastics, which deteriorate to form microplastics and nanoplastics that are invisible threats to humans as well as other organisms and the environment.

3.2 Routes of Exposure

Microplastics and nanoplastics have grown rapidly because of their presence in various ecosystems. The routes of exposure can be through the air, soil, and water. These include exposure through inhalation of small plastic debris present in the environment, oral consumption, contaminated food and water, cutaneous exposure *via* the application of soaps, lotions, scrubs, or contact with dirt [4] (Figure 3.1).

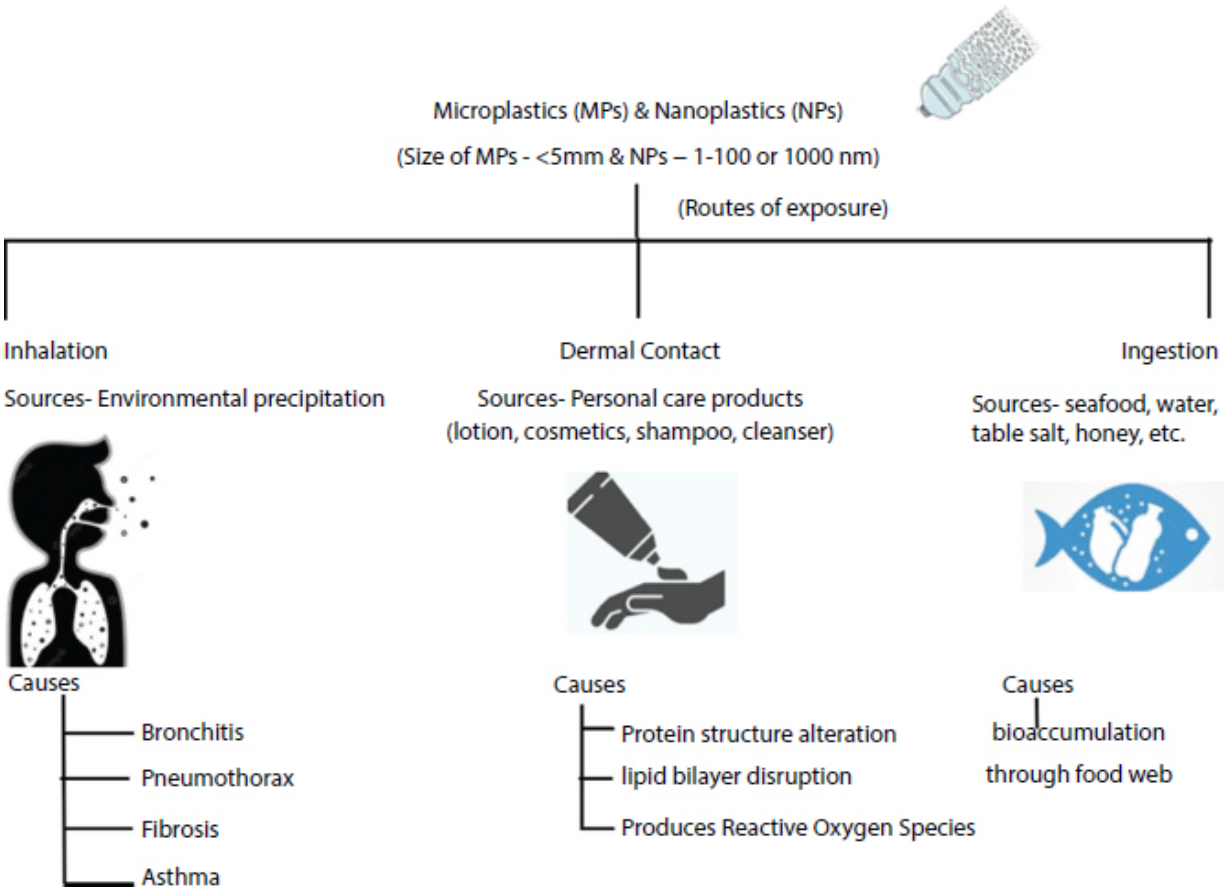


Figure 3.1 Route of exposure of microplastics and nanoplastics.

3.2.1 Inhalation

Microplastic and nanoplastic particles have been found in environmental precipitation because of the plastic breakdown in the environment. Depending on their sizes, these particles can be inhaled. Inhalable particles may enter the mouth and nose and deposit in the airway, whereas breathable particles are deposited in the lungs [4].Concerning the dissemination of microplastics and nanoplastics into the respiratory pathway of human beings, a recent study discovered that more than 50% of samples of the broncho-alveolar region of the lungs examined; they observed plastic particles in the range of 1.60–5.58 µm, explicatory of the fact that lungs and the process of

respiration is a major route of contact and it also serves as a location of microplastics and nanoplastics amassing in humans [4–6]. According to previous studies, microplastics and nanoplastics can harm human health by causing bronchitis, fibrosis, pneumothorax, and asthma. In a 14-day experimental investigation in rats, the levels of tumor necrosis factor alpha (TNF-α), transforming growth factor beta (TGF-beta), and fibrosis-related factors were modified [4, 5]. Owing to their small size, plastic nanoparticles (<1 µm) can produce multiple detrimental effects. Inhaling microplastics and nanoplastics is associated with several disorders, including a higher risk of cancer. Studies on model organisms have shown that exposure can cause granulomatous lesions. Compared to other atmospheric nanoparticles, microplastics and nanoplastics can be equally harmful, making comparisons difficult. As a result, investigations of the toxicity of various polymer sizes and their surface features are required [4].

During SARS-CoV-2 outbreak, mask wearing became mandatory, with N95 surgical masks being the most efficient in minimizing virus transfer. As masks became scarce, many people opted to use masks made of diverse materials, including cotton, nylon, clothes, and polypropylene textiles. The fabric may wear and tear during the disinfection process, thereby impairing the material and increasing the hazard of the particles reaching the respiratory system during the process of breathing [4]. Researchers have used Raman spectroscopy and infrared spectrometry to identify microplastics and nanoplastics using seven different masks: N95, activated carbon, sky surgery, cotton, ply surgery, fashionable, and non-woven. After disinfection, the N95 mask functioned effectively and decreased particle inhalation. Furthermore, nonwoven fabric, cotton, and activated carbon masks were found to mitigate the dangers of microplastic and

nanoplastic inhalation. It was observed that wearing one of these masks for a minimum of 2 h might reduce plastic particle inhalation compared to wearing a mask [4, 6].

3.2.2 Dermal Contact

Despite the fact that microplastics and microbeads are included in a variety of personal care products, including cleansers, cosmetics, and shampoos, and little emphasis has been devoted to monitoring dermal exposure even though nanoplastics may pass through the dermal layer [7]. Microplastics, specifically nanoplastics, can be incorporated into cells because of their small size, the polarity of most polymers, and their 3D structure. Moreover, the possible interaction between nanoplastics and proteins may result in alterations in the protein structure [8] and lipid bilayer disruption [7]. One of the earliest studies to examine the possible cytotoxicity of microplastics and nanoplastics examined the consequences of polyethylene (PE) and polystyrene (PS) on human epithelial cells and found that PS produced reactive oxygen species (ROS). This demonstrates the need to evaluate diverse plastic polymers rather than conflating microplastics and nanoplastics from different materials [7, 8].

3.2.3 Ingestion

Microplastics can be produced by a variety of methods and distributed across several environmental matrices, ultimately making their way into the food chain and the human body. Humans rely heavily on the sea for nutrition. Microplastics can enter fish organs actively (through prey disorientation), passively (by gill water filtering), or through the consumption of polluted prey [9]. Microplastics can be consumed by consumers or predators in the marine food chain *via* prey such as arthropods, polychaetes,

crustaceans, mollusks, and annelids. Although the gastrointestinal tract (GIT) of organisms living in the sea is removed during processing, the incidence of microplastics in the GIT of marine animals has caused global concern because seafood may serve as a significant source of microplastics for humans [2, 9].

Plastic debris has been detected in the digestive tract of fish. In a study conducted on pelagic and benthic species microplastics were found in their muscles and gills. Several marine species included as main food sources have been found to contain microplastics in their gastrointestinal tract and on their external surface, which can be a great threat to human life as they can transfer and bioaccumulate through the marine food web [10]. Small fragments of microplastic particles were introduced into drinking water during the purification process. Compared with raw water, treated water has a comparably low microplastic content. The microplastic levels found in drinkable water are not trivial, implying that potable water may be a substantial source of microplastics for individuals. Those who exclusively drink water from plastic bottles consume more microplastics than those who drink tap water alone [11, 12].

All these investigations indicate that microplastics particles are consumed in significant quantities. However, these examples are still insufficient illustrative of genuine microplastic contamination; thus, further research is needed to collect data on human consumption of microplastics from meals other than those associated with the aquatic environment [2].

3.3 Phenomenon of Microplastics in Nourishment and Nutrients

3.3.1 Sodium Chloride

It is challenging to eliminate microplastics from finished sea salt products without additional purification procedures because saltwater includes microplastics, and table salt is often made by distilling seawater. After the sea salts dissolved, hydrogen peroxide (H_2O_2) was frequently used to break down organic debris in the solution. Sodium iodide (NaI) is typically used for density separation where solution density is equal to 1.8 g per cm^{-3}. The investigational size range was smaller one-0.2–11 µm; hence, no significant variations in microplastic concentrations were found. This could be because utilizing a stereomicroscope and FT-IR spectroscopy to detect microplastics smaller than 10 µm is incredibly challenging. Therefore, more research is needed to reduce microplastic contamination when producing salt from sources other than seawater [13, 15].

3.3.2 Marine Organisms (Crawfish, Mussel, Oyster): Techniques Used for Microplastic Identification

Researchers have found microplastics in fish and shellfish that have been caught in the wild and purchased from aquaculture farms or markets in the previous 10 years [16]. Several pretreatment techniques have been used to digest and eliminate biological matrices from fish samples. To identify microplastics in bulky fish, stomach/or (GIT) was extracted and processed. Different chemicals, including nitric acid (HNO_3), potassium hydroxide (KOH), hydrogen peroxide (H_2O_2), sodium hydroxide (NaOH), Fenton's reagent (Fe^{2+} ion with H_2O_2), perchloric acid ($HClO_4$), and

digestive enzymes, have all been used to break down organic materials, in contrast to the digestion method used for sea salts. Depending on the number and type of samples used, different amounts of time are required for digestion. Increased digestion temperatures have the potential to reduce the necessary time but also increase the risk of thermoplastic loss. FT-IR spectroscopy, in combination with microscopic inspection, is the most widely used technique for chemical identification. In addition, Raman spectroscopy was used because it performs better with smaller particles. Microplastics are primarily discovered in the digestive systems of animals and are infrequently identified in edible tissues such as muscles. Human exposure to microplastics from edible fish tissue must be further studied as it is unclear how they are delivered to palatable tissues. Blue mussels, prawns, clams, and other shellfish have been closely examined for the presence of microplastics [14, 15].

3.3.3 Canned and Prepackaged Foods

Microplastics have been found in several processed meals [17] and studied in various liquids, including milk, beer, and honey. While staining followed by totaling may have inflated the quantity of particles, the significant amounts of microplastics in beer samples require further validation. 30% of H_2O_2 was used to oxidize honey; however, thousands of suspicious particles were still observed. In addition to sugar, more research is necessary because it is possible that the processing of sugar contaminates it with microplastics. Studies should be conducted to account for regional consumption trends because people frequently eat food or medicines that are susceptible to microplastic contamination. Dried seafood is typically consumed worldwide. Prominent levels of anthropogenic particles were found in hot teabag water, although not all particles

were classified as microplastics [18]. Because smaller particle sizes are more likely to have adverse effects on organ function after ingestion, it is important to identify the microplastic content in teabags and prevent exposure [15].

3.3.4 Soil Biome

Scientists have recently concentrated their efforts on the connection between the sources and diversity of microplastics. The terms “microplastic combination” and “microplastic communities” denote different shapes, colors, polymer types, and particle sizes, which are organized in environments, such as natural groups of microplastics. The pollution sources’ complexity can be inferred from the diversity of “microplastic ecosystems.” The properties of microplastics have shed light on their origin. The chemical makeup of microplastics in sewage was examined, revealing that polyester and acrylic fibers were the most abundant [19, 20]. Only one field study has demonstrated the presence of nanoplastics in soil, and this study is the first to establish the occurrence of soil nanoplastics. In addition, it predicts that there are higher chances for bigger plastics to break down into nanoplastics in the soil in 30 years. Furthermore, there are still many unanswered questions and untapped potential in the field of nanoplastic detection in various contexts, which are still in the preliminary stages of progress [20].

3.4 Impact of Microplastics and Nanoplastics on Mammalian Health

Currently, scientists have started to prognosticate the possible harmful effects of microplastics and nanoplastics on mammalian health in *in vivo* animal models. It was shown that mice that manifested photo-aged polystyrene (PS) with diameters of 5–20 µm with a time interval of 28

days exhibited their existence in the liver, gut, and kidneys. The levels of triglycerides and total cholesterol were remarkably reduced in the microplastic groups, and globules of lipids were identified in the liver, indicating that microplastics are a source of metabolic lipid disarrangement and liver tenderness in mice. Mice exposed to polystyrene had reduced interstinal mucosa and notable fluctuations in abundance and variety of enteric lives. Adult mice were used, and the exposure was in the range of 1–10 mg nanoplastics/kg body weight. The particles had a mean radius of 39.8 nm and manifested locally for more than a month [1].

The consumption of seafood has drastically increased in the coming years. In 2015, the global utilization of seafood constituted 17% of the global animal protein intake and 6.7% of the consumed protein. The GIT of mussels and fish contains microplastics. Microplastics incorporated from polypropylene (PP), polystyrene (PS), polyethylene (PE), polyesters, and polyethylene terephthalate (PET) are fed by a diversity of aquatic organisms, such as oysters, zooplankton, shrimp, mussels, and fish, which are accessible in food chains. Microplastics are consumed through food in the range of 39,000–52,000 globules per person per year [21]. Microplastics pose a significant danger to human health by modifying marine microbial groups with the help of gene transfer and increasing the spread of microbial affinity. Undiscerning the use of antibiotics has steered the emergence of antibiotic-resistant bacteria (ARB). In aquatic environments, microplastics serve as a medium for microbes, including ARB. The chronic consequence of microplastic assemblage in the tissues and cells of marine organisms causes a possible danger to living beings, as ingestion of microplastics can lead to infertility [22] and alteration of chromosomes, resulting in obesity and cancer [21].

Drinking water (1,200 to 1,600 ml per day) might be a principal source of these microplastics and nanoplastics. They can be identified in table salt, sugar, and honey, postulating other causes of microplastic and nanoplastic manifestations in humans [1]. Among the 114 lung samples collected from patients undergoing lung incision for tumor exclusion, 87% contained plastic or cellulosic fibers, indicating that these fibers are inhalable and deposited in the lung tissue. Synthetic fibers are the main cause of floating microplastics, especially in the innermost areas where assemblages are greater. It was previously reported that inhalation of fibers during work in factories causes cancer, but studies on groups of workers dealing with nylon revealed no proof of increased risk of cancer but had a higher prevalence of respiratory infuriation [23, 24].

3.5 Nanoplastics and Microplastics: Effects on Environment and Marine Life

The effects of microplastics and nanoplastics on the environment and human well-being can be considerably improved by knowing their origins and locations. Microplastics and nanoplastics have been discovered worldwide in a variety of habitats, including marine and terrestrial habitats. They are found in the air, water, deposits of clay, and in foods. For instance, microplastics are found in most oceans and are bio-absorbable and bio-accumulative by marine life. Therefore, research on the effects of microplastics on animal wellness is crucial. Furthermore, earthworms that live in the topmost layer of soil and deeper layers have been found to contain microplastics [25].

Microplastics and nanoplastics can be detected using a variety of techniques like Raman spectroscopy, energy dispersive X-ray (EDX), micro-Fourier-Transform Infrared Spectroscopy (Micro-FTIR), Scanning Electron Microscopy (SEM), Transmission Electron Microscopy (TEM), and Energy Dispersive X-ray (Micro-FTIR). The first nanoplastics were found in the North Atlantic subtropical circulation, which has sparked recent studies on nanoplastic pollution. The fact that the analytical methods used to study microplastics, including FTIR and Raman spectroscopy, cannot be utilized to study nanoplastics is mostly to blame for the paucity of research on these smaller particles. In addition, existing technologies for characterizing nanoplastics such as TEM, are unable to efficiently differentiate between plastics and organic natural materials. Nanoplastics were examined using scanning transmission X-ray microscopy (STXM) [25, 26].

Additionally, some of these creatures, such as most bivalves, are exploited as a significant source of food for people, posing a risk of direct contact with plastic particles. Oysters, clams, shellfish, mussels, and other bivalves are filter feeders. All the food they consume enters the system responsible for digestion immediately and may be employed in microplastic and nanoplastic investigations. According to research, plastic particles can build up in the gut of blue mussels and be absorbed into their circulatory system, whereas larger particles can remain in their bodies. Another study discovered that when exposed to the same quantity of plastic particles, blue mussel larvae ingested particles with a diameter of 100 nm [25, 27].

Mytilus edulis and *Crassostrea gigas*, two common bivalves that humans eat, have microplastics in their soft tissues. Based on the amount of microplastics observed in the flesh of these two marketable bivalves, it is predicted that 11,000 different types of microplastics are consumed

annually by shellfish consumers in Europe, suggesting that microplastics accumulated in bivalves may be a significant source of exposure for seafood consumers [28]. Studies on clams (*Vineries philippinarum*) from various sources have shown that there is no discernible difference amongst wild and cultivated clams in terms of the consumption and accumulation of microplastics and nanoplastics, with the range of recognized plastics between 0.07 and 5.47 particles/g. Studies have demonstrated that microplastics in shellfish exhibit certain properties. Studies have also examined potential biological alterations caused by plastic particles in other sea animals [29]. For example, the phylum Annelida has enormous numbers of sea worms called lugworms (deposit feeders). Lugworms (*Arenicola marina*) found along the Northern European coast's native ecological environment contained 1.2 to 2.8 particles/g. However, the results also indicated that neither the organisms nor the bioaccumulation of other chemicals was significantly affected by these collected plastic particles. Additional studies with *Dunaliella salina*, demonstrated that plastic micro particles of the size 200 µm (>cell size) stimulated *D. salina* development and photosynthesis at doses of 200–350 mg/l. The antagonistic effects increase with decreasing microplastic diameter [25, 30].

3.6 Conclusions

Evidence points to the fact that microplastics and nanoplastics negatively influence the health of both in a negative manner of both marine and freshwater animals and humans. No solid information is available on the effects of microplastics and nanoplastics on human health. However, it is possible for them to slowly erode through the intestinal wall, enter the bloodstream, and spread to different tissues and organs. Their toxic effects may result

in comparable harm to the human body if collected over time. The study of nanoplastics is still in the preliminary stages. DOwing to the special characteristics of nanoparticles that depend on their size and shape, nanoplastics and microplastics may have diverse effects on them. One of the most contested aspects of plastic pollution is the effect of microplastics on the bioavailability and toxicological effects of endogenous and exogenous contaminants. This subject has been covered in detail in the literature. This is understandable, given that several interconnected elements influence the effective distribution of drugs by microplastics. As a result, it is still difficult to determine whether microplastics are actual chemical transporters and, secondly, whether microplastics are pertinent to examine in this context based on modeled analyses given their relative contribution to oceanic mass compared to other sorbents. The field’s expansion is to incorporate a wider variety of particles, such as nanoplastics, tire wear particles, and “bio plastics.” In general, more physiological and environmental realism is required to close the gap between the laboratory and the real world. Toxicological assessments can be made, provided our current knowledge level increases, as the level of knowledge is the rate-limiting factor in these studies.

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Conflict of Interest

No conflict of interest to declare.

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4
Microplastics and Nanoplastics and Related Chemicals: The Physical–Chemical Interactions

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Abstract

Microplastics and nanoplastics are small plastic particles less than 5 mm in size. They can be found in the environment, including bodies of water, soil, and air. These particles are produced from the breakdown of larger plastic items or are intentionally manufactured on a small scale for industrial purposes, such as in cosmetics, cleaning products, and clothing. The accumulation of microplastics and nanoplastics in the environment can have toxic effects on organisms including fish, birds, and other wildlife. These particles can enter the food chain and potentially harm humans consuming contaminated seafood and other products. Furthermore, these particles have been found to leach harmful chemicals, such as bisphenol A (BPA) and phthalates, which are endocrine disruptors that can negatively impact human health, including reproductive and developmental problems, cancer, and other chronic diseases.

In conclusion, the toxic effects of micro- and nanoplastics on the environment, food, and human health are a growing concern and require more attention and action from governments, industries, and individuals. Reducing plastic waste and promoting the use of more sustainable alternatives can help mitigate this issue and protect our planet and health.

Keywords: Macroplastics, nanoplastics, ecological distribution, health effects, toxicological effects

4.1 Introduction to Micro- and Nanoplastics

Plastics are synthetic materials made from polymers that have become ubiquitous in modern society because of their versatility, durability, and low cost. Plastics pose significant environmental and health concerns. Some of the concerns include environmental pollution and exposure to chemicals in plastics. Plastics are a major source of environmental pollution because they are non-biodegradable and can persist in the environment for hundreds of years. Plastics can accumulate in landfills, oceans, and other ecosystems where they can harm wildlife and disrupt

food webs. Plastics can also break down into smaller particles, known as microplastics, which can be ingested by wildlife and humans and potentially harm their health. Plastics can contain chemicals such as bisphenol-A (BPA), phthalates, and flame retardants, which can leach out of the material and food, water, and the environment. These chemicals have been linked to a range of health problems including reproductive disorders, cancer, and developmental delays. Plastic production requires significant amounts of fossil fuels and other resources, contributing to climate change and environmental degradation [1]. To address these concerns, efforts are underway to reduce plastic use and improve waste management practices including recycling, composting, and plastic bans or taxes. Consumers can also take individual actions to reduce their plastic footprint, such as using reusable bags, bottles, and containers and avoiding single-use plastics [2].

Microplastics are small plastic particles less than 5 mm in length. They can be created in several ways, including the breakdown of larger plastic products, shedding of plastic fibers from textiles during washing, and the use of microbeads in personal care products such as facial scrubs and toothpaste. Microplastics are a growing concern because they are pervasive in the environment, can harm marine life and potentially enter the human food chain. Studies have shown that microplastics are found in a wide variety of environments, including oceans, rivers, lakes, and air.

Microplastics can have several negative environmental effects. For example, they can be ingested by a wide range of marine organisms, including fish, shellfish, and plankton. This can lead to physical harm or blockage in the digestive tract, as well as the absorption of harmful chemicals that may be present on the surface of microplastics.

Microplastics can accumulate in sediments and disrupt benthic ecosystems. They can also transport pathogens and other pollutants throughout the environment, potentially harming wildlife and ecosystems.

Microplastics can also affect terrestrial ecosystems. For example, they can accumulate in the soil and potentially affect plant growth and health. There is also a concern that microplastics may enter the human food chain through the consumption of contaminated seafood or other foods. Although more research is needed to fully understand the potential human health impacts of microplastics, they are a growing environmental concern that requires urgent attention and action. The health effects of microplastics on humans are not yet fully understood and are an active area of research [3]. However, there are concerns that exposure to microplastics may have negative impacts on human health. One concern is that microplastics can absorb and accumulate harmful chemicals, such as persistent organic pollutants (POPs), from the surrounding environment. These chemicals can

be toxic and may cause health problems such as endocrine disruption, developmental problems, and cancer [4].

Another concern is that microplastics may enter the human body through ingestion, inhalation, or skin contact. Microplastics have been found in a variety of foods, including seafood, drinking water, and table salts. Once inside the body, microplastics can potentially cause harm by damaging organs or triggering an immune response. Although more research is needed to fully understand the potential health impacts of microplastics on humans, reducing microplastic pollution is important for protecting both the environment and human health [5].

Microplastics have been found in various foods, including seafood, drinking water, salt, honey, and beer. These microplastics originate from several sources, including plastic packaging, processing equipment, and environmental contamination. Seafood is a particularly concerning source of microplastics, as fish and shellfish can ingest microplastics in the water in which they live, which can then accumulate in their bodies. Studies have found microplastics in a range of seafoods, including oysters, mussels, and fish. This raises concerns about the potential health effects of consuming contaminated seafood because microplastics can absorb and accumulate harmful chemicals from the surrounding environment [6]. In addition to seafood, microplastics have also been found in drinking water, including both bottled and tap water. Although the health impacts of consuming microplastics in drinking water are not yet fully understood, reducing microplastic pollution is important for protecting both the environment and human health [7].

Efforts to reduce microplastic contamination in food include improved waste management practices, reduced use of plastic packaging, and increased use of sustainable alternatives to plastic. Further research is needed to better understand the risks associated with consuming microplastics and to develop effective strategies for reducing exposure. Efforts to reduce microplastic pollution include the development of biodegradable plastics, bans on microbeads in personal care products, and increased recycling and waste reduction efforts. Nanoplastics are smaller plastic particles than microplastics, measuring less than 100 nm in size. They can be formed from the breakdown of larger plastic products or can be intentionally created for use in products such as sunscreens, cosmetics, and food packaging.

Nanoplastics are a relatively new area of research and their potential impacts on the environment and human health are not yet fully understood. However, there is a concern that their small size could allow them to penetrate cell membranes and potentially cause damage at the cellular level.

The environmental effects of nanoplastics are still not well understood and are an active area of research. However, there are concerns that nanoplastics may have negative impacts on the environment and ecosystem. One concern is that nanoplastics could potentially have harmful effects on aquatic organisms because they are small enough to penetrate cell membranes and potentially cause damage at the cellular level. This can lead to disruptions in the physiological processes, reproductive success, and survival of aquatic organisms.

Another concern is that nanoplastics can accumulate in sediments and affect benthic ecosystems. Nanoplastics can also transport pollutants throughout the environment, potentially causing harm to wildlife and ecosystems. In addition to aquatic ecosystems, there is also a concern that nanoplastics can have negative impacts on terrestrial ecosystems, including soil health and plant growth.

The health effects of nanoplastics on humans are not yet fully understood and are an active area of research. However, there are concerns that exposure to nanoplastics may negatively impact human health.

One concern is that nanoplastics may enter the human body *via* ingestion, inhalation, or skin contact. Once inside the body, nanoplastics can cause harm by damaging organs or triggering an immune response. There is also a concern that nanoplastics may accumulate in organs and tissues, potentially leading to long-term health effects.

Another concern is that nanoplastics can absorb and accumulate harmful chemicals such as persistent organic pollutants (POPs) from the surrounding environment. These chemicals can be toxic and may cause health problems such as endocrine disruption, developmental problems, and cancer.

Although more research is needed to fully understand the potential health impacts of nanoplastics on humans, reducing nanoplastic pollution is important for protecting both the environment and human health. Strategies for reducing nanoplastic pollution include improving waste management practices, reducing the use of plastic products, and developing sustainable alternatives to plastics [8]. The effects of nanoplastics on food are still not well understood and represent an active area of research. However, there are concerns that nanoplastics could potentially contaminate food and negatively impact human health.

Nanoplastics can enter the food chain through several routes, including the contamination of crops, livestock feed, and seafood. Once inside the body, nanoplastics can cause harm by damaging organs or triggering an immune response. There is also a concern that nanoplastics can accumulate in the tissues of animals and pass through the food chain, potentially leading to increased exposure and long-term health effects.

Although more research is needed to fully understand the potential impacts of nanoplastics on food, reducing nanoplastic pollution is important for protecting both the environment and human health. Strategies for reducing nanoplastic contamination in food include improving waste management practices, reducing the use of plastic products, and developing sustainable alternatives to plastics. Further research is needed to better understand the risks associated with consuming nanoplastics and to develop effective strategies for reducing exposure.

Further research is needed to better understand the environmental impacts of nanoplastics and develop effective strategies for mitigating their effects. Reducing nanoplastic pollution is important for protecting the environment and human health.

4.2 Sources and Distribution of Micro- and Nanoplastics

Microplastics and nanoplastics can originate from a variety of sources, such as plastic debris, textiles, personal care products, and road traffic. Microplastics can be generated from larger plastic debris, such as packaging, fishing gear, and disposable plastics, which break down over time owing to environmental conditions. Microplastics can be released from synthetic fabrics during washing and drying. It can be added to personal care products such as facial scrubs, toothpaste, and exfoliating shower gels as abrasives. Microplastics can be generated during industrial processes such as plastic production and processing, and from the wear and tear of tires and road surfaces. It can also be released from landfills, as plastic waste breaks down and leaches into the surrounding environment, and from wastewater treatment plants, as treatment processes are not designed to remove it from the water. It is important to note that while these are common sources of microplastics, there may be additional sources not yet fully understood [9, 10].

To reduce nanoplastic pollution, it is important to develop sustainable and eco-friendly alternatives to traditional plastics. Additionally, more research is needed to understand the potential impacts of nanoplastics and develop effective strategies for mitigating their effects on the environment and human health [11].

4.3 Ecological Impacts of Micro- and Nanoplastics

Micro- and nanoplastics have the potential to disturb ecological processes and harm species, whether they are created purposely at microscopic sizes

or as byproducts of the breakdown of larger plastic products. These minute plastic particles have the following ecological effects.

1. Aquatic ecosystem disruption: The ecological effects of micro- and nanoplastics are particularly prone to affecting aquatic ecosystems, such as rivers, lakes, and oceans. These contaminants may accumulate in water bodies, posing dangers to aquatic life at different trophic levels. Microplastics can have physical and physiological consequences on a variety of organisms, including zooplankton, fish, and marine animals. By way of illustration, the consumption of microplastics by filter-feeding species might impair the effectiveness of their eating and the flow of energy within food webs [12]. The growth and development of aquatic organisms, including mussels and coral larvae, have also been reported to be disrupted by nanoplastics, which may have an impact on population dynamics and ecological stability [13].
2. Contamination of Soil and Terrestrial Ecosystems: Microplastics can also enter terrestrial ecosystems through several different channels, such as plastic fertilizer applications or the disintegration of mulch films made of plastic. These accumulating particles have the potential to harm plant health, the soil biota, and nutrient cycling. Recent research has shown that plants can absorb microplastics, which raises questions about how they can enter the food chain and what effects they might have on terrestrial organisms [14]. Furthermore, nanoplastics may enter plant tissues because of their small size, which could have an impact on plant physiology and growth [15].
3. Alteration of Ecosystem Functions: Key ecological processes and functions may change due to the presence of tiny plastic particles in ecosystems. These particles can transport and absorb chemical contaminants, including heavy metals and persistent organic pollutants (POPs), which cause their redistribution in the environment. According to Koelmans *et al*. (2019), this mechanism can worsen the toxicological effects of these pollutants on organisms, such as decreased fertility, abnormal behavior, and weakened immune systems [16]. According to Hidalgo-Ruz *et al*. (2012), the buildup of microplastics on the exterior of natural substrates such as rocks and sediments might alter habitat structures and influence the colonization and behavior of benthic animals [17].
4. Bioaccumulation: Microplastics may be consumed by species at lower trophic levels, and their accumulation and passage through the food web may cause biomagnification, which may influence higher trophic levels [18].
5. Alteration of nutrient cycles: By adsorbing and transferring organic and inorganic contaminants, microplastics may have an impact on the

availability and cycling of nutrients in ecosystems. Microplastics have been found to directly interfere with the balance of plant chlorophyll a/chlorophyll b ratios, modify soil bulk density and water-holding capacity, and reduce the photosynthetic rate in plants. These effects have been demonstrated to affect the ability of plants to root. By changing the dominant bacterial phyla in the soil or the genes and enzymes involved in the carbon, nitrogen, and phosphorus cycles, microplastics may have an impact on the cycling of soil nutrients. Microplastics adhered to the root surface physically prevent pollutants from contacting the roots when they have combined effects on plants with other pollutants, but they are more likely to increase the harmful effects of pollutants on plants [19].

4.4 Food Contamination and Human Exposure to Micro- and Nanoplastics

Micro- and nanoplastics enter the food chain mostly through contaminated water sources. Because they filter-feed, fish and other seafood species are particularly susceptible to eating these particles. Microplastics have been discovered in a range of shellfish, including mussels, oysters, and deep-sea fish. Additionally, sewage sludge application and irrigation with contaminated water have the potential to introduce plastics into agricultural soils, which might contaminate crops and ultimately our food supply [20].

Micro- and nanoplastics have a variety of mechanisms for entering the food chain after being released into the environment. They have the potential to contaminate water sources such as rivers and oceans, and be consumed by marine life. These plastic fragments may build up in the tissues of organisms, such as shellfish, fish, and other seafood that humans ingest later. Furthermore, the use of plastic mulch films, sewage sludge, or plastic fertilizers can cause microplastics to be present in agricultural soils. Therefore, these particles may be absorbed by plants, resulting in their presence in fruits, vegetables, and other crops (Figure 4.1).

Concerns regarding the potential negative effects of micro- and nanoplastics on human health arise once they reach the body. Studies have indicated that such particles can accumulate in organs and tissues, potentially having negative effects, even if the precise processes and long-term impacts are still being studied [21]. Microplastics can interact with the digestive system, possibly resulting in inflammation and reduced nutrient absorption. Additionally, they may function as carriers for other toxicants such as organic pollutants and heavy metals, thereby enhancing the toxicological consequences on human health.

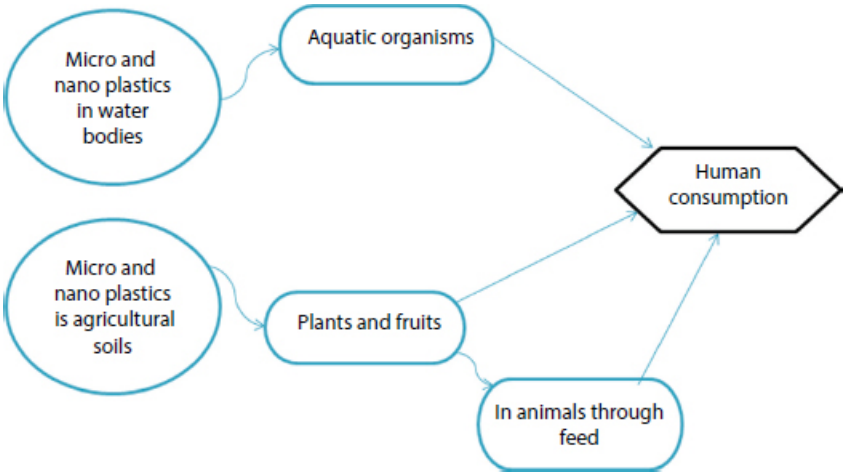


Figure 4.1 Pathway of micro- and nanoplastic contamination in food chain.

Humans are primarily exposed to micro- and nanoplastics through ingestion, whereas other routes include inhalation and skin contact. The presence of these particles in the natural environment has sparked concerns regarding possible human exposure and the resulting health effects. To evaluate the risks that these tiny plastic particles can present, it is essential to understand how they enter and interact with their bodies.

Ingestion is one of the main ways in which people are exposed to microplastics and nanoplastics. Microplastics, including seafood, salt, honey, and drinking water, are found in a range of foods and beverages. For instance, 94% of tap water samples analyzed in a study conducted in the United States contained microplastic particles [22]. Another study examining regularly eaten seafood discovered microplastics in all samples, with bivalves such as mussels and oysters having the highest amounts [23].

In addition to ingesting micro- and nanoplastics, inhalation poses a risk for human exposure. Microplastics that are airborne from external sources and indoor air pollution caused by the degradation of plastic materials can be exposed through breathing. Microplastics were found in indoor dust samples during a study in China, increasing the possibility of respiratory exposure [24]. Further studies are necessary to fully comprehend the scope and importance of inhalation exposure to microplastics.

Dermal contact is another way for people to be exposed to microplastics. Direct skin contact can result from using microbead-containing personal care products, such as exfoliating scrubs. Additionally, microplastics found in fabrics and apparel may come in contact with the skin when engaging in normal activities. Despite the paucity of research on dermal exposure, studies have demonstrated the presence of microplastics in cosmetic items

and revealed probable dermal uptake pathways [16, 25]. The additional data are presented in Table 4.1.

4.5 Toxicological Effects of Micro- and Nanoplastics on Human Health

In recent years, the pervasive presence of plastics in the environment has raised concerns regarding their potential impact on human health. While larger plastic debris, such as bottles and bags, have received considerable attention, emerging research highlights the toxicological effects of microplastics (particles smaller than 5 mm) and even smaller nanoplastics (particles smaller than 100 nm) [31].

4.5.1 Sources and Routes of Exposure

Micro- and nanoplastics can arise from various sources, including the fragmentation of larger plastic items, direct release from consumer products (such as personal care products), and breakdown of synthetic fibers [31]. Once released into the environment, these particles can contaminate air, water, and soil, ultimately reaching the human body *via* ingestion, inhalation, or dermal exposure [32, 33].

Table 4.1 Health effects and nature of macro- and nanoplastics.

Particle type	Polymer type	Common sources	Health effects	References
Microbeads	Polyethylene	Personal care products	Potential ingestion and inhalation risks	[26]
	Polypropylene	Exfoliating scrubs	Chemical and microbial interactions	
Microfibers	Polyester	Synthetic textiles	Potential ingestion and inhalation risks	[27]
	Nylon	Washing machine effluents	Bioaccumulation in organs	
Microfragments	Various plastics	Degraded larger plastics, plastic waste, packaging	Potential ingestion and inhalation risks, accumulation in tissues	[28]
Microplastic films	Polyethylene	Plastic bags, packaging	Potential ingestion and inhalation risks	[29]
	Polypropylene	Plastic films	Chemical and microbial interactions	
Nanoparticles	Various plastics	Weathering and breakdown of larger plastic particles	Potential ingestion and inhalation risks, cellular and subcellular effects, Oral exposure accounts for the alteration of communities	[30]

4.5.1.1 Ingestion

Food and water are the primary routes of exposure to microplastics and nanoplastics. Plastic particles have been detected in a variety of food and

beverage products, including seafood, bottled water, and table salts [34, 35]. Upon ingestion, these particles can accumulate in the gastrointestinal tract, potentially leading to local inflammation and the disruption of normal physiological processes [36].

4.5.1.2 Inhalation

The inhalation of airborne microplastics and nanoplastics is another concern. Particles suspended in the air can be easily inhaled, thereby entering the respiratory system. Once inside the lungs, they trigger inflammation, oxidative stress, and impaired lung function. Studies on laboratory animals have shown that inhalation of microplastics can result in pulmonary fibrosis and other respiratory disorders [37].

4.5.1.3 Dermal Exposure

Microplastics and nanoplastics can also come into contact with the skin through various personal care products and cosmetics as well as through environmental contamination. Although the absorption of plastics through the skin is generally considered lower than that through ingestion or inhalation, the potential toxic effects on the skin and underlying tissues are still a matter of concern [37].

4.5.2 Toxicological Effects

Research on the toxicological effects of micro- and nanoplastics on human health is still evolving and much remains to be understood. However, several studies have suggested the potential adverse health outcomes associated with exposure to these particles [38, 39].

4.5.2.1 Inflammation and Immune Response

Micro- and nanoplastics have been shown to trigger an inflammatory response in various cell types, leading to the release of proinflammatory cytokines. Chronic inflammation can contribute to the development and progression of several diseases including cardiovascular, respiratory, and autoimmune disorders.

4.5.2.2 Genotoxicity and Carcinogenicity

Preliminary studies have indicated that microplastics and nanoplastics can induce DNA damage and genotoxic effects. These particles can interfere with DNA repair mechanisms, increase the risk of mutations, and potentially lead to the development of cancerous cells.

4.5.2.3 Endocrine Disruption

Certain plastics contain additives, such as phthalates and bisphenols, which are known endocrine disruptors. Micro- and nanoplastics can

release these additives when ingested or come into contact with bodily fluids, potentially interfering with hormonal balance and reproductive health [40].

The future study of the toxicological effects of micro- and nanoplastics on human health is important as the understanding of this emerging issue continues to evolve. Research in this field is expected to focus on several key areas to comprehensively assess the potential risks associated with exposure to tiny plastic particles.

Hazard Identification: Future studies should aim to identify and characterize the specific toxicological hazards posed by micro- and nanoplastics to human health. This includes investigating the potential of these particles to induce inflammation, oxidative stress, genotoxicity, and endocrine disruptions. By understanding the mechanisms by which micro- and nanoplastics exert their toxic effects, researchers can better evaluate their impact on human physiology and disease development.

Exposure Assessment: Further research will be conducted to assess human exposure to microplastics and nanoplastics. This involves determining the routes of exposure, such as ingestion, inhalation, and dermal contact, and quantifying the levels of plastic particles in various human matrices including food, drinking water, and biological tissues. Improved analytical techniques and standardized protocols will contribute to accurate exposure assessment and risk evaluation.

Health Effects: Future studies should focus on understanding the potential health effects associated with exposure to micro- and nanoplastics. This includes investigating the impact on organ systems, immune function, reproductive health, and developmental outcomes. Long-term epidemiological studies will help elucidate the potential associations between microplastic and nanoplastic exposure and the incidence or prevalence of diseases, allowing for informed risk management strategies.

Vulnerable Populations: Research will likely emphasize studying the vulnerability of specific populations to the toxicological effects of micro- and nanoplastics. Certain groups, such as infants, pregnant women, and individuals with compromised health, may be more susceptible to the adverse effects of plastic particle exposure. Identifying and understanding the factors that contribute to differential susceptibility will aid targeted risk assessment and protection measures.

Risk Management and Regulation: Future research in this field will contribute to the development of risk management strategies and regulatory frameworks. The findings on the toxicological effects of micro- and nanoplastics will inform policy decisions aimed at reducing plastic pollution, promoting sustainable practices, and ensuring the safety of consumer products. Mitigation efforts, such as waste management

strategies and development of environmentally friendly materials, will also be a focus of future research.

By addressing these key areas, future research on the toxicological effects of micro- and nanoplastics on human health will enhance our understanding of their potential risks and guide the development of strategies to mitigate these risks effectively.

4.6 Conclusions and Recommendations for Mitigating the Toxic Effects of Micro- and Nanoplastics

The ubiquity of microplastics and nanoplastics in the environment and their potential health impacts on humans have raised concerns among researchers and policymakers [41]. Although research on the toxicological effects of these particles is still in its early stages, emerging evidence highlights the need for action to mitigate their impact on human health.

4.6.1 Reduce Plastic Production and Use

One of the primary approaches for mitigating the toxic effects of micro- and nanoplastics is to reduce their production and use. This can be achieved through policy interventions, such as taxation, bans, and regulations that limit the use of single-use plastics [41]. A reduction in plastic production and use limits the number of plastic products that enter the environment and, consequently, reduces the release of micro- and nanoplastics [42].

4.6.2 Improving Waste Management

Improving waste management systems is another important strategy for mitigating the toxic effects of microplastics and nanoplastics. This includes the development of innovative methods for collecting, sorting, and recycling plastic waste. Additionally, it is essential to develop and implement effective measures to prevent plastic litter from entering natural environments such as waterways and oceans [43, 44].

4.6.3 Enhance Public Awareness

Public awareness and education campaigns are essential to reduce the use of single-use plastics and promote responsible plastic waste management practices. The public needs to understand the potential health risks associated with micro- and nanoplastics, and the importance of taking proactive measures to reduce their exposure [42].

4.6.4 Develop and Implement Testing Protocols

Standardized testing protocols are required to evaluate the potential toxicological effects of micro- and nanoplastics. These protocols will help establish the mechanisms of toxicity and inform regulatory decision-making regarding safe levels of exposure to micro- and nanoplastics.

4.6.5 Future Research

Although much is still unknown about the toxicological effects of micro- and nanoplastics, further research is needed to fill critical knowledge gaps. This includes developing more sensitive methods for detecting and quantifying micro- and nanoplastics in various environmental matrices and investigating their potential effects on human health.

In conclusion, the presence of microplastics and nanoplastics in the environment and their potential health impacts on humans are growing concerns. Reducing plastic production and use, improving waste management, enhancing public awareness, developing and implementing testing protocols, and conducting further research are essential to mitigate the toxic effects of micro- and nanoplastics.

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5
Microplastics and Nanoplastics: Sources, Distribution, Behaviors, and Fate

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Abstract

Over the past few decades, plastic production and consumption have proliferated, accompanied by the release of tiny plastic particles called micro- and nanoplastics (MNPs). The fate, distribution, and behavior of these particles are becoming a cause of concern for scientists, who have found that MNPs can be introduced into the environment through a variety of sources, including industrial processes, consumer products, and waste management. Once they enter the environment, MNPs can enter in different environments, such as the ocean, lakes, rivers, soil, air, and remote areas such as the Arctic and Antarctic. The fate of these particles in each environment depends on several factors, including their size, shape, density, chemical composition, and surrounding conditions. When it comes to the behavior of MNPs, these particles can have adverse effects on marine organisms, including changes in behavior, growth, reproduction, and mortality, which is not only a consequence of their polymeric structures, but also because of the transportation of toxic chemicals and pathogenic strains that can be transferred to organisms, including humans, through the food chain and exacerbating climate change by affecting the light absorption properties of the atmosphere and creating a positive feedback loop.

Keywords: Climate change, food chain, micro- and nanoplastics, MNPs, marine, terrestrial

List of Abbreviations

BPA	Bisphenol A
DBP	Dibutyl Phthalate
DEHP	Diethylhexyl Phthalate
DOC	Dissolved Organic Carbon
HDPE	High-density Polyethylene
IUPAC	International Union of Pure and Applied Chemistry
LDPE	Long-branched low-density Polyethylene
LDPE	Low-density Polyethylene
MNPs	Micro and Nanoplastics
POPs	Persistent Organic Pollutants
PAHs	Poly Aromatic Hydrocarbons
PCL	Poly Capro Lactone
PLA	Poly Lactic Acid
PCBs	Polycarbonated Biphenyls
DDT	Dichlorodiphenyltrichloroethane
PE	Polyethylene
PET	Polyethylene Terephthalate
PP	Polypropylene
PS	Polystyrene
PVC	Polyvinyl Chloride
RBCs	Red Blood Cells
RIC	Resin Identification Code
LLDPE	Short-branched linear low-density polyethylene
NOAA	US National Oceanic and Atmospheric Administration
VOCs	Volatile Organic Compounds
WWTPs	Waste-Water Treatment Plants

5.1 Micro- and Nanoplastics: Principles and Sources

Plastic is a general term for many types of polymers widely used in different industries, such as packaging, electronics, agriculture, and automotive industries. They have also found their way into our daily lives because of their excellent physicochemical properties, including resistance to corrosion and degradation, light weight, and low cost. These features have made single-use plastics economically efficient products and have brought throwaway culture to our lifestyle [1], especially during the COVID-19 pandemic. Single-use plastic-based personal protective equipment has dramatically increased and enormous amounts of plastic waste are released into the environment, including open oceans, wastewater treatment plants, and agricultural soils [2]. High production rates, improper waste management, and degradable nature of synthetic plastic polymers have resulted in the accumulation of plastic waste. It has been estimated that 4.9 trillion metric tons of plastics have now reached the open oceans, and more than 80% of plastic waste was not recycled or even incinerated in 2015, disposed of directly to nature, or landfilled [3, 4], which has caused the accumulation of plastic waste. These post-consumption plastic products fragment into smaller particles through the effects of physicochemical factors in the environment, including UV radiation, heat, wind, abrasion, and microbial enzymes (Figure 5.1) that according to NOAA (US National Oceanic and Atmospheric Administration), are called “microplastic” when reaching the size less than 5 mm and nanoplastic when becoming less than 100 nm in diameter [5, 6] and because they are formed as a result of bigger plastics (mezoplastics) fragmentation, are referred as secondary micro- and nanoplastics (MNPs). According to some reports, heat and sunlight are conducive to the production of microplastics. In contrast, cold and anoxic environments slow the rate at which plastics degrade and fragment into smaller particles [7–9]. Another significant source of microplastics is the shedding of fibers from synthetic clothing during laundering. Almost all synthetic clothing items contain polyester, nylon, or other types of plastic fiber. Industrial pellets of plastics, also known as nurdles,

are other sources of microplastics. They are small beads that melt during manufacturing to create plastic products such as toys, bottles, and clothing that often spill from transport containers and end up in the oceans. Secondary MNPs accounted for at least 70% of the total plastic released into the environment. On the other hand, primary MNPs are originally manufactured in micro- and nano-sized polymers, mainly used in cosmetics (scrubs, toothpaste, washing powders, and other detergents) as microbeads, polyester clothes, paints, electronics, etc. [2, 6, 10]. Therefore, regardless of the polymeric chemical composition, MNPs are generally classified into primary and secondary micro- and nanoplastics based on the manner in which they are formed; however, there are other criteria to categorize these particles according to their shape, size, and polymeric composition.

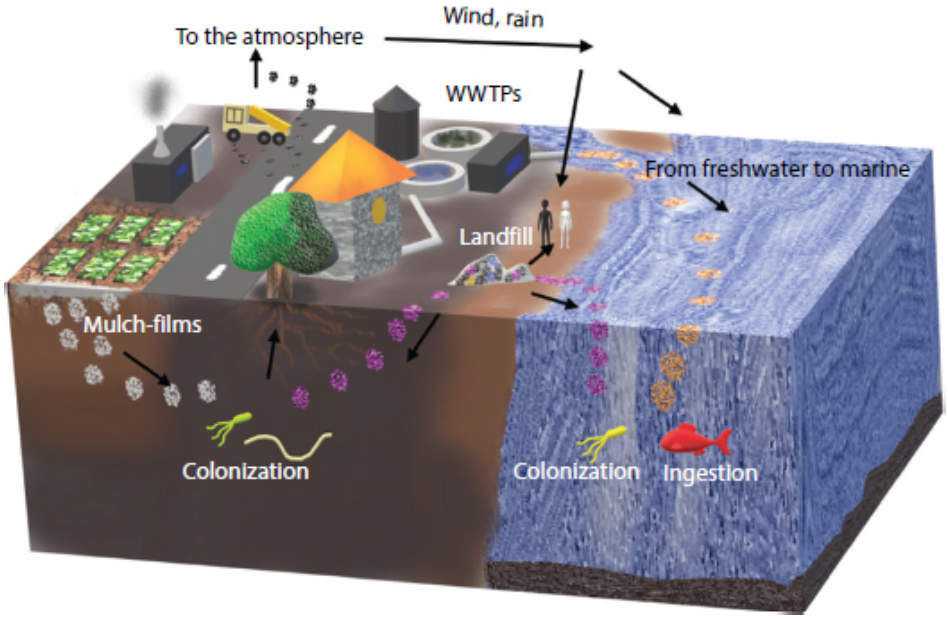


Figure 5.1 Sources and distribution of MNPs. Mulch films, landfills, WWTPs, and tire friction are among the sources of MNPs. When introduced to the environment, these particles manipulate the microbial community compositions of terrestrial and aquatic systems, causing harm to their biota and cycle between the atmosphere, soil, and water through natural forces.

Based on the shape, MNPs can be found and produced as fibers, granules, films, styrofoams, and fragments with different surface-to-volume ratios. The chemical composition of these polymers is variable but mainly includes polyethylene (PE), polystyrene (PS),

polypropylene (PP), polyethylene terephthalate (PET), and polyvinyl chloride (PVC). In this regard, some properties of MNPs are based on their polymeric composition (see *Section 5.4.2.1*), while some other features and behaviors are not chemical composition-dependent and are general in MNPs [1].

5.2 Micro- and Nanoplastic Behavior

5.2.1 Physiochemical Properties of MNPs: Toxicity and Reactivity

The most common way to categorize plastics is through the Resin Identification Code (RIC), which is a number inside a recycling triangle symbol found in many plastic products. There are seven categories of plastics, labeled 1 through 7, each with different chemical properties and recycling processes (Table 5.1). The word plastic is an ambiguous concept that could be misleading, but it should be considered that there is a plethora of polymers that could fall into this classification. According to the International Union of Pure and Applied Chemistry (IUPAC), plastic is a "polymeric material that may contain other substances to improve performance and reduce costs." In this regard, a plastic polymer can be synthetic or natural. Approximately 90% of the total manufactured plastics fall into the synthetic category, including PE (Low-density Polyethylene (LDPE) or High-density Polyethylene (HDPE)), PET, PP, PS, PVC, and PUR, which are petrochemical based. On the other hand, starch, cellulose, and proteins are among the natural polymers that could be incorporated into plastic manufacturing. From another perspective, there is another category for plastic polymers called "bioplastic," defined as a kind of polymer that is bio-based and/or biodegradable. Therefore, although they are often marketed as environmentally friendly and biodegradable polymers, meaning that they break down more quickly than traditional plastics and reduce the amount of plastic waste, according to their definition, not all bioplastics are biodegradable, even though they are from natural resources; thus, they could have downsides. One of the biggest issues associated with bioplastics is that they require specific conditions (such as high temperatures) to be fully decomposed, which are not commonly found in landfills or oceans. This means that, in reality,

bioplastics may not be completely compostable; instead, they could still contribute to environmental issues in the long term, considering that they fragment and decompose faster than petrochemical-based plastics and turn into MNPs in a smaller period.

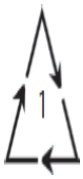
5.2.1.1 Petrochemical-Based Plastics

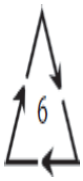
In addition to the general dangers of MNPs, which depend mostly on their physical properties, the polymeric composition of the particles can also determine the type and severity of these effects. Petrochemical-based plastics are among the most manufactured and used plastic polymers and present hazards based on their polymeric composition.

Polyethylene terephthalate (PET) is a high-molecular-weight thermoplastic polymer, consisting of terephthalic acid and ethylene glycol monomer repeats with the IUPAC name of poly(ethylbenzene-1,4-dicarboxylate). Polyethylene terephthalate is widely used in our daily life and in different industries, from food and beverage packaging to polyester fibers in clothing, asphalt composition, and some sensor electrodes. This polymer can be used as a plasticizer in other plastic manufacturing processes to improve its mechanical properties. The emission of dangerous chemicals when PET is heated is one of the key issues of this polymer, as PET can leak hormone-disrupting substances into the environment, such as phthalates and antimony, when subjected to temperatures exceeding 65 °C, which is above its glass transition temperature (Tg) (see Table 5.1). Numerous health issues, including cancer, developmental disorders, and reproductive disorders, have been associated with these chemicals. In addition to being exposed to heat, PET can release its compounds when it comes into contact with some solvents, such as acids or drinks with high alcohol content. Because of this leaching, hazardous substances may be absorbed by the plastic and then passed on to consumers. This is particularly concerning for bottled water, which is frequently stored in PET bottles for extended periods of time. Different life forms can easily access microplastics and di(2-ethylhexyl) phthalate (a PET plasticizer) for ingestion. In fact, zooplankton, and possibly higher trophic levels, are seriously endangered by MNPs and plasticizers derived from

this polymer. Terephthalic acid has been shown to reduce the ability of *Folsomia candida* to reproduce when present in the soil at concentrations above 30 $mg \cdot kg^{-1}$. This would be the cause of concern that with a higher density than water, PET-MNPs are expected to easily reach sediments and freshwater and marine beds, in which they may interact with sediment microorganisms, and other living organisms, where temperature and plasticizer types are the two main factors that determine how PET-MNPs affect them.

Table 5.1 Applications and general properties of plastic polymers.

RIC	Name	Main application (s)	Properties	Density	Tm (°C)	Tg (°C)
	PET	Water bottles, food packaging, shampoo bottles, viscose rayon for clothing	Lightweight, high tensile strength, gas, and transparency barrier	1.38	255–265	67–125
	HDPE	Milk and juice bottles, cleaning products, yogurt packaging	High strength-to-density ratio, good chemical resistance, waterproof	0.944–0.965	130.8	-120
	PVC	Pipes, window frames, credit cards	Rarely degrades, chemical resistant, hardness, high tensile strength (RPVC)	1.38	100–260	85–90

RIC	Name	Main application (s)	Properties	Density	Tm (°C)	Tg (°C)
	LDPE	Grocery bags	Low-cost, good weatherability, good chemical resistance	0.91–0.925	105–115	-100
	PP	Ketchup bottles, yogurt containers, microwave meal containers	Semi-rigid, good chemical, and heat resistance	0.895–09.2	130–171	-20
	PS	CD and cassette cases	Hard, transparent, gamma radiation resistance	0.96–1.05	240	100
	Other	Composites and laminates, baby bottles, etc.	Depends on the polymer type	Variable	Variable	Variable

Polyethylene (PE) is the most widely used and manufactured plastic polymer in the world and is produced with the repeated units of ethylene in three different densities and molecular branch numbers, including High-density Polyethylene (HDPE), long-branched Low-density Polyethylene (LDPE), and short-branched Linear Low-density Polyethylene (LLDPE). PE can degrade into potentially dangerous substances such as Bisphenol A (BPA), accumulate in the environment, poison water supplies, and eventually enter the food chain. These substances have a history of cancer, hormonal imbalances, birth deformities, and other health issues. This polymer generally has a lower density than water (Table 5.1), making it a floating polymer. Many studies have shown the presence of PE particles in the intestines and grills of marine organisms. For example, the investigation of Mak *et al*. [11] on zebrafish revealed that 89% of the intestine was taken up by PE particles, restricting feeding capacity. Aberrant behavior and atypical reproduction-related genes were also observed. Moreover, PE manufacturing can also be harmful to the environment and human health, during which toxic pollutants such as ethylene oxide and Volatile Organic Compounds (VOCs) can be emitted into the atmosphere, contributing to air pollution and respiratory issues [12, 13].

Polypropylene (PP) is FDA-approved and is usually regarded as a safe material for contact with food. However, there are some concerns about its potential health effects. Some of these include chemical leaching and the release of micro- and nanoparticles into food; both can occur when PP is subjected to heat, acidic, and salty contents, or over time due to abrasion. Although there have been no reports on its cancer-causing potential, owing to the absence of BPA in its chemical composition, some recent studies have shown that the toxicity of plastic products, including PP varies depending on the manufacturing process, and may harm the human endocrine system and cause stress responses in cells [14].

Polystyrene (PS) is made from styrene, a chemical classified as a possible carcinogen by the International Agency for Research on Cancer. Exposure to styrene can occur *via* inhalation, ingestion, or skin contact. Studies have shown that workers who are regularly exposed to styrene may be at an increased risk of certain types of cancer, including leukemia and lymphoma. In addition to the toxic

nature of the PS itself, some additives that are used in its manufacturing process may also pose a risk to living organisms. Monofunctional peroxides, benzoyl peroxides, and zeolites are such a kind that could accumulate in fatty tissues of organisms. Hwang *et al*. [15] suggested that PS is typically damaging to Red Blood Cells (RBCs), although the degree of toxicity varies depending on the dose, time of exposure, and nature of the polymer. Polystyrene microparticles (PS-MPs) have also been shown to have hemolytic activity on RBCs, indicating the cytotoxic effect of these particles. Moreover, an increase in IL-6 and TNF-α, both inflammatory factors, in the presence of PS-MPs in the same study, revealed the inflammation-inducing potential of these particles.

Polyvinyl Chloride (PVC), one of the primary constituents of PVC, contains phthalates and other plasticizers that can leach out of the material and pose a risk to human health. These chemicals have been linked to developmental and reproductive problems in animals and are suspected to have similar effects in humans. Exposure to phthalates has also been linked to asthma, allergies, and certain cancers [16]. Another concern regarding PVC is the release of dioxins and other hazardous chemicals during its production and disposal. These chemicals can contaminate soil and water, and have been linked to immune system damage, hormone disruption, and cancer. Polyvinyl chloride also releases heavy metals, such as lead, cadmium, and mercury, which can pose health risks if they accumulate in the human body. These metals are mainly used as colorants in plastic production and can be released during decomposition in natural environments. In addition, PVC-MPs can inhibit biogas production in anaerobic digesters, as effect that has been revealed to be related to the BPA leachates of these particles, inhibiting the activity of hydrolyzing and acidifying microorganisms in the process [17, 18].

5.2.1.2 Bio-MNPs as a New Cause of Concern

Despite the assumption that bioplastics are harmless and safe for humans and the environment, some risks should be considered before using them confidently as alternatives to conventional plastics. Polylactic Acid (PLA), one of the main concerns regarding the toxicity of this polymer, is the potential for the leaching of

harmful chemicals into the surrounding environment through the release of MNPs. Polylactic acid is sensitive to heat and acidic conditions, which can cause it to break down and release its additives into the environment. This can occur in various applications, such as food packaging, disposable tableware, and textiles. For example, plasticizers added to PLA can leach into food when hot or acidic substances are stored in PLA based food containers. Furthermore, PLA biodegradability can have toxic effects. When PLA products are disposed of in landfills, they can break down and release methane, a potent greenhouse gas, and produce acidic substances, which can damage the surrounding environment and soil. Although PLA is a promising and sustainable material, its potential toxicity needs to be considered. Using safe additives and implementing environmentally friendly production methods are necessary to reduce the risk of toxicity. Furthermore, proper disposal and recycling of PLA products are essential to minimize their negative impact on the environment. Polycapro Lactone (PCL) is another type of bioplastic considered to be a safe and non-toxic material, but recent studies have shown potential toxicity concerns associated with its use. One of the main concerns regarding PCL is its degradation products. As PCL breaks down in the environment or the body, it releases small molecules, such as caprolactone, which can be toxic in large quantities. Caprolactone has been shown to cause liver and kidney damage, and lung inflammation in animal models. Another concern is the use of additives in PCL products, which can leach out and cause toxicity. For example, plasticizers, such as Dibutyl Phthalate (DBP) and Diethylhexyl Phthalate (DEHP), are often added to PCL to improve its flexibility and durability. These additives have been linked to reproductive and developmental issues as well as cancer. Furthermore, some studies have suggested that PCL nanoparticles may exert toxic effects on human cells. These nanoparticles can enter the body through inhalation, ingestion, or skin contact, causing oxidative stress, inflammation, and cell death. It is important to note that the toxicity of PCL depends on its use, exposure time, and concentration. For instance, short-term exposure to low concentrations of PCL may not have any adverse effects, whereas long-term or high-concentration exposure may be harmful. To ensure the safety of PCL products, it is essential to regulate the amount and type of additive used. It is also necessary

to monitor the degradation of PCL in the environment to avoid accumulation of caprolactone and other harmful degradation products. Therefore, although PCL is a promising biodegradable polymer with various applications, it is not completely free of toxicity concerns. To ensure the safety of PCL-based products, more research must be conducted on their possible hazards and effects on the environment and human health. It is crucial to balance the advantages of using PCL with the potential risks. It has been established that some biodegradable plastic micro-films, including PCL, Poly(butylene adipate-co-terephthalate (PBAT), Polyhydroxy Alkanoate (PHA), PLA, and starch-based bioplastics, act as persistent carbon sources to boost the abundance of particular fungal taxa, including *Aspergillus*, *Fusarium*, and *Penicillium* [19, 20].

Despite the known negative impacts of the aforementioned polymer types, there is still very limited information on how the chemical composition of MNPs may have different hazardous effects on biota [21], but it has been accepted that MNPs are emerging persistent pollutants that pose a threat to aquatic and terrestrial environments around the world [2]. It has been shown that at the same diameter and concentration, PVC is more toxic to algal species than PS and PE [6]. Besides the chemical nature of the polymer, different degrees of toxicity may be due to their leachate composition, which is generated from additives, such as plasticizers, lubricants, and antioxidants, which are found in greater quantities in PVC than in other polymers, but the composition of leachate is also determined by the pH, salinity, and other environmental factors, not only by the polymeric composition.

5.2.1.3 Biological and Environmental Hazards of MNPs: The Effects on Biodiversity

Due to their tiny size and bioretention potential, MNPs disperse faster than conventional plastics [22, 23]. Second, MNPs are difficult to remove from the environment after they have been introduced, leading to continual accumulation in the environment. They are currently found in numerous organisms and in different environmental media around the world [24], as numerous living organisms at various trophic levels ingest them after they enter

the food chain [25, 26] (see Figure 5.1), causing digestive tract blockage or suffocation. This is especially true for buoyant particles floating on the surface or suspended in a water body [2, 27]. In addition, some by-products of plastic degradation, such as amines, phthalates, plasticizers, and flame retardants, have been proven to be carcinogenic or toxic [28], posing serious ecological risks to ecosystem biodiversity and health. Additionally, persistent organic pollutants, including polychlorinated biphenyls (PCBs) and polyaromatic hydrocarbons (PAHs) [2, 29–32], heavy metals [24, 33], and pathogenic strains can be absorbed by MNPs because of their large surface area and hydrophobicity, which makes MNPs a good vector for pathogenic and organic pollutants between trophic levels and different habitats. Due to these properties, inhaling and consuming microplastics may result in oxidative stress, cellular and DNA damage, and inflammatory reactions, among other things, according to recent reports [24]. It has also been shown that fish consumption of nanoplastics can induce neurotoxic effects on locomotor activity. Oysters and mussels are examples of marine benthic animals that are affected by MP residues in the deep sea [2]. In contrast to their effects on lysosomal membrane stability and long-term inflammation in mussels, they slow the growth rate, affect energy uptake, and interfere with reproduction and the development of progeny in oysters [34]. The charge of MNP particles is another factor that determines their toxicity to some organisms. For instance, positively charged particles seem to be more hazardous to phytoplankton than negatively charged ones because the cellulosic cell wall of phytoplankton is negatively charged and repels the positively charged particles through electrostatic interactions, making these organisms more attractive to these types of MNPs. Generally, reduced reproduction, growth, and fitness of living organisms, as well as biodiversity changes in microbial communities, have been linked to MPs and NPs in aquatic and terrestrial environments.

Micro- and nanoplastics provide suitable hydrophobic surfaces for microbial colonization, creating new microhabitats in the introduced environments. This surface is also called “plastisphere.” The formation of new microbial communities in the plastisphere and in close contact will increase the rate of horizontal gene transfer, in which a wide range of antibiotic and

drug-resistance genes can be exchanged, causing new drug-resistant strains to appear that will threaten the health of all higher living organisms [35, 36]. In addition to microbial associations, different types of antibiotics and biomolecules, such as allergens and pyrogen, may also be adsorbed to the plastisphere based on the polymeric nature, charge, MNPs diameter, surface properties, and environmental factors [37]. It has been revealed that different polymers are colonized by different microbial compositions compared to the surrounding and other polymeric compositions. A rough surface that is the result of weathering, is more suitable for microbial biofilm formation than smooth-surface plastics Considering the biotic and abiotic interaction between MNPs and microorganisms, changes in the behavior of these particles and the immune responses of living organisms are expected. Studies have shown that, while the nature of viruses does not alter their interaction with MNPs, their attachment to these particles may change their conventional route of entry into the target cell. However, some viruses may become inactive and interact with some types of MNPs. *Poliovirus* 1 as an example could be inactivated in contact with PP in the groundwater system according to Benoit and Christophe's study [37, 38]. Generally, the behavior of MNPs is not only affected by their polymeric structure, but their introduction to the environment is a starting point for abiotic and biotic factors that interact with these particles, altering their chemical composition-based impacts on the ecosystem and biota.

5.3 Micro- and Nanoplastics' Distribution and Fate: From Terrestrial and Aquatic Environments to the Human Body

5.3.1 Terrestrial Environments

While the issue of MNP pollution in the ocean has been receiving increasing attention in recent years, their presence in terrestrial environments is also a growing concern, as the soil is the primary and long-term sink for these particles [22]. Microplastics and nanoplastics can enter the soil through the decay of plastic products or by being directly released into the environment

through littering or improper waste disposal. Biosolids, landfilling, fertilizers, plastic mulches, atmospheric depositions, and tire friction can deposit them into the soil, but the major sources of MNPs, causing their release to terrestrial environments, are Wastewater Treatment Plants (WWTPs). Biosolids are the profitable byproducts of these systems, consisting of organic matter and residues of activated sludge that are mainly applied as agricultural fertilizers. It has been revealed that biosolids retain almost 80% of the MNPs that have found their way to WWTPs. As a result, using treated and untreated biosolids as fertilizers, MNP particles that are carried by them move vertically through the soil *via* water infiltration and undergo mechanical, chemical, and enzymatic fragmentation. Therefore, agricultural soils are among the most affected terrestrial environments [36]. Besides biosolids, mulch films, which are widely used for agricultural soil conditioning, can release millions of micro- and nanoparticles and are one of the main direct sources of MNP introduced into the soil, where the recalcitrant nature of the plastic polymers makes it a big reservoir of MNPs. The types of MNP polymers present in soil are highly dependent on their sources. For instance, some reports have revealed that PVC is the main polymer type that contaminates industrial zones [39], while MNP pollution of agricultural soils in China is mostly PE- and PP-based [40].

With a lower density than soil minerals, MNPs cause aggregation and decrease the water binding capacity by lowering soil porosity, which finally changes the integrity, physicochemical properties, productivity, and microbial composition of the soil. Water cycle alteration is another result of porosity decrement, through which water flow behavior, including vertical water movement and water evaporation, is affected [41, 42]. There are many reports on the decrease in Dissolved Organic Carbon (DOC) and phosphorus as a result of introducing PP powder into the soil [43]. On the other hand, when the amount of PP powder exceeded 28% w/w, DOC increased dramatically [44]. Electrical conductivity and pH are other factors that could be affected by MNPs in soil [23, 36, 45]. These effects are complex and depend on various factors such as the type of MNPs, their concentration, and soil type. For example, Li *et al*. have shown that LDPE microplastics at concentrations greater than 1% concentrations will decrease pH but increase the

electrical conductivity of agricultural soil [46]. These alterations in soil properties could change the composition of the soil microbiome and as microorganisms play a vital role in nutrient cycling and decomposition, alterations in the diversity and richness of the soil microbiome will affect its natural function and quality. The formation of distinct microbial populations and changes in the endogenous community of microorganisms as a result of introducing new microhabitats and plastic-based niche construction by MNPs, which act as selective substrate sources, are also expected to harm the natural ecosystem of soil [36]. In contrast, some studies claim that geographical and soil conditions are more important than the presence to shape microbial communities and changing their composition would not be possible just by MNP accumulation. This fact highlights the importance of the impact of climate change, as when changing the climate patterns concurrent with the presence of emerging compounds and MNP pollution, disturbance of the ecosystem's balance will not be surprising because changing the microbiome composition as the main organism of the trophic levels will cause top-down effects on other lower trophic levels and disrupt the balance of the ecosystem.

There is still much to be understood about the effects of microplastics in terrestrial environments, but this is a growing concern that demands more attention as we cannot ignore the effects of MNPs in soil and other places on land. One of the unknown effects of MNPs is climate change, which has not yet been fully understood, but it should be noted that these particles can increase CO_2 emissions from wetlands [47] as well as other greenhouse gases by changing microbial activities and behavior [46]. Several steps must be taken to address this issue, including reducing the use of single-use plastics and improving waste management practices. Additionally, more research should be conducted to better understand the impact of microplastics on the environment and identify potential solutions.

5.3.2 Aquatic Environments

The presence of MPs in the environment was first reported in the marine debris of the western Sargasso Sea, with an average amount of 3,500 particles/km^2 [48]. Currently, they are

widespread in the open oceans, seas, ice cores, polar regions, and freshwater systems. It has been estimated that hundreds of tons of microplastics drift onto ocean surfaces [24, 49]. Studies have found that rivers are one of the main ways for microplastics to enter the oceans, and that 80% of the microplastic pollution in the ocean originates from land (see Figure 5.1). The world's oceans are now known to be contaminated by MNPs, affecting marine species at every level of the food chain, from the North Pole to the South Pole and from the surface to the seafloor, according to more than hundred studies [22]. In freshwater systems, the main sources of MNP pollution are leachates from landfills and soil erosion, improper handling of plastic waste, sewage input, and shipping facilities and activities [1].

According to recent reports, fibers, and fragments constitute the most common MNP shape in water, accounting for up to 80% of the total. This is a significant increase compared to pellets and granules [50–52]. Based on the polymer type, three types of plastics are most frequent in surface waters: PE, PP, and PS, but the type of MNPs differs in oceans and freshwater because of differences in pollution source, location, and hydrodynamic conditions [52, 53]. Freshwater MNPs are more affected by regional sources because they are typically closer to the sources of pollution and have smaller water bodies than the ocean [52, 54]. Regardless of the polymer type, MNPs can be transported over long distances in aquatic environments and accumulate in coastal zones, where they can be ingested by marine organisms owing to their small size. These particles can also enter the food chain via various pathways, potentially posing a risk to human health. The light-absorbing characteristics of microplastics might affect the snow and ice surface in the cryosphere, thereby accelerating melting owing to rising plastic pollution and global warming [55].

5.3.3 Air and Atmosphere

Microplastics in the air are becoming an increasingly concerning environmental problem. The accumulation of microplastics in the atmosphere poses a significant risk to human health and the environment. These particles originate from various sources, including microfibers from textiles, microbeads in personal care products, microplastics generated during transportation, and

particles released from the weathering and breakdown of plastic debris. Microplastics deposited in the sea or other bodies of water can be transported by wind, sea surface spraying, and carried by dust and rainfall into the atmosphere (see Figure 5.1), where the formation of haze and fog can concentrate them. Tire friction is also among the known causes of MNPs deposition into the atmosphere in outdoor environments by carrying particles with road dust. Depending on their size, content, and shape, MNPs that are introduced to the environment by airflow experience various fates, which are influenced by the particle size. When MPs enter the atmosphere, they react swiftly with the water vapor to create aerosols. The smaller the particle size, the longer they float in the air, causing spending more time to be suspended in the atmosphere, and may finally infiltrate terrestrial or marine systems through moist deposition, while larger particles sink as a result of gravity [56].

Owing to their small size, airborne microplastics can easily penetrate indoor spaces and respiratory systems. In addition, traveling fast across long distances indicates that they have the potential to seriously harm living organisms, including humans, as revealed by the presence of microplastic fibers in lung biopsies [57]. According to a recent study, microplastic particles inhaled by city dwellers can accumulate in the lungs, liver, and other human organs, posing several health problems. Ingesting microplastics is another route of their entry into the body, which has also been related to the emergence of cancer, heart disease, and chronic inflammation [58], but the impact of airborne microplastics is not limited to human health. These particles can adversely affect the environment. MNPs can enter the soil through deposition or rainfall, which can harm the soil biota and ecosystem. Furthermore, airborne MNPs can affect aquatic life when they are deposited in water bodies, ultimately leading to ocean pollution. In fact, there is a plastic or MNP cycle among different environments, and as plastics are chemically based on carbon blocks, their release into these habitats, although at an unknown rate, will introduce a load of carbon sources apart from the photosynthetic and net primary production carbons. Consequently, a change in the carbon cycle and greenhouse gas flux is expected through these newly added carbon sources. However, the magnitude and

depth of these effects are not yet understood and should be the objective of future experiments [23].

5.3.4 Wastewater Treatment Plants

As shown in Figure 5.2, there are several steps in the wastewater treatment process, including the primary, secondary, and tertiary treatments. During the first stage of treatment, physical processes such as screening and sedimentation are used to remove large debris and solid waste. During the secondary treatment, biological processes are used to break down organic matter in the water and in the final stage, tertiary treatment, advanced filtration, and disinfection methods are used to remove any remaining contaminants. Despite these advanced treatment methods, MNPs can still make their way through the system because of their small size (Figure 5.2). These particles are produced from various land-based sources and end up in wastewater treatment plants, which are thought to be the links between pollutants and natural environments [59]. Studies have shown that wastewater treatment plants can be a major source of microplastic pollution in waterways, agricultural soils, and food chains [9, 53, 60, 61]. However, previous studies have shown that the number of MNPs in WWTPs' effluents, regardless of the design and efficacy of the system, is season-based. In this regard, Shirazi *et al*. [62], revealed that this number is the highest quantity in spring, with more than 47,000 MNP in 1 L of the effluent, and the least in summer. It has also been found that fibers and microbeads are the two most common types of microplastics found in WWTPs [2]. Among the different stages of WWTPs, the grit and grease removal units trap approximately 45% of the influent MPs, while the primary sedimentation tank holds about 34% of them. This indicates that a total of 80% has accumulated in biosolids, which are mainly used as soil fertilizers that directly release these particles into the soil matrix and present serious risks to both human health and terrestrial biota [9, 63, 64], including terrestrial plants that could absorb and transfer them to other food chain members.

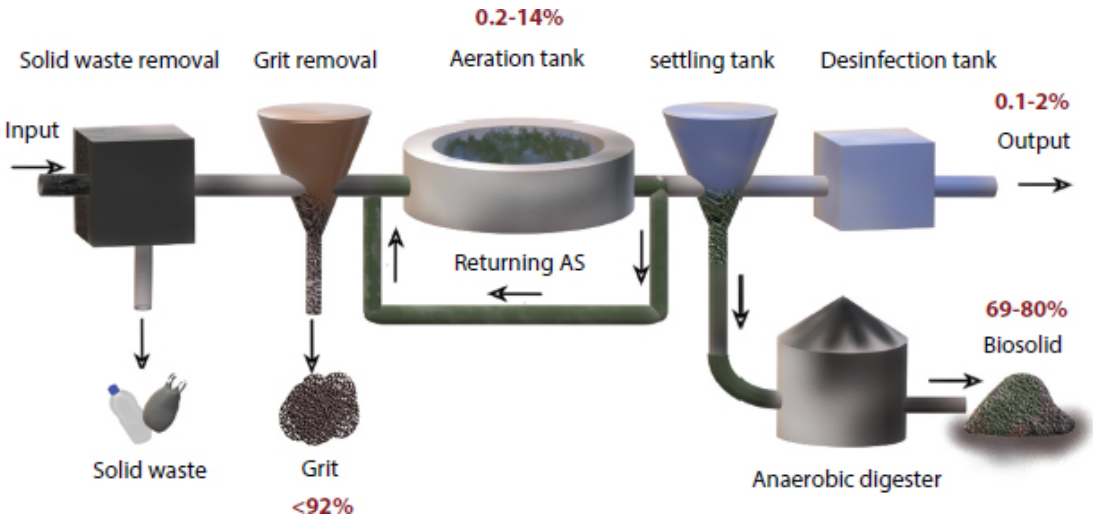


Figure 5.2 WWTPs unit operations and procedures. Each unit can trap and transfer MNPs to the next stage, and finally to the environment/waterways. Biosolids with 69%–80% MNP entrapment serve as one of the main sources of MNP deposition.

In addition to MNP deposition in the environment through WWTPs and their hazards, the toxic chemicals of plastic polymers released by MNPs can rupture the proper function of biological treatment by affecting microbial communities in activated sludge and biofilm structures by penetrating the exopolysaccharide structures. It has been revealed that these chemicals could affect the enzymatic activity of acidogenic and methanogenic bacteria and alter the hydrolysis and biogas production rates in anaerobic digesters. PVC and PS are polymers that could cause these obstacles [17, 18].

5.3.5 Cells and Organs

An increasing body of research indicates that microplastics may enter commonly consumed food products through the consumption of microplastics found in the environment [65], contamination during manufacturing [66], and leachates of plastic packaging [67]. These particles can translocate into human tissues, elicit a localized immune response, and release constituent monomers, hazardous chemicals, and pollutants that are adsorbed from the environment, such as heavy metals and Persistent Organic Pollutants (POPs), such as Polycarbonated Biphenyls (PCBs) and Dichlorodiphenyltrichloroethane (DDT) [68, 69]. In the case of nanoplastics, oral intake, inhalation, and skin exposure are the

most common routes for humans exposed to these particles. Using plastic pipes in drinking water networks and the propensity of MNPs to leak in water under prolonged fluid pressure, drinking water can act as a source of significant levels of MNPs. In addition, leaching MNPs through plastic packaging into food and beverages has been reported in several studies. Reaching MNPs through bioaccumulation is another route that indirectly faces humans for these particles. Seafood is among the most common sources of this type of exposure, and, as mentioned above, atmospheric deposition of MNPs in indoor and outdoor environments has been proven in recent years, exposing humans to these particles through inhalation. This condition becomes more serious, because MNPs can absorb toxicants and pathogenic microorganisms, causing them to enter the body. When the particles reach a size of less than 40 nm, entry through the skin barrier is also possible. This may occur when taking a shower and using cleansing products containing nanoplastics (NPs). Studies conducted *in vitro* and *in vivo* have revealed that MNPs ingested by humans may accumulate in the intestinal lumen, and some of these particles are expelled through feces [70, 71]. *In vivo* experiments have also shown that NPs dispersed in the intestinal lumen can cross the intestinal barrier and move on to enter blood vessels [72]. When NPs are absorbed into human lungs in the alveolus, they may pass through the blood–air barrier and reach the blood circulation system. Additionally, after intravenous injection, NPs can cross the blood–brain barrier, accumulate in the brain [73], and cause brain damage in fish [74]. It has also been shown that NPs can permeate the placental barrier via passive diffusion. Crossing critical biological barriers is a drastic problem that may have adverse harmful effects on humans and other living organisms [75] and the effects of this invasion should be of great concern to make more effort on investigating them.

5.4 The Effect of Abiotic and Biotic Factors on MNPs’ Behavior and Fate

Abiotic factors refer to non-living chemical and physical properties, as well as environmental factors. Temperature, pH, electrical conductivity, salinity, sunlight, seasonal changes, and

oxygen concentration, are among the most important and studied abiotic factors, while biotic factors are those features of the environment that are determined by the presence, diversity, richness, and activity of living organisms. The point is that categorizing these factors separately does not mean that they are independent of each other, but rather influence each other and together determine the properties of each habitat as a connected network. Considering this, introducing to the environment, MNPs encounter mechanical and physical factors, such as weathering, seasonal changes, freezing, animal scratching, and wave force, which cause size reduction and dispersal. Their chemical molecular bonds are influenced by various other physical factors, such as thermal oxidation during which plastics break down thermally at temperatures above 350 °C, and photo-oxidation due to UV-radiation exposure. Studies have shown that different types of microplastics can behave differently at different temperatures, but generally, higher temperatures promote the leaching of certain chemicals, such as bisphenol A and phthalates from microplastics [76, 77], while the extreme heat will break down the polymeric chain, altering its tensile strength, molecular weight, crystallinity, and color, but the point is that temperatures in natural environments rarely reach the levels required for thermal degradation and this process is not typically observed. However, some plastics undergo thermal processing during production, making them more easily biodegradable in the natural environment. Both photo and thermal oxidation introduce new functional groups into the polymeric structures of MNPs, which could act as hotspots for hydrolytic enzyme action. The presence of functional groups in the polymer, such as ester and ether groups, is essential for hydrolytic breakdown that targets the covalent bond of these groups, changing the molecular weight and subsequently the strength of the plastic [1, 78]. Hydrolytic degradation is the result of microbial activity through the secretion of different families of hydrolytic enzymes, which cleave the polymer chain in a process called biotic degradation. This process takes place in three main steps: depolymerization, mineralization, and assimilation, which change the mechanical, physical, and chemical properties of the polymers. One of the main differences between biotic degradation and abiotic processes is the weight loss of plastics which is the result of the assimilation

and consumption of mineralized polymers by microorganisms, a potential that sheds light on this process as a promising way to remediate these pollutants. Oxygen can also induce MNPs biodegradation by promoting the aging process in these particles by generating Reactive Oxygen Species (ROS) in the presence of UV radiation [79]. On the other hand, it is a factor that may induce toxicity by increasing the adsorption capacity of MNPs for pollutants, which can have significant implications for the environmental behavior of MNPs and the ecological risks they pose.

As an important abiotic factor, pH can affect the adsorption capacity of microplastics. Generally, in the pH range of 6–9, which is common in non-extreme environments, the adsorption of surrounding compounds to these particles will increase [80]; however, this is not a constant algorithm, as this effect varies depending on the target compound, polymeric type, size of the microplastics, and other abiotic factors, including salinity. For example, tetracycline adsorption on PVC microplastics decreases with increasing pH (from 5 to 11), primarily owing to the electrostatic repulsion between microplastics and tetracycline [81]. The gastrointestinal tract is also an environment where the pH value varies from extremely acidic in the stomach to alkaline in the small intestine. This means that once ingested, microplastics face a wide pH range of pH and it has been revealed that bisphenol A can rapidly desorb from microplastics, especially PP-MPs, reaching the small intestine due to the pH increase. However, the temperature and sodium concentration of the intestine directly affect the desorption process [77], indicating that abiotic effects on MNPs behavior are not defined by a single factor and should not be considered a one-dimensional effect. The mechanism by which pH influences microplastic behavior varies depending on the type of microplastic and target compound. However, some research articles have mentioned that electrostatic repulsion [81], amorphous structure and functional groups of the polymers [77], and changes in the surface charge of the particles [80] are some of the reasons for the pH-based effects on MNP behavior.

The presence of chemicals and salts may also affect the behavior of MNPs in the environment. For example, the NaCl concentration

has been proven to directly affect the hydrodynamic diameter of PS nanoparticles. In other words, with an increase in ionic strength, the PS nanoparticles have an affinity to aggregate and form larger accumulations. This indicates that PS-NPs will aggregate in seawater [15].

5.5 Conclusions and Future Perspectives

Micro- and nanoplastics are a growing concern in all environments. These tiny plastic particles, less than 5 mm in size, are now widely distributed throughout the world's oceans, rivers, lakes, agricultural and forest soils, air that we breathe, and water that we drink. The sources of MNPs are diverse, and include the breakdown of larger plastic debris, synthetic clothing, and personal care products. Although there is still much to be learned about the impact of microplastics on aquatic ecosystems, research has already shown that they could have serious consequences. One of the immediate concerns with MNPs is their impact on different forms of life. Because microplastics resemble food particles, they have been found in the digestive systems of many animals, from plankton to whales. As a result, these animals can experience physical blockage in their digestive tracts and starvation owing to their inability to consume real food. Additionally, MNPs can carry toxic chemicals that can be released when they break down in the digestive systems of animals, leading to bioaccumulation within the food chain. Another concern is its potential impact on human health. While research is ongoing, it is believed that microplastics can enter the human body through the consumption of contaminated seafood, salt, and drinking water. The long-term effects of MNPs consumption are unclear, but current research suggests that it may lead to inflammatory responses and other health problems. Preventing the release of microplastics into the environment is of paramount importance to mitigating the issues associated with these tiny particles. Some solutions include reducing plastic consumption, properly disposing plastic waste, switching to alternative fibers for clothing, and developing eco-friendly personal care products. In conclusion, MNPs pose a significant threat to all environments, their inhabitants, and humans. Their widespread presence and potential health impacts demand action to minimize their release and prevent further

contamination. As scientists continue to study the effects of MNPs, individuals and businesses must take steps to reduce their use of plastics and limit their impact on the environment. The fate, distribution, and behavior of these particles pose significant challenges that require robust environmental policies and mitigation strategies. The overwhelming evidence suggests that plastic pollution, including MNPs, is a global environmental problem that requires immediate action to address. Fortunately, there is increasing awareness and activism regarding this issue, and we can all play a role in reducing plastic usage and helping to make the world a cleaner and healthier place.

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6
Microplastics and Nanoplastics in Food

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Abstract

With the continuous increase in the use of plastic-based products, there is an increasing global issue of plastic waste generation. Improper plastic waste management causes these plastic products to remain in the environment, and over time, undergo various types of degradation processes, which convert them into small particles. Further degradation eventually leads to the formation of microplastics (MPs) and nanoplastics (NPs) that are dispersed in the environment. There are multiple routes through which these MPs and NPs can reach food products, such as through plastic food containers used for packaging or by being ingested into animals and absorbed into plants. When ingested by humans, MPs and NPs can cause several health issues, as discovered by researchers through *in vivo* studies. These micro-/nanoparticles can also affect livestock by causing many types of diseases in their bodies and can affect their growth.

Keywords: Microplastics, nanoplastics, food contamination, plastic waste, plastic degradation

6.1 Introduction

The era of plastics began with the beginning of the 20th century, and the synthetic plastic industry continued to grow along with the growth of the petroleum and petrochemical industries. This new material, known as plastic, has revolutionized many industries such as automobiles, aerospace, consumer goods, packaging, electronics, medical and surgical tools, building, and construction. The reason behind this dramatic rise in the application of plastics is their unique properties, such as high specific strength, light weight, printability, ease of processing, versatility, and high environmental stability [1–3]. This exponential global growth of plastic production approached 380 million tons per year in 2018, and is expected to triple by the end of 2050 [4, 5]. Nonetheless, governments, many private organizations, scientists, and environmentalists have realized the negative impact of plastics and raised concerns about the damage associated with the improper management of plastic waste [4, 6]. Despite the introduction of various plastic recycling and management policies, improper plastic management is becoming a global issue and a threat to human survival and the environment. Owing to improper waste management, plastic materials remain on Earth for extended periods. When plastic waste encounters various external factors, such as ultraviolet (UV) radiation, the mechanical force of water, oxidation, hydrolysis, and biological metabolism, it is subjected to repeated fragmentation, which continues until the formation of microplastics (MPs) (less than 5 mm in size) and nanoplastics (NPs) (less than 0.1 um in size) [2, 7]. The process is illustrated in Figure 6.1. Along with plastics, many different additives added to plastic products to enhance their properties also become a part of the environment once it is wasted.

These plastic particles have been found to be part of both marine and terrestrial ecosystems. MPs and NPs have also been detected in air by several researchers [8, 9]. It has been shown that MPs and NPs become part of the air, and individuals can easily inhale these particles. Other ways of entering these particles inside the human body are through food products and dermal exposure. Many personal care products, such as cosmetics, lip balm, and toothpaste, are also sources of MPs and NPs. The high consumption and occurrence of MPs and NPs have resulted in their presence in the food chain [10]. Studies have found that dietary exposure to MPs and NPs comes from two main sources: food products contaminated by existing MPs and NPs in the environment and food chain and food contaminated by MPs and NPs due to contact with plastic packaging during storage and transportation [2]. A recent study estimated the average human intake of MPs particles through food to be 39,000 to 52,000 particles per year which is 15% of the general food products consumed in the United States alone [10, 11]. Figure 6.2 shows the possible routes of MPs and NPs entering different food products.

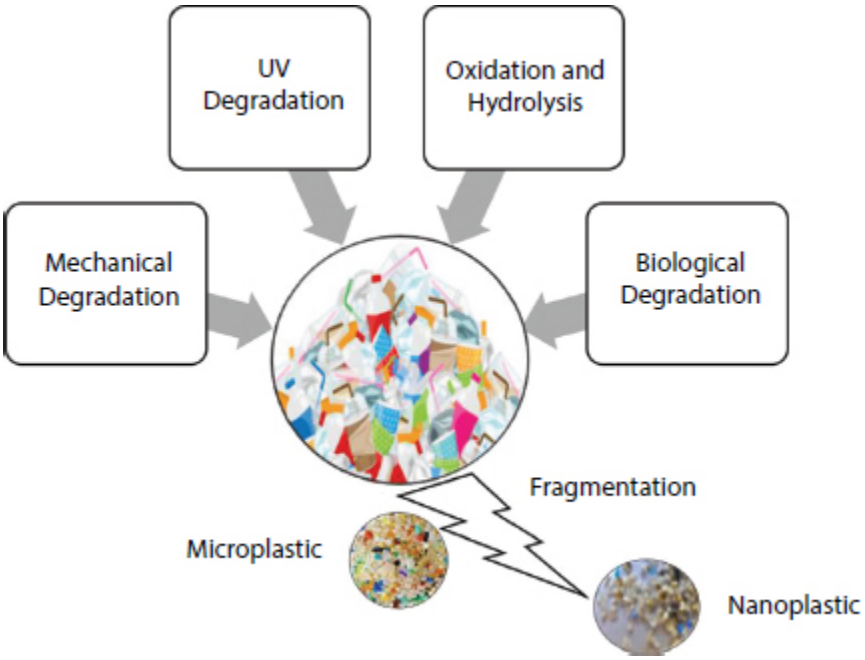


Figure 6.1 Conversion of plastic waste into microplastics and nanoplastics under several types of degradation processes.

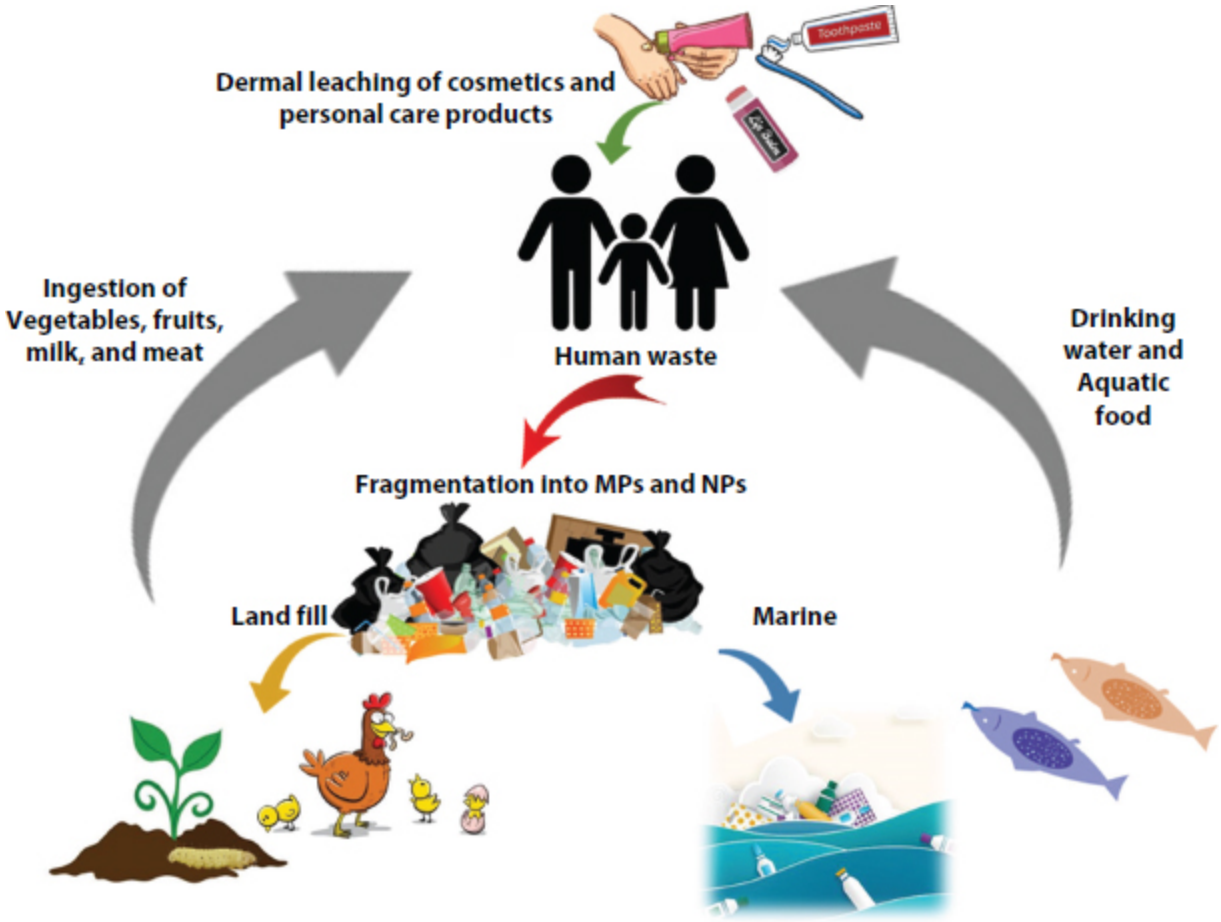


Figure 6.2 Routes of entering MPs and NPs into food and the human body.

6.2 Sources of Micro-Nanoplastics Affecting Food

The increased bioavailability and ubiquity of MPs and NPs in marine and land areas has made it possible for MPs and NPs to become part of many food products. Therefore, the huge rise in plastic waste in the environment and food chain poses a risk to human health [12–14]. Researchers have discovered many ways in which MPs and NPs can enter the human food chain, such as; the consumption of MPs and NPs by animals from their natural environment, food contamination during production processes, and/or

leaching into food and drinks from plastic packaging [15–17]. MPs and NPs have been detected in many food products, such as salt, honey, sugar, fish, shrimp, beer, water, and many fruits and vegetables [18–23], and are found in food products and water in many shapes such as fragments, films, fibers, irregular spheres, and pellets [24].

6.2.1 Micro-Nanoplastics in Seafood

In recent studies, more than five trillion plastic waste trash pieces (higher than 250,000 tons) have been found in the global marine environment. The main source of this plastic in aquatic regions comes from continental sources usually through rivers and industrial and urban sewage [25–27]. Plastic debris is converted to MPs and NPs, and owing to their small size, they are easily ingested by marine species. Apart from the ingestion of MPs and NPs, which are found as waste in rivers and sea, fishes and other sea animals also take these particles from their small preys and feed given to them on farms [28]. After ingestion of MPs and NPs, the absorption of these particles can occur in the bodies of fish and sea animals, leading to their distribution through the circulatory system. Eventually, it can enter different tissues and cells, possibly causing several adverse effects in the body, such as tissue damage, oxidative stress, growth retardation, neurotoxicity, and behavioral abnormalities [29–31].

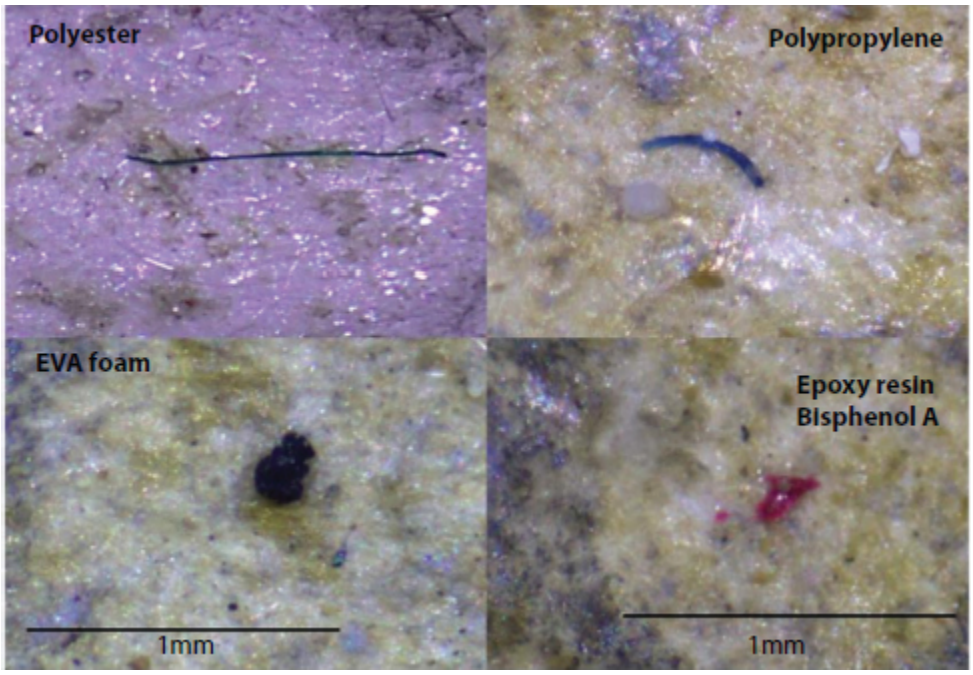


Figure 6.3 Evidence of polymer-based MPs found in blue mussels by the Norwegian Institute for Water and Research [36].

MP-based food contamination was found in many types of fish and seafood, such as shrimps, seabass, prawns, whiting, crabs, flatfish, mussels, lizardfish, grouper, oysters, mullet, tuna, barracuda, mackerel, plaice, haddock, sardines, anchovies, urchins, scampi, cockles, and periwinkles. Seafood eaten without removing the digestive tract, such as mussels, is considered a conservative source of MPs. In a previous research study, it was found that consumption of 225 g of mussels can give an exposure of a maximum of 900 MPs particles [32]. Figure 6.3 shows the MPs found in blue mussels in a study by the Norwegian Institute for Water and Research. On average, the MPs and NPs particles found per fish are between one and seven, in bivalves, 0.2–4 (median value)/g, and in shrimp, 0.75 particles/g [33]. Li *et al*. explained that the measured particle size distribution of MPs in seafood is less than 5–10 µm [34]. In another study, Nalbone *et al*. estimated the

particle size to be higher than 150 µm (up to 6,000 µm) [35]. It is evident that fisheries and marine environments are part of the global food security system. The estimated seafood consumption per capita now exceeds 20 kg/year globally and 7 kg/year alone in the United States. Hence, MPs and NPs in marine environments affect food security and food safety.

6.2.2 Micro-Nanoplastics in Water and Beverages

Water is a critical substance required by the human body for its survival. Therefore, much research is being conducted on the presence of MPs and NPs in drinking water and other beverages. Studies have confirmed the presence of plastic particles in water worldwide, including but not limited to drinking water, surface water, tap water, and mineral water. According to many scientific studies, our bodies require at least 3–4 L of water per day to maintain a healthy body and for the functioning of organs, so water is the most important source of dietary exposure. The intake of large amounts of MPs and NPs along with water is greatly increasing the potential threat to human health [2].

There are many potential sources of MPs and NPs in water, such as oil and gas extraction from the sea, damaged fish nets, litter released in water, and plastic waste from human waste during tourism. A large source of MPs and NPs in wastewater is laundering clothing, which includes synthetic fiber materials, such as polyester, nylon, acrylic, and fleece. These particles, totaling one million tons per year, are added to wastewater streams, which later become part of the environment [37, 38]. Another major source of MPs and NPs in wastewater is exfoliants and toothpaste containing microbeads (on average 4,000 per 1.6 g) [39]. In addition, the degradation of tires produces thermoset plastics in the

form of dust. These particles become shorter over time and become part of the water streams through rainwater [40–42]. Road-marking paint also contributes to 10% of the MPs entering the environment. Water from these rivers or other storage sources is usually treated at wastewater treatment plants; however, nanosized plastic particles are still found in the treated water, as shown in the SEM images of water in Figure 6.4 [43]. Rainwater or storm drains take away with them the plastic waste from the land and eventually become a part of lakes, rivers, and streams. River water and its streams also carry MPs and NPs to larger freshwater bodies, such as ponds, lakes, and wetlands, as well as to the ocean. Therefore, the presence of MPs and NPs in water is an important issue that must be resolved as soon as possible.

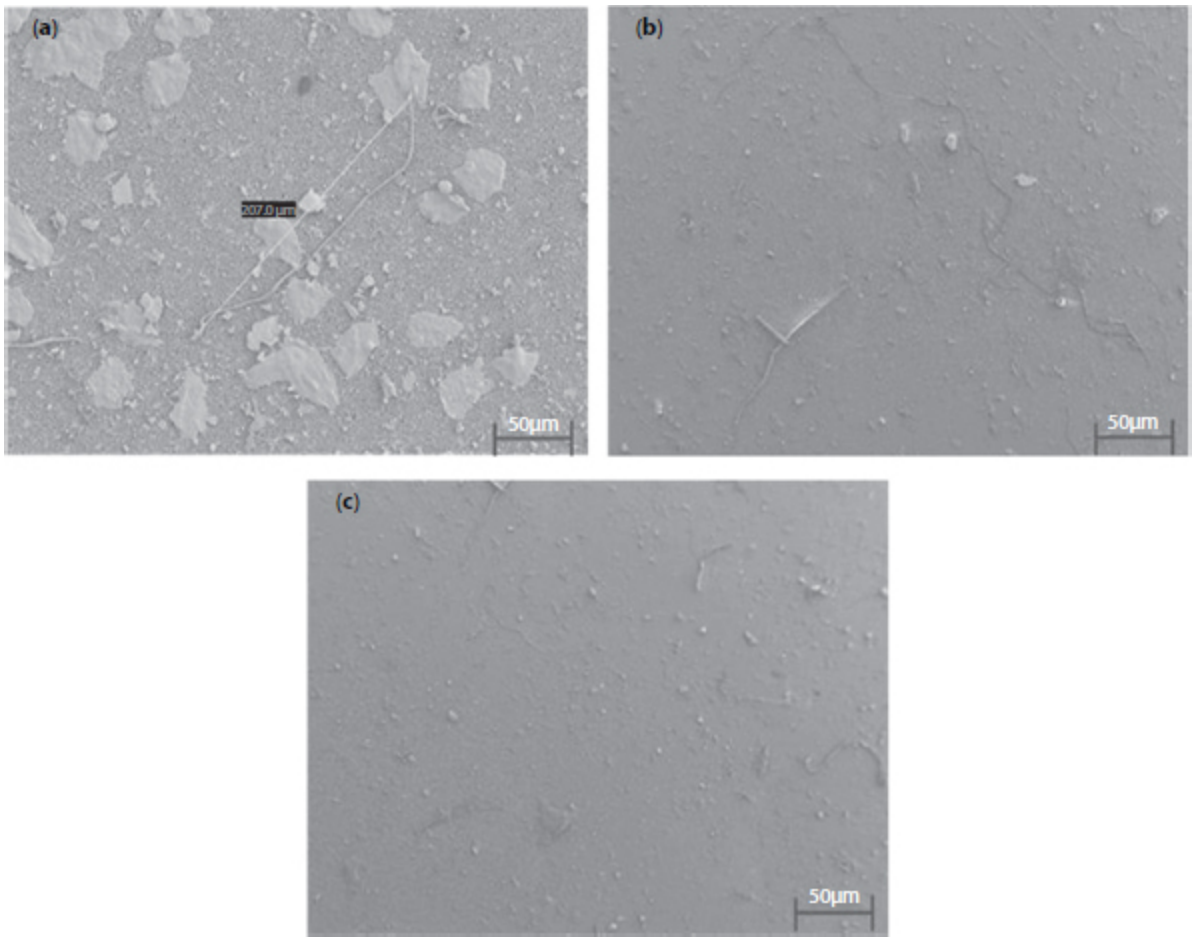


Figure 6.4 SEM images of MPs detected in (a) raw water, (b) treated water, and (c) tap water [43].

Recently, researchers have found that packed beverages (e.g., bottled drinking water, milk, beer, juice, and other refreshments) are also contaminated with MPs and NPs, and humans are orally exposed to these contaminants every day. Most of these particles originate from drinking water, which is the main ingredient in beverages [44]. Shruti *et al*. reported the number of MPs found in several beverages such as soft drinks, energy drinks, and cold tea. The average number of MPs found in soft drinks was 40 ± 24.53. A total average of 14 ± 5.79 particles were detected in the energy drink samples. Approximately 11 ± 5.26 MPs particles were identified in the cold tea samples. The detected MPs were in the shape of small fibers that were

mostly blue and brown. The particle sizes of MPs and NPs detected in different beverages were in the range of 100–3,000 µm [45].

6.2.3 Micro-Nanoplastics in Meat

Poultry and meat are the major sources of proteins and other key nutrients required by the body for good functioning. According to a rough estimation, meat is the only food available for early humans to survive. With the civilization of human beings, new food options have been introduced; however, meat intake is still quite common in humans, and new methods have been introduced for the processing and preservation of meat. In recent studies, MPs and NPs have also been detected in the meat of various animals because of MPs ingestion by animals, mainly from their feeds and forages [46, 47]. The ubiquitous availability of these plastic particles in groundwater, surface water, and wastewater in the form of fibers, fragments, and films has also become a source of MPs intake during animal drinking [48]. Most animals consume plants, which can absorb NPs through plant roots and transport them from the xylem to their edible parts [49]. The ingestion of MPs and NPs can also occur during the grazing of animals from garbage, especially in developing countries, where the problem of plastic waste is more common and livestock are not restricted to confined areas [50]. These ingested plastic particles can also release chemicals slowly in the rumen, which can become part of the bloodstream, tissue, and contaminate milk [51]. MPs were also reported to be transferred into chickens by eating earthworms from home gardens. MPs of various sizes (generally from 5,000 µm to 100 µm) have been found in the digestive system of chickens [52, 53]. Zhang *et al*. reported an average MP consumption of 3–677 mg/week in domestic animals [54].

Another route of MPs that can become part of the meat is from cutting boards. Most polyethylene-based plastics were found in meat samples that were cut on four distinct types of cutting boards with different surface textures and colors, as shown in Figure 6.5. Washing the meat before cooking or processing reduced the number of MPs but did not completely remove the particles. In such cases, the high mechanical properties of cutting boards play a major role, in avoiding the contamination of meat by MPs [55].

6.2.4 Micro-Nanoplastics in Fruits and Vegetables

As explained in previous sections, water can carry MPs and NPs while flowing on land, and this running water can help in the migration of these plastic particles. When these plastic particles become part of the soil, they begin to affect plant roots. Other than migrated MPs and NPs, agricultural films, organic fertilizers, and atmospheric deposition are other means that help in the contamination of soil with MPs and NPs [56, 57]. These MPs and NPs in the soil produce many types of biogeochemical processes that change the nutritional value of the soil. Based on their size and shape, MPs and NPs can penetrate the seeds, roots, stems, leaves, and fruits. The amount of water uptake also affected when MPs and NPs block seed pores [58, 59]. Figure 6.6 demonstrates the various mechanisms by which MPs and NPs affect plant growth and contaminate fruits and vegetables.



Figure 6.5 MPs found in meat when cutting over cutting boards differ in color and surface texture [55].

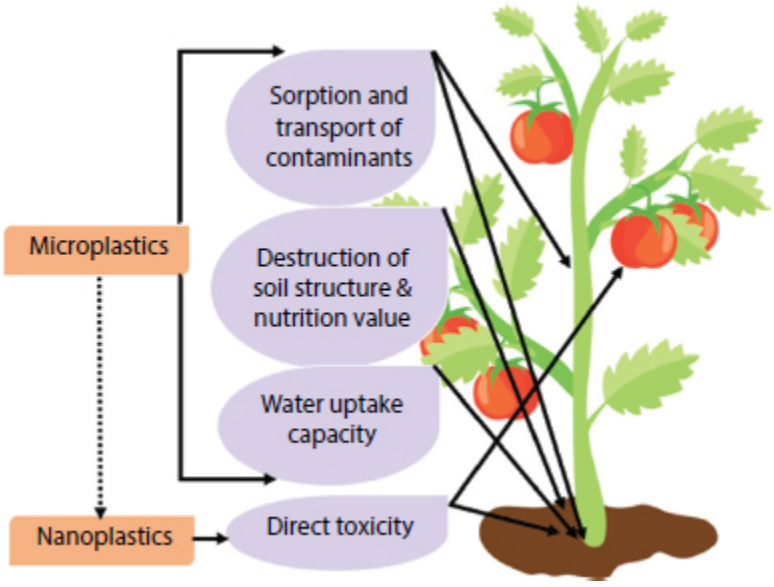


Figure 6.6 Mechanisms by which MPs and NPs affect plants.

6.2.5 Micro-Nanoplastics in Other Food Sources

In addition to the above-mentioned main food sources, MPs and NPs are also detected in exceedingly small amounts in many other foods. Most foods that are processed and packed contain MPs and NPs such as salt, honey, and sugar-refined flour. Salt is obtained from two sources: sea or lake water, or mining rock salt. As discussed in previous sections, seawater or lake water is highly polluted by MPs and NPs; hence, there is a high chance that the salt obtained from these sources is contaminated by MPs and NPs [2]. Traces of MPs and NPs were also found in honey, and the route of entry of MPs and NPs in honey was through honeybees. Honeybees have been proven to be able to gather MPs particles from the air and during their interactions with soil, plants, and water. Plastic particles and small fibers are usually stuck in the wings or the bodies of worker bees during the nectar collection process, which eventually causes honey contamination [60].

During the last decade, the growing trend of takeaway food in disposable containers has also caused a huge increase in MPs and NPs in food products. In addition to MPs and NPs, the thermal degradation of plastic due to hot food also causes contamination of food by plastic additives. The higher temperature of food or the use of food containers in microwave ovens accelerates the transfer of MPs and NPs, leading to serious potential threats to the human body [2, 61]. Du *et al.* reported that these MPs were mostly derived from plastic containers, and that the plastics detected were polypropylene, polystyrene, polyethylene, and polyethylene terephthalate. The amount of MPs and NPs detected in food ranges from three to 29 particles per container [62]. A serious issue was raised by Li *et al.*, who found

polypropylene MPs in infant feeding bottles, and the content was reported to be as high as 16,200,000 particles per liter [63].

6.3 Impact of Micro-Nanoplastics

The presence of MPs and NPs causes many hazardous effects, either directly by affecting the human body or damaging the food chain by affecting the health of animals and seafood, as well as by affecting the production of crops and edible plants. Studies on the direct effects of MPs and NPs on the human body are extremely limited. However, some diseases have been observed because of the presence of plastic particles and many plastic additive chemicals in the human body. However, MPs and NPs cause many diseases in animals and seafood, which are eventually transferred to humans during meat ingestion [27, 64, 65].

6.4 Direct Impact on Human Health

Numerous *in vitro* and *in vivo* studies have found that MPs and NPs along with plastic additives cause serious effects on the human body which are either short-term or long term such as damage to different body cells, inflammation, apoptosis, necrosis, oxidative stress, autophagy, and immune responses [7]. Below are the details of these effects on the human body.

6.4.1 Oxidative Stress and Apoptosis

Some important daily physiological activities require superoxide radicals and hydrogen; however, excess of these radicals can damage the antioxidant defense system, which results in oxidative stress and more damage to proteins, DNA, proteins, and lipids [66]. The major cause of oxidative stress is an increase in toxicity due to an increase in radical

generation in cell lines, as explained by many researchers [67–69]. PS-based MPs and NPs transferred through food containers were found to cause oxidative stress and autophagic cell death. Cationic PS-based NPs generate reactive oxygen species (ROS) and endoplasmic reticulum (ER) stress in lung cells [70, 71]. Apoptosis was observed due to the modified and unmodified/functionalized PS-NPs in several human cell types [72–75].

All the above-mentioned outcomes were obtained from *in vivo* studies, and no severe effect was observed in the in vitro studies. In the limited *in vitro* studies that have occurred until now, no severe toxicity was found in the heart, liver, kidneys, duodenum, large intestine, ileum, or spleen of mice through ingestion of different types of MPs [76]. A few authors have found that oral exposure to PS-based MPs and NPs can result in neurotoxic responses, liver inflammation [77], weight loss [78], and changes in amino acid and bile acid metabolism [79, 80].

6.4.2 Autophagy

Another impact discussed by many researchers is that of autophagy. This involves the engulfment of intracellular waste by autophagosomes, which eventually fuse with lysosomes and are digested by lysosomal enzymes [81]. Lim *et al.* [82] explained that NPs help induce autophagy, which causes the activation of endoplasmic reticulum (ER) stress. Functionalized PS particles also activate autophagy by generating ROS and ER stress [70]. Many researchers have also discussed the damage caused by amino-functionalized PS NPs to lysosomes as a result of lysosomal dysfunction, eventually leading to cell death [81].

6.4.3 Damage to Different Body Cells

The ingestion of MPs and NPs causes major damage to the gastrointestinal cells because of the exposure of MPs and NPs in the gastrointestinal tract [75]. *In vivo* studies have shown a negative impact of MPs and NPs exposure on the gastrointestinal barrier. After exposure to an exceedingly small number of NPs for just 4 h, a significant reduction in cell viability was observed. Subsequently, the NPs led to apoptosis in cells and caused damage to the cell monolayer. Hence, it was concluded that over time, NPs may damage the gastrointestinal barrier protection capability, creating an undesirable impact on the intestine. Moreover, it was found that the damage to gastrointestinal cells due to MPs and NPs depends on the size, concentration, and surface chemistry of the plastic particles [83]. Forte *et al.* discovered that at 1 and 2 lg/mL, there was no effect on cytotoxicity, but 44 nm NP particles at 10 lg/mL, caused a considerable reduction in cell viability [84]. However, there was a huge inconsistency in some toxicity results; for this reason, there is a need to conduct a factorial study design to achieve a conclusion [83].

Another impact was on immune function due to the ingestion of MPs and NPs, which include interleukin secretion, impaired expression of surface receptors, and immune cell death [5, 85–87]. Approximately 10–2,000 lg/mL of carboxylated polystyrene (PS) NPs affected cytotoxicity, cytokine release, chemotaxis, oxidative burst, phagocytosis, nitric oxide production, and myeloperoxidase generation in the tested cells [83].

The production of neurotoxicity due to the exposure of MPs and NPs to specific plastics has been reported, mainly because of oxidative stress, which damages neurons [88]. A comprehensive study has focused only on the effects of MPs and NPs of functionalized PS on various human cells.

In an *in vitro* study, carboxyl-functionalized PS NPs and polyethylene glycol with sizes in the range of 45–70 nm caused toxicity and affected metabolic activity [89]. NPs can easily enter the blood–brain barrier, leading to neurotoxicity. There is a need to further investigate the ways in which plastic particles affect human nerve toxicity and the mechanisms underlying the development of neurodegenerative diseases.

6.4.4 Inflammation

Inflammation is an important cause of internal damage that results in the development of various diseases [90]. Many authors have reported that MPs and NPs can alter inflammatory factors in various cells [91–93]. Ingestion of large MPs and NPs (202 nm–535 nm) may produce inflammatory effects on human lung cells. In addition, the same concentrations of amino-functionalized MPs and NPs (i.e.,1, 50, and 100 lg/mL), in size 50 nm may mark the release of pleiotropic cytokines (IL-6 and IL-8) [83].

6.5 Affecting the Food Chain

In *Section 6.3*, a detailed explanation of the effects of MPs and NPs effect on human health is mostly based on *in vivo* studies, and no major health effects were noted directly on human health and organs. The proportion of these MPs and NPs is not very high, which can cause a considerable directly and rapid impact on human health. Most of the time, the effects we observe on the human body occur because of the ingestion of food from infected animals and seafood. It was observed that all these health conditions explained above because of the exposure to MPs and NPs also occur in animals. Animals when ingesting MPs can cause inflammation in their bodies and cytotoxicity, including cell damage, oxidative stress, genotoxicity, and

immunotoxicity [94]. These health impacts in animals are more severe because of the high intake of MPs and NPs, as mentioned in *Section 6.2*. In addition to the effect is on livestock and sea animals, another major effect was also found in edible plants. When the number of MPs and NPs increases in the environment, an interaction occurs between these particles and plants, which causes oxidative stress that adversely affects plant metabolism, photosynthesis, and other growth parameters. Furthermore, the phytotoxicity caused by MPs and NPs particles is based on the characteristics of the plastic particles, such as size, amount, shape, chemical functionality, and type of plastic. Other factors from the plant end were the species, tissues, and stage of growth. MPs NPs also indirectly affect plant growth by altering the soil properties, nutritional value, and microbial community [95]. This eventually directly affects the food chain and indirectly affects human health.

6.6 Detection of Micro-Nanoplastics in Food

As discussed in detail, MPs and NPs are now part of many food products that are either transferred through packaging materials or from animal and plant sources. Currently, the challenge is to detect the amount of these particles in food sources, and standardized testing techniques are required to follow a mechanism for the detection of plastic particles in food items. Various research methods have been used to identify plastics in food products. Before starting the quantitative analysis of MPs and NPs, it is necessary to prepare a sample for testing. In most cases, it is necessary to separate the material of interest from the matrix or the rest of the material. Techniques that can be utilized for this purpose

are filtration, flotation or sedimentation, and matrix dissolution. There are three ways to eliminate organic materials from MPs: acidic/alkaline chemical digestion, and enzymatic digestion. Each instrument and analytical method offers a unique method of detection [96]. Table 6.1 summarizes the analytical techniques used by researchers for the detection of MPs and NPs in food items.

Table 6.1 Characteristics of various detection methods used for identifying MPs and NPs in food [96].

Analytical technique	Method benefits and limitations
Micro-IR Spectroscopy	• Small amount of sample required • Nondestructive • Polymer type can identify • FTIR-microscopy can provide data about the micron-sized MP particles and give information on the number of particles of polymer type. • Expensive equipment and cannot identity degraded samples due to weathering
Micro-Raman Spectroscopy	• Small amount of sample required • Partial Nondestructive • Polymer type can identify • FTIR-microscopy can provide data about the micron-sized MP particles and give information on the number of particles of polymer type.

Analytical technique	Method benefits and limitations
	• Expensive equipment and cannot identity degraded samples due to weathering • Not reliable for degraded samples • Time-consuming for broad imaging • Sample may degrade due to long-time exposure to laser
Pyrolysis gas chromatographymass spectrometry (Py-GC-MS)	• Detect plastics and organic additives • Measures small concentration (<0.5 mg) • Limited sample pre-treatment is required • Time-consuming • Data on particle number, shape, and size is not provided • Expert operator is needed
Secondary Ion Mass Spectrometry (SIMS)	• Good spatial-resolution imaging • Can provide data on mixtures of plastic particles

Analytical technique	Method benefits and limitations
	• low µm range detection is possible • Can detect inorganic and organic chemical contaminants (heavy metals) • Costly • Complex working principle and difficult to operate • Vacuum compatible sample is required
Differential scanning calorimetry (DSC)	• Quick and easy to operate • Unknown material can be identified by comparing it with reference material • Not able to detect a mixture of plastic particles

6.7 Conclusion

There is an increased concern about the huge rise of plastic waste in the environment and its impact on human lives and the food chain, and many researchers have proven that small plastic particles can accumulate, transfer, and contaminate the human population. However, contrary to the visible massive plastic waste items and their side effects, another growing concern is the huge increase in environmental contamination due to microplastics (MPs)

and nanoplastics (NPs). These MPs and NPs may have severe toxicological effects on human lives, which have not been studied directly in human bodies, but *in vivo* studies have confirmed many health issues that may occur in humans as well. Many characterization tools and analytical techniques are used to detect the type, size, and shape of MPs and NPs in food. Overall, the side effects of MPs and NPs on human life depend on numerous factors such as plastic chemical functionality and biocompatibility, size, and amount of plastic ingested.

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7
Microplastics: Properties, Effect on the Environment and Removal Methods

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Abstract

Microplastics have spread worldwide, contaminating the ecology as they do so. They are tiny particles measuring less than 5 mm and have emerged as a pervasive and alarming environmental issue. Their widespread presence in various ecosystems, from oceans to freshwater bodies, and even the atmosphere, has raised concerns about their potential adverse effects on both the environment and human health. Industrial, domestic, agricultural, and livestock effluents contain microplastics that cannot be eliminated by sewage treatment plants. These wastewaters also contain micropellets from cosmetic products, especially facial scrubs and textile fibers; the latter result from the shedding of particles while washing textiles. Municipal wastewater treatment facilities remove large plastics from wastewater; however, after multiple water treatment steps, some microplastic or nanoplastic particles are still present and are released into the effluent. In this chapter, the authors have focused on the sources,

distribution, and effects of microplastics on the environment and the goals and initiatives set up by the United Nations Sustainable Development Goal (SDG 14).

Keywords: Impact and removal of microplastics, microbeads, microplastic

7.1 An Insight Into Microplastics (MPs)

Owing to the adaptability of plastics, they permeate all aspects of our lives in today's society, including workplaces, homes, cars, and other items we use. Pens, devices, computers, toys, phones, and just much all of the other significant items we use every day can be included. Plastics are utilized instead of natural materials, such as paper, leather, or glasses because of their long-lasting, convenient, and affordable qualities.

Currently, there is no indication of a decrease in the manufacture of plastic materials. The continuous use of this artificial plastic has entailed considerable litter in the environment. Improper disposal systems have led to the widespread use of plastic waste. Plastic debris is a well-known pollutant in the ocean that has received much attention since it was first reported in the 1970s. MPs are easily visible and are known to have negative impacts on the environment. The main sources of plastics are land-based, originating from urban and storm runoff, beach visitors, sewer overflows, construction, and illegal dumping. A considerable amount of plastic waste is added to marine ecosystems, which pollutes water resources for several years or millennia. Therefore, this topic has become a serious concern in research and is of utmost importance for the abundance of MP particles [1].

Fragmentation of plastic debris can be classified into four categories: i) MP, ii) Mesoplastic, iii) Nanoplastic, and iv) Macroplastic. MPs are composed of plastic polymers of “micro” size. They are categorized as solid man-made particles with a specified size range of less than 5 mm. A new goal that poses difficulties for traditional sampling and reporting procedures is to determine their impact on biota and human health. MPs have been found in the surface waters of remote areas of the Arctic and Antarctic oceans. As MP have a lower density than water, they remain on the upper layer of water where other marine fauna survive and other photocatalytic activities occur; as a result, they are ingested by marine invertebrates, fishes, sharks, whales, turtles, and other mammals. This ingestion has various effects on retention in the gut, stomach storage capacity, and impaired sanitation. In addition, toxic chemicals, additives, and by-product plastic stabilizers are carried by these particles, releasing ecotoxicological effects in the environment. According to these studies, plastic waste accumulates pollutants such as polychlorinated biphenyls (PCB), polypropylene (PP), and polyvinyl chloride (PVC), up to a million times the amount observed in seawater. To obtain a clear view of MP concentrations in the marine ecosystem and the risks posed to marine life and linked ecological processes, it is important to know the measures of MP sediments [2].

7.2 Microplastic Definitions

According to the UN Environment, MPs are the “solid phase materials, ranges <5 mm, water-insoluble, non-biodegradable and plastic composition.” The upper range of MP size was defined by the National Oceanic and Atmospheric Administration (NOAA). According to them, MPs are plastic fragments of less than 5 mm in length.

They can be found in the form of pellets, foams, fibers, films, and microbeads in many colors. None of the prohibition proposals on MP also covers the smaller size category. With a typical netting size of 330 µm, several studies have defined sizes of up to 330 µm based on plankton. According to the European Commission, MPs should be sampled in seawater between 20 µm and 1 mm in size [3].

The International Union for Pure and Applied Chemistry (IUPAC) guidelines recommend MP dimensions between 100 µm and 1 µm. MP can absorb organic pollutants, trap heavy metals, and simultaneously release harmful additives that are incorporated into the plastic during production. Biodegradable MPs that degrade in the form of smaller fragments mostly have bio-based products, and this biodegradability level in the environment has not yet been estimated. Nanoparticles range from 1 to 100 nm in at least two directions [4].

7.3 Properties of MPs

MPs are durable, lightweight, and flexible, making them widely used in products. As of their micro size, many of them have similar physiochemical properties but have hazardous impacts. MPs are often known as “mermaid tears,” as they are heterogeneous in characters with different shapes, colors, sizes, specific densities, and polymer types [1]. As their sources of plastics are different, the physiochemical properties also vary: 1) Shape MPs have complex structures, including short or long fibers, polygonal, circular, and triangular for 2D structures, and pellet-shaped, cone-shaped, cuboidal, spherical, and pyramidal for other irregular shapes; 2) The color of MPs is different: some are red, yellowish, white, and brown, as detected microscopically; and 3) complicated chemical

composition in PP, polyethene (PE), polytetrafluoroethylene (PTFE), PVC, polystyrene (PS), and many others. as they polymerize structures that make them persistent in environment with non-biodegradable properties [5]. They undergo biotic and abiotic degradation, depending on the chemical and physical characteristics of the polymer. The general method of degradation is microbial degradation, which disintegrates the plastic and photo, or thermal degradation leads to the oxidation of MPs under favorable conditions; however, the degradation process is so slow that it seems negligible in marine ecosystems. This can lead to the small fragmentation of MPs, but complete breakdown was not examined. They come in several varying densities, and their buoyancy depends on their composition. Therefore, they both float or sink and accumulate in the environment and are difficult to separate from other matter. As many of them have a lower density than water, they float in the upper layer of oceans and have different colors, sizes, and shapes. Diverse marine organisms, ranging from zooplankton to higher organisms, consume MPs, which propagate the food chain. Hence, this contamination in oceans has become a considerable threat globally and now the oceans are notoriously called “plastic soup.” The MPs are so small that they cannot be filtered by sewage or wastewater treatment plants. MPs found their way to reach every corner of the globe and pollute the ecosystem. Once they contaminate the environment, MPs become too vast and excessively dispersed, making them difficult to remove from the environment. Any other remediation can affect marine organisms, leading to ecological damage owing to their small size [6, 7].

MPs are products made from petroleum, which have a large surface area, making them easily reactive with other hydrophobic pollutants, such as plasticizers, colorants, polycyclic aromatic hydrocarbons (PAHs), and retardants.

As soon as they reach the marine environment, they become accessible to marine organisms, introduce toxins into the food chain, and integrate into the food web, greatly exacerbating issues that are already present. Many measures have been developed to identify and quantify MPs, but there is still a wide gap in the knowledge regarding sampling and other impacts [8].

7.4 Primary and Secondary Microplastics

Depending on their origin, MPs are differentiated into primary and secondary MP. The primary MPs were in the millimeter range. They are mostly used for regular purposes, such as in offices and household items such as soft toys, pens, fillers in furniture, air blasting media, personal hygiene products, and even in face-cleansers and medicines as vectors for drugs, toothpaste, and thermacol. The major primary MPs enter the environment through land-based activities. Primary MP can enter in the environment through the scraping of tires; the main pathway is road runoff, which is introduced in water bodies by industries through the runoff/effluent of plastic. Laundering of synthetic textiles was the main source. According to previous studies, a synthetic company releases 1,900 MPs in a single wash into water bodies. Pharmaceutical industries have manufactured MP-associated drugs for better results that drain down in water bodies. Cosmetics such as face wash, gel, foam, scrub, body wash, and soaps contain many micro- and nano-sized MPs that drain directly down into the water system [9]. They rely on routine filtration systems in sewage water treatment or wastewater treatment plants, and there is no way back once they enter the natural world through waterways. The environment audit committee in UK

parliament says, “A single shower results in thousands of MP in the drain.” Secondary MPs are created by the decomposition of bulkier plastic particles, primarily plastic waste, rather than by being manufacturing in factories or synthetic textile businesses. Owing to structural dissolution, collapse can be chemical, physical, or biological [10].

Physical processes involve photo- or thermal degradation to reduce the plastic into small fragments, but it takes a long time under favorable conditions.

The chemical process mainly involves the degradation of MP with the help of chemicals, but it can also affect the marine environment.

Biological processes involve the degradation of plastic debris *via* microbial action. These processes ensure that the plastic is decomposed into smaller fragments from the unmanaged plastic. They remain in the environment for an indefinite time and are available for marine organisms to get confused between food particles and MP and enter the food chain. When biodegradable material is subjected to decomposition, only starch content breaks down, leaving behind MPs. MPs obtained from paints, synthetic textiles, truck tires, and cars are classified as primary MP [11, 12].

7.5 Microbeads

Plastics used in personal care and cosmetics products, mainly for improving skin color, cleansing action, or other functions, are termed microbeads. The microbeads were 1 mm in size. They are mostly spherical and amorphous in nature. Microbeads are also referred to as microcapsules, nanocapsules, nanospheres, and microspheres. Most microbeads used in toothpaste, cleansers, scrubs, face washes, sunscreens, shaving creams, lotions, and skin

exfoliators are smaller in size, making them abundant in such products [13]. However, some of these microbeads were visible to the naked eye. Their reactivity, characteristics, and activity are extremely similar to those of primary and secondary MP, and they have the propensity to attract and hold onto contaminated or harmful compounds when they come into contact [14]. Microbeads trap toxic chemicals that are carcinogenic, and most skincare products that contain microbeads in abundant amounts enter the skin and cause cancer [15]. Owing to several instances of “down the drain,” there is a cause for concern regarding cosmetic items that include microbeads. These microbeads immediately leak into the water system after washing off the user’s hands or face [16].

7.6 Impacts of MPs

The versatile property of plastics due to their chemical building block structure makes them a growing concern for the environment. Substitution of natural products with plastics leads to an array of environmental and health problems. The use of plastic has created a boom in many industries, but it has also become a threat to the environment owing to its non-degradability property for thousands of years. Plastic travels in many ways, is un-recycled, ends up in the oceans through water channels and remains in the form of MPs for a long time. MPs have a variety of negative environmental effects. The concern associated with these minute particles is that they may harm marine ecology and adversely affect anthropogenic activities. Due to its socioeconomic implications, achieving sustainability is difficult for many development targets [17, 18].

7.6.1 Ecological Impacts

Fish-eating birds such as osprey and sea eagles are among the diverse aquatic biota, which also includes animals such as phytoplankton, zooplankton, and fish. The marine food chain is initiated by phytoplankton, zooplankton, and other chain initiators. They are millimetric in a size range similar to that of MP [19]. Therefore, these MPs are available as food for marine flora and fauna. Ingestion of MP by these zooplankton leads to the bioaccumulation of MP in the food chain. Studies have shown that Daphnia is a marine organism (zooplankton) that is very sensitive to changes, such as pollution and toxins and acts as the main component of the food chain. The downfall of these main planktons poses a major threat to the entire food chain. Ingested MP can cause starvation of an organism due to physical impacts such as a damaged stomach, blockage in the digestive tract, and reduced reproductive efficiency. Accumulation of MPs in the gut of organisms reduces their lifespan and immunity, making them prone to diseases [20, 21]. They not only remain in the gut of the organism but also dissimilate from other adjoining tissues or lead to organ damage. Studies have shown that MPs can translocate from the stomach to the liver, hepatopancreases, and circulatory system, and can be retained in the tissues. Like in one of the marine organism mussels that humans also consume. MPs act as a source of energy for algae and smell similar to the food consumed by fish. Many fishes consume MPs in the form of phytoplankton and have become addicted to consuming plastic. When male and female fish consume MPs, their genes are altered, resulting in behavioral abnormalities, disability, and eventual extinction of the species. MP consumption also causes stress and necrosis of the liver. Consuming MP poses a serious hazard to ecology because it alters organism structures, which might have a

detrimental impact on secondary producer growth or photosynthesis [22].

7.6.2 Chemical Impacts

High reactivity, large surface area, and adsorbed contaminants lead to the exposure to chemicals to the environment. Chemically contaminated MPs contain additives such as plasticizers, stabilizers, fillers, and fire retardants with toxic groups such as alkyl phenols, phthalates, and polybrominated diphenyl ethers that cause endocrine disruptors. Endocrine glands produce hormones that regulate metabolism, growth, and sexual function in the body [23]. They release toxic chemicals into the body, which affects the system. Various studies have shown that PE consists of toxic additives that are responsible for the decrease in impaired reproduction, and that this impaired synthesis is responsible for egg formation in fish. The intake of Persistent Organic Pollutant (POPs) that are known as “forever pollutant” are transferred from food to organisms and thus causes respiratory problems [24]. Anisakis disease is commonly found in humans and is caused by the consumption of parasite-infected fish. The leaching of metal particles caused by MPs is found in the gut of ingested organisms and poses a risk to diseases. The most known disease caused by this factor is “Minamata disease,” which is a neurological syndrome caused by a severe intake of mercury. The chemicals released by MPs are detected across sea species, and their impacts pose a risk to a decline in seafood sources [25]. These MPs do not contain harmful chemicals and provide a larger surface area for the growth of marine insects that are not found in seawater. These microbes lay eggs on MPs and give rise to different microbes that are not part of the marine ecosystem, leading to potential consequences [26].

7.6.3 Socio-Economic Impact

Plastic havoc is a global concern, yet its social and economic impacts are still in the forecast stage. The marine system is the main source of livelihoods in and around coastal areas. Tourism, fisheries, and agriculture were the main economic and social stages. Fisheries and aquaculture are directly affected by plastic debris, which leads to the loss of targets. Another sector that is getting highly affected by marine litter is tourism, and the consequences of this visible litter are decreasing in the regional and local economies. Significantly, there is a reduction of the number of visitors resulting in the reduction of revenue. A human and marine ecosystem has an intangible relationship; our contribution towards the Earth's climate and the water cycle is also essential. Sadly because of human activities, the world's ocean now becomes a sink for plastic, and there is no research work that has been carried out to evaluate its socio-economic impacts [26–28].

7.6.4 Removal of MPs

Sewage treatment plants are unable to completely remove all MPs found in industrial, household, agricultural, and livestock wastewaters. Large plastics are removed from wastewater at a municipal wastewater treatment facility, but some MP or nanoplastic particles are still present after numerous water treatment steps and are still discharged into the final effluent. Many researchers have discussed various methods for removing MPs or nanoplastics from wastewater as shown in Figure 7.1.

7.6.5 Chemical Method

The process of employing chemicals to remove MPs has been investigated in several methods, but generally

speaking, the most frequently employed approach is water treatment coagulation/precipitation. There are significant variations in the number of MPs removed according to the type, quantity, and retention period of the coagulant [30]. Therefore, numerous investigations are being conducted to determine the ideal coagulant type and circumstances, and more precise studies on the removal of MP are required in the future. The elimination of MPs using the same technique was investigated but with a coagulation agent, $FeCl_3 \bullet 6H_2O$. They discovered that removal at a neutral pH improved with increasing coagulant concentration, with this tendency being more pronounced for small MPs (<0.5 mm). Similarly, the removal efficiency increased at high pH, high coagulant concentrations, and smaller MP particle sizes [31]. However, anionic polyacrylamide performed significantly better than cationic polyacrylamide in this situation and at a high dose of the coagulation agent (2 mM), to increase the clearance rate of the PE MPs.

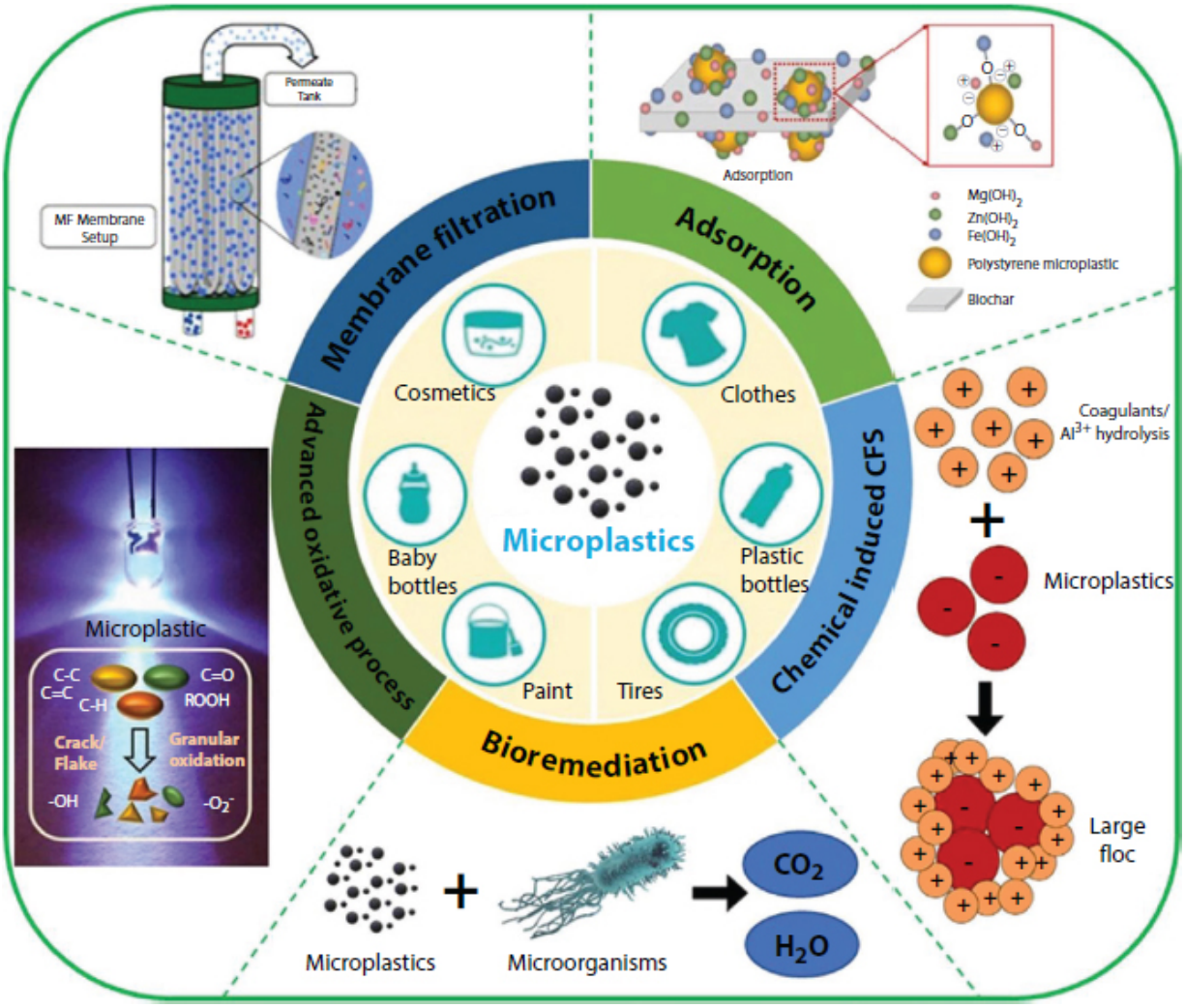


Figure 7.1 Overview of different methods for eliminating MPs

[Taken from copyright@ 2023, Elsevier, Ltd., Science of The Total Environment] [29].

It may be possible to remove MPs using dynamic membrane technology. This method involves depositing a layer of nanoparticles by penetration and dragging them onto a conventional membrane to reduce transmembrane pressure and prevent fouling of the primary membrane. It looks like a cake layer on top of the backing membrane. One of its key benefits is the lack of chemicals and the generation of byproducts. This method has the potential to

be widely employed because it can be used to maintain membranes with affordable porous materials.

7.6.6 Absorption and Filtration

MPs are projected to be more dangerous than other contaminants in aquatic ecosystems because they can entangle and bioaccumulate in organisms, killing fish, marine birds, mammals, and reptiles. Removal techniques are necessary owing to their persistence and low degradability. MPs are typically thought of as persistent materials; however, depending on their nature and chemical composition, they can disintegrate. Recently, significant progress has been made in our understanding of how plants absorb tiny plastic particles. For instance, 2.0 mm polymethylmethacrylate microbeads and 200 nm PS nanobeads can be internalized by the cereal plants *Triticum aestivum* (wheat) and *Lactuca sativa* (lettuce) [32]. The authors claimed that these beads could pass through the root stele and then move from the root to the shoot with the help of transpirational pull. Plastic internalization appears to be influenced by the particle charge. Overall, the surface charge of the particles had a significant impact on the sorption of MPs on algal surfaces. An anionic polysaccharide in the algal chemical structure explains why positively charged MPs tend to be adsorbed more effectively (Figure 7.2) [33].

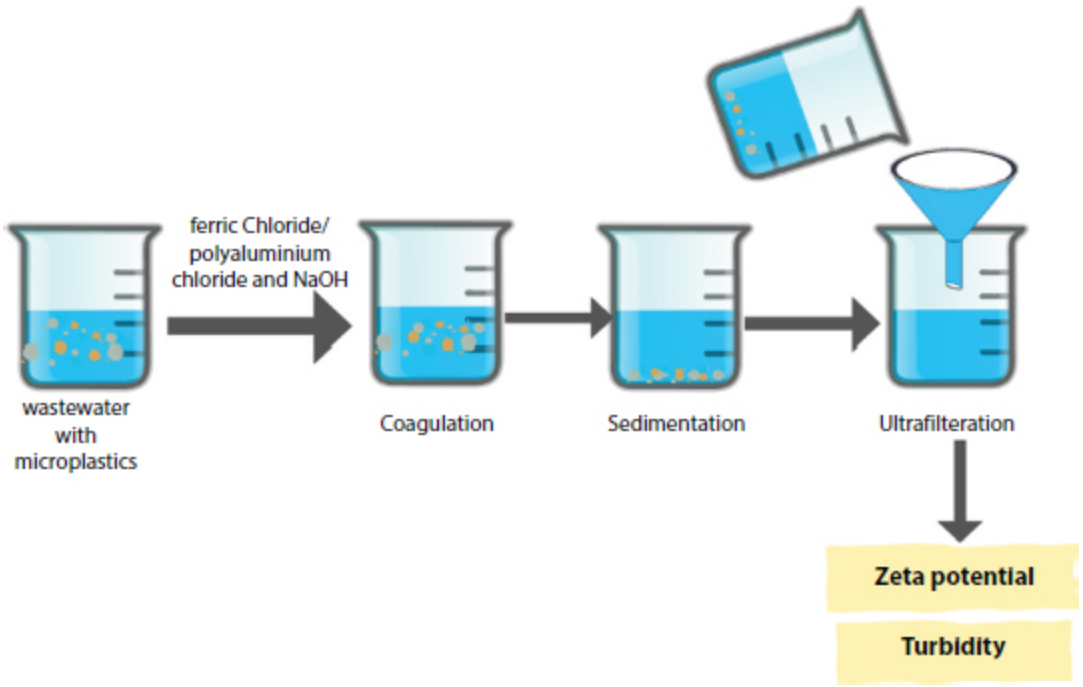


Figure 7.2 Chemical method for removal of MPs [34].

7.6.7 Biological Method of Removal of MPs

Enhancing biodegradation by using microbes to break down MPs does not harm the environment. As a result, identifying bacteria that can break down MPs is a potential and environmentally friendly method for promoting bioremediation and influencing ecosystem cleansing without having negative effects (Table 7.1). Significant microbial diversity exists in mangrove forests, and this diversity is important for a variety of environmental processes and uses. Low aeration and moisture levels, high temperatures, pH, and organic matter content enhance substrate conditions that favor the growth of microbial communities. Mangroves along the shore have historically been used as landfills for solid trash. *Bacillus cereus* and *Bacillus gottheilii* are two different types of bacterial strains isolated from mangrove sediments that have been employed to remove various MPs made of PE, PET, PS, and PP [35].

The initial findings of a four-month study using two types of indigenous marine communities, namely the *Agios consortium* and *Souda consortia*, to remove high-density polyethylene (HDPE) secondary MPs from saltwater [36].

Table 7.1 Different microorganisms used for the bioremediation of microplastics.

Microorganisms	**Microplastic**	**Method of analysis**	**Ref.**
Sphingomonadaceae and *Methylophilaceae*	PS	Weight loss	[37]
Bacillus, *Brevibacillus*, *Cellulosimicrobium*, *Lysinibacillus*, *Ochrobactrum*, and *Pseudomonas genera*	Low Density PE	FTIR	[38]
Pseudomonas aeruginosa and *Microbacterium paraoxydans*	LDPE	Weight loss and FTIR	[39]
Pseudomonas azotoformans, *Pseudomonas stutzeri*, *Bacillus subtilis*, and *Bacillus flexus*	PP	Weight Loss	[40]
Phanerochaete chrysosporium	PVC	FTIR, SEM and NMR	[41]
Bacillus sp. and *Paenibacillus* sp.	PE	Weight loss, TGA and SEM	[42]
Pseudomonas and *Bacillus* strains	PS	HPLC, NMR, FTIR	[43]
Tepidimicrobium, *Sporanaerobacter*, *Lutispora*,	PE	SEM	[44]

Microorganisms	Microplastic	Method of analysis	Ref.
Caldicoprobacter, and *Methanothermobacter*			
Tenebrio molitor	LDPE	IR spectroscopy and gravimetry	[45]
U. vulgaris	PS	Fluorescent	[46]

7.7 Global Initiatives

Microplastics have become a major concern for society. A few efforts have been made to achieve a better society.

7.7.1 United Nations Sustainable Development Goal (SDG 14)

The United Nations approved 17 Sustainability Targets in 2015, with the intention of making the world a healthier place to live. By 2025, the first goal of SDG-14, which aims to “conserve and sustainably utilize the oceans, seas, and marine resources,” is to “avoid and considerably reduce marine pollution, caused by land-based activities, including marine debris.” The UN also acknowledged the significance of talking about marine plastic pollution, and ingestion throughout the food chain as the biggest concern. The UN organized an ocean conference with 193 member countries to strategize on the reduction of marine pollution and for the voluntary implementation of SDG14 [46, 47]. The agenda is as follows:

i. Depletion of plastic and microplastic pollution.

ii. Accomplishing long-term and strong planning for a reduction in plastic usage in public and calling other countries to act on an urgent basis.

A resolution asking for the phase-out of primary microplastics from trash, industrial products, and personal care packaged products (PCCP) was approved by the Environment Assembly in 2017. This is the initial concrete step involving national and local governments, non-government organizations, the private sector, and citizens. International awareness is highly energetic, and the International Convention for the Prevention of Pollution from ships and the dumping of waste and other matters have been present for several years. Recently, the national level has been established. For example, in the United States, government-prohibited microbeads have been launched in the United States House of Representatives. Similarly, a law-making committee was launched by the government in New York and later passed in California. The European Government has also participated in removing plastic bags that accumulated rather than degraded in the oceans [48, 49].

7.8 Conclusion

This chapter reviews how MP migration and conversion in the marine environment are affected by biological intake, digestion, colonization, mobility, and adherence. In seawater– sediment–organism reservoirs, biological ingestion and adherence to MPs can change their distribution pattern. In contrast to the drag force exerted by dynamic activities on MPs, the effect of biological processes on the vertical migration of MPs in water is mostly achieved by altering the density of environmental MPs, such as the production and decomposition of fecal pellets and biofilm. MPs may migrate independently of

water currents due to their biological mobility, eventually settling in maritime regions. However, rather than seeing it as a standalone problem, it is crucial to see plastic pollution as another man-made component eroding planetary boundaries. Therefore, when implementing upstream or downstream measures to address plastic pollution, it is essential to consider climate change, land usage, and cost efficiency at every stage.

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Notes

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8
Identification, Quantification, and Presence of Microplastics and Nanoplastics in Beverages Around the World

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Abstract

Plastic waste mismanagement has led to environmental pollution, affecting natural elements such as water resources. Microplastics present in water bodies pose a threat to human health when the water is destined for human consumption. Therefore, studies on their presence in beverages have been reviewed in literature. Research articles from 2016 to 2024 were reviewed in April and March 2023, using the main keyword “microplastics,” followed by beverage type. Microplastics were omnipresent in beverages, although identification and classification techniques differed between studies, with concentrations ranging from 1.15 to 3,769 pieces/L. Their main characteristics included a size of 1 to 6,742.48 µm, a predominance of fragments and fibers, transparent particles as the most common color, and a typical chemical

composition of polyethylene, polyamide, polypropylene, and polyethylene terephthalate. The most commonly reported origins of microplastics are water sources, packaging, and work surface area. The presence of plastic particles in beverages poses a risk to human health and produces concern about how we deal with our waste and how the measures that have been currently taken have not been sufficient. Therefore, preventing plastic waste generation must be prioritized, and it is necessary to explore remediation strategies.

Keywords: Fibers, fragments, drinking water, bottled water, soft drinks, milk

8.1 Introduction

Water plays a crucial role in social development, life, and human health. It is the most abundant component of the human body and is essential for its functioning [1]. The most important functions of water in human bodies include transportation of nutrients to the cell, regulation of body temperature, proper functioning of the brain and nerves, and proper functioning of the digestive process, respiration, and blood circulation. Water is also an important resource for daily intake of essential minerals [2].

8.1.1 Water Consumption Around the World

The Spanish Society of Community Nutrition (SENC) recommends drinking approximately four to six glasses of water daily to maintain optimal hydration [1]. According to the United Nations (UN), almost 900 million people worldwide live without drinking water [3]. It has been estimated that over two billion people live in countries that experience water stress [4]. Bottled water consumption in high- and low-income countries is driven by different

reasons. In high-income countries, consumers find bottled water to taste better than tap water (due to its mineral composition), and consider it to be healthier, practical to purchase, carry, and dispose of the bottles. However, bottled water consumption in mid- and low-income countries is associated with low-quality tap water and untrustworthy public water-supply systems [5].

In the Global North countries, an important percentage of inhabitants consume bottled water, even with well-developed domestic water supply systems. Bottled water is the main source of drinking water for 31% and 38% of Canadians and Americans [5].

8.1.2 Water in the Beverage Industry

Liquid ingestion by humans involves beverages other than water. The total alcohol per capita consumption in the world's population (older than 15 years) drinks an average of 6.4 L of pure alcohol per year. According to the World Health Organization (WHO) data [6], the average world consumption of soft drinks is 77 L per capita per year, while Mexicans are the first consumers of soft drinks in the world with a consumption of 160 L per capita every year [7].

Water is the main ingredient in the products of the beverage industry, but it is also used in the production processes, including glass bottle washing, cooling, and floor scouring and washing. For instance, in the soft drink industry, only 20%–30% of the total water consumed forms part of the product [8]. To produce 1 L of soft drink, approximately 3–4 L of fresh water is required [9].

8.1.3 Water Quality and Water Pollution

Water is not pure in its natural state and its mineral composition depends on its origin. Water quality affects

both the safety of beverages and the processes used to obtain them. Water quality is affected by natural and anthropogenic contaminants, such as pesticides, agricultural products, industrial waste, heavy metals, pathogenic microorganisms, and microplastics (emerging contaminants derived from the degradation of plastics) [3, 5].

Plastic production has reached approximately 400 million tons per year, and the bottled water industry is responsible for up to 5.5% of global plastic production. Although there are several packaging options (glass bottles, multilayer cartons, aluminum), polyethylene terephthalate (PET) is the most common material used in bottled water containers (over 80%), and most of these bottles become waste after one use [5]. This massive production and use of plastics, combined with poor management of plastic waste disposal is conducive to microplastic contamination in aquatic water bodies, including rivers, lakes, and marine ecosystems [10].

Microplastics (MPs) are contained in some self-care products, and the fibers shed from synthetic fabrics during the washing process commonly end up in wastewater [10, 11]. Some MPs from wastewater may be removed when they go through wastewater treatment plants (WWTPs), depending on the treatment units. Nevertheless, a portion of them can pass through the WWTP, enter aquatic water bodies, and finally accumulate in the environment [10]. Some studies have shown the behavior of microplastics during different treatment processes. They commonly show that large microplastics are easily removed during physical treatments, whereas smaller microplastics are removed during biological treatment and secondary clarifiers [12]. However, this technology does not eliminate all of them. The smaller the microplastics, the greater the complexity of their removal, resulting in higher costs, maintenance requirements, and energy consumption [13].

In recent years, it has been shown that humans are exposed to microplastics through the ingestion of food and drinks. Because microplastics are also found in freshwater, this is being pointed to as a potential route of entry for contamination in beverages. However, some studies indicate that plastic beverage containers can wear out and thus contaminate the liquid with microplastics [9, 14]. Microplastics can act as vectors of other environmental pollutants, such as polycyclic aromatic hydrocarbons (PAH) and heavy metals [5, 14]. For this reason, when ingested by humans, they can cause potential damage to health. However, there is still very limited information on the negative effects of microplastics on human health, as studies have been conducted in model organisms. The potential damage inflicted on human health includes alterations in immune and stress responses and reproductive and developmental toxicity [15].

The objective of this research is to carry out a bibliographic review of the findings on microplastics in beverages, in terms of their presence, size, shape, color, and type found in different products.

8.2 Methodology

This section describes the steps followed in the search and selection of scientific articles related to the presence of microplastics in beverages. It also explains the management of information regarding the sampling and the results reported in the investigations.

The search for articles was conducted in March and April 2023. ScienceDirect and Google Scholar databases were used to look for suitable papers. The main keyword for the article search was “microplastics,” followed by one of the following words: beverages, drinks, bottled water, soft drinks, juice, carbonated drinks, energy drinks, isotonic

drinks, hydrating drinks, dairy beverages, or alcoholic beverages. The Boolean operator “+” was used.

The search was restricted to applying criteria such as the period from 2016 to 2024. Only research articles and short communications were analyzed (congresses, review articles, books, and book chapters were excluded). Once the search yielded all articles, they were ordered by relevance, and the first 100 listed for every combination of keywords were reviewed.

Subsequently, an inclusion criterion was applied to verify that the papers referred to drinks packaged in a liquid presentation, bottled drinking water, or beverages from refill kiosks suitable for human consumption. Breast milk, powders, concentrates, syrups, and tea bags were excluded from the study. Finally, repeated articles were discarded, resulting in 31 articles from which relevant information was extracted, registered in spreadsheets, and analyzed. Figure 8.1 presents a flowchart of the search process and the selection of articles.

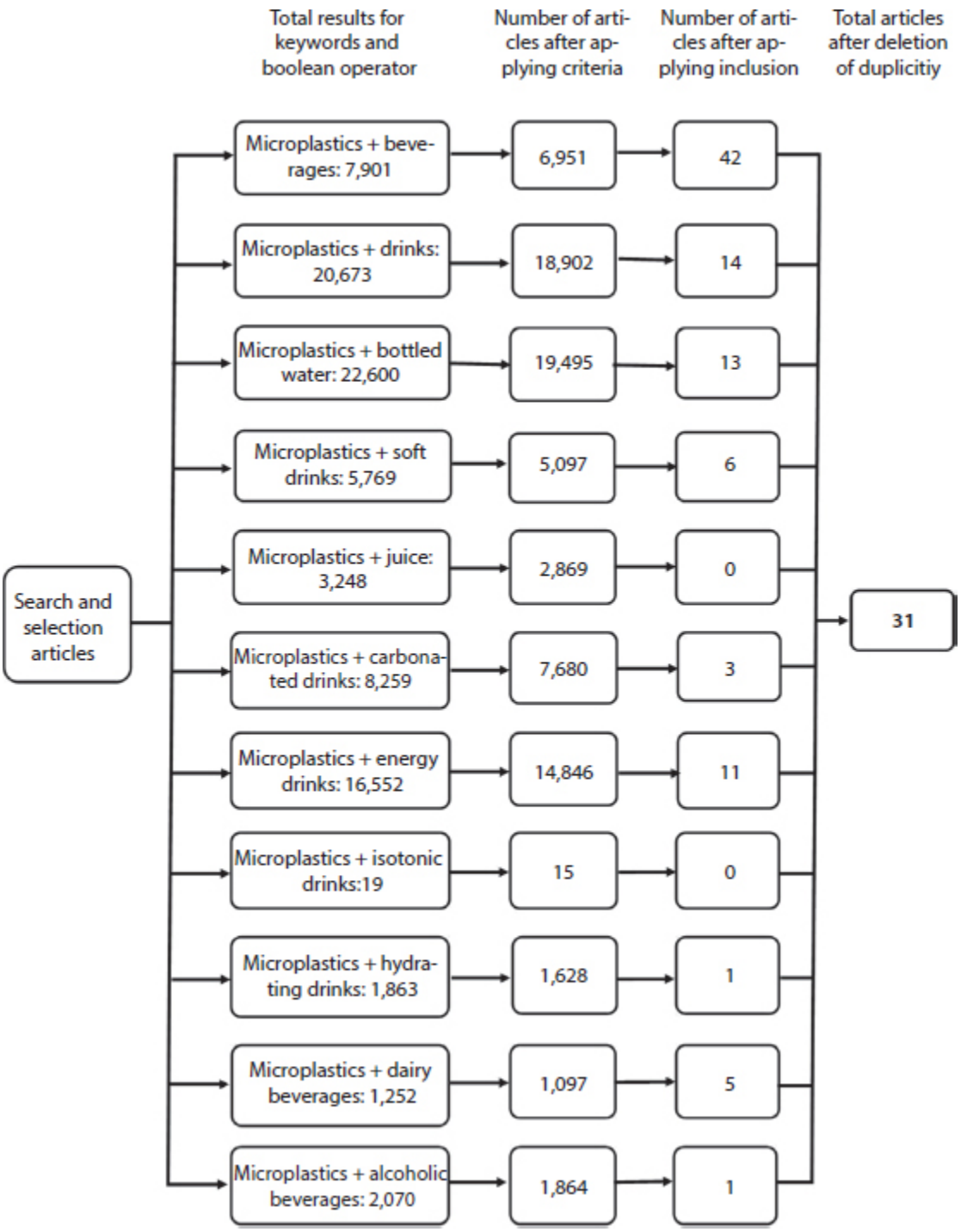


Figure 8.1 Article research and selection process.

8.3 Results

The results are divided and discussed in four sections:

microplastic identification and extraction techniques, microplastic concentration, microplastic classification, and population exposure.

8.3.1 Countries where Studies were Performed

This research was conducted in 19 different countries. As shown in Figure 8.2, the countries with the highest percentage of participation were Mexico (16.13%), Germany (12.90%), and the Republic of Korea (9.68%), followed by China, Italy, and the Netherlands (6.45%).

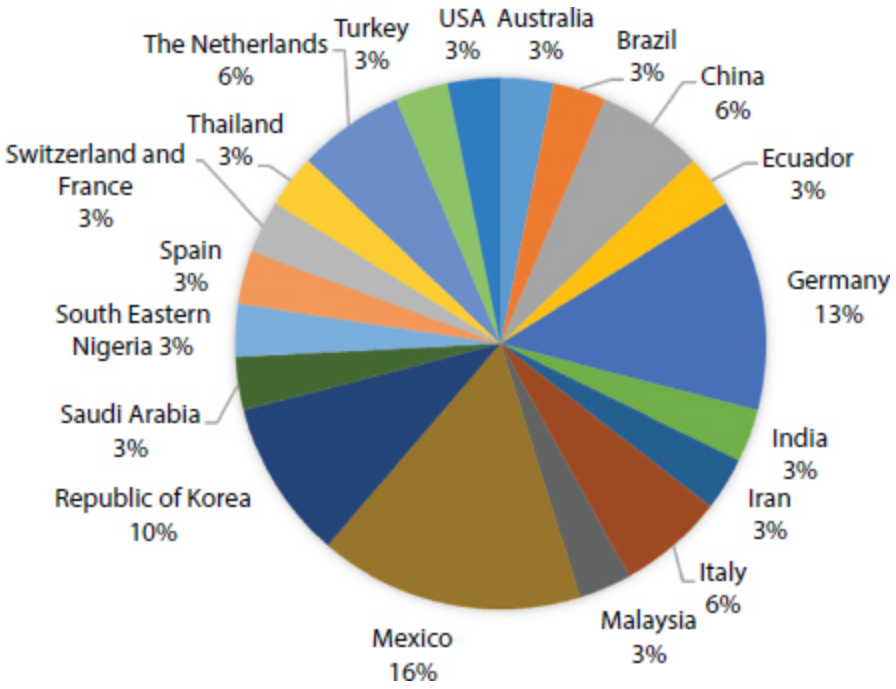


Figure 8.2 Countries where microplastic studies were performed.

8.3.2 Techniques for Identification and Extraction of Microplastics

The occurrence and distribution of microplastics has become a topic of great interest in recent decades, especially in ecosystems such as the water column [16], beaches [17], mangroves [18], and even marine fauna [19].

This last research topic is the one that has the greatest relationship with human exposure to microplastics through the ingestion of fish and shellfish. Humans are exposed to the ingestion of microplastics not only through seafood [20], but also by consuming food and beverages that are prepared from terrestrial sources [21]. The sources of these microplastics can be external, such as the atmosphere and the manufacturing and packaging processes [22], or internal, that is, the foods or beverages that originally contained microplastics.

Considering the above, the presence of microplastics in food and beverages has become a topic of great interest today, which allows us to not only understand how the presence of these contaminants has spread in different media globally but also how humans are exposed through ingestion.

As it is a recent research topic, there is no standardized procedure for the extraction and identification of microplastics in this type of media. Each investigation reports different methodologies depending on the type of beverage and its characteristics, the equipment and reagents available in the laboratory, the size of the particles to be studied, and the objective of the study. In general, these methodologies are divided into five stages, from the selection of the type of beverage to the identification and chemical classification, as shown in Figure 8.3.



Figure 8.3 Common steps for the analysis of microplastics in food and beverages.

8.3.2.1 Selection of the Type of Beverages

Research related to microplastics in beverages has focused on the most commonly consumed beverages, such as bottled water, milk, mineral water, beer, wine, soft drinks, iced tea, and energy drinks, as these beverages could represent one of the main sources of human exposure to microplastics [9, 21]. It is important to mention that although most of the research reviewed in this chapter (70.96%) focused only on the study of a single type of beverage, some of them studied more than two types (29.03%).

According to the reviewed investigations, the beverages studied in the highest proportion were bottled drinking water from the purification source and mineral water, with 20.45% for each. This could be related to their greater scope and consumption by the global population. The other types of beverages studied were soft drinks (11.36%), milk (9.09%), and beer (9.09%) (Figure 8.4).

Bottled water has become an important point of interest because of possible external contamination sources during the packaging process; however, to assess the influence of packaging as a main source of microplastic contamination [23, 24], water obtained directly from effluents from water treatment plants [25, 26],underground sources [27], rivers [28], and distributed through retail purification plants has also been studied [25, 29].

Nine main packaging materials were identified for the analyzed beverages were identified in the articles (Figure 8.5). Among these, the most common were PET bottles (31%), which included single-use and reusability. The second most used material was glass (25%), followed by a multilayer carton (14%), which is a flexible or rigid packaging containing more than one layer of different overlapping materials [30]. Canned beverages and beverages packaged in polycarbonate were the least common (2%). In three studies, the packaging material was not specified (2%), in another three studies, the drinking water samples were stored in glass containers (glass-sample containers) (6%). In one case, the sample was not collected in a container (2%) and was filtered ot the sampling point.

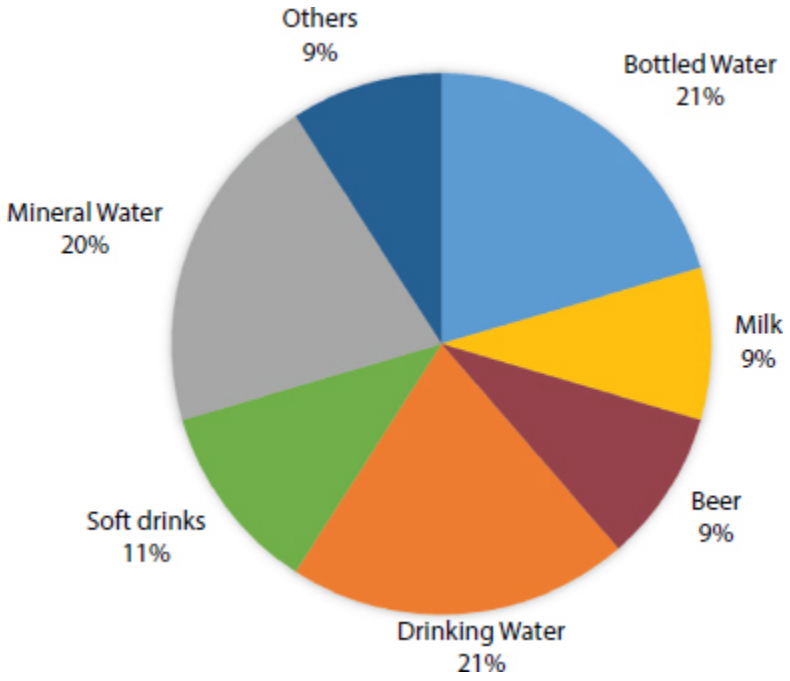


Figure 8.4 Distribution of the types of drinks studied. Note: The "Others" category includes research carried out on energy drinks, white wine, and iced tea.

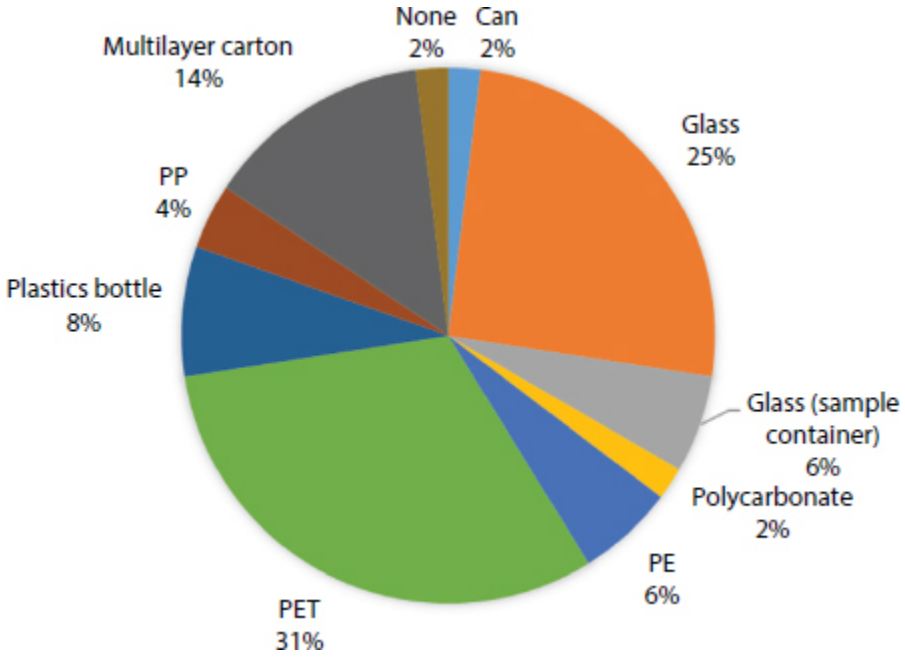


Figure 8.5 Material of the containers of the analyzed beverages.

In the case of the studies carried out in this type of

research, not only the number of samples and replicates were considered, but also the number of sampling points, which are usually located at different points of interest, such as different water treatment plants of the same region or country [31] or the water obtained from public drinking fountains located in different locations [32]. The number of sampling points ranged from 1 (effluent from a water treatment plant) [26] to 63 (different drinking water filling kiosks in a city) [29].

8.3.2.2 Sample Preparation

In this phase of the process, the number of samples, their volume, and dilution solution used were considered. The number of samples varied from 1 [26, 33] to more than 100 [29, 34], depending on the type of beverage and the objective of the study. For example, in some cases, we sought to analyze the occurrence of microplastics in beverages of different brands and containers; therefore, one or more brands were evaluated, but with a different number of samples of each one (in containers of different materials such as plastic, glass, or multilayer carton packaging) [33, 35, 36]. Replicates were performed for each sample to obtain more accurate and reliable results. In most cases, only three replicates per sample were used. However, there were some exceptions in which duplicates [28, 29] and four replicates per sample [37] were used.

The sample volume also depended on the type of beverage studied, type of packaging, and volume in which it is commonly sold. Most of the investigations used a sample volume between 0.25 to 2 L (77.42%). For effluents from water treatment plants, the sample volume was much higher, ranging between 965 L [26] and 2,500 L [27].

Finally, not all the samples required the use of a dilution solution; only three papers reported its use. This medium

was used in mineral water with a high content of calcium or magnesium carbonate to reduce or make an equivalence with these ions [36], or to dilute some other types of beverages (such as juices, soft drinks, milk, and beer) [14, 37] to facilitate the filtration process of the samples. The dilution solutions included distilled water (at a 1:1 ratio with 500 ml of water) [14], EDTA solution sodium dodecyl sulfate [36], and ultrapure water [37].

8.3.2.3 Digestion

To facilitate the separation and identification of microplastics in the samples, as well as to eliminate particles of organic and inorganic matter that could be wrongly considered as microplastics (false positives), a digestion process was carried out before the filtration of the samples. This phase of the process considers three important aspects: the digestion solution, time, and temperature. In the majority (77.42%) of the investigations reviewed in this chapter, digestion of the samples was not carried out.

In the case of the investigations that reported this step, the most commonly used digestion solution was Hydrogen Peroxide (concentration between 15 and 35%). Other solutions used in this phase of the process were Potassium Hydroxide (KOH) (15%) [28], multienzymatic detergent, EDTA 0.5 M, alkaline tetramethylammonium hydroxide solution [37], and Fe (II) solution (0.5 M) [29]. The digestion process was carried out at different times and temperatures, from 6 min [37] to 72 h [14] and from 20 °C [38] to 75 °C [29], respectively.

8.3.2.4 Filtration

Filtration is a technique that allows microplastics present in samples of different types of beverages to be separated through a porous material (filter), which facilitates the

identification and final characterization of the retained particles. At this stage of the process, the type of filter (the material from which it was made) and the size of its pores (in µm) were the main factors considered.

These characteristics depend on the objective of the study and the size of the particles that were identified. The most commonly used filters were the Whatman filter, polycarbonate filter, and cellulose filter, with 19.35%, 16.13%, and 12.9%, respectively. Other types of filters that are used to a lesser extent include nylon [14], silicon [37], and silver membranes [39]. Filtration is usually carried out with a single type of filter for all samples; however, the combination of different types of filters (and even different pore sizes) has also been reported, such as glass microfiber filters (0.7 µm) in conjunction with aluminum oxide filters (0.1 µm) [38], stainless brass sieves (250 µm) with nylon membranes (10 µm), and polytetrafluoroethylene (PTFE) (1 µm) [14], and the combination of stainless steel filters (3 µm) with aluminum oxide filters (0.2 µm) [27].

Regarding the pore size, as in the case of the types of membranes, different sizes used were reported, from 0.1 µm [38] to 0.22 mm [21], with most of them between 0.1 and 0.45 µm (45.16%), followed by filters with pore size between 10 and 11 µm with 29.03%.

In some cases, a separation process was carried out before sample filtration. Only two of the reviewed papers reported this type of process, which mainly consists of separating microplastics by density using zinc chloride (ρ = 1.6 g/L) [26] and sodium solutions [40]. This type of solution was used so that particles with a lower density than the solution would float on the surface, and it would be easier to extract them.

8.3.2.5 Visual Identification and Characterization

Once the beverage samples were filtered, the identification and characterization of the particles (color, size, and chemical composition) retained in the filters were carried out. To carry out this phase of the process, two main techniques were considered: identification using spectroscopic methods and identification through staining of the particles with different dyes.

Unlike visible microplastics (0.5–5 mm), whose color, size, and shape can be easily identified, for the type of particles present in beverages, this characterization was carried out with equipment such as microscopy coupled with X-ray spectroscopy SEM-EDX [29, 32], stereomicroscopy [22], and scanning electron microscopy (SEM) [9, 41] due to the smaller size of the analyzed particles.

Bengal rose [22] and Nile red [28, 34, 42, 43] dyes were used to identify of the filtered particles. For the identification and characterization of the stained particles, fluorescence microscopy [28, 42, 43], optical and scanning microscopy (SEM), Energy Dispersion X-ray Spectroscopy (EDS) [34], and stereomicroscopy [22] were used.

The determination of chemical composition of the particles that were not stained with any dye was determined mainly with the following techniques: Fourier Transform Infrared Spectroscopy (FTIR and μFTIR) (45.16%), Raman and μRaman Spectroscopy (25.81%), and Infrared Laser Indirect (LDIR) (12.90%).

8.3.2.6 Quality Control and Contamination Prevention

Different quality control and contamination prevention measures were employed before the samples were processed in the laboratory to avoid the interference of external particles from different sources, such as the

surrounding air or materials used during the experimental process.

These measures vary depending on the type of beverage studied, the type of laboratory, and the size of the microplastics analyzed. However, some of them remained constant for all the investigations, such as the use of a cotton lab coat and nitrile gloves, cleaning of the work area, use of a laminar flow hood during the experimental procedures, use of laboratory material made exclusively of glass and metal, as well as its previous cleaning with ultrapure water and ethanol, to remove any trace of plastic particles.

Other measures used to a lesser extent were the inspection of filtration equipment and filters before use [29, 35, 43], previous filtration of the solutions used in the experimental procedure [24, 25, 36, 37, 40], cleaning of the equipment in an ultrasonic bath [31, 36], and aspiration of the work surface and floors of the laboratory with HEPA filters [44].

For some beverages, such as bottled water (carbonated and non-carbonated), an additional contamination control measure was to remove the label from the surface and clean the outside of the bottles with water and ethanol, as well as leave them in a hood laminar flow 24 h before sample processing [34–36, 45].

In addition to these measures, a blank test was carried out, which, according to the research analyzed in this chapter, was mainly ultrapure water (38.71%), and deionized water was also used to a lesser extent (9.68%). The purpose of this blank was to avoid altered results or to obtain false positives as well as to show the efficiency of contamination control measures. These blanks were subjected to the same procedure as the beverage samples (from preparation to identification and characterization).

The number of blanks used for each experimental procedure was changed according to the type of sample (as well as the number of samples and their respective replicates). In some cases, only one blank was inspected [46], whereas in other investigations, up to 18 were evaluated [35]. In some other investigations, the use and analysis of a blank was mentioned, but the number or type of blank used was not reported [14, 26, 28].

8.4 Microplastic Concentrations in Beverages

In the case of water, the analysis of MP is highly relevant because it is the main ingredient of any drink. If the water is contaminated by microplastics, the drinks in which the water is used for its production will also be contaminated.

Among the seven types of beverages in which the reviewed studies were grouped, a high proportion of investigations (from 40% in soft drinks to 80% in mineral water) reported that all the samples studied contained microplastics. However, there are also cases in which this information is not included, for example, some studies that analyze purified water, mineral water, soft drinks, milk, tea, and other types of beverages (Table 8.1). In general, these data allow us to determine whether microplastics are omnipresent in products for human consumption, the exposure of consumers to these products, and to identify the origin of microplastics depending on the type of beverage in question.

The concentrations of microplastics in different beverages vary, and one of the great challenges is that there is no consensus as to whether the total, minimum, maximum, interval, mean, or median concentration should be reported. In this study, the average concentrations are

presented because they were the most reported by the studies. These, in turn, were grouped into six categories, of which the group of 1–25 pieces/L presented the highest average proportion with 35%, from 10% in mineral water to 67% in tea (Table 8.1). The lowest concentration of MP was reported in purified water, stored in a glass bottle with an aluminum and plastic lid, from Saudi Arabia, which was an average of 1.15 pieces/L [45], while the highest concentration of MP was found in mineral water, packaged in PET bottles, marketed in Germany, whose average was 3,769 pieces/L [35]. This finding supports raise the hypothesis that beverages stored in plastic containers have higher MP concentrations than those packaged in bottles of other materials. The number of pieces of MP present in a specific drink volume is the base data that allows the calculation of daily, monthly, or annual intake of humans.

Contamination by microplastics in beverages can be attributed to several factors closely related to their sources; however, these factors may vary depending on the type of beverage. In this study, the sources from which the MP was present in the studied beverages could be identified based on what was reported by the articles (Figure 8.6), and it was found that the number of reported sources was 12, 11, 9, 8, 7, 7, and 6 sources for mineral water, soft drinks, tea, purified water, beer, milk, and others, respectively.

The main sources of MP were water (22%), packaging (16%), and work areas (9%). In the case of water, it is used during the entire production process of a particular beverage, either as an ingredient in the beverage or as part of the washing or cleaning of materials, equipment, containers, and work areas [31]. The water used may contain microplastics because of the source from which it comes, especially if the body of water is contaminated by solid waste or if treated wastewater is discharged. The

water used as an ingredient or as drinking water goes through treatment to eliminate contaminants and impurities, for which filtration systems usually consist of membrane filters or activated carbon filters [47]; however, if the plastic particles present are smaller than the opening of the membrane filter or the activated carbon particle size (especially microfibers), they will not be retained and will be present in purified or drinking water.

Table 8.1 Proportion of beverages with MP and mean concentrations (pieces/L).

Type of beverage	Proportion of beverages with MP (%)	Mean MP concentration (pieces/L)	References
Purified water	All (65%) 90%–99% (14%) <90% (14%) Not specified (7%)	1–25 (60%) 51–75 (14%) >100 (13%) Not specified (13%)	[23–25, 27, 28, 32, 34, 41–45, 53]
Mineral water	All (80%) <90% (10%) Not specified (10%)	1–25 (10%) 26–50 (10%) 51–75 (20%) 76–100 (10%) >100 (20%) Not specified (30%)	[22, 31, 33, 35, 36, 39, 43, 50, 53]
Beer	All (50%) 90%–99% (25%) <90% (25%)	1–25 (20%) 26–50 (40%) >100 (20%) Not specified (20%)	[9, 14, 21, 33]
Soft drink	All (40%) 90%–99% (20%) <90% (20%) Not specified (20%)	1–25 (40%) 26–50 (40%) Not specified (20%)	[9, 14, 21, 40, 52]
Milk	All (75%) Not specified (25%)	1–25 (25%) 26–50 (25%) >100 (25%)	[14, 37, 46, 51]

Type of beverage	Proportion of beverages with MP (%)	Mean MP concentration (pieces/L)	References
		Not specified (25%)	
Tea	All (67%) Not specified (33%)	1–25 (67%) Not specified (33%)	[9, 21, 40]
Others	All (60%) <90% (20%) Not specified (20%)	1–25 (20%) 26–50 (20%) >100 (20%) Not specified (40%)	[9, 21, 29, 55, 56]

L, liter.

Membrane filters, as well as water treatment plants, can also release MP due to abrasion when filtering the water or passing the water through containers or plastic distribution lines; these three aspects were also considered sources of MP in the reviewed studies (mean of 8%, 5%, and 2% for water treatment plants, membrane filters, and water distribution pipes, grouped in others). Therefore, it is important to use membranes, equipment, or materials that are not made of plastics. Filters are problematic when treating water and processing other beverages such as milk, where different filtrations (macro, micro, and ultrafiltration) are used to retain microorganisms and particulate matter [46]. The lack of substitution of the filters during the processes reduces their efficiency because continuous filtering can cause abrasion, and therefore, the detachment of particles.

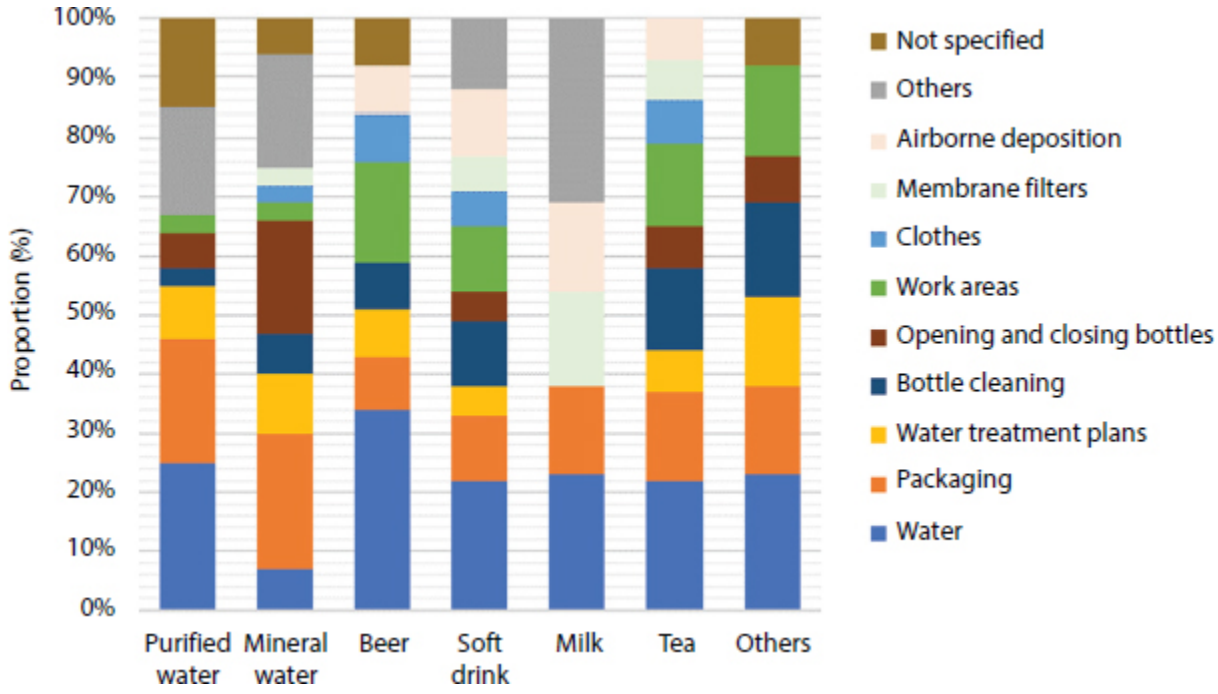


Figure 8.6 Sources of MP by type of beverage.

The second most reported source was packaging, including the container in which the drink was stored and its lid. Packaging made from plastic materials (PET, polyethylene, PE, and polypropylene, PP) can cause significant leakage of MP. In some studies, in which beverages were packaged in PET and PE containers, a high proportion of MP from both polymers was observed [35, 36, 43]. Another source of MP that is closely linked to packaging is the mechanical process of opening and closing a bottle cap made of plastic due to the friction generated during the process [22, 39, 42]; this hypothesis has been raised regardless of whether the bottle is plastic or glass [36].

The third most addressed source of MP was work areas where cleaning is not performed efficiently, which would cause the accumulation of particles found in the air, or cleaning with synthetic textiles, which can release fibers [9, 31, 40]. Furthermore, if air currents or continuous movements are common in both cases, this would cause the particles present on the surfaces of the work areas to

resuspend and subsequently fall into beverages during the production process. In addition, the type of work clothing can also influence the production of beverages; synthetic clothes were reported by 3% of studies as a source of plastic pollution.

8.5 Microplastic Characterization in Beverages

Microplastics can be characterized according to their size, type, color, and/or chemical composition. It is important to note that differences in the characteristics of microplastics are influenced by sample handling, quality control measures (including contamination prevention and the use of blanks), sample processing, and identification methods. In this section, the characteristics of MP within the reviewed literature are summarized.

8.5.1 Microplastic Sizes

The reviewed articles contemplate a wide interval of microplastic sizes, as shown in Table 8.2, going from as small as 1 µm to 6,742.48 µm, the larger MP probably being present within the samples because of airborne contamination during sample handling. The size classification type is quite common, although a few articles do not include the precise size interval of the studied MP; three articles did not include any information on this parameter [33, 39, 42].

A few studies did not separate the size intervals for each type of beverage studied. One of the articles for bottled and mineral water reported MP sizes in the interval of 5–100 µm [43], while another distinguished their results for Australian bottled water (68 ± 52 on average) and imported bottled water (80 ± 62) [44]. For beer and soft

drinks reported size intervals have been 2.48–6,742.48 µm [14], 100–3,000 µm [9], and 45–99 µm [21].

Only 41.94% of the articles also categorized MP into different groups based on their size, but the selected size ranges usually did not coincide, which made it difficult to compare information; however, all of them reported that MP were mainly in the smaller size ranges that were tested.

A few articles categorized the majority of MP as less than 1,000 µm (69–90%) in bottled water [41], soft drinks, cold tea, energy drinks, and beer [9], and there was a case where MP in the size range of 1,000–5,000 µm in mineral water were significantly low (10%) [22]. An important proportion of MP was also smaller than 500 µm in drinking water (50%) [32] and milk (40%) [46].

The smallest size ranges reported in the literature are less than 300 µm in drinking water [48] and 10–50 µm (65.55% and 71.13%) in bottled and mineral water [48]. Some of the smallest particles were reported by Schymanski *et al.* [35], who categorized them according to beverage container type and found particles in the size range of 10–20 µm and 5–10 µm in samples contained in single-use bottles (30% and 41%, respectively), returnable bottles (29% and 56%), glass bottles (32% and 45%), and cartons (28% and 39%). Kankanige and Babel [43] also found particles in the last size range (41%) in bottled drinking and mineral water.

The size reported was less than 5 µm for 95% of the MP in mineral water tested by Oßmann *et al.* [36]. Such tiny particles are of concern to human health because they are completely unnoticed and unaffected by conventional drinking water treatment.

Particle size determination is limited by the size of filtration membranes, capacity of the equipment, visualization

techniques (such as the use of dyes), and by the operator. Hence the detection of even smaller particles is difficult, and they can be present in beverages as a part of the raw materials (like water and other ingredients) and the production process, where they could also break into smaller pieces through abrasion; thus, the smallest particles could have been underestimated. If they reach the nanometric scale, they could even pass through the cells and reach beyond the gastrointestinal system, as suggested by a study on MP contamination in marine and terrestrial ecosystems [49].

Table 8.2 Microplastic range sizes reported in the reviewed literature.

Beverage type	Microplastic size (μm)	Reference
Drinking water	50–150 200–1,000 100–5,000 25–500 20–5,000	[27] [28] [32] [45] [25], [24] [26], [48]
Bottled water	25–5,000 20–100 10–1000	[23], [41] [34] [53]
Mineral water	<1,000–<5,000 5–>100 5–100 1–<10 11–500 40–535	[22] [35] [43] [36] [31] [50]
Soft drink	36–2,228 50–100	[40] [52]
Milk	≥5 <500–5,000 <500	[37] [46] [51]
Alcohol	7–475	[38]

8.5.2 Microplastic Types

Among the reviewed articles, 77.42% reported MP classification in terms of types, the most common being fragments (including irregular and square), fibers, and films, although pellets/granules and spherical shapes were also identified, as shown in Figure 8.7. Only two authors reported circular-shaped MP [42, 50], denominating them

as beads and rounded MP, and one also identified rod shapes.

For bottled water, mineral water, and drinking water, 75%, 40%, and 85.71% of the reviewed articles included microplastic types, the most common being fibers and fragments for these beverage types. In the case of milk, 75% of the articles included the MP type classification, and the most frequent types were also the same.

For alcoholic beverages, only 40% reported MP types, and identification was only specifically performed for beer, with a prevalence of fibers and fragments. Soft drinks also contained greater amounts of the same MP, as stated by 60% of the articles. Diaz-Basantes *et al*. [14] did not differentiate between MP types for beer, skim milk, and soft drinks, regardless indicating frequency of fragments and fibers. Ultimately, the tested energy drinks [9] contained a wide range of fibers.

It is worth noting that although fibers are the most commonly reported source of pollution, some articles suggest that fibers might also indicate some degree of sample contamination [42], and one [27] even dismissed these MP types in their samples because they had a parallel treated blank that revealed that these particles originated during sample handling; hence, the importance of applying contamination prevention measures when handling samples, and visual and chemical identification techniques of microplastics, to differentiate between organic and synthetic materials.

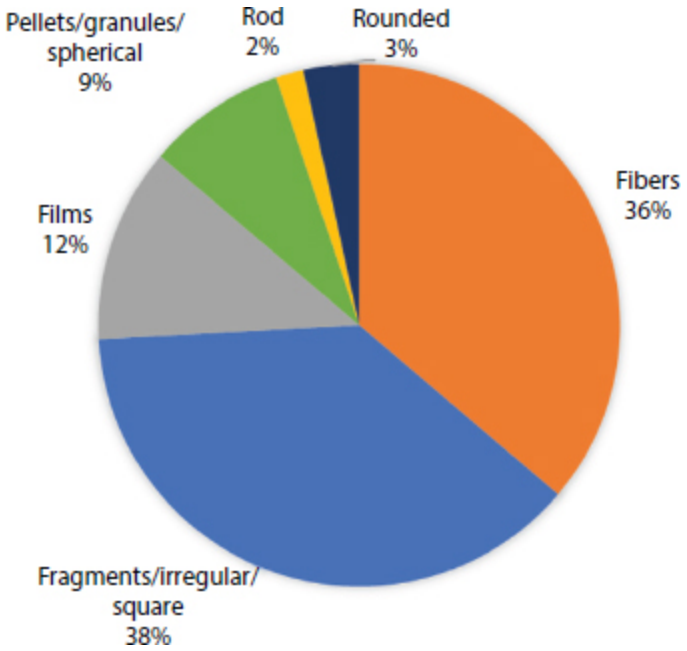


Figure 8.7 Common reported microplastic types in the reviewed literature.

8.5.3 Microplastic Colors

Of the reviewed articles, only 38.71% reported microplastic classification in terms of color, with the most common types being white/transparent, blue, black/dark, and red, as shown in Figure 8.8.

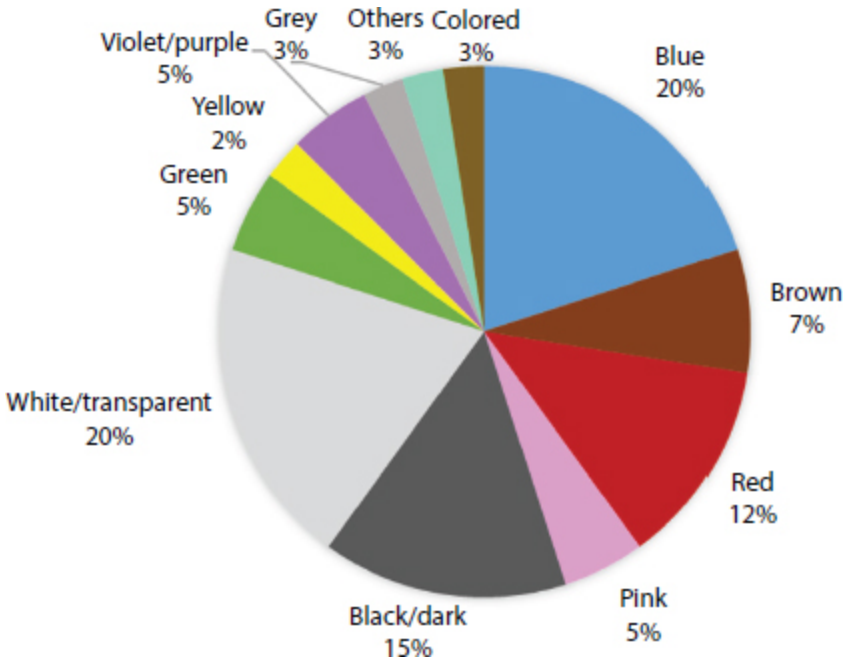


Figure 8.8 Common reported microplastic colors in reviewed literature.

Only 25% and 20% of the bottled and carbonated water articles, respectively, included MP color information. Bottled water contained mainly transparent MP (96.5%) [23], while black MP (95%) were abundant in mineral water [22]. In contrast, one study [43] did not distinguish between the two types of beverages and reported blue, green, and red as the most common MP colors. Of the drinking water articles, 57.14% included color classification of MP, with the most frequent colors being white/transparent and black/dark [24, 27, 32, 25].

In the case of milk, 50% of the reviewed articles mentioned color classification, reporting blue and pink as common colors [46, 51], and it is also suggested that these characteristics are influenced by the packaging material of the beverages.

For alcoholic beverages, 40% of articles mentioned common colors only for beer, registering the presence of MP colors such as blue, red, and green [9]. For soft drinks

(fruit drinks, cold and liquid tea, and others), 80% of the articles indicate MP colors, pointing at blue and transparent as frequent colors [9, 40, 52]. One articles did not distinguish between MP colors for beer, skim milk, and soft drinks, finding mainly green, yellow, and red [14]. Finally, the most common colors for energy drinks are blue and red [9].

Color determination of MP is relevant to explain their possible origin as part of the packaging, and/or the equipment and materials used in beverage production. The wide presence of transparent particles suggests that they have originated from the beverage bottles themselves, as supported by the highest concentration in bottled water [35], as mentioned before. The color of these particles can also hinder their visualization, blending them with the background of the filtration membrane and even organic particles, which could lead to their underestimation.

8.5.4 Microplastic Chemical Composition

A total of 80.65% of the reviewed articles included the analysis of MP chemical composition, detecting a wide range of polymers, as shown in Figure 8.9. The most common types were PE, polyamide (PA), PP, and PET. The category for “others” was used in three articles [35, 53, 45]. In addition to plastics, cellulose has also been detected in beverages, although some percentages have not been reported, being prevalent in bottled water (71.13%–88.99%) [53, 54], mineral water (59.48%) [35, 53], and a few in drinking water [55].

A total of 87.5% of the articles for bottled water and 88.89% for mineral water reported chemical compositions, where the most prevalent chemical compositions were PET, PP, and PA in the first case [23, 34, 43–45, 53, 54], and PP, PE, PET, and PA in the second one [22, 31, 33, 35, 36, 39,

43, 53]. Among drinking water articles (85.71%), the most frequent particles were PET and PE [24, 26, 27, 29]. For milk, 60% of the papers report the chemical composition of MP, the most common of which are polysulfone (PSU), PA, PE, and PP [37, 46, 51].

Among the reviewed articles on alcoholic beverages, 60% specified the MP chemical composition, reporting a different variety, such as PE in wine [56], polystyrene (PS), PP [33], polyacrylamide (PAM), polyester amide (PEA), and PET in beer. In 50% of articles on soft drinks that report MP composition, there is also a chemical diversity of particles such as PA, PET, PE [52], PAM, PEA, and acrylonitrile-butadiene-syrene (ABS) [9]. One study [14] did not differentiate between beer, skim milk, and soft drinks and reported the currency of HDPE/LDPE, PAA, and PP in general. In the case of energy drinks, PA, PEA, and ABS were found [9].

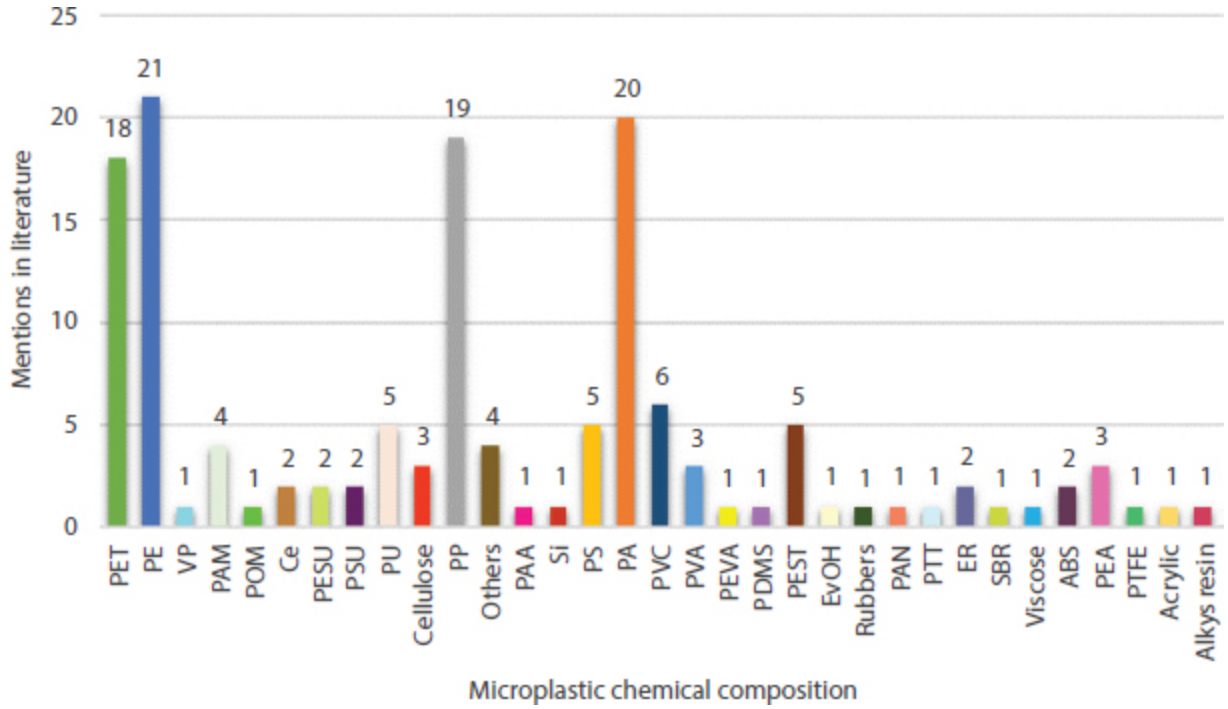


Figure 8.9 Common reported microplastic chemical composition in the reviewed literature, where PET, polyethylene terephthalate; PE, polyethylene; VP, vinyl polymers; POM, polyacetal; Ce, cellophane, PESU, polyethersulfone; PSU, polysulfone; PU, polyurethane; PP, polypropylene; PAA, polyacrylic acid; Si, silicone; PS, polystyrene; PA, polyamide; PVC, polyvinyl chloride; PVA, polyvinyl alcohol; PEVA, polyethylene vinyl acetate; PDMS, polydimethylsiloxane; PEST, polyester; EvOH, etilen-vinil-alcohol; PAN, polyacrylonitrile; PTT, poly-trimethylene terephthalate; ER, epoxy resin; SBR, styrene-butadiene-copolymer; ABS, acrylonitrile-butadiene-styrene; PEA, polyesteramid; PTFE, polytetrafluoroethylene.

It has been proposed that a significant part of MP pollution originates from packaging materials that suffer abrasion or wear off, such as PET in bottles, PE in caps [54], wine stoppers [56], carton coating foils [35], and PP and PVA in milk bottles [37]. Older reusable containers could provide a greater MP concentration in beverages than newer ones, as found in an article on mineral water with PET particles

[36]. This is probably due to abrasion during cleaning, capping, transportation, storage, and opening of the containers, which is performed multiple times during the lifecycle of the container.

If plastic materials are used during beverage production as a part of tools, containers, and equipment, for example, these could also contribute to pollution. An investigation of MP in milk linked the presence of PSU particles to abrasion from membrane filters used during milk processing [37].

8.6 Human Exposure

Consumption of water by humans is a biological matter of survival. It can be consumed in its natural form or by adding other ingredients such as sugars, flavorings, and coloring, among others, or indirectly in milk, wine, or beer. However, the risk associated with the intake of these beverages may be affected by the presence of microplastics. This type of pollution can originate from different sources.

1. Microplastics can be in the used water during production [44].
2. Contamination can originate from various industrial processes [57].
3. Ingredients such as sugars, dyes, and preservatives, among others, could be contaminated with microplastics [58].
4. Beverages can contaminated by the plastic containers used for them, including lids [44].

The number of microplastics that enter the human body from drinking beverages depends on the consumption of each person, which is subject to various factors, such as the

region or country of the study [59], age and sex of the consumers [60], socioeconomic factors [61], cultural identity [62], and type of beverage [21].

However, in beverage studies, the estimated daily intake (EDI) tends to be calculated, which is an empirical value that represents the amount of a substance or contaminant in beverages or food that can be consumed in a period [63]. The ingestion of microplastics in beverages can be calculated as follows [22]:

$$\text{EDI (MP/kg/time)} = (C \times IR) / BW$$

Where:

IR, ingestion rate of studied beverage (L/d);

C, concentration of microplastics particle in bottled beverages (particles/L);

BW, average body weight (kg).

Table 8.3 shows the estimation of exposure on studies on microplastics.

Of the 14 studies cited in Table 8.3, only eight present the EDI value, while the rest were reported in concentration units. Although the EDI value has been presented in some articles, its comparison is complicated because of three factors: the microplastic size measurement range, the microplastic detection technique, and the five factors mentioned above.

The risks to humans related to the ingestion of microplastics have not been fully clarified; however, there is evidence that shows that they can accumulate in living tissues, causing oxidative stress [64], in addition to the fact that certain types of microplastics could be carcinogenic endocrine disruptors [65].

8.7 Conclusions

The present research demonstrates the omnipresence of microplastics in different types of packaged beverages, which is a relatively new subject. While it is true that microplastics were reported in all beverages examined in this study, the concentrations vary, and a significant part of this variation is due to the techniques employed, as they are neither standardized nor the units in which the information is reported, making comparisons difficult.

Table 8.3 Human exposure to microplastics through the reviewed beverages.

Country	Beverage type	Human exposure	Reference
Mexico	Water (refill kiosks)	1.47×10^4 MP/adult/year 6.73×10^3 MP/child/year	[29]
Mexico	Milks	858 MP/y (does not distinguish by type of population)	[66]
China	Bottled water	Plastic bottle (MP/kg/y): infants 1,127.9, children 744.6, adults 456.3 Glass bottle (MP/kg/y): infants 1489.2, children 985.5, adults 730.0	[53]
Iran	Bottled mineral water	Adults 5.35 MP/kg/y Children 23.43 MP/kg/y	[22]
China	Bottled water	Adults 100.0 MP/kg/y Children 219 MP/kg/y	[54]
Nigeria	Bottled water	Adults 0.019–0.193 MP/kg/y Children 0.120–0.800 MP/kg/y	[34]
Italy	Soft drinks	389.5 MP/y (does not distinguish by type of population)	[40]
Saudi Arabia	Water	Men 36.5–90.2 MP/kg/y	[45]

Country	Beverage type	Human exposure	Reference
		Women 36.5–73.0 MP/kg/y	
India	Milk	12,775–29,200 MP/y (does not distinguish by type of population)	[67]
Korea	Beer	223.6 MP/kg/y	[21]
	Soft drinks	707.2 MP/kg/y	
	Imported bottled water	1,300.0 MP/kg/y	
Mexico	Water	Adults 368.7 MP/kg/y Children 675.3 MP/kg/y	[25]
Turkey	Soft drinks	Adults 0.7–2.2 MP/kg/y Children 2.2–6.6 MP/kg/y	[52]
Malaysia	Bottled water	Adults 24.8–69.4 MP/kg/y Children 32.5–91.3 MP/kg/y	[68]

MP, microplastics; kg, kilogram; L, liter; y, year.

Although the different types of beverages studied share certain similarities in the techniques for the identification and classification of microplastics, having standardized techniques will not only allow result comparison but also ensure that methodologies are replicable in different parts of the world and under different study contexts. However, it is important to recognize that the different characteristics of beverages and the resources available to

research groups influence the methods used, so this must be considered when designing them.

There are different hypotheses regarding the origin of microplastics in beverages; however, studies agree that they are mainly derived from the water used in their production, followed by packaging materials and workplace areas. Therefore, sampling throughout the production process for microplastic analysis could be useful in determining their origin and preventing beverage contamination.

Among the most notable characteristics gathered from these studies, the size of microplastics was predominantly recorded in the smallest studied intervals (up to less than 5 µm). The most common types of microplastics were fragments and fibers, with transparent being the most frequent color (corresponding to the color of most current packaging). The typical chemical composition of microplastics consists of PE, PA, PP, and PET, which are regularly used in packaging fabrication, whether as bottles, bottle caps, or laminated carton coatings.

Regarding human exposure to microplastics through beverage consumption, estimations are difficult to compare among studies, even those examining the same type of beverage. This is because of the lack of consensus regarding whether calculations should be performed in a general manner or by specific population groups (e.g., adults, children, infants, or women and men) and whether individuals' body weight should be considered.

Microplastics present a significant global challenge. Plastics have become essential materials in fields ranging from the medical sector to everyday life. On the other hand, waste management has not been sufficient to prevent most plastics from entering the environment through treated and untreated water discharges, air deposition, and improper

disposal, where they contaminate terrestrial and aquatic ecosystems and can subsequently impact human health when present in water, which is vital for bodily functions. Urgent action is needed to prevent their release into the environment, reduce their impact, and implement microplastic removal in conventional water treatment processes for both drinking water and wastewater.

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9
Microplastics and Nanoplastics in Terrestrial Systems

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Abstract

Micro- and nanoplastics are generated by the breakup of massive amounts of plastic trash or by environmental release. Most of these plastics end up in terrestrial systems, without proper treatment. Community health is one of the major fields of study, and the possible discharge of plastics into the surroundings is a major concern. Over the last decade, the occurrence and toxicity of micro/nanoplastics in aquatic habitats have been broadly studied and reviewed, whereas terrestrial ecosystems have received less attention. Microplastics can modify the microbial, enzymatic, and physical/chemical characteristics of soil, which can have an impact on plant and soil fauna. Upon entry into plants, these contaminants accumulate and help as possible pathways for incorporation into the food chain and biomagnify alongside the food web, causing major health risks. This chapter aims to provide an overview of the interaction of micro- and nanoplastics, translocation, and their impact on terrestrial systems, including soil, plants, and terrestrial species.

Keywords: Nanoplastics, microplastics, terrestrial system, translocation, accumulation

9.1 Introduction

Plastics are a diverse group of malleable, synthetic, and semi-synthetic materials, mostly made of polymers. Plastics can be molded into various forms and sizes of solid objects depending on their application. Polymers are composed of certain elements such as carbon, hydrogen, oxygen, nitrogen, chlorine, and sulfur [1]. According to the International Union of Pure and Applied Chemistry (IUPAC), plastics are “polymeric materials that contain additional compounds to increase performance and/or lower costs” [2]. They are often synthetic, have a high molecular mass, and are frequently made from petrochemicals. Owing to their easy production, water resistance, chemical resistance, and temperature resistance, plastics can be utilized in a wide range of applications. Consequently, they are widely used in modern society, from spacecraft to stationaries [3, 4]. Over 6,300 million tons of plastic are produced globally, 79% of which are dumped in landfills and other ecological areas [5]. Plastics have become a part of our daily lives, and millions of plastic bottles and trillions of plastic bags are consumed annually. These plastic bags or bottles are only utilized for approximately 12 min before being thrown away, and require years to decompose [6]. Owing to inappropriate waste management and unauthorized disposal, environmental plastics are unrestricted. They have an extended lifespan and can easily accumulate in various environmental matrices [7]. Plastic pollution has been ranked the second most critical ecological problem worldwide after global warming [8–11].

Microplastics (1 mm to 5 mm) and nanoplastics (1 nm to 1,000 nm) are two forms of weathered plastics in the environment with different environmental implications and potential effects [12, 13]. In terms of size, quantity, ecological characteristics, and bioavailability, nanoplastics are different from microplastics and present a higher hazard

to the environment and public health, although their threat is still not fully understood [12, 14, 15]. Depending on the exposure method and the method of interaction with other pollutants, microplastics and nanoplastics have different harmful consequences in territorial environments. Plastic surface characteristics might alter significantly owing to interactions with these pollutants, and the agroecosystem may absorb these microplastics, nanoplastics, and/or contaminants [9, 16–18]. Primary microplastic particles produced for various applications, including industrial abrasives and cosmetics items, can enter the environment directly [19]. In contrast, secondary microplastics are created when the environment degrades larger amounts of plastic. Wearing plastic trash on soil or deep below the soil can produce secondary microplastics [20].

Materials that contain at least one nanoscale dimension and are not larger than 100 nm are considered nanoparticles [21]. The function of cells is affected by nanoplastics, as they are more prone to penetrate biological membranes [22]. Pollution on land due to plastics is 423 times greater than that in the ocean [23]. Nanoplastics can enter the soil in several ways, including the breakdown of larger plastic products (such as vehicle tires, packaging, and mulch made of plastic film), direct input of nanoplastic-containing products (such as personal care items), wastewater irrigation, sewage-derived fertilizer and landfills, atmospheric deposition [24–26], cleaning supplies [27], synthetic fibers from washing clothing and textiles, [28–30] coatings, paint, and abrasive blasting [31–33]. Direct littering and improperly managed garbage, such as industrial spills and landfill site releases, also add plastic to terrestrial ecosystems [27, 34–37]. Figure 9.1 shows different sources of micro and nanoplastics in terrestrial system.

This is also related to the deliberate or accidental burning of plastics, such as the negligent burning of plastics or accidental wildfires that discharge plastic particles into the air and surrounding environment, which are subsequently taken into neighboring rivers [34, 38, 39]. Owing to their small size, these micro- or nanosized particles can float in air, travel through water, and eventually build up for a very long time in soil. Agricultural activities may also release plastics into the atmosphere through inappropriate bale twine and wrapping disposal or by the application of sewage sludge to agricultural fields [6]. In addition to inadequately managed trash, the plastic stock present in terrestrial ecosystems either has a significant buildup or serves as a source for other ecosystem compartments [11]. It has been established that the primary sources of plastics deposited *via* wet deposition are urban areas and resuspension of plastic particles in soil [4].

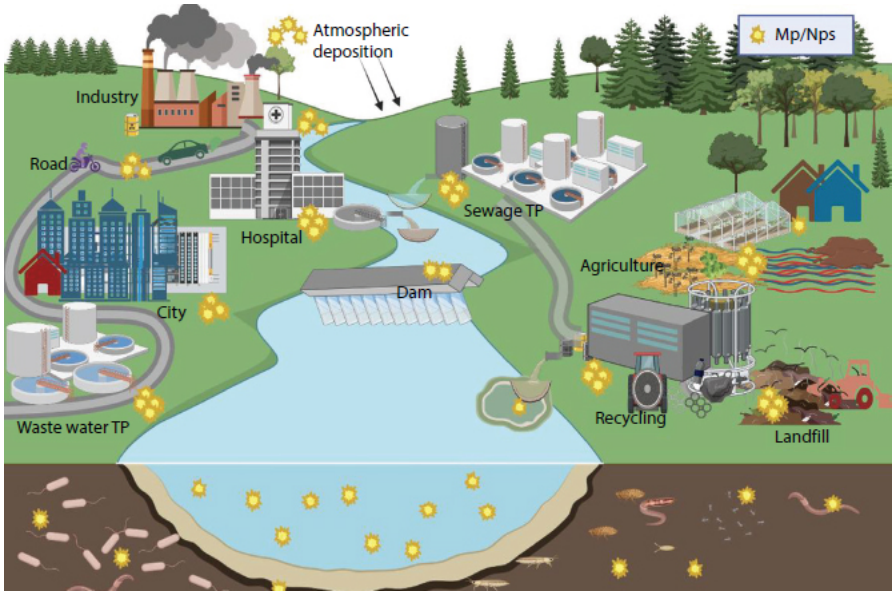


Figure 9.1 Conceptual figure of various sources of micro- and nanoplastics in terrestrial ecosystem.

9.2 Micro/Nanoplastics in Soil

Microplastic buildup in soil has several negative effects on soil ecosystems. Microplastics can infiltrate soil through several different pathways, such as trash, irrigation, flooding, composted soil, sewage sludge, littering, plastic mulching, silage films, boxes, plastic reservoirs, packaging components, films, used personal care product containers [40], and detergents [41, 42]. Microplastics can impair soil quality and nutrient cycling by changing the soil's physical qualities, reducing fertility, and upsetting the local microbial community [42–44]. Over time, plastic waste degrades into macroplastics (more than 25 mm), mesoplastics (5–25 mm), microplastics (less than 5 mm), and nanoplastics (<1 µm) [45–47].

Mesoplastics (5–25 mm) and macroplastics (>25 mm): Mesoplastics and microplastics coexist in farming and floodplain soils [48–50]. Mesoplastics have a higher mass concentration than microplastics because they are larger microplastics. Because macroplastics degrade and act as a source of microplastics, it is possible that soils with high levels of macroplastics also contain microplastics.

Microplastics (<5 mm): Globally, microplastic pollution has been identified in various soil types, including agriculture and forestry, greenhouses, home gardens, coastal environments, and floodplains [42]. Microplastic pollution is typically more concentrated when treated sewage sludge and wastewater are used as substitutes for fertilizer, and its repeated use can cause contamination of soil with microplastics [26]. Applying sewage sludge is not the only way microplastics eventually end up in agricultural land [51], and the breakdown or fragmentation of plastic trash is another key source for microplastics to end up in soil [52]. Organic fertilizers are also a route for microplastics to reach terrestrial ecosystems.

Nanoplastics (<1 µm): Nanoplastics are plastic particles <1 µm in size [53]. The formation of nanoplastics, the smallest plastic particles, is mostly due to the fragmentation of larger plastic to micro- and nanometer-sized particles [54]. They can be easily internalized into tissues and translocated further owing to their small size. Thus, there is a significant implication for the accurate assessment of their fate.

9.2.1 Source of Micro/Nano Plastics in Soils

Due to their widespread distribution, agricultural soil is the main dump yard for plastic waste. A key entry point for micro- or nanoplastics in the terrestrial environment appears to be the disposal of wastewater sludge as compost [55, 56]. There are four primary sources of micro/nanoplastic emissions into the ecosystem: the production of raw materials, the usage of nanotechnology products, the consumption of commodities, and waste management services [21]. Nanoparticles are commonly found in agricultural soils because of their extensive usage in products including nano-fertilizers, insecticides, seeding treatments, hydroponic solutions, and agro-films [57–62]. Other sources of micro/nanoplastics include mulching using plastics, irrigation, flooding, air deposition, trash, and roadway runoff [63, 64]. Nanoplastics are also released by waste treatment facilities as concentrated sludge or water and are found as artificial fibers incorporated into organic wastewater sludge [21, 56]. Solid waste landfills and unintentional sludge fertilizer discharge are the main propagators of micro/nanoplastic contaminants. The diffuse source of nanoplastics in soils is also due to their application as cosmetic and culinary additives. The deliberate discharge of nanoparticles into the environment can also result from remediation technology.

9.2.2 Effect of Micro/Nanoplastics

Terrestrial ecosystems may be at risk from micro/nanoplastics that contaminate the soil. These artificial materials may lead to changes in the soil, causing soil fauna to be under stress [43, 65]. Recent research indicates that the type of micro/nanoplastic determines how it affects soil characteristics [43, 66].

9.2.2.1 Effect of Micro/Nanoplastics on the Physical and Chemical Properties of Soil

The type of microplastic will influence its impact on soil properties. The effect on soil structure and the water cycle is less significant when the plastic particles size and shape are similar to soil particles. When compared to other types of microplastics like polyester or polyamide (PA) beads, which have impacts of lesser magnitude, polyester fibers, for instance, reduce the bulk density of soil and water-stable aggregates [43, 66]. Micro- or nanoplastics may potentially impact the chemical composition of the soil. In a study, the level of dissolved organic carbon, inorganic nitrogen, and total phosphorus in sandy loam soil considerably rose when the microplastics were added to the soil at a concentration of 28% (w/w) for 30 days. However, similar changes were not seen when the concentration was reduced to 7% (w/w) [67]. Many studies indicate that mulching using plastic residues reduces the fertility of the soil and adversely affects plant growth. The impacts of microplastics on the ecosystem, however, are not completely understood.

9.2.2.2 Effect of Micro/Nanoplastics on Soil Microorganisms

Micro/nanoparticles are a major factor influencing the toxicity of both pathogenic and beneficial microbes. They are lethal to native soil bacteria [68]. Invertase, phosphatase, urease, microbial community structure, bacterial diversity, dehydrogenase activity, nutrient cycling, alterations in humic compounds, and biological nitrogen fixation are all significantly impacted by micro/nanoplastics [69]. Micro/nanoplastics can transport a variety of bacterial, fungal, archaeal, algal, and other microbial species that negatively impact on soil quality and human health [63]. Microbial evolution is likely to be affected by the presence of micro/nanoplastics and microfibers during soil formation, stability, and dissolution [70]. The cornerstone of food production and climate regulation is biogeochemical cycling in which soil microorganisms play a significant role. Micro/nanoplastics may significantly affect the microbial populations and biophysical conditions of the soil [70]. Microplastics can also offer new microbial niches that

encourage the growth of particular groups of microbes, which could have unanticipated effects on ecosystem processes [71].

The impact of particles of various sizes on microbial populations depends on changes in the surface-to-volume ratios of micro/nanoplastic particles [72]. Due to their bioaccumulation in yeast and filamentous fungal cells, nanoplastics (<0.1 µm) can permeate the cell membranes and have lethal effects [43, 73–75]. Nanoplastics enter the soil-detritus food chain and build up there, causing a biological impact on bacteria. Since the water-stable aggregates are regarded as crucial habitations for soil organisms and hotspots for microbial activity, a reduction in these aggregates brought on by polyester fibers has significant effects on soil and plant health [76].

9.2.2.3 Effect of Micro/Nanoplastics on Soil Fauna

Certain anthropogenic contaminants can be the leading cause of environmental variations in soil that have an impact on soil organisms [43, 65]. The decay of organic substances, nitrogen cycling, bioturbation, and the control of soil-borne infections and pests are just a few of the ecosystem services that soil biota perform as part of soil ecosystem processes. The soil fauna is divided mainly into three groups: macrofauna (greater than 2 mm), mesofauna (0.1–2 mm), and microfauna (less than 0.1 mm) [77].

Earthworm development and survival are unaffected by low concentrations of microplastics (0%–0.5% w/w) but are suppressed by high concentrations (1%–2% w/w), which increases earthworm mortality [78]. Hence, depending on the concentration of micro/nanoplastics, the soil fauna may be directly poisoned. Earthworms function as vectors in the vertical and horizontal transport of micro/nanoplastics, increasing the risk of exposure to other soil species by dispersing micro/nanoplastics into the soil through their burrows, casts, and adhesion to the outer body surface of the worm [65]. Micro/nanoplastic ingestion is critical for regulating the toxicity of soil organisms and their potential for bioaccumulation in the food web [44, 65]. Moreover, micro/nanoplastics can concentrate and adsorb dangerous substances on their surfaces, thereby increasing the risk to soil fauna [79, 80].

According to previous studies [81], earthworms can improve the soil environment by increasing soil absorptivity, soil enzyme activity, and the availability of nutrients (particularly phosphorus). They greatly enrich the soil with high concentrations of readily digested nutrients such as P, N, K, and Ca. Numerous studies [82–84] have demonstrated that microplastics affect earthworm survival, growth rate, vermicast production, digging, and feeding behavior. As a result, microplastics may also indirectly affect plant growth by affecting soil creatures, such as earthworms [85].

9.2.3 Degradation and Transport of Micro/Nanoplastics

Micro/nanoplastics naturally deteriorate owing to biodegradation, thermal oxidation, UV radiation, and physical abrasion. These processes, which include chain cleavage, disproportionation, and an increase in oxygen-containing functional groups, change the chemical structure of the polymer chains that constitute microplastics [86]. However, owing to the nature of micro/nanoplastics, these processes are extremely slow (particularly in soil). For example, in one study, the degradation of polypropylene (PP) in soil was negligible (0.4%) after one year [87]. However, polyvinyl chloride (PVC) and polystyrene (PS) placed under soil for more than 32 years do not degrade [88]. Microplastics in the soil can travel, similar to other contaminants. They can migrate short distances through agricultural methods (such as plowing) and bioturbation. Concerning bioturbation, some earthworm and collembolan, [89] species play a critical role in determining the mobility of microplastics. They can move microscopic plastic particles from the soil surface to their depth. Additionally, some data suggest that microplastics

can move far distances through soil erosion and runoff, where they can infiltrate bodies of water and even the ocean [23].

9.3 Micro/Nanoplastics in Plants

Plants are an essential living component of terrestrial ecosystems. Because they are very common in soil, terrestrial plants will inevitably come into contact with micro/nanoplastics. Studies have reported that micro- or nanoplastics can accumulate in flora and have adverse effects on plant growth. For example, reduced biomass has been observed in *Triticum aestivum* and *Lolium perenne* following exposure to films and fibers. Additionally, microplastics have several effects on plant growth, including direct toxicity, altering the structure of the soil, immobilizing nutrients, and affecting root symbionts and microbial communities in the soil [90, 91]. As plants are crucial active constituents of terrestrial ecosystems, it is essential to comprehend the effect of micro- or nanoplastics on plant communities [92]. Soil ecosystems worldwide, including agricultural fields, urban and industrialized zones, and rural areas, have now been reported to contain microplastics. The current state of knowledge regarding the absorption, transportation, and biological impact of micro- or nanoplastics on terrestrial flora is discussed in this section. This article also discusses the effects of exposure to microplastics or nanoplastics on the morphological, physiological, and biochemical characteristics of plants.

9.3.1 Source of Micro/Nanoplastics

Micro/nanoplastics can intrude the environment, both directly and indirectly. Microplastics penetrate plant communities in terrestrial habitats, mostly through the soil. When plastic is utilized for a purpose, such as in soil conditioners, building materials, and plastic mulch [53, 63], it results in direct contamination. Even after treatment in a wastewater treatment plant (WWTP), micro/nanoplastics remain in wastewater, and those not retained finally end up in the sludge. Irrigation using wastewater from urban and industrial areas and the application of sludge on land are two important indirect routes taken by micro/nanoplastics [54]. Various types of anthropogenic activities release plastics into the land that later reach the ocean and are regarded as indirect sources of microplastics in the oceans. Microplastics that accumulate in the ocean are transported to other land areas *via* ocean currents. Oceanic movements can consequently cause coastal land to become contaminated with micro- or nanoplastics from sources other than those in the vicinity [54]. The main sources of micro/nanoplastics introduced in plant communities include agricultural irrigation water, agricultural mulch film and plastic debris, municipal sludge on land, and microplastics from the atmosphere [24–26]. Among these, airborne microplastics can reach the soil and directly affect roots and aboveground plant parts [93].

9.3.1.1 Plastic Mulching

Mulching is a method of protecting the top layer of the soil to retain soil moisture by reducing evaporation, and black plastic films are commonly used for this purpose. Mulching is a significant method by which microplastics are introduced into soil systems, impacting agricultural plant communities [53, 94]. Plastic mulching is widely used in areas where plastic leftovers cause severe environmental issues. The presence of plastic film remnants can change the structure and porosity of soil, which has a substantial effect on enzyme activity and microbial diversity of the soil [67, 95]. These alterations have long-term, substantial consequences on plant biomass and community structure. For instance, the addition of plastic film causes fragmentation, and these fragments in the soil enhance the evaporation of soil water by forming desiccation cracks, which causes water loss and encourages the emergence of drought-tolerant plants in the neighborhood [96]. The widespread use of plastic mulch in agricultural areas is a significant contributor to soil

erosion. However, owing to its various advantages, this type of mulching is now commonly used in agriculture worldwide. Regrettably, soil nutrient and carbon stocks are decreased by plastic mulch [97]. The carbonyl groups in the buried mulch films harm microorganisms by preventing their growth. Micro/nanoplastics are a type of physical soil contamination, and preliminary studies indicate that microfibers may reduce soil bulk density [43].

9.3.1.2 Packaging

Plastics have been used more frequently in packaging, thereby significantly improving daily life. The primary cause of plastics in the market is packaging [98], and as plastic resin production develops, consumer marketplaces acquire more plastics daily [99]. The widespread use of plastics, also referred to as “white pollution,” negatively impacts the environment [100]. All disposed plastic waste eventually reaches terrestrial and marine environments [9].

9.3.1.3 Irrigation Water

The primary sources of irrigation water include ground water, surface water, and treated water from sewage systems, all of which contain microplastics in a variety of shapes, material compositions, and concentrations [101–103].

9.3.1.4 Sewage Treatment Plants (STPs)

The principal sources of micro- and nanoplastics in sewage wastewater treatment facilities include clothing fibers, personal care items, microbeads, cleaning products, and plastic debris. [9]. Microplastics cannot be completely removed by the standard sludge extraction methods employed in sewage treatment plants. Sludge compost is a leading source of microor nanoplastics in the soil as it spreads in the soil as fertilizer [104–106]. Washing clothing is the main source of sewage micro/nanoplastics, and N1900 fibers are produced from each wash in household washing machines [28]. Common wastewater treatment practices release 300 million pieces of plastic garbage daily into adjacent waterways [107].

9.3.1.5 Wastewater Treatment Plant (WWTP)

Even after treatment in a wastewater treatment plant (WWTP), many micro/nanoplastics remain in wastewater, and those not retained eventually reach in the sludge [54]. To manage aquifer recharge, industrial, and wastewater treatment plants (WWTPs), effluents are frequently discharged into surface waters or even directly administered into aquifers, creating another pathway for micro/nanoplastic contamination [54, 108].

9.3.1.6 Air-Borne

Very minute particles or fibers distributed through air (coming from surface dumps or landfills) enter terrestrial systems by atmospheric deposition [20, 109]. The most recent studies report that the deposition flux of microplastics in the atmosphere can reach upto 1.46×10^5 particles per square meter of an area, of which 95% constitutes fibers and 50% are microplastics below the size of 0.5 mm with polyesters, polyvinyl chloride, polyethylene, and polystyrene as their main constituents [110]. Atmospheric deposition of microplastics has a major impact on plant communities in urban cluster areas [111].

9.3.1.7 Others

Other typical indirect pathways include unintentional release, careless discards, and improper disposal techniques [112]. The accidental release of micro/nanoplastics during manufacture, transport, and usage occurs more frequently in heavily industrialized areas [113].

9.3.2 Migration or Uptake of Micro/Nanoplastics From Soil and Atmosphere

Some microplastics cannot permeate cellulose-rich plant cell walls and stomata because of their size. However, when microplastics break down into nanoplastics, they can enter plant cells. In addition, the transport processes of microplastics into cells (i.e., through plasmodesmata), the kinetics of transport, and their accumulation and toxicity are size-dependent [114]. The charge of the nanoparticles has a direct impact on how well they attach to the plant cell surface because of the negative charge on the cell wall. The uptake and translocation of microplastics are influenced by the hydrophobicity on the plant surface [115, 116]. The first *in vitro* study of the uptake of micro/nanoplastics by plants conducted on tobacco reported that the plant cells did not take up 100 nm nano polystyrene beads, but they did take up 20–40 nm beads [9, 117]. Micro/nanoplastic uptake has been observed in the root hair of plants. The roots of plants are most affected by microplastics, followed by shoots and leaves. This is due to the ease of absorption of microplastics by plant roots present in polluted soil and deposition from the air to plant parts [26]. However, nanoparticles (<0.1 μm) can pass through biological membranes into plant cells, thereby incorporating them into the food chain [118, 119]. Smaller size is an essential factor that facilitates the entry of nanoplastics through cell wall pores.

9.3.2.1 Uptake Pathways of Micro/Nanoplastics

The two different exposure modes of plants to micro- or nanoplastics are root exposure and foliar exposure.

9.3.2.1.1 Root Uptake

Plants mainly bioaccumulate micro/nanoplastics through their roots. For uptake and translocation, nanoplastics must pass through physiological root barriers, comprising the cuticle on the root surface, followed by the epidermis, cortex, endodermis, and Casparian strip, before moving up towards the shoots through the xylem. The function and make-up of the cuticle on the root surface were similar to those of the cuticle on the leaves. The potential of nanoplastics to penetrate the cuticle of the root surface is uncertain. However, nanoplastics can directly penetrate the epidermis in areas where the cuticle is not fully developed, such as the development of root hairs and tips of primary and secondary roots [120, 121].

The apoplastic and symplastic pathways are the two primary routes by which roots absorb nanoplastics and transport them to other plant tissues (Figure 9.2). The apoplastic pathway is also known to be a nonselective direction with the least resistance. The apoplastic channel for translocation is favored by various nutrients and unwanted metal complexes that dissolve in water [122]. After entering the cell wall pores, the nanoplastics disperse inside the intercellular space or pass through it without entering the cell membrane. Studies have reported the presence of nanomaterials in the apoplastic space of plant roots, including 20 nm ZnO in ryegrass roots, 12 nm Au, 20–80 nm Ag, and 43 nm CuO in *Elsholtzia splendens*. Ma *et al.* [123] reported that the cell wall pores diameter ranges from 5 to 20 nm, but an intriguing finding was that the presence of nanoplastics larger than 20 nm had been detected in the intercellular space. Nanoplastics may break down cell walls, by expanding the pore size. Another possible hypothesis is that nanoplastics penetrate the space between cells or even xylem *via* dead roots or physical wounds caused by herbivores or mechanical injuries [120, 124]. The physical barriers present in plant tissues restrict the internalization of large sized microplastics from being internalized [121].

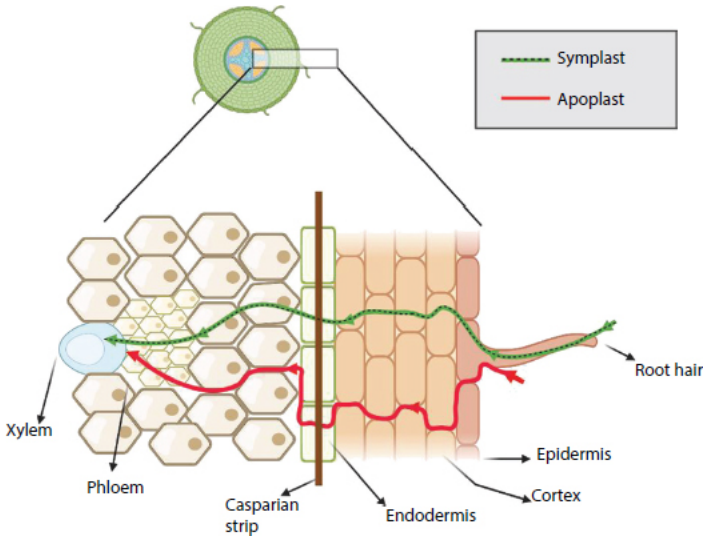


Figure 9.2 Transport mechanism of micro/nanoplastics.

The Casparian strip in the root blocks apoplastic passage of substances and liquids into the root stele. The casparian band of roots is a discontinuous section that may allow for unrestricted transit in the apoplast. Both the areas of secondary root initiation and the root tip, where the endodermal cells are immature, are home to these zones. These openings, which have submicrometer widths and lengths that can range up to several micrometers (crack-entry mode), are known mechanisms of infection by bacteria or plant diseases. These may serve as pathways for the intake of micro- or nanoplastic particles [125]. Cells move *via* a mechanism called the symplastic route. Two barriers are involved in the transport of nanoplastics *via* the symplastic pathway. First, the cell membrane is penetrated by nanoplastics, allowing access to the cytoplasm. The second involves the transport of nanoplastics to nearby cells through the plasmodesmata [120]. Plasmodesmata are intercellular conduits used for symplastic transfer. Plasmodesmata are channels that cross a plant's cell wall and connect the cytoplasm of nearby cells, allowing intercellular communication [126]. Numerous studies of macromolecular transport through plasmodesmata have shown that biological macromolecules such as nucleic acids, RNA, and proteins are transported between cells through these channels [127].

9.3.2.1.2 Foliar Uptake

Because of their broad and uneven surfaces, plant leaves contribute significantly to the interception of some airborne microplastics and serve as their main sink. The waxy cuticle of higher plants serves as the initial barrier for nanoplastics to enter the tissue and protect the leaves from water loss and unrestricted interchange of certain solutes [128]. Pathways that support solute uptake across the cuticle (cuticular pathways) are the lipophilic pathway that transports nonpolar solutes by diffusion and penetration, and the hydrophilic pathway that transports polar solutes through polar aqueous pores [129, 130]. Accordingly, nanoplastics larger than 5 nm were absorbed by the leaves and accumulated, whereas nanoplastics smaller than 4.8 nm could easily pass through the cuticle *via* the cuticular pathway. A characteristic of nanoplastics that might alter foliar uptake pathways is lipophilicity, which has been previously observed in molecules [130]. Different leaf properties (such as trichomes, stomata, or cuticle) give a wide spectrum of lipophilicity, of which the guard cell of stomata is hydrophilic and protein like whereas leaf cuticle is lipophilic. Therefore, it is probable that the amphiphilicity and lipophilicity of nanoplastic surfaces affect the extent of absorption of nanoplastics through the cuticle [131]. The

cuticle serves as the main leaf barrier, restraining the passage of nanoplastics smaller than 5 nm. Nanoplastics (>10 nm) enter the stomata, and their cellular transit into plant vascular systems occur via apoplastic or symplastic routes [115, 132].

Studies have shown that hydrophilic chemicals can also be absorbed through the stomatal pores (stomatal pathway) and cuticular channels. Stomatal apertures are roughly 25 µm in length and 3–10 µm wide in morphology [129]. Thus, the only known route for nanoplastics to enter the internal tissues from the leaf surface is through the stomata. Numerous plant species, including *Arabidopsis thaliana*, *Citrullus lanatus*, *Cucurbita pepo*, *Allium porrum*, and *Lactuca sativa*, have been detected using TEM, CLSM, or XRF in their leaf stomata and deeper tissues, thus supporting this pathway of nanoplastic uptake [120].

9.3.3 Accumulation and Translocation

Microplastic buildup in plants has direct ecological effects and impacts on food safety and agricultural sustainability [133]. Because of contemporary farming methods, agroecosystems are more susceptible to plastic contamination. Despite the potential for these pollutants to enter the agricultural system, less information is currently available on the micro- and nanoplastics that disturb soil biota, particularly plants. The effects of micro/nanoplastics on plant growth and biomass production have recently been documented [9]. In-depth research on nanoplastics on wheat and broad beans has been published by Jiang *et al.* and Lian *et al.* [102, 134]. Studies have also been conducted to understand the toxicity of nanoplastics applied to the foliage of lettuce and maize plants [134, 135]. They can accumulate in these either by adsorbing on surface of the roots or by being absorbed by roots, fruits, and vegetables [102, 136, 137]. Different types, shapes, sizes, and concentrations of microplastics affect plants differently [93]. Once micro- or nanoplastics enter plant roots, they can move upward through the vascular stream to the aboveground regions.

In a study using fluorescent tags in lettuce, wheat, and *A. thaliana*, microplastics were visible. Submicron sized particles were trapped and adsorbed on the exterior of root owing to the mucus on the root cap. They are unable to penetrate the cortex of the root or extend to the stellar region by crossing apoplastic barriers and the free intercellular space in root cells [138]. However, submicron or nanosized microplastics can enter plants *via* vascular tissue along with water and nutrients by means of transpirational pull [90].

A report, for the first time, clearly showed how pumpkin (*C. maxima*) roots absorb iron oxide nanoparticles (Fe_3O_4) and transport them through plant tissues [44]. Plant transpirational pull significantly influences the absorption and translocation of microplastics and nanoplastics [123].

Some microplastics also enter the plant’s stele through cracks at the locations from which lateral root emergence, passes through xylem vessels in the roots and then moves to the stem and leaf areas [93]. In a study of root uptake by wheat and lettuce, the sub-micro-sized polystyrene (PS) and polymethyl-methacrylate particles (PMMA) were observed to penetrate the stele and were transported to the shoot area [90]. Plastic beads as small as 200 nm and larger beads as large as 2 µm may enter the root xylem vessels through cracks formed at regions of lateral root initiation. PS nanoplastics with particle sizes between 50 and 150 nm also enter carrot roots and reach the leaves [135, 139].

In an exposure study of PS nanoparticles in rice roots, it was observed that PS nanoparticles (~20 nm) were distributed in the intercellular spaces, whereas in the case of onion seeds cultivated in PS (~50 nm), plastic particles were found in the central, cortical, and epidermis of the primary roots after 72 h [140]. This finding shows that polystyrene nanoparticles can pass through various biological barriers and enter root cells [141].

In another study, it was found that cells other than the root cap mucilage, normally visible to the naked eye, had fluorescent labeled 0.2 µm PS microbeads trapped on them [125]. Negatively charged mucilage and exudates, which serve as the first line of defense in plants, have been found to prevent metal nanoparticles with positive charges from adhering to the outside of the cell wall. The 0.2 µm PS luminescence signal was found on the root cortical tissue cell wall. They were mostly found in the vascular system, demonstrating that the apoplastic transport system transports the PS beads through intercellular channels.

PS beads were observed inside the apical meristem and throughout the lateral root cap of wheat (*T. aestivum*) and lettuce (*L. sativa*) roots. The apical meristem tissue undergoes active cell division and permits particles from the root cap to penetrate the cell wall. While the Casparian is still developing, only the epidermal layers of the apical root section allow PS beads to enter the apical meristem. The PS beads entered the cortex by openings in the epidermal cells but could not pass through the endo-epidermis owing to the permanence of the Casparian band. Strong PS luminescence signals were eliminated in the lateral root apex cracks through which these particles entered the epidermis, endodermis, and cortex. Small PS beads enter through cracks into the root xylem of wheat and lettuce [9, 125].

The cytoplasm of adjacent cells is favors the transfer of nanoparticles with sizes between 10 and 50 nm (symplastic pathway). Thus, the apoplastic pathway allows the translocation of larger nanoparticles (50–200 nm). Internalized nanoparticles are transported through phloem sieve tubes along with the sugars. As these organs act as powerful sap sinks, nanoparticles can flow in both directions and accumulate in roots, stems, fruits, grains, and young leaves to various degrees due to vascular transport through the phloem [114, 115, 142, 143].

Aggregates of micro- and nanoparticles can be absorbed or adsorbed by plants [144]. The absorption and transit of plastic particles by plants are significantly controlled by the transpiration pull. Plastic particles penetrate the epidermal tissue in the primary and secondary roots of wheat, stimulate through the pericycle, and are then transported into the xylem [125]. These particles migrate to the aboveground portions of the plant *via* the xylem. The vascular system transpires plastic particles from the roots to shoots. Confocal images demonstrated that the vascular system of the stem is the primary location of plastic luminescence signals. Stems, leaves, and fruits can swiftly absorb water transport systems that support nanoplastics.

Plants with higher transpiration rates have a greater capacity to absorb plastic particles. Once embedded in plant roots, microscopic plastic particles can follow the transpiration stream *via* xylem to the aboveground regions of the plant. Thus, higher transpiration rates might act as more potent inducers of micro (nano)plastic particle uptake and translocation inside plants [141].

Only a small percentage of micro- and nanosized plastics enter plants through the intercellular region of root cells, and these particles require a specific amount of transpiration pulling force to reach the aboveground portion of the plant [93]. The uptake and transport of micro- or nanoplastics in plants is a passive method that is predominantly powered through the vascular stream by transpiration pull. Therefore, factors that disturb transpiration frequency influence the migration of micro- or nanoplastics in terrestrial plants [125].

9.3.4 Effect of Micro- and Nano Plastics

The impact of micro/nanoplastics on higher plants differs depending on the microplastic characteristics, plant type, and environment. Certain studies have shown that under specific experimental conditions, microplastics have no effect on higher plants or have a

beneficial impact. However, other studies have shown that micro/nanoplastics have direct and indirect effects on plants. Research has revealed that microplastics can impede more advanced plant growth and performance [85]. Polyethylene (PE), polypropylene (PP), polystyrene (PS), polyvinyl chloride (PVC), polyamide (PA), and polyethylene terephthalate (PET) are among the microplastics frequently found in the environment [88]. PS is the most commonly chosen microplastic for investigation. One potential explanation is that the fundamental PS spheres have modest and homogenous particle sizes (≤1 µm) [43]. Many studies have used hydroponic techniques to examine the effects of microplastics on higher plants. This was mostly due to the even distribution of micro/nanoplastics in water [140]. As a result, microplastics have greater effects on plants when grown hydroponically than on soil [145]. The following Table 9.1 provides brief data on the diverse effects of micro- or nanoplastics on terrestrial plants.

9.3.4.1 Inhibitory Effect

Most investigations have demonstrated that microplastics inhibit higher plants. The inhibitory effects include decreasing the rate of seed germination [146, 148, 193, 201], inhibition of root activity and growth [66, 91, 137, 160, 171, 190, 194, 202, 203], oxidative stress [160, 181, 190, 191], disturbing antioxidant enzyme activity [102, 135, 173, 176], affecting photosynthetic efficiency [145, 155, 171, 196, 202], inhibiting gene expression [42, 160, 175], affecting normal metabolism [125, 140, 173, 191], changing tissue elemental composition [66, 158, 192], and reducing biomass [41, 91, 137, 146, 150, 151, 167, 192, 198, 204].

9.3.4.2 Blocking Pores or Light

In some plants, root uptake and transport of minerals, such as potassium and iron, are negatively affected by PS. This also interferes with the spread and reuse of these mineral elements. Micro- or nanoplastic particles accumulate in the vessels of leaves, blocking cell junctions or cell wall pores. After exposure to microplastics, plant growth is inevitably impeded owing to a lack of nutrients and water [42, 90, 102, 167]. Microplastics can build up on seed capsule pores and root surfaces, inhibiting water absorption and lowering the rate at which plant seeds germinate [176].

Table 9.1 Effects of micro/nanoplastics on higher plants.

	Micro/nanoplastics					
Plant sps.	**Type**	**Size**	**Shape**	**Concentration**	**Effect**	**Refere**
Lolium perenne	HDPE	102.6 µm	particle	0.1% (w/w)	An increase in root/shoot ratio and root biomass.	[146]
	PLA	65.6 µm	particle	0.1% (w/w)	Fewer seeds germinated, reduced biomass and shoot height.	[146]
Lepidium sativum L.	PC	3 ± 1 mm	Fragment	0.1–10% (w/w)	Fewer seeds sprouted, resulting in a drop in biomass and shoot height.	[147]
	PS	50–4,800 nm	sphere	10^6–10^{10} Particles/L	Accumulation on the root hairs, decreased germination rate and blocked seed capsule pores.	[148]
	PET, PP, PE, and PVC	0.125 mm	Fragment	0.02% (w/w)	On exposure in first 8hr reduced germination rate was observed. After 24 h, no significant difference was observed. After 24 h, variations in root growth were seen, however the effect was negligible after 48–72 h.	[149, 1
Festuca brevipila, *Calamagrostis epigejos*, *Holcus lanatus*, *Hieracium pilosella*,	PET	1.28 ± 0.03 mm	Fibers	4 gkg^{-1}	Plant community structure was affected by PES fibers.	[151]

	Micro/nanoplastics					
Plant sps.	**Type**	**Size**	**Shape**	**Concentration**	**Effect**	**Refere**
Achillea millefolium, *Plantago lanceolata*, and *Potentilla argentea*						

	Micro/nanoplastics					
Plant sps.	**Type**	**Size**	**Shape**	**Concentration**	**Effect**	**Refere**
Lactuca sativa L.	PS	0.2 and 1.0 µm	Microbeads	–	Microplastics adhere, uptake, buildup, and translocate in vascular tissue.	[152]

	Micro/nanoplastics					
Plant sps.	**Type**	**Size**	**Shape**	**Concentration**	**Effect**	**Refere**

	Micro/nanoplastics					
Plant sps.	**Type**	**Size**	**Shape**	**Concentration**	**Effect**	**Refere**
	PMMA	0.2–10 µm	Microbeads	150–500 mg/kg	Beads of 0.2 and 2 µm were observed in root, xylem sap and leaves, while beads with size more than 5 µm were rarely present.	[90]
	PE	~23 µm	Fragments	0.25, 0.5, 1.0 gL^{-1}	Lettuce root activity and development were restrained. The concentration of soluble protein, sugar, and vitamin C in leaves decreased and cellular structure was distorted.	[153]
	PS	0.1–>10 µm	Sphere	0.25–1 gL^{-1}	The biomass and height of lettuce decreased, and the effect of oxidative stress was caused by PS exposure in plant tissues.	[153]
	PVC	100 nm–150 µm	Fragment	0.5%, 1%, 2% (w/w)	Small particle sized PVC was taken up by leaves increasing the physiological toxicity of lettuce leaves.	[154]
	PMFs	2.55 mm	Fiber	0.1%, 0.2% (w/w)	Plant shoot length, photosynthetic capacity, and chlorophyll content were affected	[155]

	Micro/nanoplastics					
Plant sps.	**Type**	**Size**	**Shape**	**Concentration**	**Effect**	**Refere**
					negatively by PMFs. However, PMFs increased lettuce's roots. PMFs changed the metabolic pathways connected to nitrogen and carbohydrates.	
	Macro EPS	8.3 ± 0.5 mm	Sphere	-	Low degrees of contact with the crop's water absorption rate had no negative effect on the crop.	[156]
	PS	0.23 ± 0.04 µm	spherical	10 (mg/mL)	The vascular assembly transports the polystyrene to the stem and leaves.	[125]
	PS	93.6 nm	spherical	0, 0.1, 1 mgL^{-1}	The lettuce plant showed decrease in height, dry weight, leaf area and pigment content. This decrease of micronutrients and essential amino acids indicates decline of its nutritional value.	[134]
	LDPE	4–10 mm	Film	1% (w/w)	A decrease in biomass. Changed composition of bacterial community	[91]

	Micro/nanoplastics					
Plant sps.	**Type**	**Size**	**Shape**	**Concentration**	**Effect**	**Refere**
					near rhizosphere.	
	Bio	50 µm–1 mm	film	1% (w/w)	A change in bacterial community in rhizosphere affecting mixing of volatile organic compounds	[91]
Tobacco BY-2 cells	PS	20–1,000 nm	Sphere	1:1,000 v/v	The cells were able to quickly internalize 20–40 nm sized PS nanobeads through endocytosis, but they were unable to do so for 100 nm particles. Tobacco protoplasts, on the other hand, internalized PS particles as large as 1,000 nm.	[117]
	PS	20 nm	sphere	30 and 10 µM	Endocytosis of 20 nm-diameter beads by BY-2 cells was observed while larger beads (100 nm) were not internalized. Protoplasts internalized beads as large as 1,000 nm diameter.	[117]
Murraya exotica	SMA	12 ± 4.5 nm	–	55 mgL^{-1}	Plant stems had SMA nanoparticles, which were continuously enriching over time.	[157]

	Micro/nanoplastics					
Plant sps.	**Type**	**Size**	**Shape**	**Concentration**	**Effect**	**Refere**
Arabidopsis thaliana	PS-NH_2 and PS-COOH	56–71 nm	Sphere	10–100 mgL^{-1} 10–50 mg	Epidermis and hairs of root showed PS-NH_2 accumulation whereas stellar region and maturation zone had PS-COOH which was more selectively absorbed by plants than PS-NH_2.	[133]
	PS	1 µm	Sphere	0.029 gL^{-1}	The root surface, especially the tip had PS accumulation. However, there was no evidence that the internal root structure contained plastic particles.	[133]
Cucumber (*Cucumis sativus* L.)	PS	100–700 nm	sphere	50 mgL^{-1}	PS is transferred through the stem and its degradation releases benzene, thus altering sugar metabolism and chlorophyll content.	[158]
	CuO	40 nm	–	800 mgL^{-1}	Altered fruit nutritional supply, especially metabolites.	[159]

	Micro/nanoplastics					
Plant sps.	**Type**	**Size**	**Shape**	**Concentration**	**Effect**	**Refere**
Onion (*Allium cepa* L.)	PS	100 nm	Sphere	25–400 mgL^{-1}	Significant reduction in root length and induced cytogenetic toxicity by producing ROS and inhibiting the expression of *cdc*2.	[160]
	PS	20–190 nm	Particle	0.01–1.0 gL^{-1}	Polystyrene particles enter in cells of root and gets accumulated in cytoplasm and vacuoles. This accumulates in epidermis, root central and cortical regions. In the cell nucleus, smaller polystyrene particles i.e., 25 nm were found.	[161]

	Micro/nanoplastics					
Plant sps.	**Type**	**Size**	**Shape**	**Concentration**	**Effect**	**Refere**
Spring onion (*Allium fistulosum*)	PA, PS, HDPE, PP, and PET	15–5,000 µm	Fiber	0.2–2.0% (w/w)	Both the total and root biomass increased, along with mean diameter, root length, and microbial activity.	[66]
	PA, PFs, PEHD and PP virgin pellet	8–5,000 µm	spheres	PES 0.2%, PF, PP, and PEHD 2.0% (w/w)	Change in soil microbial activity, plant biomass, root characteristics, and tissue elemental composition.	[43]
	PS	547–555 µm	particle	2% (w/w)	Total root length, mean diameter, and root biomass all considerably increased.	[66]

	Micro/nanoplastics					
Plant sps.	**Type**	**Size**	**Shape**	**Concentration**	**Effect**	**Refere**
Wheat *(Triticum aestivum* L.)	Bio	1.61 mm	film	1% (w/w)	Root/shoot ratio increased, decreased fruit biomass, stem biomass reduced, and plant height growth was inhibited.	[91]

	Micro/nanoplastics					
Plant sps.	**Type**	**Size**	**Shape**	**Concentration**	**Effect**	**Refere**

	Micro/nanoplastics					
Plant sps.	**Type**	**Size**	**Shape**	**Concentration**	**Effect**	**Refere**
	LDPE	11.3 µm	spherical	100, 500, 1,000 (mgL^{-1})	The germination rate was reduced, but no changes in the wheat biomass, root or bud length.	[162]
	PS	100 nm	spherical	0.01–10 (mgL^{-1})	PS had no impact on seed germination rate but increased the growth and chlorophyll content of wheat seedlings. PS decreased the number of micronutrients and the shoot to root biomass ratio.	[163]
	PA	15–20 µm	Microsphere	2% (w/w)	The total root length, mean diameter, and biomass all increased.	[163]
	PMMA and PS	0.2–10 µm	Spheres	0.5–50 mgL^{-1} 150–500 mg/kg	The stem, leaf and xylem portions of the exposed plants contained 0.2 mm and 2 mm beads; microbeads smaller than 5 mm were rarely taken up.	[118]
	PVC and PE	125 µm	Fragment	1–20% (w/w)	At low exposure doses, shoot and root biomasses dropped, whereas at high exposure	[164]

	Micro/nanoplastics					
Plant sps.	**Type**	**Size**	**Shape**	**Concentration**	**Effect**	**Refere**
					doses, these increased.	
	CuO	~40 nm	–	500 mg/kg	Reduction in root length, plant-induced changes in rhizosphere.	[165]
	PS	0.5 µm	Fragment	100 mgL^{-1}	PS had a moderating effect on the toxicity of heavy metals but had no apparent impact on the growth, photosynthesis, or ROS concentration in wheat seedlings.	[162]
	PE	200 µm	Fragment	800 mgL^{-1}	Root length of wheat increased.	[166]
	PE	200 µm	Fragment	0.2% (w/w)	Wheat growth was not significantly affected by PE.	[166]
	PS	100 nm, 5 µm	Sphere	Hydroponics: 0, 10, 20, 50, 100, and 200 mgL^{-1}) Soil culture: 0, 1, 10, 50, and 100 mg/kg	In the hydroponics test, high concentrations drastically decreased the elongation of wheat roots and stems. In soil culture, damaged photosynthetic system and altered protein synthesis was observed.	[145]
	PE	250 µm	Fragment	0.5%, 1%, 2%, 5%, 8% (w/w)	PE caused elongation in wheat root, but no increase in	[118]

	Micro/nanoplastics					
Plant sps.	**Type**	**Size**	**Shape**	**Concentration**	**Effect**	**Refere**
					aboveground biomass.	

	Micro/nanoplastics					
Plant sps.	**Type**	**Size**	**Shape**	**Concentration**	**Effect**	**Refere**
Maize (*Zea mays* L.)	PE	3 µm	sphere	100 mgL^{-1}	Reduced plant root and shoot biomass after exposure to PE; Root carbon content increased while shoot nitrogen content decreased; Significantly lower plant transpiration was observed.	[167]

	Micro/nanoplastics					
Plant sps.	**Type**	**Size**	**Shape**	**Concentration**	**Effect**	**Refere**
	PE	2 cm × 2 cm	Fragment	150–600 kg/ha	Root and shoot growth suppressed, and yield and water consumption were dramatically reduced.	[168]
	HDPE	100–154 µm	particle	0.1%, 1%, 10% (w/w)	Plant biomass did not show significant changes.	[169]
	PS	100–154 µm	particle	0.1%, 1%, 10% (w/w)	The biomass of the plant decreased, particularly the root system.	[169]
	PLA	100–154 µm	particle	0.1%, 1%, 10% (w/w)	Low concentrations (0.1%, 1%) exhibited no significant impact, but high concentrations (10%) drastically reduced plant biomass.	[169]
	PCF	3.79 ± 0.60 mm	Fragment	0%, 0.01%, 0.1%, 1% (w/w)	It was seen that addition of 1% microplastics increases chlorophyll *a*, chlorophyll *b* and carotenoid content. There was no significant effect on root.	[134]
	PS	22.0 ± 1.5 nm, 24.0 ± 2.2 nm	Spherical	100 mgL^{-1}	In comparison to PS-COOH, PS-NH2 exhibits greater photosynthesis inhibition and	[135]

	Micro/nanoplastics					
Plant sps.	**Type**	**Size**	**Shape**	**Concentration**	**Effect**	**Refere**
					more antioxidant system stimulation.	
	PE	212–300 µm	Spherical	0.1% (w/w)	Soil and bacterial community were negatively affected by PE. It also affected the expression of antioxidant gene.	[41]

	Micro/nanoplastics					
Plant sps.	**Type**	**Size**	**Shape**	**Concentration**	**Effect**	**Refere**
Rice (*Oryza sativa* L.)	PS	19 ± 0.16 nm	Spherical	10–100 mgL^{-1}	The rice roots absorbed PS particles, which subsequently accumulated in the intercellular spaces.	[170]

	Micro/nanoplastics					
Plant sps.	**Type**	**Size**	**Shape**	**Concentration**	**Effect**	**Refere**
	PTFE and PS	10 µm	Fragment	0.04–0.2 gL^{-1}	Highly concentrated microplastics reduced grain biomass, chlorophyll content, and net photosynthetic rate. In rice tissues, microplastics caused an oxidative burst.	[171]
	PVC	0.5 cm × 0.5 cm	Fragment	2–6 g/kg	Both the biomass and quantity of rice tillers dropped.	[172]
	PS	8.5–30.7 µm	Fragment	50–500 mgL^{-1} 50–250 mg/kg	PS treatment decreased the biomass and length of the shoots but had not much impact on root growth. It also lowered MDA and POD level as well as SOD and POD activity in rice leaves. At lower exposure doses, leaf CAT activity and ROS increased, whereas at higher exposure the activity decreased. In leaves, there was a reduction in the levels of organic acids, saccharides, and amino acid metabolites.	[173]

	Micro/nanoplastics					
Plant sps.	**Type**	**Size**	**Shape**	**Concentration**	**Effect**	**Refere**
	PS	1 µm and 100 nm	Sphere	0, 0.1, 1, 10 mgL^{-1}	PS cause oxidative stress and alters the properties of amino acid metabolism, which reduces primary roots and nutritional absorption.	[174]
	PS	135.9–530 µm	Fragment	0.01%, 0.5% (w/w)	Rice's metabolite synthesis and energy expenditure were affected by PS exposure.	[175]
	PS	200 nm	Sphere	0.1, 10, 1,000 mgL^{-1}	PS showed no effect on rice seed germination. However, PS markedly lengthened the roots and lowered the activity of antioxidant enzymes.	[176]

	Micro/nanoplastics					
Plant sps.	**Type**	**Size**	**Shape**	**Concentration**	**Effect**	**Refere**
Carrot (*Daucus carota* L.)	PS	0.1–5 µm	particles	10–20 mgL^{-1}	Carrot roots can absorb the PS particles (≤1 µm) which may then build up in the intercellular space. Particles having size ≥1 µm were not seen in root tissues whereas ≤0.2 µm PS particles migrated to the leaves. Under cell pressure, the intercellular polystyrene particles were distorted.	[171]
	PP PA PES	5 mm	Fiber	0.1%, 0.2%, 0.3%, 0.4% (w/w)	When the concentration reached 0.4, root mass increased along with the aboveground biomass.	[177]
	LDPE, PET, PP	5 mm	Film	0.1%, 0.2%, 0.3%, 0.4% (w/w)	Root mass and above-ground biomass decreased with increase in the concentration.	[177]
Broad bean (*Vicia faba*)	PS	0.1–5 µm	Sphere	10-100 mgL^{-1}	PS reduced rice biomass by suppressing rice root activity, which had an effect on transpiration and stomata.	[102]

	Micro/nanoplastics					
Plant sps.	**Type**	**Size**	**Shape**	**Concentration**	**Effect**	**Refere**
Phaseolus vulgaris	LDPE	53–1,000 µm	Particle	0.5–2.5% (w/w)	The biomass of the aboveground and root systems remained same.	[137]
	PLA	53–1,000 µm	Particle	0.5–2.5% (w/w)	Significant reduction in biomass of root and shoot on treatment with 1.5, 2, and 2.5 concentrations.	[137]
Chinese cabbage (*Brassica rapa* var. *chinensis*)	PBAT	2 cm × 0.5 cm	Film	6 g/kg wet soil	A small increase in plant dry weight was observed.	[178]
	GPPS HDPE	<25 25–48 48–150 150–850 µm	Fragment	2.5, 5, 10, 20 g/kg	Chinese cabbage's fresh weight was reduced by GPPS, however adding HDPE had little effect on the fresh weight. Microplastics altered the levels of soluble sugar, starch, and chlorophyll in the leaves.	[179]
Brassica napus L.	PE	293 µm	Fragment	0.001% 0.01% 0.1% (w/w)	A decrease in total chlorophyll content affecting the sugar synthesis was observed. There was a reduced nutritional component in plants and an increased	[180]

	Micro/nanoplastics					
Plant sps.	**Type**	**Size**	**Shape**	**Concentration**	**Effect**	**Refere**
					accumulation of lead.	
Brassica oleracea Raphanus sativus	PE	<1 mm	Sphere	0.01 0.1 1 10 100 1,000 10,000 mg/L	In broccoli and radish buds, PE promotes oxidative stress, which increases the levels of ROS. Low concentration of PE will block the nutrient absorption and increases anthocyanin content.	[181]
Tomato (*Lycopersicon esculentum*)	PET PP LDPE HDPE	0.31–2.11 mm	fragment	17,870–47,130 particles/kg soil	High concentration of micro or nanoplastics lowered plant biomass, but soil supplemented with sludge containing microplastics have shown to enhance plant growth. Long exposure of these plastics delayed fruit maturity and reduced fruit production.	[136]
Calamagrostis epigejos	PES	30 µm	Fiber	0.4% (w/w)	The total biomass showed no significant change, but root biomass increased.	[151]
Raphanus sativus	ZnO CuO	ZnO (<100 nm) CuO (<50nm)	Particles	10–1,000 mg/kg	Accumulated Cu and Zn resulted in shorter roots, shorter shoots,	[182]

	Micro/nanoplastics					
Plant sps.	**Type**	**Size**	**Shape**	**Concentration**	**Effect**	**Refere**
					and lighter F1 seeds weight.	
Arachis hypogaea	CuO	5, 20, and 40 nm	Sphere	50–500 mg kg^{-1}	Grain weight was reduced by 10–30%.	[183]
Capsicum annum L	CuO	20–100 nm	particles	500 mg kg^{-1}	Reduction in zinc content by 55% in leaves and 47% in fruits thereby affecting nutritional quality.	[184]
Solanum lycopersicon	CuO	18 nm	particles	0.2–2 µg g^{-1}	Shoot and root showed copper accumulation.	[185]
Glycine max	ZnO	10 nm	particles	0.05–0.5 µg g^{-1}	Leaf chlorosis, necrosis.	[186]
Spinacia oleracea	ZnO, CuO	15 nm	particles	1,000 g kg^{-1}	Root and shoot length reduced.	[187]
Lemna minor (Duckweed)	PE	10–45 µm	Fragment	50,000 $particles^{-1}$	Polyethylene has no effect on plants photosynthesis and growth even though PE was adsorbed.	[188]
		4–12 µm	Fragment	0, 10, 50, 100 mgL^{-1}	PE inhibited root length and damaged root tissues and cells but there was no effect on chlorophyll content.	[189]
Lactuva sativa L.	PVC-a and PVC-b	100 nm–18 µm	Fragment	0.5, 1, 2%	No prominent impact on root activity or MDA content in leaves. PVC-a concentration at 0.5 and 1% enhance root length, surface area, and	[125]

	Micro/nanoplastics					
Plant sps.	**Type**	**Size**	**Shape**	**Concentration**	**Effect**	**Refere**
					diameter. An increase in SOD activity at 1% PVC-a concentration and promoted carotenoid synthesis while PVC-b inhibited this.	
Salvinia cucullata	PS	1 µm	Fragment	3, 15, 75 mgL^{-1}	Reduction in root activity and relative growth rate at high concentration of PS. Excessive accumulation of these microplastics caused activation of ROS.	[190]
Hordeum vulgare L.	PS, PMMA PS	PS (5.64 ± 0.07 µm), PMMA (96.75 ± 0.58 nm)	Spherical	2 gmL^{-1}	Barley absorbed and accumulated PMMA microplastics, which inhibited the growth of rootlets. PS caused changes in ROS metabolism activity in plant by increasing ROS content.	[191]
Cucurbita pepo	PET PVC PE PP	40–50 µm	Fragment	0.02, 0.1, 0.2%	The root and shoot growth were hindered by microplastics. Various microplastics had different concentration-dependent	[192]

	Micro/nanoplastics					
Plant sps.	**Type**	**Size**	**Shape**	**Concentration**	**Effect**	**Refere**
					effects, which includes decrease in chlorophyll content, leaf size and photosynthetic efficiency. It also causes changes to the micro- and macro-element profile.	
Kurodagosun	PS	0.1–1 µm and 5 µm	Sphere	10, 20 mgL^{-1}	Carrot roots exposed to 1 µm PS microplastics build up in the intercellular layer. More PS enters the carrot tissue because of arsenic exposure, increasing health risk.	[139]
Nelumbo nucifera	PS PP HDPE PVC PET PUR	4 mm	Fragment	20% (w/w)	Reduction in germination rate. A significant germination and growth inhibition by PVC and PUR.	[193]
Wheat Lettuce Corn Radish	PS	100 nm and 5 µm	Sphere	1 and 10 mgL^{-1}	The exposure treatment considerably reduced the root length, root dry weight and root/shoot ratio of maize and lettuce but had no effect on the root growth of radish and wheat.	[194]

	Micro/nanoplastics					
Plant sps.	**Type**	**Size**	**Shape**	**Concentration**	**Effect**	**Refere**
Impatiens balsamina, Trifolium repens, Orychophragmus violaceus	PS	80 nm 2 µm	Fragment	0, 10, 50, 100, 500 mgL^{-1}	As PS concentration increased, the plants' germination potential and rates considerably decreased.	[195]
Vigna radiata	Shoe sole fragments	57–229 µm	Fragment	0% 0.1% 1% (w/w)	Plant development and photosynthetic processes were negatively affected by the leachates and microplastics.	[196]
Roraima sp.	PS	3 µm	Sphere	0.05 0.25 1.25 6 mgL^{-1}	Adsorbed PS had no effect on plant growth.	[197]
Myriophyllum spicatum, *Elodea* sp.	PS	50–190 nm, 20–500 µm	Fragment	0.03%, 0.1%, 0.3%, 1%, 3% (w/w) *(Myriophyllum Spicatum)* 0.1%, 0.3%, 1%, 3%, 10% (w/w) *Elodea* sp.	Shoot length decreased with increasing concentration of PS. In both the species the nanoplastics reduced root shoot ratio.	[198]
Glycine max L. Merrill	PS	100 nm, 1 µm, 10 µm, 100 µm	Sphere	1 mg/kg	Phenanthrene absorption was decreased. PS also increased oxidative damage and changed the expression of certain genes.	[42]
Uttricularia aurea	PE PVC	5 µm	Sphere	50 mgL^{-1}	PE showed no significant effect. PVC on the other hand decreased length, macrophyte fluorescence	[199]

	Micro/nanoplastics					
Plant sps.	**Type**	**Size**	**Shape**	**Concentration**	**Effect**	**Refere**
					and chlorophyll content.	
Utricularia vulgaris	PS	1 µm 2 µm 5 µm	Sphere	15, 70, 140 mgL^{-1}	High concentration of PS altered the protein and pigment composition inducing ecotoxicity and oxidative damage.	[200]

9.3.4.3 Mechanical Damage to Roots

Even though microplastics of larger size are not absorbed by plant roots, they can inhibit root activity and growth, particularly those with rough surfaces and sharp edges [153, 203, 205]. PS and polytetrafluoroethylene (PTFE) microplastics may mechanically harm rice roots and generate ROS [171]. The level of ROS is above what the cell can tolerate, and the expression of antioxidant enzyme is inhibited.

9.3.4.4 Hindering Gene Expression

Microplastics may have harmful consequences by altering gene expression. For instance, PS decreases the disease resistance of plants (specifically, *A. thaliana*) by downregulating disease resistance genes [133, 160]. Comparison with control studies showed that PS microplastic-treated samples significantly reduced cdc2 expression in a dose-dependent manner. Plant cells cannot undergo mitosis because of PS-induced inhibition of cdc2 expression, since cdc2 is crucial for the G2/M transition. PS considerably decreased antioxidant enzyme activity in rice roots [176]. PS particles of a particular size (500 or 700 nm) inhibited the relative expression of CAT [90].

PS microplastics inhibit the gene expression of proteins required for the tricarboxylic acid cycle during transcriptome studies in rice grains [175]. After exposure to PS, a substantial decrease in soluble protein was observed in *U. vulgaris* [200]. The most likely explanation for this is that PS microplastics alter gene expression during protein synthesis. PS prevents the production of jasmonic acid and lignin by altering the gene expression in *O. sativa* [140]. Additionally, it has been hypothesized that PS can selectively modulate the expression of genes involved in metal absorption and translocation, inhibiting these ion transport processes [206].

9.3.4.5 Release of Additives

Different additives, such as flame retardants, plasticizers, antioxidants, lubricants, pigments, antistatic agents, acid scavengers, and thermal stabilizers, were typically incorporated into the basic polymer to improve its performance [79, 146, 205]. The degradation of polylactic acid or PLA caused an unfavorable effect on the shoot length of *L. perenne* owing to the weak bond between the additives and the basic polymers, making it a favorable condition for leaching and discharging, leading to toxic effects on plants. *Vigna radiate* photosynthesis iss impaired by leachate from shoe sole fragments [196].

9.3.4.6 Adsorption or Transport of Contaminant

Micro/nanoplastics act as a surface for soil with various qualities to acquire contaminants with various traits (such as hydrophobicity) and remain stable for prolonged periods. Micro/nanoplastic particles may carry phytotoxic compounds before entering the soil (for example, when introduced during manufacturing). Toxic compounds that are either already present in or have been adsorbed onto the surfaces (and within the particle "eco-corona") [207] of micro/nanoplastics in the soil could harm plant roots or their symbionts, which could adversely affect plant growth.

9.3.4.7 Alteration of Soil Properties

Micro- or nanoplastics may indirectly affect plant growth by modifying soil properties, because soil conditions influence the growth of higher plants. According to recent studies, microplastics can alter the physical characteristics of soils, such as their structure, porosity, water retention ability, and water conductivity [43, 151]. The ultrafine PES microfibers (d <5 µm) drastically decreased the soil water retention capacity and volumes of 30 m pores [97]. Microfibers and microplastic particles might greatly lower the content of macroaggregates (i.e., >2,000 µm), but they could also increase the content of microaggregates (250– 63 µm) [146]. Film residue affects the germination of crop seeds and seedling growth by reducing soil porosity and air circulation, altering microbial populations, and decreasing soil fertility [208].

9.3.4.8 Effect on Soil Microbes

Micro/nanoplastic contamination of soils may alter soil properties and the geochemical and biophysical settings. This significantly affects the structural and functional diversity of soil microbial communities and can seriously harm the soil ecosystem [209]. This also involves the functional diversity of soil enzymes and microorganisms [210]. This shift in soil structure and composition may also impair soil fertility by slowing down the expansion of microbial communities in the rhizosphere [211]. The rhizosphere is a crucial contact where the advantageous and unfavorable microbial groups of the soil interact with plants. However, microplastic particles have different surface-to-volume ratios depending on the particle size; hence, their interactions with microbial populations and plant communities will be different [212].

Nanoparticles undergo numerous changes, including translocation, after being released into the environment. The most significant metal oxide nanoparticles, ZnO and CuO, are highly hazardous to various organisms [21]. Although metal nanoparticles are frequently thought to be insoluble, organic acids in soil, such as citric and oxalic acids, improve the dissolution of these nanoparticles, increasing their mobility and availability to the soil flora and fauna [21].

CuO nanoparticles reduced the microbial biomass, disrupted enzymatic activities, and changed the community architecture in flooded paddy soils [213]. The enzymatic activities of bacterial populations in saline-alkali and black soils were also affected by ZnO nanoparticles [214]. Additionally, CuO nanoparticles were found to be harmful to native soil bacteria [68], and the development of cavities, holes, membrane disintegration, cellular collapse, blebs, and lysis in soil bacterial isolates.

It is well known that soil characteristics significantly impact the performance of soil bacteria. The ingestion of PVC microplastics by *U. aurea* through bladders produces toxic effects on growth and physiological parameters [41, 199]. PE microplastics harm soil bacterial communities. This alters the population and functionality of microorganisms in the bladder and disrupts regular interactions between plants and microbes, impeding the growth of *U. aurea* [85].

In agricultural soils, microplastics from high-density polyethylene mulch films can be a unique habitat for bacteria [215]. PE microplastics may have a detrimental [216], positive [217], or negligible [17] impact on the richness and diversity of bacterial communities.

Two microplastics, biodegradable polylactic acid (PLA) and conventional non-degradable high-density polyethylene (HDPE), have also been studied for their effects on the growth of maize and arbuscular mycorrhizal (AM) fungal communities in soil treated with or without ZnO micro/nanoplastics. Contamination with micro/nanoplastics can have significant ecological effects on plant fitness, quality composition, and variations in the soil microbial community, with unknown repercussions on agroecosystems [218].

Microplastics can support unusual bacterial species more than nearby soil particles can. Flakes and pits on the surfaces of microplastic particles imply active hydrolysis sites where different microorganisms colonize [212]. Consequently, soil bacteria have distinct ecological niches provided by micro/nanoplastics [219]. Chloroflexi, Acidobacteria, Bacteroidetes, and Gemmatimonadetes bacteria were found to aggressively colonize the microplastic particles in agricultural soil. These bacteria are referred to as “KeyStone Species” within bacterial ecosystems, and microplastics operate as “special microbial accumulators” [215].

The carbon content of plastic particles is very high [209], and as the material does not easily dissolve, most of this carbon is comparatively inert. Given the high C:N ratio, this material will eventually begin to deteriorate slowly, causing microbial immobilization [188].

9.3.4.9 Stimulatory Effect

Despite being classified as contaminants, microplastics encouraged the growth of higher plants. A few plastics include nutrients, such as nitrogen, which is the fundamental building block of proteins. It is possible for nitrogen to seep into the soil and have effects similar to those of fertilization. According to Lozano and Rillig [151], when *Hieracium* was subjected to drought conditions, polyester (PES) microfibers increased shoot mass by 85% and Calamagrostis by 66%, respectively. They attributed this to microfiber-induced increased aeration, greater root penetration, and decreased soil bulk density [85].

9.3.4.10 Soil Microbial Community and Root Symbionts

The richness of soil biota and its impact on root-colonizing bacteria, such as N-fixers, pathogens, and mycorrhizal fungi, are essential for plant performance [220]. Changes in soil structure brought on by microplastics influence microbial communities and disturb mineralization rates and the communities of symbionts that colonize roots [221]. Nanoplastics may potentially harm symbionts that live in the soil, such as arbuscular mycorrhizal fungi. Other nanoparticles can also have an impact on mycorrhizal activity. However, this has not been demonstrated for nanoplastics [222]. The repercussions of such alterations are still unknown because of the lack of understanding, and scientific emphasis must be given to understanding how microplastics or nanoplastics affect the makeup of soil communities or bacteria that colonize roots [70].

Plant growth is directly influenced by different microbes in the soil such as nitrogen-fixing bacteria and arbuscular mycorrhizal fungi. If microplastics suppress microbial activity, they may have an immediate impact on plants and crops [220]. Micro/nanoplastics affect relative abundance and community structure of arbuscular mycorrhizal fungi [204]. PLA microplastics were found to affect interactions between soil microbial species [223]. This may have affected the rate of microbial-assisted mineral absorption and nitrogen fixation.

9.4 Micro/Nanoplastics in Terrestrial Organism

Microplastic pollution poses an emerging and ever-rising threat that has substantially increased over the past few decades. This is consistent with the escalating use of plastics, particularly one-time-use plastics. Approximately 460 Mt of plastics were consumed globally in 2019 [224]. Plastic build-up in rivers, lakes, oceans, and soil is increasing alarmingly in the modern age. Although most of the discarded plastic debris are macroplastics, micro- and nanoplastics are the successors of these macroplastics [225, 226]. Contrary to popular beliefs, plastics no longer exist as inert and problem-free in any ecosystem. Micro/nanoplastics accumulate in local flora and fauna through biomagnification. These accumulated plastics find their way to the human body, causing immune system disruption, cytotoxicity, and neurotoxicity, and may even be transferred to other tissues upon exposure [227]. Since humans consume meat, fish, and other animals such as snails [228, 229], understanding the mechanism of action of these plastics helps mitigate the overall effect of these hazards effectively.

9.4.1 Effect of Micro/Nanoplastics on Terrestrial Living Things

The rate at which the world consumes and deposits plastics is alarmingly increasing. The rate of increase in the use of this seemingly inert material was 8.4% in 2015 [230]. Studies and surveys conducted worldwide have demonstrated the negative effects of plastics on the health of the environment and organisms. It is worth noting that only 9% of the generated plastic was recycled [230]. The widespread use of plastics will cause various effects on terrestrial organisms. It can be argued that microplastics are more dangerous than macroplastics [43, 231, 232].

9.4.1.1 Ingestion

Microplastics directly or indirectly reach the body of an organism and cause deterioration. Primary microplastics (plastics produced by industries in the micro range, such as beads and pellets) and secondary microplastics (plastics that are made due to the degradation of bigger plastics, such as bottles and packing cases) both follow almost a similar pattern of ingestion among animals [233, 234]. Wild animals and birds consume plastics that resemble the shape and size of their prey [235]. These micro- or nanoplastics also enter the organisms at high trophic levels when they consume animals that have microplastics in their systems.

Ingestion of microplastics in animals can be broadly classified as direct or indirect ingestion [236]. In direct ingestion, the microplastics enter the animal system through food or airborne particulates. In airborne ingestion, the particulates generated through incomplete combustion or grinding are inhaled by animals, whereas micro- and nanoplastics enter the system as contaminants in food. In food, plastics occur owing to improper packing or may be present as an indirect mode of ingestion. For example, studies have shown that earthworms ingest microplastics in the soil. Studies indicate that earthworms burrow deep into soil strata, thus taking microplastics with them [237]. Once they die, microplastics, which are further broken down in their bodies, are deposited in the lower strata of the earth. This deposition of plastics with a reduced size will ultimately allow the same to be mixed with groundwater due to rain or surface runoff. Once microplastics reach the groundwater, they have a direct path to the human body [238].

9.4.1.2 Gastrointestinal Tract

The ingestion of microplastics by terrestrial organisms occurs mainly through the mouth. It takes microplastics directly into the stomach and intestine. Fake satiety is one of the significant effects of plastic intake, i.e., the animal feels a false sense of fullness with the plastics filling its stomach [239]. Smaller plastics pass through the stomach into the

intestines. Studies on mice and zebrafish have shown that intestinal permeability increases owing to the passage of plastics into the body, causing further complications [240]. Spikes in IL-1α, IL-1β, TNF-α, IFN-γ, and IL-6 levels indicated an immune response in the intestines associated with inflammation. When fed to mouse models, polyethylene terephthalate, which is widely used in many plastics, was found to have many detrimental effects [241]. Shunted growth with a 40% mortality rate was observed. There was significant blockage of intestines; organ damage; upregulation of cholesterol, propionic, glycocholic acid, niacinamide, xanthine, and ectoine; and downregulation of anserine, arachidonic acid, and histamine. Neurological damage has been observed in many animals. As ingestion through the mouth is the most significant mode by which micro/nanoplastics enter the system, all the above symptoms are directly associated with the uptake of plastics through the intestines [242].

9.4.1.3 Microplastics on Respiratory Pathways

The widespread use of plastics in the modern world has resulted in a new category of air pollutants, namely, micro- and nanoplastics suspended in air. These plastics are mainly formed by industrial processes such as grinding. However, chemical weathering, biological decomposition, and photooxidation are crucial for the accumulation of plastics in air [243]. Micro- and nanoparticles suspended in the air are inhaled into the lungs of organisms [244]. A549 lung alveolar cells were used to study the effect of microplastics (polystyrene) under *in vitro* conditions. It was observed that, with an increase in particle size, cell proliferation decreased. However, no cytotoxicity was observed [245]. This lack of cytotoxicity may be due to an *in vitro* trial. However, studies using normal lung epithelial tissue and polystyrene have shown cytotoxic and inflammatory responses owing to the formation of reactive oxygen species [246]. In the natural system, inhaled nanoparticles can accumulate in the alveoli, bronchioles, and alveolar ducts [247]. Once microplastics accumulate in the lungs, immunological responses are generated [248], which can lead to the formation of granulomas or fibrosis [247]. The formation of reactive oxygen species and free radicals is another reason for cell damage in the respiratory system [249]. Although studies on the human body using microplastics are insufficient, inhalation of micro- or nanoplastics can cause ailments such as dyspnea, asthma, chronic airway disease, interstitial lung disease, pulmonary fibrosis, and cancer [250].

9.4.1.4 Interaction of Microplastics on Gut Microbiota

The importance of the gut microbiota in almost all living organisms has been the focus of the past few decades [251]. Dysbiosis in the human body can cause inflammatory bowel disease, autism, malnutrition, obesity, immune deficiency, and cancer [252]. The ingestion of microplastics tends to interact with the gut microbiome either by bringing in chemicals that adhere to the microplastics or directly by the chemicals that make up the plastics [253]. Extensive studies performed on zebrafish with exposure to various microplastics showed significant changes in the microbiome, among other effects such as mechanical damage, inflammation, increased permeability, metabolism disruption, oxidative stress, and toxicity [254–257]. Similar studies have been performed on various fish species, and the conditions obtained were similar to those for zebrafish [258, 259]. Non-vertebrates have also been studied for microbiome dysbiosis by using microplastics [260–262]. Interestingly, 16S rRNA sequencing of the microbiome in mussels showed an altered gut microbiota, so that harmful microorganisms gained importance in the organisms, thus affecting them [261]. The altered gut microbiota in animals also directly affects the environment because virulent organisms from the gut come in contact with the biosphere through feces. Dysbiosis among mammals such as rats and mice, has shown oxidative stress [263], metabolic disorder [255, 264–266], and reproductive toxicity [267].

9.4.1.5 Endocrine System

Most plastics are considered inert and falsely considered to co-exist with natural resources. Although regular plastics are misconstrued as safe, there is more danger with plastics containing chemical additives such as bisphenol A (BPA), polybrominated diphenyl ether (PBDE), bisphenol S (BPS), tetrabromobisphenol A (TBBPA), and phthalates [268]. BPA has various effects, such as disruption of fertility and changes in sexual maturation and reproduction [269]. It also affects genes related to thyroid hormone production, resulting in thyroid impairment and hypothyroidism. It acts as an anti-estrogen or anti-androgen, which might affect sperm production and gonadal development. BPA can imitate and antagonize estrogen, indicating that it is a partial agonist of the ER or selective estrogen receptor modulator (SERM). BPA binds to the androgen receptor (AR) and acts as an AR antagonist at high concentrations. Bisphenol S (BPS) is an organic molecule with the chemical formula $C_{12}H_{10}O_4S$, which contains two phenol functional groups on either side of the sulfonyl group. BPS and BPA are endocrine disruptors due to the presence of hydroxy group on the benzene ring. Because of this phenol moiety, BPS and BPA can mimic estradiol [270]. Phthalates are esters of the dialkyl or alkyl aryl of phthalic acid. These are known endocrine inhibitors. Phthalate exposure to the fetus may cause problems during endocrine system development. This also reduces male fertility. Other polymers such as TBBPA and PBDE affect the thyroid gland through various mechanisms.

9.5 Conclusion

Microplastic and nanoplastic pollution are inherently complex. Because of the extensive use of plastics, there are several primary and secondary sources of microplastics in the environment. This is an important global implication and provides a vast area of research. In recent years, research on micro(nano) plastics in environmental systems has gained momentum. This chapter provides a good outlook on micro(nano) plastics in terrestrial ecosystems and their sources, effects, and accumulation in soil, plants, and organisms. To avoid reaching critical threshold concentrations, measures to address the issue of macro-, micro-, and nanoplastics in the environment on a broad and global scale should be devised as the abundance of microplastics is expected to increase further.

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10
Microplastics in Cosmetics and Personal Care Products

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Abstract

Microplastics in personal care products and cosmetics have been identified as a relevant source of plastic pollution that reaches the aquatic environment mainly through wastewater discharge. Regardless of the existence of water treatment plants, these particles, also known as microbeads, can be leached into water bodies, where they interact with different species and other pollutants. In this chapter, a literature review is performed to assess the current state and problems associated with the presence of these particles as ingredients in the mentioned products. The analysis of 67 research papers showed that three main lines of study have been developed: the characterization of microplastics extracted from personal care products and cosmetics, the study of the interactions between microplastics and other ingredients on those products, and the assessment of the toxicity of product-derived microplastics to different species. The results regarding these three topics show the prevalence of white polyethylene microbeads, which have been analyzed mainly in scrubbers. The studied interactions included UV-filter adsorption and stabilizers on the surface and microbeads. Regarding toxicity, the results varied depending on the species and general condition of the tests. Cosmetics and personal care products will always be in our lives; therefore, we should assess the risks involved in their use.

Keywords: Microbeads, toxicity, adsorption, bans, leachate

10.1 Introduction

There is no universally accepted definition of microplastics in the literature, as their defining characteristics change from author to author. The most accepted description is that they are small solid particles or fragments of plastic that are less than 5 mm in size. According to the European Chemicals Agency (ECHA), microplastics are solid plastic particles smaller than 5 mm with a composition of mixed polymers and functional additives [1].

A way can be classified as: primary and secondary. Primary microplastics are intentionally and microscopically synthesized for use in different products and industries, such as pharmaceuticals, food, beauty, cosmetics, and personal care products. Primary microplastics can be used in products such as toothpaste, shower gels, liquid makeup, nail polish, and sunscreens [2]. Secondary microplastics are generated by larger plastic fragmentation due to environmental exposure, time, and structural damage caused by biological, physical, and chemical factors.

10.1.1 Personal Care Products (PCPs) and Cosmetics

10.1.1.1 Consumption and Categorization

Personal care products (PCPs) are classified as contaminants of emergent concern (CECs) [3], which are found in trace concentrations within environmental compartments [4]. PCPs are considered harmful to biota metabolism, given their relative toxicity, environmental persistence, bioaccumulation potential, and liberation through seemingly domestic activities [4].

PCPs are defined as a “category of self-care [items] generally used for personal hygiene, cleaning and grooming” [5] and thus include a wide variety of products. Nevertheless, a lack of international consensus regarding its definition might have favored deficiencies in the regulation of PCPs [5], especially considering that some of them might contain endocrine disruptors [6].

According to Cosmetics Europe [7], PCPs are classified into seven categories: 1) oral, 2) skin, 3) sun, 4) hair care, 5) body care, 6) decorative cosmetics, and 7) perfumes. Cosmetics (Category 6) are discussed in detail in the following sections. It is also noteworthy that in this chapter other PCPs classifications will not be assessed, as some authors include baby and adult diapers or sanitarian pads within this category [8], whose items will not be included in the present review.

Although the consumption of PCPs is widespread, their primary market is Pacific Asia, followed closely by North America and Europe [9], particularly China and the US [10]. It has been estimated that China will consume approximately 80 billion USD [11], accounting for at least 30% of the global market value [9]. The US market for PCPs consumption has been growing, and it is estimated that it will reach a 3.17 billion USD market value by 2025 [12].

PCPs pathways may vary, but they usually reach environmental compartments through urban waste waters, skin volatilization leading to airborne deposition by users, or direct contact with environmental media, such as the use of UV blockers in marine and freshwater environments while swimming and sunbathing [4, 13]

Cosmetics were defined by the Food, Drugs and Cosmetics Act (USA legislation) in 2007 as “articles intended to be rubbed, poured, sprinkled or sprayed on, introduced into, or otherwise applied to the human body for cleansing, beautifying, promoting attractiveness, or altering the appearance” [14].

10.1.1.2 Microbeads in PCPs and Cosmetics

PCPs contain a wide variety of substances; however, in this chapter, the focus will be placed on intentionally added plastic microbeads. Microbeads in cosmetics and PCPs have similar purposes, mainly to improve their cleaning and exfoliation properties [15]. These beads are usually added by producers for scrubbing or cleansing purposes [16], as bulking substances, as a conservative for extending the life span of PCPs, or even by gradually releasing active ingredients along with usage [17]. They are used on rinse-off products, which are designed to be washed and removed immediately after application, such as exfoliants or scrubs, but they are also used on leave-on products, which are intended to be kept on the skin, such as creams of sunscreens.

The Soup Foundation found that every nine out of 10 PCPs in the European market contain microplastics [18], and it is crucial to note that the definition of microplastics is not the same as the most accepted in the literature. Cases that are usually excluded are water-soluble polymers (WSPSs), engineered nanoplastics, and biodegradable polymers. In this chapter, we considered only solid particles that agreed with the most accepted microplastic definition. Nonetheless, it is essential to emphasize the importance of

studying excluded cases because of their abundance in commercially available products and little information about them in the literature.

Microbeads are, by definition, synthetic, water insoluble and non-biodegradable plastic particles that are intentionally added to cosmetics, especially to products that rinse off, to help clean the face, body, baths, and teeth of users, as well as improve the removal of skin cells [15]; there is also the usage only for decorative purposes.

Microbeads are classified as primary microplastics, and even with high removal efficiency rates in treated wastewater, they can reach the environment [19]. It has been estimated that at least 8 trillion particles are released daily through wastewater plants [20], without considering untreated wastewater. These microplastics are hard, if not impossible, to efficiently recover from the environment once they have reached it, and thus end up being a plastic pollution source, given that they do not biodegrade [15].

In this sense, most of the actions proposed as solutions are preventive, and bans are enacted worldwide. However, as mentioned before, the lack of an international consensus and broader guidelines hinders containment efforts. In Europe, a critical decrease in plastic microbeads contained in PCPs has been detected, given its production bans [21]. In other leading markets, such as China, as far as 70% of surveyed PCP had plastic microbeads [22].

10.1.1.3 Environmental Effects of Microplastics

Most microbeads are released to water bodies and especially to the marine environment because even if there are water treatment plants, they do not always remove the microbeads because of their size [23]. It is important to note that, among the studies performed on marine environments, approximately 80.5%, do not consider microplastics smaller than 300 µm. Because most of the microbeads are smaller than 350 µm [24], most approaches to evaluate microbead pollution in marine environments would have to be broadened to include these smaller-sized particles [25].

Several estimates of microplastic release into the environment have been calculated. In a report published by the United Nations Environment Programme (UNEP) in 2015, it was estimated that approximately 0.01 Mt of microplastics from PCPs and cosmetics are released each year, with an estimated input by region distributed as follows: North America 10%, Western Europe 3%, Japan 1%, Central Europe 6%, China 22%, India 9%, Middle East 6%, Latin American and the Caribbean 8%, Oceania 1%, Africa 16%, and Asia (except Japan, China, and India) 17% [26]. Another study reported that approximately 12,000 tons of microplastics from PCPs and cosmetics are released each year, and approximately 1,500 tons escape from wastewater treatment plants and reach the marine environment [24]. These numbers represent approximately 0.1% and 0.8% of the total global emissions of primary microplastics into the ocean, which are approximately 1.5×10^6 tons each year. It was also reported that between 1970 and 2019, approximately 300,000 tons of microplastics derived from PCPs accumulated in the environment [24].

The half-life of plastics is estimated to reach hundreds of years, which is more than that of any persistent organic contaminant [25]. Nonetheless, it is known that their toxicity depends on their form and size, but it also depends on their chemical structure, their composition, and additives, which in turn might affect the type of contaminants adsorbed [25, 27].

The removal of microplastics differs from region to region. Different concentrations end in superficial waters or agricultural soil because of the use of wastewater treatment sludge as fertilizer [28]. When they are already in the marine environment, microplastics can travel large distances floating or as sediments and form aggregates with other particles. It is known that these particles are present even in the Arctic [29]. In addition to forming aggregates and traveling long distances, microplastics are eaten by aquatic organisms,

and thus can enter the trophic chain. If microplastics have other contaminants associated with or adsorbed, their consumption at specific concentrations can cause toxic effects. The ocean is already polluted by many substances that can cause negative effects if consumed alone. However, when they are adsorbed by microplastics, they can act as another chemical additive source, contributing further to adverse effects in organisms. Studies have demonstrated that chemicals associated with microplastics can be found even in tissues; however, there is little evidence that this occurs under natural conditions [30].

Microplastics can reach humans through the trophic chain when they are consumed by organisms, such as fish or other marine organisms, but they can also enter the human body through the water we drink [31]. One relevant way trat microplastics from PCPs can enter the human body is orally. However, it is more likely that most of these particles are liberated into the environment without entering organisms [25]. Therefore, most of the attention must be paid to the interactions and consequences of these particles when they are already in the environment or reach us through indirect ingestion or inhalation. As with other organisms, human toxicity can occur because of exposure to microplastics with associated contaminants [28].

10.2 Methodology

The information search was performed on the Science Direct, Springer, and Web of Science libraries in March 2023. The following keywords were used for article selection:

- microplastics + personal care products
- microplastics + cosmetics
- microplastics + glitter
- microplastics + exfoliants
- microplastics + sunscreen
- microplastics + scrubs
- microbeads + characterization.

These keywords were written in the section title, abstract, or author-specified keywords in Science Direct, and a strict search was performed in Springer and Web of Science. Only published research articles were selected, excluding publications such as reviews or book chapters. The search period was from 2011 to 2023.

From all the articles found using the criteria already mentioned, only articles in which microplastics were contained and extracted from PCPs were selected. A total of 67 research articles were selected, and their information was later registered in a database.

10.3 Results

The study of microplastics in PCPs and cosmetics is of recent interest, as shown in Figure 10.1. An increasing trend in research article publications related to the topic can be observed, with a decrease in 2021, likely due to the COVID-19 pandemic. It is noteworthy that although the search period began in 2011, only articles from 2015 onward were found.

A total of 50.7% of the selected articles were published in Europe (including Turkey), followed by Asia (35.8%) and America (10.4%). In Africa and Oceania, only one article has been published in each region. The countries with the most published articles were China, with ten and Spain, with nine research articles (Figure 10.2). Even though interest in the

study of microplastics has increased, this has mainly been reflected in research developed in the European continent.

The selected articles were not abundant despite the flexibility of the selection criteria, given that even though there are many microplastic research articles, those that study primary microplastics used in PCPs are not common.

Three articles referred to the use of microplastics in PCPs. The first type included articles in which extraction from the products and characterization of microplastics were performed. The second type studies the interaction of microplastics with other substances, such as those included in PCPs as ingredients or other contaminants, and the third type studies the toxicity of microplastics from PCPs in different organisms. The Results section will be divided into three types of articles.

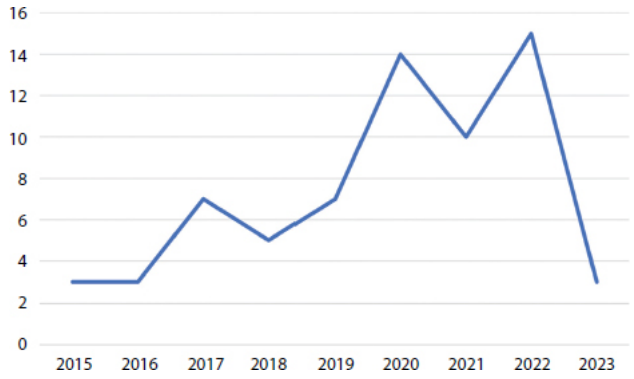


Figure 10.1 Publication years of the selected articles.

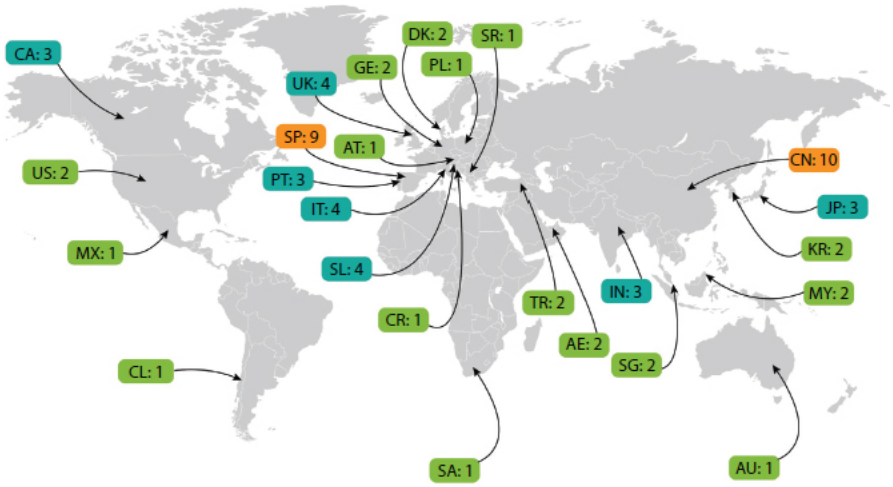


Figure 10.2 Country of origin from the institutions of the selected articles. CA, Canada; US, United States; MX, Mexico; CL, Chile; UK, United Kingdom; GE, Germany; DK, Denmark; PL, Poland; SR, Serbia; SP, Spain; AT, Austria; PT, Portugal; IT, Italy; SL, Slovenia; CR, Croatia; TR, Turkey; SA, South Africa; AE, United Arab Emirates; IN, India; SG, Singapore; AU, Australia; MY, Malaysia; KR, Korea; JP, Japan; CN, China.

10.4 Characterization of Microplastics in PCPs and Cosmetics

10.4.1 Types of Samples

Of the 32 articles that focused on the characterization of MPs in PCPs and cosmetics, 183

registries (samples) were reported. Articles reported between one and 21 samples, and most samples were cosmetics, as can be seen in Figure 10.3.

More than half of the samples were cosmetics, followed by the PCPs. After classifying the samples, almost half were scrubs (48%), followed by different types of gels (12%). Glitter represented 5% of the samples, as shown in Figure 10.4.

Samples such as plastic bags and textiles were included, as shown in Figure 10.4, as they were used in some studies to compare detection techniques for PCPs or cosmetics, or when they were also found within environmental samples that also contained PCPs and cosmetics.

10.4.2 Per Country of Origin

Most articles did not include the country of origin of the PCPs and cosmetics. These data are important given that, as will be further discussed in the upcoming sections, several countries have enacted bans on adding microplastics to PCPs and cosmetics manufactured within their borders. Of the 183 sampled reported products, 156 did not specify the origin of the product sample, and from the remaining 27, most of the products were said to be made in India (26%), followed by Russia (19%) (Figure 10.5).

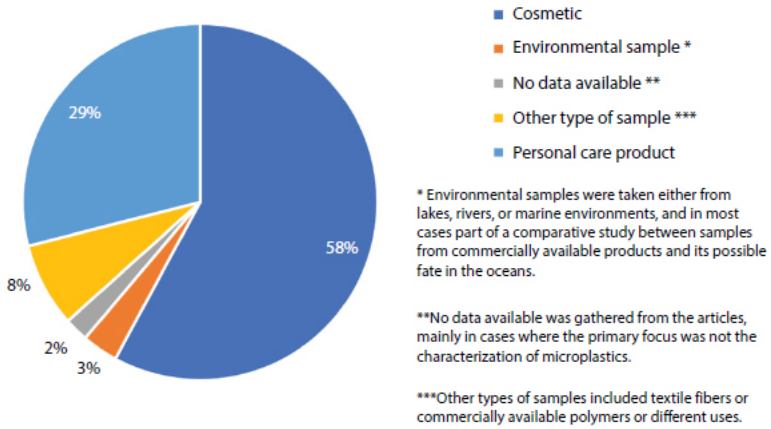


Figure 10.3 Percentual distribution of samples by type of sample.

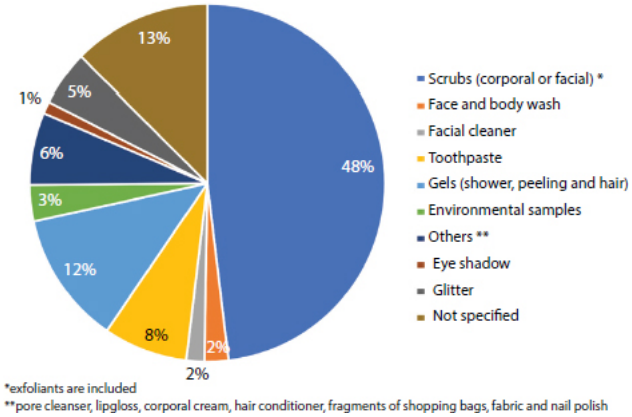


Figure 10.4 Percentual distribution of samples by type of product.

10.4.3 Forms of Microplastics

In 15% of the 183 reported samples, more than two different types of microplastic shapes were found, with the highest number of microplastic forms in the same two samples [32] in 23% of the reported samples, no form was specified. From the rest (141 samples), the

most common form was fragments or irregularly shaped particles (37%), followed by spheres (20%) and glitter, as can be seen in Figure 10.6.

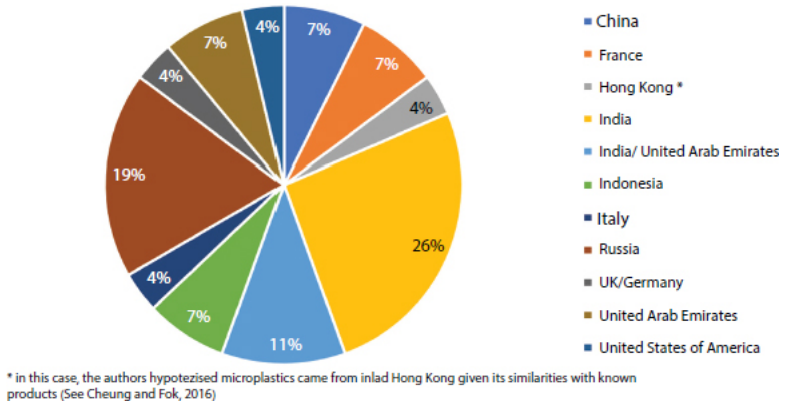


Figure 10.5 Countries of origin of products which included this information (n = 27).

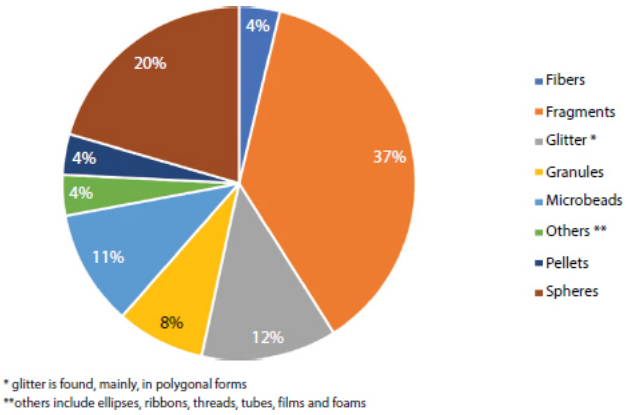


Figure 10.6 Percentage distribution of the most common forms of microplastics in PCPs and cosmetics (n = 161). Note: A total of 141 samples are specified. The sum might not coincide with the number of samples (141 vs. 161 reported in the graph) because, in some samples, more than one form was reported.

10.4.4 Colors of Microplastics Found in PCPs and Cosmetics

No color was specified for 33% of the reported samples. From the rest of the samples (122), the most common colors were white (35%), blue (16%), and transparent (13%). In some cases, individual samples reported multiple colors; thus, Figure 10.7 registered the occurrences of color, which caused some samples to be added more than once.

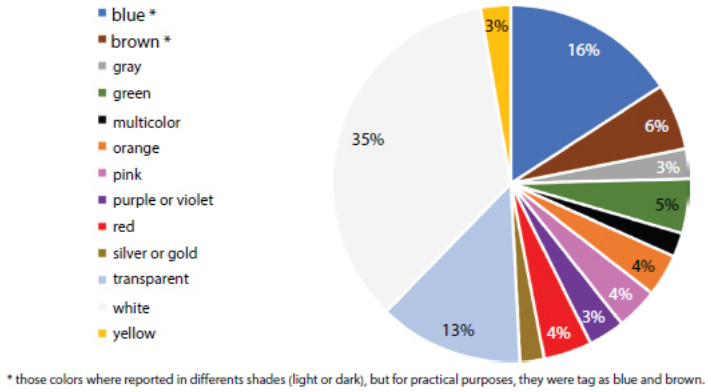


Figure 10.7 Percentage distribution of color in microplastics from PCPs and cosmetics.

10.4.5 Sizes of Microplastics

The reported sizes vary widely among the studies. Thus, to better illustrate the dimensions, the range, mean values, and standard deviations are reported in Table 10.1. This table describes the number of instances from the available reported samples (183), dimensions of microplastics were reported by using ranges (lower and highest found values), and in the number of cases, other statistical descriptors were used, such as mean and standard deviation.

Table 10.1 then shows that, in most cases, microplastic sizes are described by their largest available data or the mean size of the studied particles. This is relevant because, as stated before, most studies need to broaden their detection limits to include particles smaller than 350 µm and avoid mischaracterization or underestimation of the microplastic contributions of PCPs and cosmetics to the environment.

For the samples with reported ranges, the minimum value was 0.9 µm and maximum value was 580 µm. The distribution is shown in Figure 10.8 and most of the samples were concentrated below 101 µm.

The same can be done for the highest values reported by range, where the minimum value was 1.3 µm and the maximum 2,188 µm, as can be seen in Figure 10.9. In the case of the highest range value, the observed behavior is slightly different, and most sizes occur only once. When arranged by frequency in a histogram, the highest values are between 1 and 1,000 µm, and the most common is between 100 and 200 µm (Figure 10.9).

Table 10.1 Number of samples and reported dimensions (n = 183).

Reported value?	Lowest range	Highest range	Mean	Standard deviation
Yes	108	111	111	76
No [a]	75	72	72	107

Note [a]: Not reported.

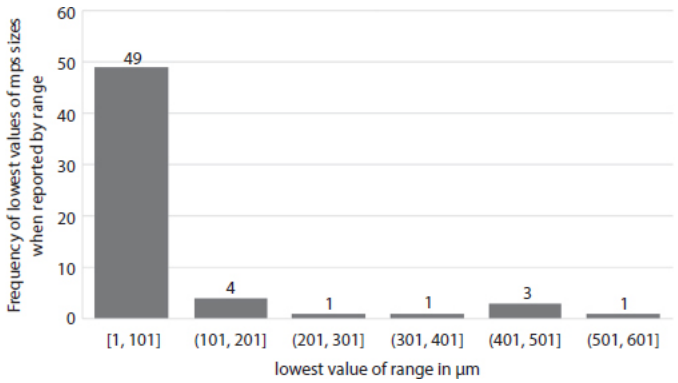


Figure 10.8 Histogram of the frequency of lowest range values reported for sampled microplastics in PCPs and cosmetics (n = 183, range size = 100).

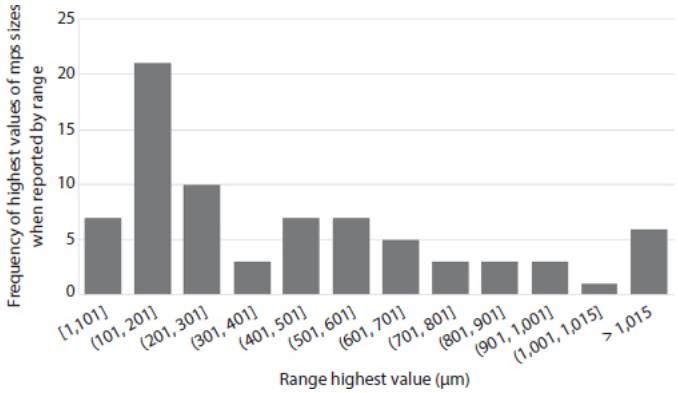


Figure 10.9 Histogram of frequency of lowest range values reported for sampled microplastics in PCPs and cosmetics (n = 183, range size = 100).

10.4.6 Types of Polymers

Most samples determined the type of polymers, and as expected, the majority were made of polyethylene (62%), which has been reported as one of the most commonly added microbeads in PCPs and cosmetics. Other reported polymers include polypropylene, polyethylene terephthalate, polymethyl methacrylate, and nylon [17].

The percentage distribution of polymers in the available samples is shown in Figure 10.10. Some samples contained more than one type of polymer, and in these cases, the polymers were counted as many times as they appeared. Therefore, the final sum of the graphed items is greater than that of the available samples. The samples that did not report polymer characterization are shown as “ND.”

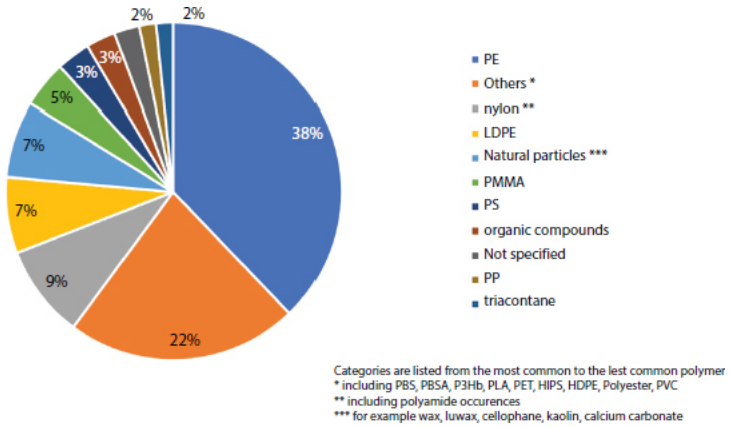


Figure 10.10 Distribution of microbeads materials (natural, polymers, and biodegradable polymers). PE, polyethylene; LDPE, low-density polyethylene; PMMA, polymethyl methacrylate; PS, polystyrene; PP, polypropylene; PBS, polybutylene succinate; PBSA, poly(butylene succinate-co-butylene adipate; P3Hb, Poly(3-hydroxybutyrate); PLA, polylactic acid; PET, polyethylene terephthalate; HIPS, high-impact ground polystyrene; HDPE, high-density polyethylene; PVC, polyvinyl chloride.

10.4.7 Experimental Methods Used to Extract and Analyze Microplastics

Even if different procedures are carried to extract and characterize microplastics from PCPs and cosmetics, some common features can be identified, which are described in the following sections.

10.4.7.1 Extraction Method

Figure 10.11 shows that filtration is the most common method for microplastic extraction from PCPs and cosmetics samples, owing to its robustness and availability (73% of the samples). Cellulose membranes, which vary in pore size, are commonly used. Nonetheless, it has been reported that there is a significant loss when this method is used because some microplastics remain trapped in membrane pores [33]. In some cases, the method was complemented by heating [34], vacuum [35], dissolution of the ingredients [36], and drying [37], among other techniques. The second most common method was density separation with saline solutions of 1.6 g/L density, where some plastics tend to float [38]. The third method involves wet oxidation with hydrogen peroxide, which removes organic ingredients from which microplastics are extracted [32].

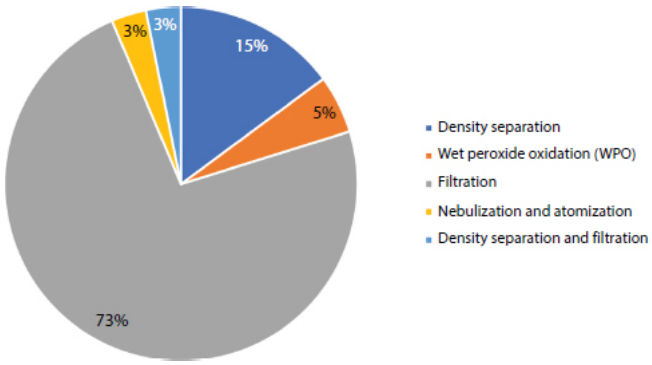


Figure 10.11 Distribution of extraction methods from PCPs and cosmetics.

10.4.7.2 Particle Size Analysis Method

Most studies use microscopes to determine microplastic dimensions (Figure 10.12). Typically, they pair the use of a microscope with software. One of the most widely used is ImageJ, which is an open-access image analysis software [39]. Microscopes are used because they are the cheapest method, while scanning electron microscopy (SEM) is a technique that allows the study of smaller microplastics.

Many of the microplastics contained in PCPs are 1 µm (0.001 mm) or smaller in size [40], which limits the scope and usefulness of the microscope to study them. Nonetheless, in the reviewed articles, many of the analyzed products were scrubs and gels, which have microplastics with larger particle sizes and smaller sizes. For example, SEM was used to analyze particles from a facial scrub with a mean size of 0.024 ± 0.006 µm [41], whereas the particles from a body scrub ranged from 200 to 500 µm in size and were analyzed using an optical microscope [32].

10.4.7.3 Polymer Type Analysis Methods

The Fourier transform infrared (FTIR) technique is the most widely utilized (Figure 10.13) owing to its practicality, speed, and price [42]. However, the limitation of FTIR is the particle size that can be analyzed. Currently, there are other options, such as Raman spectroscopy or µFTIR, which allow for the identification of microplastics with smaller particle sizes. Standard FTIR spectrophotometers are generally cheaper than Raman devices because multiple excitation wavelengths and laser sources are sometimes required for different sample types [43].

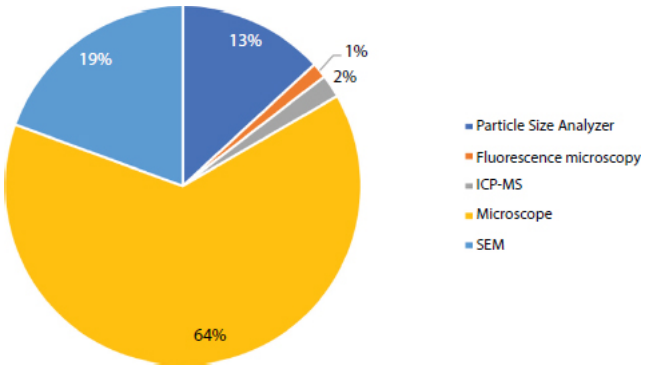


Figure 10.12 Distribution of particle size analysis methods for microplastics extracted from PCPs and cosmetics. ICP-MS (Inductively coupled plasma mass spectrometry); SEM (Scanning Electron Microscope).

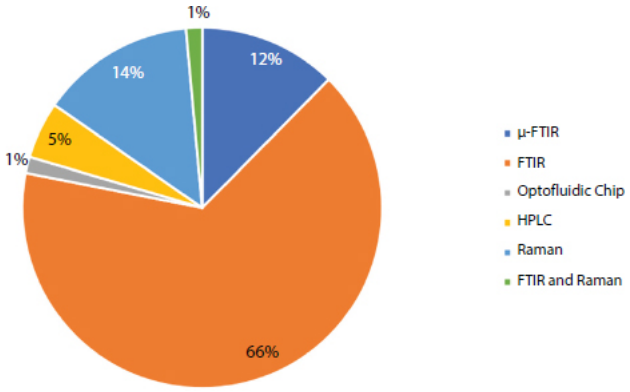


Figure 10.13 Distribution of polymer-type analysis methods for microplastics in PCPs and cosmetics.

A novel technique is the optofluidic chip; with this technique, microplastics are trapped and classified in a liquid sample. The images were obtained and identified using spectroscopic methods. The advantage of this technique is time savings [44].

10.5 Interaction Between Microplastics from PCPs and Other Substances

Microplastics have been commonly used in PCPs and cosmetics [24], however, they may also include other ingredients. One example is the use of organic and inorganic substances added as ultraviolet filters or stabilizers [45].

Organic ultraviolet filters (UV filters) are used to provide radiation protection to PCPs and other materials (e.g., textiles and plastic products). Sometimes, they are considered emerging contaminants for aquatic ecosystems since it has been shown that significant concentrations can negatively affect organisms in freshwater bodies [46]. In contrast, stabilizers in care products are used to extend shelf life and stability over time, although they have been reported to be toxic, recalcitrant and bioaccumulative [47].

Microplastics in aquatic environments can sorb these substances [48], causing their entry into trophic chains [49]. Adsorption processes have been reported to be affected by factors such as salinity, microplastic size, dose, and the degree of weathering the polymer has endure [50].

In a study of 13 beaches in the Canary Islands (Spain), microplastics were collected and subsequently analyzed for the presence of filters and UV stabilizers. The results confirmed the presence of 10 filters and stabilizers on all beaches, at concentrations ranging from 1 to 4,031 ng/g; octocrylene was the most frequently found compound, and its presence was attributed to the high demand for tourism [45].

Under laboratory conditions, the adsorption of organic filters oxybenzone (BP-3), camphor 3-(4-methylbenzylidene) (4-MBC), 2-ethylhexyl 4-methoxycinnamate (EHMC) and octocrylene (OCT) on virgin microplastics has been evaluated for polyethylene and a high adsorption capacity of 990 µg/g has been found; the main factor affect in the sorption process was salinity [50].

10.6 Toxicity of Microplastics from Personal Care Products and Cosmetics

Of the 35 tests performed in organisms or cells that were registered (Table 10.2), 23 used microplastics extracted from scrubs. These were followed by microplastics obtained from hair gel and glitter in 12 cases. The use of scrubs as a source of microplastics makes sense because, as mentioned in the characterization section, scrubs are the most studied type of products. However, there is a gap in the research because these are not the only cosmetics with microplastics. There is also little research on the effects on different organisms of microplastics contained in products such as toothpaste [32], gel showers [51], and liquid eyeshadows [52].

Other tests have been performed with microplastics, which could be part of the ingredients of PCPs and cosmetics, such as polyethylene [53] or polystyrene [54] microbeads. However, these studies were not considered for the methodological review of the chapter because microplastics were not extracted directly from the PCPs, and their use was not strictly specified. In one of these studies, the toxicity of purchased polyethylene microbeads was tested on the marine copepod, Acartia tonsa. The test results showed no mortality in the organisms after the 48-hour exposition to any of the concentrations (0–25,000 MP/mL); however, when microbeads with adsorbed triclosan (TCS) were used for the tests, it resulted in an increment in the toxicity because of a synergic effect compared to the toxicity by triclosan alone [53].

10.6.1 Toxicity of Different Types of MP

The most common polymer type among the PCPs used in the tests was polyethylene. Other polymers were tested only once in the reviewed articles: PLA [55], PMMA and PA extracted from the gel glitter [56]. It should be mentioned that toxicity does not depend only on the polymer type; other factors, such as the morphology of the particles or the concentration, have a strong influence as well [25].

10.6.2 Effects in Different Organism’s Groups

Tests were performed on different organisms, most of which were aquatic, and some insights are presented in the following sections.

Table 10.2 Toxicity tests of microplastics from PCPs.

	Test organism type	Test organisms	Product	Polymer type	Particle size	Time	Concentration
[57]	Crustacean	*Daphnia magna*	Facial scrub	PE.	Mean value 140.6 ± 80.0 µ	48 h	100 mg/L
[57]	Fish	*Danio rerio*				96 h	100 mg/L
[57]	Plant (weed)	*Lemna minor*				7 days	100 mg/L
[59]	Cell	*Human blood cells (erythrocytes)*	Facial scrub	PE	~100 nm	24 h	2.5, 5, and 25 µg/mL
[59]	Cell	*Human blood cells (lymphocytes)*				24 h	2.5, 5, and 25 µg/mL
[59]	Plant	*Allium cepa (bulbs) (Onions)*				24 h	5, 10, 15, 20, and 25 µg/mL
[58]	Plant (freshwater floating plant)	*Lemna minor*	Scrub	PE	30 µm–600 µm (mean particle size 71.30 ±	7 days	0, 10, 50, and 100 mg/L

	Test organism type	Test organisms	Product	Polymer type	Particle size	Time	Concentration
					34.29 mm)		
[58]	Plant (freshwater floating plant)	*Lemna minor*	Scrub	PE	40 µm–400 µm (mean particle size 96.00 ± 69.99 mm)	7 days	0, 10, 50, and 100 mg/L
[55]	Lake phytoplankton	*Phytoplankton*	Body wash scrub	PLA	~60 µm–~1,010 µm	7 days	67 mg/L

	Test organism type	Test organisms	Product	Polymer type	Particle size	Time	Concentration
[37]	Phytoplankton	*Thalassiosira weissflogii (Bacillariophyceae)*	Scrub	NS	Max: 608 µm Min: 22 µm Media: 173 ± 131 µm	72 h	1,000–24,000 MP/L

	Test organism type	Test organisms	Product	Polymer type	Particle size	Time	Concentration
[64]	Human skin cells	*Keratinocytes*	Facial scrub	PE (low PS content)	30 to 300 nm 100 ± 20 nm	2–3 min	100 µg/mL
						6 days	
						NS	25–50 µg/mL
						6 days	250 µg/mL
			Facial scrub	PE (low PS content)	90 to 230 nm	2–3 min	100 µg/mL
						6 days	
						NS	25–50 µg/mL
						6 days	250 µg/mL
[62]	Freshwater snail (Gastropod)	*Biomphalaria glabrata*	Facial scrub	PE	83 µm–789 µm mean: 346 µm ± 88 µm	48 h	5 mg/L

	Test organism type	Test organisms	Product	Polymer type	Particle size	Time	Concentration
[62]						21 days	0.5 mg/L
[60]	Fish	*Melanotaenia fluviatilis*	Face Scrub Soap	PE	10-700 µm	63 days	10 mg of MBs spiked with 200 ng/g of target PBDEs

	Test organism type	Test organisms	Product	Polymer type	Particle size	Time	Concentration
[56]	Bacteria Freshwater	*Aliivibrio fischeri*	Glitter for hair gel	PMMA	954.09 ± 315.24 µm	15–30 min	Leachates of 3, 90, and 180 days
[56]	Algae Freshwater	*Raphidocelis subcapitata*				72 h	
[56]	Crustaceans Freshwater	*Daphnia magna*				48 h	
[56]	Marine Bacteria	*Aliivibrio fischeri*				15-30	Leachates of 3, 90, and 180

	Test organism type	Test organisms	Product	Polymer type	Particle size	Time	Concentration
						min	days
[56]	Marine Algae	*Phaeodactylum tricornutum*				72 h	
[56]	Marine Echinoderms	*Paracentrotus lividus*				NS	

	Test organism type	Test organisms	Product	Polymer type	Particle size	Time	Concentration

	Test organism type	Test organisms	Product	Polymer type	Particle size	Time	Concentration
[56]	Bacteria Freshwater	*Aliivibrio fischeri*	Glitter for hair gel	PA	3,047.83 ± 63.08 µm	15–30 min	Leachates of 3, 90, and 180 days

	Test organism type	Test organisms	Product	Polymer type	Particle size	Time	Concentration
[56]	Algae Freshwater	*Raphidocelis subcapitata*				72 h	
[56]	Crustaceans Freshwater	*Daphnia magna*				48 h	

	Test organism type	Test organisms	Product	Polymer type	Particle size	Time	Concentration
[56]	Marine Bacteria	Aliivibrio fischeri				15–30 min	Leachates of 3, 90, and 180 days

	Test organism type	Test organisms	Product	Polymer type	Particle size	Time	Concentration
[56]	Marine Algae	*Phaeodactylum tricornutum*				72 h	
[56]	Marine Echinoderms	*Paracentrotus lividus*				NS	
[56]	Fish	*Danio rerio*	Facial scrub	PE	100–400 µm	14 days	0.5 and 5 mg/L

	Test organism type	Test organisms	Product	Polymer type	Particle size	Time	Concentration
[56]	Terrestrial Isopod	*Porcellio scaber*	Facial scrub	PE	80-700 µm	14 days	MP via feeding on food pellets (4 mg/g) 0.4% w/w

10.6.2.1 Bacteria

A study was found in which a test was performed following the norm ISO 11348-3:2019. Luminescence inhibition was evaluated in the bacteria *Aliivibrio fischeri* in freshwater and in a marine environment with 3-, 90-, and 180-day leachates from two different glitter particles. Both PMMA and PA glitter leachates, led to bioluminescence inhibition, but the threshold was not exceeded in any case. No ecotoxicological risk was found. According to the analysis, the toxicity of the plastic leachates is not linked to the type of polymer or the color, but to the sensibility of the species, and only in some cases to the medium and the soaking time used for the leachate [56].

10.6.2.2 Plants

Three of the four studies on plants used Lemna minor, a freshwater floating plant. In one of these studies [57], the number of fronds and root growth were evaluated after exposure to microplastics soaked previously for 3 weeks in water with five different organic loads. The observed effects were on root growth when the microplastics were exposed to high organic loads. However, when microplastics were exposed to low organic loads, the toxic effects were mitigated. In another study, the impact of exposure to two different microplastics extracted from two scrubs was compared; one had sharp particles and the other had smooth surfaces. The sharp particles affected the root cells, reducing their viability; with the smooth particles, negative effects were not observed [58]. No changes were observed in the number of roots, leaf growth rate, or photosynthetic pigments.

In the Allium cepa (onion) plant, exposure to macroplastics of approximately 0.1 µm for 24 h affected root growth and suppressed mitotic activity [59].

10.6.2.3 Phytoplankton

Tests were also performed on the phytoplankton. In one study carried with PLA microplastics extracted from a scrub, there were no significant effects on the chlorophyll concentration, cell count, or total biomass, but there was inhibition in one of the main taxonomic groups (cryptophytes), and the count of another taxonomic group (chrysophyte) increased [55].

For the species Thalassiosira weissflogii, the aggregates resulting from the exposure of a defined number of cells were analyzed, and these were found to be larger than the aggregates formed by only microplastics and only phytoplankton [37].

10.6.2.4 Algae

In algae, growth inhibition was analyzed in two species, one from freshwater (Raphidocelis subcapitata) and another from the marine environment (Phaeodactylum tricornutum), both with leachates generated after soaking for 3-, 90-, and 180-days of two types of hair gel glitter, one in PMMA and the other in PA. The results differed for every species type, glitter, and soaking time. In freshwater algae, growth inhibition that exceeded the threshold of 10% was recorded for the 3- and 180-day leachate of the PMMA glitter, whereas the 90-day leachate stimulated growth. With PA leachates, growth inhibition was observed for the 3-day leachate [56].

In marine algae, there was inhibition with the PMMA 18-day leachate and the PA 90- and 180-day leachates. All other cases showed growth stimulation, except for the PMMA 90-day leachate, which did not cause any change compared to the controls. After analyzing the data, a moderate toxicological risk was found for the 90-day leachate of the PMMA glitter [56].

10.6.2.5 Animals

Tests on aquatic animals accounted for a large proportion of the articles. These tests included both freshwater and marine organisms. In some cases, there was a significant effect, while in others, no effect was observed.

For Melanotaenia fluviatilis, the rate of PBDEs increased during the exposure time, resulting in increased accumulation in the fish. It was also identified that microplastics by themselves do not reduce contaminants in organisms, as suggested by some authors [60].

In Danio rerio fish, when microplastics were aged in water with different organic loads, no effect was observed, as there was no significant pollutant absorption [57]. In another study with the same organisms, the antioxidative stress response induced by PE microbeads with tetrabromobisphenol A (TBBPA) was increased compared to that induced by TBBPA or PE microplastics alone. It is important to note that, in this study, PE microbeads had a better capacity for absorbing TBBPA than the reference PE microplastics [61].

Tests in the crustacean Daphnia magna showed no effect when the organisms were exposed to microplastics aged in water with different organic loads [57]. When they were exposed to leachates from two different gel hair glitters (PMMA and PA), no ecotoxicological risk was found, but the mortality threshold increased with the PMMA 3-day leachate and with the PA 180-day leachate [56]. In the same study, tests were performed on Paracentrotus lividus, a marine echinoderm, and larval development abnormalities were observed with PMMA and PA leachates. Only an ecotoxicological risk was found for PMMA 90-day leachates [56].

The freshwater snails Biomphalaria glabrata were fed with microplastics, and no mortality was recorded, although they ate and defecated the microplastics. These ingestions did not have any effect on body length or weight [62].

Only one study has examined the effects of terrestrial organisms. In this, the species Porcellio scaber was fed with food pellets containing microplastics. In the 14 days of the experiment, although the microorganisms consumed and defecated microplastics, there was no observed change in the feeding and defecation rate, body mass, or energy reserves [63].

10.6.2.6 Humans (Cells)

In humans, tests have been performed on blood and skin cells. After 24 h of exposure to microplastics approximately 0.1 µm in size extracted from a facial scrub, mitotic activity suppression was observed in erythrocytes, indicating the inhibition of DNA synthesis and an increase in hemolysis. In lymphocytes, DNA damage occurs as genetic material migrates from the nucleus to the tail [59].

Various tests have been carried out on skin cells (keratinocytes), exposing them to two types of microplastics from scrubs. It was observed that there was a significant attachment after 3 min of contact and that they were not completely removed, even after two washes. In addition, low reactive oxidative stress was observed, which was higher when the microplastics were amorphous, sharp, and at high concentrations [64]. Previous results have shown that there are effects on human cells owing to their interaction with microplastics, even affecting DNA.

10.7 Worldwide Bans on Microbeads in PCPs and Cosmetics

One of the many ways to control microplastic release through PCPs and cosmetics has been the enactment of bans on microbeads added to such products [15]. Some countries have made pledges and let voluntary phase-out be the main tool to deal with the issue, while others have enacted bans on specific products and domains.

The first countries to decree bans were Europeans, starting in 2014. Figure 10.14 shows the evolution of bans on microbeads within the European continent. The first countries to “rally” in favor of controlling microbeads in PCPs and cosmetics through bans were The Netherlands, Austria, and Sweden, which addressed microbeads concerns with the 28 environmental countries of the European Union members [65].

It is important to note that, although talks at the European Level have sounded alarms on the problem of plastic pollution, including the addition of microplastics to PCPs and cosmetics, some critics say not enough has been made [18].

The EU-based ban was introduced in 2018. The crafted instrument contains some exceptions that might be working against the main goal of the ban itself [18], which is facing out completely purposely added microplastics in products that, then, might escape into the environment and pollute. The main exception is that the European ban does not include three specific types of microbeads: water-soluble polymers, engineered nanoplastics, and biodegradable polymers, which might also be associated with environmental pollution risks and ecotoxicological effects [18].

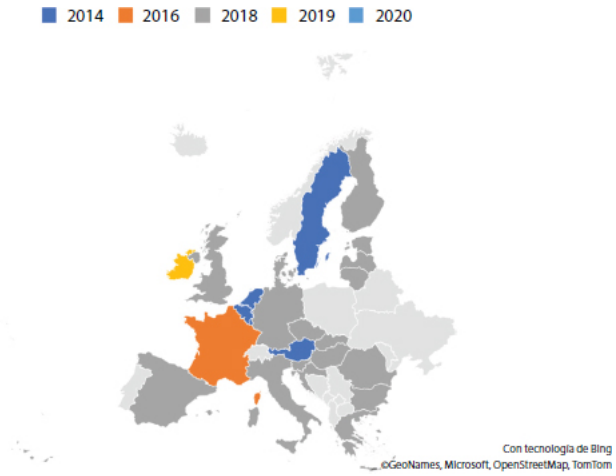


Figure 10.14 Countries within the European region with legal or administrative dispositions related to microbeads in PCPs and cosmetics.

In the rest of the world (Figure 10.15), important laws have been passed in several countries, such as the Microbead-Free Waters Act of 2015 (USA) or the Microbeads in Toiletries Regulations in Canada.

Almost all bans focused on rinse-off products and planned incremental phases for each coming year. Prohibitions like the Australian [66] go beyond and state design requirements and oversight for manufacturers [66].

Most bans also focus on selling, manufacturing, and importing PCPs and cosmetics that have microbeads, and in some cases, the exceptions to the ban are quite broad, as in the case of France. For example, the Law for the recovery of biodiversity from 2016 banned microbeads for exfoliating and cleaning purposes, excluding other particles deemed safe, not persistent, or included in other products, such as household cleaning items [67].

In Asia, one of the main markets and manufacturers of PCPs and cosmetics, the first country to address the issue of microbeads with a ban was Taiwan [67]. China, which according to this review, is the fourth largest producer of PCPs and cosmetics in the registered samples, has, so far, no disposition regarding a ban on microbeads, but several domestic societal organizations have been working to bring attention to the problem to governmental authorities [67]. In fact, by 2020, some incremental steps and domestic regulations, especially those focused on monitoring and classifying microbeads, have been created, and a nationwide ban is expected soon [68].

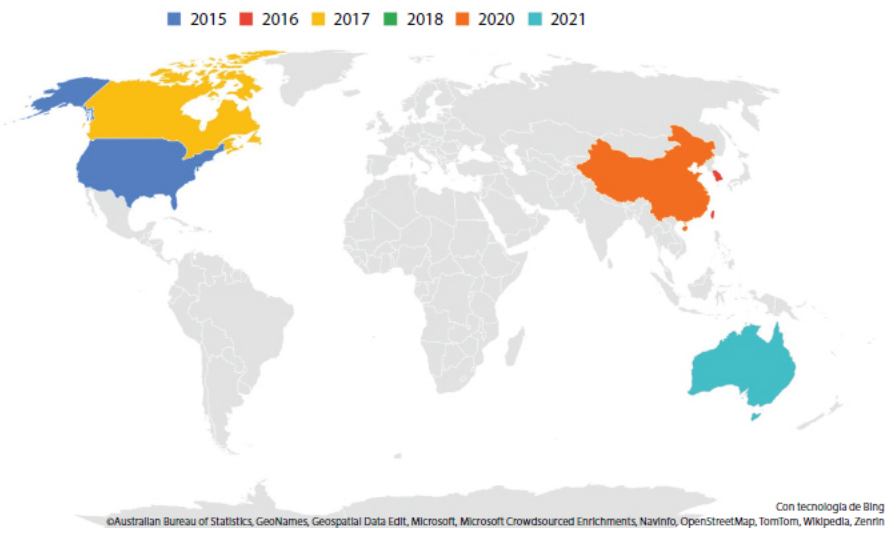


Figure 10.15 Countries outside the European region with legal or administrative dispositions related to microbeads in PCPs and cosmetics.

10.8 Conclusions

In recent years, the use of personal care products and cosmetics has increased. However, the consequences of this phenomenon in the environment have not been studied thoroughly studied. One aspect that needs to be studied is the impact of microplastics, such as microbeads, because the wide variety of ingredients in these products include microplastics.

Research on this topic has focused on products, such as scrubs. Nonetheless, it must be noted that the use of microplastics in them has already been banned in several countries. However, the use of plastic glitter particles, mainly in cosmetics, is less regulated. The presence of microplastics in these widely used products is concerning, as the inherent toxicity of plastic particles can increase due to interactions with other pollutants in aquatic ecosystems. Hence, research efforts must reinforce the study of these interactions to better reflect real conditions. Another field that must be approached is the use of novel types of materials, such as nanoplastics, biodegradable, and soluble plastics.

Controlling the use of microplastics on PCPs has been performed in different countries without significant disturbances in the markets or the way consumers use these products. This differs from the reactions in which other bans (such as single-use plastics) have been implemented. Given this situation, it seems that bans on microbeads are a feasible strategy to decrease plastic pollution in oceans by controlling their release from their origin point.

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Note

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11
Study on Microplastic Content in Cosmetic Products and Their Detrimental Effect on Human Health

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Abstract

The term ‘cosmetics’ was born from a Greek word ‘Kosmeticos’ which means to beautify. A large quantity of cosmetics is used worldwide to beautify one’s own body and look beautiful. A variety of chemical compositions, including microplastics, are mixed in the desired concentration as preservatives, stabilizers, surfactants, and shines when formulating cosmetics owing to their enhanced their properties, quality, and effects. Most chemicals are bioactive and cause severe health issues in humans. Synthetic microplastics with different particle sizes of less than 5 nm have received significant attention because of their prospective effects on the social economy, ecological environment, and human health. Moreover, microplastics are used for manufacturing soap, shampoo, toothpaste, moisturizers, sunscreen lotions, facial masks, eye shadows, lipsticks, and children bubble baths. This chapter reveals the cosmetic product market in India,

where the sources of plastics and microplastics are polyethylene, polypropylene, polyethylene terephthalate, poly (methyl acrylate), and microbeads are using when formulating the cosmetics. Moreover, cosmetics used with plastics, microplastics as preservatives, and alternative natural products have a significant effect on human health.

Keywords: Cosmetics, beautify, microplastics, preservatives, human health, synthetic plastics, formulation

11.1 Introduction

Among all the fastest growing industries globally, the cosmetic market position is nearly third place and its economic market is predicted to reach approximately USD 5.25 billion by 2023 based on the growth rate [1]. Cosmetic consumers, mostly women, can use about 79% of products for beauty, makeup, hygiene and care purposes, moisturizers, and deodorants [2]. The growth of the cosmetic industry has dramatically increased with increases in consumerism and demand, which has the environmental and health consequences. Figure 11.1 illustrates the schematic representation and classification of cosmetics that are generally used by the Food and Drug Administration (FDA) identified.

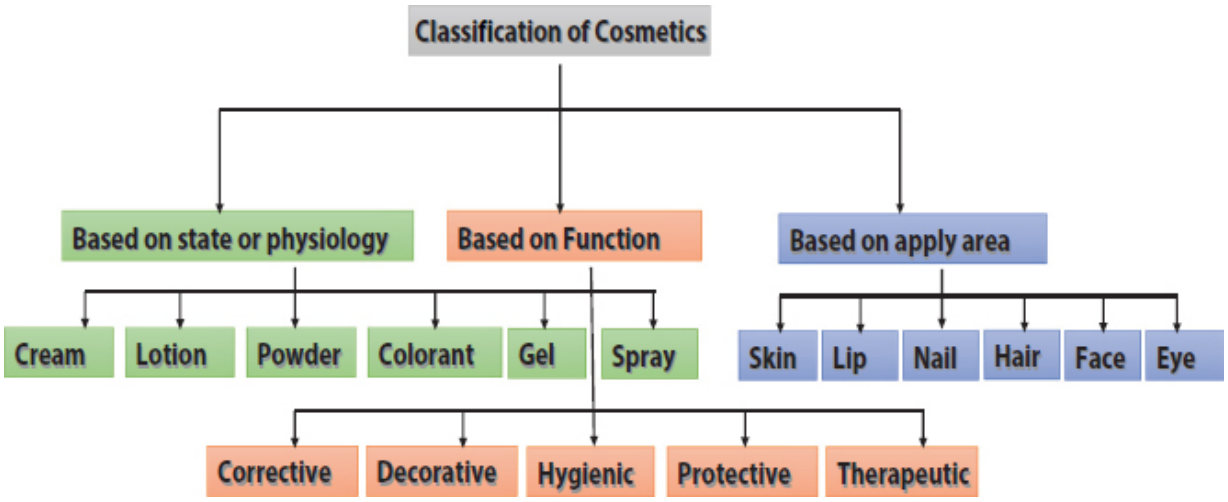


Figure 11.1 Classification of cosmetics by Food and Drug Administration (FDA).

However, over the past few years there has been an increased use of plastics and microplastics in personal care and cosmetic products; hence, the cosmetic manufacturing process is associated with the plastic and microplastic industries owing to their unique properties and specific applications. The word microplastic was announced by Thompson *et al*. in 2004, which covers all types of plastic material sizes of particles between 1 nm and <5 mm. When the particle size is greater than 5 mm, it is termed macroplastics [3]. The estimation of microplastic compositions in various sources globally is very difficult owing to its smaller size. Microplastics are generally manufactured from primary plastics and are used in the manufacture of various cosmetic products such as soap, toothpaste, shampoo, lipstick, lotions, and moisturizer products [4].

In general, all types of plastic materials used as ingredients in cosmetic products during formulation are made from petroleum carbon sources, termed thermoplastics and thermosets. Similarly, polyamide, polystyrene, polyethylene, poly(methyl methylacrylate), and polytetrafluoroethylene are the best examples of

thermoplastics, while polyurethanes and polyester are in the thermoset category. Another type of plastic is silicone in the form of polymerized silicones or inorganic and organic polymers with a silicone and oxygen backbone, which are widely used in the cosmetic industry because they are water-soluble or water-dispersible. Moreover, cosmetics such as foundation makeup substances, particularly lipstick, have solid silicone resins owing to their film formation property because they can add structure to products [5].

The function of plastics and microplastics in cosmetic products depends entirely on their type, size, shape, and particle composition. These polymer blenders can regulate a large number of cosmetic functions such as film formation, viscosity, liquid absorption tendency, emulsification, bulking agents, glitters, skin conditioning, and abrasives. Some oral care properties include tooth polishing, gallants in denaturing adhesives, delivery of fragments such as oils, vitamins, insect repellents, moisturizers, sun filter, and other active ingredients, and degradable active ingredients in the porous particle matrix. The functions of these plastics and microplastics clearly go beyond the well-known, well-publicized scrubbing of microbeads [6].

Cosmetic products may contain more than 10,000 chemicals and polymers, of which approximately 20% are recognized as safe. Some chemicals, which are major ingredients in cosmetics, cause various diseases, such as birth defects, cancer, and developmental and reproductive impairments [7]. Cosmetics that are absorbed into the human body through dermal penetration or other routes those can act as reproductive toxins, endocrine disruptors, carcinogens, and mutagens [8]. Similarly, the micropastics can flow through the bloodstream and are associated with serious health problems, such as cancer related to

hormones, infertility, and neurodevelopmental disorders. Nevertheless, the Food and Drug Administration (FDA, USA) has identified nine dangerous ingredients in cosmetic products, including formaldehyde, glycol ether, coal tar colors, lead, mercury, phthalates, parabens, and phenylenediamine [9]. The present work focuses on toxic chemicals such as plastics and microplastics blended with cosmetics during formulation, their effects on human health complications, and alternative natural sources predicted in the scientific literature.

11.2 Cosmetic Products in India

A person's look, hygiene, general well-being, and sense of confidence can be improved by utilizing beauty and personal care products and services. They include a variety of products and services such as cosmetics, skincare, hair care, perfumes, and grooming tools. Lipstick, foundation, mascara, and eyeshadow are common examples of cosmetics, while moisturizers, anti-aging creams, cleansers, and toners are examples of skincare items. Numerous cosmetics and personal care items, including bath salts, face masks, and aromatherapy, are also intended to promote relaxation and enjoyment. Indian customers incorporate these goods into their everyday routines, as they become more aware of their appearance. Figure 11.2 illustrates the different types of cosmetic products available in the Indian market. Similarly, Figure 11.3 disclose the shares of various clusters of cosmetic products available in Indian market for 2021. The data was collected from google by using the key word like "India cosmetic market".

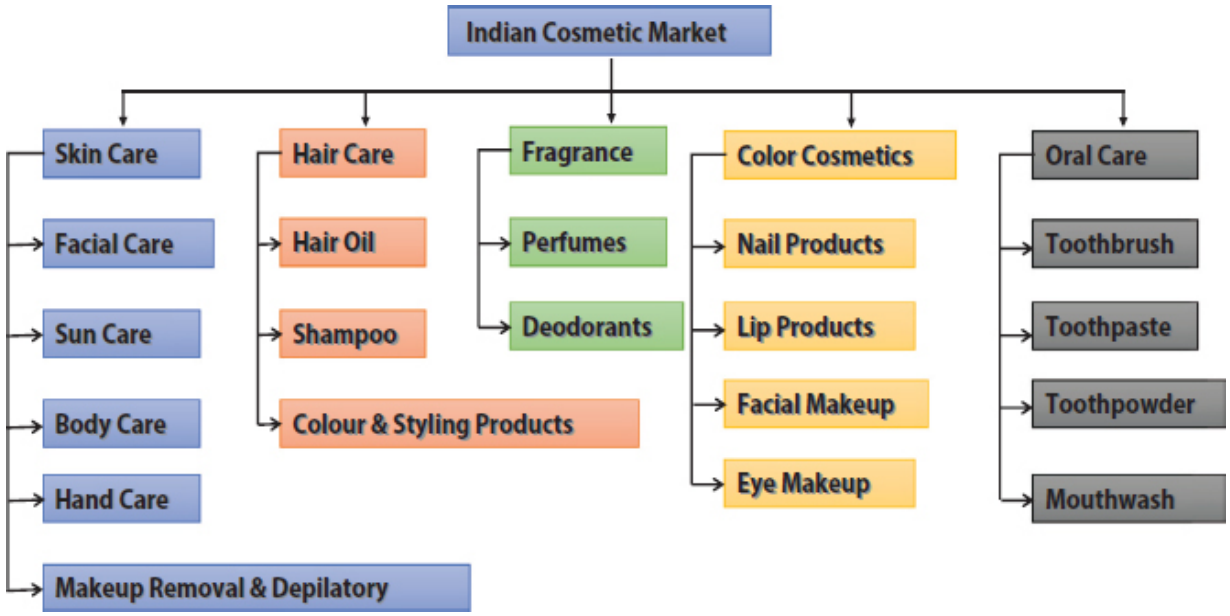


Figure 11.2 Flow chart of different cosmetic products widely available in the Indian market.

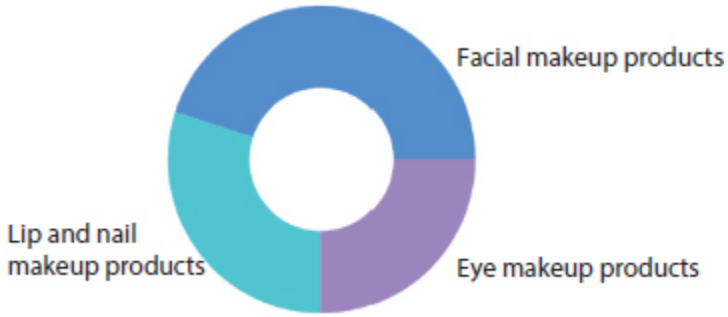


Figure 11.3 Cosmetic product market share in India for 2021. The data collected from Google by the keyword "India cosmetic market."

As customers become more aware of the possible harm caused by chemicals and choose products with natural ingredients, there is a growing trend towards natural and organic products. Numerous market competitors provide vegan, paraben-free, fragrance-free, halal, and organic products in response to this demand, which are thought to be safer for skin and general health. Additionally, there is a trend towards goods that promote health and wellness, such as anti-aging and skin-brightening goods. Additionally,

consumers gravitate to online shopping because of the ease and affordable prices that go along with it, as well as the accessibility of a variety of goods from both domestic and foreign suppliers. India has a sizable and expanding young population, which is an important demographic.

Another significant trend is an increasing emphasis on sustainability in the personal care and cosmetics sectors, with consumers choosing goods that are ethical and environmentally favorable. Additionally, customers are turning to anti-pollution skincare and hair care products because of the rising pollution levels in Indian cities. With the emergence of personalized skincare products and smart beauty equipment, technology has become increasingly significant. Accordingly, market participants are launching goods that use cutting-edge technology to provide customized solutions and convenience. The increased need for inclusive products that accommodate all skin tones and hair types is another new market trend.

11.3 Source of Plastics and Microplastics

Plastic and microplastic materials are produced directly from industrial (1° source) sectors and break down or disintegrate into small pieces of plastic materials (2° source) [10]. Plastic materials break down into microplastics in different ways such as biodegradation and non-biodegradation processes. In general, processes such as physical degradation, photodegradation, thermal degradation, hydrolysis, and thermal oxidative degradation are non-biodegradation methods. The polymeric structure can be decomposed by a non-biodegradation process and changing its structural and mechanical properties, including the specific surface area, which increases the interaction with microorganisms [11]. Moreover, plastics

and microplastics are generally utilized in the beauty products industry for the manufacture of a variety of beauty care products, such as sunscreen lotions, facial masks, eye shadows, moisturizers, shampoos, and toothpaste. The functions of plastics and microplastics in cosmetics include film formation, viscosity regulation, skin conditioning, and as emulsion-stabilizing agents [5]. Table 11.1 summarizes the various microplastic ingredients and their significant functions in cosmetics.

Table 11.1 Functions of different plastics and microplastics as ingredients in cosmetic products.

Cosmetic ingredient	Function of ingredient
Polypropylene	Viscosity regulatory function
Polystyrene	Light stabilizer and viscosity controller
Polyquaternium	Antistatic and film formation
Silsesquioxane	Skin conditioning
Polyurethane	Film formation
Siloxane	Skin conditioning
Polyethylenterephthalate	Film formation
Methicone	Antifoaming and lubricant
Dimethiconol	Antistatic, antifoaming, film formation, and emollient
Polythelene	Film formation, Viscosity regulation, and abrasive
Polyacrylate	Solvent
Polythelene glycol	Penetration enhancement
Polypropeline glycol	Plasticizer, lubricant, antifoaming, and abrasive
Acrylate copolymer	Film formation and antistatic function
Acrylate cross polymer	Viscosity regulation, emulsion stabilization, and film formation
Polyamide	Viscosity regulation, abrasive, opacifying, and bulking
Polymethylmethacrylate	Foam formation

11.4 Uptake and Bio-Accumulation of Microplastics

There are three important pathways for microplastics to leach into the human tissues: breath, digestion, and skin interaction, as shown in Figure 11.4. Nevertheless, inhaled airborne microplastics reach the human body owing to surrounding dust, textile industries, and rubber tires. The food chain and water supply are highly damaged when ingested by microplastics. Skin membranes are a very comfortable way to allow the plastics and microplastic materials into the body organs in terms of cosmetics, and it is possible to enter through injured parts, sweat glands, and hair follicles [12]. The total amount of microplastics and plastics accumulated in the human body can be increased in three ways [13].

The present study describes microplastics in beauty care substances and their significant effects on human health. Body and facial washes are the most common cosmetics applied to the skin surfaces. The top layer of the skin, namely the carenum, shields the skin and forms a barrier against various injuries, chemical substances, and microbial agents. Plastics and microplastics are deposited on the skin through cosmetic products, which are hydrophobic absorbed through the stratum carenum. Finally, the plastic and microplastic units arrive at the body via sweat glands, skin wounds, and hair follicles [14]. A study of UV irradiation in mouse skin revealed that carboxylate quantum dots can improve skin permeability. Intercellular adhesion can be compressed by the altered expression of tight injection-related proteins, such as zonula occludens-1, claudin-1, and occludin. Some chemical compounds are used to promote the penetration of drugs and formulations through the skin barrier. These compounds include alcohols (both short chain and long

chain), esters, glycols, fatty acids, sulfoxides, surfactants, and terpenes. However, the common chemical ingredients used in body lotion, such as glycerol, hydroxyl acid, and urea, also increase the ability of plastics and microplastics to permeate the skin barrier [15]. According to this study's findings, microplastics can enter the human body through the skin barrier and cause a number of hazardous diseases.

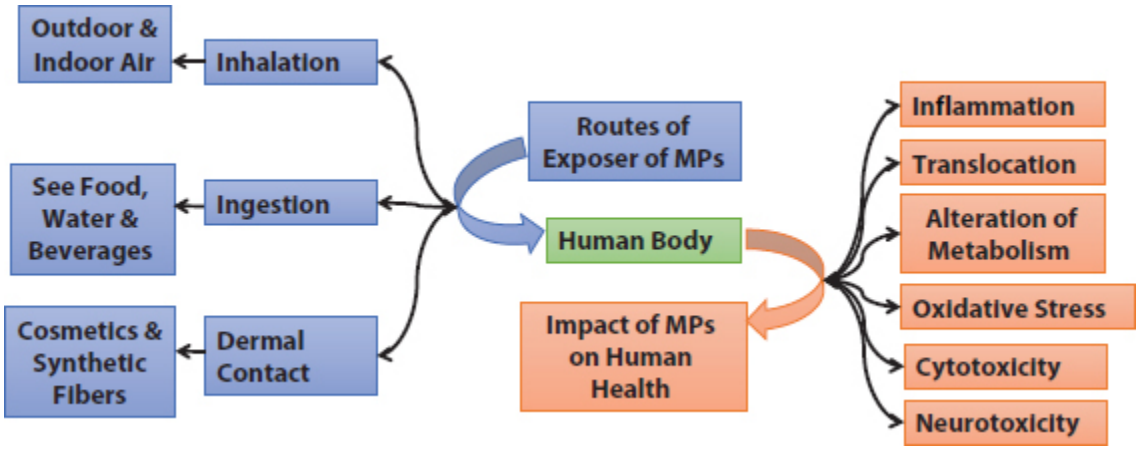


Figure 11.4 Schematic representation of three key routes for microplastics entering the human body, modified from reference [13].

11.5 Effect of Microplastic Exposure on Human Health

Numerous *in vivo* and *in vitro* experiments have provided information about plastics and microplastics, which are responsible for serious health issues in humans, including oxidative stress, immune response, physical stress and damage, necrosis, and apoptosis (Table 11.2).

11.5.1 Oxidative Stress and Apoptosis

Numerous *in vitro* studies have confirmed that different plastic and microplastic particles can enhance oxidative stress, apoptosis, and autophagic cell death, depending on

the cell environment. For example, amine-modified polystyrene microplastic particles, which may interact strongly and form aggregates with mucin- and non-mucin-secreting intestinal epithelial cells, can cause mucin- and non-mucin-secreting intestinal epithelial cells to undergo apoptosis. Cationic polystyrene microplastics are altered after reactive oxygen species formation and endoplasmic reticulum stress in mouse macrophages and lung epithelial cells, which is similar to how misfolded protein aggregation results in autophagic cell death [16].

Table 11.2 The significant toxic effects of plastics and microplastics on human health were modified from reference [13].

Toxic effect	Type of microplastic particle	Function on human body
Oxidative stress and apoptosis	• Amine altered polystyrene microplastic particles. • Poly vinyl chloride and polymethyl methacrylate.	• Mucin-induced aggregation and strong interaction. • Caused all intestinal epithelial cells to undergo apoptosis. • Caused lung epithelial cells and mice macrophages to undergo autophagic cell death. • Caused multiple human cell types to undergo apoptosis.
Inflammation	• Polystyrene microplastic with carboxylate and amino group modified particles. • Polyethylene microplastic	• Modified scavenger receptor expression. • Enhanced energy metabolism (M2) and TGF-1 (M1). • Impact on bone resorption around prosthetics.

Toxic effect	Type of microplastic particle	Function on human body
	particles generated from plastic prosthetic implants • Polystyrene microplastic particles	• Impact on the inflammatory reaction near the implant. • Activated liver inflammation • Produced negative outcomes for neurotransmission.
Metabolic homeostasis	• Pristine and fluorescent polystyrene microplastics • Pristine polystyrene microparticle. • Anionic carboxylated polystyrene microplastics	• Modifications in bile acid and amino acid metabolism. • Produced intestinal barrier failure and dysbiosis of the gut microbiota. • Ionic homeostasis and altered ion channel function • A reduction in energy metabolism. • Metabolic disease linked to dysbiosis of the gut microbiota and impaired gut barrier.

Toxic effect	Type of microplastic particle	Function on human body
		• Ionic homeostasis and altered ion channel function • K^+ channels basolaterally activated. • Induced ion outflow of Cl^- and HCO_3^-.

11.5.2 Inflammation

One type of microplastic, particularly large polystyrene particles with a size of approximately 200–500 nm, can cause major problems for human lung cells. Compared to the same cells exposed to 65 nm particles. In the human body, the organ interleukin-8 can be injured more by larger microplastic particles than by cells of the same size, which are visible to particles of approximately 65 nm. The larger polystyrene particles (approximately 202 nm and 535 nm) produced inflammatory effects on human A549 lung cells, according to an *in vitro* investigation using different diameters of the substance [16]. Additionally, compared to lung cells exposed to 64 nm micro particles, those treated with larger particles showed more interleukin-8. In human leukemia, histiocytic lymphoma, and gastric adenocarcinoma cells, carboxylated or unaltered polystyrene microplastics or nanoparticles strongly increase the expression of interleukin-6 and interleukin-8 genes. This suggests that the increase in inflammatory retorts by polystyrene microplastic particles is probably responsible for the formation of makeup particles rather

than the particles themselves. This finding suggests that rather than the particle charge, the higher inflammatory reactions to polystyrene particles are probably driven by the particle composition or simply by the presence of particles.

Additionally, it was discovered in several prior studies that the polyethylene parts used in prostheses could fracture under the impact of damage, producing waste in the joints. The function of ligand NF-B is to activate the receptor system, which promotes osteoclastic factors such as periprosthetic bone resorption and may ultimately lead to the loss of prosthesis, which is triggered by polyethylene wear particles. The periprosthetic tissue of implants made of polyethylene with ultrahigh molecular weight has been found to contain significant amounts of microplastic substances, which can measure sizes between 0.2 m and 10 m [17]. A further indication that the inflammatory response was triggered is the presence of macrophages in the tissues near the implant area. In a study of patients who underwent failed total hip arthroplasty with titanium alloy, polyethylene particles with an average diameter of 530 nm made up the majority of the wear debris in the interfacial membranes. To counteract these detrimental effects, surgeons are routinely adopting metal-on-metal joint replacements [18].

11.5.3 Metabolic Homeostasis

The distribution of microplastics in various parts of the body, such as the kidney, liver, and gut of mice given pristine polystyrene microparticles, with sizes of approximately 5 m and 20 m for 28 days, revealed that the bigger particles were evenly distributed throughout all muscles, whereas small sizes of substances were detected even at high concentrations in the gut [19]. Strong evidence suggests that the accumulation of microplastics in

mouse tissue impairs lipid and energy metabolism, produces oxidative stress, and triggers neurotoxic reactions. Furthermore, intestinal barrier failure, metabolic abnormalities, and gut microbiota dysbiosis were observed in pregnant mice that had ingested microplastics. The metabolisms of the F1 and F2 generations were permanently changed as a result of the maternal-level impacts of microplastic exposure [20].

11.6 Alternatives of Microplastics in Cosmetic Products

The cosmetic industry primarily faces problems related to the disposal of microplastics through beauty products and their packing materials with some kind of coal byproduct. However, some problems arise in terms of clinical testing with animals, three-dimensional skin-frame development, and the appearance of organ-chip transformation technology [21]. The cosmetic industry must focused on the development of sustainable cosmetic products, termed ecological, green, and natural products. Such ecological cosmetics can protect the skin when using naturally available products, such as herbs, plants, and essential oils. These natural constituents can function as transporters, additives, emulsifiers, and natural humectants [22].

The cosmetic industry, generally used as a packing polymer, uses polyethylene, which is highly responsible for human health and environmental pollution. To overcome these difficulties, biopolymers with excellent properties have been developed from plant derivatives, biomass, cellulose and microorganisms. The common aliphatic polyester is polylactic acid, which is manufactured from sugar through the fermentation process, and can exhibit biodegradability, biocompatibility, high mechanical resistance, cost–benefit ratio, and nontoxicity. Various

industrial applications of polylactic acid are food packing, tissue engineering, and cosmetic and drug distribution [23]. Moreover, starch is prepared from plants and their byproducts such as potatoes, corn, rice, and wheat. These materials show excellent biodegradability, low toxicity, and outstanding mechanical properties when bound to plasticizers such as citric acid, sorbitol, and glycerol. Polyhydroxyalkanoates are polyesters that are the best alternative biodegradable polymers to traditional petroleum-derived plastics synthesized from microorganisms, such as carbon and fatty acid reserves, metabolized in appropriate reaction environments [24].

Moreover, some plants and their products have anomalous antibacterial and antifungal properties, which are directly involve in enhancing the resistance and conservation of products. For example, medicinal plant products such as thyme, turmeric, garlic, tea, and rosemary contain sufficient metabolites, including alkaloids, terpenoids, polypeptides, polyphenols, and polyacetelenes [25]. Some essential oils, such as monoterpenes, diterpenes, phenolic compounds, and phenylpropanoids, are excellent alternative products for preservatives and antioxidants. The antimicrobial stability of essential oils is moderately lower, about 60–90 days, than the normal preservation of petrochemicals, particularly parabens. Enhancing the stability, function, and durability of essential oils is imperative for the industry.

Moreover, new types of sustainable development have been combined with various properties of microalgae in cosmetic formulations, particularly as stabilizers, emulsifiers, excipients, foams, dyes, and thickeners. The residues of *Arthrospira* and *Chlorella* microalgae substances widely are used as alternative preservatives in regenerating products, sun protection creams, peels, emollients, anti-aging products, and making colorful dyes because of the presence

of carotenoids, chlorophyll, and phycobilins. Similarly, red microalgae, such as palmaria and porphyra, are extensively used in cosmetics owing to the presence of a high amount of arginine and precursor to urea [26].

Another cosmetic preservative, bacterial cellulose, is obtained from microorganisms such as Gluconacetobacter, Acetobacter, Agrobacterium, Achromobacter, Aerobacter, Sarcina, Azobacter, Rhizobium, Pseudomonas, Salmonella, and Alcaligenes through carbon consumption and biosynthesis processes [27]. Bacterial cellulose is usually applied in various sectors, including transportation of active substances into the skin, emulsion stabilizers, emulsifiers, thicker creams, and the food packing industry, owing to its biodegradable and biocompatible nature. Therefore, biological products are more important than traditional cosmetics in terms of environmental and health issues. At every level of development, green products, such as eco-friendly cosmetic products, must be effective, safe for the environment, and created with the least amount of environmental harm possible [28].

11.7 Conclusions

Currently, people are widely using care and beauty products such as cosmetics, skincare, hair care, perfumes, and grooming tools. Lipstick, foundation, mascara, and eyeshadow are common cosmetics, while moisturizers, anti-aging creams, cleansers, and toners are skincare items. However, some chemical substances in the form of plastics and microplastics have been added as preservatives in cosmetics. These polymer blenders can regulate many cosmetic functions such as film formation, viscosity, liquid absorption tendency, emulsification, bulking agents, glitters, skin conditioning, and abrasives. The functions of the plastics and microplastics in cosmetic products totally

depend on plastic type, particle size, particle shape, and composition. Some oral care properties include tooth polishing, gallants in denaturing adhesives, delivery of fragments such as oils, vitamins, insect repellents, moisturizers, sun filters, and other active ingredients, and degradable active ingredients in the porous particle matrix. The top layer of the skin is the carenum which protects the skin and forms a barrier against various injuries, chemicals, and microbial agents. Plastics and microplastics are deposited on the skin through cosmetic products, which are hydrophobic and absorbed through the stratum carenum. Finally, the plastic and microplastic particles arrive at the body *via* sweat glands, skin wounds, and hair follicles. In the cosmetic industry, the most commonly used packing polymer is polyethylene, which is highly responsible for human health and environmental pollution. To overcome these difficulties, biopolymers with excellent properties have been developed from plant derivatives, biomass, cellulose and microorganisms. The common aliphatic polyester is polylactic acid, which is manufactured from sugar through the fermentation process, and can exhibit biodegradability, biocompatibility, high mechanical resistance, cost–benefit ratio, and nontoxicity.

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12
Effects of Micro- and Nanoplastics on Human Genome

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Abstract

The “Plasticene Era,” as it has been properly termed by experts, is the new geological era that humanity is experiencing now. The manufacture and use of plastics have increased to unprecedented levels over the past few decades, causing a global plastic pollution problem with far-reaching effects on ecosystems and human health. Plastic materials can degrade over time into tiny pieces known as microplastics (sizes ranging between 1 µm and 5 mm) and nanoplastics (size of 1 µm), which have aroused concern owing to their potential impact on the environment and human health. These particles have been observed to build up in several organs and tissues. Microplastics and nanoplastics have been shown to cause oxidative stress, inflammation, and cellular damage, which may have negative health effects such as tissue damage, organ dysfunction, and the emergence of chronic diseases. Furthermore, new data suggests that micro- and nanoplastics may influence the human genome. Plastic particles have been shown to damage DNA, alter gene expression, and obstruct vital biological functions such as

DNA repair mechanisms. Such genomic changes may increase the likelihood of genetic mutations and chromosomal abnormalities, as well as accelerate the onset of cancer and other genetic illnesses. To limit the possible dangers associated with microplastics and nanoplastics and to protect human health and the environment, methods for minimizing plastic pollution and enhancing waste management practices are essential.

Keywords: DNA, genome, human health, microplastics, nanoplastics, plasticene era

12.1 Introduction

Around the 1950s, when the commercial development of polymers such as polyethylene, polyolefins, and polypropylene started, plastic production was underway [1]. Since then, plastics have become an essential material in many industries and everyday products owing to their versatility, low cost, and durability. However, the widespread use of plastics has also led to environmental concerns, as many types of plastics are not biodegradable and can persist in the environment for hundreds of years. The global production of plastics has increased by 9% every year [1]. A total of 5.25 trillion plastic bits are estimated to be present in water bodies [2]. Although most plastic garbage is impervious to biodegradation, it is extremely susceptible to natural processes such as oxidation, ultraviolet radiation, and the hydrolytic properties of water bodies [3]. As a result, weathering of this collected plastic debris produces microscopic particles, known as microplastics, that range in size from 5 mm to 100 nm [3]. The open ocean is the most well-known sink for microplastics [1]. Even ice blocks discovered in the Antarctic region contain microplastics, indicating how pervasive their presence is in all environmental media [3].

Inadvertent ingestion of microplastics by aquatic species, ranging from fish to zooplankton, leads to the transmission of microplastic to higher trophic levels [3]. Microplastics can be converted to nanoplastics through mechanical breakdowns such as abrasion and erosion. For example, waves and currents in the ocean can cause microplastics to rub against each other and break down into smaller particles. It has been proven through various studies and experiments that synthetic marine ecosystems can act as a probable source for the conversion of macroplastics to micro- and nanoplastics [4]. Nanoplastics are very small, with less than 100 nm-sized plastic fragments [5]. Most micro- and nanoplastics that impact on human health are correlated with the size of the bacteria and viruses that infect a healthy human body [6]. Micro- and nanoplastics produce alterations at the cellular level in the human body, akin to infections that are quickly internalized by human cells and affect cellular functions [6]. Because of their exceedingly small size, which allows them to enter living organisms and collect them, nanoplastics are currently a major topic of concern because they can potentially impair both human health and the environment. Studies are still being conducted to determine how to prevent these incredibly small plastic materials from harming living biota. The potential effects of micro- and nanoplastics on the human genome are an area of active research that is not yet fully understood. However, some studies have suggested that exposure to microplastics and nanoplastics may have genotoxic effects on human cells, which could lead to DNA damage and potentially increase the risk of cancer. If said precisely, here are some instances where scientists have discovered how nano- and microplastics can damage DNA in the human body. A previous study showed that exposure to microplastics leads to DNA damage in human lung cells. Researchers have observed an increase in the frequency of DNA double-strand breaks, which can

lead to genomic instability and potentially increase the risk of cancer [7]. Another study by Prasad *et al.* (2020) investigated the effects of nanoplastics on human lung cells and reported that exposure to nanoplastics caused oxidative stress and DNA damage. Researchers have found that nanoplastics induce the formation of reactive oxygen species (ROS), which can cause DNA damage and cell death [8]. In this review article, we aim to provide all the hazardous effects of nanoplastics and microplastics are having on human health, more precisely the human genome.

12.2 Source of Micro- and Nanoplastics

One of the most pressing global issues is plastic trash. Data that have recently undergone analysis predict that by 2022, approximately 300 million tons of plastic will be consumed annually. Regarding visibility, plastic trash can be roughly divided into two categories, which are mostly:

Visible Plastics: Plastics, such as water bottles and polythene bags, which have become a standard amenity for the populace, are separated into a) macro plastics (particles greater than 2.5 cm in diameter); b) meso plastics (particle size ranging from 5 mm to 2.5 cm).

Invisible Plastics: Invisible Plastics, or more accurately microplastics, are plastic contaminants that are invisible to the naked eye. Granules and filaments are two types of microplastics [2]. Invisible forms of plastics can be divided into two groups: a) microplastics (5 mm to 100 nm) and b) nanoplastics (1 nm to 100 nm).

After the formation of microplastics, further degradation results in the formation of nanoplastics. The classification of plastics into different forms and the formation of

nanoplastics are shown in Figure 12.1. A good feature of visible plastics is that they are easily detectable in most cases, which results in ease of removal. This invisible form remains undetected in nature and causes pollution in the form of micro- and nanoplastics. In recent practices, there are certain detectable measures for microplastics, but there is no such advancement in the case of nanoplastics, which have proven to be a greater threat to the environment.

Nanoparticles can be further divided into: a) primary and b) secondary particles. Primary particles are formed mainly from different industrial products that are generally used by the public.

Tire Wear: The intense friction that a moving car's tires constantly experience causes tire fragments to be released into the air as particles. This method releases almost 30% of the total bulk of the tires. It is important to examine the negative effects of this source of nanoplastics and devise the necessary measures [5].

Textile Wastes: During the laundry of different clothing nylon, polyester microfibers from synthetic clothing are released and mixed with laundry wastewater; subsequently, water is mixed in a larger water source such as oceans and rivers. Nanoparticles are taken up in different food chains, causing damage are being done [5].

Marine coatings (performed on ships) and personal care products such as cosmetics are industrial nanoplastic suppliers [2].

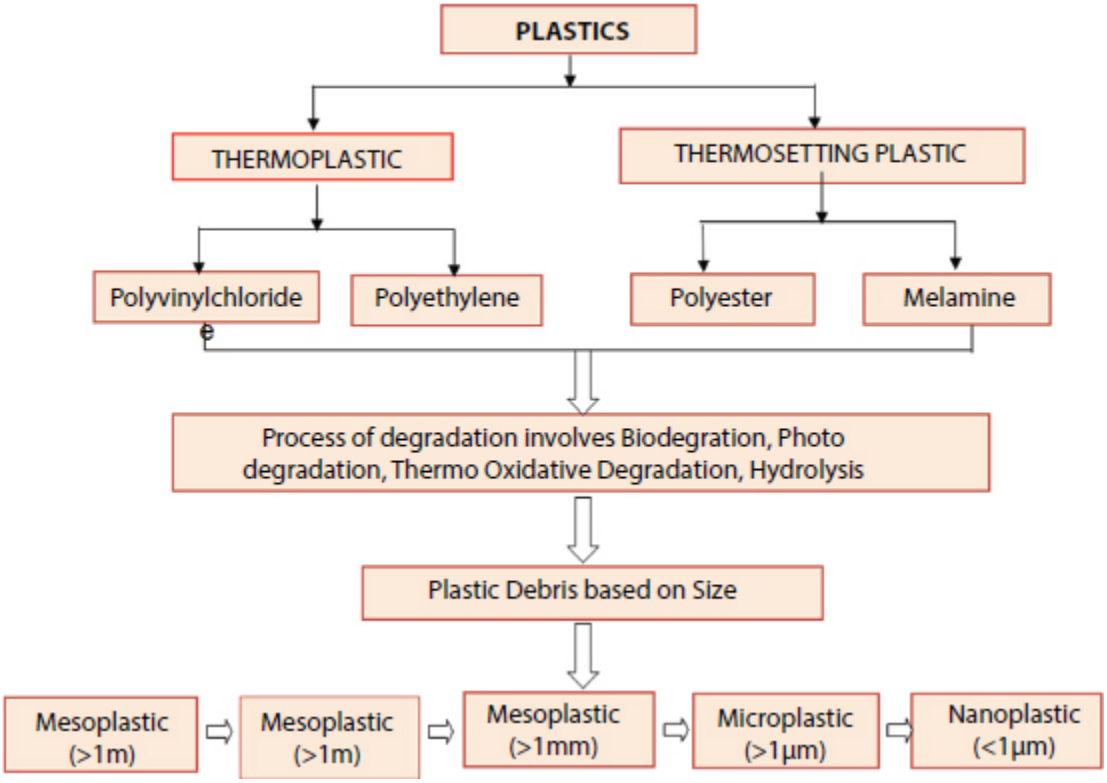


Figure 12.1 Classification of plastics and their derived products due to environmental effects.

Microplastics have been found in different rare locations, such as the Arctic, in the middle of the ocean, where human activity is very rare. Different experiments were performed to determine the possible reasons. The major reasons for this are the involvement of UV rays from the sun, waves, wind speed, and other similar factors that are responsible for the formation of nanoparticles by breaking down the particles only if the aforementioned industrial pollutants are somehow present. This cycle cannot be controlled by any means because the cycle of plastic degradation is a natural phenomenon.

12.2.1 Source of Micro- and Nanoplastics From the Marine Ecosystem

Plastic waste can be found at any level in the oceans. This property is attributed to the lower density of plastic waste

compared with that of water. Different microorganisms accumulate over the material over time; as a result, they sink deep into the ocean. Different chemicals secreted by microorganisms and certain environmental factors break down particles into smaller pieces. These smaller particles are more vulnerable to crossing cell walls and membranes of different marine organisms, resulting in a decrease in the population density of some green algae species, such as *Scenedesmus obliquus* which significantly reduces the chlorophyll level [2]. The degradation of plastics and their transport to the marine ecosystem are shown in Figure 12.2. The reduction in particle size did not have any effect on the density, as the chemical composition remained the same. The accumulation density of the nanoparticles depends on several factors: the speed of water flow; the slower the flow of water, the greater the chance of higher density accumulation; and for the opposite situation, when the water flow is faster, there is a higher chance of overturning the particles. Another important aspect is the speed of the wind; the higher the speed of the wind, the lower the chance of higher density accumulation, as they can flow to greater distances. In different textile industries, microfibers that accumulate in municipal wastewater are treated at water treatment plants. In most cases, these microparticles pass through the filter and accumulate in coastal and marine environments.

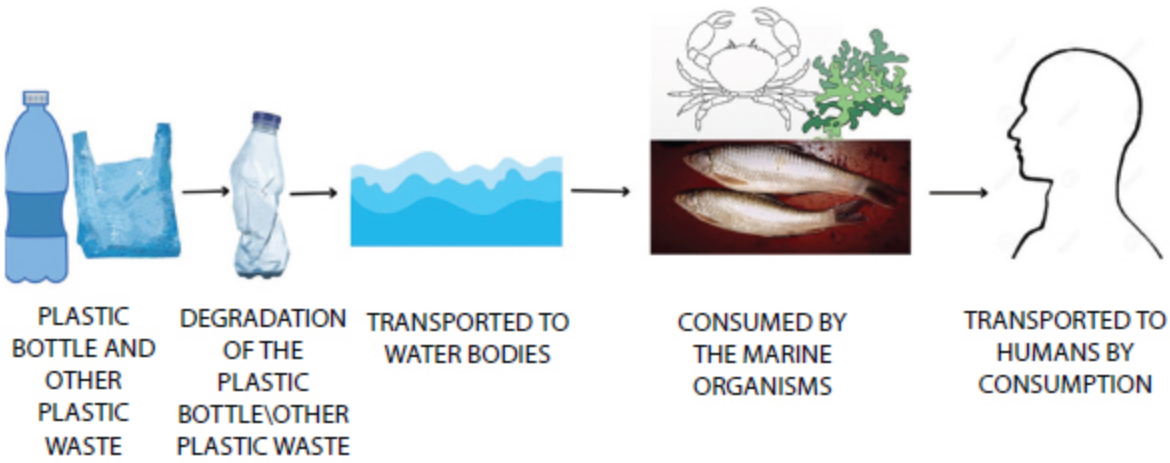


Figure 12.2 Route of microplastic and nanoplastic transport to humans through marine ecosystems.

12.2.2 Source of Micro- and Nanoplastics From Soil Ecosystem

Nanoplastic particles have a higher surface-to-volume ratio, many of which serve as colloidal particles [9]. This property is a major reason for the high mobility of nanoparticles across the soil bed. Different physicochemical properties of soil, such as changes in pH, organic matter content, porosity, and water holding capacity, and certain physicochemical properties of nanoparticles, such as size, shape, colloidal properties, and different human activities such as digging, have contributed to the effect of nanoparticles on the soil. Nanoparticles are free to move in comparison to other particles such as metal oxides present in the soil. This property is very scary to plants as the probability of taking up these particles increases. During different experiments, the formation of cracks and pores in the root head was observed during the uptake of plastic particles [9]. Because soil porosity on surface charge soil pH plays a key role in the transportation of nanoparticles, the higher the pH, the higher the distance travelled. The use of different composts, plastic mulching, and sewage sludge is a prime source of plastic pollution in soil [10].

After uptake by plants, nanoparticles are added to the food chain, which has various detrimental effects on the ecology and health of different organisms.

12.2.3 Other Sources of Micro- and Nanoplastics

Microplastics can be ingested through the consumption of contaminated foods and beverages. It is found in seafood, drinking water, salt, honey, and other food products. Microplastics can enter the food chain through multiple pathways, including pollution of oceans, rivers, lakes, agricultural runoff, and contamination during food processing and packaging [11]. Microplastics are inhaled from the air that we breathe. They are usually released into the atmosphere through activities such as burning fossil fuels, tire wear from vehicles, and microplastic fibers released from textiles during laundering or outdoor activities. Once in air, they can travel long distances and be inhaled into the lungs [5]. Microplastics can be found in personal care products, such as toothpaste, exfoliating scrubs, and cosmetics, as well as household products, such as cleaning agents and detergents. Microplastics are often added intentionally as ingredients, such as microbeads or microfibers, for their functional properties or as fillers [12]. Microplastic fibers can be released from clothing and textiles during laundering, wearing, and disposal. Synthetic fibers, such as polyester, nylon, and acrylic, are major contributors to microplastic pollution in the environment, including microplastic fibers that can end up in the human surroundings. Microplastics can also be released from plastic packaging, especially during the production, use, and disposal of single-use plastics. Weathering, degradation, and fragmentation of plastic products can result in the generation of microplastics that can enter the environment and potentially cause human exposure [13,

14]. Microplastics can be generated from various industrial processes such as plastic manufacturing, processing, and waste management. These microplastics can be released into the environment and pose risks to human exposure through inhalation, ingestion, or dermal contact.

12.3 Pathways Through Which Micro- and Nanoplastics Enter the Food Chain

Although plastics have a humongous social advantage, they have become a major social concern as they have a hazardous effect on the health of living organisms [15]. Although humans are responsible for the introduction of plastics, which is essential for daily use, it is now quite evident that they are affected to a great extent by these polymer-based particles. When plastic waste is incinerated, it has been known for decades that it releases noxious gases such as dioxins, furan, and polycyclic aromatic hydrocarbons. Alongside the release of these harmful substances, the burning of plastic wastes can also leave residues of two metals that are very harmful to human health: cadmium and lead [16]. Over the past six decades, the exposure of humans to microplastics and nanoplastics has increased extensively, as the production of plastics has increased by 6 tons [17]. Normally humans are subjected to very low concentrations of nanoplastics for long periods of time [5]. Microplastics and nanoplastics have become ubiquitous in the environment, with potential exposure routes including air, water, food, and even consumer products. Recent studies on microplastics and nanoplastics have found an abundance of microplastics in ecosystems that are dwelled by marine organisms and that these marine organisms are then consumed by humans, which makes this a very probable pathway for these miniature

plastic materials to enter the human system [16]. One of the unpropitious effects of microplastic is the transport of harmful chemicals to organisms [18]. There are two predominant ways through which microplastics and nanoplastics can enter the human system: ingestion and inhalation. Dermal exposure can also be a pathway through which these extremely small particles can enter the human body system [17]. The transport of micro- and nanoplastics to humans by various exposures is shown in Figure 12.3.

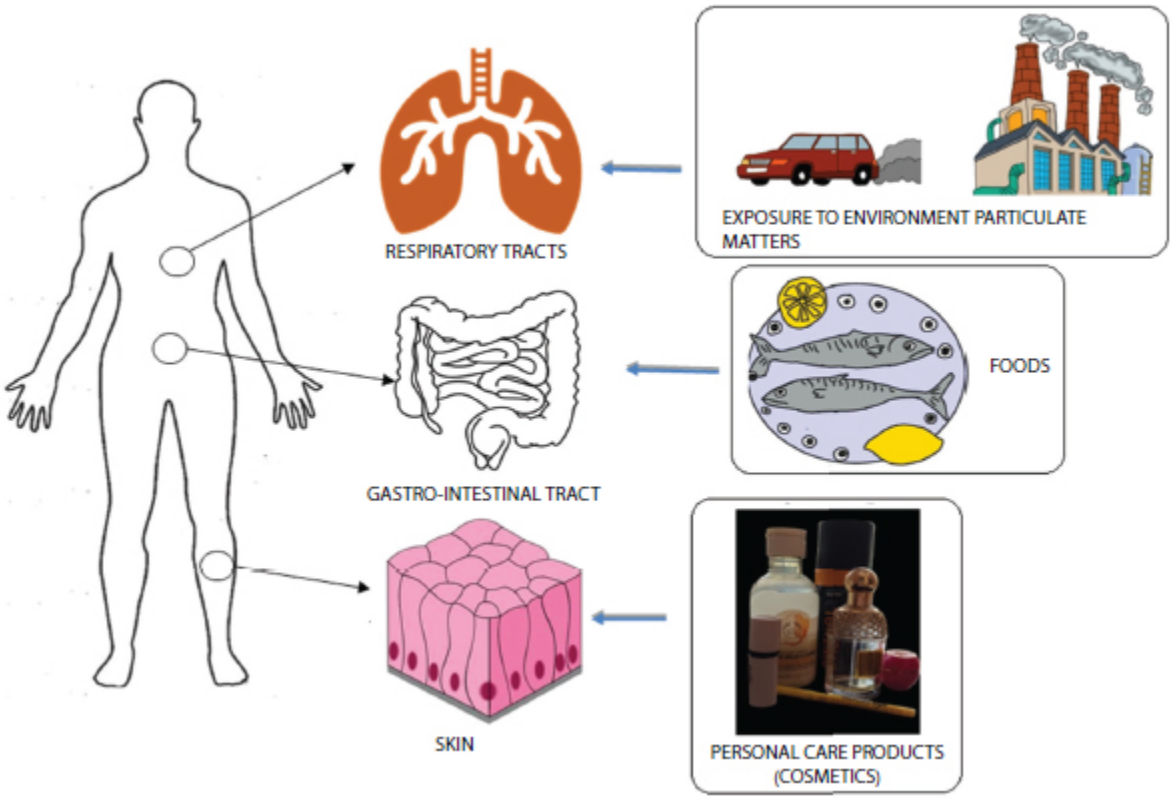


Figure 12.3 Pathways through which micro- and nanoplastics get exposed to tissues and organs

(adapted from [77, 78]).

12.3.1 Transports by Inhalation

Humans can inhale microplastics and nanoplastics after they become airborne. These specks have been identified in dust as well as in indoor and outdoor air. According to a

Chinese study, urban dwellers may breathe up to 700,000 microplastic particles daily [19]. Workers working in sectors that manufacture or utilize plastics, such as the textile, rubber, and plastic industries, may inhale microplastics and nanoplastics. Studies have shown that the air at these sites contains microplastics and nanoplastics [20]. Common household goods, including synthetic textiles, carpets, and upholstery, can leak microplastics and nanoplastics into air. Members of the home may breathe in these particles after they become airborne [21]. Exposure to microplastics and nanoplastics in the environment, such as walking or cycling close to busy roadways, can be introduced into the human body. According to a French study, bicyclists on congested roads may breathe up to 90,000 microplastic particles daily [22]. It is well recognized that nanoscale particles, including various varieties, can breathe and enter the respiratory system. It is conceivable that nanoplastics can be breathed and enter the deeper layers of the lungs, including the alveoli, owing to their small size [23].

12.3.2 Transports by Ingestion

Microplastics and nanoplastics are now more common found in meals and drinks. According to recent investigations, plastic particles are found in popular food items such as brine, dairy products, honey, vegetables, fruits, mineral-enriched water, and crustaceans. Therefore, it is necessary to study human exposure through ingestion while considering the full food manufacturing chain, from agriculture to consumption [7, 24]. Even though micro- and nanoplastics are a topic of research, there are still many challenges regarding the methods used to find them in beverages and foods, as well as description factors such as application dose. The diversity of these particles, which might have various chemical compositions as well as

varying densities, sizes, and formats, is the first problem that makes it challenging to standardize the procedures. Additionally, there are no specific definitions or reference values [24]. Foods of aquatic origin are among the most researched foods [25]. Micro- and nanoplastics typically accumulate in the gills, liver, and gut of fish, which are typically not consumed by humans [26]. However, a disproportionately high concentration of micro- and nanoplastics has been found in some filter-feeding species, such as shrimp. An estimated 11,000 particles are ingested annually by animal consumers [27]. Commercial mineral water also contains plastic particles, even water that was bottled in containers made of glass, in addition to meals of aquatic origin [28], whereas several kinds of table salt have been shown to contain polypropylene particles [29]. Plastic fragments have been discovered in a variety of crops and fruits, including lettuce, broccoli, potatoes, and carrots [30]. The discovery of microplastics in human fecal samples supports the notion that humans swallow particles. The effects of this form of oral exposure remain unknown [31]. Particles can interact with the digestive system and fluids after swallowing. Plastics can adsorb chemicals, heavy metals, protein molecules, or even microbes on their surface, which can increase their toxicity, even though they are thought to be inert in chemical terms. Micro- and nanoplastics function like Trojan horses in this circumstance, bringing several environmental toxins with them. When swallowed, particulates may come into contact with the intestinal microbiota, the fluid that lines the lumen of the gut, and even the cells of the epithelium themselves, leading to cellular reactions and a variety of physiological changes [32–35]. According to recent studies, particles can either travel across cells in the intestine (paracellular transport) or be absorbed by epithelial cells in the gut (endocytosis) depending on their size [36]. Research on animals has demonstrated the presence of micro- and

nanoplastics in systems/organs such as the lymph nodes, liver, and kidneys [37–39].

12.3.3 Transport by Dermal Exposure

In particular, body and facial washes used directly on the skin are a major source of nanoplastics [40]. Another significant method of exposure is the dermal application of nanocarriers for medication delivery. Small particle size and stressed skin conditions are crucial for the penetration of nanocarriers into the skin, despite the lack of conclusive data on their effects [41]. Currently, no research has explicitly examined how well nanoplastics can penetrate the oter layer of the skin. The possibility of nanoparticles from textiles penetrating the skin barrier in extremely small quantities was only recorded in one investigation [42]. The stratum corneum, the outermost layer of the skin, serves as a barrier that protects the skin from harm, toxins, and microorganisms. Corneocytes make up the stratum corneum, which is encased in lamellae of hydrophilic lipids, such as ceramide, long-chain free fatty acids, and cholesterol [43]. The skin may become polluted with plastic particles through contact with contaminated water or by using beauty and health items. Plastic particles can enter the body through glands that sweat, wounds in the skin, or hair shafts because micro- and nanoplastics are hydrophobic, making it unlikely that they would be absorbed through the stratum corneum through polluted water [41]. The number of 20 nm polystyrene nanoplastics concentrated in the follicles that produce hair was higher than that of 200 nm nano -plastics, according to a laser confocal scanning micrograph of the skin. However, neither of the particles could penetrate the corneal layer or anchor themselves to the underlying tissues of the skin. Polystyrene nanoparticles with sizes of 20–200 nm can penetrate only the outer levels of the dermis to an extent of

2–3 µm [44]. Dermal exposure hass also been explored in the medical context. Plastics, for instance, are known to cause modest inflammatory reactions during suturing as well as foreign body reactions with fibrous encapsulation. The effects of polyvinylchloride and polyethylene were examined in a mouse study, and the results demonstrated that polyethylene was associated with less inflammation than polyvinylchloride. However, owing to variations in the surface characteristics, micro- and nanoplastics can potentially cause irritation and foreign body reactions. Micro- and nanoplastics cause oxidative stress in human epithelial cells of the epithelium, demonstrating the necessity for studies examining the impact of exposure to micro- and nanoplastics [75].

12.4 Harmful Impacts of Micro- and Nanoplastics on Human Health

Microplastics and nanoplastics, which are tiny particles of plastic measuring less than 5 mm and less than 100 nm, respectively, have raised concerns owing to their potential toxic effects on human health. These small plastic particles can originate from various sources, such as the fragmentation of larger plastic items, degradation of plastic waste, or intentional production of nanoplastics. Microplastics and nanoplastics can contain or adsorb toxic chemicals, including additives used in plastics and environmental pollutants [45]. When ingested, these particles can release toxic chemicals into the digestive system, potentially leading to adverse health effects [46]. Research has shown that microplastics and nanoplastics can induce cellular toxicity, disrupt normal cellular functions, and cause oxidative stress [19, 47]. The small size and large surface-area-to-volume ratio of microplastics and nanoplastics enhance their interactions with biological

systems. These particles may induce physical damage, inflammation, and oxidative stress in cells and tissues, potentially leading to various health problems [9]. Microplastics and nanoplastics can be absorbed through ingestion, inhalation, or dermal exposure [17]. Microplastics and nanoplastics can affect various organs in the human body. While research on the specific impacts of microplastics and nanoplastics on organ systems is still limited, some studies have provided insights into potential target organs. Microplastics and nanoplastics can interact with biological systems including cells, tissues, and organs. They can trigger immune responses, disrupt cellular processes, and interfere with normal physiological functions. These interactions can result in a range of toxic effects such as inflammation, cellular damage, genotoxicity, endocrine disruption, and even potential carcinogenic effects [13].

Precipitation from the atmosphere has been found to contain micro- and nanoparticles that result from plastic degradation [48]. Particles that can enter the nose and mouth and deposit themselves in the upper part of the respiratory tract are referred to as inhalable, whereas those that are deposited in the lungs are referred to as breathable [49]. These particles might be vulnerable to general host defense systems that clear mucus through a mucociliary process. These particles might be taken in by alveolar macrophages, which would then carry them to the colon for elimination [50, 51]. Research has shown that exposure to microplastics and nanoplastics may adversely affect human health, leading to ailments such as pneumothorax, bronchitis, fibrosis, and asthma [48]. These microplastics, owing to their extraordinary minuscule size, result in chronic inflammation as they get stuck in the lung tissue, which later leads to pathology [52]. Studies utilizing epithelial basal cells of the alveoli (A549) have

demonstrated that nanoplastics derived from polystyrene can alter the gene expression, leading to inflammatory reactions and the activation of apoptotic pathways when smaller polystyrene nanoplastics are utilized. According to the findings of one study, human and mammalian respiratory systems can suffer real harm and functional disruption from nano plastics [53]. In the most recent research, plastic particles spanning in size from 1.59 to 5.58 µm had been identified in the bronchoalveolar region of more than 50% of the evaluated lung samples, confirming for the first time that micro- and nanoplastics exist in the respiratory tract of humans and that the lungs act as a site of accumulation of micro- and nanoplastics in people [54]. Polystyrene-derived nanoplastics promote pulmonary cytotoxicity, which worsens respiratory symptoms [55].

Microplastics may cross human tissues; however, there is no clear proof for this. However, oral medication formulations, including polyvinylchloride and polystyrene particulates (<150 µm), may penetrate the gut tissue through the endocytic process, which is then discovered in the nearby circulatory system and lymph tissue [6]. Blood cells can stick to plastic particles, producing a protein ring that makes it easier for particles to be taken up [6]. Studies on nanoplastics and microplastics have shown that these particles can change the biological composition and structure of hepatocytes, gut cells, macrophages, epithelial cells, and lung cells [6, 52, 53, 56, 57]. Following oral administration of microplastics, these particles were localized in the intestines and deposited in the spleen, liver, testis, and kidneys [58, 59]. The liver, placenta, brain, spleen, and kidney all contain these particles after intravenous administration of nanoplastics directly into the bloodstream [60]. Concerns over female fertility and harmful consequences on the growing embryo would arise

if nanoplastics are found in the placenta. First-generation (F1) children experienced significant changes in blood metabolite and lipid profiles because of maternal exposure to polystyrene microplastics ranging from size of 0.5 to 50 µm [43]. Fatty acid metabolic problems may be linked to metabolic alterations. Additionally, sex-specific variations in lipid profiles and blood metabolites have been observed in the offspring [61]. A different study that used microplastics derived from polyethylene as a source, found that pup body weight and lymphocytes were affected [62]. It has been demonstrated that populations of the gut microbiota are altered by polystyrene microplastics, which are linked to reduced mucin secretion along with the expression of genes related to mucin [61]. Studies have suggested that various intestinal microbial communities may play a role in the energy balance, which may alter body weight and problems linked to obesity. Gut microbiota affects many aspects of human health. Blood pressure control, management of chronic kidney disorders, prevention of cardiovascular disease, and boosting immunological function are further health advantages [63]. In recent years, micro- and nanoplastics have emerged as a very harmful threat to all organisms dwelling in various ecosystems, such as marine, soil, and terrestrial ecosystems. These particles are so small that they can travel within various cells and tissues in the human body, thus inducing toxic effects.

12.5 Impacts of Micro- and Nanoplastics on the Genome of Humans

12.5.1 DNA Damage

Microplastics can damage DNA in cells, leading to mutations that can have negative effects on human health. When microplastics are ingested, they can enter cells and interfere with the normal functioning of DNA, potentially leading to breaks or alterations in the DNA sequence. These changes can affect gene expression and potentially lead to the development of diseases, such as cancer. Additionally, some microplastics contain chemicals that can damage DNA via oxidative stress or other mechanisms. Figure 12.4 shows the cellular interaction of microplastics and their action on biomolecules (DNA and proteins). Overall, microplastic DNA damage is a serious concern for human health and underscores the importance of reducing exposure to these pollutants [16].

12.5.2 Epigenetic Changes

Microplastics have been found to induce epigenetic changes, which are alterations in gene expression that are not caused by changes in the DNA sequence. Epigenetic changes can be caused by a variety of environmental factors, including exposure to pollutants such as microplastics. Some studies have suggested that exposure to microplastics can lead to changes in DNA methylation, a process that can affect gene expression. Other epigenetic changes linked to microplastic exposure include alterations in histone modifications, which can affect the way DNA is packaged within cells, and changes in microRNA expression, which can regulate gene expression post-transcriptionally. Overall, epigenetic changes caused by

microplastics are a cause for concern, as they can have lasting effects on gene expression and potentially lead to adverse health outcomes [38].

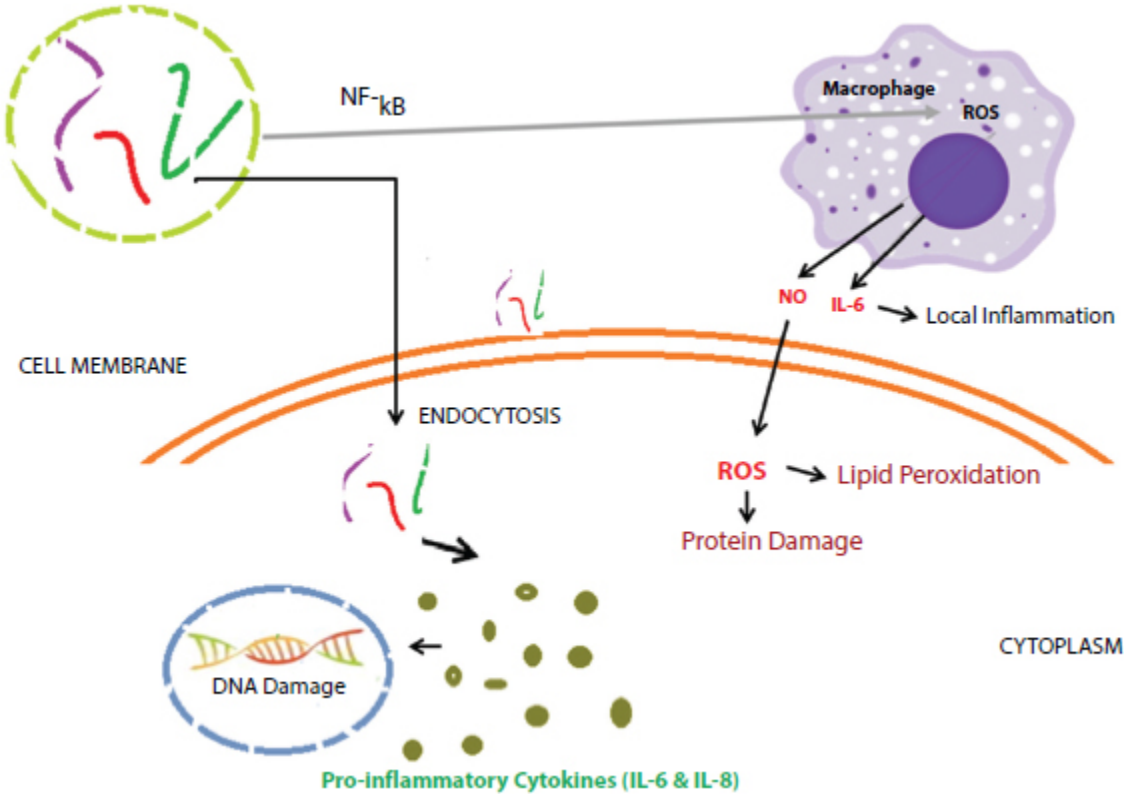


Figure 12.4 Interaction of microplastics (MPs) with cellular biomolecules (DNA and protein) and their effect on cellular biochemical processes

(adapted from Tagorti and Kaya, 2022).

12.5.3 Inflammation

Exposure to microplastics can have a range of effects on the body, including triggering an inflammatory response.

Studies have found that exposure to microplastics can trigger inflammation in various tissues, including the lungs, liver, and intestines. One mechanism by which microplastics may cause inflammation is the activation of the innate immune system. The innate immune system is the first line of defense against pathogens and is

responsible for initiating an inflammatory response. When microplastics enter the body, they activate the innate immune system, triggering an inflammatory response.

Another mechanism by which microplastics may cause inflammation is the generation of reactive oxygen species (ROS). ROS are highly reactive chemicals that can damage tissues and cells and cause inflammation. Microplastics can generate ROS through various mechanisms, including binding to and activating immune cells known as macrophages.

Inflammation can affect gene expression in several manners. For example, it can lead to changes in DNA methylation, a process that affects gene expression. Inflammation can also activate signaling pathways that lead to changes in gene expression, particularly in genes related to immune function and oxidative stress.

Overall, the link between microplastic exposure, inflammation, and gene expression is an area of active research [51].

12.5.4 Reproductive Health

Microplastics have been found to have a negative impact on reproductive health by affecting hormone levels and disrupting reproductive processes in both males and females. In males, exposure to microplastics can lead to reduced sperm count, altered sperm morphology, and decreased fertility. In females, exposure to microplastics can lead to disrupted ovarian development and menstrual cycles as well as an increased risk of miscarriage and birth defects. Microplastics have been found to disrupt the endocrine system, which is responsible for producing and regulating hormones in the body. Specifically, microplastics can interfere with the function of certain hormones such as estrogen, androgen, and thyroid hormones. This can lead to

hormonal imbalances, which can cause a variety of health problems including reproductive issues, obesity, diabetes, and even certain types of cancer. In addition to disrupting hormone levels, microplastics can mimic the action of hormones, leading to abnormal growth and development of reproductive organs in both males and females. The exact mechanisms by which microplastics disrupt the endocrine system are still being studied, but evidence suggests that they can have significant impacts on human health and the environment [64, 65, 73].

12.5.5 Neurological Effects

Microplastics have been found to affect brain function and development by crossing the blood–brain barrier and interacting with neurons. This can lead to neurological disorders, such as autism, ADHD, and anxiety. The small size and unique surface properties of microplastics make them particularly capable of penetrating biological barriers and interacting with cellular components, including those in the brain. Once inside the brain, microplastics can trigger inflammation, oxidative stress, and other cellular processes that can damage neurons and impair cognitive function. The long-term effects of microplastic exposure on the brain are still being studied, but evidence suggests that they can have significant impacts on human health and development. To mitigate the potential impacts of microplastics on health and environment, it is important to reduce the use of single-use plastics and properly dispose of plastic waste [66].

12.6 Toxic Effects of Micro- and Nanoplastics

Toxicity refers to the presence of chemical substances that disrupt normal bodily functions. It has been observed that

high concentrations of nano- and microplastics can have toxic effects. At extremely high concentrations, micro- and nanoplastics can cause cell death due to necrotic plasma membrane rupture, as well as hinder cell signaling and disrupt extracellular and surface structures, such as proteoglycans. Endocytosed nanoplastics are particularly problematic as they can be released into the cytosol and interact with different organelles, such as the mitochondria and nucleus. Some studies have shown that nanoplastics can also disrupt cell division by disrupting mitotic spindle formation [67]. Even if they end up in the lysosome, they are not easily digested and can reduce the capability of the organelle to degrade certain substances. Microplastics are known to exhibit autophagic characteristics within their organelles. Different blood proteins, such as albumin, form protein–plastic complexes; if these compounds exceed a specific limit the blood vessels can be blocked [11]. The most likely reason for the toxic effect of plastic particles is the leachable additives, which are used to enhance properties such as softness and UV stabilization. The most widely used plastic additives include phthalates, organotins, and bisphenols. The actual theory behind the movement of additives inside the polymer lies in a correlation between these two; as the additives lie inside the three-dimensional porous structure of the plastic polymer, the lower the size of the additive would be its capability to move across the polymer, and temperature and pH also play a crucial role here. Owing to the hydrophobicity and large surface area of plastic polymers, it is easier for organic pollutants to attach to their surface [24]. There could be two types of attachments on the surface of the nanoplastic polymer: absorption and adsorption [72, 74].

1. Absorption is defined by the penetration of the substance inside the system, which involves weak

forces such as Van der Waal’s force, as seen in the case of Polycyclic Aromatic Hydrocarbons.

2. Adsorption is the strong attachment of hydrophobic substances at the surface of a polymer, which is much stronger than that of absorption. Polyvinylchloride can be described as an example of this case.

12.6.1 Several Toxic Effects of Nano- and Microplastics

i. Chemical contaminants, such as persistent organic pollutants (POPs) and heavy metals, can be carried or stored in microplastics. These toxins may increase the toxicity of microplastics by adhering to their surfaces [68]. Microplastics can be consumed by organisms and released into the environment, where they can enter the food chain and accumulate at higher trophic levels.

ii. Physical harm and blockage: Microplastics can physically harm organisms by abrasion, lacerations, or blockage of the digestive system. Microplastics can hinder growth, cause energy depletion, and restrict feeding in marine species, which can ultimately influence their survival and ability to reproduce.

iii. Feeding and behavioral disruption: Microplastics can affect how different organisms feed. For instance, filter-feeding species such as mussels and zooplankton might mistake microplastics for food and eat them, which would impair their ability to feed effectively and absorb nutrients [69]. Malnutrition and reduction in general fitness can result from this.

12.6.2 Effects on the Immune System and Inflammatory Responses

Research has indicated that microplastics may cause immune system reactions and inflammatory responses in several organisms. Oxidative stress, DNA damage, and physiological disturbances are all effects of inflammation and immune system activation, which may influence an organism's general health and ability to survive [34].

iv. Transfer across the food chain: Microplastics can move up the food chain to higher trophic levels after being consumed by small creatures. The cumulative effects of microplastic exposure may be felt by predatory creatures that eat plastic-tainted prey, thereby magnifying toxicity and negative effects.

12.7 Case Study on a Small Regional Place of India (Puducherry)

A small district called Puducherry is in the Union territory, Pondicherry, which is in India. The fishing community in this area depends on the Pondicherry coast as its primary source of income. The shoreline measures 23.94 km. Thirteen fishing communities can be found along this stretch of shore, and fishing is the main source of income for these communities. In Puducherry, the selected beaches were six in total [70]. Six beaches were sampled from a high tideline. High tideline was marked by late detritus and shells. Depending on the length of the beaches, samples were taken from six different locations: Pondicherry University, Paradise Beaches, Veerampattinam, and Serenity [70]. Approximately 24 h were spent in the oven to dry the bulk sample at 343.15 K [76]. Following the removal of the bulk, a 100 g sample was sieved twice, once through a 5 mm sieve and once through a 300 m sieve. The

particles that remained after passing through the 300 µm sieve were then moved to a glass container and packed in aluminum foil to prevent damage [70]. One liter of distilled water was used to dissolve 360 g of sodium chloride to create a supersaturated NaCl solution. Each beaker containing 100 g of the material was added to 300 ml of a solution containing saturated salt. The collected material was then agitated for 2 min with a glass rod to remove any potential microplastic particles that may have become stuck among the sand grains. The solution was then allowed to stand for 24 h [70]. The density separation supernatant was carefully transferred to a beaker and screened using a pump that produces vacuum and filtering equipment [71]. The microplastics were numbered and documented based on their color and shape. The study's findings were then used to conclude that fishing and leisure activities are the two factors that best explain the prevalence of microplastics on Puducherry's beaches.

12.8 Conclusion

Concerns over the potential negative effects of micro- and nanoplastic particles on human health have risen in recent years. The information that is now available suggests that micro- and nanoplastic pollution may have negative consequences on human health; however, research in this field is still in its early stages, and additional research is required to completely understand the impact. The ingestion of these micro- and nanoplastics can occur through food, water, or breathing. Microplastics can accumulate in several organs, including the lungs, liver, and digestive system. These particles can accumulate and cause oxidative stress, tissue and organ damage, and inflammation. According to a growing body of evidence, nanoplastics and microplastics are detrimental to human

health. These tiny particles, which can enter the body through a variety of sources, can accumulate in organs and cause tissue damage, oxidative stress, and inflammation. Additionally, the danger of endocrine disruption, carcinogenicity, and developmental abnormalities is increased by the release of hazardous compounds and additives from these particles as well as by their capacity to adsorb and concentrate environmental pollutants. Micro- and nanoplastic disturbances of the gut microbiota raise additional concerns because they can lead to gastrointestinal diseases, metabolic problems, and immunological dysregulation. The capacity of these particles to transcend biological boundaries prompts concerns about systemic effects, which could have an impact on growing fetuses and other key organs, including the brain. However, it is important to recognize that the study of micro- and nanoplastics is still in its early stages, and there are many questions about how they will affect people's health. To better understand the long-term impacts, dose–response connections, and potential interactions with other environmental stressors, more research is required. Additionally, to reduce the release of micro- and nanoplastics into the environment, efforts should be made to reduce plastic pollution, increase recycling, and implement sustainable practices. Policymakers, businesses, and individuals must act quickly to address micro- and nanoplastic pollution, considering the mounting evidence and possible consequences associated with this problem. Strict regulations, public awareness campaigns, and creative solutions for plastic waste disposal are crucial for protecting human health and the environment from the negative impacts of micro- and nanoplastics. We can work towards a cleaner, healthier future for future generations if we collaboratively address this issue.

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Note

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13
Harmful Effects of Plastics, Microplastics, and Nanoplastics

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Abstract

Plastics were invented to replace the existing source of natural reserves of metals and because of their advantageous properties such as flexibility, light weight, compressibility, ease of portability, low cost, and moldability. However, in subsequent years, it was observed that these plastics are becoming a major threat to the environment. The reason for their threat is the disposal of plastics as waste after their usage in electronics, food packaging, transportation, clothing, sanitary purposes, gift wrappers, birthday decorations, and toy materials. In the present article, plastics, microplastics, and nanoplastics and their harmful effects are discussed. Plastics that are further degraded by various agents, such as microorganisms or ultraviolet radiation, may produce microplastics, while further degradation produces nanoplastics. Although plastics may not affect the health of organisms, microplastics (MPs) and nanoplastics (NPs) affect the nature, and in turn, destroy the health of living organisms. Therefore, the use of plastics to produce nanoplastics and microplastics should be minimized. In this chapter, the generation of MPs and NPs is discussed,

together with the techniques for MP and NP measurement. Various processes for the degradation of plastics are briefly discussed. As MPs and NPs are potential threats to the environment, their harmful effects of MPs and NPs have been discussed.

Keywords: Plastics, microplastics, nanoplastics, human health, microorganisms, aquatic organisms

13.1 Introduction

Plastics are typically made of artificial polymers that are not created by nature. They are also known as man-made polymers, because their synthesis is man-made. Plastics can damage the environment and living organism in their surroundings. These are not naturally generated and are not fully biodegradable. In addition to harming land-based life, plastics have negative effects on marine life, terrestrial animals, and aerial birds. In the past, no plastic remnants were discovered during investigations of sediments, glaciers, or beneath oceans. However, the amount of plastic in air, water, and some places is now enormous [1]. Plastics are widely employed in almost every industry, particularly in the packaging of cooked food and raw food products, infrastructure, cosmetics, agriculture, and transportation. Compared to other naturally abundant materials, plastics are more durable, corrosion-free, ductile, lightweight, transportable, and inexpensive [2]. Various materials are available for plastics. Plastics have no rust, making them ideal for use in construction and infrastructure. Although the cost of producing plastics is minimal, the implications of their use are destructive as they are harmful to the environment. Polymers are difficult to destroy; however, they can be converted into microplastics or nanoplastics after undergoing through numerous physical, chemical, and biological processes [3].

Microplastics (MPs) have dimensions of less than 1 mm and nanoplastics (NPs) have dimensions of less than 100 nm. Microplastics and nanoplastics are primary sources of environmental risks associated with plastics [4]. The negative consequences of MPs and NPs have been explored using various physical, analytical, chemical, and biological approaches. MP remnants have been discovered by researchers in coastal regions, oceans, equatorial seas, and north and south poles.

13.2 Generation of MPs and NPs

COVID-19, a pandemic, has spread the use of plastics and plastic waste in the environment. This is due to the use of surgical masks, hand gloves, and personal protective equipment (PPE) and their disposal in large amounts results in the generation of plastic waste and hence MPs and NPs [5]. Ultraviolet rays are one of the factors responsible for the generation and photodegradation of microplastics (MPs). Mechanical agitation and UV irradiation can generate MPs and NPs *via* hydrolytic and photodegradation processes, respectively. This was also observed for low-density polyethylene [6]. Dissolving organic matter in water promotes the photoaging of aliphatic MPs/NPs while stopping or having a minor effect on the photoaging of aromatic MPs and NPs [7]. Fragmentation of MPs occurs from marine debris of polyethylene and polypropylene, and NPs are produced from MPs through irradiation [8]. MPs can form sub-micron-scale secondary MPs through mechanical, ultraviolet, ozone, and thermal degradation processes. Processes such as thermal drying of sludge and lime stabilization have been found to lead to the fragmentation of MPs, which increases the specific surface area and number of MPs, leading to an increase in the potential

ecological threat posed by MPs in the sludge [9]. Microplastics are generated by the opening and closing of water bottles [10]. Figure 13.1 shows the factors that may lead to the formation of MPs. MPs can be generated by scissoring with scissors, tearing with hands, cutting with knives, or manually twisting to open plastic containers/bags/tapes/caps. These processes can generate approximately 0.46–250 microplastics/cm [11]. Weathering degradation of plastics on beaches results in surface embrittlement and microcracking, yielding microparticles that are carried into water by wind or wave action [12]. Mechanical stress caused due to wind or sand can fragment the plastic like plastic water bottle or plastic bags etc. to produce MPs. This was observed through simulations in high-density polyethylene [13]. Hydrolysis is also responsible for the generation of MPs, as observed through micro-FTIR analysis. In polycarbonate films, degradation occurs due to hydrolysis as indicated by the cleavage of the C−O bond from the carbonate group of the polycarbonate backbone and detachment of micro-sized plastic particles from the PC films [14].

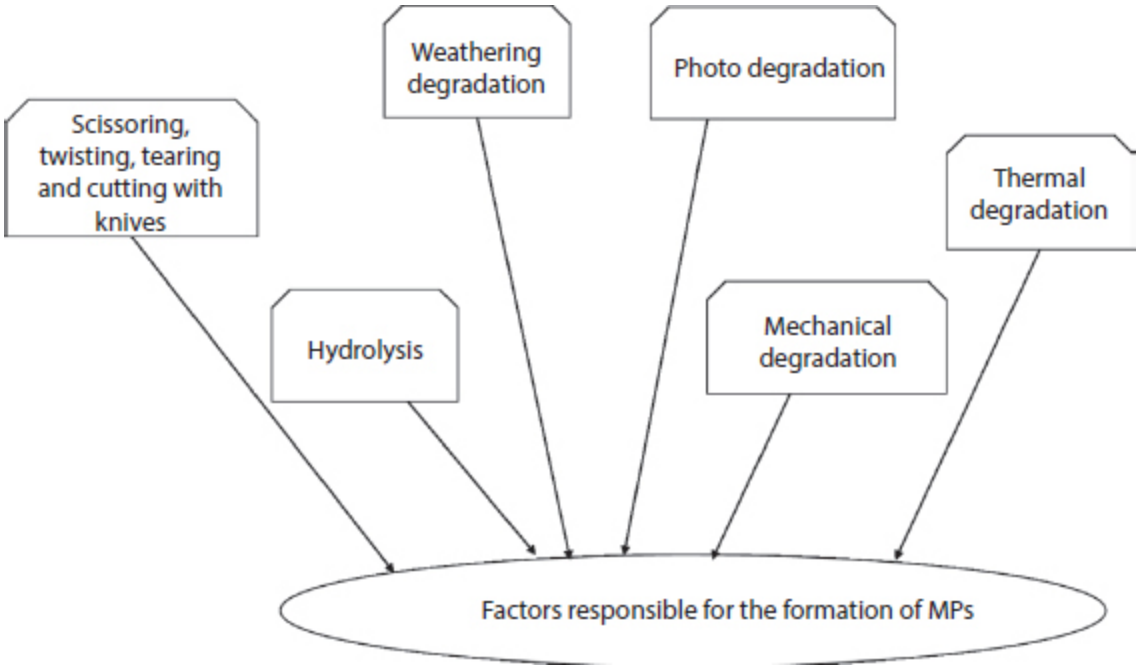


Figure 13.1 Factors responsible for the formation of MPs.

13.3 Techniques for MP and NP Measurement

Approaches recommended by the NOAA have been used most frequently for MP and NP measurements [15]. It includes sampling, preservation, digestion, and identification in various types of environments such as water, air, soil, and microorganisms [16]. The separation of MPs and NPs in different environments such as air, water, soil, and microbes, and analytical steps generally involve sample collection, processing, and filtration, followed by physical and chemical methods for characterization.

13.3.1 Sampling

Trawls or neuston sets are generally used for water sample collection. Stainless steel is generally used in sediment soil and sludge. Atmospheric wet and dry depositions and

pumps were used to collect the samples in air. Planktonic nets are used for aquatic organisms.

13.3.2 Flotation

Flotation is an effective method for separating MPs from the collected samples. Sodium chloride and sodium iodide were used to separate low- and high-density plastics, respectively.

In complex samples, digestion is essential for the separation of MPs. Most commonly, 30% hydrogen peroxide solution is used for digestion, and potassium hydroxide is used mainly for biological samples [17], and Fenton's reagent for sediments [18]. Digestion can be acidic, alkaline, or enzymatic, depending on the digestion solution used.

13.3.3 Filtration

After flotation, filtration was performed using a membrane made of nylon, cellulose nitrate, and polycarbonate [19].

13.3.4 Identification of MPs and NPs in Different Environmental Conditions

Scanning electron microscopy (SEM), visual and optical microscopy, Raman mapping, and spectroscopy have been used to identify MPs and NPs in different environments [20–22]. Visual microscopy can be used to study the color, shape, or size. SEM can be used to obtain high-resolution images of surface structures. SEM images can be used to study changes in the microscopic structure before and after the aging/degradation of MPs. Micro—FTIR or ATR—FTIR was used to identify the composition of the samples by analyzing the functional groups. A support vector machine is used for steady detection of various types of plastics [23, 24]. An optical sensor was also used to detect transparent

and translucent MPs in water. Gas chromatography–mass spectrometry (GC–MS) and liquid chromatography–mass spectrometry (LC–MS) are recommended, in addition to visual and spectral verification [24, 25]. Surface enhanced Raman spectroscopy, pyrolysis gas chromatography, and mass spectroscopy are some of the most efficient methods for identifying NPs in a sample [26–28]. The efficiency and accuracy of the detection need to be further explored and improved. IR microscopy is an important tool for identifying MPs and NPs [29]. Atomic form microscopy is also an efficient method for detecting MPs and NPs [30]. Microscopy and thermal analysis were used to identify the samples [31]. Electrochemical techniques have been used to identify and removes plastic particulates from water [32].

Rapid advancements in infrastructure have led to an increase in the number of micro/nanoplastics in the environment. Its production, threshold level, toxicity, and harmful effects have been explored by researchers. Table 13.1 shows the quantity of MPs taken from various areas such as coastal waters of China, sediments of Chang Jiang Estuary, China, Fuhe River Estuary sediments in Northern China, Bayhead sediments of Durban Harbour, South Africa, seawater, and sediments of the Marina trench.

Table 13.1 The quantity of MPs taken from various areas.

Area of sample taken	MPs	Reference
Coastal Waters of China	0.13–545 pieces/m^3	[33]
Sediments of ChangJiang Estuary, China	20–340 pieces/m^3	[34]
Fuhe River Estuary sediments in Northern China	1,040 ± 462 item/kg	[35]
Bayhead sediments of Durban Harbour, South Africa	745.4 ± 129.7 particles per 500 ml	[36]
Seawater at a depth of 2,673–10,908 m in the Marina trench	2.06–13.51 pieces/L	[37]
Sediment at a depth of 5,108–10,908 m in the Mariana Trench	200–2,200 pieces/kg	[37]

The presence of MPs in the ocean is mainly due to land, and the fresh water of the river enters the ocean, which is adulterated due to landfilled degradable MPs.

Sequential isolation of MPs and NPs by membrane filtration and cloud point extraction was performed using polystyrene (PS) and polymethymethacrylate (PMMA). The trapped MPs together with the glass membranes were above 90%, and in filtrate, NPs were approximately 90%. The collected MPs together with the glass membrane were frozen in liquid nitrogen, ground, suspended in water, and subjected to pyrolysis-gas chromatography-mass spectrometry (Py-GC/MS). The NPs in the filtrate were concentrated by cloud-point extraction, heated at 190 °C to

degrade the extractant, and then analyzed by Py-GC/MS [38].

13.4 Various Methods for the Degradation of Plastics

Plastics based on polymers such as polyethylene, polypropylene, and polyethylene terephthalate (PET) degrade through photo, thermal, and biodegradation mechanisms [39].

13.4.1 Chemical Degradation

This refers to degradation at the molecular level, such as breaking of bond, or oxidation of long polymer chains to create new molecules, usually with significantly shorter chain lengths [40]. Hydrolysis breaks MPs into tinier particles [41].

13.4.2 Thermal Degradation

It is an unnatural and commercial method for degrading plastics into smaller pieces under the effect of temperature. The thermal decomposition behavior of plastics can be investigated using dynamic thermogravimetric analysis [42].

13.4.3 Physical Degradation

Physical degradation can be due to factors such as weather, climatic conditions, marine conditions, and stimulated land fill conditions [43–45].

13.4.4 Biodegradation

Bacteria present in the environment can assist bacterial biodegradation [46]. Similarly, fungal-assisted

biodegradation, algae-assisted biodegradation, integrated enzymatic biodegradation, and combined biodegradation are various types of biodegradation [47–49]. This biodegradation of plastics, especially MPs, leads to a large production of NPs, which can affect the environment and, in turn, the lives of living beings. *In situ* experiments on polypropylene and polyethylene have been performed in marine water and studied using Pseudomonas spp., Clostridium spp., anaerobic bacteria, and fungi, showing that the degradation depends on the material site, material type, and weather conditions [50]. Climatic conditions are the main factors that affect the growth of microorganisms. The degradation of MPs occurs through microbes such as *Bacillus* strains [51]. *Bacillus sphericus* has been used for marine microbe-mediated biodegradation [52].

13.4.5 Photodegradation and Photoaging

The photodegradation or photoaging of MPs affects the alteration and interaction of pollutants in water and soil [53, 54]. Photoaged MPs are a potential hazard to the environment. MPs and NPs are also produced in the process of photoaging. The effect of photoaging on the degradation of MPs is of great deal concern. Photoaging leads to an increase in the surface area of MPs and more oxygenated functional groups, which further leads to more adsorption of heavy metals on their surface [55]. The adsorption of heavy metals is greater in photoaged MPs than in those without photoaging MPs. Polymers such as photoaged PP and PS affect the enzymatic activity of microbes in soil [56]. The organic matter leached from the PS after UV treatment increased its toxicity to the bacteria. PVC aged after UV irradiation inhibits the growth rate of freshwater algae [57]. The presence of inorganic substances, such as iron oxide and iron red pigment, in the natural environment also inhibits the photoaging of MPs

[58]. Different types of water environments also affect the photoaging of MPs, depending on their salinity and ions [59]. A high salt concentration implies many ions, a higher refractive index, and low absorption of light, and thereby inhibiting the photodegradation of MPs [60]. The sulfide present in the environment promotes oxidation of the surface groups of the polymer and results in chain breaking of the polymer [61]. Low-molecular-weight organic substances such as anthraquinone, benzoyl peroxide, and benzoyl phenol increase the rate of photodegradation and oxidation of polymers such as polystyrene [62].

13.5 Harmful Effects of Plastics, Microplastics, and Nanoplastics

Plastics are made of natural polymers *via* polymerization; however, the reversible process is not possible. If these plastics are disposed of, they break down into a huge amount of microplastics and nanoplastics. Nanoplastics are more harmful because of their small size, which allows them to permeate biological membranes. The primary sources of MPs and NPs are consumer and industrial uses. Macroplastic products that disintegrate into micron-sized smaller particles are a secondary source of micro- and nanoplastics in both terrestrial and aquatic environments. The traces of microplastics observed were pellets, fragments, film, fiber, foamed plastic, etc. Both MPs and NPs are present in terrestrial as well as aquatic environments, through which they enter the human body and adversely affect human health.

Plastics, microplastics, and nanoplastics are all synthetic polymers manufactured for the profitability of human lives. As they do not exist in nature, their resolution or decomposition into organic matter is not easy for living organisms. After use, these plastics are completely

hazardous to the environment. Depending on their size, plastics can be broken down into microplastics and nanoplastics. These plastics are not only hazardous to the environment but also to living beings associated them, including wild animals, domestic animals, reptiles, amphibians, aerial birds, sea animals, microorganisms, and humans. Therefore, the creators (humans) of plastics ultimately suffer. In the following section, the harmful effects of plastics on various classes of organisms are discussed.

MPs and NPs can enter the body through three ways of ingestion, inhalation, and skin contact. MPs are mostly generated from industries or through the physical, chemical, or biological degradation of large plastic wastes [11]. Daphnid lifespan was decreased by exposure to high concentrations of micro plastics (10^2 to 10^5 particles mL^{-1}), which led to extinction within one and four generations, respectively. Microplastics also affect reproduction and growth [63]. The concentrations of microplastic particles inside the air were found to be larger than those outside the building.

Not only humans but also animals are at greater risk as they consume plastics from the environment, mostly due to the consumption of food from the environment and leaching of chemicals from plastic packaging of food and drinks. Microplastic and nanoplastic fragments have been detected in honey, beer, salt, sugar, fish, shrimps, and bivalves. Fourier-transform infrared spectroscopy (FTIR) studies have shown that MPs are present in tap, bottled, and spring water sources [64].

13.5.1 Effect on Gastric Health

Analysis of human stool samples indicated that humans frequently consume MPs and NPs [65]. Therefore, it is

necessary to investigate the movement of these MPs and NPs through the gut epithelia and lumen. Because the functional size of tight junction holes may vary to a maximum of 1.5 nm, MPs cannot penetrate at this level. It is possible that MPs and NPs can enter the lymphatic tissue and infiltrate the microfold (M) cells in Peyer's patches through phagocytosis or endocytosis [66].

Because of the interaction of nanoparticles with other substances, it is important to understand their effect in the gastrointestinal (GI) tract. It is found that the NPs interact differently with proteins, lipids, carbohydrates, nucleic acids, ions, and water [67]. The nanoparticles change after being eaten, which affects their absorption capacity and rates.

13.5.2 Effect on Pulmonary Exposure

MPs and NPs effect the health of the pulmonary parts of the body. Some researchers have shown that the ingestion of MPs and NPs can affect health through their toxins and direct inhalation of NPs. The alveolar surface area of the lungs is substantial (approximately 150 m^2) and the tissue barrier is very thin (less than 1 m). Nanoparticles can be distributed throughout the human body because this barrier is permeable enough to allow them to enter the capillary blood system [68].

The ingestion of plastic particles, especially micro- and nanoplastics, raises several adverse health issues, including particle toxicity and chemical toxicity, and can restore infection and parasite vectors. The range or size of the particles determines their penetration inside the lungs, alveolar surface, or other body regions that can damage the organs. Various parameters such as hydrophobicity, surface charge, surface functionalization, surrounding protein coronas, and particle size influence the absorption

and expulsion of microplastics and nanoplastics in the lungs [69]. Absorption of MPs and NPs can also increase the rate of pulmonary cancer and inflammation.

13.5.3 Effect on Derma

The health and beauty industry is one of the creators of nanoplastics. In this way, nanoparticles of plastics come into direct contact with the skin. The penetration of NPs into the outer or inner layers of the skin is unclear. It is heard or some of us have experienced that the longterm use of beauty products can harm the skin. The outermost layer of the skin, known as the startum coreneum, protects the skin from external agents, especially harmful agents such as microorganisms and toxins, and is improbable for the plastic particles to enter through the stratum corneum; however, they can enter through sweat ducts, skin wounds, or hair follicles as MPs and NPs are hydrophobic. NPs smaller than 20 nm can penetrate the top layer of the skin or hair follicles [70]. In face washes with microbeads, NPs with diameters of approximately 20–50 nm are also present. UV radiation can damage the skin layer and allow the penetration of NPs into the skin. In conclusion, the investigations stated above, both *in vitro* and *in vivo*, have proven that micro- and nanoplastics can enter the human body through the skin barrier.

13.5.4 Harmful Effects on Human Health

Plastics such as polycarbonates are made of a monomer called bisphenol A (BPA). This monomer has harmful effects on human health, including damage, sperm decrement, type-2 diabetes, cardiovascular disorders, changes in the immune system, increased obesity, menstrual pain, infertility, cancer pain, and recurrent miscarriages [71]. Di (2-ethylhexyl) phthalate (DEHP) is also produced in millions of tons every year, and these

phthalate groups can enter the body through medical means such as dialysis, ingestion of contaminated materials, and house dust. It was observed in markets that on plastic containers, especially for babies' written BPA-free, BPA is a harmful component for human health. As babies have a poor immune system, their milk bottles or water sippers should be free of BPA. Leachates from the polyurethrane, polyvinylchloride, and PVS are toxic to the environment [72]. Through oral ingestion, the gut mucosa acts as an important barrier for harmful particles to enter the body. The persorption of particles through the gut entering the blood vessels is detectable and can directly pass through the urine [73].

13.5.5 Harmful Effects on Microorganisms

Microplastics are affected by microorganisms and vice versa. Microorganisms have both positive and negative effects on plastics. Microplastics can affect microorganisms, as well as the gut microbiota, which ultimately affects host health. Microplastics also provide a medium or substrate for the growth of microorganisms. Microorganisms can form biofilms that interact with the microplastic particles [74, 75]. Temperature, humidity, and habitat are the factors that primarily affect biofilm formation by microorganisms. Low quantities of microplastics do not affect the growth of soil microorganisms, whereas their high content, especially of microglass, affects the growth of protozoa [76].

13.5.6 Harmful Effects on Sea Animals

A large amount of plastic waste is produced in every year and every day and enters the marine environment, such as rivers, lakes, ponds, sea, and oceans. When exposed to ultraviolet light, these plastics break down into smaller pieces, such as microplastics, and then further into

nanoplastics. The content of microplastics in urban areas and the aquatic animals in urban areas were found to have more microplastics affecting life. These microplastics attract dyes, chemicals, and heavy metals on their surfaces and can enter the bodies of aquatic organisms, ultimately affecting their lives and marine environment [77]. Microplastics affect the gut microbiota of aquatic organisms, which depends on biofilm formation and the overall effect of chemicals [78]. A large amount of plastic usage would ultimately make this planet a plastic planet. Plastics have various adverse effects on aquatic organisms as they attract hydrophobic pollutants and unwanted unnatural products entering the body, ultimately affecting aquatic organisms [79]. The effects of microplastics have been investigated on zooplankton. Zooplankton ingest microplastics in them and these when consumed by other organisms can enter the food chain and can transfer to trophic level too [80]. During the production of plastics, various chemicals such as plasticizers UV inhibitors are used, which can leach out of the plastic and affect the marine environment [81]. Microplastics and polyacryclic hydrocarbons pose a serious threat on hematological parameters. It can also elevate the intracellular contents of reactive oxygen species (ROS), resulting in lipid peroxidation (LPO) and DNA damage [82]. It has also been observed that microplastics affect gametogenesis in oysters, which may impact offspring and adult fertility [83]. Therefore, these microplastics are also responsible for infertility in aquatic organisms [84]. The polymers found in water resources are mostly polypropylene, polystyrene and polyethylene [85]. Approximately 80% of the polymer (polyethylene and polypropylene) was found in the Baltic Sea. This content is higher in the European coastal regions [86, 87]. Microplastics are also found in seabirds, such as albatross, fulmars, and penguins that are upper-level trophic predators that also affect wildlife [88].

13.5.7 Harmful Effects on Animals

Polymers such as polvinylchloride and polyurethane leach chemicals that are harmful chemicals to the environment. Polyethylene releases a higher amount of chemicals into the environment. Sea birds and birds are the wild animals most affected by macroplastics [89]. Various microscopic techniques, such as Raman microscopy, FTIR microscopy, optical microscopy, and electron microscopy have been used to study samples from wild animals [90]. Approximately 84% of the fibers have been found in dead birds in northeastern Poland [91]. Microplastics affect domestic animals, such as chickens, mice, buffaloes, cows, and goats, and they widely affect the plastic and microplastic content of the land. Microplastics can enter the gastrointestinal tract and affect the metabolic activities of animals. Gas chromatography–mass spectrometry can be used to detect low levels of targeted plasticizers.

13.6 Measures to Avoid the Further Extension of Harmful Effects of Plastics, Microplastics, and Nanoplastics

To prevent marine litter, waste recycling, waste reuse, conversion of waste to energy, biodegradable or eco-polymers, gear marking, filtration at the entry of waste into the water, and waste management activities at the land scale. Waste can be dumped from a very large distance from oceans or water containing sources, such as sea, rivers, lakes, or beaches. Nonharmful wastes can be passed on to the marine world, beach cleaning tasks, fishing retrieval programs, and marine debris monitoring. Educational awareness campaigns can be conducted to avoid disposal and limit the use of plastics. Figure 13.2

shows the replacement of degradable plastics with degradable plastics or eco-friendly resources. The figure shows that plastics in various fields can be replaced with other low-cost, hygienic, and biodegradable materials. For example, plastic water bottles can be replaced by steel or metallic water bottles. Disposable plastic crockery is not only an agent of microplastic and nanoplastic production; however, it produces unhygienic food for human health. Therefore, it is essential to replace them with disposable completely biodegradable crockery. Most of the times carry bags used are plastics that can be replaced with paper or clothes bags, as they are easily biodegradable. Plastic food packaging is not harmful for later issues, such as disposal as waste, but also for the instant food is packed in them. Hot food can contaminate food quality and affect human health. Plastic food packaging can be replaced by eco-friendly packaging. The next measure we can take is to minimize or curb the production of such plastic materials, which can be replaced with eco-friendly, biodegradable, reused, or recycled materials.

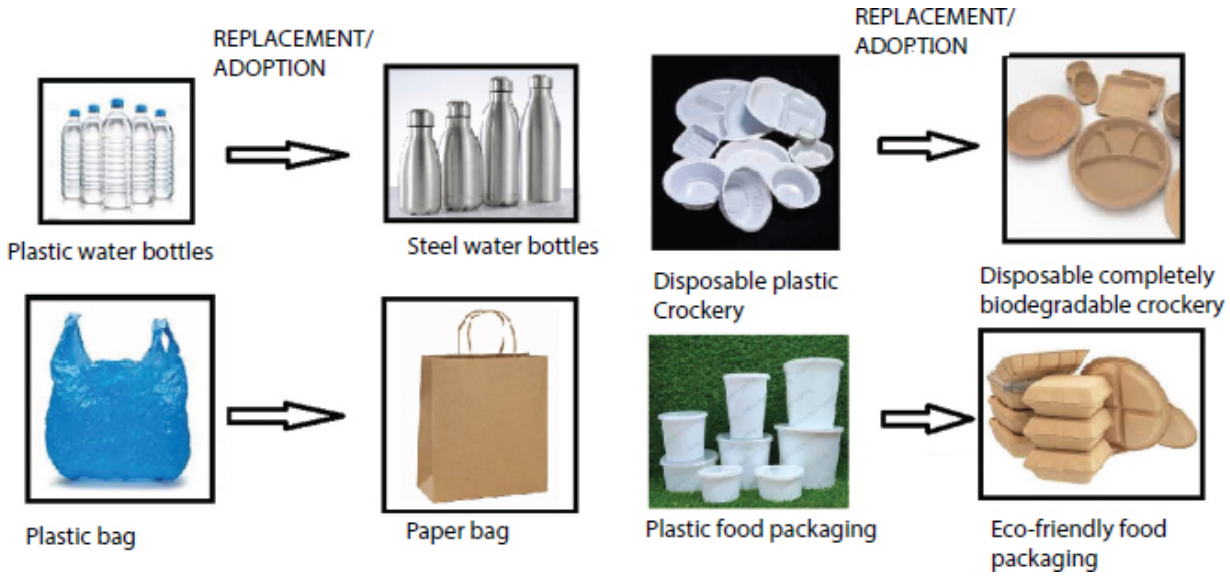


Figure 13.2 Replacement of plastic materials to biodegradable or eco-friendly materials.

13.7 Conclusions

In this chapter, plastics are discussed as potential threats to the environment and living organisms in the environment. Plastics, through various processes such as photodegradation, thermal degradation, biodegradation, mechanical agitation, weathering conditions, and human handling factors, can lead to the formation of MPs and NPs in the environment. The detection methods for MPs and NPs have been studied using sampling, floatation, and filtration. Techniques such as scanning electron microscopy (SEM), optical microscopy, Raman mapping, and atomic force microscopy (AFM) have been used to detect MPs and NPs under different environmental conditions. Chemical, physical, thermal, bio-, and photodegradation methods are the main methods for the formation of MPs and NPs from plastics. The study of the harmful effects of MPs and NPs on gastric health, pulmonary exposure, and derma showed that NPs, owing to their smaller size can penetrate lymphatic tissues in the gastrointestinal tract and penetrate the skin more frequently than MPs. Moreover, the harmful effects of MPs on different organisms such as humans, animals, aquatic organisms, and microorganisms have also been discussed, and it has been shown that direct inhalation of NPs is possible. MPs can enter the body of organisms through the intake of food from plastic waste by animals, or by the lower quality of food packaging plastic can enter the human body and affect human health. Plastics can be replaced by low-cost, recyclable, or biodegradable materials.

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14
Hazardous Effects of Microplastics and Nanoplastics in Marine Environment

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Abstract

Plastics, including microplastics (MPs) and nanoplastics (NPs), cause havoc in all habitats, including marine environments. The low degradation and impossibility of recovering all the plastic particles have led to the concept of considering plastic as an irreversible threat that could ultimately disrupt earth system processes. Although we have clear-cut information on the primary sources of plastics, secondary sources arising from the degradation of primary sources throw very little light on which particular source has a major contribution. Bioremediation and transformation have advanced technological approaches in addition to prevailing methods, such as incineration, recycling, or landfills, to minimize microplastic pollution. Studies have revealed that polyethylene (PET) is successfully degraded by *Brevibacillus borstelensis*, *Rhodococcus rubber*, and *Pseudomonas chlororaphis*. *Pseudomonas putida*, *Ochrobactrum*, and *Pseudomonas fluorescens* break down the polymer chain of polyvinyl chloride (PVC). Studies have shown that, rather than using single microorganisms, the use of consortia isolated from marine habitats will offer greater results in degrading MPs and NPs of marine origin.

Keywords: Microplastics, environment effects, marine pollution, polyethylene, polyvinyl chloride

14.1 Introduction

The term plastics arose from the Greek term “Plastikos.” The basic ingredients in making current day plastics include organic and inorganic substances. Oil, coal, and natural gas are the sources of raw materials required to create plastics [1]. Plastics have had an unavoidable impact on our daily lives since 1950 [2]. Since then, biodegradable items have been supplanted by non-biodegradable plastic materials for several uses [3]. Plastic materials are increasingly being used in the food, apparel, housing, construction, medical, and recreational industries. The global production of plastic garbage is estimated to be approximately 57 million tons per year [4].

The most problematic plastic in any environment is undoubtedly polyethylene (PE), one of the most inert manmade polymers due to its resistance to microbial attack [3]. PE, a thermoplastic polymer composed of lengthy chains, was created by mixing chemical ethylene with other ingredients. When ethylene enters a polymer, it transforms into ethane, and the straight polymer portions have the same structure as simple chain hydrocarbons [5]. The most widely used synthetic polymer, polyethylene, is a very inert and nearly non-biodegradable material whose annual global output is currently approximately140 million tons. Plastic garbage accumulates in the environment when appropriate disposal methods are not available, creating an increasing ecological hazard to both terrestrial and aquatic life [3].

Plastic usage has been increasing due to its importance in aesthetics, financial situations, and the fields of health. Microplastics have been a major form of contamination in different compartments of the environment, such as aquatic, terrestrial, and atmospheric [6, 7]. The size of the particles ranges from 1 µm to 5 mm and is reported in different polymer compositions, types, sizes, and shapes. The source of origin is either primary or secondary. Manufactured pellets

used for feedstock and plastic production [8]. Textiles, personal products such as scrubs for the face or body [9, 10] and medicines fall under primary sources. While secondary sources include the breakdown of larger garbage articles like tires, plastic containers, line fibers, and nets [11]. This breakdown occurs through aging processes and weathering like UV radiation, hydrolysis, microbial degradation, wave and wind abrasion, and hydrolysis. Commonly, these primary microplastics are released into wastewater from industries and household wastewater entering estuaries, rivers, and finally oceans [12], while secondary microplastics are a major source of microplastic contamination in the marine environment. As the accumulation of plastic debris from different origins continues in the aquatic surroundings, there will be an enormous increase in the abundance of plastic pollution [13, 12]. Because of their bioaccumulation and persistence properties in the environment, microplastics are alleged to be detrimental to all forms of life, including humans. On the other hand, microplastics contain toxic substances that are added in industrial processes (e.g., flame retardants, colorants, and plasticizers) or collect other harmful substances during their passage through the environment, including microorganisms, metals, and polyaromatic hydrocarbon compounds [14].

Due to the low degradation and impossibility of recovering all the plastic particles, plastic pollution has been traditionally considered an irreversible threat that could disruption of Earth system processes. Considering all these, certain measures have been passed to reduce plastic pollution, starting from production, manufacture, consumption, and disposal [15]. However, the researchers have come up with biodegradable plastics that are developed from renewable natural resources, which can reduce the use of conservative polymers. Since plastics are designed to last longer, their degradation in the environment will be delayed, and by the time it does, it will affect wildlife to a greater extent [15].

Anyhow, these biopolymers generate microplastics as well [16].

The major source of plastic—around 80% of it—is from land. Improper disposal and poor management of plastics from landfills enter the sewage system and are carried through rivers and other water bodies to the ocean. The other 20% of the pollution in the oceanic surroundings is generated through recreational activities, fishing nets, and litter discharge from boats and ships [17]. It is estimated that around 5.25 trillion micro- and nanoplastics are causing sea surface contamination across the globe [18]. During physical treatment methods such as sedimentation and filtration, microplastics are just separated out and remediated, which then remain in the sludge portion for the duration of the treatment of wastewater in wastewater treatment plants (WWTPs). Furthermore, when the municipal sludge is used as organic fertilizer, the microplastics present in the sludge are released into the ground, making microplastic pollution inevitable [16]. Hence, there is a definite need to have knowledge of microplastic manifestation and dissemination in various environmental sections and the normal description and procedures for abstraction, categorization, and quantification [12, 15].

Since 1970, several studies have been conducted to identify potential bacteria that could breakdown plastics [12], and microbial attack-resistant biodegradable polymers such as natural polymers, polyesters, and polyamides have also been produced [19]. The two most significant pollutants in maritime environments are polyethylene and hydrocarbons [20]. Nearly 170 marine organisms have been recorded to ingest artificial polymers, which can result in hazardous complications for their health such as intestinal impaction and perforation, diminished food uptake, and the exchange of toxic mixtures with these marine species [21]. At disposal sites, polythene waste degrades both chemically and physically due to weathering, although mineralization takes a

long time and it may continue to exist in microscopic form for a lengthy period [2]. Bioremediation using microbes has proven to be an eco-friendly as well as an effective alternate method to degrade and reduce the plastic pollutants in the environment. Microbial degradation of microplastics often results in the production of non-toxic compounds such as H2O, CH4, and CO2 as their end products [22]. Researchers have been exploring the potential microbes that are competent to degrade plastic since 1970 [12]. The screening of microbial consortia from a highly contaminated area will lead to the identification of microorganisms with a higher ability to degrade plastics.

The current advancement in the field of biotechnology will ease the screening and selection of microorganisms that produce plastic-degrading enzymes. Hydrolysis of plastic through enzymatic means is a two-stage process; first, the enzyme catalyzes a hydrolytic cleavage and binds to the substrate. Enzyme-substrate binding is necessary to sustain the structural stability of an enzyme biocatalyst [23]. These microorganisms are then classified based on their degradation efficiency [24]. Biopolymers, or biodegradable polymers, are being developed that are more amenable to microbial attack, and these could be either natural or synthetic polymers. However, due to the the molecular complexity of microplastics, they remain recalcitrant in the environment. But the high levels of heterogeneity of plastics regarding size, complexity, and chemical nature make it difficult for researchers to find a one-stop solution for this plastic menace, including the ones in the marine environment [25].

One of the challenges in the bioremediation of microplastics is augmenting the enzymatic degradation of the plastics and the translucent assemblies that inhibit this process by microbes. Furthermore, studies are requisite to characterize this degradation of plastics with monomers varying from long-chain aliphatic polyethylene (PE), which is chemically simple,

to complex aromatics derived from polyethylene terephthalate (PET) and polyurethane (PU) [25]. Many aquatic eukaryotes have been recently tried as bioremediants for microplastics using the phenomena of biosorption and bioaccumulation, where the biomass will be immobilized and used to biodegrade [26]. The practicability of using microbial consortiums that have the potential to retain microplastics has not yet been reconnoitered. Biosorption, i.e., biofilm formation on the plastic film, leads to the bioaccumulation of microbes. Biofilm formation by microbes enables a stable surface and a better environment for the microbes to attach and reproduce on the plastic film. This will further aid in deteriorating the plastic much more efficiently [27]. The marine environment being contaminated by intense and permanent accumulation of plastics, especially at higher depths, becomes a great source for bioprospection of potential microbes in degrading the plastic [28, 29]. Microbial strains such as *Lysinibacillus* and *Salinibacterium*, *Aspergillus* sp., *Brevibacillus borstelensis*, *Pseudomonas* sp., *Arthrobacter* sp., *Vibrio furnisii*, *Bacillus sphericus*, *Brevundimonas vesicularis*, *Pseudomonas* sp., *Clonostachys rosea*, *Trichoderma* sp., and *Rhodococcus* sp., are known to degrade plastics [4, 20, 30–33]. In this chapter, we are trying to get a bird's eye view of the existing knowledge that might come in handy to other researchers in the future.

14.2 Fate and Sources of Microplastics

Plastic breaks down into tiny particles known as microplastics, measuring 0.001 to 5 mm in size, or even nanoplastics, measuring less than 0.001 mm, when exposed to sunlight, wind, waves, and other environmental variables [34]. Microplastics can be found in a variety of places, including bigger plastic waste that breaks down into ever-tinier fragments. Additionally, microbeads, a subset of microplastics, are little bits of synthetic polyethylene that are added as defoliants to several toothpastes and cleansers in

the health and beauty industries. These minute particles are so small that they can effortlessly get through even the most sophisticated water filtering systems and wind up in the Great Lakes and oceans, potentially endangering aquatic life [35].

These days, microplastic particles have been found everywhere in an extensive variety of settings, including freshwater, marine water, agroecosystems, the atmosphere, food and drinking water, and other remote sites [36]. They are firm and dense like boulders, or they can be as thin as tiny palls and carried by the wind to great distances [37]. Their widespread distribution makes them a significant geological indicator of the Anthropocene, according to many scientists [38]. Microplastics may contain two different kinds of chemicals: (i) additives and polymeric raw materials such as monomers or oligomers derived from plastics; and (ii) chemicals absorbed from the environment [36]. Additives are added to plastic during the manufacturing process to give it color, transparency, and improve its resistance to ozone, temperature, light radiation, mold, bacteria, and humidity, as well as its mechanical, thermal, and electrical resistance [39]. To find an answer to reduce the accumulation of microplastics in the environment, it is extremely necessary to understand their source, their destination, and the mechanisms and factors that could possibly lead to their dissemination. The most commonly reported primary microplastics are exfoliants, which are used to remove dead skin cells in cosmetic products. The drilling fluid, air blasting technology, and abrasives in the petroleum industry are the foremost contributors to primary microplastics [12, 40].

Considering secondary sources of microplastics, there is no reliable information on which particular source has made a major contribution [41]. Then, after the River Clyde sediment was visually and chemically characterized, it was observed that it was most likely to have originated from fishing, like the nylon fibers in the the fishing net or the fishing thread and a few other sources, such as WWTP discharge [14].

Microplastics that reach the environment undergo weathering and are transported to other environmental sectors. The physiological and biological mechanisms will drive the microplastics from the water surface to the sediment [42]. Coastal regions being high energy areas, such as heavy winds, will cause vertical mixing or surface drifting. Further studies of how the microplastics are transported in the riverine systems are studied through models [43]. Although there is inadequate evidence on microplastics degradation, there are few studies that talk about the mechanisms of the degradation [15].

Estimating and measuring the quantity of microplastics released into the environment is difficult. Due to the abundance of primary and secondary sources of microplastics and the absence of standardized sampling and measurement techniques, estimations of the quantity of microplastics released and created are quite unclear. According to research, there are currently at least 1.4 crore tons of microplastics on the ocean floor around the world [44], and every year, an additional 15 lakh tons are thought to enter the water [45]. Depending on the release process, microplastics are categorized into primary and secondary.

14.2.1 Primary Microplastics

The environment is immediately exposed to primary microplastics as plastic particles. They are included in products as stabilizers, glitter, or granular components in artificial turf sports fields, for example. Primary microplastics can also be produced because of production-related spills, use-related degradation of plastic products such as tire abrasion, paint, and coating flaking, and washing or wearing synthetic clothing [46]. According to some estimates, homes and commercial enterprises discharge 32 lakh tons of primary microplastics annually, of which 15 lakh tons end up in the marine environment [45].

14.2.2 Secondary Microplastics

Larger plastic items in the environment break down from improperly disposed plastic waste, such as abandoned fishing gear, scattered plastic packaging, or material lost from open landfills, resulting in the formation of secondary microplastics. Up to a distance of 10 km, rivers can reach by wind-carrying plastic garbage from open landfills or dump sites [47]. According to a study by a group of researchers, more than 1,000 rivers dump plastic into the ocean worldwide, with tiny rivers in heavily populated urban areas, particularly in Asia, being the most heavily contaminated [48].

14.2.2.1 Microplastics from Textiles

Synthetic textiles account for about 8% of the microplastics released into the water in Europe; this proportion is thought to be between 16% and 35% worldwide. Every year, between 200,000 and 500,000 tons of microplastics from textiles enter the world's oceans [46]. In the last few years, additional studies have been conducted on the subject of textiles losing micro-sized polymeric particles, also known as microplastics. However, there has been a significant range in the methodologies used to measure the shedding of microplastics from textiles to date [49]. When synthetic textiles are abraded, the fibers are mechanically stressed and shed as a result. This can happen during production as well as during the use of these textile materials, including washing and drying methods [50]. This ultimately resulted in the ability of microplastics to penetrate aquatic systems and wastewater, which was early identified as one of the sources [51].

The size of the global textile market exceeded 100 million metric tons in 2016, and a large portion of that production—65%—is made of synthetic polymers, of which 53.4 million tons are made of polyester alone. Being the primary component of synthetic fabrics, polyester-based materials—primarily PET—are a source of microplastic pollution and, as such, require appropriate controls during both the production

and usage of clothing and other textile end products. Being the primary components of synthetic fabrics, polyester-based materials—primarily-PET)—are a source of microplastic pollution and, as such, require appropriate controls during both the production and usage of clothing and other textile end products [49].

14.2.2.2 Microplastics from Fishing Gears

Marine-based causes of plastic pollution are less thoroughly studied than land-based sources, despite the latter receiving more attention [52]. It is believed that the primary marine-based cause of plastic contamination is industrial fishing [53]. The abandonment, loss, or otherwise discarded fishing gear (ALDFG) in the oceans and seas [54] has been acknowledged as a significant issue. Historically, non-synthetic materials like cotton, flax, or hemp fibers were used to manufacture fishing gear (such as nets, ropes, and lines) [55]. Once the large-scale production of plastics started in the 1950s, synthetic fibers became a preferred material for manufacturing fishing gear owing to their features such as lower cost, high tensile strength, and resilience to degradation on exposure to salt water compared to other natural fibers [56].

Today’s fishing gear is made of a variety of synthetic polymers, such as nylon, polyethylene, and polypropylene, which greatly contribute to plastic waste [57]. Trawls as well as fixed and drift nets pose the utmost risk for ALDFG to harm marine megafauna by ingestion [58] and entanglement [59]. Recent research found several freshwater turtles, including *Batagur dhongoka, Geoclemys hamiltonii*, and *Batagur baska*, and the endangered Ganges River dolphin (*Platanista gangetica gangetica*), are at increased peril of entanglement. These researchers classified and quantified ALDFG in the Ganges River and recognized hefty expanses of fishing rope and net [57].

14.2.2.3 Microplastics from Food Packaging

The accumulation of microplastics in the environment and their adulteration of food items has turned into a universal menace to both human and environmental well-being. But in modern times, there is a problem with direct contact and the release of plastic flaking into food items due to the widespread use of plastic food containers, plastic packaging, plastic bottles, disposable cups, newborn feeders, plastic-coated metal, and paper cartons [61]. Due to their ability to undergo mechanical recycling, thermoplastic resins such as PET, high-density polyethylene (HDPE), low-density polyethylene (LDPE), polypropylene (PP), polyvinyl chloride (PVC), and PS are frequently used to make plastic packaging materials and containers [62]. Multiple environmental conditions, including sunlight, water, temperature, and physical stress, affect how plastic materials degrade to produce microplastics [62]. Microplastics are present in different types of foods and drinks, including milk, honey, sugar, table salt, German beer, crustaceans, fish, and tinned fish. They are also present in bivalves, crustaceans, fish, and other types of seafood [60, 61].

14.2.2.4 Microplastics from Landfills

The leachate from landfills is developing into a large microplastic pool because of the massive buildup of plastic waste from city and manufacturing sources in landfills. Humans and biota may be adversely affected by the environmental release of microplastics from landfill leachate [63]. Plastic trash surrounding both active and closed landfills continues to be a source of microplastics. Leachate and gases produced by an amalgamation of biological, chemical, and physical processes comprise the landfill's output. Small specks and/or fibers, such as microplastics, are therefore dispersed into the environment through the air and leachate [64]. After being dumped, plastics begin to degrade in aerobic conditions (at first), but rapidly switch to anaerobic

conditions, where acid production and methane formation *via* the process of fermentation from organic solid wastes have an adverse effect on plastic decomposition. Landfilled microplastics will continue to form nanoplastics even in the absence of light and oxygen because of the varying temperatures, possibly reaching around 60 to 90 °C and pH (4.5–9). The presence of deep-seated fires, physical pressure, and compaction, along with reduced microbial activity, are also additional factors for the formation of nanoplastics from microplastics. The varied microplastic concentrations in leachate may be area-specific or, to some extent, the result of the absence of identical credentials [65]. Leachate's microplastic composition is a direct reflection of the use of plastic products. For instance, there is a close correlation between PP and the consumer goods, medical applications, and automotive industries [66]. The plastic particles in the leachate might be round and ovate, foamy, spherical, tablet-like, wires, strips, bits, and other shapes. Particles with ragged edges are called fragments. Granules are round objects. Plastic particles with a spongy feel are called foam [67]. All these varied plastic particles reach nearby water bodies and eventually the oceans.

14.2.2.5 Damages caused by Plastic Pollution

Plastic litter has become a significant environmental problem worldwide. When plastic waste accumulates in soil, it results in various negative effects on soil health, plant growth, and overall ecosystem functioning. Under the stimulus of several environmental variables, plastics that are released into the environment will stay behind, continue to deteriorate, and fragment into small particles. Plastic pollution of the environment on a global scale produces trash and is an acknowledged hazard to soil ecosystems, which serve as a major sink for microplastics [68]. As they have larger surface area to volume ratios, these microplastics and nanoplastics are expected to have a deleterious impact on the environment and are more likely to adsorb organic pollutants and

pathogens from their surroundings [69]. Metal can leak from microplastics, which can also harm crop quality [70]. Plastic micro- and nanoparticles are the source of soil plastic pollution. It has been discovered that soil MP pollution is worse than that in the marine environment [71]. Plastic pollution can harm soil in several ways, changing its biological, chemical, and physical characteristics. The following are some of the major damages that plastic causes to soil.

14.2.3 Plastic Contamination in Agricultural Soils

Different agricultural practices cause plastic pollutants in the soil to take on a variety of shapes and compositions [72]. Because of the disintegration of PVC pipes, plastic containers used for agrochemicals, and other microplastic sources, microplastic pieces are found in agricultural soils. After fertilizers with plastic coatings are applied directly to the soil, MP microcapsules also develop in the soil [73]. Because plastic microbeads can be found in cosmetic and cleaning goods as well as in wastewater, they can be detected in the soil environment. Due to the use of wastewater and sewage sludge as manure, which contain microplastic foams, which are also detected in soils from agricultural fields [74].

Another source of microplastic contamination in agricultural soils is fertilization with controlled-release fertilizers, which is responsible for 3% of the soil’s plastic sources. This technology is envisioned to encapsulate the nutrient (e.g., NPK) in a polymer covering to accomplish the controlled release of nutrients. But the enclosed ampoules linger in the soil, fouling the environment since they do not break down after the discharge of the nutrient [74, 75]. Yet another source of microplastics are compost manure sources that contain the feces of domestic animals that are used in agricultural fields and horticulture [76, 77], and usually direct intake through feed results in microplastic ingestion in animals. ending up in their feces [78].

14.2.3.1 Physico-Chemical Properties

When plastic debris gets into the ground, it can physically pollute the soil. Large plastic waste, like bags or bottles, can create barriers that prevent root penetration and growth. Because of this, there may be less room for plants to grow and develop, as well as establish and grow their root systems. In the experiment conducted by [68]. The soil's pH dropped as the number of microplastics increased. This could explain why the addition of microplastics to the soil increased soil aggregate size, soil porosity, and soil aeration while also altering soil microorganisms and soil pH [79, 80]. The amount of nitrate-N and ammonium-N in the soil increased dramatically because of the addition of microplastics. According to certain studies [81], it is expected that adding microplastics to soil greatly raises the amount of nitrogen in the soil based on the certain studies [81]. Therefore, the buildup of microplastics may have an impact on the nutrient cycling mechanisms in soil ecosystems.

14.2.3.2 Microbial Activity

In the soil, plastics can change the microbial communities. According to several studies, plastic pollution can lower microbial diversity and decrease soil microbial activity. Disruptions to microbial populations will have a significant bearing on soil health, biologic matter decomposition, and the cycling of nutrients. Additionally, plastic prevents the development of beneficial organisms like earthworms, which makes the soil less fertile. A complicated discrepancy in the environmental nature and response of other pollutants in soil is formed by the emergence of plastics as ecosystem trauma-causing agents, which impacts soil well-being and alters soil properties—both biological and physical [11]. Microplastics can stimulate soil enzymes when they are added to the environment [82].

14.2.3.3 Soil Erosion

Plastic waste on the soil's surface can prevent natural water infiltration and increase surface runoff, which contributes to soil erosion. Because the topsoil's protective layer may be washed away more easily, this can speed up soil erosion. In addition to decreasing agricultural output and increasing sedimentation in water bodies, soil erosion can result in the loss of fertile soil, which may eventually reach the nearby water bodies. These petite plastic particles can amend the physical makeup of the soil below us and diminish its water-holding capacity. They can also have an effect on plants by restraining root growth and nutrient uptake [83].

14.2.4 Living Beings and Environment

Microplastics are created when plastic is dumped in open landfills and reacts to the unparalleled conditions. They are easily transferred, combined with the environment, and stored in the body through a variety of exposures, including eating food, breathing in dust, and drinking contaminated water. MP soil contamination has become a severe issue that requires due attention since it poses a threat to human health and the environment [82]. Many marine species, including crab, oyster, mussel, and sea cucumber, consume microplastics that later become part of human food. As all these plastic products contain chemicals like formaldehyde, benzene, and dioxins, the microplastics in these seafoods ruin the taste of the food and make it more unsafe to consume.

14.2.5 Marine Environment

Microplastics have a big effect on the marine ecosystem. The marine fish and marine food chain will be impacted in many ways by marine microplastics. The microplastics may reduce food intake, delay growth, result in oxidative damage, and cause aberrant behavior in fish and other aquatic species [84]. Marine life's growth and development may be hampered by microplastics. When marine life ingests microplastics, they

build up in the digestive tract and block it, which causes satiety, a fall in food intake, and lower energy reserves in the body. This has an impact on the growth of marine life. In addition to creating aesthetic issues, plastic waste puts maritime pursuits like fishing and tourism at risk [85, 86]. Ghost fishing is caused by abandoned fishing nets, which can cause losses to commercial fisheries. Since floating plastic garbage can remain at the sea's surface for extended periods of time and can quickly be colonized by marine animals, it may eventually make it easier for 'alien' or non-native species to spread [86, 87].

14.3 Minimizing the Microplastics in the Environment

Bioremediation and transformation have been an advanced technological approach along with existing approaches such as incineration, recycling, and landfills in minimalizing the microplastics pollution [7].

14.3.1 Recovery/Alteration of Microplastics

Commercially, companies follow the methods of chemically recycling plastics that have been demonstrated at the laboratory level [88]. Recent studies have reported that plastics can be converted into tarry and charcoal phases by pyrolysis at different temperatures [89]. By applying certain physical processes, such as oxidation and post-breaking-down methods, to the separation of microplastics from the sludge and using biofilters for filtration in the wastewater, microplastics can be collected. These are then processed for reusable products and fuel, but it must be studied further to minimize the release of resins or any other kinds of microplastics further [90]. Few studies focused mostly on the pyrolysis of polyolefins (PP, PS, and PE) to recover heat energy [91], and those that cannot be used for recycling can be used for waste-to-energy strategy processes [92].

14.3.2 Biology Mediated Approaches

Bioremediation and material valorization (i.e., enzyme-mediated biodegradation, phytoextraction, and composting) are considered efficient strategies to monitor microplastic pollution [92]. Different strategies, such as physio-chemical ones have been applied to control the microplastic pollution. However, one checkpoint in applying these methods is the expense associated with most of these techniques, and here is where bioremediation comes in as a cost-efficient alternative. WWTPs, on the other hand, are not typically designed for removing microplastics; hence, they still release the microplastics through wastewater effluent and sludge. In consideration of this, scientists have worked with marine fauna, aquatic flora, and macrophytes for the reduction of microplastics from WWTPs [26]. A similar approach like phytoextraction in the solid and liquid phases was applied by researchers who worked with eukaryotes such as annelids (sandworms) and echinoderms (sea cucumbers) for bioremediation of microplastics in WWTPs [26].

Microbial degradation is a recent development in disposing of and degrading microplastics and plastic debris, which are almost non-degradable. Plastics of petroleum origin can be used as substrates for microbial degradation and valorization [94]. This is comparable to enzyme remediation technology, where the first step is to screen the plastic-degrading microbes and isolate depolymerases and other required enzymes. During the microbial valorization of microplastic wastes, the products at the end of the depolymerization can be applied at the final stage (e.g., depolymerization products can be exposed to bio-fermentation). Post-separation these depolymerization products are exploited for the biosynthesis of high-value substances via diverse metabolic pathways [93, 94]. All these microbial degradations occur through enzymes, and therefore it is vital to understand the principle behind the degradation of these microplastics [95, 96].

14.4 Severance of Microplastics from Water and Sediments

For optimum degradation of plastics to occur, it is a must to gather degrading microbes or enzymes and the plastics. By providing the optimum temperature, pH, humidity, and nutrients, the degradation rate might speed up. Thus, the microplastics and plastic debris should be isolated or filtered from soil, water, ornd sediments and used for obtaining valuable chemicals by applying biological depolymerization and subsequent fermentation [97]. One of the major sources of secondary microplastics is WWTPs, where the effluent is discharged into the riverbanks as sludge when used as organic fertilizer. They distribute the microplastics into the soil and water. To date, chemical treatments, filtration, ingestion, and physical sorption have been used as water remediation approaches to separate the microplastics [98, 99]. In one of the studies, it was reported that biodegradation of the debris by marine and terrestrial microorganisms was very low due to the unavailability of oxygen and light [100]. While certain challenges in biotechnological advancement, such as the toxicity impact on microorganisms can be fully understood, the intermediate compounds that could block the metabolism of the microorganism are still unknown [101]. Apart from filtration and physical sorption, researchers are trying to devise novel methods to separate out the microplastics from the sediments or polluted water. One of those methods includes using iron oxide-manganese dioxide core-shell micromotors under the exterior magnetic field. This method has proven to remove 10% more microplastics from the suspended wastewater in a period of 2 h [98, 102]. If the microplastics are concentrated, then their accessibility of microplastics to the microorganisms will increase which can result in efficient degradation.

14.5 Marine Microbial Strains Associated in Degrading Microplastics

A microorganism is a minute living thing that might have one cell or many cells. Microbiology is the study of microbes. Antonie van Leeuwenhoek first identified microorganisms in 1674 while using a custom-built microscope [103]. Most protozoa, all bacteria, and all archaea are microorganisms. They also comprise several types of fungus, algae, and even animals, such as rotifers. There are juvenile stages in many macroanimals and plants that are also microbes. Viruses (and viroids) are classified as microorganisms by some microbiologists but as non-living entities by others. *Aspergillus flavus*, *Aspergillus niger*, *Penicillium funiculosum*, and *Streptomyces* are examples of fungal species that may break down different types of plastics, according to [104]. Additionally, *Rhizopus delemar*, *Firmicutes*, and *Rhizopus arrizus* were mentioned as bacteria [105]. It was discovered that the fungus *Pestalotiopsis microspora* can successfully break down PE. Table 14.1 shows some bacterial and fungal strains that biodegrade plastics.

According to a recent study, a sizable number of microorganisms, particularly some bacteria and fungi, possess the ability to degrade these synthetic polymers much more swiftly than the natural process by utilizing specific exoenzymes when under stress. The effectiveness of the degradation of microbial consortia from various habitats was a key focus of the experiments. The phyla *Proteobacteria* (48%), *Firmicute*s (37.4%), and *Actinobacteri*a (9.8%) contain most of the bacteria that have been found. The biodegradation of human wastewater already makes extensive use of bacteria from many phyla [106, 107]. Hydrocarbon contamination brought on by petrochemical sector actions is one of the focal environmental issues of the day. Environmental concerns are especially high when petroleum products are accidentally released. It is well established that organic pollutants with

hydrocarbon components are both neurotoxic and carcinogenic [108]. Contaminants at oil spill sites are also degraded using these microorganisms. *Proteobacteria* are the most plentiful and metabolically varied phylum of bacteria, and they play a significant role in industry, agriculture, and medicine [109].

Table 14.1 Bacterial and fungal strains depicting the biodegradation of plastics.

Source	Microorganisms	Plastic act as substrate	Reference
Fungus	*Aspergillus flavus*	Polycaprolactone (PCL)	[118]
	Penicillium funiculosum	Polyhydroxybutyrate (PHB)	[118]
	Aspergillus niger	PCL	[118]
	Streptomyces	PHB,PCL	[118]
	Fusarium	PCL	[119]
	Amycolaptosis sp.	Polylactic acid (PLA)	[119]
	Pestalotiopsis microspore	Polyurethane	[105]
	Phanerochaete chrysosporium	Polyethylene	[119]
Bacteria	*Rhizopus delemar*	PCL	[118]
	Firmicutes	PHB, PCL and PBS	[118]
	Proteobacteria	PHB, PCL and PBS	[118]
	Rhizopus arrizus	Polyethylene adipate (PEA), PBS, PCL	[118]
	Pseudomonas astutzeri	Polyhydroxyalkanoate (PHA)	[119]

Pseudomonas, a genus of ecological significance in soil and water habitats that currently includes 216 known valid species [110], was the most often mentioned genus in the evaluated studies. Similarly, several Pseudomonas strains that breakdown PE were identified by [27]. Most of the members of the phylum *Firmicutes* are gram-positive bacteria that may produce endospores and are mainly found in soil [109]. The *Bacillus* genus, which at present encompasses 282 known legitimate species, was the group of organisms that was referenced the most. The bacteria in this genus can produce either oval or cylindrical endospores and can either be facultative aerobes or true aerobes. *Pseudomonas*, *Bacillus*, and mixes of these two species were found in 21.0% of the experiments as the bacteria that efficiently started the biodegradation of the synthetic polymers [111]. Table 14.2 represents the list of microbial strains and the types of plastic that they degrade. Based on Table 14.2, it was identified that polyethylene (PET) was successfully degraded by *Brevibacillus borstelensis*, *Rhodococcus rubber*, and *Pseudomonas chlororaphis*. Meanwhile, *Pseudomonas putida*, *Ochrobactrum*, and *Pseudomonas fluorescens* broke down the polymer chain of polyvinyl chloride (PVC).

The unique capabilities of marine microbial enzymes have gained attention for degrading microplastics. Researchers have identified many microbial strains that are capable of degrading microplastics. A bacterial strain, *Pseudomonas aeruginosa* MCMK2, demonstrated the ability to degrade LDPE, which is one of the most popular types of microplastics. The enzymes produced by these strains were found to be lipases and esterases that facilitated the degradation of microplastics [112]. Later in 2021, Paco *et al.* identified a fungal strain, *Zalerion martimum*, which degraded polystyrene, another prevalent MP polymer. The extracellular enzymes released by the fungus, *viz.*, peroxidases and laccases, effectively depolymerized and degraded the polystyrene particles [113]. Studies have found that bacterial

strains such as *Proteopbacteria*, *Firmicutes*, *Bacteroidetes*, and *Cyanobacteria* have greater potential to colonize microplastics than other groups of bacteria [114]. A few species of Vibrio that dominated plastic samples were found to be opportunistic pathogens. It also suggests that the hydrophobicity of the plastic facilitates biofilm formation on the plastic surfaces [115]. Bacterial groups such as *Pseudomonas*, *Cornybacterium*, *Streptomyces*, *Arthobacter*, *Rhodococcus*, and *Micrococcus* happen to utilize plastic as the sole carbon source under laboratory conditions [116]. From an application perspective, microbial degradation of microplastics on a large scale note is still possible. Recently, scientists have been working on a marine microbial consortium to degrade polyethylene terephthalate (PET) microplastics. It was observed in their study that, within 28 days, the consortium was capable of degrading the PET up to 67%, indicating its potential for bioremediation [117].

Table 14.2 List of microbial strains and the types of plastic which they degrade.

Plastic	Microorganism	Reference
Polyethylene (PET)	*Brevibacillus borstelensis*	[120]
	Rhodococcus rubber	[121]
	Pseudomonas chlororaphis	[122]
Polyvinyl chloride (PVC)	*Pseudomonas putida*	[123]
	Ochrobactrum	[3]
	Pseudomonas fluorescens	[3]

14.6 Work Done in our Laboratory

Work has been initiated in our laboratory to isolate and characterize microplastic-degrading microorganisms from the marine environment. We collected plastic samples from various beaches in south India and were able to successfully isolate two microorganisms and characterize them using 16S RNA sequencing. Further work is in progress to find more isolates and work on a consortium of microplastic-degrading bacteria.

14.7 Conclusions

In conclusion, we can understand that the primary and secondary microplastics and nanoplastics from various sources, ranging from agricultural fields, textile factories, and landfills, also find their way eventually to the oceans. In the marine environment, unique sources such as the wear and tear of fishing nets and other gear and the industrial effluents also contribute to marine pollution with micro- and nanoplastics. Among the various solutions for this problem, it is evident that enzymatic degradation using suitable microbial consortia is an eco-friendly and economic solution.

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15
Human Toxicity of Nano- and Microplastics

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Abstract

The toxicity of microplastics to human beings has raised global concern because of the ubiquity of these emergent pollutants in the environment and their entrance into trophic chains. This chapter reviews the advances and challenges in assessing toxicity in humans based on the analysis of basic concepts of toxicology and recent research results. The challenges and scope of applying animal studies, cell- and tissue-based toxicology assays, *in silico* or computational models, tissue engineering, organ-in-chip engineering, and human dosing studies are discussed. The analysis of the results of toxicity studies with nano- and microplastics, based on the revision of 85 research articles, showed that toxicity tests were mainly performed with mice and human cells, while polystyrene, polyethylene, polyethylene terephthalate, and polyvinyl chloride were the most common plastics tested. The main results regarding particle uptake, cell viability, stress response, cancer, genomic effects, and toxicity in the reproductive system are described. While the results show, in most cases, a negative effect associated with the presence of micro- and

nanoplastics, differences in methodologies and research approaches make it difficult to establish clear cause–effect relationships.

Keywords: Animal studies, human cells, mice, plastic pollution, polystyrene

15.1 Introduction

Toxicity is defined as the capacity to cause injury in living organisms [1]; this can be done by damaging organ systems, disrupting biochemical processes, or affecting enzyme systems [2]. In the current scenario, where plastic pollution has emerged as a global threat to the environment, the toxic potential of plastics has raised the concerns of society, governments, and the academic sector [3]. The ease of plastic transport through ecosystems, the possibility of fragmentation, and evidence of ingestion and inhalation by different species [4] could maximize the toxic effects of plastics and increase their bioavailability [5].

When discussing the toxicity of plastics, toxic effects need to be distinguished from the physical damage caused by the entanglement or ingestion of these materials. Although plastic particles and wastes can cause injuries owing to tearing, accumulation, translocation, internal abrasion, and blockages [5], their toxicity is mainly related to the molecular structure of specific chemical compounds, which define their activity and capacity to bind to individual biological targets and produce effects [6].

Plastics are complex materials produced by the combination of polymers and additives that are added to improve processability, resistance, flame retardancy, and other functional properties. Assessment of their toxicity can focus on plastic materials, monomers, and byproducts that could remain in plastics during production, or on additives

that could leachate during disposal and use. Scientific evidence has been gathered regarding the toxicity of specific additives such as flame retardants [7, 8], phthalates [9, 10], and bisphenol-A used to produce polycarbonate [11, 12].

This chapter reviews the basic concepts and recent advances in the assessment of plastic toxicity to humans. It includes a description of the basic concepts of toxicology, the technical and conceptual challenges found in its evaluation, and the strategies developed for testing chemicals in human beings. Finally, a summary of the research results in the last few years describes the main findings and remaining knowledge gaps.

15.2 Basic Toxicology Concepts

The adverse effects due to the action of toxic substances are determined by various factors [1, 2, 6]:

a. Intrinsic toxicity: related to molecular structure, presence of certain functional groups, reactivity, and easiness of transport in the environment, defined by solubility, and volatility

b. Dose: the amount of a substance administered at one time

c. Exposure: defined by the route of contact to the organism, frequency, duration, and environmental circumstances

d. Individual response of the affected organism: depending on age, sex, health conditions, species

The effect or toxicity can be classified according to the conditions of exposure or the characteristics of the toxic effects, as shown in Table 15.1.

Table 15.1 Toxicity types based on exposure conditions and toxic effects.

According to frequency and duration of exposure	According to time of development and duration of induced effects
• **Acute:** effects of a single dose or multiple doses in a 24-h period • **Subacute:** repeated exposure for 30 days • **Subchronic:** repeated exposure for 1 to 3 months • **Chronic:** exposure for more than 3 months	• **Temporary:** it lasts for a short duration of time • **Irreversible:** persistent throughout the life of the individual • **Immediate:** development soon after a single exposure • **Delayed:** it has a delayed onset of action • **Cumulative:** produced by the incremental injury due to successive exposures

15.3 Challenges and Opportunities for Evaluation of Toxicity in Humans

Toxicity has traditionally been evaluated in animals. However, significant efforts have been made to minimize, and eventually eliminate, animal testing, not only due to increasing concern for animal suffering, but also due to the challenges of extrapolating toxicity data from animal models to humans. International organizations are working on producing and validating alternative methods such as *in vitro*, *in silico*, and tissue- or organ-in-chip studies.

The study of the toxicity of nano- and microplastics (NPs/MPs) also presents challenges related to the physical and chemical characteristics of these particles, including size-dependent effects, adsorption of a variety of substances on the surface of NPs/MPs and surface charges on the particles, among others. In addition, there are issues related to the need to conduct toxicity assays in controlled conditions, most of which are carried out with virgin particles of defined shape (i.e., spheres), and real NPs/MPs are not virgin and have different and irregular shapes. In this section, we explore toxicity studies conducted using animals, alternative methods, and human dosing studies.

15.3.1 Animal Studies

The use of animals in toxicology studies has become widespread with the development of the Median Lethal Dose (LD_{50}) test by Trevan [13]. Toxicological studies in animals have been conducted to predict toxicity in humans, including the development of pharmaceuticals and legislation to protect humans from exposure to toxicants. Evaluation of toxicity in animals is considered a cornerstone of human safety [14]. Since 1938, the United States Federal Food, Drug, and Cosmetic Act mandates animal toxicity testing [15].

15.3.1.1 Animal Welfare and Prediction of Human Toxicity from Animal Models

In 1831, British physiologist Marshall Hall, proposed five welfare principles for animal experimentation [16]. Later, in 1959, Russel and Burch presented the concept of three Rs [17, 18]:

- Reducing the number of animals and maximizing the information obtained from the used animals

- Refinement of experiments to minimize pain and distress and to promote the welfare of animals
- Replacement methods to avoid or replace animal use.

Currently, many countries have legislations regarding the welfare of animals used for scientific or academic purposes. Examples of these are Directive 2010/63/EU of the European Parliament and Council of 22 September 2010 on the protection of animals used for scientific purposes [19], and the Animal Welfare Act of the United States of America [20]. Additionally, international organizations have made significant efforts to improve animal welfare and reduce the number of animals used in research. For instance, the International Council for Harmonisation (ICH) of Technical Requirements for Pharmaceuticals for Human Use [21] and the Organisation for Economic Co-operation and Development (OECD) [22].

Some of the advantages of using animals to predict human toxicity include the possibility of evaluating a wide range of doses and endpoints, availability of background information on some strains of animals (genetic makeup, metabolism of xenobiotics, and other responses to chemical compounds), and conducting research in a controlled environment with a group of uniform animals [14]. However, some studies have questioned the value of animal studies in predicting human toxicity. A study published by Olson *et al.* [23] is frequently cited to support the prediction of the toxicity and therapeutic effects of pharmaceuticals in humans using animal models. This study analyzed 150 pharmaceutical compounds and found a true positive concordance rate in human and animal toxicity of 71% for rodent and non-rodent species, which were predictive for 63% of human toxicities and rodents (43%). Shanks *et al*. [24] discussed some limitations of the Olson *et al*. [23] study. Several studies have reported that there are significant differences

in animal and human toxicity [24–28]. For instance, Bailey *et al.* [25] analyzed a dataset of 2,366 drugs with available animal (rat, mouse, and rabbit) and human toxicity data. This study found that the absence of toxicity in animal models provides little evidence for the absence of drug reactions in humans.

15.3.1.2 Characteristics of Animal Studies

The animals used in scientific research include mice, rats, guinea pigs, dogs, cats, and non-human primates. The selection of the animal model used in each study depends on several factors such as the prediction of human response, animal welfare cost, lifespan, and knowledge of their physiology. The main administration routes for the test substances are oral (diet, drinking water, gavage, or encapsulation), inhalation (nose-only or whole body), dermal (substances that penetrate the dermal layer), intravenous (bolus injection, slow intravenous injection, or intravenous infusion), intraperitoneal, intramuscular, and subcutaneous [16]. A summary of the types of animal studies is presented in Table 15.2.

15.3.2 Alternative Methods to Animal Testing

Therefore, there is a need to develop alternative toxicity methods for animal testing. In addition to the welfare of animals and prediction of human toxicity in animal models, other reasons include the cost of animal research, the large and increasing number of chemical compounds and mixtures of chemical compounds that require testing, and the need to produce new chemicals (for pharmaceutical purposes, for example) [17]. Governmental organizations have established research projects to develop and validate alternative methods. Examples include the European Centre for the Validation of Alternative Methods (ECVAM) [29], The New Approach Methods Work Plan: Reducing Use

of Vertebrate Animals in Chemical Testing from the United States Environmental Protection Agency [30], and The Guidance Document on Good *In Vitro* Method Practices by the OECD [31].

15.3.2.1 Cell- and Tissue-Based Toxicity Assays

In vitro methods are performed in a controlled environment outside a living organism, using cells or tissues cultured under controlled conditions in flasks or plates [32]. *In vitro* chemical toxicity models are typically associated with perfused organ preparations, isolated tissue preparations, single-cell suspensions, and cell culture systems. Table 15.3 presents a summary of the types of cells or tissues used in the toxicity studies.

Table 15.2 Types of animal studies. Adapted from [16].

Type	Description	Examples
Acute toxicity	• The test studies the effects of a single dose of a substance in a short period of exposition • The test substance is normally administered orally • The test is used to determine susceptible species and target organs • Mortality and morphological, histological, and pathological changes are recorded • Determination of lethal doses is carried out	• Fixed dose procedure • Acute toxicity class method • Up and down procedure • Acute toxicity for inhalation • Acute eye irritation
Subacute toxicity or repeated dose toxicity	• The test studies the effects of exposure in the medium term • Exposition to the test substance should not last more than 10% of the life	• Genotoxicity testing • Mammalian erythrocyte micronucleus test

Type	Description	Examples
	span of the animal, normally between 14 and 90 days • Exposure to the test substance occurs daily • The highest dose used in the test should not cause lingering signs or lethality • The lowest dose used in the test should not show toxicity	• Neurotoxicity studies
Subchronic toxicity	• The test studies the effects of exposure in the medium term • Exposition to the test substance normally last more than 90 days • Body weight variations, biochemical and cardiovascular indicators, and behavioral changes are recorded	

Type	Description	Examples
	• Histopathological analyses are conducted	
Chronic toxicity	• The test studies the effects of exposure in the long term • Exposition to the test substance should last long enough to allow the development of cumulative toxicity. Normally the study period last from three to 24 months • The test should include one rodent (20 animals per sex) and one non-rodent species (four animals per sex) • The highest dose used in the test should present toxicity effects without animal suffering, morbidity, or death	• Carcinogenic toxicity • Reproduction toxicity • Developmental toxicity or embryotoxicity studies

Type	Description	Examples
Toxicokinetic studies	• The test studies the absorption, distribution, biotransformation, and excretion of the test substance	• Maximal concentration in blood or maximal excretion after administration • Time to reach the maximal concentration or excretion • Half-life of the test substance • Total amount of the test substance absorbed in a period

Owing to the lack of availability and difficulties associated with the isolation and purification of primary human cells to assess the toxicity of chemicals and materials [33], testing in cell lines, which are well characterized and easily cultured, has become a feasible alternative [34]. Cancer-derived cell lines are widely used to test the toxicity of MPs in humans [35], their advantages include their ability to provide an indefinite source of biological material for experimental work [36], resistance to apoptosis, and standardization. Cancer cells used to assess plastic toxicity in humans include A549 [37–43], Caco-2, HeLa [41], JEG-3 [44], and Raji cell lines [45]. These cell lines have been used as models to test cell viability, formation of reactive

oxygen species, uptake, and apoptosis, among other toxic effects.

Several tests are used to evaluate *in vitro* cellular function or cell death, and to obtain information on the mechanism of action of a chemical. Cytotoxicity tests were used to measure the potential of a compound to cause cell death (necrosis and apoptosis). Necrosis is a class of irreversible cell death resulting from acute cellular injury, and apoptosis is a type of programmed cell death. Several *in vitro* tests have evaluated their effects on reproduction and cancer. In addition, there are specialized *in vitro* tests for environmental toxicology (aquatic short-term toxicology, genotoxicity, mutagenicity, hematotoxicity, hepatotoxicity, and immunotoxicity) [32]. There is a global effort to develop and validate *in vitro* test methods, the EURL ECVAM has validated methods for acute toxicity, aquatic toxicity, aquatic bioconcentration/bioaccumulation, biologicals, carcinogenicity, eye irritation/serious eye damage, genotoxicity, phototoxicity, repeated dose toxicity, skin corrosion, skin irritation, skin sensitization, and toxicokinetics. Methods validated by ECVAM can be accessed through the website of the Joint Research Centre [48].

The *in vitro* toxicity of NPs/MPs is currently tested using methods such as cell viability (detection of mitochondrial activity, LDH release upon necrosis, necrosis, or apoptosis, intact lysosomes via neural red uptake, Caspase-3), stress response (detection of reactive oxygen species), inflammatory response (inflammatory marker production *via* ELISA), DNA damage, and genotoxic effects, among others. However, the properties of NPs/MPs, such as hydrophobicity and surface charge, may interfere with assay components or detection systems. In addition, most toxicity assays are carried out with virgin materials that are

well-characterized with a specific shape, in contrast to the many different forms that NPs/MPs may have in the environment.

Table 15.3 Types of cells or tissues used in toxicity studies. Adapted from [32, 46, 47].

Type of cell or tissue	Description
Slices of tissues	• Used tissues include the liver, lung, kidney, and brain [32] • These tissues retain to some extent the structural and functional features [32]
Two- or three-dimensional cell cultures	• The cell culture can be carried out by isolation and reaggregation of cells from different organs. Cell cultures retain some functional properties of the original organ or tissue [32] • Three-dimensional cell cultures maintain their differentiated behavior and expression profiles much better than twodimensional cultures. They can be derived from biological, synthetic, biologically inspired synthetic matrices or hybrids [46]
Cells from body fluids	• This type of cells can be maintained *in vitro* for several days, or used to produce stem cells [32]
Primary cultures and early passage cultures	• Primary cultures are carried out by removing the pieces from tissue or organs in aseptic conditions [47]

Type of cell or tissue	Description
	• Cells are obtained by tissue explant culture and chemical, or enzymatic digestion [47] • In this type of culture, the cells are in their closest form in the tissues [47] • Primary cell cultures usually are diploid cells [46] • Some examples are cortical or cerebellar neurons [32]
Finite cell lines	• These are cell lines that can be subcultured for an extended period, about 50-60 passages before dying [32] • These cells are genetically stable and remain diploid for many passages
Continuous cell lines	• Continuous cell lines are often derived from malignant tissues and tend to be aneuploid [46]. They can be also derived from normal embryonic tissues [32] • These cell lines can be subcultured indefinitely and do not reach senescence [32] • These cell lines are karyotypically abnormal and accumulate mutations [47]

Type of cell or tissue	Description
Stem cell lines	• Stem cell lines are cells that can renew themselves, differentiate, and be colonized [47] • This type of cell lines can be classified according to their origin in embryo-derived and adult stem cells; or to their potency as pluripotent (can differentiate into any cell type in the body [167]), multipotent (differentiates to only two cell types), and unipotent (differentiates to only one cell type) [47] • The CTX0E16 is an example of this type of cell line [32]

15.3.2.2 In Silico or Computational Modeling

Computer models use mathematics, computer science, and physical simulations to understand various systems. The results obtained from the simulation increases the probability of solving a problem. The application of this technology in the biological field makes it possible to predict the development of a disease by monitoring cellular interactions, molecular processes, and effects on tissues and organs [49]. To carry out a simulation, it is important to consider four variables: the simulation and its confidence level, parameter estimation, model comparison, and the interference of latent variables. Simulations must also be simple, interpretable, and consider all possible hypotheses [50].

In silico modeling was similar to *in vitro* simulation. However, this allows for an unlimited variety of parameters. It can also be considered an *in vivo* test, in which pharmacological and physiological processes are developed such that this technology is more representative of an organism [51]. The most common applications of *in silico* modeling are drug detection, drug reaction prediction, PCR sequencing, and cell simulation [52].

In silico modeling lacks control of *in vivo* experiments, but it is possible to include parameters that allow better adjustment of a model without the ethical considerations of animal studies [51]. The test is inexpensive compared to animal studies, and results can be obtained in a much shorter time [52]. However, modeling cell dynamics, which depend on the size of the modeled system, can be a complex process [53].

15.3.2.3 Tissue Engineering

Tissue engineering builds three-dimensional structures to restore, preserve, or improve the function of organs and tissues. This is achieved through a combination of cells, biologically active molecules, and scaffolds [54]. Some of the most common clinical applications include the skin, cardiovascular system, osseous system, and nervous system, among others [55].

Scaffolds are structures that replicate the shape of the tissue, either *in vitro* or *in vivo* tests, and function as the basis for cell adhesion and can carry out their process [56]. The main manufacturing methods are solution casting, electrospinning, phase separation, additive manufacturing, self-assembly, extrusion, and foaming [57].

Decellularization, organoid technologies, and 3D bioprinting are the main methods used to manufacture scaffolds for tissue engineering [58]. In decellularization,

cells and their components are removed from the extracellular matrix to produce a natural matrix with mechanical integrity. Water, proteins, and polysaccharides are the main components of the ECM [59]. Organoid technologies are 3D stem cell cultures that reproduce human tissues, either in physiological functions or structures, allowing for personalized treatment options [58]. Finally, 3D printing can be performed *via* bonding or direct printing. The binding technique is performed on a powder base, where a binder inkjet liquid base is printed. In direct printing, waxes or plastic polymers are dispensed to solidify the formation of consecutive layers of the object. The main printing techniques are bioprinting, inkjet bioprinting, laser-assisted bioprinting, extrusion bioprinting, and stereolithography [57].

Some clinical applications of tissue engineering have been *in vitro* spindle models, cardiac tissue, pancreas, liver, skin regeneration, cartilage, heart, and even the bladder, the latter is the only one that has been able to be completely modified [56, 58].

15.3.2.4 Tissue- and Organ-in-Chip Engineering

Technological advances have enabled the development of innovative processes applied in medicine such as organ-on-chips. Organ engineering on chip mimics *in vitro* assays and was developed for application in tissue engineering and microfluidics [60].

Organ-on-chips are devices that are manufactured to micrometer size, to perform or maintain the function of a living organ, and to analyze the structure, which is made of a polymer covered with cells. Applications include the study of cells and tissue morphology, metabolism, and oxidative stress, among other functions. This technology has also

been used to investigate diseases, drug metabolism, cancer cell yield, and drug side effects [61].

The human organs to which this technology has been most applied are the brain, kidney, liver, heart, intestine, lungs, and bones. The main cell types used are human primary, immortalized, embryonic, adult, induced pluripotent, and gene-edited stem cells [62]. The most widely used biomaterials for the manufacture of organs-on-a-chip are collagen, gelatin, fibrin, hyaluronic acid, chitosan, alginate, polycarbonate, polylactic acid, polydimethylsiloxane, and synthetic biomaterials [62, 63].

In the organ-on-chip database, it is possible to find study data regarding medicines, devices, organs, patents, cancer research, toxins, mathematical and 3D models, laboratory data, and publication statistics [64]. This initiative allows for the exchange of information for future improvements.

This technology may allow higher-quality results than cell culture or animal experimentation. However, it is necessary to conduct more studies to identify the biological differences between humans and organ-in-chip models to validate either the mechanism of action or the response in humans. These issues are primarily associated with the operating conditions of the organ-on-chip and its building materials [65]. In addition, the manufacturing cost is high; therefore, commercialization and applications of this technology are limited [62].

Organ-on-chip technology has become widely accepted as a human-specific experimental platform for preclinical research and therapeutic testing. This shift is largely based on the results of *in vitro* studies using these platforms, which are increasingly predictive of clinical data for integrated human physiology.

15.3.3 Human Dosing Studies

In medicine, intentional human dosing studies are a well-established practice, with regulations to protect and respect research participants [66]. Drug development can take between 10 and 15 years and includes steps in which the drug is administered to humans (Figure 15.1).

The study of accidental human exposure to chemicals has also yielded valuable evidence for the risk assessment and safety evaluation of substances [66]. Additionally, the US EPA has considered accepting intentional human toxicity studies conducted by third parties. The EPA's interest in this type of study was promoted by the implementation of the 1996 Food Quality Protection Act on agricultural pesticides. Human toxicity studies of pesticides are controversial because they are conducted by third parties with an economic interest [66].

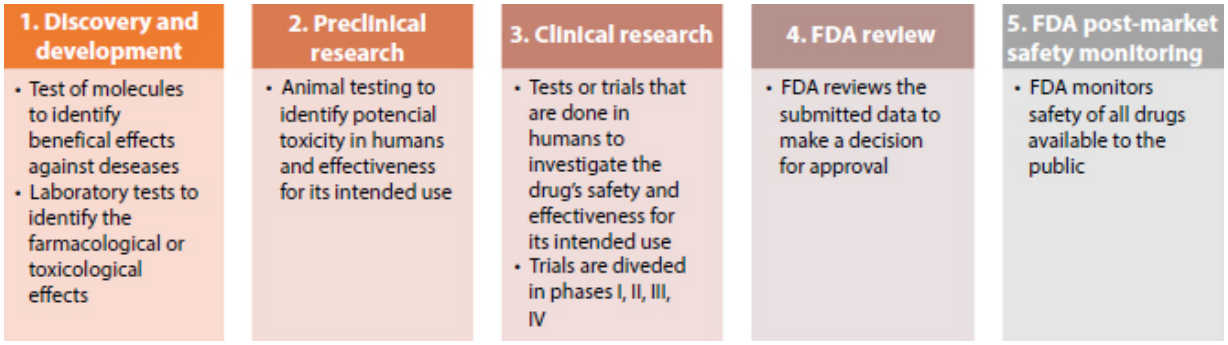


Figure 15.1 Drug development process of the United States Food and Drug Administration (FDA) [67].

15.4 Toxicity Studies With Nano- and Microplastics

A search was conducted for research articles published in ScienceDirect and Scopus from 2022 to date. The search was restricted to toxicity studies in human cells, fluids or tissues, and mammals, and the keywords used for the

search on these databases were “human toxicity” and “nanoplastics”; and “human toxicity” and “microplastics.” Review articles and papers published in conference proceedings were excluded. Only articles written in English were included in this analysis. A total of 85 articles were selected, and these studies reported the results of 442 toxicity assays. Toxicity tests were mainly performed using mice [68–101] and human cells [37–39, 42–44, 74, 97, 100, 102–123] (Figure 15.2). However, research has been conducted with rats [124–128], human fluids [45, 101, 129–132], and mouse cells [76, 77, 94, 95, 133–137]. Finally, other studies have included enzymes [138–140], computational modeling [101, 141, 142], tissue- or organ-in-chips [118], and porcine organs and cells [143].

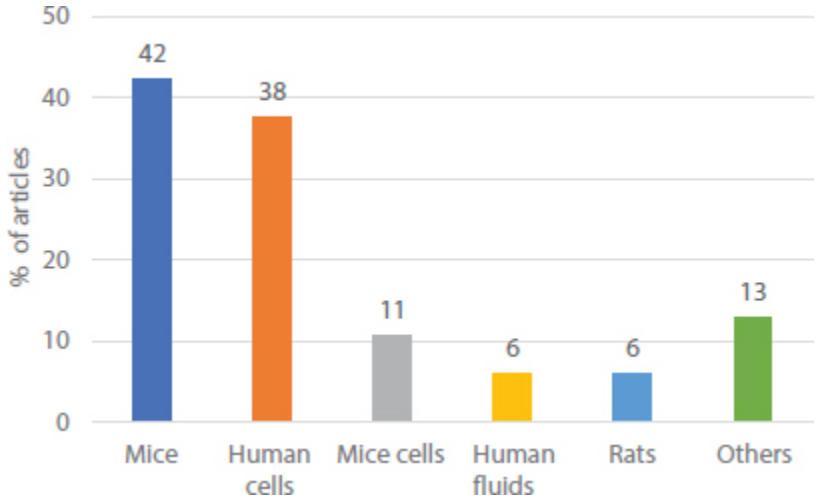


Figure 15.2 Samples used to perform toxicity tests with nano- and microplastics. Percentages were calculated based on the total number of studies analyzed (n = 85).

The abundant use of mice in toxicological research is because, like humans, mice are mammals, they are presumed to have similar immune responses to diseases and infections, their endocrine hormonal systems are similar, their bodies go through processes similar to those of humans (e.g., aging), it is available plenty of background information (genome), and they share about 80% of their

genes with the human species. Although there are technological advances today, and cell cultures and organoids can be used to assess toxicity, there are processes which occurs in living organisms that cannot be replicated in cell culture. Diseases involve more than one organ, and to assess the health effects of a particular substance or toxin, the whole body must be tested to assess its response to these new agents; the same occurs when trying new medications. These points indicate that mice play a special and important role in toxicology [144]. Although, *Section 15.3.1* of this chapter, highlights some of the concerns of animal studies.

In contrast, 67% of the studies that evaluated toxicity in mice used male mice, followed by 22% that used female mice, whereas the remaining studies did not report the sex of the mice used. In the case of the female mice 6% of the investigations used pregnant females to study the toxicity in female reproduction, for example, in ovules, embryos, and their offspring [68]. The study of offspring allows us to understand the types of transgenerational effects and how they occur [136]. In addition to the sex of the organism, studies can also focus on age, which is related to the stage of its life cycle. Regarding tests with human cells, it was found that the two most commonly used are Caco-2 and A549. The former are epithelial cells isolated from colon tissue with colorectal adenocarcinoma (colon or rectal cancer) [145], whereas the latter are lung epithelial cells isolated from cancerous lung tissue [37, 146].

Regarding the particles used for the toxicity tests, 56% of the investigations studied NPs, 21% studied MPs, and 21% studied both types of particles (Figure 15.3). Some studies investigate the effects of NPs/MPs in a wide range of particle sizes [70, 120, 135]. In this chapter, particles with a size smaller than 1 µm were considered NPs, whereas MPs were considered plastic particles with sizes between 1

µm and 5 mm [147–149]. It was also found that just over half of the studies (58%) investigated spherical MPs/NPs; however, a significant percentage of the investigations (33%) did not report this information, 6% reported irregular particles, and 4% studies mentioned other types.

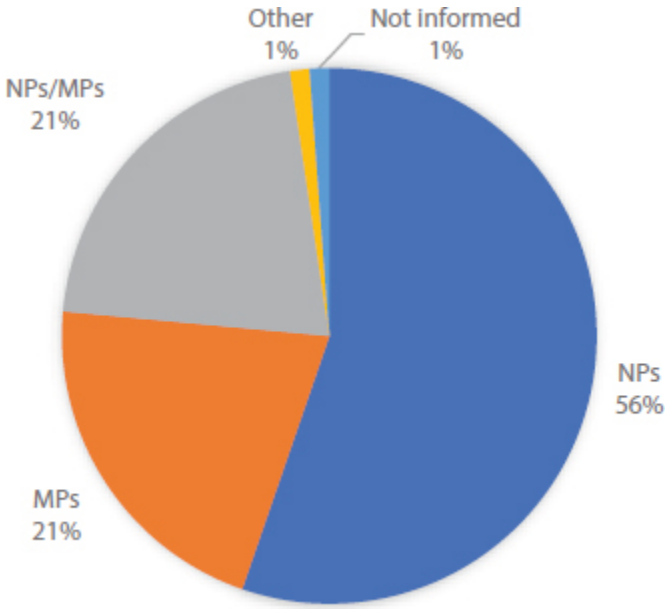


Figure 15.3 Percentage of studies that investigated nano- and microplastics analyzed in this study. Percentages were calculated based on the total number of studies analyzed (n = 85).

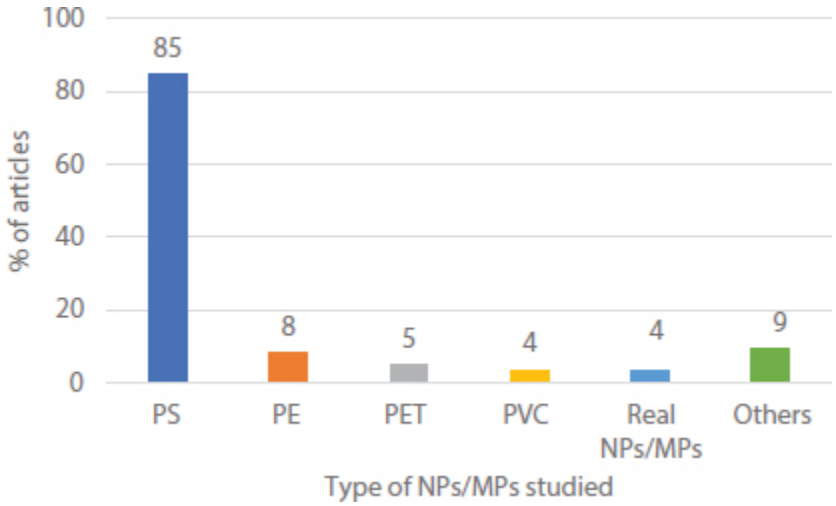


Figure 15.4 Types of nano- and microplastics used in toxicological studies analyzed in this study. Percentages were calculated based on the total number of studies analyzed (n = 85).

The most studied plastic type is polystyrene (PS), followed by polyethylene (PE), polyethylene terephthalate (PET),

and polyvinyl chloride (PVC) (Figure 15.4). The high use of PS in toxicity studies could be related to the fact that this type of plastic is one of the six most widely used commodity plastics worldwide, which, although it only represents 7% of global consumption [150], is found in a wide variety of single-use plastics, some of which (such as containers, cutlery, cups, plates, straws, and stirrers) come into direct contact with humans through the consumption of food and beverages [43, 73, 151] posing a health risk. In addition, other reasons for PS use could be that these are also one of the types of small waste that have been found the most in marine and coastal ecosystems in various parts of the world, which is why they have environmental relevance [152] and that present an easy suspension when evaluating biological effects [43].

15.5 Toxicity of Nano- and Microplastic Reported in the Literature

The literature review conducted during this research showed that there is increasing interest in investigating the effects of NPs/MPs in the human body and other biological systems. The research included *in vivo*, *in vitro*, *in silico*, tissue-, organ-in-chip, and human studies. However, it is evident that *in vivo* and *in vitro* tests were predominant. In this section, we report some of the toxicities identified in the articles that were analyzed, including particle uptake in cells or accumulation in tissues or organs, cell viability assays, stress response assays, cancer-related studies, genotoxicity, and reproductive assays.

15.5.1 Particle Uptake or Accumulation Studies

One of the tests commonly performed in NPs/MPs toxicity studies is particle uptake or particle accumulation in cells, tissues, or organs. Uptake or accumulation of NPs/MPs has been reported in human cells [42, 102, 103, 107, 108, 111, 121–123], human fluids [132], mice [73, 75, 76, 96, 97, 101], and mouse cells [76, 133, 137]. This type of test evaluates the effects of particles in the receptor systems (cells, organs, or tissues) and measures the transport or internalization of particles previously stained with a fluorescent dye, such as red Nile [102], to identify their location in the cells.

15.5.1.1 Human Cells

Some of the human cells studied include vascular endothelial EA.hy926, a somatic hybrid cell used for the investigation of cardiovascular diseases [153], colonic epithelial CCD841CoN from normal colon tissue [154], HepG2 isolated from a 15-year-old white male with liver cancer [155], embryonic kidney 293T cells with epithelial morphology from the kidney of an embryo [156], intestinal CCD-18Co with fibroblast morphology isolated from normal colonic tissue of a 2.5-month-old black female, macrophage mTHP-1 from the peripheral blood of a patient with acute monocytic leukemia, and Caco-2 and A549 cells [157, 158].

Studies that analyzed CCD841CoN, CCD-18Co, mTHP-1, A549, and Caco-2 cells found that internalization of small particles was higher than that of large particles. In colon cells (CCD841CoN, CCD-18Co), accumulation increased with exposure time. For particles with sizes of 0.1, 0.5, and 1 µm, the average number of particles at 0.5 h was 32, 27, and 21, respectively, while at 24 h, the number increased to 40, 32, and 22, respectively [103]. In addition, low uptake was observed for polystyrene particles with a size of

5 µm at exposure times of 0.5 and 24 h, with an average of 11 and 14 particles, respectively [103]. CCD-18Co cells showed 70%–90% internalization of NPs and 10%–20% internalization of MPs after 48 h of exposure, which suggests that rapid NPs and slow MPs internalization is produced through dependent and independent processes of adenosine triphosphate (ATP). Therefore, NPs/MPs aggregation mechanisms could contribute to a general increase in the particle size, which could affect their uptake and release [122]. In MTHP-1 cells, the kinetics of NPs and MPs uptake, indicated that the internalization process was rapid, with 21% and 27% of MPs and NPs uptake, respectively, after only 1 h of exposure to the highest dose. At 24 h, these percentages increased to 52% and 58%, corresponding to 2.08 and 2.32 µg per well of NPs and MPs, respectively [121].

In Caco-2 cancer cells, simultaneous exposure to MPs or NPs with perfluorooctane sulfonate (PFOS) was evaluated. PFOS significantly increased the absorption of polystyrene NPs particles by cells, while the cellular uptake of MPs was not significantly affected. However, the combined effect of MPs and PFOS treatment causes membrane damage, disrupting the integrity and fluidity of cell membranes and reducing their ability to transport substances [123]. A549 cells took up more fluorescent particle (flash red) MPs after 24 h than 72 h, and they did not show effects on cellular metabolism after 24 h and 72 h. MPs (fluorescent or non-fluorescent) showed a slight decrease in the viability of A549 cells compared to the control groups, with a strong negative correlation to dose [108].

15.5.1.2 Human Fluids

Regarding the toxicity of NPs/MPs in human fluids, only one study on gastric fluids was found, which reported that the kinetics of polystyrene NPs aggregation in gastric

environments are governed by complex interactions between the properties of plastic particles, the conditions of the gastric solution, and its components. The adsorbed gastric constituents alter the properties of the particles, which may also affect the food digestion process. Regarding pH, the results suggest that when the NPs are ingested during the fasting state (pH 2), they would probably form large aggregates (with a hydrodynamic diameter of approximately 1,500 nm in 20 min), while NPs ingested during food digestion (pH 5) could remain dispersed for more than 6 h. The authors concluded that it is very important to consider the digestive state when evaluating the toxicity of NPs in human gastric fluid [132].

15.5.1.3 Mice

Toxicity studies have focused on mice, mainly male subjects, and all the tests found in the literature review involved polystyrene particles. These studies detected the presence of particles in different organs, especially the kidneys. Meng *et al.* [101] reported that particles with sizes of 0.050, 0.300, 0.600, and 4 µm participated in the systemic circulation and accumulated in the kidney of mice. This effect was especially observed with 0.600 µm particles, which in turn could accumulate in an aggregated fashion and induce a severe biological response [101]. Xiong *et al.* [75] detected particles with sizes of 0.080, 0.5, and 5 µm in the kidney and intestine of mice; chronic exposure to these particles could induce kidney damage, which ultimately promotes renal fibrosis.

Sun *et al.* [73] studied the combined effect of MPs (size of 5 µm, concentrations of 0.012 or 0.120 mg/kg) with 0.080 mg/kg epoxiconazole in the liver and kidney. The results showed that epoxiconazole enhanced accumulation in tissues, tissue damage, dysfunction, oxidative stress, and metabolic disorders caused by MPs [73]. In contrast, the *in*

vivo studies by Shan *et al.* [97] showed that NPs (at a concentration of 0.5 to 50 mg/kg body weight for seven days) significantly induced increased blood–brain barrier permeability and accumulation in a dose-dependent manner in the brain of mice. In addition, NPs have been identified in microglia (central nervous system cells that function as immune system elements), which induce microglial activation and neuronal damage in the mice brain [97]. Finally, the reproductive toxicity of MPs was evaluated in female mice after continuous exposure for 35 days. The results showed the presence of MPs and their possible accumulation in the heart, liver, spleen, lung, kidney, brain, large intestine, small intestine, uterus, ovary, and blood. Concentrations ranged from 9.95 ± 5.74 µg/g·mL in the large intestine to 135.86 ± 7.23 µg/g·mL in blood [76].

15.5.1.4 Mice Cells

Studies found in the literature review on mouse cells have included polystyrene, polyethylene terephthalate, low-density polyethylene, and polypropylene particles. Liu *et al.* [159] showed that MPs in cells decreased ATP content, reduced mitochondrial membrane potential, damaged mitochondrial genome integrity, and caused a homeostasis imbalance between mitochondrial division and fusion [159]. In addition, Li *et al.* [133] found that NPs penetrated splenic lymphocytes, and that NPs with a diameter of 0.050 µm were more rapidly absorbed by cells. NPs also induce cell apoptosis, increase the expression of apoptosis-related proteins, cause the production of reactive oxygen species, alter mitochondrial membrane potential, impair mitochondrial function, and decrease cell viability [133]. Finally, a study by Deng *et al.* [137] with RAW264.7 cells from mouse macrophages, revealed that MPs and NPs originating from PET and PE plastic containers used for

food packaging could be immediately accumulated by cells and that short-term exposure to MPs caused inflammation, whereas MPs exposure substantially suppressed lysosomal activities and macrophage cell viability.

15.5.2 Cell Viability

The cytotoxicity of polystyrene and polyethylene NPs/MPs has been reported in *in vitro* studies: the exposure times in these studies varied from 2 to 72 h. The results showed a decrease in cellular viability [43, 95, 100, 106, 107, 116, 119, 120, 160] and a reduction in the protein levels of ZO-2 genes [95]. Halimu *et al*. [115] reported membrane potential changes and impaired cellular energy metabolism in mitochondria. In addition, exposure to polystyrene NPs was found to increase the risk of pulmonary diseases such as fibrosis [115]. Other factors that affect cell viability are simultaneous exposure to other chemicals, size of NPs/MPs, and surface charge of NPs/MPs [102, 115].

15.5.3 Stress Response

The cellular stress response is often studied through the detection of reactive oxygen species (ROS). ROS are unstable molecules that contain oxygen and react with other cellular molecules. ROS accumulation may damage DNA, RNA, and proteins, leading to cell death [161]. Increments in the Levels of ROS due to NPs/MPs exposure have been reported in human cells, tissues, and fluids (A549, U937, THP-1, Jurkat, HaCaT, Caco-2, trophoblast HTR-8/Svneo, murine microglia BV2, placenta choriocarcinoma JEG-3, small intestinal epithelial HIEC-6, and vascular endothelial EA. hy926 cells and Raji-B), mouse cells (NCM460, murine splenic lymphocytes, hippocampal neuronal HT22, spermatocyte line GC-2, and bladder epithelial cells), mice (C57BL/6, Kunming, offspring of Kunming pregnant female mice exposed to NPs/MPs), and

porcine coronary arteries [37, 38, 44, 45, 68, 72, 74, 75, 77, 88, 93, 94, 99, 102, 103, 112, 121–123, 128, 131, 133, 135, 143, 160, 162]. Upregulation of ROS production has been reported from NPs/MPs concentrations from 4×10^{-6} to 1,000 µg/mL in *in vitro* experiments and from 0.4 to 50 mg/kg body weight/day in *in vivo* experiments. Increased ROS production has been reported for polystyrene, polyethylene, polyethylene terephthalate, polypropylene, and polyvinyl chloride. Studies have shown dose- [37] and size-dependent responses [44, 123, 135]. ROS production also depends on the surface charge of NPs/MPs [44, 133]. Finally, the upregulation of ROS may be attenuated or enhanced by simultaneous exposure to other compounds [123, 128].

15.5.4 Cancer

The relationship between plastic pollution and cancer has mainly been associated with the presence of additives and sorbed hydrophobic pollutants. Flame retardants have been associated with an increase in thyroid cancer [163], and the risk of cancer development has been associated with the presence of phenanthrene and other polycyclic aromatic hydrocarbons sorbed on the surface of MPs [164, 165].

Regarding the toxicity of plastics by themselves, it has been reported that the inhalation of NPs could increase the risk of lung cancer [166], and that polystyrene NPs could increase the size of tumorous cells [160]. In contrast, a study on polystyrene NPs with cancer-related signaling pathways showed no significant effect [118].

15.5.5 Genotoxic Effects

Several genotoxic effects have been found to be associated with prolonged exposure to NPs/MPs (Figure 15.5), which

were the result of tests carried out mainly in mice, both *in vitro* and *in vivo*.

In vivo studies have observed synaptic dysfunction, mitochondrial decline, loss of neurons, energy metabolism disorders, ATP metabolic alterations, synaptic function, and neurodegeneration [83, 93, 99], whereas the effects identified *in vitro* experiments include increased oxidative stress in cells, loss of lysosomes, and damage to their mitochondrial structure and function [121, 123, 143].

Prolonged exposure to NPs/MPs affects cell structure and function, damages DNA, and in some cases, significantly increases cytotoxicity. It is important to highlight that current studies that have been carried out focused mainly on exposure to polystyrene; however, it is possible that there are undiscovered effects associated with exposure to other plastic materials.

15.5.6 Reproductive System

Sperm motility, malformations in the offspring, dysregulation of testosterone production, decreased fertility, and oocyte dysfunction have been associated with exposure to NPs/MPs [98, 162]. Exposure to MPs can be transported to the testicles through the bloodstream, modify the synthesis of testosterone, decrease the size of the litter, and decrease the survival of pups [68, 69, 124].

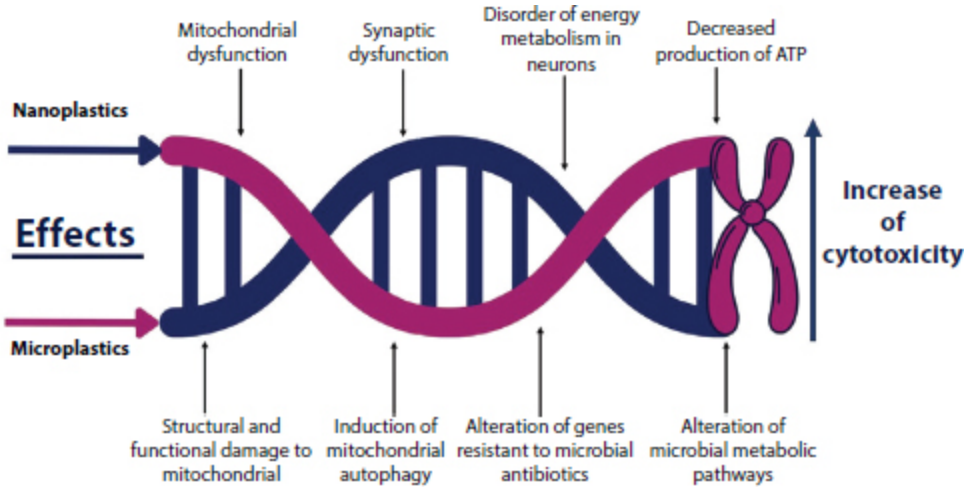


Figure 15.5 Genotoxic effects of nano- and microplastics.

As technology advances, it is possible to work on computational models that can predict toxic effects due to these materials with confidence levels of 95%, such as affectations in the placenta due to the inhibition of some enzymes [138], however, it is still necessary to perform physical tests to verify the prediction of the models.

15.6 Conclusions

Plastic pollution is an emerging environmental problem, and the ubiquitous presence of NPs/MPs in the environment and their accumulation in biological systems have raised concerns about the toxicity of these particles. Evaluation of the toxicity of chemical substances has traditionally been carried out through *in vivo* studies (with animal subjects). However, concerns about the value of animal studies in predicting human toxicity and international efforts to reduce the number of animals and increase the welfare of animals used in research have led to the development of alternative methods, such as *in vitro*, *in silico*, tissue engineering, and tissue- or organ-in chip assays. These alternative methods show promising results and present advantages, such as cost reduction (for *in vitro*

and *in silico* assays) and the time required to obtain toxicity data. However, in the foreseeable future animal tests will continue to be an essential part of toxicity evaluations.

Toxicity studies of NPs/MPs reported in the scientific literature include *in vivo*, *in vitro*, *in silico*, and tissue- or organ-in-chip assays. In mammals, the toxicity of NPs/MPs has predominantly been studied in mice and human cells. Research has focused on NPs (plastic particles smaller than 1 µm), virgin materials made of polystyrene, and spherical particles.

Toxic effects of NPs/MPs include uptake and accumulation of particles in cells, organs, and tissues, decrease in cell viability, increased production of reactive oxygen species, increased risk of lung cancer due to inhalation of NPs/MPs, and genotoxic and reproductive effects. Toxicity data for NPs/MPs show size- and dose-dependent responses in many cases.

As has been shown toxicity research is mainly conducted with virgin materials of spherical form, but real NPs/MPs are of irregular form, and their properties have changed due to weathering. Other factors that affect the toxicity of NPs/MPs are substances sorbed on their surface, simultaneous exposure to other chemicals, and the surface charge of NPs/MPs.

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16
Plastic-Related Chemicals: Occurrence in Environment and Ecotoxicological Impacts

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Abstract

The presence of plastic additives such as plasticizers and flame retardants, in different environments raises concerns about their impact on organisms and health. Plastic additives are found indoors, in water, in the soil, and in biota entering the body through dietary and non-dietary sources, potentially causing health issues. Studies have revealed significant findings, including increased oxidative stress, altered antioxidant enzyme activities, DNA damage, gene expression changes, and a modified gut microbiome in plasticizer-exposed earthworms. These effects occur even at low concentrations, indicating potential soil toxicity and the need for further investigation. Additionally, wheat plants can absorb and metabolize plasticizers, posing concerns about their bioaccumulation in the food chain and associated health risks. Flame retardants used for fire safety raise concerns owing to their persistence in the environment, ability to bioaccumulate, and associated health risks. These additives harm organisms such as marine mussels, fish, and soil inhabitants, affecting their immune, reproductive, endocrine, and neurological systems. Research has linked exposure to developmental, reproductive, endocrine, and neurotoxic effects, especially in

infants and children. Although progress has been made in understanding the toxicological aspects of plastic additives, knowledge gaps remain regarding their basic toxicological data, metabolic pathways, and potentially harmful effects, particularly for non-prioritized additives.

Keywords: Additives, plasticizers, PAEs, flame retardants, OPEs, bioaccumulation, oxidative stress, endocrine-disrupting chemicals

16.1 Introduction

The pervasive presence of plastic debris in natural ecosystems has made plastic pollution a prominent concern as plastic materials have become an indispensable part of modern society, serving a wide range of applications [1, 2]. Plastic production incorporates a diverse range of additives, classified into the following categories: functional additives, fillers, reinforcements, and colorants (Figure 16.1). Functional additives include plasticizers, flame retardants, stabilizers, antistatic agents, curing agents, lubricants, foaming agents, and biocides. The colorants include soluble azo dyes and pigments. Fillers are composed of materials such as kaolin, mica, talc, clay, calcium carbonate, and barium sulfate. Finally, reinforcements involve substances such as carbon and glass fibers [2].

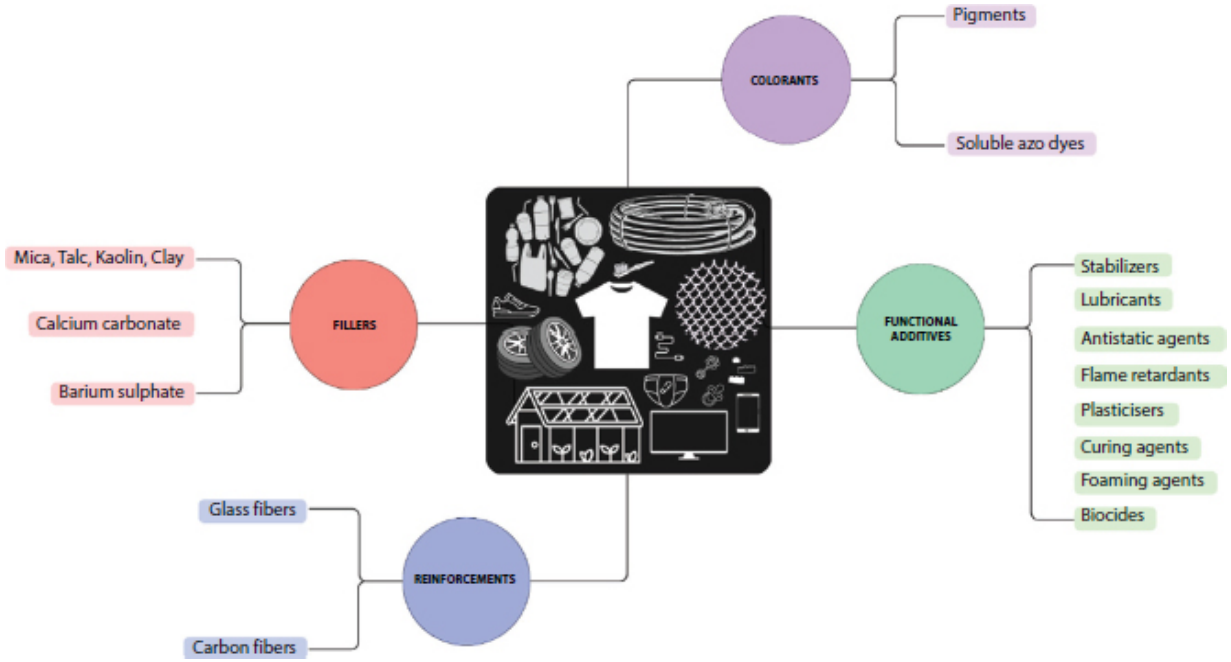


Figure 16.1 Four main categories of additives are used to maximize the performance and versatility of everyday plastics.

The incorporation of additives is essential in the production of plastics to meet various requirements for specific purposes, such as increasing the bulk, improving mechanical strength and flexibility, enhancing electrical conductivity, adding color, delaying oxidation, and preventing ignition [1, 3]. However, as additives are incorporated into plastics to modify the polymer structure, they might also affect the sorption capacity of the polymer [1]. Moreover, additives have a weak bond with polymers and therefore have the potential to leach out over time, influenced by the physicochemical properties of the additive and its surroundings, and the size of both the polymer pores and additive molecules [1, 3, 4]. This leaching process can be accelerated when plastic waste is exposed to UV degradation and/or high temperature [3, 5].

Within the environment, several plastic additives have been identified as endocrine disruptors or toxic, raising concerns about their potential impact. Examples are phthalates, bisphenol A, nonylphenols, polybrominated diphenyl ethers, and metals [1, 3, 5, 6]. These additives have been found in various environments, highlighting the need for a

comprehensive understanding of their ecotoxicological effects. In 2019, the European Chemicals Agency (ECHA) collaborated with PlasticEurope, Cefic, European Plastics Converters (EuPC), academics, and representatives from European Union (EU) member states to publish a comprehensive inventory of plastic additives used in the EU under REACH regulation. The initiative registered additives with an annual production exceeding 100 tons. The resulting inventory (available at https://echa.europa.eu/pt/mapping-exercise-plastic-additives-initiative) comprises 418 functional additives and provides information on commonly associated polymers and typical concentration ranges. The additives include stabilizers, antioxidants, nucleating agents, pigments, antistatic agents, plasticizers, and flame retardants [7].

The ecotoxicological consequences of plastic-related chemicals can be deepened by acknowledging the diverse range of additives used in plastic production and their potential environmental impacts. In the following sections, we explore the specific categories of plastic-related chemicals, namely plasticizers and flame retardants, and examine their effects on both the environment and living organisms.

16.2 Plasticizers

Plasticizers improve the processability of polymers, enhance the flexibility and toughness of the final products, and are among the most commonly used additives in the industry [2, 8]. They comprise different chemical families such as phthalate acid esters (PAEs) (also known as ortho-phthalates esters or phthalates), benzoate esters, aliphatic dibasic acid esters (adipates, azelates, sebacates, and citrates), terephthalate esters (including bisphenol A), and trimellitate esters, among others [2, 7, 9].

Annually, 7.5 million tons of plasticizers are consumed worldwide. Within Europe, the yearly utilization of additives amounts to approximately 1.35 million tons, with over 85% employed in flexible PVC applications such as electric cables,

flooring, wall coverings, and coatings [2, 7, 9, 10]. Among plasticizers, PAEs such as dibutyl phthalate (DBP) and bis (2-ethylhexyl) phthalate (DEHP) are the most widely consumed because of their exceptional compatibility with PVC [1, 11, 12].

PAEs have emerged as significant environmental pollutants and are classified as endocrine-disrupting chemicals with antiandrogenic properties, posing a significant risk to the human endocrine, reproductive, and respiratory systems [10, 13–17]. The United States Environmental Protection Agency (US EPA) includes DEHP, diethyl phthalate (DEP), di-n-butyl phthalate (DnBP), butyl benzyl phthalate (BBP), dimethyl phthalate (DMP), and di-n-octyl phthalate (DnOP) as priority pollutants [18]. The EU restricts the marketing of toys and childcare articles containing DEHP, BBP, DBP, and di-isobutyl phthalate (DiBP) at a concentration by weight, individually or in combination, greater than or equal to 0.1% [19]. Moreover, non-phthalate plasticizers (e.g., di(2-ethylhexyl) adipate (DEHA), trioctyl trimellitate (TOTM), acetyl tributyl citrate (ATBC)), and di-(iso-nonyl)-cyclohexane-1,2-dicarboxylate (DINCH)) have been used to replace phthalate plasticizers in various industrial applications, such as fabric coatings, food-packing materials, children's toys, and electrical cables [10, 11, 20]. However, due to the scarcity of information on the toxicity of these replacement plasticizers, there is a lack of understanding of the potential human health implications associated with chronic exposure [21].

16.2.1 Presence and Accumulation of PAEs in the Environment

PAEs have been extensively reported in air [18], wastewater [13], soil [10, 18], sludge [22], food [23], drinking water [14], and sediment [11] (Table 16.1). The presence of PAEs in soil has received significant attention because of the intensive application of plastics in agriculture, such as mulching and greenhouse cultivation, as well as improper plastic disposal methods, atmospheric deposition, sludge disposal, irrigation with reclaimed water, and landfilling [10, 13, 16, 22]. However,

the extent to which PAEs are released from the use and aging of agricultural plastic film remains uncertain, as it is influenced by their concentration, film material, and thickness [15]. Furthermore, it is worth mentioning that a quick search yields numerous Chinese studies in this field. China's dominance as the primary global producer of vegetables is closely linked to its position as the leading producer and consumer of agricultural films such as mulch and shed. In fact, China boasts an extensive agricultural film coverage area of over 253,333 km^2, which accounts for about 90% of the world's total greenhouse vegetable production [24]. Consequently, it comes as no surprise that soils from China exhibit some of the highest levels of PAEs globally, posing a significant human health risk through soil ingestion and the consumption of contaminated food [17].

Table 16.1 Occurrence of plasticizers in the environment, concentration ranges, and matrix sources (time span: 2017–2023).

PAE	Concentration	Matrix, Location	Ref.
Dimethyl phthalate (DMP)	64.6–126 pg m^{-3}	Outdoor air, China	[23]

PAE	Concentration	Matrix, Location	Ref.
	135–176 pg g^{-3}	Indoor air, China	
	<0.01 µg m^{-3}	(Greenhouse) air, China	[18]
	<0.01 µg kg^{-1}	(Greenhouse) soil, China	
	2.9–16.7 ng g^{-1} dw	Soil, China	[10]
	0.05–197 ng g^{-1} dw	Sediment, Korea	[11]
	14.31 ± 26.28 ng L^{-1}	Drinking water, China	[14]
	6.8–400 ng L^{-1}	Freshwater, Uganda	[8]
	16.7–37.4 ng L^{-1}	River water, China	[23]
	0.0006–1.12 µg g^{-1} dw	Sewage sludge, China	[22]
	<LOD–4.93 µg L^{-1}	Influent WWTP, Poland	[13]
	<LOD–6.50 µg L^{-1}	Effluent WWTP, Poland	
	2.46–24.6 µg L^{-1}	(Active) landfill leachates, Poland	
	7.77–26.3 µg L^{-1}	(Close) landfill leachates, Poland	

PAE	Concentration	Matrix, Location	Ref.
	2.66–3.67 ng g^{-1} dw	Fish, China	[23]
	4.18–5.96 ng g^{--} dw	Pig, China	
	3.36–3.68 ng g^{-1} dw	Chicken, China	
	3.53–3.82 ng g^{-1} dw	Cattle, China	
	21.0–38.4 ng g^{-1} dw	Vegetables, China	

PAE	Concentration	Matrix, Location	Ref.
Diethyl phthalate (DEP)	400–966 pg m^{-3}	Outdoor air, China	

PAE	Concentration	Matrix, Location	Ref.
	297–549 pg g^{-3}	Indoor air, China	
	<0.01 µg m^{-3}	(Greenhouse) air, China	[18]
	0.01–0.02 µg kg^{-1}	(Greenhouse) soil, China	
	1.9–7.2 ng g^{-1} dw	Soil, China	[10]
	0.07–10.3 ng g^{-1} dw	Sediment, Korea	[11]
	5.905 ± 11.57 ng L^{-1}	Drinking water, China	[14]
	38–1,100 ng L^{-1}	Freshwater, Uganda	[8]
	8.63–57.4 ng L^{-1}	River water, China	[23]
	0.0003–1.41 µg g^{-1} dw	Sewage sludge, China	[22]
	<LOD–4.05 µg L^{-1}	Influent WWTP, Poland	[13]
	<LOD–3.26 µg L^{-1}	Effluent WWTP, Poland	
	<LOD–14.5 µg L^{-} –	(Active) landfill leachates, Poland	
	0.19–9.48 µg L^{-1}	(Close) landfill leachates, Poland	

PAE	Concentration	Matrix, Location	Ref.
	0.57–1.64 ng g^{-1} dw	Fish, China	[23]
	1.80–16.7 ng g^{-1} dw	Pig, China	
	0.33–1.44 ng g^{-1} dw	Chicken, China	
	0.35–0.81 ng g^{-1} dw	Cattle, China	
	2.42–69.1 ng g^{-1} dw	Vegetables, China	

PAE	Concentration	Matrix, Location	Ref.
Di-isobutyl phthalate (DiBP)	172–708 pg m^{-3}	Outdoor air, China	

PAE	Concentration	Matrix, Location	Ref.
	90.1–152 pg g^{-3}	Indoor air, China	
	0.01–0.05 µg m^{-3}	(Greenhouse) air, China	[18]
	0.16–0.58 µg kg^{-1}	(Greenhouse) soil, China	
	9.4–123.2 ng g^{-1} dw	Soil, China	[10]
	<0.02–11.2 ng g^{-1} dw	Sediment, Korea	[11]
	103.8 ± 310.5 ng L^{-1}	Drinking water, China	[14]
	77.4–205 ng L^{-1}	River water, China	[23]
	0.0003–5.92 µg g^{-1} dw	Sewage sludge, China	[22]
	<LOD–93.1 µg L^{-1}	Influent WWTP, Poland	[13]
	<LOD–76.2 µg L^{-1}	Effluent WWTP, Poland	
	<LOD–7.39 µg L^{-1}	(Active) landfill leachates, Poland	
	<LOD–4.33 µg L^{-1}	(Close) landfill leachates, Poland	

PAE	Concentration	Matrix, Location	Ref.
	2.00–2.40 ng g^{-1} dw	Fish, China	[23]
	4.20–12.9 ng g^{-1} dw	Pig, China	
	1.30–3.73 ng g^{-1} dw	Chicken, China	
	1.57–2.36 ng g^{-1} dw	Cattle, China	
	0.63–77.8 ng g^{-1} dw	Vegetables, China	

PAE	Concentration	Matrix, Location	Ref.
Di-n-butyl phthalate (DnBP)	319–666 pg m^{-3}	Outdoor air, China	
	146–334 pg g^{-3}	Indoor air, China	
	0.01–0.03 µg m^{-3}	(Greenhouse) air, China	[18]
	0.14–0.55 µg kg^{-1}	(Greenhouse) soil, China	
	9.2–100.7 ng g^{-1} dw	Soil, China	[10]
	0.48–56.2 ng g^{-1} dw	Sediment, Korea	[11]
	595.9 ± 1794 ng L^{-1}	Drinking water, China	[14]
	135–801 ng L^{-1}	River water, China	[23]
	0.0004–5.31 µg g^{-1} dw	Sewage sludge, China	[22]

PAE	Concentration	Matrix, Location	Ref.
	21.0–44.5 ng g^{-1} dw	Fish, China	[23]
	26.4–34.0 ng g^{-1} dw	Pig, China	
	17.9–24.7 ng g^{-1} dw	Chicken, China	
	19.6–22.3 ng g^{-1} dw	Cattle, China	
	155–338 ng g^{-1} dw	Vegetables, China	

PAE	Concentration	Matrix, Location	Ref.
Benzyl butyl phthalate (BBP)	N.D.—209 pg m^{-3}	Outdoor air, China	
	32.4–64.7 pg g^{-3}	Indoor air, China	
	<0.01 µg m^{-3}	(Greenhouse) air, China	[18]
	<0.01–0.02 µg kg^{-1}	(Greenhouse) soil, China	
	<0.15–1,322 ng g^{-1} dw	Sediment, Korea	[11]
	0.0011–0.0149 µg g^{-1} dw	Sewage sludge, China	[22]
	0.53–1.35 ng L^{-1}	River water, China	[23]
	0.12–1.19 ng g^{-1} dw	Fish, China	
	0.41–1.16 ng g^{-1} dw	Pig, China	
	0.17–0.42 ng g^{-1} dw	Chicken, China	
	0.17–0.92 ng g^{-1} dw	Cattle, China	
	1.03–1.68 ng g^{-1} dw	Vegetables, China	

PAE	Concentration	Matrix, Location	Ref.
Dibutyl phthalate (DBP)	3.5–16,000 ng L^{-1}	Freshwater, Uganda	[8]
	<LOD–65.6 µg L^{-1}	Influent WWTP, Poland	[13]
	<LOD–365.8 µg L^{-1}	Effluent WWTP, Poland	
	<LOD–6.34 µg L^{-1}	(Active) landfill leachates, Poland	
	0.27–2.41 µg L^{-1}	(Close) landfill leachates, Poland	

PAE	Concentration	Matrix, Location	Ref.
Di(2-ethylhexyl) phthalate (DEHP)	713–2,600 pg m^{-3}	Outdoor air, China	[23]

PAE	Concentration	Matrix, Location	Ref.
	1,550–2,300 pg g^{-3}	Indoor air, China	
	0.03–1.32 µg m^{-3}	(Greenhouse) air, China	[18]
	0.04–1.04 µg kg^{-1}	(Greenhouse) soil, China	
	36.9–1142.2 ng g^{-1} dw	Soil, China	[10]
	1.00–2740 ng g^{-1} dw	Sediment, Korea	[11]
	178.2 ± 422.0 ng L^{-1}	Drinking water, China	[14]
	210–23,000 ng L^{-1}	Freshwater, Uganda	[8]
	73.3–261 ng L^{-1}	River water, China	[23]
	0.326–67.8 µg g^{-1} dw	Sewage sludge, China	[22]
	<LOD–143.3 µg L^{-1}	Influent WWTP, Poland	[13]
	<LOD–27.1 µg L^{-1}	Effluent WWTP, Poland	
	<LOD–249 µg L^{-1}	(Active) landfill leachates, Poland	
	<LOD–143 µg L^{-1}	(Close) landfill leachates, Poland	

PAE	Concentration	Matrix, Location	Ref.
	2.05–52.4 ng g^{-1} dw	Fish, China	[23]
	1.00–9.09 ng g^{-1} dw	Pig, China	
	1.83–30.5 ng g^{-1} dw	Chicken, China	
	N.D.–6.78 ng g^{-1} dw	Cattle, China	
	12.6–75.9 ng g^{-1} dw	Vegetables, China	
Di-isononyl phthalate (DINP)	21.3–553 ng g^{-1} dw	Sediment, Korea	[11]
	<LOD–142.8 µg L^{-1}	Influent WWTP, Poland	[13]
	<LOD–25.2 µg L^{-1}	Effluent WWTP, Poland	
Diisodecyl phthalate (DiDP)	<3.92–169 ng g^{-1} dw	Sediment, Korea	[11]
Dicyclohexyl phthalate (DCHP)	6.9 ng g^{-1} dw	Soil, China	[10]
	0.0014–0.0836 µg g^{-1} dw	Sewage sludge, China	[22]
Dihexyl phthalate (DNHP)	1.5 ng g^{-1} dw	Soil, China	[10]
Hexyl 2-ethylhexyl phthalate (HEHP)	8.7 ng g^{-1} dw		

PAE	Concentration	Matrix, Location	Ref.
Dipropyl phthalate (DPP)	<LOD–80.3 µg L^{-1}	Influent WWTP, Poland	[13]
	<LOD–31.2 µg L^{-1}	Effluent WWTP, Poland	
	<LOD–0.96 µg L^{-1}	(Active) landfill leachates, Poland	
	<LOD–11.4 µg L^{-1}	(Close) landfill leachates, Poland	
Diisoheptyl phthalate (DiHP)	<LOD–61.7 µg L^{-1}	Influent WWTP, Poland	
	<LOD–17.7 µg L^{-1}	Effluent WWTP, Poland	
Di-n-octyl Phthalate (DNOP)	0.50–1.06 ng L^{-1}	River water, China	[23]
	0.0036–0.0687 µg g^{-1} dw	Effluent WWTP, Poland	[22]
	N.D.–0.22 ng g^{-1} dw	Fish, China	[23]
	N.D.–0.13 ng g^{-1} dw	Pig, China	
	0.13–0.15 ng g^{-1} dw	Chicken, China	
	N.D.–0.14 ng g^{-1} dw	Cattle, China	
	N.D.–1.37 ng g^{-1} dw	Vegetables, China	

PAE	Concentration	Matrix, Location	Ref.
Di(2-ethylhexyl) adipate (DEHA)	12–6,100 ng L^{-1}	Sewage sludge, China	[8]
N-butyl benzenesulfonamide (NBBS)	7.5–200 ng L^{-1}	Freshwater, Uganda	

N.D., not detected; LOD, limit of detection; dw, dry weight.

Li *et al.* (2021) carried out a study investigating the distribution of 17 PAEs congeners at the soil-air interface in plastic-covered greenhouses from autumn to spring. The highest concentrations of PAEs, whether in the air or soil, were recorded during spring [18]. DEHP was identified as the dominant congener detected in the soil during spring and in the air during both seasons, whereas DiBP was the major congener detected in the soil during autumn. The researchers concluded that the concentrations of both Σ17 PAEs and individual congeners in the greenhouse system are influenced by the partition coefficient of each congener and seasonal variation, which in turn are affected by temperature and farming practices [18]. Furthermore, the study demonstrated that the soil acts as a sink, adsorbing leached PAEs from plastic debris, as even restricted compounds such as DEHP or DnBP, were still detected [18].

Zhu *et al.* (2019) conducted another study to assess the presence of nine PAE congeners in sludge from 46 WWTPs. The study identified DEHP as the predominant congener, followed by DBP and DiBP. Researchers estimated the daily intake of DBP and DEHP from soil amended with sludge, suggesting minor exposure to humans [22]. Kotowska *et al.* (2020) concluded that the efficiency of PAE removal from wastewater through activated sludge depends on the chemical properties of each congener and the amount of coagulant used during the purification process in WWTPs [13]. Moreover, phthalate monoesters (MPAEs), the primary degradation products of

PAEs, are typically excreted in urine and frequently detected in wastewater and drinking water [10, 14]. The presence of MPAEs in these environmental matrices highlights the possibility of human exposure to PAEs, leading to concerns regarding their potential impact on human health. For example, the overall intake of DEHP can be estimated from the concentration of mono (2-ethyl-5-hydroxyhexyl) phthalate (MEHHP) in urine [14].

He *et al.* (2020) collected samples from various sources, including indoor and outdoor air, river water, fish, vegetables, and domestic livestock (pigs, chickens, and cattle), in an agricultural region. In all sampled matrices, DEHP, DMP, DEP, BBP, DiBP, and DnBP were detected. Among these PAEs, DnBP is the major congener detected in animal and plant tissues. Vegetable samples exhibited, highest levels of PAEs (330 ng g^{-1} dw), pig samples (51.3 ng g^{-1} dw), fish samples (49.2 ng g^{-1} dw), chicken samples (44.8 ng g^{-1} dw), and cattle samples (30.7 ng g^{-1} dw) [23]. Edwards *et al.* (2021) searched for PAEs and replacement plasticizers in 64 food samples, including hamburgers and chicken burritos, obtained from fast food restaurants in the USA. The results indicated that all tested plasticizers were present in all food samples analyzed. DnBP exhibited the highest incidence, detected in 81% of the samples, followed by DEHP at 70%. In addition, the replacement plasticizers DINCH and DEHA were detected in 14% and 41% of the analyzed samples, respectively [21]. The prevalence of DnBP detected in both studies may be related to its extensive use and may indicate a risk of bioaccumulation within organisms.

16.2.2 Ecotoxicological Effects of PAEs

Given the widespread occurrence of plasticizers in distinct environmental settings, as stated in Table 16.1, it is expected that organisms in terrestrial and aquatic ecosystems encounter and interact with these compounds, which can also be associated with humans. Despite advancements in understanding the toxicological aspects of PAEs, there are still

gaps in knowledge regarding their basic toxicological data, metabolic pathways, and potentially harmful effects. This is particularly true for PAEs that are currently not included in the US EPA list [25].

In the ecotoxicology assessment of soil, invertebrates are commonly employed due to their high reproductive capacity, widespread presence, easy collection, and strong interaction with soil [26]. Earthworms, such as *Eisenia fetida*, play a crucial role as indicators of soil pollution due to their importance in soil processes, their sensitivity to environmental contaminants, and their contribution of up to 60% of the total biomass of soil-dwelling animals. These organisms effectively assess the impact of contaminants on soil inhabitants, serving as a dependable benchmark for measuring toxicity and ensuring the safety of the soil environment [27–29].

Studies on the effects of specific plasticizers, such as DEHP and DBP, on earthworms have revealed significant findings. In the case of DEHP, increasing concentrations led to increased malondialdehyde (MDA) and metallothionein (MT) content, elevated superoxide dismutase (SOD) activity, heat shock protein 70 (HSP 70) expression, and genotoxicity parameters of *E. fetida*. Nonetheless, at a low DEHP concentration of 1 mg kg^{-1}, noticeable decreases in peroxidase (POD) activity and neutral red retention time (NRRT) were observed, along with alterations in mitochondrial membrane potential difference values [28]. Similarly, earthworms exposed to various concentrations of DBP exhibited increased antioxidant enzyme activities, apart from the lowest concentration tested, which was 0.1 mg kg^{-1}. Elevated concentrations of DBP led to the excessive production of reactive oxygen species (ROS), causing DNA damage in earthworm coelomocytes and significant increments in MDA content [27]. Furthermore, the exposure of *E. fetida* to di-isononyl phthalate (DiNP) affected the gut microbiome, leading to a decrease in the Chao index and alterations in the abundance of *Chloroflexi*, *Proteobacteria*, and *Ottowia* in the surrounding soil [30]. Additionally, a dose–response correlation was observed between DNA damage and

the plasticizers DMP, DnOP, and BBP. Among them, DMP exhibited the highest Olive Tail Moment (OTM) value, indicating the most significant DNA damage, followed by DnOP and BBP [26]. These findings emphasize the potential of PAEs to induce oxidative stress, cytotoxicity, and DNA damage in soil organisms. Moreover, Wu *et al.* (2022) investigated the impact of three PAEs (DEHP, DBP, and DMP) on soil metabolic functions and revealed their disruptive effects on the complexity of the symbiotic network among soil bacterial communities, resulting in a reduction of bacterial interactions. Notably, exposure to DBP caused substantial alterations in the soil metabolite profile, with 100 metabolites showing upregulation and 72 showing downregulation among 172 differential metabolites. Furthermore, the analysis of pathway enrichment provided insight into the effects of DBP, as it was found to interfere with 43 metabolic pathways, including fundamental processes such as residual steroid metabolism and plasticizer metabolism [31].

In a study reported by Gao *et al.* (2019), it was found that wheat plants can absorb and metabolize PAEs, with translocation of the compounds from roots to aboveground tissues influenced by soil type and PAES concentrations. Higher stress concentrations resulted in increased DBP content in wheat tissues, although the content decreased as the plants continued to grow. The presence of DBP metabolites in both soil and plant tissues indicates that wheat plants can efficiently metabolize DBP into monoesters. This metabolic process was facilitated by the increased content of glutathione S-transferase (GST) and cytochrome P450 (CYP450), suggesting the existence of similar pathways for DBP metabolism and detoxification between plants and mammals [32]. Moreover, DBP adversely affects the germination and growth of *Brassica rapa var. chinensis*, disrupting antioxidant defense and phytohormone balance. It also leads to shifts in microbial populations, particularly in rhizosphere soil and roots, which can impact nutrient availability and ecosystem functions [33]. The ecotoxicological impacts of DBP on vegetables, coupled

with the accumulation of DBP or its metabolites in tissues, can potentially increase health risks through the process of bioaccumulation in the food web [26]. These findings underscore the potential hazards linked to PAE exposure through the intake of vegetables cultivated in contaminated soils.

Although soil is commonly identified as a significant reservoir of PAEs, it is important to acknowledge that these compounds have also been found in the aquatic environment. Therefore, it is to be expected that aquatic organisms are affected by the presence of PAEs. Indeed, DBP has been found to induce lipid peroxidation, oxidative stress, and DNA damage in zebra fish (*Dania rerio*) [34]. Similarly, DEHP has been shown to disrupt thyroid hormone activity, reducing the eye size and total body length of Japanese medaka (*Oryzias latipes*) [35]. These findings suggest that exposure to PAEs can lead to cellular damage, developmental abnormalities, and potential health risks to aquatic organisms.

16.2.3 Human Exposure to PAEs

There is a recognized association between phthalate exposure and adverse effects on the endocrine, reproductive, and cardiovascular systems in humans. The concern surrounding human exposure to PAEs is significant because these compounds can be absorbed by a variety of routes, including inhalation, ingestion, and dermal absorption by a variety of sources, including food (e.g., fish, livestock, and vegetables), water, air, dust, and soil [14, 17, 23]. This concern is supported by the high detection frequencies of PAE metabolites in human matrices [36]. It is important to note that human exposure pathways may vary in regions with different PAE sources [23].

Concerning PAE-contaminated soil, both the food chain and dermal adsorption are pointed out as the primary routes of human exposure to these pollutants. The utilization of shed films, specifically PVC or metallocene polyethylene (mPE), which have DEHP concentrations above 140,000 mg kg^{-1} and

total PAE concentrations ranging from 2.59 to 359 mg kg^{-1}, illustrates the potential ecological impact of PAEs in agricultural environments [15]. DEHP is readily released into the environment from films at high concentrations and can be absorbed by plant leaves, thus presenting a potential risk of PAE exposure through ingestion. Wang *et al.* (2021) concluded that the DEHP content in the air of plastic greenhouses, particularly those with mPE and PVC films, exceeds the acceptable levels set by the US EPA [15]. Furthermore, Zhou *et al.* (2021) evaluated the non-carcinogenic and carcinogenic risks associated with exposure to PAEs in Huang-Huai-Hai (China). The researchers focused on both non-dietary and dietary exposures to determine the major pathways impacting human health. The study found that the non-carcinogenic risk from dietary DBP was higher in soils used for vegetable production than in soils used for other crops due to the higher concentration of this substance. For DEHP, the relationship was reversed, except for children in Tianjin Province. The differences in DEHP risk among provinces were influenced by various factors, including the DEHP concentration in each cropping system and the dietary habits of the population. In provinces where DEHP concentrations in vegetable and cropland soils were comparable, population dietary habits emerged as the most important factor influencing the non-carcinogenic risk associated with DEHP. However, in regions with significant differences in DEHP concentrations, soil concentrations were found to be the most important determinant of non-carcinogenic risk from DEHP. Children are more susceptible to non-carcinogenic risks from exposure to PAEs that are not ingested in the diet. Therefore, to accurately assess the health risks associated with PAEs, it is essential to fully consider the type and concentration of PAEs and the dietary habits of the population [25]. This comprehensive approach ensures a more accurate estimation of the potential health effects of PAE exposure.

16.3 Flame Retardants

Flame retardants are used to prevent the ignition of a particular material or to slow the development of a fire. Rather than being chemically bonded, flame retardants are typically blended with polymeric materials. This property facilitates the release of these compounds into the environment through processes such as leaching, volatilization, and abrasion, which can affect various environmental compartments [37–39]. For several decades, halogenated flame retardants, including polybrominated biphenyls (PBBs), polybrominated diphenyl ethers (PBDEs), hexabromocyclododecane (HBCDD), and tetrabromobisphenol A (TBBPA), have been extensively used in various materials such as plastics in electronics, printed circuit boards, and insulation [40, 41]. These compounds exhibit resistance to degradation by chemical, biological, and photolytic processes, making them persistent organic pollutants (POPs) [40, 42].

Owing to their lipophilic nature, halogenated flame retardants have the potential to bioaccumulate in human, animal, and plant tissues, which may lead to adverse environmental and human health effects [42–45]. In fact, exposure to PBDEs has been associated with several human health effects, including diabetes, thyroid disorders, and impairments in executive functions and attention control [41, 44, 46, 47]. Consequently, these chemical compounds have become a matter of international concern, leading to restrictions and legislative bans. For instance, in the EU, the use of BDEs and PBDEs in electrical and electronic devices, systems, and toys is regulated by the European Restriction of Hazardous Substances (RoHS) Directive (2002/95/EC). Moreover, commercial formulations of PBDEs, specifically octabromodiphenyl ether and pentabromodiphenyl ether, were classified as persistent organic pollutants (POPs) and listed for elimination in Annex A of the Stockholm Convention in 2009. In 2013, HBCDD was also added to the list, and in 2017, decabromodiphenyl ether

(BDE 209) was added, leading to restrictions on its production, use, or marketing in 2019 [40, 42].

The restrictions have resulted in an increase in the manufacture and use of substitute products such as novel brominated flame retardants (NBFRs) and organophosphorus flame retardants (OPEs) [41, 48]. The recognition of NBFRs as environmental pollutants has emerged relatively recently, as they have replaced PBDEs in many products [41, 49]. For instance, decabromodiphenyl ethane (DBDPE) is being employed as a replacement for BDE 209 [47]. Recent studies have demonstrated that NBFRs follow the same release mechanisms as PBDEs and exhibit comparable hazardous characteristics when released into the environment, raising concerns about potential environmental and human health impacts [47, 49].

OPEs, which are used not only as flame retardants but also as plasticizers and defoamers, are extensively used in several industries such as plastics, furniture, textiles, construction, electronics, vehicles, and petroleum [37, 48]. OPEs have been hypothesized to be less harmful to human and environmental health due to their shorter half-lives and less persistence than older flame retardants; however, it is worth noting that OPEs have already been detected in water, air, soil, sediments, biota, plants, and biological samples (e.g., urine and breast milk) [38, 45, 50–53]. Despite this, there are limited regulations in place for OPEs [53–55]. Nevertheless, recent toxicological and epidemiological studies suggest that certain OPEs, including tris(2-chloroethyl)phosphate (TCEP), are suspected to be linked to reproductive toxicity, neurotoxicity, endocrine disruption, and carcinogenicity [51, 55]. Therefore, the presence of OPEs in the environment may pose risks to human health via multiple pathways, including dermal contact, inhalation, and both dietary and non-dietary ingestion.

16.3.1 Presence and Accumulation of Flame Retardants in the Environment

In response to the recognition of adverse health effects, the production of PBDE formulations has been phased out in many countries due to guidelines restricting these substances. However, PBDEs can still be detected in several environmental compartments, either due to poor waste management or because products containing them are still in use. Thus, despite the cessation of production, the legacy of PBDEs remains a concern because they can persist in the environment for long periods of time, bioaccumulate, and affect health. For example, a study by Falandysz *et al.* (2019) examined PBDE concentrations in oils and canned livers from the Baltic Sea and North Atlantic from 1972 to 2017. Total concentrations of BDEs varied between 9.9 and 415 ng g^{-1} in oils, while they ranged between 10.5 and 13 ng g^{-1} in canned livers. In all samples, 4,4′-tetrabromodiphenyl ether (BDE-47) was the predominant PBDE, with the highest concentration of 308 ng g^{-1} measured in a Baltic cod liver oil sample from the year 1993 [56].

NBFRs have been reported in atmospheric samples from various regions, including Europe, America, Asia, and Africa. The concentrations of NBFRs in these samples were higher than or comparable to those of PBDEs [49]. A study conducted in Australia analyzed 30 soil samples from different industrial and non-industrial areas. Six NBFRs were detected in 24 of the samples, namely 2-ethylhexyl-2,3,4,5-tetrabromobenzoate (EH-TBB), pentabromoethylbenzene (PBEB), 1,2-bis(2,4,6-tribromophenoxy) ethane (BTBPE), pentabromotoluene (PBT), hexabromobenzene (HBB), and decabromodiphenylethane (DBDPE), with concentrations ranging from non-detectable levels to 384 ng g^{-1} dw. HBB was the predominant compound, while DBDPE had the highest concentrations. The study highlights that electronic polymer manufacturing and waste recycling activities are significant sources of soil pollution in NBFRs. In contrast, non-industrial areas showed minimal impacts. These findings suggest that, similar to PBDEs, NBFRs

have the potential to contaminate soil [49]. In a study by Xiong *et al.* (2021), nine NBFRs were detected in soil moss (*Sanionia uncinata*) and lichen (*Usnea aurantiaco-atra*) from Antarctica. The dominant NBFR found was decabromodiphenylethane (DBDPE). Although NBFR concentrations were low, their presence suggests long-range atmospheric transport and deposition into soil, as with the banned BFRs. Moreover, NBFR concentrations were found to be higher in plant samples than in soil samples, indicating a potential for bioaccumulation in wildlife [39]. Furthermore, it should be noted that as the different environmental compartments are interconnected, translocation of contaminants is also expected to occur in aquatic environments.

Regarding OPEs, their concentrations in the environment have shown a gradual increase over time, which is believed to be directly linked to their growing use following the phaseout of halogenated flame retardants. OPEs have higher vapor pressures than PBDEs and can be more readily released into the atmosphere by processes such as volatilization or combustion of products and materials containing them. They can then become airborne and be deposited in water bodies or on land. Environmental measurements have provided clear evidence that OPEs exhibit long-range environmental transport and accumulation comparable to PBDEs, despite initial expectations, classifying them as persistent mobile organic compounds (PMOCs) [55]. Studies indicate that OPEs can indeed bioaccumulate in aquatic environments, including fish, where these compounds gradually accumulate in the tissues over time. Fish are exposed to OPEs primarily through the ingestion of contaminated food and water [38, 57]. In a study conducted by He *et al.* [38] in an agricultural area in Chongqing, China, the concentrations of OPEs in various biotic samples were investigated. The results showed that fish samples with a concentration of 960 ng g^{-1} lw had the highest OPE levels. This was followed by chicken samples with a concentration of 676 ng g^{-1} lw, while samples from cattle had a concentration of 545 ng g^{-1} lw, and samples from pigs

exhibited the lowest concentration of 535 ng g^{-1} lw. This bioaccumulation process raises concerns that OPEs may enter the food chain.

High concentrations of triphenyl phosphate (TPhP), tricresyl phosphate (TCP), tri(2-ethylhexyl) phosphate (TEHP), and tris(2-chloroethyl) phosphate (TCEP) have been detected in Lake Victoria, Uganda. The concentrations found were N.D.–8,100 ng L^{-1} for TCP, 24–6,500 ng L^{-1} for TCEP, 54–4,300 ng L^{-1} for TPhP, and N.D.–4,300 ng L^{-1} for TEHP. The high concentrations may be attributed to the lack of strict regulations for the disposal of OPE-containing industrial waste, but also to atmospheric deposition [8]. In a study by Wang *et al.* [48], 75 sewage sludge samples from 67 WWTPs were analyzed and a mean concentration of 1,290 ng g^{-1} dw ($\sum$20 OPEs) was found, with tributoxyethyl phosphate (TBEP) and TEHP being the most abundant compounds. These findings highlight that OPEs can accumulate in sewage sludge and can transfer from soil to crops when applied as fertilizers in agriculture.

In plants, the accumulation of OPEs varies among different cultivars, as observed by Hu *et al.* [45], with carrots (*Daucus carota L.*) exhibiting a higher preference for OPE enrichment compared to maize (*Zea mays L.*). The behavior of OPEs in the soil-plant system is affected by the characteristics of both plant (lipid content) and OPES (hydrophobicity and molecular geometry). OPEs exhibit a broad range of K_{OW}, indicating diverse environmental behavior within congeners. Plants with a high lipid content tend to accumulate hydrophobic chemicals, and this accumulation is influenced by root lipid substances, primarily affecting the distribution of organic compounds near the root epidermis. Although the specific role of plant lipid content in the uptake and movement of OPEs remains unclear, it may contribute to the overall fate of OPEs in the soil-plant system. Additionally, OPEs with electron-ring substituents such as tris(methylphenyl) phosphate (TMPP), TPhP, tris(1,3-dichloro-2-propyl) phosphate (TDCPP), and TCPP have higher

potential for uptake by plant roots compared to OPEs with open-chain substituents such as tri-n-butyl phosphate (TnBP), 2-ethylhexyl diphenyl phosphate (EHDPP), TEP, TEHP, TCEP, and TBOEP. The variance in uptake ability can be attributed to the electron-rich structure of OPEs, which enables stronger non-covalent interactions and promotes their enhanced sorption onto charged surfaces of roots [45]. Furthermore, hydrolysis of organophosphate triesters into their metabolites was observed in the cell wall of wheat plants (*Triticum aestivum L.*), suggesting that organophosphate diesters have higher stability in plants compared to organophosphate triesters [58].

A study conducted in China investigated the occurrence and distribution of 14 OPEs in 65 road dust samples. Among the OPEs analyzed, tris(2-chloro-1-methylethyl) phosphate (TCPP) was the most abundant, with a concentration of 384 mg kg^{-1} [59]. Another study conducted at a recycling facility concluded that primitive outdoor recycling operations and open storage of waste were the primary factors for the release and dispersion of OPEs into the soil and surrounding dust. The levels of TCEP, TPhP, and TMPP in the soil indicated a significant ecotoxicological risk at the outdoor waste recycling facilities. In addition to the risk of environmental contamination, the study also highlighted the occupational risk associated with working at these recycling sites. Workers may be exposed through dermal contact, ingestion, or inhalation, which may pose health risks [60]. Table 16.2 provides some examples of the presence of NBFR and OPE in the environment. The table includes information on concentration ranges and the sources of these flame retardants in several matrices.

16.3.2 Ecotoxicological Effects of Flame Retardants

Owing to their lipophilic nature, high octanol-water partition coefficients (K_{OW}) values, and poor water solubility, numerous flame retardants tend to persist in water bodies and sediments. The persistence of flame retardants in aquatic environments

raises substantial concerns due to their potential to bioaccumulate and trigger harmful effects in exposed organisms [57]. Pollutants can have a dual effect on organisms, directly impacting their immune, reproductive, or endocrine systems while also leading to diverse changes in bioenergetics and the development of tissues and specimens [61].

Marine mussels such as *Mytilus galloprovincialis* are commonly employed as bioindicators owing to their ability to accumulate and withstand high concentrations of environmental pollutants. The exposure of *M. galloprovincialis* to environmentally relevant concentrations of BDE-47 was found to significantly affect cell activity within the digestive gland. The limited elimination of BDE-47 through biotransformation, observed after a 30-day period, may have long-term effects on the growth, survival, and reproduction of *M. galloprovincialis* [61]. In juvenile crucian carp (*Carassius auratus*), exposure to environmentally relevant doses of HBCDD resulted in the accumulation of the substance primarily in liver and muscle tissue. The decrease in thyroxine and total triiodothyronine levels indicated a disruption in the thyroid hormone function of *C. auratus* [62]. Thyroid function plays an essential role in regulating various physiological processes, including cell development, thermoregulation, metabolism, growth, and reproduction. TPhP also affects thyroid function in fish, mammals, and birds. In Japanese quail (*Coturnix japonica*), TPhP exposure did not affect mRNA expression of thyroid-related genes, but it had differential effects on thyroid structure and activity in males and females. Males showed increased thyroid structure at high TPhP exposure, whereas females had suppressed thyroid structure and activity across all TPhP exposure levels. In addition, highly-TPhP-exposed females showed a decrease in circulating free triiodothyronine levels [63].

Table 16.2 Occurrence of NBFRs and OPES in the environment, concentration ranges, and matrix sources (time span: 2017–2023). N.D., not detected; MDL, method detection limit; LOD, limit of detection; dw, dry weight; lw, lipid weight.

Substances	**Concentration**	**Matrix, Location**	**Ref.**
Decabromodiphenyl ethane (DBDPE)	N.D.–384 ng g^{-1} dw	Soil manufacturing industries, Australia	[49]

Substances	Concentration	Matrix, Location	Ref.
	N.D.–295 ng g^{-1} dw	Soil waste disposal facilities, Australia	
1,2-Bis(2,4,6-tribromophenoxy) ethane (BTBPE)	N.D.–63.8 ng g^{-1} dw	Soil, manufacturing industries, Australia	
	N.D.–11.4 ng g^{-1} dw	Soil, waste disposal facilities, Australia	
2-Ethylhexyl-2,3,4,5-tetrabromobenzoate (EH-TBB)	N.D.–1.79 ng g^{-1} dw	Soil, waste disposal facilities, Australia	
Pentabromotoluene (PBT)	N.D.–0.10 ng g^{-1} dw	Soil, waste disposal facilities, Australia	
Hexabromobenzene (HBB)	N.D.–1.37 ng g^{-1} dw	Soil, manufacturing industries, Australia	
	N.D.–90.9 ng g^{-1} dw	Soil, waste disposal facilities, Australia	

Substances	Concentration	Matrix, Location	Ref.
Trimethyl phosphate (TMP)	3.03 µg kg^{-1} dw	Road dust, China	[59]
	3.81–242 ng g^{-1} lw	Fish, China	[38]
	0.69–50.0 ng g^{-1} lw	Pig, China	
	1.59–15.6 ng g^{-1} lw	Chicken, China	
	N.D.–18.2 ng g^{-1} lw	Cattle, China	
	N.D.–1.39 ng g^{-1} lw	Vegetables, China	
Tris(methylphenyl) phosphate (TMPP)	61.6 µg kg^{-1} dw	Road dust, China	[59]
	35.2–100 ng g^{-1} dw	Sludge cake, USA	[71]
	20.1–405 ng L^{-1}	WW influent, USA	
	11.5–181 ng g^{-1} lw	Fish, China	35]
	4.59–315 ng g^{-1} lw	Pig, China	
	8.00–1,090 ng g^{-1} lw	Chicken, China	
	10.8–97.0 ng g^{-1} lw	Cattle, China	
	0.81–20.8 ng g^{-1} lw	Vegetables, China	

Substances	Concentration	Matrix, Location	Ref.
Triethyl phosphate (TEP)	<LOD–29 ng L^{-1}	Freshwater, Uganda	[8]
	53.7 µg kg^{-1} dw	Road dust, China	[59]
	2.99–189 ng g^{-1} dw	Sludge cake, USA	[71]
	184–775 ng L^{-1}	WW influent, USA	
	N.D.–0.42 ng g^{-1} dw	Farmland soil, China	[60]
	1.90–18.1 ng g^{-1} lw	Fish, China	[38]
	N.D.–8.02 ng g^{-1} lw	Pig, China	
	0.94–9.52 ng g^{-1} lw	Chicken, China	
	2.39–14.8 ng g^{-1} lw	Cattle, China	
	N.D.–0.95 ng g^{-1} lw	Vegetables, China	

Substances	Concentration	Matrix, Location	Ref.
Tripropyl phosphate (TPrP)	<LOD	Road dust, China	[59]
	4.57–14.0 ng g^{-1} lw	Fish, China	35]
	N.D.–15.4 ng g^{-1} lw	Pig, China	
	2.61–55.6 ng g^{-1} lw	Chicken, China	
	4.44–24.2 ng g^{-1} lw	Cattle, China	
	N.D.–2.04 ng g^{-1} lw	Vegetables, China	

Substances	Concentration	Matrix, Location	Ref.
Tri-n-butylphosphate (TnBP)	41.1 µg kg^{-1} dw	Road dust, China	[59]
	13.8–106 ng g^{-1} dw	Sludge cake, USA	[71]
	50.8–649 ng L^{-1}	WW influent, USA	
	<MDL–4.59 ng g^{-1} dw	Farmland soil, China	[60]
	37.0–2,380 ng g^{-1} lw	Fish, China	[38]
	24.7–217 ng g^{-1} lw	Pig, China	
	30.2–1,260 ng g^{-1} lw	Chicken, China	
	49.7–461 ng g^{-1} lw	Cattle, China	
	N.D.–2.04 ng g^{-} lw	Vegetables, China	
Tributyl phosphate (TBP)	9.6–500 ng L^{-1}	Freshwater, Uganda	[8]

Substances	Concentration	Matrix, Location	Ref.
Tri-iso-butylphosphate (TiBP)	20.3 µg kg^{-1} dw	Road dust, China	[59]
	0.608–131 ng g^{-1} dw	Sludge cake, USA	[71]
	57.9–102 ng L^{-1}	WW influent, USA	
	<MDL–4.68 ng g^{-1} dw	Farmland soil, China	[60]
	N.D.–312 ng g^{-1} lw	Fish, China	[38]
	20.1–141 ng g^{-1} lw	Pig, China	
	28.6–262 ng g^{-1} lw	Chicken, China	
	23.9–370 ng g^{-1} lw	Cattle, China	
	1.18–9.55 ng g^{-1} lw	Vegetables, China	
Tris(2-chloro-1-methylethyl) phosphate (TCPP)	25–600 ng L^{-1}	Freshwater, Uganda	[8]
	1,010 µg kg^{-1} dw	Road dust, China	[59]
Tris(1-chloro-2-propyl)phosphate (TCIPP)	132–661 ng g^{-1} dw	Sludge cake, USA	[71]
	820–145,000 ng L^{-1}	WW influent, USA	
	<MDL–16.8 ng g^{-1} dw	Farmland soil, China	[60]
Tributoxyethylphosphate (TBEP)	73.8 µg kg^{-1}	Road dust, China	[59]

Substances	Concentration	Matrix, Location	Ref.
	dw		
Tripropyl phosphate (TPP)	3.07–10.3 ng g^{-1} dw	Sludge cake, USA	[71]
	5.65–45.8 ng L^{-1}	WW influent, USA	
Tri(2-ethylhexyl) phosphate (TEHP)	<LOD–4,300 ng L^{-1}	Freshwater, Uganda	[8]
	5.97 µg kg^{-1} dw	Road dust, China	[59]
	607–2,750 ng g^{-1} dw	Sludge cake, USA	[71]
	13.1–1,850 ng L^{-1}	WW influent, USA	
	1.26–45.7 ng g^{-1} dw	Farmland soil, China	[60]

Substances	Concentration	Matrix, Location	Ref.
Tris(2-chloroethyl) phosphate (TCEP)	24–6,500 ng L^{-1}	Freshwater, Uganda	[8]
	397 µg kg^{-1} dw	Road dust, China	[59]
	48.6–109 ng g^{--} dw	Sludge cake, USA	[71]
	195–8,450 ng L^{-1}	WW influent, USA	
	1.72–3.79 ng g^{-1} dw	Farmland soil, China	[60]
	17.1–159 ng g^{-1} lw	Fish, China	[38]
	11.1–116 ng g^{-1} lw	Pig, China	
	9.31–135 ng g^{-1} lw	Chicken, China	
	15.3–123 ng g^{-1} lw	Cattle, China	
	1.54–5.26 ng g^{-1} lw	Vegetables, China	
Tricresyl phosphate (TCP)	<LOD–8,100 ng L^{-1}	Freshwater, Uganda	[8]

Substances	Concentration	Matrix, Location	Ref.
Triphenyl phosphate (TPhP)	54–4,300 ng $^{-1}$	Freshwater, Uganda	[8]
	103 µg kg^{-1} dw	Road dust, China	[59]
	223–688 ng g^{-1} dw	Sludge cake, USA	[71]
	187–845 ng L^{-1}	WW influent, USA	
	0.53–8.33 ng g^{-1} dw	Farmland soil, China	[60]
	211–18,912 ng g^{-1}	Indoor air (homes and offices), Spain	[72]
	172–12,853 ng g^{-1}	Indoor air (homes and offices), Netherlands	
	3.49–59.6 ng g^{-1} lw	Fish, China	[38]
	2.64–36.4 ng g^{-1} lw	Pig, China	
	3.76–92.1 ng g^{-1} lw	Chicken, China	
	3.82–4.44 ng g^{-1} lw	Cattle, China	
	N.D.–2.21 ng g^{-1} lw	Vegetables, China	

Substances	Concentration	Matrix, Location	Ref.
Cresyldiphenylphosphate (CDPP)	33.0 µg kg^{-1} dw	Road dust, China	[59]
2-Ethylhexyl diphenyl phosphate (EHDPP)	22 µg kg^{-1} dw		
	22.2 µg kg^{-1} dw		
	7.7–730 ng L^{-1}	Freshwater, Uganda	[8]
	121–427 ng g^{-1} dw	Sludge cake, USA	[71]
	1.12–4.58 ng g^{-} $^{-}$ dw	Farmland soil, China	[60]
Tris (2-butoxyethyl)phosphate (TBOEP)	21.1–59.9 ng g^{-1} dw		
	N.D.–112 ng g^{-1} lw	Fish, China	[38]
	3.49–38.9 ng g^{-1} lw	Pig, China	
	N.D.–119 ng g^{-1} lw	Chicken, China	
	9.56–62.6 ng g^{-1} lw	Cattle, China	
	0.62–4.45 ng g^{-1} lw	Vegetables, China	

Substances	Concentration	Matrix, Location	Ref.
	2,490–13,500 ng g^{-1} dw	Sludge cake, USA	[71]
	129–69,500 ng L^{-1}	WW influent, USA	
Tris(2,3-dibromopropyl) phosphate (TDBPP)	288–397 ng g^{-1} dw	Sludge cake, USA	
	370–530 ng L^{-1}	WW influent, USA	
Tris(1,3-dichloro-2-propyl) phosphate (TDCPP)	431–1,700 ng g^{-1} dw	Sludge cake, USA	
	259–3,450 ng L^{-1}	WW influent, USA	
	37.2 µg kg^{-1} dw	Road dust, China	[59]
	<MDL–28.3 ng g^{-1} dw	Farmland soil, China	[60]
	N.D.–381 ng g^{-1} lw	Fish, China	[38]
	16.0–163 ng g^{-1} lw	Pig, China	
	8.82–110 ng g^{-1} lw	Chicken, China	
	N.D.–117 ng g^{-1} lw	Cattle, China	
	N.D.–5.86 ng g^{-1} lw	Vegetables, China	

Substances	Concentration	Matrix, Location	Ref.
p,p′-1,3-Phenylenep,p,p′,p′-tetraphenylesterphosphate (PBDPP)	54.1–555 ng g^{-1} dw	Sludge cake, USA	[71]
	99.0–2,660 ng L^{-1}	WW influent, USA	
Diphenylphosphate (DPHP)	10.9–210 ng g^{-1} dw	Sludge cake, USA	
	30.0–1,290 ng L^{-1}	WW influent, USA	
	106–1,251 ng g^{-1}	Indoor air (homes and offices), Spain	[72]
	151–4,189 ng g^{-1}	Indoor air (homes and offices), Netherlands	

The presence of flame retardants can also lead to ecotoxicological effects on soil inhabitants, including earthworms. Exposure to BDE-209 and BDE-47 can induce toxic effects in *E. fetida*, affecting various biological processes including nerve activities, osmoregulation, the tricarboxylic acid cycle, energy metabolism, and amino acid metabolism [64]. Moreover, when exposed to soil contaminated with tris(2-butoxyethyl) phosphate (TBOEP), *E. fetida* experienced various detrimental effects. These included the inhibition of acetylcholinesterase activity, leading to neurotoxicity. Moreover, there was a decrease in the rate of weight gain, indicating developmental toxicity, and a reduction in the number of juveniles, indicating reproductive toxicity [65]. Additionally, it has been observed that TnBP can accumulate in earthworm cerebral ganglions, reducing glutamate levels and increasing Ca^{2+} concentration, resulting in an osmotic imbalance in the nervous tissue and leading to apoptotic cell death. Moreover, exposure to 1,2-dibromo-4-(1,2 dibromoethyl)

cyclohexane (TBECH), a NBFR, was found to induce toxic effects on *Z. mays L.*, specifically inducing or inhibiting GST and CYP activities and gene expressions, suggesting its bioaccumulation and biotransformation within the plant [66]. These toxic effects on important physiological pathways of terrestrial organisms highlight the potential harm that flame retardants can cause to the environment and human health.

16.4 Human Exposure to Flame Retardants

As a result of the widespread use of flame retardants in the environment, flame retardants can be encountered by humans through multiple pathways, including inhalation of airborne compounds, ingestion of contaminated water and food, or dermal contact with products containing flame retardants [8, 38, 45, 67].

The frequent hand-to-mouth and object-to-mouth behaviors of children increase their susceptibility to bioaccumulating lipophilic compounds, including flame retardants, from indoor dust in comparison to adults [67]. Although accumulation through dietary exposure and breastfeeding is possible, non-dietary pathways, particularly exposure through house dust, play a more substantial role in the accumulation process among children [53, 67, 68]. In a multi-country European study, the presence of PBDEs in the blood of children from four countries and OPEs in the urine of children from eight countries was examined. This study revealed that the levels of BDE-47 in blood samples recently collected from children were similar to or lower than those found in adult samples from previous years, indicating that regulations on PBDEs have effectively reduced human exposure to these compounds. This means that the concentrations of OPEs require careful evaluation due to their extensive occurrence and potential endocrine-disrupting effects, which in turn impact cognitive ability, reproduction, and metabolism. Bis(1,3-dichloro-2-propyl) phosphate (BDCIPP) and diphenyl phosphate (DPHP), which are metabolites of

TDCIPP and TPhP, respectively, were found to be widely distributed across Europe with minimal variations between countries, possibly influenced by open market conditions. The authors of this study faced difficulties in identifying specific lifestyle factors that could be attributed to flame retardant concentrations. However, they observed correlations between flame retardant concentrations and factors such as socioeconomic status, cleaning practices, and the physical characteristics of residences [53].

The liver is widely acknowledged as the primary site of toxicity for xenobiotic compounds in mammals [68, 69]. In a recent study by Saquib *et al.* (2022), transcriptomic changes caused by TBEP were demonstrated in human liver cells (HepG2). This study revealed the activation of cancer pathway genes following exposure to TBEP. Furthermore, TBEP induced DNA damage and disrupted the cell cycle, resulting in apoptosis of HepG2 cells. Additionally, TBEP was found to increase oxidative stress levels (triggered by reactive oxygen species (ROS) and nitric oxide), induce the influx of Ca^{2+}, and disrupt mitochondrial function in liver cells. These findings provide valuable information regarding the potential cytotoxic effects of TBEP on liver cells and are supported by the study conducted by Zhou *et al.* [69]. Nevertheless, depending on whether the OPEs are aryl, halogenated, or alkyl, their effects on ROS production, DNA damage, and endoplasmic reticulum (ER) stress can vary. For instance, elevated ER stress was observed after exposure to aryl-OPEs, while alkyl-OPEs contributed to the overproduction of ROS and DNA damage, and induced cytotoxicity and apoptosis [68].

Metabolic processes mediated by CYP450 and phase II enzymes such as GST in human hepatic fractions significantly contribute to the metabolism of OPEs, resulting in the formation of hydroxylated and dealkylated metabolites. Consequently, renal excretion is the primary pathway for the elimination of OPEs such as TCEP [52, 70]. A Chinese study estimated the renal (CL_{renal}) and hepatic (CL_h) clearance rates of OPEs and their corresponding diester metabolites in the human body using 30

paired samples of human plasma urine. CL_{renal} values were estimated based on the concentration of organophosphate diester metabolites in urine, whereas CL_h was estimated by extrapolating the *in vitro* intrinsic clearance data of organophosphate diester metabolites in human liver microsomes (CL_{HLM}). The study reported that the median total concentration of OPEs in plasma varied from 0.53 to 6.52 ng mL^{-1}. Additionally, the total concentration of metabolites in urine ranged from 0.35 to 5.95 ng mL^{-1}. The CL_{renal} of TPhP, TCEP, and TDCIPP were estimated to be 68.9, 50.9, and 33.3 mL/kg/day, respectively. Similarly, the CL_h presented the same order, with TPhP having the highest clearance rate (166 mL/kg/day), followed by TCEP (35.3 mL/kg/day), and TDCIPP (20.6 mL/kg/day). The CL_{HLM} of these three OPEs decreased in the following order: TPhP (0.0308 mL/mg protein/min), TDCIPP (0.0034 mL/mg protein/min), and TCEP (0.0006 mL/mg protein/min). The discrepancies between the CL_{renal} and CL_{HLM} trends, as observed for TDCIPP, suggest that the clearance of OPEs in humans is influenced by both their intrinsic metabolic characteristics and their affinity for plasma proteins. The binding of OPEs to plasma proteins can reduce CL_{renal}, thereby explaining the observed differences. This study suggests that the half-lives of chlorinated OPEs (TCEP and TDCIPP) in humans, estimated based on CL_h, are longer than those of aryl-OPEs (TPhP and EHDPP) and alkyl-OPEs (TnBP and TBOEP) [70]. Furthermore, BDCIPP has been detected in the urine of more than 90% of Americans, indicating widespread exposure to TDCIPP in the population. Epidemiological studies have revealed sex-specific associations between metabolic syndrome and urinary BDCPP levels, a known risk factor for conditions such as heart disease, stroke type, and type 2 diabetes. The potential effect of TDCIPP exposure on glucose regulation was studied using a mouse model [52].

16.5 Conclusion

Plastic additives play a key role in improving the properties and performance of polymers. However, caution is needed in their use, as their occurrence and persistence in the environment have raised concerns regarding their potential ecotoxicological effects. Studies focusing on earthworms have provided significant insights into the effects of specific plasticizers, such as DEHP and DBP, suggesting the potential toxicity of PAEs in soils. Furthermore, wheat plants have been found to absorb and metabolize PAEs, indicating the possibility of bioaccumulation through the food web. Despite advancements in the understanding of the toxicology of PAEs, there are still gaps in knowledge regarding their toxicological data, metabolic pathways, and harmful effects, especially for non-prioritized PAEs. Flame retardants are known to persist in the environment and accumulate in organisms, posing a risk to ecosystems. They have been detected in indoor environments, water bodies, and food items, with potential adverse health effects as ongoing research suggests associations with developmental, reproductive, endocrine, and neurotoxic effects, particularly in infants and children. While certain additives have been linked to adverse effects, more studies are needed to understand their long-term consequences and establish comprehensive regulatory frameworks. It is vital to continue exploring alternatives to unsafe additives and promote plastic waste management practices to mitigate potential risks to the environment and human health.

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Note

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17
The Invisible Threat: Micronanoplastic Materials

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Abstract

The increased use of micro- and nanoplastics (MiNaPs) has been reported by various researchers. Plastic materials that are sustained in the environment disintegrate into smaller parts and thus pose diverse challenges for different forms of life with which they come in contact with. The release of these particles cannot be attributed only to industries or households. The repercussions are borne by the environment in general and by receptors in particular. Recently, much has been reported about how MiNaPs have been affecting the aquatic environment and disturbing the ecology. They are persistent environmental pollutants with more than 90% of the plastic particle load in an environment composed of microplastic fragments. The high surface-area-to-volume ratio of microplastics makes them effective sorbents for harmful compounds. The presence of MiNaPs in the atmosphere will eventually impact all forms of life, with varied impacts depending upon concentration, time of exposure, etc. Airborne MiNaPs can travel long distances from their source and are deposited in terrestrial and aquatic environments, thereby causing varied damage to the receptors. An attempt was made to study the sources, origins, and impacts of MiNaPs. All major reasons, after the effects and scope of research regarding MiNaPs, are discussed here.

Keywords: Micro and nanoplastics, plastic pollution, environmental impact, biocides, poly aromatic hydrocarbon, inorganic pigments

17.1 Introduction

The breakdown of large plastic contaminants into smaller particles that are difficult to trace leads to aquatic and terrestrial pollution. These tiny grains of plastics are called microplastics (MPs). The size range of microplastics lies in between 0.1 µm and 5 mm whereas that of nanoplastics (NPs) lies in between 0.001 µm and 0.1 µm [1, 2]. Many studies have primarily focused on microplastics, and more attention should be paid to plastic pollution caused by NPs. The soil system is significantly affected by NP accumulation [3].

MiNaPs can attach to microorganisms and eventually affect the ecosystem. Here, microorganisms act as carriers which transport micro- and nanoplastics (MiNaPs) throughout the plant and animal kingdom [4]. It has been reported that researchers tend to incline their studies on MiNaPs in marine environments, while the soil system is a significant reservoir of MiNaPs. The interaction between MiNaPs and organic chemicals in the vicinity of the soil environment can cause severe soil toxicity. In addition, prolonged exposure can lead to greater threats to living habitats [5–8]. Owing to their significant and extended area-to-proportion, MiNaPs can effectively carry various harmful pathogens and organic pollutants [9, 10]. Living beings that accidentally consume MiNaPs pose a

high risk of life-threatening ailments, such as cancer, infertility, delayed growth and organ failure [11, 12].

The detection of quantities of microplastics and nanoplastics in soil is an emerging field for environmental scientists to explore [13, 14]. There is a requirement for management of MiNaPs through public awareness and the replacement of non-biodegradable plastics with biodegradable ones to effectively diminish the impact of MiNaPs in the soil system. The source, or the base root cause, of MP pollution resulting in the environment comes out to be agricultural practices, large plastic waste fragmentation, and indirect disposal. These scattered categories find their way into systematic categorization into direct and indirect sources of MiNaPs [15]. As per research, MiNaPs find wide applications in electrical appliances, plastic goods, personal care, cosmetics, and textiles. Some toothpaste is also a base composition for MiNaPs [16]. Through these particular goods and services, they leach out into the soil environment and affect it adversely. The eventual deposition of these MPs would result in marine environmental pollution [17].

The annual growth of the global population continues at a mean value of 1.78% from 1957 to 2018. The population has reached 7.7 billion as of 2019 [18]. Nowadays, the way of life needs reusable goods, like cold drinks or water bottles. On the other hand, the increase in these products has led to accumulative quantities of plastic pollution around the world. Plastic fragments from the cosmetics, cleaning agents, textile, and plastic manufacturing industries are the primary sources of MPs and NPs, while weathering-produced fragmented plastic particles are referred to as secondary MPs and NPs [19]. The prime origins of MiNaPs in the soil surrounding them are plastic mulching, landfilling plastic wastes, sludge, irrigation of wastewater, and atmospheric deposition. These sources eventually have an impact on biodiversity as well as the environment [20–23].

Concerns about the capacity of microplastics to absorb ostensibly a wide range of ecological substances, such as massive materials, various contaminants, and medicinal substances, have also been raised by recent investigations [24–27]. MiNaPs fall into two categories: primary MiNaPs and secondary MiNaPs. Primary MiNaPs are microscopic-scale polymers produced for commercial or domestic applications. Pellets of pure or virgin plastic are likewise regarded as primary MiNaPs. These major MPs have numerous uses in a variety of sectors, including the cosmetics (such as skin care products, dental cream, or body lotion), fabrics (such as hosiery,) and medicative industries. Furthermore, the main sources of production of this type of MPs are the washing of synthetic textiles and the huge quantities of primary MPs produced in the abrasion of car tires while driving [28, 29]. Therefore, land-based activities result in harmful effects produced by MiNaPs, as shown in Figure 17.1.

The three kinds of environmental effects are caused by micronanoplastic material. They are physical, chemical, and biological. They have been shown in Figure 17.2 [30–34].

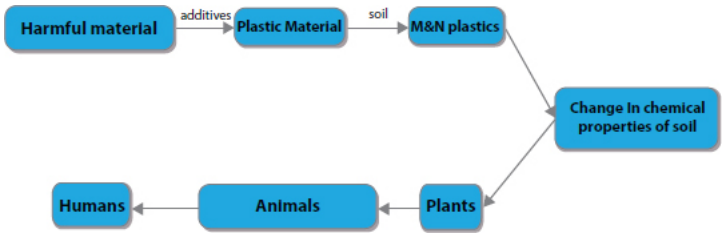


Figure 17.1 Harmful effects of MiNaPs.

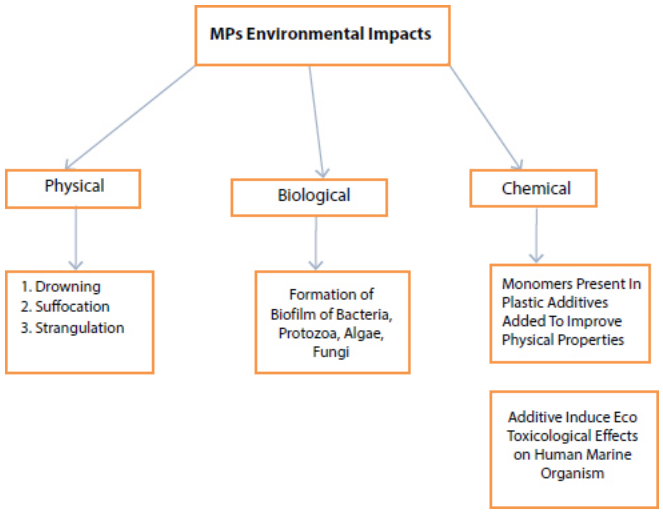


Figure 17.2 Impact of microplastics on the environment [30–34].

17.2 Microplastic and Toxic Chemicals

Microplastics are effective sorbents for harmful substances like toxic metals, organic compounds, or POPs that load on their surface due to their high surface-to-volume ratio. As a result, it transports pollutants to organisms and among environmental media. More than 300 plastic supplements liberate the pollutants in the surrounding area [35].

17.2.1 Heavy Metals

Heavy metals, being a universal component, trace their path towards the environment by either human activities or natural means [36, 37]. Sewage discharge, storm drain pollution, contaminated material from industries, pesticides, soil erosion, and numerous other things are major root causes of heavy metals in the environment. In a similar vein, harbors and marinas are notorious for having a lot of microplastics that were produced by human activity [38].

As per literature, the surface of microplastics has significant amounts of heavy metals. Data on harmful metal' adsorption onto microplastics and their prevalence in soil or the air, where they are airborne, are scarce [39, 40].

17.3 Organic Pollutants

In nature, organic contaminants tend to be organic. They are composed of carbon atoms that are covalently bound to other substances. They are classified as persistent organic pollutants (POPs), viz., organo chlorine pesticide (OCP) and polychlorinated biphenyls (PCBs), when they are present in the atmosphere [41]. POPs are not easily broken down by biochemical or photolytic mechanisms [42]. Some chemical substances are exploited to

improve the quality of plastic products, but they are mostly toxic to the environment and ecosystem and cause contamination of soil, water, and air [43–47]. These contaminants can also enter the human body through heavy metals and various chemicals with plastic additives and can cause harmful effects on human bodies, as shown in Table 17.1 [44, 45].

17.3.1 Organic Contaminants

They contain a carbon backbone, and different kinds of contaminants are bonded together with covalent bonds. Few organic contaminants are persistent in nature, viz., OCPs and PCBs. Persistent organic contaminants (POCs) do not degrade through photolytic and biochemical processes [48]. They can undergo bioaccumulation with an unfavorable influence on the environment. They can be easily transferred from their sources and can be adapted in the environment to significant toxic levels. POC sources can be biosynthetic routes, volcanoes, or manmade means through the preparation of chemicals in labs [49]. POCs are generally lipophilic in nature and have little solubility. The fine fractions of suspended solids carry significant amounts of organic contaminants, which adsorb on nano- and microplastics.

Table 17.1 Harmful effect of plastic additives on human health.

Polymers	Additives	Heavy metals	Diseases in humans
PET, PVC, PP, PE Full form	Inorganic pigments and fire retardants	Aluminum	Cancer in Breast
Various Plastics	Biocides	Antimony	Cancer
PVC, LDPE	Biocides	Arsenic	Lungs, Kidney, Liver, and skin problem
PVC	Inorganic pigments	Barium	Cardiovascular and kidney diseases
PET Bottles	Inorganic pigment	Cobalt	Neurological, Hearing, and Visual impairment

PBT, polybutylene ierephthalate; PET, polyethylene-terephthalate; PE, polyethylene, PVC, polyvinyl chloride; LDPE, low-density polythene; PP, polypropylene; PU, polyurethane.

Table 17.2 Study on different kinds of contaminants present in toxic chemicals at different concentration.

Toxic chemical	Name of contaminants	Concentration (ng/g)	Area/location
General Organic Pollutants	Hexane	5,000	Southern Brazil
Poly aromatic hydrocarbon	-hexane and methylene chloride [1:1 ratio v/v],	243–491	Southern Brazil
Phenanthrene	Methanol, LSC	600–6,100	During laboratory studies
Dichlorodiphenyltrichloroethan	Hexane/Dichloromethane	0.16–3.1	Coastal sites of Japan
Dichlorodiphenyltrichloroethane	Hexane/Dichloromethane	1.69–276	International pellet watch; 30 locations across (North America, Europe, Asia, Africa and Oceania
Hexachlorobenzene	*n*-Hexane and methylene chloride (1:1, v/v)	12.4–17.5	Southern Brazil
Aliphatic hydrocarbon	Dichloromethane	50–330	California, USA

In recent years, environmentalists have been highly concerned regarding loads of organic contaminants adsorbed on MiNaPs. As per literature, chemicals that have received attention are organic pollutants such as insecticide (DDT) and its degradation product, polychlorinated biphenyls (PCBs), and poly-aromatic hydrocarbons such as phenanthrene and fluoranthene [50]. A study on the presence of different kinds of contaminants in toxic chemicals on MiNaPs along with their concentrations has been tabulated in Table 17.2 [51–57].

17.3.1.1 Polychlorinated Biphenyls (PCBs)

PCBs can contain up to 209 chlorinated compounds. Out of these, 113 chlorinated compounds have been found in the environment [58, 59]. The first report of PCB adsorption on microplastics was studied in seawater in 1978 with a concentration of 5,000 $ng \cdot g^{-1}$ [60]. However, little literature could be observed until early 2001 regarding PCB adsorption on microplastics. Mato *et al*. in 2001 found resin pellets of polypropylene from microplastics present in coastal areas of Japan [61]. It is found in the range of 97 to 117 $ng \cdot g^{-1}$. 55 resin pellets could be analyzed in a concentration range of 28–2,300 $ng \cdot g^{-1}$ in the coastal region of Japan. Researchers collected plastic samples from the coastal regions of California, Hawaii, and the North Pacific Ocean. They were extracted, and PCB concentrations were found to be in the range of 27–980 $ng \cdot g^{-1}$ [61].

17.3.1.2 Polyaromatic Hydrocarbons (PAHs)

PAHs are also called polynuclear-aromatic hydrocarbons. They are made up of numerous aromatic rings. They have two rings like naphthalene and three rings like phenanthrene or anthracene. Literature has reported their occurrence in microplastics at different

concentrations [62, 63]. A study was done by Mato on different microplastics from California and Hawaii, and it was observed that PAH concentrations were found to be in the range of 39 to 1,200 ng g^{-1} [64]. The samples were collected from the river San Gabriel, and high concentrations of PAHs, i.e., 6,200–9,200 ng g^{-1} could be observed [61]. A few discolored microplastics could also be found in the same areas. As per the literature, discolored microplastics have larger amounts of PAHs than new ones.

Studies by Teuten *et al*. reported that PAH concentrations were 10^6 times higher on polystyrene than in adjoining water. The quantities of PAHs in the pellet samples from the Kato Achaia area of Greece vary from 66 to 637 ng g^{-1} [65]. The concentration of polycyclic aromatic hydrocarbons in the Gulf of Aliveri, Central Greece, was also studied by scientists, and its harmful impact has been reported [66]. In addition, a higher concentration of PAH was observed in microplastics on Portuguese beaches [67]. Literature revealed that plastic fragments collected from Portuguese beaches had PAH values ranging from 1 to 10 ng g. Further, concentrations of PAHs at different depths on sandy beaches in the coastal area of São Paulo state, Brazil, were also found [68]. It was observed that the highest number of PAHs was found on the surface layer of 0 to 10 cm depth with 1,996 ng g^{-1}, with smaller values of 1,336 ng g^{-1} and 1,004 ng g^{-1} recorded at 60 to 70 cm and 30 to 40 cm, respectively.

17.3.1.3 Organochlorine Pesticides (OCPs)

OCPs are manmade and are included in the category of chlorinated hydrocarbon derivatives. They are mostly employed in agriculture and chemical industries [69]. Even though OCPs are restricted by the law of the nation by most countries, they are still present in the air everywhere in the world. Organochlorine pesticides (OCPs) are bioaccumulative and highly stable micro- and nanoplastics [70]. These chemicals can travel at high speeds from one place to another and are easily adsorbed. To understand the fatal problems related to the occurrence of plastic pellets in water bodies, it is necessary to develop different methodologies that allow for rapid determination of contaminant levels. As per the literature, hexachlorocyclohexane (HCH) isomers, dichlorodiphenyltrichloroethane and its related compounds (DDTs), hexachloro benzene chlordanes, cyclodienesand mirex are few examples of OCPs found on MiNaPs [71].

17.3.1.4 Dichloro-diphenyl-trichloroethane (DDT) and its Related Compounds

Dichlorodiphenyl dichloroethylene (DDE) concentrations were observed in microplastics collected from coastal areas of Japan. The amount was ranging from 0.16 to 3.1 ng g^{-1} [72]. Ogata studied the presence of DDTs in plastic pellets collected from different beaches on five continents (Asia, Africa, North America, Europe, and Oceania) [73]. However, the highest concentration of DDTs was observed in Hermosa Beach, California, USA, at 267 ngg. In 2012, Bakir *et al*. studied the behavior of DDT on MiNaPs and observed that MiNaPs have a higher affinity for DDT than phenanthrene [74].

The literature reveals that organic pollutants adsorbed on microplastics are generally aromatic hydrocarbons, phenanthrene, sulfamethoxazole, chlorobenzenes (CBs), aliphatic hydrocarbons, and pesticides [75, 76]. Studies have also revealed that the adsorption of organic contaminants on nano- and microplastics is achieved through electrostatic interactions, partition effects, van der Waals forces, halogen bonding, hydrogen bonding, and hydrophobic effects, as shown in Figure 17.3. Resin is the most important constituent of plastics and has a nonwetting surface. POCs are easily adsorbed on the surface of plastics *via* hydrophobic interactions because of their elevated fat solubility and low water solubility [74]. They are also attached mainly by van der Waals forces and partition effects [77]. Two different types of POCs were adsorbed on MiNaPs *via* the partition effect [78]. In addition, an electrostatic force of attraction is formed at a specific pH. The electrostatic

force of attraction depends on the electrification of MiNaPs and the quantity of charge involved. Hydrogen and halogen bonding are generally formed between hydrophilic organic pollutants and plastics [32]. Hydrogen bonding generally occurs between microplastics with halogen atoms and POCs. Organic compounds also exhibit having non-specific van der Waals forces of attraction with hexadecane compounds, such as DDT [79, 80].

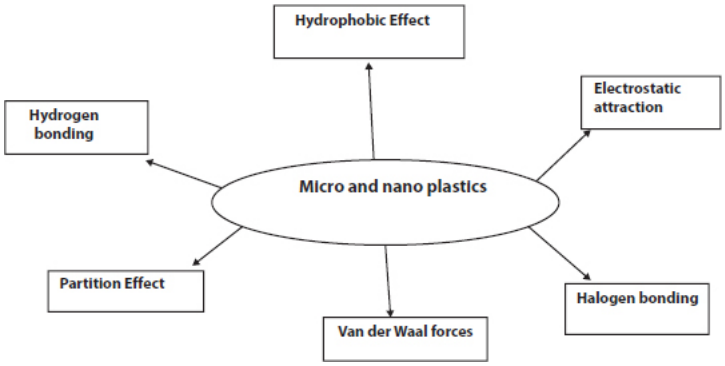


Figure 17.3 Adsorption of persistent organic contaminants on micro- and nanoplastics by different types of forces.

17.4 Microplastic Toxic Chemical Interaction

The significant surface-area-to-volume ratio of microplastics contributes to their effectiveness as sorbents for harmful compounds. This property allows them to absorb and accumulate toxins from their surrounding environment. Microplastics can act as carriers of these toxins, facilitating their transfer to various life forms and environmental media. Spectroscopic methods are mostly used to quantify harmful compounds on microplastic surfaces even with very low concentration. Local contamination is influenced by a variety of variables, i.e., the extract ant and the analytical tools utilized, concentrations, and the presence of harmful substances. Further research is necessary to ascertain the source of noxious compounds found in microplastics, whether they originate from additives or the surrounding environment.

A specific toxic chemical may have higher initial concentrations if added even as a plastic additive [81].

17.4.1 Environment

Microplastics and nanoplastics are persistent environmental pollutants, with more than 85% of the plastic persisting in the environment composed of microplastic fragments. They are of particular concern because of the increase in their surface A/V ratio and their potential for ingestion by marine organisms [82]. The release of greenhouse gases during the breakdown of environmental plastics has been confirmed by a recent study. This is a constant process, and as plastic manufacturing and trash rise, so too will the impact. Oceanic flora and fauna are important contributors to the microbial carbon pump, and plastic pollution can reduce phytoplankton’s capacity to fix carbon through photosynthesis, as well as zooplankton survival and reproduction rates, metabolic rates, and the functioning of the aqua ecosystem’s primary food chain [83]. The products of biotic and abiotic factors affect the biochemical life cycle in benthic and pelagic habitats, which is a great danger for human health [84].

For better comprehension of the effects of plastics, it is essential to determine the size, amount, and composition of plastics that stay in the ocean, as this may assist in

identifying possible dangers for either surface-feeding or seafloor-feeding animals. Polyethylene was the type of plastic that was found to be most common in the sea.

Polyethylene has been identified as the most commonly found form of plastic in the ocean; additional research is required to understand whether lighter polymers are more easily carried by winds and currents than heavier plastics [85].

17.4.1.1 Soil

Although soils are threatened by high levels of plastic pollution, they are crucial for ecosystem sustainability. Plastics possess the capacity to modify the physicochemical properties of soils, besides affecting biota. Nanoplastics have a pronounced influence owing to their small size and large surface area. Their movement through the soil environment is governed by the physicochemical characteristics of both the soil and plastic particles. While sampling soils and conducting subsequent procedures for their identification and quantification, their small size is a principal drawback. Their absorption by plant roots and ingestion by soil-dwelling animals can result in morphological deformation, organ damage, and physiological malfunctions [86].

Microorganisms are essential for soil function. Microbial florae is a signal of contamination. MiNaPs can affect the soil properties and microbial communities. The application of plastic compost can have a substantial impact on the soil microbial population, leading to an increase in what is known as the “plastisphere.” Perez *et al*. reported in his study that nanoplastic material also affects the pH, physicochemical parameter, toxicity, and reduced fertility in soil plant system [87]. The sources of MiNaPs addition to the soil are grouped into three types:

i. Agricultural activities
ii. Smash of runoff and deposition
iii. The fissure of the larger plastic waste.

Another cause of microplastics in the surroundings is thought to be the solicitation of garbage sludge to farm soils. [88]. The varied impacts of MiNaPs on soil are dependent on the shape, polymer type, and time of exposure, which tend to increase soil pH, impact soil respiration after long incubation times, and affect soil enzymes, eventually affecting the carbon, nitrogen, and phosphorus cycles [89]. Microplastics could, in fact, change the microbial ecology in the soil [90–92]. Therefore, it impacts the activity of enzymes [93]. A common agricultural technique used to increase soil fertility is the application of compost. A study conducted on topsoil (0–30 cm) to assess whether it could be a potential pathway of MiNaPs in soil confirmed the same. It has also been reported that littering is another source through which they are added to soil [94].

Utilization of sewage sludge as fertilizer for crops is common, and in North America and Europe, approximately 50% of the sludge is recycled in this manner. Several studies have explored the fate of MiNaPs within WWTPs and concluded that they have high trapping efficiency [88, 95]. The MiNaPs reaching WWTPs have been reported to be of cosmetic origin or fibers from washing of garments, tier debris, and fragmented plastics generated by road runoff [96].

Recently emerging contaminants known as MiNaps have garnered significant attention from the scientific community [97]. It has been reported that microplastics are abundantly distributed worldwide and have often been used by scientists as key indicators of the recent and contemporary periods. The impact of microplastics, owing to their complex physicochemical properties, is convoluted. They not only discharge toxic chemicals into the environment but are also amalgams of hazardous chemicals [98].

MiNaPs are further divided into primary and secondary types based on their source of origin. The primary ones are a repercussion of the particulate emissions of industrial origin; they find way into the environment in small sizes, correlated with specific applications and consumer products. Consumer products reported in the literature include toothpaste, plastic manufacturing raw materials, cosmetic products, and textile fibers [99]. Secondary MiNaPs are by-products of the degradation of plastic debris as they are exposed to microbes, physical deterioration, or when they come in contact with animals [100]. The impact of MiNaPs on different forms of life is discussed below.

17.4.1.2 Atmosphere

Atmosphere has been reported both as a source and reservoir of MiNaPs; their presence has not been limited to urban and semi-urban areas, but rural areas have also not been left behind. Airborne MiNaPs can travel long distances from their source and are deposited in terrestrial and aquatic environments, thereby causing varied damage to the environs [101].

In contrast to microplastics in other ecosystems, microplastics dispersed in the air can potentially be continually and directly absorbed, which is extremely dangerous for human health. Therefore, it is crucial to have a better perception of the amount, begetters, and dangers of MiNaPs in the surroundings [102]. Limited studies have been conducted to assessing airborne microplastics. A study conducted in two streets of Surabaya city in Indonesia concluded their presence to the tune of 132.75–174.97 particles/m^3, which indicates a symbolic proportion of MiNaPs pollution [103]. More contrasting studies exhibiting 4.18 particles/m^3 MiNaP pollution in Shanghai, China and 0.3–1.1 particles/m^3 [104]. The dispersion and likely health effects of MPs and microrubbers in air and dust from the streets of Asaluyeh County, Iran, were studied and assessed [105]. A survey conducted in the Forni Glacier of the Italian Alps reported 74.4 ± 28.3 MiNaPs/kg of sediment dry weight [106]. The presence of MiNaPs has also been reported in fresh snow fall in Canada and Austria where the values have been reported as 21,900 ng/l and 23,600 ng/l, respectively [107, 108]. An analysis of town soot from Tehran metropolis, Iran, reported 2,933–20,167 particles/kg of dry dust [109]. The presence of MiNaPs in the atmosphere will eventually impact all forms of life, with varied impacts depending upon concentration, time of exposure, etc.

17.4.1.3 Aquatic Environment

Microplastics are found in all aquatic environments, and thus pose a threat to important biota [110, 111]. Fish are a vital source of proteins for humans, and are thus required for the growth of the human body. Contamination of aquatic life by MiNaPs is a serious risk that needs special attention. Fish may suffer from drastic health ailments following exposure to MPs, either single-handedly or in combination with other pollutants. Fish exposed to MiNaPs may experience tissue damage, oxidative stress, genetic alterations, and reduced antioxidant levels. Problems such as neurotoxicity, growth deterioration, and unexpected behavior have also been reported in the event of exposure to MiNaPs [112].

It has been reported that open dumping and burning can cause MiNaPs to be deposited in groundwater and soil. Microplastics play a biological role in the marine environment because of their ability to be colonized by microorganisms. This colonization results in changes in physical properties such as increased density and decreased hydrophobicity, which can be mistaken for food by other aquatic organisms. The biological effects of microplastic ingestion include an imbalance in oxidative pressure, metabolism, and antioxidant capacity, and DNA alteration, immunity-related, and neurological impairment. Microplastics can cause biological responses such as inflammation, decreased feeding, weight, and mortality. Studies have shown that PS can alter the heart rate and decrease the hatching rate of the embryos.

Polystyrene has also been isolated from the mussel circulatory system, reducing citrate dehydrogenase-IDH, and killing brine shrimps. Alimi *et al.* [113] demonstrated that MiNaPs have both direct and indirect biological impacts on the environment. The GIT-neuro-endocrine pathways of micro- and nanoplastic-induced deleterious effects linked to ADME of MiNaPs by freshwater creatures need to be better understood. In addition, precise, transparent, and reliable statistics on the prevalence and plastic ingestion/egestion rate of MiNaPs, and the uptake of water repelling micropollutants by marine animals are needed. To better understand the mechanisms underlying size-dependent toxicity, further research should focus on nanoplastics. In summary, it can be concluded that the biological mechanism through which MiNaPs affect marine life remains unclear (Figure 17.4).

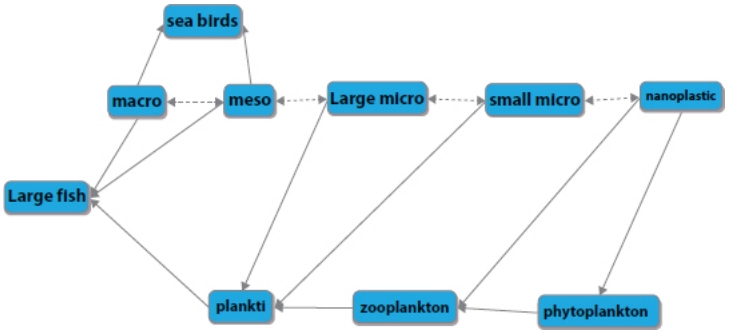


Figure 17.4 Aquatic environment affected by MiNaPs.

17.5 Toxicity to Human

At lower concentrations, humans are affected by nanoplastics for almost their entire life-span. The detection of nanoplastics is far more difficult than that of microplastics; thus, their impact has not been extensively investigated [113]. Once living organisms are exposed to nanoplastics in the short term, their elimination is challenging [114].

Researchers have reported that smaller nanoplastics gain entrance and accumulate in cells and tissues, eventually affecting their physiological activities [115]. Nanoplastics attack living organisms, cross biological barriers, and are devolved to their offspring. Those found in living organisms accumulate as they are excreted out of the body at very low concentrations. They enter the human system *via* drinking water and consuming materials that they are exposed to [116]. An investigation of 80% bottled water brands in Iran reported the presence of MiNaPs [117]. It has also been estimated earlier that people who drink bottled water might take 90,000 additional MPs every year, as opposed to people who drink tap water, where its consumption has been reported as 4,000 microplastics. According to a previous study, drinking alcohol, sugar, and tap water might result in humans ingesting MiNaPs directly because varied concentrations of MiNaPs were found in these samples [118]. Presence of MiNaPs has also been reported in beer, milk, honey and other beverages [119, 120].

A high concentration of MiNaPs has been reported in studies conducted in private homes and offices, in comparison to their outdoor concentrations, which is suggestive of the high amount of MiNaPs that humans are exposed to [121]. The entry of 40 nm nanoplastics through the epidermal route has also been reported [122]. This happens when close contact is established between humans and the environment, e.g., when taking showers with water or making use of personal care products carrying them [123].

MiNaPs are absorbed into the human pileup in the intestinal lumen; however, some of them are excreted through the stool [124]. Animal experiments have also revealed that nanoparticles can infiltrate and pave the way into blood vessels [125]. The alveolus, which

is responsible for liberating O_2 to the blood and receiving CO_2 from the blood, has a large pore size; it plays a vital role in creating a blood–air barrier [126]. Thus, once inhaled, MiNaPs may cross the blood–air barrier and eventually find way into the blood circulatory system, leading to various problems. Animal feed is another alternative route through which MiNaPs enter the human body [127]. Sea salt has also been reported to be a carrier of MiNaPs in humans [128].

17.6 Toxicity to Environment

The pollution created by MPs/NPs is responsible for human health risks and environmental issues and leads to increased accumulation in the food chain of the ecosystem, as reported by Barboza [129]. Many sources that comprise components related to MiNaPs are drug delivery materials, coatings, and electronics. 3D printing is one of the many ways to produce nanoplastics [130, 131]. Therefore, it is important to examine the effects of microplastic consumption on human health.

In humans, prolonged consumption of microplastics leads to abrasion, oxidative stress, body ulcers, and reduced growth rate. In addition, there are many adverse effects on reproductive abilities [132, 133]. The problems arising in the ocean due to plastic accumulation are not restricted only to the aquatic ecosystem. The long-term exposure leads to difficulty in estimating MiNaPs and eventually affects human commodities [134]. Mamun *et al.* in 2023 studied tissue obstruction owing to the unalienable amplitude of toxic micronanoplastics [135]. The food chain is also affected by the absorption of POPs by MiNaPs [135, 136]. Owing to the uptake of toxins such as polynuclear hydrocarbons (PNHC), organochlorine (OC) pesticides, and polychlorinated biphenyls (PCBPs) from their surroundings, the venomous properties of micronanoplastics are enhanced and turn over to the food chain [137–140].

Bergmann showed in his study that green plants were also affected by composition and physicochemical changes in soil due to MiNaPs [141]. Microplastic nanoparticles soil pollution can also affect the soil composition, texture, composition, and activity [142]. Humans are also affected by MPs throughout the food chain [143–145]. Amobonye reported a very serious environmental problem in aquatic/marine ecosystems and human communities through nanoplastic materials in 2021 [146]. Frydkjaer *et al.* showed that plankton, zooplankton, vertebrates, and invertebrates were unpredictable over an enormous number of micronanoplastic materials [147].

Various fish and other organisms in the underwater environment can supplement micronanoplastics. These MPs are dispersed in tissues and influence the [148, 149] rate of growth, biological enzyme activity, reproduction, and oxidative damage illustrated by various scientists [150, 151]. Few chemical materials are used to enhance the aspects of plastic products, but they are generally harmful to the surrounding environment as well as the ecosystem. Some chemical substances are used to improve the quality of plastic products, but they are mostly toxic to the environment and ecosystem and cause soil, water, and air contamination. This contamination can also enter the human body through exposure to various chemicals, and heavy metals with plastic additives can have harmful effects on human bodies (as shown in Table 17.1).

17.7 Impact of MiNaPs on Marine Environment and Terrestrial Habitat

Kossuth *et al.* and Stephens *et al.* showed that electronics, paints, adhesives, drugs, and 3D printing can release polymer nanoparticles that can cause serious problems to human health [152, 153]. Uncontrolled consumption of MPs can cause stress, ulcers, abrasion,

and reduced reproduction and growth rates in humans [154, 155]. The incapable quantity properties of MPs can cause disadvantageous tissue hindrance in humans [156]. The toxicity of the food chain also increases because of the absorption of POPs alongside MPs/NPs [157]. The noxious nature of nanoplastics increases due to the ingestion of toxins, polynuclear aromatic hydrocarbon (PNAHC), organo chlorine (OC) pesticides, and PCBP from the surroundings, which are transmitted to the food chain [158–161].

The physicochemical composition of soil, water cycling, and ecosystems in terrestrial surroundings reduce the growth of plants through MiNaPs [162]. Micronanoplastics have one of the photophysical territories of nanoparticles that have a perfect influence [163]. MPs are also imparted to human beings across the food chain [164, 165]. It is stipulated that individuals face problems of noxious MPs. The main environmental issue in aquatic ecosystems is plastic sea pollution (Figure 17.5). Small nanoplastics are also a challenge for the community [166, 167].

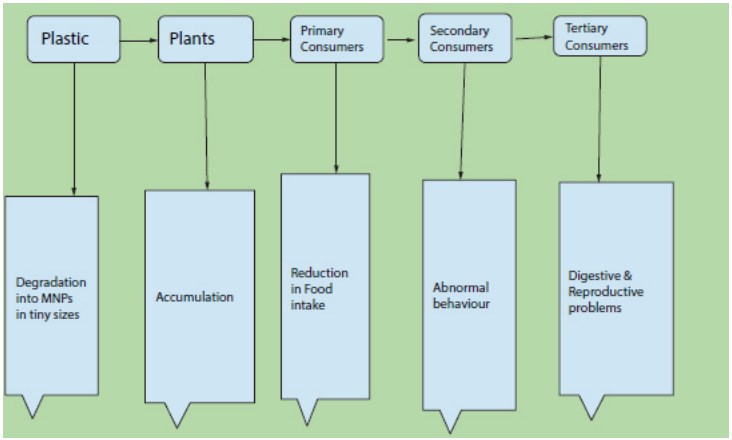


Figure 17.5 Ecosystem affected by MiNaPs.

17.8 Conclusion

The hazardous impact of microplastics is so high that they can travel up to 95 km in the air from their originating sources and migrate into new surroundings. Although there are several sources of MiNaPs, anthropogenic activities have shown a dominant effect.

- MPs are often sustained in indoor atmospheres as well, which eventually affects the inhabitants while they inhale it. They can be directly injected *in vivo* through consumption of contaminated water, or trophic transfer can be an indirect method (seafoods and plants). Inhalation or direct epidermal contact with airborne MiNaPs.
- It is evident that microplastics may interact with marine and soil species through consumption or entanglement, and laboratory studies have demonstrated that this might have negative consequences.
- The effects of microplastics on the microbial community in soil, animals, and plants depend on the type, concentration, and dimensions of the particles.
- Microplastics in the soil can accumulate in plants, and laboratory studies have shown that this can cause plants to grow poorly. When comparing biodegradable microplastics to conventional microplastics, plant growth was more adversely affected.
- Human consumption of microplastics may have consequences that are currently unknown, although they can be particle-based (by particle localization), chemical-

based (with accompanying hazardous compounds), or microbial-based (*via* biofilm formation).

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18
Comparative Analysis of the Toxicity of Micro- and Nanoplastics along with Nanoparticles on the Ecosystem

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Abstract

In today's advancing era of science and technology, several chemical groups offer significant benefits for various applications, but they can also pose severe hazards if not handled carefully. The two most prevalent lethal groups in this study were plastics and nanoparticles (NPs). Based on the specific properties of chemical composition, morphology, size, surface charge, oral exposure, particle fate, and alternation in particle behavior through ingestion, one can determine whether a particle is safe or not. The emerging fields of nanotechnology and plastics have been introduced to living beings, and their surroundings are becoming a threat to their well-being. As the particle size decreases, the toxicity increases. Therefore, micro- and nanoscales particles can easily enter the system, taking advantage of this factor, and cause harm to humans, aquatic and terrestrial systems, and the atmosphere through air, water, or food. Owing to their increasing production and ubiquitous application, the risk of NPs and micro- and nanoplastic pollution is significant. The goal is to emphasize the endangerments and offer a fair overview of the critical issues and priorities for upcoming attention.

***Keywords*:** Toxicity, micro- and nanoplastics, nanoparticles, ecosystem

18.1 Introduction

For a system to exist, it needs the surroundings to satisfy living conditions. Therefore, the environment matters the most when considering its effects. An ecosystem can be described as a community of living organisms (biotic) that interact with each other and their

physical (abiotic) environments within a specific geographical area. This includes all living organisms, such as plants, animals, fungi, and bacteria, as well as non-living components, such as air, water, soil, and rocks. The biotic and abiotic components of an ecosystem are interconnected and influence each other. For example, plants use carbon dioxide from the air to carry out photosynthesis, which produces oxygen and glucose that are used as food by other organisms. Animals, in turn, breathe in oxygen and exhale carbon dioxide, contributing to the carbon and oxygen cycle in the ecosystem. The interactions between the biotic and abiotic constituents of an ecosystem result in the cycling of nutrients, energy flow, and the maintenance of various ecosystem services, such as water purification, carbon sequestration, and pollination. These processes occur at different levels of the organization, from individual organisms to the entire ecosystem. Scientists have studied ecosystems to understand their dynamics, how they change over time, and how they are affected by human activities, such as pollution, deforestation, and climate change. The study of ecosystems is a multidisciplinary field involving biology, ecology, geology, chemistry, and physics.

How the ecosystem sustains itself depends on its exposure to certain aspects, which may strengthen or destroy it. Toxicity can affect it physically, chemically, or biologically; at the same time, it can be classified into physical toxins including radiation, coal dust, silicon dioxide, etc., chemical toxins such as substances containing carbon monoxide, bleach, various acids, and biological toxins involving snake venom, or toxic substances produced from animals, plants, or microorganisms. Toxicity is a scientific term that refers to the degree to which a substance or chemical can harm living organisms. It measures the extent to which a substance or chemical can damage biological processes or systems. Toxicity is determined by various factors, such as the type of substance or chemical, its concentration, and the path of contact (e.g., inhalation, ingestion, dermal contact, and period of interaction). Toxic substances can cause a range of negative effects in living organisms, including damage to organs, tissues, and cells, disruption of biological processes, and even death. Toxicity can be acute, occur immediately after exposure, or chronic, and occur over a longer period. Acute toxicity is typically measured using tests that assess the lethal dose or concentration of a substance that causes death in a specific percentage of test animals, such as LD50 (lethal dose 50%). Chronic toxicity is typically measured using tests that evaluate the long-term effects of exposure to various biological processes or systems. Toxicity can be expressed in different ways,

depending on the intended audience and purpose. For example, the toxicological effects of a substance can be communicated to the general public in terms of safe exposure limits or warning labels on products, while toxicologists may use more quantitative measures, such as the dose–response curve or the no-observed-adverse-effect level (NOAEL), to assess the toxicity of a substance.

NPs are defined as tiny particles with dimensions ranging from 1 to 100 nanometers (nm) and they exhibit certain physicochemical properties, e.g., particle morphology, surface charge, surface area, surface coating, and degree of agglomeration [1]. They can be synthesized from various materials, such as carbon-based substances, metals, metal oxides, and polymers. Owing to their small size, NPs have specific chemical, physical, and biological properties, resulting in their applications for social and economic benefits, from industrial to pharmaceutical applications [2]. Their use is significant in electronics owing to their good electrical and optical properties; in energy storage systems such as efficient batteries, fuel cells, and solar cells; in medicines and healthcare; in the environment; in remediation to remove pollutants from water or soil through adsorption, catalysis, or other mechanisms; and in cosmetics to improve product efficiency and provide UV protection. According to several studies, NPs can penetrate cell membranes and interact with biological systems owing to their high oxidative activity [3]. At the same time, this small size is one of the reasons why most NPs are more hazardous than bulk materials. Hence, the same distinctive features of NPs have raised concerns that interactions with these materials will cause physiological responses in biological systems. Many studies and evidence have suggested their toxicity to ecology [4]. Therefore, as shown in Figure 18.1, nanotechnology also has its own benefits and liabilities.

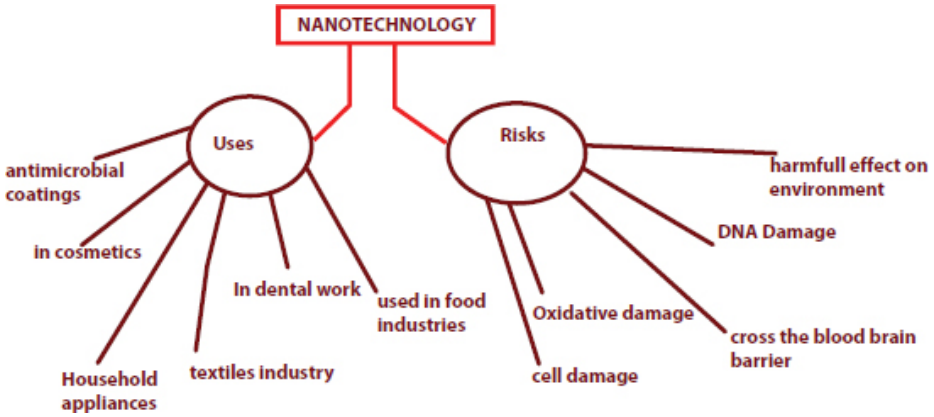


Figure 18.1 Advantages and disadvantages of nanotechnology [3].

Plastics are synthetic or semi-synthetic polymers, consisting of long chains of molecules linked together. They can be molded into different shapes and forms, making them useful for various applications. Plastics have become ubiquitous in our daily lives owing to their versatility, durability, and affordability. They are used in numerous industries including packaging, construction, transportation, electronics, and healthcare. Some common types of plastics include polyethylene (PE), polypropylene (PP), polyvinyl chloride (PVC), polystyrene (PS), and polyethylene terephthalate (PET) [5]. Despite their several benefits, they pose significant environmental and health challenges. These include environmental pollution as they are non-biodegradable, harming human health by leaching through food, water, or air, and most importantly, climate change as plastics contribute to greenhouse gas emissions. Microplastics and nanoplastics are generally smaller fragments of plastics that have become a growing concern because of their extensive presence in the environment and their possible impacts on ecosystems and human health. Microplastics are plastic particles with diameter ≤0.5 mm, whereas nanoplastics are much smaller with sizes of 100 nm or less. They can originate from various sources, including the breakdown or fragmentation of larger plastic items such as bottles and packaging, as well as from the shedding of synthetic fibers from textiles during washing. Microplastics can also be deliberately manufactured for specific purposes such as microbeads added to personal care products. Owing to their small size, they can easily enter cells or tissues or, on the other hand, they may be ingested into organisms very smoothly, leading to more significant biological interactions. This results in adverse effects on cellular functions, oxidative stress, and inflammation. Therefore, it is a significant challenge to reduce the use of plastics, promote recycling, and develop new sustainable and

biodegradable alternatives. Ultimately, comprehensive risk assessments and holistic approaches are necessary to mitigate the impacts of NPs and micro- and nanoplastics and ensure the protection and preservation of the environment for future generations.

18.2 Literature Survey

Sajid *et al.* evaluated the effects of NPs on human health and the environment, despite their eccentric advantages and applications [6]. Accordingly, the inherent features of NPs, such as nature, shape, size, charge, medium of synthesis, surface chemistry, stability, reactivity, mobility, and storage period, influence their toxicity; the three main pathways of NPs exposure to humans and wildlife are digestion, inhalation, and dermal contact. Therefore, many studies have focused on the cytotoxicity and genotoxicity of NPs. They also suggested that this can be avoided by maintaining appropriate laboratory conditions. Figure 18.2 shows the effect of NPs on human health and environment involving some solutions for it.

Xu *et al.* reviewed the literature on the health relevance of micro- and nanoplastics, comprising 133 important published papers [7]. Most of these articles (105 numbers) were based on the effects of PS micro- and nanoplastics because of their large abundance, well-characterized nature, and ability to manufacture them in various particle sizes, with fluorescence labeling and different surface modifications. A total of 117 articles reported adverse health impacts resulting from exposure to micro- and nanoplastics. Several biological effects were identified in mammalian *in vitro* analyses, including cytotoxicity (toxicity to cells), oxidative stress, inflammatory response, genotoxicity (damage to DNA), embryotoxicity (toxicity to developing embryos), hepatotoxicity (toxicity to the liver), neurotoxicity (toxicity to the nervous system), renal toxicity (toxicity to the kidneys), and even carcinogenicity (potential to cause cancer). Furthermore, rodent *in vivo* models have established their bioaccumulation in various organs, including the liver, spleen, kidney, brain, lungs, and gut. These studies revealed adverse effects at various levels, namely reproductive toxicity and transgenerational toxicity (effects passed on to subsequent generations). Therefore, overcoming these diadvantages is a challenge.

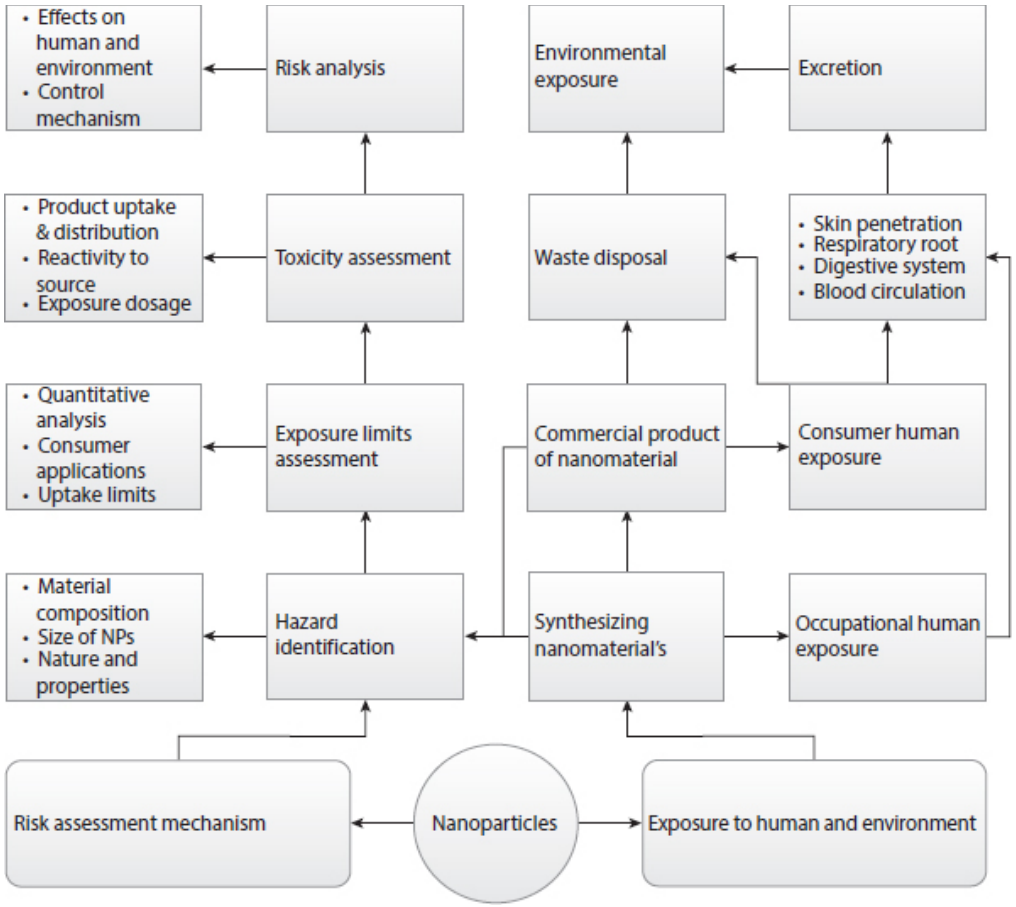


Figure 18.2 Human exposure mechanism and risk evaluation of NPs [6].

Singh *et al*. concluded that although nanotechnology is a revolutionary multidisciplinary field spanning a wide range of basic and applied sciences for manufacturing and deploying nanosized materials for new solutions, its unintentional release to the environment eventually makes its way to the aquatic environment, where it can affect live animals and sensitive ecosystems [8]. NPs enter aquatic ecosystems *via* industrial discharge, wastewater treatment effluent dumping, and indirect runoff from soil. After entering the aquatic environment, the NPs are heavily influenced by their surroundings and undergo a variety of conversions such as agglomeration, aggregation, dissolution, and sulfidation. To fill this vast information gap, a comprehensive investigation of the interactions and effects of NPs on numerous types of aquatic ecosystems, along the food chain and food web of ecosystems, is required. Examples of NPs affecting the microbial community, invertebrates, and fish species at various concentrations are shown in Table 18.1.

According to Chae *et al.*, aquatic ecosystems currently face significant environmental challenges in the form of contamination caused by bulk plastics and plastic debris [15]. This has become a prominent factor among others that affects these ecosystems. Specific concerns are tiny plastic fragments, including micro- and nanoplastics, which have emerged as major contributors to the pollution of marine and freshwater environments. They have diverse effects on the growth, behavior, reproduction, and death of aquatic organisms.

18.3 Exposure to Ecosystem and Translocation

18.3.1 Sources and Distribution

18.3.1.1 Nanoparticles

The toxicity of NPs can vary depending on their specific composition, stability, mobility, reactivity, size, shape, surface properties, and exposure conditions [6]. Regulatory bodies and researchers continue to study the potential risks associated with different types of NPs to ensure their safe use and to minimize their impact on human health and the environment. Toxic NPs can originate from various sources and are distributed through different pathways, as shown in Table 18.2.

Some common sources and distribution pathways of toxic NPs are as follows [6, 22]:

Nanomaterial Manufacturing: Synthesis of NPs can result in the release of toxic particles during the manufacturing process. These particles can be dispersed in air or water if not properly contained or treated.

Consumer Products: Certain consumer products, such as cosmetics, sunscreens, and cleaning agents, may contain NPs as ingredients. Although not all NPs used in consumer products are toxic, some may have adverse effects on health and the environment if released into the ecosystem.

Industrial Processes: Industrial activities such as manufacturing, mining, and metal refining can generate toxic NPs. These particles can be released into the environment *via* air emissions, wastewater discharge, or improper waste disposal.

Nanoparticle Contamination: NPs can contaminate soil, water bodies, and natural resources through direct spills, leaks, or runoff from industrial facilities or waste sites.

Air Pollution: Vehicle exhaust, fossil-fuel combustion, and other industrial processes contribute to airborne NPs. The particulate matter emitted during these activities can contain toxic NPs, which can be inhaled and pose risks to human health.

Waste and Landfills: Improper disposal of products containing NPs or nanomaterials can lead to the release of toxic particles into the environment. Landfills and wastewater treatment plants may not effectively capture or remove NPs, thus allowing them to enter the ecosystem.

Table 18.1 Examples of microbial community, invertebrates, and fish species to evaluate how it affects the internal organs [8].

Species/Organisms	NPs	Conc.	Impact	Ref.
Nitrifying bacterial species	Ag NPs	0.10 ppm	No variation in membrane fouling rate but enhanced amount of extracellular polymeric materials.	[9]
N-related microbial species	CuO NPs	1 ppm	Degradation in bacterial genera was related to biosynthesis and metabolism of carbohydrate and glycan and biosynthesis of other secondary metabolites.	[10]
Mesocosms (phytoplankton, zooplankton, macroinvertes, macrophytes, and fish)	TiO_2 NPs	0.025 and 0.25 ppm	Biomass of Rotifera was noticeably decreased while biomass of Cladocera, Copepoda, phytoplankton, macrophytes, chironomids, and fish was unaltered.	[11]
Caenorhabditis elegans	TiO_2 NPs	0.001 ppm	Diminish in movement of wild nematodes attributed to the increase in formation of intestinal reactive oxygen species (ROS).	[12]
Oreochromis niloticus and *Tilapia zillii*	Ag NPs	2 and 4 ppm	Harmful effects on the brain antioxidant system.	[13]
Prochilodus lineatus	TiO_2 NPs	1–50 ppm	Deposited in liver, muscle, and brain and reduced muscular acetylcholinesterage (AChE) activity.	[14]

Table 18.2 Sources, targets, factors, and toxicity of different NPs [6].

NPs	Sources	Factors	Toxic effects	Ref.
Ag NPs	In domestic appliances as antimicrobial agent, plastics, paints, varnish, and textile	Toxicity depends on surface properties, release of Ag^+, and coating of Ag particles	Damage to cell membrane, disrupt adenosine triphosphate (ATP), deoxyribonucleic acid (DNA) replication, and release oxidative ROS	[16]
Au NPs	Drug delivery, high quality imaging, gene therapy, cancer therapy, and wastewater treatment	Functionalization greatly affects toxicity; acute and chronic exposure depends on ligands attached	Affects reproduction in daphnia, mammals, algae, and fishes, accumulates in liver, and causing oxidative stress	[17], [18]
CNTs	Water treatments, lubricants, electronics, sport equipment, and bioaccumulation in food chain to human health	Depends on particle size, shape, surface charge, functionalization, degradability, protein corona, and surface modifications	Human lungs, epithelial cells, Rat glioma cells, Embryos toxicity, cell proliferation, Gill accumulation, liver injury, brain, intestine pathologies, respiratory toxicity, phytotoxicity, seeds germination, and inflammation	[19]

NPs	Sources	Factors	Toxic effects	Ref.
Quantum dots (QDs)	Photoluminescence, medical applications, use in plant bioimaging, tracking different particles, drug delivery, and therapy	Depend on surface modifications, bioactivity, size, concentration, stability, and environmental factors	Oxidative stress in plants, endothelial toxicity through activation of mitochondrial cells, and cell viability decreased with increasing concentration of QDs	[20, 2]
TiO_2 NPs	Sunscreen products, food products, paints, toothpaste, medicine, agriculture products, effluent, sludge, and wastewater	Depends on surface chemistry, stability, residence, and release of TiO_2 in wastewater	Negatively effects on wheat growth and soil enzyme actives, embryonic toxicity to fishes, phytotoxic effects, and DNA fragmentation	[21]

Once released into the environment, toxic NPs can be distributed through various pathways, including

Airborne Transport: NPs can be carried by wind over long distances and deposited onto land or bodies of water. They can also be inhaled by humans and animals, potentially causing respiratory and other health problems.

Water Dispersal: NPs can be transported through water bodies such as rivers and oceans, where they can accumulate and contaminate aquatic ecosystems. They can also be ingested by marine organisms and can enter the food chain.

Soil and Sediment Deposition: NPs can settle in soils and sediments, where they can persist and potentially impact soil fertility, plant growth, and the health of soil-dwelling organisms.

Biological Uptake: NPs can be taken up by plants, animals, and microorganisms. They can be transferred through the food chain, potentially accumulating in higher organisms and causing toxicity.

18.3.1.2 Micro- and Nanoplastics

These small plastic particles are of concern because of their extensive occurrence in the environment and their possible impacts on ecosystems and human health. The sources and distribution pathways of microplastics and nanoplastics are as follows [5]:

The different sources of microplastics and nanoplastics are described below:

Fragmentation of Larger Plastic Items: Larger plastic items, such as bottles, bags, and packaging materials, can break down over time owing to weathering, UV radiation, and physical forces such as wave action. This fragmentation process generates smaller microplastic particles.

Synthetic Fibers: Synthetic textiles, such as polyester, nylon, and acrylic, shed microplastic fibers during washing and use. These fibers enter wastewater systems and ultimately reach aquatic environments.

Industrial Processes: Industrial activities, including plastic manufacturing, recycling, and disposal, can release microplastics and nanoplastics into the environment. These particles may be generated during production or as by-products of plastic degradation processes.

Personal Care Products: Certain personal care and cosmetic products, such as exfoliating scrubs and toothpaste, contain microplastic beads or particles. When these products are used and washed off, the microplastics can enter the wastewater system.

Atmospheric Deposition: Microplastics and nanoplastics can be transported through the atmosphere. They may be generated by the degradation of larger plastic debris or by atmospheric fallout, where airborne particles settle on land or water surfaces.

The distribution pathways of microplastics and nanoplastics are depicted below.

Water Bodies: Microplastics and nanoplastics can enter rivers, lakes, and oceans *via* various pathways, including wastewater discharge, runoff from land, and atmospheric deposition. Once in water bodies, they can be relocated by currents, deposited in sediments, or ingested by aquatic organisms.

Soil: Microplastics can be transported to soils through land runoff or the application of plastic-containing materials, such as mulch or compost. They can also be introduced through the disposal of plastic waste in landfills or through illegal dumping [23].

Marine Organisms: Microplastics can be ingested by marine organisms such as fish, shellfish, and marine mammals. These particles can accumulate in the digestive tract of these organisms, potentially causing harm and entering the food chain.

Airborne Transport: Microplastics and nanoplastics can be transported through air over long distances and deposited onto land or bodies of water. They can also be inhaled by humans and animals, potentially causing health concerns.

Human Exposure: Humans can be exposed to microplastics and nanoplastics through various pathways, including ingestion of contaminated food and water, inhalation of airborne particles, and skin contact with products containing microplastics.

Micro- and nanoplastics persist in the environment and can take hundreds of years to degrade. Their small size and ability to travel through different pathways make them a widespread pollutant. Efforts are being made to mitigate plastic pollution, including improved waste management, reduction in plastic usage, and the development of eco-friendly alternatives. Figure 18.3 schematically represents the transfer of plastic litter in marine ecosystems, through various trophic levels within the food web, and between tissues within polluted animals. Figure 18.4 represents the toxic effects of micro- and nanoplastics on various organisms.

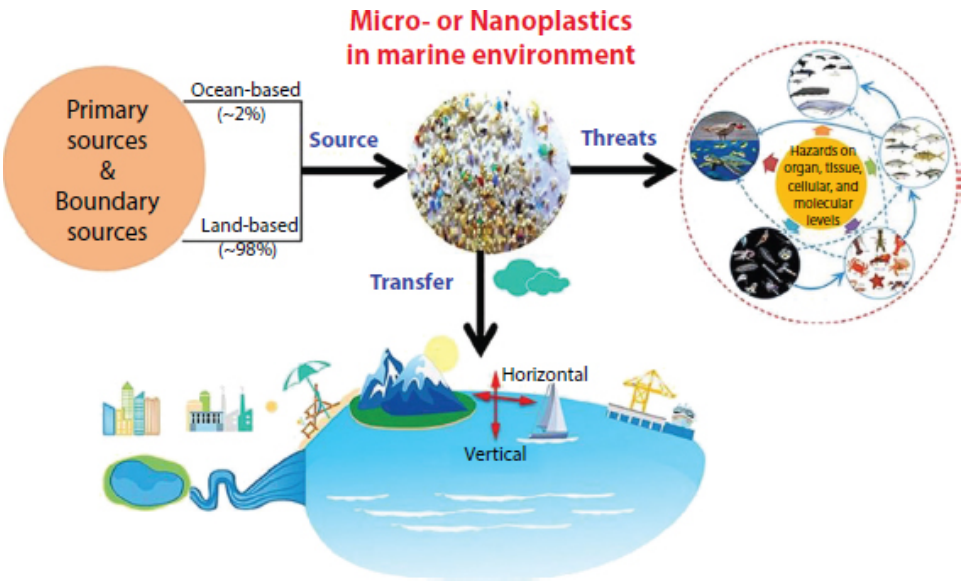


Figure 18.3 Schematic representation of transfer of plastic litters in marine ecosystem [24].

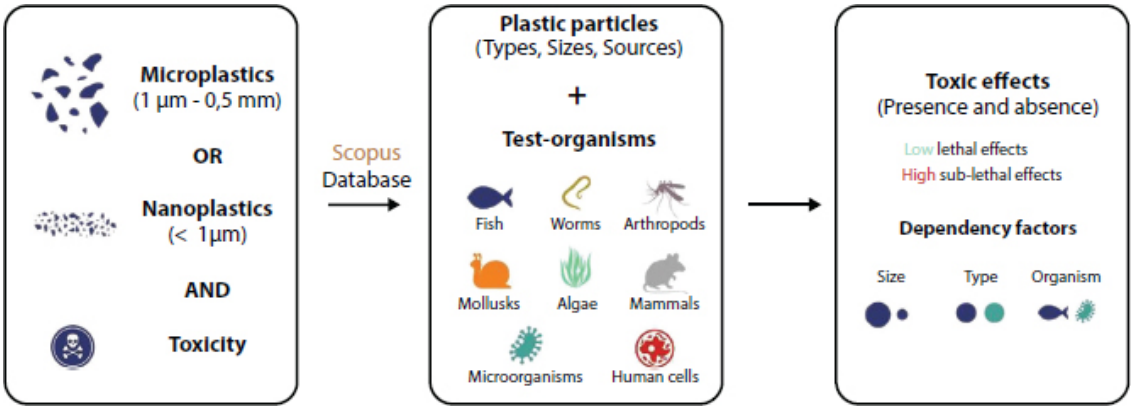


Figure 18.4 Schematic representation of toxic effect of micro- and nanoplastics on different organisms [25].

18.3.2 Toxicity in Human Health

Both NPs [24] and micro- and nanoplastics [26, 27] have the potential to impact human health. NPs have been extensively studied, and their toxicity is influenced by various factors. Microplastics and nanoplastics are an emerging area of research, and further studies are needed to better understand their health effects and establish appropriate regulations. The following is a comparison of their effects:

Size and Composition:

> *NPs:* The NPs have dimensions ranging from 1 to 100 nm. They can be composed of various materials including metals, metal

oxides, carbon-based materials, and polymers.

Micro- and Nanoplastics: Microplastics are small plastic particles measuring less than 5 ml, whereas nanoplastics are even smaller, typically less than 100 nm. They result from the breakdown of larger plastic debris and can be composed of diverse polymers, such as polyethylene, polypropylene, and polystyrene.

Exposure Routes:

NPs: Exposure to NPs can occur *via* inhalation, ingestion, or dermal contact. They can be released into the environment through industrial processes, combustion, or nanoparticle-containing products.

Micro- and Nanoplastics: Contact between micro- and nanoplastics primarily arises from ingestion, as they are present in food and water sources. They can also be inhaled or come in contact with the skin through environmental exposure.

Potential Health Effects:

NPs: The toxicity of NPs arises from various factors, e.g., their size, shape, surface chemistry, and composition. Some NPs have been associated with adverse effects including oxidative stress, inflammation, genotoxicity, and potential organ-specific toxicity. However, not all NPs exhibit the same level of toxicity, as some are designed for medical applications and engineered to be biocompatible.

Micro- and Nanoplastics: Research on the health effects of micro- and nanoplastics is still emerging, but their toxicity is not yet well understood. Studies have suggested that micro- and nanoplastics could potentially cause inflammation, disrupt cellular processes, and accumulate in organs and tissues. They may also act as carriers of other toxic substances or pathogens.

Research and Regulation:

NPs: NPs have received considerable research attention owing to their diverse applications and potential risks. Several regulatory bodies, such as the United States Environmental Protection Agency (USEPA) and the European Chemicals Agency (ECHA), have issued guidelines and regulations regarding the safe use of NPs.

Micro- and Nanoplastics: Research on micro- and nanoplastics is still in its early stages, and there is a limited understanding of their long-term effects on human health. Regulatory measures specific to micro- and nanoplastics have not been well established, although some countries and organizations have implemented bans or restrictions on certain types of plastic products.

18.3.3 Toxicity in Environment: Land, Water, and Air

NPs [6] as well as micro- and nanoplastics [5] can adversely effect the environment, including land, water, and air. The following is a comparison of their environmental toxicities:

Environmental Presence:

NPs: NPs can be released into the environment through industrial processes, waste disposal, and the use of nanoparticle-containing products. They can enter various environmental compartments, including the soil, water bodies, and atmosphere, depending on their release pathways.

Micro- and Nanoplastics: They result from the breakdown of larger plastic debris and are prevalent in various environmental settings. They are found in marine and freshwater systems, terrestrial ecosystems, and the atmosphere as airborne microplastics.

Persistence and Accumulation:

NPs: The persistence of NPs in the environment depends on their characteristics and surrounding conditions. Some NPs can remain in the environment for extended periods, potentially accumulating in the soil and water systems. However, their mobility and fate can vary widely, depending on factors such as particle size, surface chemistry, and interactions with other environmental components.

Micro- and Nanoplastics: Micro- and nanoplastics are highly persistent in the environment owing to their resistance to degradation. They can accumulate in the soil, sediments, and aquatic ecosystems, posing long-term risks to organisms and ecosystems. Microplastics are found in various organisms, including aquatic species, birds, and terrestrial animals.

Ecological Impacts:

NPs: NPs can have both direct and indirect effects on ecosystems. They may be toxic to various organisms including bacteria, algae, plants, and aquatic organisms. Additionally, NPs can potentially affect nutrient cycling, microbial communities, and the overall ecosystem structure and function.

Micro- and Nanoplastics: Micro- and nanoplastics can cause physical harm, such as ingestion and entanglement, to a wide range of organisms. They can also serve as substrates for the adsorption of other pollutants and chemicals, potentially facilitating their transfer into organisms and leading to bioaccumulation and biomagnification.

Research and Regulation:

NPs: NPs have received significant research attention owing to their environmental impact. Regulatory frameworks, such as the registration, evaluation, authorization, and restriction of chemicals (REACH) in Europe, include provisions for nanoparticle safety assessment.

Micro-and Nanoplastics: Research on micro- and nanoplastics is rapidly evolving, and there is growing concern about their environmental impacts. International efforts and regulations are being developed to address microplastic pollution, including the Basel Convention, the European Union's single-use plastics directive, and bans on microbeads in personal care products.

Both NPs and micro- and nanoplastics have detrimental effects on the environment. NPs can impact ecosystems through their toxicity and potential disruption of ecosystem processes. Microplastics and nanoplastics, owing to their persistence and accumulation, can harm organisms and contribute to the transfer of other pollutants. Ongoing research and regulatory measures aim to better understand and mitigate the environmental risks associated with both nanoparticle and microplastic and nanoplastic pollution.

18.3.4 Ecotoxicity in Aquatic Organisms

NPs [1, 8] and micro- and nanoplastics [15, 26] can have adverse effects on aquatic organisms, as shown below:

NPs:

Toxicity Mechanisms: They induce toxicity in aquatic organisms *via* various mechanisms. They may generate ROS, leading to oxidative stress and damage to cells and tissues. NPs can also interfere with cellular processes, disrupt enzymatic activity, and cause inflammation in aquatic organisms.

Bioavailability and Uptake: They are taken up by aquatic organisms *via* different routes, including ingestion, inhalation, and dermal contact. Their small size and surface properties can enhance bioavailability and facilitate their accumulation in tissues and organs.

Species Sensitivity: The toxicity of NPs can vary among aquatic organisms. Some species, such as fish, crustaceans, and algae, have been extensively studied for nanoparticle toxicity. The response to NPs depends on factors such as species-specific physiological characteristics, life stage, and exposure duration.

Micro- and Nanoplastics:

Physical Harm: They can physically harm aquatic organisms through ingestion, entanglement, and internal injuries. Ingested microplastics and nanoplastics can cause damage to the digestive tract, leading to reduced feeding, nutrient uptake, and overall fitness.

Chemical Effects: They act as carriers of other chemical pollutants, including organic pollutants and heavy metals. These pollutants can be adsorbed onto the surfaces of micro- and nanoplastics, potentially transferring into organisms upon ingestion and causing toxic effects.

Ecological Interactions: They interact with aquatic organisms at different trophic levels, thereby affecting food webs and ecosystem dynamics. They may affect the behavior, reproduction, and survival of organisms, with potential cascading effects on ecosystem functioning.

18.4 Challenges and Precautions

The mitigation of NPs [28, 29] and micro- and nanoplastics in the environment is an ongoing challenge. However, there are several future scopes and strategies for addressing and minimizing their

impact. There are some potential approaches for saving the environment from NPs and micro- and nanoplastics [30, 31]:

Regulation and Policy: Strengthening regulations and policies regarding the use, disposal, and release of NPs and micro- and nanoplastics can help to reduce their environmental impact. Governments can implement restrictions on certain types of plastics, promote sustainable alternatives, and enforce appropriate waste management practices.

Improved Waste Management: Advanced waste management systems can prevent the release of NPs and micro- and nanoplastics into the environment. This includes enhancing recycling, implementing efficient treatment methods for plastic waste, preventing littering, and encouraging responsible disposal practices.

Sustainable Materials and Alternatives: Promoting the use of sustainable materials and eco-friendly alternatives to plastics can significantly reduce the generation of micro- and nanoplastics. Research and development efforts have focused on biodegradable plastics, bio-based materials, and innovative packaging solutions that minimize environmental harm.

Public Awareness and Education: Raising public awareness of the impacts of NPs and micro- and nanoplastics is crucial for driving behavioural changes. Educational campaigns, informative initiatives, and targeted outreach can help individuals to understand the importance of reducing plastic consumption, proper waste disposal, and the adoption of sustainable practices.

Filtration and Treatment Technologies: Develop effective filtration and treatment technologies that can help to remove NPs and micro- and nanoplastics from various environmental matrices. This includes the improvement of waste-water treatment systems, filtration methods for drinking water, and implementation of advanced air filtration technologies.

Research and Monitoring: Continued research and monitoring efforts are essential to understand the behavior, fate, and potential risks associated with NPs and micro- and nanoplastics. This knowledge can inform the development of targeted mitigation strategies and assist in the assessment of their environmental impact.

Innovative Clean-up Technologies: Clean-up technologies, such as specialized filters, sensors, and nanomaterial-based remediation approaches, can aid the removal and remediation of NPs and micro- and nanoplastics from the environment.

International Cooperation: Collaboration between governments, organizations, and industries at the international level is crucial for addressing the global issue of NPs and micro- and nanoplastics. Sharing knowledge, best practices, and resources can accelerate progress for finding effective solutions and implementing mitigation measures.

Continued efforts in research, innovation, policymaking, and education are necessary to protect the environment from the harmful effects of these pollutants.

18.5 Conclusions

A toxicity comparison between NPs and micro- and nanoplastics and their impact on the environment is a complex issue that requires careful consideration. While all these pose potential risks, their toxicity and impacts can vary depending on factors such as, composition, size, shape, surface properties, and exposure conditions. NPs, including those of synthetic origin, can exhibit unique chemical and physical properties that may contribute to their toxicity. They have the potential to interact with living organisms and ecosystems, leading to harmful effects on health and the environment. However, the toxicity of NPs is highly dependent on their specific characteristics and context of exposure. Microplastics and nanoplastics, on the other hand, are primarily derived from plastic pollution. These particles can persist in the environment for long periods and have been found in various ecosystems, including freshwater systems, oceans, soil, and the atmosphere. Although microplastics and nanoplastics can cause physical harm by ingestion or entanglement, their chemical toxicity is still a topic of ongoing research. Comparing the toxicity of NPs, micro-, and nanoplastics is challenging because of their different characteristics and potential effects. Each type of particle has its own set of risks and environmental implications, which require comprehensive evaluation and understanding. To address the environmental impacts of NPs and micro- and nanoplastics, it is important to continue research on their toxicity, fate, and transport in different ecosystems. Implementing effective regulations, promoting sustainable materials and waste management practices, raising

awareness, and fostering international cooperation are key steps toward minimizing their negative effects on the environment and human health.

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19
Methods for Micro- and Nanoplastics Analysis

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Abstract

The high consumption of plastic materials, especially packages, and inefficient recycling processes are the main factors that have contributed to the pollution of terrestrial, aquatic, and atmospheric ecosystems by plastic fragments, which can be found in a large range of sizes. However, the biggest problems are generated by micro- and nanoplastics because these materials can be air transported and bioaccumulate and contaminate all living beings. Thus, consumption reduction, recycling, or the correct destination of these materials are strictly necessary; however, contamination by micro- and nanoplastics is not easy to solve, so efficient remediation methods are necessary. However, before considering the treatment methods, the best way to identify and quantify these materials should be found considering the great diversity of polymers and contaminated matrices. In this sense, this review chapter presents a description of the sources and occurrences of micro- and nanoplastics, pre-treatment of the samples before the analysis, and provides a deep review of the several techniques that can be used to characterize, identify, and quantify micro- and nanoplastics. Thus, this chapter provides an overview of the countless possibilities of techniques and their advantages, and limitations, contributing to the knowledge of this global problem.

***Keywords*:** Pollution, microscopy, spectroscopy, chromatography, spectrometry, thermal analyses

19.1 Introduction

The Industrial Revolution occurred in England in the 19th century and was marked by the transition from artisanal production processes to the fabrication of products using machines. This transition also changed the human lifestyle, which, due to the global acceleration of productive capacity, was more explored in the industrial context with high daily workloads. With lifestyle changes, there is a need to adopt habits and objects that could offer more practicalities during daily activities. Another effect of the technological advancement caused by the Industrial Revolution was the rampant use of natural resources, such as the exploitation of rubber and animal products, such as elephant ivory. In this sense, plastics developed from non-natural resources meet the needs that have arisen over time [1–3].

With advances in industrial innovation, plastics have been being developed from synthetic polymers, mixtures of organic and inorganic compounds, and additives. These innovations have enabled plastics to present unique characteristics, including different colors, mechanical and sunlight resistance, high flexibility, low thermal and electrical conductivity, high durability, resistance to corrosion, and low cost, making plastics a part of our quotidian [3–6]. From plastics, several indispensable objects have been developed for our daily lives and are part of the automotive, textile, agriculture, electrical and electronics, construction, sports and leisure, hygiene, packaging, health, and medicine industries [1, 3, 7].

For example, advances in medicine and odontology have enabled the substitution of metals and glass for polymers, improving protheses, fillings, and dental implants, as well as the production of disposable items such as syringes, catheters, blood bags, hemodialysis tubes, gloves, and surgical masks, thus making our lives easier, safer, and more pleasant [8, 9]. Considering these applications, the use of plastics in packages is highlighted, corresponding to 44% of all plastics produced worldwide, followed by civil construction (18%) and automotive areas (8%) [10]. Among all the plastic packages, 85% are not recycled and end up in landfills or the environment. Because the recycling rate is low, an annual loss of US$ 80 to 120 billion is estimated, in addition to the high pollution of ecosystems, especially oceans, with an estimated loss of 23–37 million tons of plastics in the oceans by 2040 [11].

The demand for plastic materials has driven polymer production, such that the global annual plastic production was estimated at 390.7 million metric tons in 2021, with 90.2% fossil-based plastics. Asia is the largest plastic producer, especially China, which is responsible for 32% of global production, followed by North America (18%) and Europe (15%) [10, 12]. Despite the unquestionable fact that plastic has revolutionized the industrial sectors over the centuries, thinking about the destination given to it after its utilization makes it a villain due to direct contamination caused by aquatic, terrestrial (sediment), and atmospheric ecosystems in which biota and humans are exposed. In addition, plastics can be vectors for various chemical contaminants present in the environment owing to the sorption process [7, 13–15].

Plastics vary concerning their type, composition, chemical structure, and shape and can be classified into primary and secondary microplastics; however, the degradation suffered by the microplastics can transform them into another class, the nanoplastics [3, 4, 10, 16–18]. Micro- and nanoplastics have already been identified in blood, breast milk, and placentas. Zhu *et al.* [19] evaluated 17 human placentas and found microplastics in all of them, especially polyvinyl, polypropylene, and polybutylene succinate, in a mean concentration of 2.70 ± 2.65 particles/g. Leslie *et al.* [14] evaluated blood collected from 22 healthy anonymous human volunteers that lived in the Netherlands and were maintained by the VU Medical Center Medical Ethical Committee and identified polyethylene terephthalate (PET), polyethylene (PE), and styrene polymers in mean concentrations of 1.6 µg/mL. The study did not conclude whether this concentration poses life-threatening risks to humans. In this sense, the World Health Organization (WHO) emphasized with urgency the need to have more precise evaluations concerning the risks of exposure to micro- and nanoplastics, especially for human health [20].

In addition to the risks mentioned to the ecosystems, another problem is the limitation of the analytical methods used to collect the sample, including the identification, characterization, and especially quantification of micro- and nanoplastics. Up to now, some directions concerning physical, chemical, and biological characterizations and recommendations that can be used as parameters exist; however, official measurement methods for these pollutants do not [16, 21]. To determine the contamination levels by micro- and nanoplastics with higher precision in the different ecosystems, factors such as variation in the concentrations, degradation stages, sorption capacity of other pollutants such as heavy metals and pathogens, and identification of punctual and diffuse sources need to be observed so wrong conclusions will not be reached concerning the presence and quantification of these materials [7]. Thus, several challenges and difficulties must be overcome by researchers that desire laws and environmental actions to reduce the consumption, and consequently, the discharge, of plastics, with wide cooperation among governmental and non-governmental organizations.

In this sense, the meeting that occurred in Paris from 29 May to 2 June 2023, involving 175 countries, ONGs, and companies from the plastic sector, brings hope for changes in laws that control the contamination caused by micro- and nanoplastics. In this meeting,

five debates and negotiations were performed, aiming to reach a new deal to minimize or finish the pollution caused by plastics and to incentivize the production of new plastics that have a higher biodegradability and offer less risk to the health of human beings. In the meeting opening, French President Emmanuel Macron said that the unsustainable and globalized worlds adopted up until now need to finish and that we are living in a time bomb, as a reference to the presence of micro- and nanoplastics in the different ecosystems. In this meeting, Rwanda, Norway, the European Union, Canada, Mexico, Peru, and Chile defended the idea of ending plastic pollution by 2040, while countries like China, the United States, and Saudi Arabia, in an attempt to defend their petrochemical industries, defended the idea of encouraging recycling and waste management [22–26].

Thus, this chapter aims to present the main sources, occurrences, and risks that involve the presence of micro- and nanoplastics, and which qualitative, semi-quantitative, and quantitative analytical techniques can be used to identify them in the ecosystem. The way of collecting and preparing the samples is presented, followed by the detailing of technologies such as light scattering with good results to identify nanoplastics; optical microscopy techniques that use conventional, polarized, and fluorescent light; electron and scanning microscopy that can extract information about the morphology, chemical composition, and crystalline structure; spectroscopy with electromagnetic radiation techniques; thermoanalytical techniques involving thermal analysis, gas chromatography, and mass spectrometry which are used to identify and quantify samples, and lastly more simple methods that give an idea if microplastics are present or not, such as counting, weighing, and absorbance and turbidity determination. Together with the description of these techniques, a literature review that includes published scientific studies and the results obtained with the techniques mentioned in this chapter is presented. Thus, at the end of this chapter, the reader will be able to identify the most appropriate technique to be used in the identification and quantification of micro- and nanoplastics because this chapter provides a very wide review of the several techniques that can be used to study these materials.

19.2 Micro- and Nanoplastics: Source, Occurrence, and Risks

Due to inadequate management, a low recycling rate, durability, and non-degradability, plastics are present on all of Earth, from the most populated to the most remote areas. This presence of plastics affects terrestrial (urban and rural areas), aquatic (seawater and freshwater that encompass rivers, streams, ditches, lakes, and ponds), and atmospheric ecosystems [5, 6].

Plastics can be found in a large range of sizes (from macro to nano); however, their classification regarding particle size is still not well defined. For example, when plastic materials present sizes between 2.5 and 100 cm, they are defined as plastics or macroplastics [3]. Fragments or particles with sizes between 10 mm and 1 mm [5] or 5 mm and 1 mm [3, 17, 27] are denominated microplastics, despite Gigault *et al.* [4] bringing the definition of small microplastic (25 mm to 1 mm). Nanoplastics present particle sizes lower than 1 mm (1,000 nm) [17, 28], which is the most commonly used definition; however, nanotechnology and nanomaterials areas use the higher limit of 0.1 mm (100 nm) [4, 27].

Microplastics can be divided into two categories depending on their source: 1) primary microplastics are those originally synthesized in microscopic size and are used in the cosmetics, personal hygiene, pharmaceutical, food, paints, and sandblasting areas; 2) secondary microplastics are those produced from degradation, transformation, and deterioration of big plastic materials (clothes, tires, brakes, paint, asphalt, synthetic grass, and boats) that lose the integrity of their structures and disintegrate until reaching

the micro- and nanometric scale. This occurs due to sunlight exposure, an oxidative atmosphere, and mechanical stress. Secondary microplastics are considered the biggest seawater contamination problem; however, they also affect freshwater ecosystems, streams, and estuaries, mainly via rivers, which end up flowing into the oceans and seas. Therefore, micro- and nanoplastics are considered emerging pollutants [3, 5, 6, 18, 27, 29].

Micro- and nanoplastics can be different in terms of type and composition, shape, and size. Concerning type and composition, the most common polymeric ingredients include low- and high-density polyethylene (LD/HD-PE), polypropylene (PP), polystyrene (PS), polyester (PES), polyvinyl chloride (PVC), polyethylene terephthalate (PET), polyamide (PA, nylon), polyurethane (PUR), and acrylate polymer. Among those, the polymers LD/HD-PE, PP, and PS represented 26.9, 19.3, and 5.3% of the global production in 2021, respectively, and are the main constituents of the primary microplastics, while PES, PA, and acrylate polymers constitute predominantly the secondary microplastics. Concerning the shape, microplastics can present very irregular shapes and geometries, highlighting spherical, rectangular, cylindrical, and disc formats, or in fibers, pellets, fragments, films, granules, microbeads, foam, particle, ellipse, line, and flake shapes. The particle sizes of microplastics are very varied because several factors influence them, especially when it comes to secondary microplastics [3, 10, 17, 18]. Nanoplastics result from the degradation of microplastics; thus, they are are highly polydisperse in terms of physical properties and have a heterogeneous composition, forming hetero-aggregates along with other types of materials [4].

Due to this diversity of micro- and nanoplastics, the contamination of ecosystems can occur in several ways. Soil contamination by micro- and nanoplastics is due to municipal solid waste, illegal or inappropriate dumping of industrial and hospital waste, recreational practices, runoff from wastewater treatment plants, factory and tire waste, application of sewage sludge as fertilizer, atmospheric precipitation, winds, flooding, and irrigation. Practically all these factors also contribute to the contamination of the aquatic environment, including freshwater and seawater, since the sources of micro- and nanoplastics can be dragged from the ground to freshwater and end up in oceans and seas. The atmospheric contamination by micro- and nanoplastics occurs *via* multiplicity of transport owing to their low density and size, these materials are easily air transported, especially nanoplastics, in addition to contaminations that occur from industrial and aerosol emissions. After atmospheric precipitation, the micro- and nanoplastics end up reaching the soil and water bodies, contributing to terrestrial and hydric pollution. In addition, parameters such as wind intensity, wave action, sunlight intensity, demographic density, anthropic activities, size of water reservoirs, and inadequate management practices for urban and hospital solid waste and sewage effluents affect the quantity and distribution of micro- and nanoplastic particles in ecosystems [6, 17, 18, 28].

When micro- and nanoplastics reach the ecosystems of the planet, contaminating living beings, they cause several problems: (1) due to their small size, micro- and nanoplastics accumulate in aquatic organisms because they are ingested, inhaled, and digested, affecting the entire food chain and causing several public health, ecological, and commercial problems associated with biochemical and reproductive damage and bioaccumulation. Among these, nanoplastics are more dangerous, because of their high potential to penetrate biological members; (2) microplastics act as a deposit source (carrier) to adsorb inorganic and organic toxic pollutants, such as heavy metals (Pb, Cd, Zn, Cu, and metalloids), polycyclic aromatic hydrocarbons (PAHs), polychlorinated biphenols (PCBs), and organochlorine pesticides (OCPs), which cause adverse effects, including chronic toxicity; and (3) microplastics favor the attachment and colonization of microorganisms (bacteria, fungi, viruses, algae, and protozoa) through the formation of biofilms, as well as the development of resistant microorganisms and pathogenic bacteria,

owing to their specific characteristics (small size, high surface area, and strong hydrophobicity), in addition to promoting genetic material change [5, 6, 17, 28, 30].

Thus, special attention must be given to plastic materials to reduce their consumption, recycle them, and/or give them an adequate destination. However, plastics produced decades ago are still contaminating our planet owing to their biodegradability resistance. Therefore, efficient remediation and bioremediation methods should be considered to eliminate or reduce the problem of micro- and nanoplastics. For that, several techniques can be used to pre-treat, characterize, identify, and quantify micro- and nanoplastics.

19.3 Pre-Treatment of Micro- and Nanoplastic Samples

Micro- and nanoplastics are found in all ecosystems, contaminating several matrices, such as freshwater, seawater, sediments, soils, and residual waters [31–33]. Due to the vast nature of these samples, pre-treatment stages may be necessary to remove the micro- and nanoplastics from the original material before performing its identification and quantification. In this sense, the pre-treatment can be divided into three main stages: digestion, preconcentration, and separation. Among each one of these stages, different isolated techniques can be employed, but also, and mainly, combined techniques, such as acid, alkaline, and enzymatic digestion, chemical oxidation, filtration/ultrafiltration, centrifugation/ultracentrifugation, sieving, field flow fractionation, and size exclusion chromatography, among others [34]. The pre-treatment stages aim to remove impurities and extract the micro- and nanoplastics from the matrix in such a way that their characteristics and quantities are preserved [35]. To avoid contamination or loss of characteristics, some careful steps are essential. For example, all the materials used must be made from non-polymeric materials, and laminar flow benches and air filtration systems should also be used to reduce the contamination risk from particles in the air [28]. After performing the pre-treatment stage, the micro- and nanoplastics are ready to be quantified and identified [36].

19.3.1 Sample Digestion

Depending on the nature of the sample, the separation or extraction of the organic matter from the sample or that constitutes the matrix is necessary to allow the analysis of micro- and nanoplastic particles; these may include tissue, organisms, or natural organic matter [37]. Samples of residual water, organisms, soil, sewage sludge, and sediments usually present a high organic load, so they need to be digested because the presence of organic matter can interfere with the detection of micro- and nanoplastics, especially when the microscopy technique is employed [28, 38]. However, relatively clean water samples do not need the digestion process to allow the analysis of micro- and nanoplastic particles, as is the case with seawater and potable water [39–41]. Digestion can be performed in an acid medium, for example, with 65% nitric acid [42, 43], in an alkaline medium with sodium or potassium hydroxide [44, 45], or with 30% hydrogen peroxide as an oxidant agent [46, 47]. Using enzymes is also an alternative to decomposing tissues and organic matter [42, 48]. However, these procedures should be smooth and non-destructive to the compounds that are desired to be analyzed [49, 50]. The choice of extraction method should consider the kind of sample because some protocols can damage the matrix. For example, the use of nitric acid degrades some kinds of PA, causing an underestimation or masking of the actual values of the material [51]. Depending on the matrix, more than one technique can be used to prepare the sample for identification and quantification steps.

Other factors to evaluate when choosing the protocol are cost and execution time. Alkaline digestion using potassium hydroxide is a quick and economical method that causes minimal damage to the polymeric structures compared to enzymatic digestion. The use of enzymatic digestion requires several processing steps, which increase the risk of

contaminating the air in addition to being more expensive and less efficient [52]. However, the disadvantage of alkaline digestion is the possibility of damaging the structure or decoloring the plastics [53], besides being able to leave fish bone fragments and oily residues [51]. Acid digestion, in turn, is the least indicated method, referring to possible damages in the microplastic matrix and not being adequate to extract polymers such as nylon and PET [54]. Concerning the removal of organic matter, hydrogen peroxide is commonly used due to its low cost and ease of application [55].

Piarulli *et al.* [52] investigated alkaline digestion with potassium hydroxide (KOH) and enzymatic digestion (Biozym F (lipase) and Biozym SE (protease and amylase)) to evaluate the presence of microplastics in crabs. The authors verified that digestion with KOH is a lower-cost and quicker alternative compared to enzymatic digestion, and decreases the contamination risk. The use of KOH was suggested for samples that require a short digestion time, because a digestion time longer than 4 days caused morphological changes in the PES microfibers. Such behavior was not noted for enzymatic digestion; thus, its use is recommended for samples that require a longer time. Rist *et al.* [42] also observed in their study, using optical microscopy and dynamic light scattering (DLS), that the acid, alkaline, and hydrogen peroxide treatments caused particle aggregation, which can be attributed to a change in the solution's ionic strength. In addition, these treatments can negatively interfere with the fluorescence signal of labeled plastic particles, especially in toxicological studies [42, 56]. Correia and Loeschner [48] investigated the detection of nanoplastics in fish by evaluating two pre-treatment methods: acid digestion (nitric acid) and enzymatic digestion (proteinase K). The authors verified that both methods were adequate to degrade the organic matrix. However, the acid digestion resulted in agglomerates of PS nanoparticles, which did not happen in enzymatic digestion. Dehaut *et al.* [51] evaluated the impact of different digestion methods on the quantification of microplastics in seafood samples. The methods evaluated were alkaline digestion (KOH 10% and NaOH 10 mol/L), acid digestion (65% nitric acid and 65% perchloric acid), enzymatic digestion (0.5% (w/v) pepsin diluted in 0.063 M HCl), and digestion with an oxidative solution (0.27 M potassium peroxydisulfate). The authors concluded that the best protocol was using KOH solution with incubation at 60 °C for 24 h. In addition, they discarded the use of the enzymatic method because of its implementation difficulty and digestion inefficiency. The protocol with the oxidant solution presented incomplete digestion, while the use of NaOH damaged the integrity of the polymers (cellulose acetate, polycarbonate, and PET).

19.3.2 Sample Pre-Concentration

The pre-concentration of the sample is necessary when a low mass of micro- and nanoplastic particles is present in the sample, aiming to improve the detection and quantification limits of the methods used to identify and quantify them [39, 57, 58]. Several methods are available to pre-concentrate the sample, and the choice of the best method depends on the characteristics of the sample and the subsequent steps of characterization and identification. The Raman tweezers technique, for example, requires a high concentration of micro- and nanoplastics for a good operation [6], while inductively coupled plasma mass spectrometry (ICP-MS) is a detection technique with a higher sensibility and the capacity to detect very small concentrations in samples [59]. Techniques such as membrane filtration, ultrafiltration (UF), and ultracentrifugation are commonly employed to concentrate the samples but can also be used to separate micro- and nanoplastics.

When employing the membrane filtration technique, the material of the membrane should be considered, and the most indicated ones are aluminum oxide, ceramic, or polycarbonate. Care should be taken when using polymeric membranes because they can be a source of contamination for the sample. In addition, the pore sizes should also be

considered, since pores smaller than 0.1 mm present low flow rates, affecting the filtrate volume sample [37]. Thus, for high volumes of environmental samples with a high organic load, UF can be more suitable because it can be used to concentrate and separate micro- and nanoplastics. In UF, membranes with molecular cut weights of 10 to 100 kDa are used, corresponding to approximately 5 to 50 nm [37]. UF can be performed in different configurations, such as stirred cells [39], centrifugal fields [60], and tangential flows [61]. Researchers used cross-flow UF with a polyethersulfone membrane (PES/PESU) to concentrate potable water samples and had a microplastic recovery of 50.2%. In addition, this technique identified PS in the original sample with a concentration higher than 20 µg/L [61]. Hernandez *et al.* [62] used the approach of sequential membrane filtration with pore sizes of 25, 2.5, 0.45, and 0.1 µm (twice used), being able to concentrate or separate PE nanoplastic particles in facial exfoliant cosmetics.

Centrifugation or ultracentrifugation (UC) is another strategy to concentrate samples, which allows suspended particles to sediment in a pellet. The UC is a simple technique, and it can be used when lower plastic particles are present because their density is near that of water, hampering the sedimentation process. However, this technique has limitations concerning the processed volume, which ranges from 10 to 100 mL, hampering its application in environmental water samples. Another additional disadvantage of this technique is the possibility of forming aggregates or causing damage to plastic particles due to high centrifugal forces or the redispersion of the particles in the pellet [61, 63, 64]. However, these points can be irrelevant if the subsequent analysis does not depend on the sample morphology, as is the case with pyrolysis–gas chromatography–mass spectrometry (Py–GC–MS), which promotes sample destruction anyway [37].

19.3.3 Separation of Micro- and Nanoplastics

Some separation techniques are also commonly used in the sample pre-treatment stage to remove impurities and obtain cohesive results in detection and quantification. Size exclusion separation is a widely used approach that involves different techniques, such as membrane filtration (presented in *Section 19.3.2*), field flux fractionation (FFF), and size exclusion chromatography (SEC or SEC-HPLC) [28].

FFF occurs through the application of a strength perpendicular to the particle's entrance flux [37] and is employed to fractionate biopolymers, proteins, polymers, and nanoparticles using molecular mass [65]. Several variations of FFF can use different kinds of separation fields, such as thermal, electric, gravitational, magnetic, and cross-flow fields, to attend both the micrometer and nanometer ranges [66, 67]. In addition, the FFF techniques can be combined with mass, ultraviolet radiation, refraction index, or fluorescence spectrometer detectors [68, 69]. To separate nanoplastics in water samples, the most common FFF used is asymmetrical flow field-flow fractionation (AF4); however, this technique can also be applied to evaluate nanoplastics in other kinds of matrices [61, 69, 70]. Wahl *et al.* [71] used AF4 to evaluate soil samples that contained nanoplastics. The authors verified that the use of AF4 promoted signal amplification in Py–GC–MS, demonstrating that this technique is efficient in removing organic and dissolved matter that causes interference in the detection and quantification of nanoplastics.

Chromatography is a consolidated separation and quantification method. However, for particulate samples, its application can be limited since interaction with the stationary phase may occur and the pore size might not be enough [37]. Nevertheless, its ease of application without the need for too much optimization makes chromatography an attractive technique. Concerning the use of chromatography to separate, especially nanoplastics, the most employed techniques are reversed-phase high-performance liquid chromatography (RP-HPLC) [72], size exclusion chromatography (SEC or SEC-HPLC) [73], and hydrodynamic chromatography (HDC) [74]. In the RP-HPLC technique, the separation of the molecules occurs by hydrophobicity difference, while SEC and HDC are

based on the separation of the molecules concerning their size in solution by gel filtration and the velocity difference of the mobile phase, respectively [72–74].

Mintenig *et al.* [61] proposed a methodology to determine a wide range of spectra of particle size in aquatic samples. Nanoplastic particles were evaluated by AF4-MALS (Multi-Angle Light Scattering—MALS) and Py–GC–MS. The AF4 was optimized to separate particles between 50 and 500 nm. Then, the MALS detector was able to detect PS particles in concentrations of 1 to 10 mg/L. The contribution of this study is that the AF4-MALS fractionation does not depend on the concentration but on the elution time, thus making it a promising technique to be used in the detection and quantification of nanoplastics. Castelvetro *et al.* [75] used RP-HPLC to quantify the PET micro- and nanoparticle content in aquatic sediments from a sandy beach in Tuscany (Italy). They performed the PET depolymerization using alkaline hydrolysis, then oxidation and fractionation of the hydrolysate to remove interfering compounds and pre-concentrate the monomer of terephthalic acid (TPA). The RP-HPLC quantification resulted in a recovery of 98.2% of TPA in the sediment enriched with 800 mg/L of PET micropowder, with a detection limit of 17.2 µg/kg and a quantification limit of 57.0 µg/kg. The sandy sediments showed contamination of 370 to 460 µg/kg, demonstrating the expressive results of PET being released in surface waters.

Another technique to separate nanoparticles, including nanoplastics, is optical tweezers, which allow to trap and manipulate micro- and e nanoparticles dispersed in a liquid using the minimum strength that light exerts under the matter [76, 77]. Gillibert *et al.* [78] used optical tweezers combined with Raman spectroscopy to identify micro- and nanoplastics in seawater in sizes that ranged from <20 µm to 50 nm. The authors verified that Raman tweezers enabled the capture and detection of nanoplastics in seawater, like poly(methyl methacrylate) at 300 nm.

Considering what has been shown in *Section 19.3*, the pre-treatment (digestion, pre-concentration, and separation) of the micro- and nanoplastics samples is essential to improving the sensibility of the detection methods. Those stages depend on the sample characteristics and the subsequent analyses. Each technique has its advantages and specific considerations, such as the membrane choice of the material, the pore sizes, and the possibility of forming aggregates or causing damage to the plastic particles. The selection of the most adequate method is important considering the analysis necessities and the certainty of the integrity and representativeness of the samples. However, the biggest challenge concerning the pre-treatment stages for later detection and quantification of micro- and nanoplastics is the lack of standardization in the treatments employed. Due to the large variety of samples and polymers, each stage should be adapted to the sample that is being analyzed, thus hampering the large-scale analysis of diverse materials.

19.4 Methods for Characterization, Identification, and Quantification of Micro- and Nanoplastics

After performing an adequate pre-treatment of the samples containing the micro- and nanoplastics, considering their characteristics and posterior analysis techniques, the characterization, identification, and quantification of the micro- and nanoplastics should be performed. For that, several techniques can be employed, among which light scattering, optical microscopy, electron and scanning microscopy, spectroscopy, thermal analysis, gas chromatography, and mass spectrometry can be mentioned. Each one of these techniques presents its own application specificity, advantages, and disadvantages. Thus, this item summarizes the main techniques and approaches, as well as application examples, as illustrated by Figure 19.1.

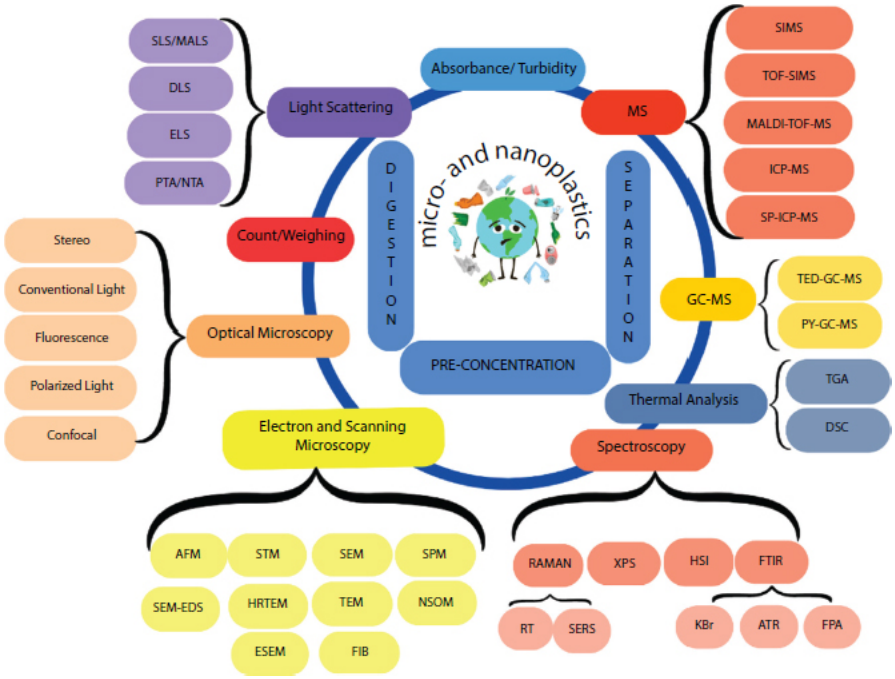


Figure 19.1 Characterization, identification, and quantification methods for micro- and nanoplastics

(source: authors).

19.4.1 Light Scattering Characterization Methods

Different light scattering (LS) techniques can be used to characterize different kinds of particles. The LS technique has been used over the years in polymers' science to characterize several plastic structures on micro- and nanometric scales [79, 80]. Among the LS techniques, static light scattering (SLS), dynamic light scattering (DLS), particle tracking analysis (PTA), and nanoparticle tracking analysis (NTA) can be highlighted. These techniques have been explored especially to monitor and characterize the formation of nanoplastics resulting from the plastic and microplastic fragmentation in simple matrices, such as saltwater [79, 81].

The SLS method, also denominated Multi-Angle Light Scattering (MALS), registers the light intensity scattered at different angles due to light interaction with the particle to obtain information about the particle sizes. The SLS can measure sub-micro- and nanoplastic polydisperse particles, including a wide range of sizes from 10 nm to 10 mm [68, 82]. Correia and Loeschner [48] used asymmetrical flow field-flow fractionation (AF4) coupled to MALS to detect PS nanoparticles (100 nm, PSNPs) in samples of fish enriched with nanoplastics (1.3 mg/g of fish). The authors verified that the AF4 method could be used to separate the PSNPs from the fish, which was previously digested by the enzymatic method, and to determine its size using MALS. The detection limit was 52 µg/g of fish.

The Dynamic Light Scattering (DLS) method is used to obtain information about the size and particle size distribution, especially in the range of 1 nm to 3 µm. Its functioning mechanism is based on the variation of a light beam intensity that crosses the suspension. This variation is associated with the Brownian movement of the particles, which, in turn, is associated with the hydrodynamic diameter of the particles. This technique is commonly employed to characterize nanoplastics; however, it is limited to detecting big particles and suffers interference from contaminations such as residues from the matrix, aggregates, or dust [82–84]. Hernandez *et al.* [62] used the DLS method to determine the particle size distribution of nanoplastics (PE) in commercial exfoliating cosmetics. The

analysis indicated the presence of particles smaller than 100 nm in the samples. The authors used another technique (scanning electron microscopy) to verify the size and shape of the nanoplastics and verified that the DLS measures resulted in a higher mean referring to the particle size in samples from a polydisperse suspension, thus justifying the differences between the techniques used. Ter Halle *et al*. [39] studied the presence of polymeric particles in seawater. For that, the colloidal fraction was isolated from the seawater samples, and after the UF process, the detection of the particles was performed using DLS. The authors verified that the polymer particles presented variations in sizes on the micro- and nanoscales. The following polymers were identified using the Py–GC–MS technique: PVC, PET, PS, and PE.

In most cases, digesting the matrix is first necessary so that micro- and nanoplastics can be separated from the organic residues and other particles, and lastly, the adequate identification of the nanoplastics can be performed. Thus, hyphenated techniques are used, that is, different techniques are coupled, for example, AF4-HDC-UV-MALS-DLS, which is attractive because it allows the separation of small molecules, particles, and macromolecules in a wide range of sizes (approximately 1–1,000 nm) before detection by light scattering, absorbance, or chromatography quantification [48, 85].

A variation from DLS that has been gaining attention is the electrophoretic light scattering (ELS) technique, which measures the fluctuation of the light intensity caused by the movement of particles in an electric field, allowing the determination of zeta potential and the characterization of the suspensions' stability, including plastic particles oxidized in the environment [82]. Lim *et al*. [86] evaluated how different kinds of microplastics (PS, PET, HD-PE, and lLD-PE) can adsorb chemical contaminants, such as dyes (BB9 and RR120) and heavy metals (Cd(II), Pb(II), As(III), and As(V)). The ELS (laser Doppler method) was employed to evaluate the surface charge of microplastics using a zeta potential analyzer in different pH conditions. The authors verified through this analysis the influence of pH in the adsorption process and verified a gradual decrease of zeta potential with a pH increase. The cationic dye adsorption is favored when pH > pH_{pzc}, while for pH < pH_{pzc}, the anionic dye adsorption is favored.

The particle tracking analysis (PTA) technique, also called nanoparticle tracking analysis (NTA), uses a microscope and a digital camera to register the light scattered through the particles. Through the Brownian movement of the particles, the processing software tracks their movement and determines their hydrodynamic diameter. Thus, with this technique, the size and the particle size distribution are possible to be determined, as is the particle concentration from the counting of the number of particles present in a certain sample volume. The NTA is less affected by big particles in polydisperse samples, making it an alternative for DLS to measure the size and size distribution of submicroplastics (0.1–1 μm) and nanoplastics (1–1,000 nm), including secondary fragmented plastic, since it can detect samples 10–1,000 times more diluted than DLS does [69, 81, 84, 87, 88]. The fluorescence combined with the particles' coloration can be used in the NTA, although it is necessary to evaluate possible interferences in the sample digestion [37]. The main disadvantage of NTA is that this technique does not provide images of the particles visualized, and the structure of the particles cannot be evaluated [89]. Lambert and Wagner [88] studied the fragmentation of PS in nanoplastics from a coffee cup lid. The lid was cut into squares of 1 cm, which were immersed in 20 mL of demineralized water and put in a weathering chamber at 30 °C with 24 h of light exposure (in the visible and ultraviolet ranges). Samples were collected within 7, 14, 28, and 56 days and characterized by NTA. This experiment demonstrated the increase in the formation of nanoplastics with time. At the end of 56 days, the control (water only) presented a nanoplastics concentration of 0.41×10^8 particles/mL, while the sample had 1.26×10^8 particles/mL, with a mean size of 224 nm.

19.4.2 Characterization, Identification, and Quantification Methods by Optical Microscopy

Structural observations of the morphology and size of particles are limited by human vision. Thus, optical microscopy is a tool that aids observation using visible light to form an image. The microscope has two lenses and a source of light. Each one of the lenses presents different functionalities; the objective lenses absorb the light emitted by the sample, forming an inverted image, and the final image is produced by the ocular lenses with direct observation and perceived by the vision [90]. The qualitative and quantitative characterizations of micro- and nanoplastics are performed by several optical microscopy techniques, among which stereoscopic microscopy and conventional light, fluorescence, polarized light, and confocal microscopy can be highlighted. Since familiarity with each one of these techniques is crucial to obtaining a more efficient and precise result, a brief description of each one and its application will be presented, containing its advantages and disadvantages.

19.4.2.1 Stereoscopic Microscope

The stereomicroscope (stereo microscope) offers tridimensional vision of the sample owing to its lenses with two optical paths and superior illumination. The equipment allows for the classification and visualization of the sample surface, thus allowing more complete mapping [27]. Most stereomicroscopes have objective lenses projected to visualize objects at low amplitudes (up to ×40), enabling microscopic vision without the need to prepare samples [91, 92]. This equipment is a valuable tool for identifying and classifying microplastics as well as evaluating their distribution and abundance in water, sediment, and soil samples. It can be employed in combination with other techniques, such as Fourier and Raman spectroscopy, to determine the composition and source of microplastics [27, 93]. Stereoscopic microscopy is expensive because of the need for special accessories to visualize tridimensional images. It is also not recommended to analyze a sample with a high organic load, which hinders the characterization of macro- and microplastics, as well as transparent particles of smaller sizes to avoid providing data from a kind of polymer [27, 94]. In the visual analysis of microplastics, the results can be discrepant owing to the individual experience of each researcher. To obtain more precise results, amplification should be considered, which ranges from ×6 to ×100 [27]. The technical details of the equipment influence the analysis regarding particle size, with sizes of 1 to 100 µm being the most common. It is important to highlight the speed and ease of this technique to identify the form, size, and colors of the samples compared to other techniques [27, 93].

Wang *et al*. [95] evaluated the presence of microplastics in freshwater in urban regions of China. The samples were collected, treated with 30% H_2O_2 to digest the organic matter, diluted in distilled water, filtered with a fiberglass filter (0.45 µm), and then visually analyzed using a stereomicroscope. Mean particle concentrations of 1,660 ± 639 to 8,925 ± 1,591 number/m^3 were visualized. Concerning the polymer kind, colored plastics were also found. Yoganandhan *et al*. [96] used a stereomicroscope in the ampliation ×40x to identify microplastics in a wastewater treatment plant. In the coagulation stage, aluminum sulfate (1, 2, and 3 g/L) was used as the coagulant and the separation of the samples was performed by sieving (4.75 mm) and then filtering (1 µm), followed by direct analysis under a microscope. The raw and treated samples presented maximum lengths of 18.35 and 4.2 mm and minimum thicknesses of 0.0933 and 0.0833 mm, respectively. Limbago *et al.* [93] used the stereomicroscope to quantify and identify the form of microplastics in surface sediments collected in upstream and downstream stations of the Molawin River in the Molawin Watershed. The sediment samples were weighed, dried at 60 °C for 48 h, and sieved (2 mm to 0.63 µm). With the amplification of ×40, microfragments and microfibers were found with particle sizes between 100 and 200 µm

downstream, 97 ± 12 items/100 g, the highest mean concentration in dry weight, confirming the presence of microplastics in the river. The previously mentioned studies and others can be found in Table 19.1.

19.4.2.2 Conventional Light Microscopy

The conventional light microscopy technique is a laboratory tool and is necessary for microplastic analysis. The principle of this technique is based on transmitted light, in which the light passes through the sample, thus generating the amplification result, allowing the identification of particles from 10 µm to 1 mm [27]. This technique presents simple and easy sample preparation, aiding its study; however, it is dependent on the researcher, whose interpretation may vary, resulting in low precision. The characterization of colored microplastics is easier than that of transparent microplastics because the latter are hard to identify with a size lower than 100 µm [97]. In addition, inadequate separation of the particles may interfere with characterization by assuming fiber and fragment forms, complicating the observation of microplastics. Visualization quality can be improved by combining it with other microscopy techniques, such as stereomicroscopy and fluorescence [98].

Masura *et al.* [21] analyzed samples of microplastic debris in seabed sediments. To prepare the sample, the sediment was dried (90 °C), sieved (0.3 mm), and subjected to a digestion process at 75 °C with the addition of 20 mL of Fe II (0.05 mg/L), 20 mL of H_2O_2 30%, and 6 g of sodium chloride. The microplastics were separated using tweezers and analyzed using conventional light microscopy with an amplification of ×40. This technique is commonly used to separate and visually identify microplastics, and thus the results in this study were qualitative; that is, the microplastics in the marine sediments analyzed were identified. Valine *et al*. [99] analyzed the presence of microplastics in river water from the Washington region in the United States, covering both rural and urban areas. In the initial characterization, the conventional light microscopy technique was used and the presence of 2.9 million microfibers per hour was identified. However, the authors observed that this technique presented low data reliability, particularly when the particles were small or transparent.

Table 19.1 Characterization and identification methods of micro- and nanoplastics by optical microscopy.

Technique/Equipment	Origin or source of sample	With or without pre-treatment	Range	Consideration/Results	Refere
Stereoscopic microscope	Microplastics in surface sediments	The samples were dried (60 °C/48 h) and sieved (2 to 0.63 mm)	100 to 200 µm	In the ×40 ampliation, microfragments and microfibers were identified	Limba *al.* [93
Stereoscopic microscope	Microplastics in effluent treatment plant	Coagulation using aluminum sulfate, sieving separation, and filtration (1 µm).	18.35 and 4.2 mm	In the ×40 ampliation, particles with thickness of 0.0933 and 0.0833 mm were identified.	Yogan *et al.* [
Stereoscopic microscope	Micro- and nanoplastics in surface fresh water	Chemical oxidation with H_2O_2 30%, dilution with H_2O, and filtration with a fiberglass filter (0.45 µm)	0.5 to 5 µm	Colored polymers, fibers, PE, and PP were detected with mean particle concentrations of 1,660 to 8,925 number/m^3.	Wang [95]
Conventional Light Microscopy	Microplastics in marine sediments	Drying (90 °C), sieving (0.3 mm), digestion with Fe II solution, 30% H_2O_2, NaCl, and tweezers separation	<0.3 mm	In the ×40 ampliation, easy visual identification was possible with qualitative results	Masur [21]
Conventional Light Microscopy	Microplastics in fresh river water	Digestion with KOH and filtration (200 µm).	0.5 mm to 200 µm	In the ×40 ampliation, 2.9 million microfibers/h were found	Valine [99]

Technique/Equipment	Origin or source of sample	With or without pre-treatment	Range	Consideration/Results	Refer
Fluorescence Microscopy	Microplastics in three matrices of soil samples	1) untreated quartz sand; 2) sandy soil treated at 550 °C/4 h, 3) sandy soil without treatment	1 to 2 µm	Identification of fluorescent PE added to the sample.	Nick *e* [105]
Polarized Light Microscopy	In situ microplastics	Sample stirring at 100 rpm	10 µm	Identification of PS with polarized light scattered at 120°	Liu *et* [104]
Polarized Light Microscopy	Microplastics in residential water	Digestion at 75 °C/30 min with $FeSO_4$ and H_2O_2, filtration (20 µm), and density separation	70 to 600 µm	Identification of PE, PP, and PET at ×4 and ×10 magnifications	Sierra [106]
Confocal Microscopy	Microplastics in surface water	Extraction fiber films fragments using 90% alcohol, and distilled water	0.5 to 5 mm	Identification of PE and PP	Melo-Augus *al.* [30
Confocal Microscopy	Micro- and nanoplastics in plant tissues	Root, stem and leaf samples between 40 and 100 mm thick	0.2 to 2 µm	Identification of PS spheres	Li *et a* [109]

19.4.2.3 Fluorescence Microscopy

Fluorescence microscopy is based on the fluorescence phenomenon and uses an ultraviolet light source, and a dye that interacts with the sample that absorbs and emits light at specific wavelengths inside the visible spectrum. The light emitted by the illumination system retains radiation at a longer wavelength than that produced. Thus, the excitation filter transmits waves of shorter lengths, whereas the barrier filters emit longer wavelengths, allowing a higher-quality contrast. This technique allows the identification of images from which it is possible to determine the number, volume, and morphological characteristics of microplastics present in the samples [94, 100]. Fluorescence microscopy is an important technique, particularly for identifying white and transparent

polymeric particles with inherent fluorescence [94, 101]. Nile Red dye is used as a substitute to interact with colored microplastics, but other dyes can also be used, such as iDye pink, iDye blue, Rit 94 DyeMore Kentucky Sky, iDye Poly, and Rit DyeMore. These dyes fluoresce synthetic polymers in environmental samples. However, the sample coloration required a pre-treatment that can be performed by coloration in filter paper or a solvent such as methanol, chloroform, or acetone. Environmental factors, such as pH, dissolved oxygen concentration, ionic strength, and temperature, affect the interaction between the sample and dye [102, 103]. The disadvantages of this technique are: (1) the UV laser causes bleaching of the sample and toxicity when a sample with organic matter are analyzed, which reduces the observation time of the sample; (2) the technique does not detect some kinds of polymers, such as PVC, PA, and polyethersulfone (PES), not providing information about the chemical bonds of these polymers; (3) the technique needs to be coupled to other analytical techniques, commonly Infrared absorption spectroscopy or Raman spectroscopy to present more precise and complete results [94, 102, 104].

Nick *et al*. [105] used fluorescence microscopy to identify microplastics in three soil matrices, which were prepared with five microplastic concentrations. The first matrix was pure quartz sand, the second matrix was clayey sandy soil, where the organic matter was removed by thermal treatment (550 °C for 4 h), and the third matrix was clayey sandy soil without treatment. These matrices were enriched with fluorescent PS microparticles (0.1, 0.03, 0.01, 0.003, and 0.001%) from an aqueous solution at 2.5%. Fluorescence microscopy allowed the visualization of microplastic particles of 1–2 µm in the untreated samples, and in the treated samples, the technique was even more precise. The previously mentioned studies and others can be found in Table 19.1.

19.4.2.4 Polarized Light Microscopy

Polarized light microscopy is a sensitive analytical method in which the microscope presents two reflector prisms, where the polarizing reflector is placed between the light source and the sample and the analyzer reflector is located between the objective lenses and the visualization tube. The polarized beam is divided into two parts, generating vibrations and emitting a light source, which allows the analysis of samples with different mechanical properties without plane symmetry. This technique provides more information than conventional light microscopy because it is more sensitive to the sample microstructure; therefore, it is an economically viable method as an alternative to analyzing microplastics. The main application of this technique is the characterization of biological tissues, microalgae, and microplastics *in situ* with particle sizes of <50 µm [97, 104, 106, 107]. Polymer molecules are anisotropic and have the capacity to change the polarization through the polarization plane at an angle of 90°, thus resulting in birefringence, the optical property of a material that has different indices of refraction for different directions of light propagation. The waves of double refraction follow different paths and vibrate in perpendicular planes, which allows the polarized light to be used to study the form and crystals of the crystalline polymeric materials, making it possible to relate the morphology and sample properties [105–107]. The properties of the polymeric structure can affect the results owing to polarization transmission. Thus, the samples must not be opaque for better passage of polarized light. In microplastic studies, an efficient application is the characterization of PE particles in toxicological analyses [97].

Liu *et al*. [104] performed the *in situ* characterization of microplastics by scattered polarized light microscopy at 120°, keeping the stirring of the particles in suspension at 100 rpm. The authors used three samples, microalgae, river water, and commercial microplastics, which were stored at room temperature (20 ± 2 °C) with preservation times of 4, 4, and 48 h, respectively. This method allowed the identification of different particles in water and other samples as well as the distinction of PS microplastics with

different sizes and shapes. Sierra *et al.* [106] evidenced the presence of microplastics in residential waters in Latin America. The samples were digested at 75 °C for 30 min with 80 mL of H_2O_2 and 40 mL of $FeSO_4$ (0.05 mg/L) and filtered (20 µm). Then, the remaining solid fraction was separated by density using a saturated solution of NaI, and the solids retained were washed with isopropanol alcohol and water, dried until constant weight, and analyzed by polarized light microscopy at amplifications of ×4x to ×10. The authors identified particles between 70 and 600 µm in PE, PP, and PET. This result provides evidence that this technique is efficient and effective for characterizing microplastics. Other studies are listed in Table 19.1.

19.4.2.5 Confocal Microscopy

In confocal microscopy, a continuous laser source irradiates only one point of the sample, reaching a predetermined depth. It contains a set of lenses and filter systems that capture the fluorescence emitted in the focal plane, obtain high-resolution and high-contrast images through optical cuts, and are posteriorly grouped to form a sample topography [108]. This technique can be used to analyze samples with sizes and thicknesses of 0.2 µm, allowing 3D images of micro- and nanoplastics [108]. Confocal microscopy is easy to implement, allowing the characterization of micro- and nanoplastics as a sensible method to detect these particles in their aquatic environment or the interaction with live microorganisms. Information regarding the surface composition of polymeric particles is also impossible to obtain [27, 100]. As with other techniques, confocal microscopy also has limitations, and its main disadvantage is the longer image acquisition time owing to the excitation of the confocal point. However, this technique is efficient for visualizing structures of interest with high resolution using a laser to excite the fluorophores (functional groups), allowing the observation of better images [108].

Melo-Augustín *et al.* [30] evaluated different microscopy and spectroscopy techniques to analyze microplastics in samples of surface waters collected in the Manzanillo and Santiago Bays. Microplastics were extracted from fragments, fibers, and films using alcohol 90% and distilled water, and the particles with sizes of 0.5 to 5 mm were selected for the study. The confocal microscopy technique determined organic pollutants present in surface waters, PE, PP, and heavy metals (arsenic and chromium), which are sources of economic activities and sewage disposal. Li *et al.* [109] determined microplastic particles present in vegetable tissues by confocal microscopy in the excitation and emission wavelengths of 488 and 515 nm or 635 and 680 nm, respectively. Root, stem, and leaf blade samples were sectioned into 40 and 100-mm thick sections. Untreated sections of the plants (control) did not show detectable autofluorescence under specific excitation. When the roots were treated with fluorescent PS microspheres, the spheres were identified. Few luminous signs were observed in the vascular system. Other studies are listed in Table 19.1.

Based on what was discussed in *Section 19.4.2*, it is important to highlight that each technique has its limitations and advantages according to its functioning principle. Stereoscopic microscopy is usually employed in samples with larger sizes (1–100 µm), while fluorescence microscopy allows the analysis of white and transparent samples using radiation and dye. On the other hand, polarized light microscopy is adequate for characterizing anisotropic polymers that form two refracted waves. Confocal microscopy allows obtaining higher-quality images with topography and 3D formation.

19.4.3 Characterization and Identification Methods by Electron and Scanning Microscopy

Electronic microscopy techniques use a beam of electrons to generate a signal from the interaction of these high-energy electrons with the sample. From these signals, information about the sample can be obtained, such as morphology, chemical

composition, and crystalline structure. Scanning electron microscopy and transmission electron microscopy are the two main types of electronic microscopy. In scanning electron microscopy, the electrons scan the surface of the sample, producing high-resolution images and enlarging the sample surface; in transmission electron microscopy, the beam is focused on the ultrathin sample interacting with it as it passes through it, forming a high-resolution image with amplification and analyzing the interior of this sample cut into thin layers [110, 111]. In this section, the main scanning and transmission electronic microscopy techniques are discussed, together with their advantages and limitations, to gain specific knowledge about the analysis of micro- and nanoplastics using these techniques.

19.4.3.1 Scanning Electron Microscopy (SEM)

The Scanning Electron Microscopy (SEM) is based on an electron beam emitted by a tungsten lamp, which scans the sample surface from point to point, interacting with the sample and exciting the electrons from the most external layers of the sample. The electronic signals generated by this interaction were captured by the detectors, forming a high-resolution image of the surface of the sample. The main components of SEM are the source of electrons (usually a tungsten lamp) and electromagnetic lenses that compress the electron beam focused on the sample surface at 10 nm, projecting the image in a special plane (3D). X-ray detectors can also be associated with scanning electronic microscopes, which are the most common energy-dispersive and wavelength-dispersive detectors [27, 111]. In the SEM analysis of polymeric samples, the samples must not present any moisture, which can interfere with the analysis and result in a poor-quality image. The methodology requires that the sample be a good conductor so that the image is formed; however, polymeric materials have a low capacity to conduct electrons. Therefore, the sample needs to be covered with gold powder so that it becomes a conductor [27, 112]. SEM monitors the presence of polymers with an amplification of up to ×500,000 and allows the evaluation of the interaction of the polymeric particles with other materials [112, 113]. Depending on the sample, the methodology to prepare is defined. For example, the sediment samples are separated by density because it is the most common technique; however, they can also be prepared by ozonation, a less reactive technique with polymers [114].

Lee and Jung [115] quantitatively analyzed the presence of microplastics in the production of marine salt. Coagulation was used as the separation method, with aluminum chloride, aluminum sulfate, iron chloride, n-butyltrichlorosilane, or n-octicchlorochlorosilane as the coagulants. The morphology of the flocs was observed by SEM, where the aggregated flocs were collected using a 1 mm filter and dried at 50 °C. Poly(n-butyl methacrylate) (PBMA) was dispersed in simulated seawater at 3% concentration and stored for 24 h without any alteration in its chemical or physical properties. The particle size distribution of PBMA ranged from 200 µm to 1,500 µm, and white flocs were observed in all samples, regardless of the type of coagulant and salinity. The SEM analysis provided results consistent with the evolution of the removal efficiency. Fries *et al*. [116] identified different types of polymers in sediment samples. For sample pretreatment, separation by density using sodium chloride was employed. To perform SEM analysis, the microplastics were transferred to a conductor carbon tape set up in a sample holder. Different fragments of PE, PP, PS, PA, chlorinated polyethylene, and chlorosulfonated polyethylene polymers were identified, which were blue, white, and transparent in color, with rounded and sharp edges of below 350 mg. Other studies are listed in Table 19.2.

19.4.3.2 Scanning Electron Microscopy With Energy Dispersive Spectroscopy (SEM-EDS)

Scanning electron microscopy together with the energy dispersive detector (SEM-EDS) allows one to qualitatively and semi-quantitatively characterize the chemical composition of the sample surface based on the photon energy principle [27]. The EDS resolution is about 120 eV, depending on the interaction zone with the electrons. The signal force is high because the detector is near the sample. Thus, the intensity of the electron beam is lower, causing less damage and contamination of the plastic surface of the sample. The polymers are identified from the spectra and compared with reference values. However, the sample surfaces must be treated to reduce quantification errors and to avoid rough surfaces [117]. The SEM-EDS analyzes only solid samples, providing quick information during its triage concerning the presence or decomposition of polymer particles to form micro- and nanoplastics with an amplification of up to 500,000 times. Those two techniques together are fundamental to characterizing the particles; while SEM provides information about the morphology and identification, EDS provides a qualitative and semiquantitative microanalysis. The SEM-EDS method also identifies polymeric additives, whether of plastic origin or not, present in microplastics. Thus, this technique is a fundamental tool to characterize micro- and nanoplastics [112, 113].

Blair *et al.* [117] used SEM-EDS to characterize microplastics in sediments from a Scotland River after extraction by density separation in sodium chloride solution. The particles were identified and enumerated at a tension of 20 keV with a mean size of 2.8 mm to 11 µm. This technique showed that the suspense granules were mainly metallic, indicating that the pellets are confused with microplastics by visual inspection. Dąbrowska *et al.* [118] analyzed microplastics in samples of cosmetics in Poland using the SEM-EDS technique to obtain information about the morphology of the microplastic particles. The preparation consisted of diluting 150 mg of the samples in 10 mL of ultra-pure water, filtering with a glass filter, and washing with ultra-pure hexane to remove apolar cosmetic components. In cosmetics, agglomerates of particles were found in layers or spheres, in addition to some products also presenting round silica particles. The analysis confirmed the presence of metal and organic and inorganic compounds, even in the purified material. PE was dominant among the synthetic microspheres in all kinds of products, while abrasive PP particles are currently almost absent in all products on the market. Ding *et al.* [119] used SEM-EDS to analyze microplastics in the soft tissue and digestive tract of shellfish and fish collected from Qingdao and Dongying markets. The shellfish had their soft tissue removed, and the microplastics were separated from the organisms using 10 mL of KOH 10%, ultrasound for 5 min, and incubation at 90 rpm at 60 °C for 24 h to remove organic matter. The fish preparation process was similar to that of shellfish; however, the digestion time ranged from 24 to 48 h. Fish were more contaminated with microplastics in lower sizes (1 mm) in the form of fibers, fragments, granules, and films compared to bivalves. Other studies are listed in Table 19.2.

19.4.3.3 Environmental Scanning Electron Microscopy (ESEM)

Environmental scanning electron microscopy (ESEM) is different from other conventional techniques because it allows the analysis of conductors and insulating materials without the need to recover the sample. The analysis of insulating materials is possible due to the acceleration column, which has a pressure control system that allows the existence of a pressure gradient in its interior. In addition, ESEM enables the characterization of samples with higher moisture content in a low-pressure nitrogen atmosphere or with volatile liquids in their composition. However, moisture should be considered in some cases. Thus, this technique is quick and does not need special sample preparation since it can characterize isolating and hydrated materials. As a disadvantage, the energy of the

beam can dehydrate and damage the samples, harming the analysis and the visualization of the image on the monitor [37, 120].

Herbort *et al.* [121] analyzed the removal of plastic particles in freshwater samples. Two kinds of samples were prepared with different concentrations. In the first method, 100 g of PE or PP and 1 L of deionized water, n-alkyltrichlorosilane, methyltrichlorosilane, n-butyltrichlorosilane, n-octyltrichlorosilane, or n-dodecyltrichlorosilane were mixed with posterior homogenization, filtration, and drying of the solids at 60 °C for 24 h. The second sample was prepared similarly but with double the volume of deionized water (2 L). The ESEM technique was used to verify the agglomeration of the molecules in images with a scale of 1 mm, and it was identified that PE formed an aggregate derivative of n-alkyltrichlorosilane. The n-alkyl substituted chlorosilanes were used due to their high reactivity in contact with water, forming silanols and leading to the formation of the macromolecular hybrid silica in the form of a gel, which includes the PE particles. Vianello *et al.* [122] analyzed, using ESEM, the morphological surface and the nature of microplastic particles present in sediments from Venice lagoon. The following polymers were identified in the sediments: PE, PP, PES, polyacrylonitrile, PS, alkyd resin, PVC, polyvinyl alcohol, and nylon. The PE, PP, and PS fragments presented irregular shapes and similar surface textures, while the microfibers presented amorphous properties and regular surfaces. Schuhen *et al.* [123] evaluated the reduction of microplastics in desalination plants to produce marine salt, thus improving the yield of the membranes. In the sample preparation, 0.1 g of PE, 0.1 g of PP, 1 L of saltwater, and 0.15 mL of binding reagent were added, and then the sample was filtered and the filters dried in an oven to be observed under a microscope. The ESEM images of the particles showed the formation of aggregates bounded by the reagent, leading to a considerable size increase. Other studies are listed in Table 19.2.

19.4.3.4 Focused Ion Beam Microscopy (FIB)

In the focused ion beam (FIB) microscope, the beam is directed to the sample, usually using electrostatic or magnetic lenses, providing high precision and the capacity to control where the ions are deposited or detected. Double ion beams (Ga^+) can be generated at energies of 10–49 keV from a plasma source or accelerated through a particle accelerator before being focused. FIB can be used coupled to the scanning electron microscope (FIB-SEM) to evaluate the morphology of the samples, or FIB can also be used to prepare samples for posterior analysis by TEM because it allows the selection of specific regions, such as grain contours and Raman spectroscopy [124]. The FIB and SEM techniques are very similar; both are used to acquire high-resolution images by collecting the secondary electrons. However, they present basic differences, for example, in the type of samples that can be analyzed. The FIBM technique is used to characterize semiconductors and insulating samples. The advantages of this technique are the high resolution and image detailing because the beams can reach micro and nano sizes, as well as the fact that this is a non-destructive technique. The disadvantage is the size of the sample, which is limited, especially in bigger samples [124, 125].

Bullard *et al.* [126] published a study about the decomposition of microplastics by abrasion. The FIBM provided information about the chemical composition of the microplastic surface samples. The authors verified that microplastics can be decomposed during air transportation and have the potential to be transported over long distances, in addition to being within the size range for respiration. The sphere microplastics of lower and higher size (640 nm) had similar composition before and after the abrasion process, with 85.3% carbon, 3.11% oxygen, 4% aluminum, and 0.5% iron. The other particles had 15.8% of titanium, in addition to sodium, magnesium, sulfur, and potassium, with concentrations lower than 1.5%. After the abrasion process, both suffered breakage of carbon composition between 30 and 40%, in addition to the decrease of other compounds.

Thus, the FIBM analysis enabled us to verify that the characteristics of the microplastic surface can be chemically changed during abrasion. Valsesia *et al.* [127] characterized the nanoplastics from seawater mussel fabric. The samples were subjected to digestion, filtration, and FIB to eliminate the organic matter and possible contaminants, thus avoiding the formation of an oxidative layer and improving the efficiency of the nanoplastics detection by the confocal Raman microscopy analysis. The particles had their sizes reduced through the process, and the FIB surface treatment showed itself to be selective for biological matrices that presented different densities and chemical compositions to the polymers. Other studies are listed in Table 19.2.

19.4.3.5 Scanning Probe Microscopy (SPM)

Scanning probe microscopy (SPM) provides high-resolution images, allowing visualization at an atomic level in several mediums (liquid, gaseous, and ultra-vacuum), so it is an important technique in the nanotechnology area. Thus, the main advantage of this technique is the high nanometric resolution of the images produced, preserving the sample surface with simple sample preparation. The strength emitted by the tip is absorbed by the sample without damage, depending on the kind of sample. The SPM investigates mainly rigid surfaces and conductors' materials; however, it can also be used to evaluate the surface of non-conductor polymers, such as microplastics, by providing 3D structure images. The disadvantage is that it cannot characterize external contamination factors [94, 128, 129].

The SPM can be divided into two classes: scanning tunneling microscopy (STM) and atomic force microscopy (AFM). The STM allows mapping the atoms' surfaces in conductor samples at high resolution, atom by atom, without the use of electrons or light beams. A scanning tunneling microscope consists of scanning by a conductor probe near the surface of the conductor sample. When the tip comes near the surface, usually less than 1 nm, the electron cloud penetrates the first atom, and the surfaces begin to overlap. In this configuration, the application of a polarization tension between the tip and the surface conducts the electrons to a tunnel through the potential barrier from the tip to the surface through an overlying electron cloud, thus generating a current. This tunneling current is very sensitive to the space between the probe tip and the surface and exponentially ranges with the distance between the tip and the sample [128–130].

Atomic force microscopy (AFM) allows the identification of the topography of insulating materials, such as polymers, but also provides the identification of the physical-chemical, electrical, magnetic, and mechanical properties of the surface. The technique is not influenced by the interaction of the matter with light, and the probe percolates the surface of the sample at a very small distance, resulting in Van der Waals interactions [128, 129].

Akhatova *et al.* [128] published a study about the use of AFM in the Nanomechanical PeakForce Tapping mode to visualize the absorption and distribution of spherical PS microplastics in human skin fibroblasts. The samples were collected from adult skin biopsies, and the cells were cultivated and supplemented with 10% bovine fetal serum, 100 IU/mL of penicillin, and 100 µg/mL of streptomycin. For the AFM analysis, the sample was put in a colloidal suspension diluted with ultrapure water, and 5 µL of the solution was deposited on the clean glass slide. PS particle images of different sizes (100, 200, 500, and 1,000 nm) were observed, which presented a spherical shape. The technique was unsatisfactory for analyzing particles smaller than 200 nm in topographical vision. For other particles, the technique was the most precise, identifying the absorption of bigger particles by the cell. Vitali *et al.* [129] described the main techniques of characterization of micro- and nanoplastics in food products and ensured that the AFM analysis allows the evaluation of the size and shape of the particles, in addition to providing 3D images and topographical information with subnanometric resolution,

depending on the probe size of its environment. The authors also used AFM to study the topography and thickness of fragments released from PP bottles during the preparation of infantile nutrition. The AFM images allowed them to identify that most of the isolated particles come from baby bottle degradation because the shape of the floc and the morphology in layers were comparable to PP desquamation. Other studies are listed in Table 19.2.

19.4.3.6 Near Field Scanning Optical Microscopy (NSOM)

Near-field scanning optical microscopy (NSOM) combines scanning techniques with conventional microscopy, projecting and allowing the examination of nanostructures. The conventional techniques of microscopy are limited by light diffraction, and the resolution is in the range of 75 to 150 nm. The basic principle of this technique consists of passing the light through a punctual source by the entrance, illuminating the sample inside the near field. The light reflection generates the evanescent field phenomenon, which occurs only near the surface that reflects the light, receiving the name near field [131, 132]. Dielectric or metallic nanotips are fundamental to developing the technique because they scan the sample from point to point where the field is propagated, causing the plasmon phenomenon on the surface. This allows the investigation to have a higher resolution limit and allows the visualization of structures with an atom in the extremity. NSOM provides signal amplification in magnetic measurements through optical plasmon resonance, implementing techniques of higher proximity such as photoluminescence [132, 133].

Nguyen *et al*. [132] evaluated the interaction between the polymers' chains through NSOM and analyzed poly(2-methoxy-5-(2'-ethylhexyloxy)-1,4-phenylene vinylene) films, also known as MEH-PPV. The authors verified a correlation between the nanoscale topography and optical properties of the MEH-PPV films because the results of the NSOM analysis provided information about how the morphology of the polymeric film can improve the optical properties and performance of electroluminescent devices based on this class of materials. Other studies are listed in Table 19.2.

19.4.3.7 Transmission Electronic Microscopy (TEM)

Transmission electronic microscopy (TEM) is another analytical methodology to identify the shape, crystalline structure, lamellar structures, size, and internal morphology of microplastics. This technique allows the analysis of materials with a dimension of 0.1 mm to 0.1 nm, applying ampliations of 100x to 500,000x, which gives TEM a better resolution compared to conventional microscopes. Because the short wavelength is inversely proportional to the incident beam in the sample, this technique uses an electron source and can observe structures with atomic resolution. The main TEM constituents are like the SEM, with a high-voltage electron gun of 10 to 120 kV, a set of electromagnetic lenses, and a signal detector [110, 112]. TEM detects the electrons that go through the plastic particle, providing internal information about the sample. The particle needs a thin thickness to allow the acceleration tension to be high, thus creating an image on the monitor. When the image is transferred to the phosphorescent monitor, the dark areas that diffract electrons are not transferred as well as the other bright areas [110, 112]. The main limitation of this technique is the sample preparation, where the following characteristics must be acquired for better analysis: the surface needs to be polished with mechanical polishing of 0.10 to 0.20 mm and cleaned on both sides with a thickness between 500 and 5,000 Å [27, 112].

Table 19.2 Characterization and identification methods of micro- and nanoplastics using electron and scanning microscopy.

Technique	Origin or source of the sample	With or without pre-treatment	Range	Consideration/results	Reference
SEM	Microplastics in sediments	Density separation with NaCl	500 µm	Identification of fragments of PE, PP, PS, PA, chlorinated polyethylene and chlorosulfonated polyethylene with blue, white and transparent coloring	Fries *et al.* [116]
SEM	Microplastics in sea salt production	Coagulation with aluminum chloride	200 to 1,500 µm	Identification of white PBMA flakes	Lee and Jung [115]
SEM-EDS	Microplastics in river sediments	Extraction and separation by density	2.8 mm to 11 µm	Identification of pellet granules	Blair *et al.* [117]
SEM-EDS	Microplastics in cosmetics	Vacuum filtration and hexane wash	120 µm	Identification of spherical PP particles	Dąbrowska *et al.* [118]
SEM-EDS	Microplastics in soft tissues and in the digestive tract of bivalves and fish	Separation with KOH and incubation at 90 rpm, 60 °C for 24 at 48 h.	<1 mm	Identification of particles in the form of fibers, fragments, granules and films	Ding *et al.* [119]
ESEM	Microplastics in sediments	Without treatment	30 to 500 µm	Identification of PE, PP, and PS with irregular shape and microfibers with regular shape.	Vianello *et al.* [122]
ESEM	Microplastics in fresh water	Coagulation with n-alkyltrichlorosilan and filtration	<1 mm	Identification of PE agglomerates	Herbort *et al.* [121]
ESEM	Micro- and nanoplastics in sea salt	Coagulation	100 nm	Identification of PP clusters	Schuhen *et al.* [123]
FIB	Microplastics	The surface was platinum coated and cleaned	640 nm	Identification of the abrasive surface of microplastics, which can be chemically altered	Bullard *et al.* [126]
FIB	Microplastics in seawater mussel tissues	Digestion	10 to 20 µm	Identification of the irregular surface of polymers	Valsesia *et al.* [127]

Technique	Origin or source of the sample	With or without pre-treatment	Range	Consideration/results	Reference
SPM	Nanoplastics in human skin fibroblasts	Supplementation with bovine fetal serum	100 to 1,000 nm	Identification of PS	Akhatova *et al.* [128]
NSOM	Polymeric film	Molding with chlorobenzene and tetrahydrofuran, and baking the film	50 to 100 nm	Evaluation of the MEH-PPV morphology	Nguyen *et al.* [132]
TEM	Intracellular microplastics in the human placenta	Conservation in agar at 60 °C/48 h	1 and 2 mm	Particles easily mistaken with organelles	Ragusa *et al.* [135]
TEM	Nanoplastics in soil	Preservation with osmium tetroxide and dehydration with acetone	80 to 100 nm	Identification of fragments, fiber, and membrane with smooth surface and irregular shape	Watteau *et al.* [134]
TEM	Micro- and nanoplastics in vegetal tissues	Root, stem and leaf samples between 40 and 100 mm thick	0.2 to 2 µm.	Identification of fluorescence in wheat roots with concentrations of 0.5 to 50 mg/L	Li *et al.* [136]
HRTEM	Nanoparticles in atmospheric environments	Addition of tetrahydrofuran to obtain plastic hydrosol with stirring for 48 h at 100 rpm	0.345 nm	Crystalline plastic particles similar to graphitic fringe	Ganguly and Ariya [139]
HRTEM	PS micro- and nanoplastics	Dilution of PS in deionized water with different concentrations	0.5 to 1 µm	Characterization with voltage of 200 kV in 10 nm scale	Akram *et al.* [142]

Watteau *et al.* [134] evaluated the detection of microplastics in soil altered with municipal solid waste compounds. The pre-treatment of the samples to preserve the organominerals was the addition of osmium tetroxide and dehydration with acetone. Lastly, the samples were cut into sections of 80 to 100 nm with a diamond blade and stained with uranyl acetate and lead citrate. The technique allowed the identification of fragments, fibers, and colored membranes; in addition, it was identified that in the bigger sections, the surfaces were smooth and heterogeneous. Ragusa *et al.* [135] evidenced the presence of microplastics in the intracellular sharing of the human placenta using TEM. From the cut TEM images of placenta samples (1–2 mm, kept in agar at 60 °C for 48 h), the presence of microplastics within human placental villi, within the layer of syncytiotrophoblasts free in the cytoplasm or encapsulated in structures beneath the plasma membrane, easily mistaken for organelles, could be observed. Li *et al.* [136] determined the presence of microplastics in vegetable tissues by TEM. Wheat roots grown in a microsphere suspension of 0.2 µm showed fluorescence dependent on a wide range of concentrations

(0.5–50 mg/L). The presence of aggregates in the xylem and the cell walls of the cortex tissue in the root of wheat indicated that the granules passed through the intercellular space through the transport system, since inside the central cylinder, the particles can move toward the aerial parts of a plant. Other studies are listed in Table 19.2.

19.4.3.8 High Resolution Transmission Electron Microscopy (HRTEM)

High-resolution transmission electron microscopy (HRTEM) allows the characterization at the atomic level of ultra-structural nanoparticles with 1 Å resolution. The formation of the image is a result of the interaction of interference patterns between transmitted and diffracted electron waves in thin samples. In the HRTEM contrast, the interference between the electrons is observed, forming the high-resolution image. However, to form a high-resolution image, objective lenses need a polar piece, generating a magnetic field. When the interaction occurs in different columns, Fraunhofer diffraction occurs [137, 138]. The main advantage of the technique to analyze polymeric particles is the formation of 3D high-resolution images that allow the observation of the structure on a nanometric scale and at an atomic level. The disadvantage of this technique is the sample preparation, which must be carefully cut into thin slices [137].

Ganguly and Ariya [139] analyzed the influence of plastic particles in atmospheric environments and showed that the emissions of micro- and nanoplastics with diameters lower than 100 nm can be inhaled, causing health problems. From the HRTEM images, a prominent network fringe of 0.345 nm plastic particles was observed; thus, the crystalline plastic micro- and nanoparticles synthesized in this study were comparable to graphitic network fringes. Lukina *et al*. [140] investigated the characteristics of a polymeric compound used as an adhesive by HRTEM, among other analyses. The authors observed nanopores in the structure of the superficial layers of the polymeric fiber, as well as packets of basal planes in the fibril composition with a curvature radius of 10 nm or more. Rius-Ayra *et al.* [141] analyzed the removal of microplastics from the combination of electrodeposition and electrophoresis to produce durable superhydrophobic coatings on an aluminum substrate. Field emission scanning electron microscopy (SEM-FEG) and high-resolution transmission electron microscopy (HRTEM) were used to characterize the samples. The images show the presence of a structure with thin layers (70 nm-thick nanolayers) superimposed on each other, corresponding to morphological units in the shape of petals, randomly arranged. Other studies are listed in Table 19.2.

19.4.4 Identification and Quantification Methods by Spectroscopy

When the micro- and nanoplastic particles are prepared, separated, and purified, they should be identified and quantified, which can be done using different techniques. Among them are the spectroscopy techniques, which are based on the emission of electromagnetic radiation under a sample and the evaluation of the interaction between the light and the constituents of this sample based on the quantification of the radiation absorbed or emitted by the elements [28, 143]. The spectroscopic techniques are considered qualitative, semi-quantitative, or quantitative methods, depending on the kind of technique. The qualitative and semi-quantitative methods are called this way because they do not cover all the nanometric range due to the detection limit of the particle size and usually promote quantification without confirmation of the exact chemical composition of all kinds of compounds or polymers [28]. A way to overcome this difficulty is to couple the spectroscopic techniques with microscopic ones, allowing the simultaneous acquisition of optical images and chemical compositions [143].

Among the spectroscopic techniques, Fourier transform infrared spectroscopy (FTIR) and Raman spectroscopy are the most commonly used ones to identify microplastics; however, imaging spectroscopy and X-ray photoelectron spectroscopy (XPS) can also be applied for this purpose.

19.4.4.1 Fourier Transform Infrared Spectroscopy (FTIR)

Fourier transform infrared spectroscopy (FTIR) is a technique used to obtain an infrared spectrum (IR) of absorption or emission that is characteristic of each sample [28] because when the IR radiation is focused on the sample, part of this radiation is absorbed, and the transmitted part is registered. This technique evaluates the molecular vibrations from the interaction between the IR radiation and the sample, resulting in characteristic infrared spectra, which are digital impressions of the sample and can be compared to the spectra from reference samples [30, 143–145]. During the analysis, simultaneous high-resolution spectral data are collected over a wide spectral range, all of which can be collected at the same time, and then these data are processed by Fourier transform [143], allowing a semi-quantitative determination of the compounds in the samples [62]. Thus, the FTIR allows the identification of the organic compounds, polymers, and functional groups in the sample through their molecular vibrations [30].

The FTIR allows a valid and precise analysis of bigger particles that are directly manipulated with clamps (>500 µm), or even smaller particles using filters. As the analysis of the whole filter is time- and work-consuming, usually a subgroup of the filter area is examined, which provides information about the filter employed [146]. Thus, FTIR has been employed to identify microplastics in facial exfoliants, plastic teabag leachate, and biodegradable plastic leachate [28, 30]. The lower microplastic particles spectroscopically identified in seawater and freshwater ecosystems were about 10 µm [143], because the resolution of IR equipment is limited by diffraction, covering a wavelength range of 1.7 µm to 13 µm (wavenumber of 4,000 cm^{-1} to 500 cm^{-1}) [147]. Thus, according to Ivleva *et al.* [143] and Zarfi *et al.* [148], particles smaller than 10 and 20 µm, respectively, would be difficult to analyze with this technique; however, according to Hernandez *et al.* [62] and Hanif *et al.* [6], this technique can also be employed to evaluate thin nanoplastic films and quantify particles with polymer identification. Thus, FTIR determinations occurred for both micro- and nanoplastics.

The FTIR has three different operating modes: the transmission mode using the potassium bromate (KBr) tablet and focal plane array (FPA), and the reflection mode (KBr and FPA), and the attenuated total reflectance mode (ATR). The FTIR using potassium bromate tablets (KBr-FTIR) allows the qualitative determination of micrometric samples and the mapping of particles with small areas, around 20 × 20 µm, through molecular vibrations from the acquisition and processing of the data obtained in the spectrum analysis [149]. This analysis presents the following advantages: it is a non-destructive method [94, 149], easy to operate [6], and provides reliable results [29, 149]. But the KBr-FTIR also presents limitations concerning the high cost of the equipment [6, 149], restriction to moisture because the presence of water in the samples causes reading interferences in the nanoplastics [62], measurement errors and the presence of noises, or even IR band overlap. In addition, the samples need to have a very low granulometry (carefully grounded) and must be homogeneously dispersed in the KBr tablet [149]. Lu *et al.* [150] studied the presence of PS microspheres in aquatic environments using KBr-FTIR. The authors observed H−OH vibrations of the molecules of water absorbed in 3,430 cm^{-1}, C−H aliphatic deformation in 2,930 cm^{-1}, C=C stirring from the aromatic ring in 1,600 cm^{-1}, and carbon peaks in the range 3,030, 755, and 700 cm^{-1}, corresponding to aromatic carbons from the PS polymer. The study pointed out that the magnesium ions action (Mg^{2+}) aided the aggregation and stabilization of the PS microsphere, helping its monitoring in aquatic samples.

Attenuated total reflectance spectroscopy (ATR) is a sampling technique used together with FTIR, constituting the ATR-FTIR mode [28, 143]. During the equipment operation in the transmission mode, the light goes through the sample, being posteriorly collected; thus, the sample surface is irradiated like an evanescent wave, allowing its identification

[144]. The morphology of the microplastic particle influences the transmitted signal; thus, it can cause disturbances in light dispersion, causing reflection errors. For that, special filters are used, which are composed of materials that allow the free passage of light [122]. ATR-FTIR is efficient in identifying microplastic particles present in different matrices. To prepare the samples, the microplastic particles should be bigger than 500 µm, because then they are adequate to be collected with tweezers and put in an ATR to develop the technique. For smaller particles, the combination of FTIR and an IR microscope (µ-FTIR or micro-FTIR) is used [144]. This combination will be later discussed.

The advantages of ATR-FTIR are [6, 29, 94, 144]: high reliability of the results, quick and precise identification, non-destructive technique, reliable monitoring, and applicability to plastics analysis. However, its limitation is the high instrumentation cost; very thick particles are only detected when compared to colored or dark particles; depending on the morphology, they should be evaluated by other techniques, such as µ-ATR-FTIR [94, 122, 143]. In addition, contact with inorganic rigid particles may result in damage to the instrumentation [151], and depending on the filter employed, the analysis time is enlarged, becoming too time-consuming [151].

Käpper *et al.* [152] evaluated the presence and characterized microplastics in water samples by ATR-FTIR. The authors reported that the technique is reliable despite interferences in the identification of colored particles. For particles >500 µm, it was verified that vibrations in the range of 1,601 and 1,492 cm^{-1} that correspond to the carbonic chains with aromatic rings; typical bands from PE at 1,061, 1,127, 1,297, 1,448, 2,845, and 2,886 cm^{-1}, and intense bands around 3,200 cm^{-1} (stretching vibration of the group OH-) and under 1,000 cm^{-1} they are attributed to the presence of inorganic oxides. For filtered and purified samples <400 µm, elongation vibrations of the $CH/CH_2/CH_3$ groups were observed in the spectral range 2,980–2,780 cm^{-1}, CH_2 bending vibrations in 1,480–1,400 cm^{-1}, C=O elongation vibration in 1,760–1,660 cm^{-1} and in 1,800–1,740 cm^{-1}, and CF_2 stretching vibration at 1,174–1,087 cm^{-1}. The presence of the polymers PE and PP (2,980–2,780 cm^{-1}), PVC (1,480–1,400 cm^{-1}), PES and PS (1,760–1,670 cm^{-1}), polycarbonate, and polytetrafluoroethylene (1,174–1,087 cm^{-1}) was verified in the samples. Melo-Augustín *et al.* [30] evaluated the composition of microplastics present in surface water from the Manzanillo and Santiago Bays by ATR-FTIR. The authors identified the plastics PE (48%) and PP (52%), the presence of elemental compounds (arsenic, cadmium, and aluminum), and organic compounds on the surface of the microplastics. The spectra presented bands corresponding to the bounding C−O (ketones, carboxyls, aldehydes, and olefins) in the range 1,600–1,750 cm^{-1}, stretching bands (O−H) in 3,000–3,800 cm^{-1}, the alcohol group between 1,520 and 1,750 cm^{-1}, and the benzene group (aromatic) at 2,400 cm^{-1}. These results evidenced the efficiency of the technique employed to identify microplastics in the marine environment.

The FTIR using the focal plane array (FPA) detector constitutes the FPA-FTIR mode, which allows the simultaneous acquisition of several spectra in minutes because each pixel provides an independent IR spectrum, providing information about the identification and concentration of materials such as microplastics. This analysis can be performed on samples that did or did not go through a pre-treatment; however, the sample pre-treatment with hydrogen peroxide (H_2O_2) is common [144, 145, 153]. FPA-FTIR presents several advantages, such as the rapid acquisition of several spectra inside an area at a unique measurement and individually, allowing the analysis of whole filters, spectra with high resolution, ease of operation (simple), non-destructive technique, and reliable monitoring [6, 29, 62, 94, 143, 145, 153]. However, FPA-FTIR has disadvantages such as the high operating cost, time-consuming analysis, wavelength, and radiation, which can

be limiting factors, moisture sensibility because the presence of water causes interferences, and the need for a competent person to operate it [6, 29, 62].

Käpper *et al.* [152] aimed to identify microplastics in environmental samples with particles of ~100 µm, employing FPA-FTIR in the range 400–4,000 cm^{-1}. The authors concluded that the technique was adequate to identify polymers, and the samples presented higher quantities of PE and PP. Although the FPA detector has limitations, such as failure detection in samples that present a high amount of microplastics, it was possible to observe the vibration band of the C−Cl stretch for the PVC in 690 cm^{-1} and the aromatic rings present for the PET in the range of 1,580–1,640 cm^{-1}. Cincinelli *et al.* [153] evaluated microplastics in sediments from the Black Sea collected at different depths in the range of 22–2,131 m by the FPA-FTIR technique. The authors verified the spectral bands that are characteristic of PE, being CH_2 and CH_3, approximately 1,470 and 1,370 cm^{-1}, respectively. The PP samples were identified in distinct bands: in 2,972 (δas CH_3), 1,469 (δas CH_2), and 1,386 cm^{-1} (δs CH_3). An increase in the PP particle absorptions was noted in the range of 3,550–3,100 cm^{-1} (OH stretching region) and 3,000–2,800 cm^{-1} (CH stretching region). The identification of band characteristics of esters was not possible. However, the polymers PE and PP were the most abundant, at around 44.5%.

When particles smaller than the limit range for FTIR are desired to be analyzed or it is desired to improve the identification, distribution, and abundance knowledge of plastic particles, a combination of techniques should be used. One possibility is to couple the spectrophotometer FTIR to the IR microscope, constituting the Fourier-Transform infrared micro-spectroscopy (µ-FTIR or micro-FTIR), which provides the molecular chemical mapping of the microplastics through IR spectra and images, in addition to comparison with the data bank to identify the sample [122]. For Harrison *et al.* [151], the µ-FTIR represents an ideal method to detect microplastics in several sediments. This technique is considered a green tool because it analyzes polymers using few reagents and solvents and requires little energy expenditure [27]. The µ-FTIR method presents sensibility and spatial resolution that can be improved to visualize all the filters with high-yield detection techniques, for example, obtaining images based on FPA [151], in addition to the simplicity of the sampling mapping and the lower detection limit compared to other techniques [27]. However, µ-FTIR has the disadvantages of a long execution time [154], a reflexive process that can contain errors, and a high particle thickness demand. This analysis can be performed in the transmission mode with too-thin particles (<15 µm) or ATR-m-FTIR or using a KBr tablet [27].

Harrison *et al.* [151] evaluated the presence of microplastics in retentates of vacuum-filtered sediments by µ-FTIR and compared it with ATR-FTIR. Concerning the spectral bands, regions of C−H (δC−H) elongation were observed at 1,500–1,450 cm^{-1}, C−H bending (δC−H) at 1,500–1,450 cm^{-1}, C−H bond bending at 1,500–1,450 cm^{-1}, and scissoring bands of C−H bonds in the region of 750–700 cm^{-1} for PE. PE was the polymer with the highest reflectance, with no PE fragments larger than 150 µm. The authors emphasized the need to improve the techniques to make them more efficient and reproductive in separating microplastics from sediments. Löder *et al.* [144] showed that plastic particles of up to 20 µm, depending on the kind of plastic, can be analyzed by FPA-µ-FTIR, identifying the C−H stretch in the region of 2,980–2,780 cm^{-1} and bend regions in 1,480–1,440 cm^{-1}, indicating PE, while the absorption in 2,980–2,780 cm^{-1} was attributed to the presence of polycarbonates. The authors informed us that not enough absorbance was produced to interpret the spectra for samples with thicknesses lower than 5 µm, for example, fibers, possibly due to the high content of non-plastics in the sample. Vianello *et al.* [122] evaluated, in the Venice lagoon, the abundance of

microplastic particles with a size ≤1 mm by the μ-FTIR technique. The authors detected microplastics in all samples evaluated, with higher concentrations in locals near the continent and the most abundant polymers being PE and PP. They verified by scanning electron microscopy (SEM) that in 93% of the samples analyzed, the particle sizes were in the range of 30 to 500 μm, and using μ-FTIR, the fiber absorption of PP was in the range of 1,000–800 cm^{-1} and the aromatic CH stretching and aliphatic of $CH/CH_2/CH_3$ was in the range of 3,200–2,700 cm^{-1}. Those studies are better explained in Table 19.3, which presents studies aiming for comparison among the methods, and indeed, it can be noted from the table the lack of standardization between the conditions used by the authors to evaluate plastic materials by the same technique.

Table 19.3 Identification and quantification methods of micro- and nanoplastics by spectroscopy.

Technique	Range	Origin or source of samples	With or without pre-treatment	Description or consideration about the technique and/or the research	Reference
KBr-FTIR	Microplastics	Aquatic environments	Synthesis by polymerization with surfactant-free emulsion	PS identification	Lu *et al.* [150]
FPA-FTIR	Microplastics	Black Sea Sediments	Extraction in saturated NaCl solution	Identification of PE, PP, acrylonitrile, PAs, and cellulose-based fibers	Cincinelli *et al.* [153]
ATR-FTIR	Microplastics	Surface waters of Manzanillo and Santiago bays	Extraction, separation and classification according to particle size	Identification of PE and PP, associated with contaminants	Melo-Augustín *et al.* [30]
ATR-FTIR	Microplastics	Fresh water from lakes and rivers	Without treatment	Evaluation of the composition of plastic matrices	Dris *et al.* [155]
ATR-μFTIR	Microplastics	Environmental samples	Extraction	Identification of particles from 5 to 20 μm	Löder *et al.* [144]
ATR-FTIR	Microplastics	Waters from aquatic ecosystems	Samples >500 μm were extracted. Samples <400 μm were purified and filtered	Particles of 50–500 μm, identification of PE, PP, PVC, polycarbonate, PS, PET, and PES	Käpper *et al.* [152]
μFTIR	Microplastics	Sediments	Vacuum filtration	Absence of PE in fragments of 150 μm	Harrison *et al.* [151]
μFTIR	Microplastics	Venice lagoon water	Sedimentation	Detection of PE and PP in the range of 30 to 500 μm	Vianello *et al.* [122]
RAMAN	Nanoplastics	Distilled water, sea salt and amniotic fluid	Distillation	Suggestion of standardization of particles of 200 nm	Schmidt *et al.* [157]
RAMAN	Microplastic	Waters from aquatic ecosystems	Samples >500 μm were extracted;	1–50 μm particles, identification of	Käpper *et al.* [152]

Technique	Range	Origin or source of samples	With or without pre-treatment	Description or consideration about the technique and/or the research	Reference
			Samples <400 µm were purified and filtered	PE and PP in greater abundance	
SERS	Microplastic	Mineral water	CuO nanorod usage	Detection of PE particles	Lin *et al.* [160]
SERS	Plastic	Food packages	Photo initiators usage	Detection of PE and PET	Peng *et al.* [161]
RAMAN TWEEZER	Micro- and nanoplastics	Seawater	Production of thermoplastics enriched with PA-6 particles	Presence of PE and PP in a concentration ~10^4–10^5 particles/mL	Gillibert *et al.* [78]
RAMAN TWEEZER	Micro- and nanoplastics	Seawater	Without treatment	Identification of polystyrene particles from 50 nm	Bernatova *et al.* [159]
XPS	Nanoplastics	Water packages	Fractioning and filtering	Presence of PE from personal hygiene products in the water	Hernandez *et al.* [62]
XPS	Microplastics	Surface waters of Manzanillo and Santiago bays	Extraction, separation, and classification	Identification of metals (arsenic, cadmium and aluminum) present on the surface of microplastics	Melo-Augustín *et al.* [30]

19.4.4.2 Raman Spectroscopy (RS or RAMAN)

Raman spectroscopy (RS or RAMAN) is another adequate method to characterize polymers [145, 155, 156]. It is a photonics technique in which a monochromatic light source, usually a laser, is irradiated under molecules and atoms from the sample, providing information about the vibrational and rotational mode and other interactions of low frequency that result in what is called Raman displacement [143, 144]. Thus, the molecular vibrations observed in the Raman technique provided the identification of the chemical structure and the functional groups of the polymers evaluated [27, 30]. This technique can be employed to analyze microplastics from different kinds of matrices, such as seawater, fresh water [143], distilled water, and marine salt [157]. The polymer is identified by the comparison of the highly specific digital impression spectra with references or a commercially available database [143]. This technique, as well as FTIR, is among the most employed methods to identify polymers [156]. However, the Raman image presents better digital visualization and higher spectral coverage, which is highlighted compared to FTIR [145]. When Raman is coupled to the MALS, it detects polymers with higher reliability [48].

The Raman technique is reliable and successful in identifying microplastics [144] because it obtains their spectra with spatial resolutions 1 to 20 µm [143]. In addition to these advantages, this technique presents measurements with a high spatial resolution (up to 1 µm), insensitivity to water [154], high sensitivity to non-polar functional groups, being suitable for analysis of non-transparent or dark microplastics [6], and the possibility to detect nanoplastics (<1 µm) in a detection limit of 2.10–3 µg/L to 200 µg/L [157]. Still, it is a non-destructive technique, and it does not give false positive or false negative results [94]. The disadvantages of the Raman technique are strong interference of fluorescence in environmental samples; partly automated identification; the performance is influenced by measurement parameters (for example, number of waves); easy interference of pigments and contaminants; time-consuming procedure; hard to monitor; high operating cost [6, 29, 94, 143, 157]. The Raman technique cannot cover all the range of nanoplastics lower than the spatial resolution of the spectroscope [158]. Therefore, when integrated into SEM, the Raman technique can be used to efficiently identify nanoplastics [28, 143].

Schmidt *et al.* [157] evaluated the presence of nanoplastics in distilled water, marine salt, and amniotic liquid by Raman. In this study, the particle sizes were standardized (Ø 200 nm). The tests did not present limitations regarding the identification reliability of the nanoplastics, as strong background signals from the environment were identified. In the marine salt samples, C−H bonds were identified, attributed to the presence of nanoplastics. Käpper *et al.* [152] identified microplastic particles in water samples by m-Raman in a clear and reliable form, indicating the chemical composition of some polymeric plastics, such as PE and PP, among others. For samples >500 µm, chain vibrations of aromatic rings were observed at 1,000 and 1,600 cm^{-1}, and strong and wide bands in the region of 3,400–3,000 cm^{-1} and 1,100–1,000 cm^{-1}, due to the presence of adsorbed water or inorganic contaminants in the samples that contained PP fibers. The presence of inorganic compounds was confirmed by the combination of the ATR-FTIR and m-Raman techniques. For purified and filtered samples <400 µm, vibrations of the $CH/CH_2/CH_3$ groups in 2,780–2,980 cm^{-1}, aromatic bending vibration in 1,580–1,640 cm^{-1}, and CF_2 stretching vibration in 709–759 cm^{-1} were observed. The PE and PVC were verified at 2,780–2,980 cm^{-1}, while the polycarbonate, PS, and PET belonged to the spectral range of 709–759 cm^{-1}.

Other variations of the Raman spectroscopy used to identify micro- and nanoplastics are surface-enhanced Raman spectroscopy (SERS) and Raman tweezer (RT). The SERS is an approach to increasing the Raman response intensity of a sample when interacting with the surface plasmon of metals such as Au, Ag, or Cu [143], which can be used to detect and analyze plastic particles in the range of micro- and nanometers with size ≥50 nm, in a highly sensible and quantitative manner [159, 160]. This technique acts through molecular vibrations, intending to also promote the identification of functional groups of polymers in a non-destructive form [30, 160]. The advantages of SERS are the high sensibility, the quantitative detection possibility, the fast monitoring, the unique molecular specificity, and the fact that it is not affected by complex compounds in solution [160, 161]. However, SERS has the disadvantage of the possibility of presenting overlap spectra in the case of multicomponent nanoplastics due to low lateral resolution and low reproducibility [6].

Lin *et al.* [160] employed SERS with copper and silver oxide nanoparticles to monitor the release of plastic particles in mineral water samples stored under irradiation according to the exposition time. The authors verified that in both the cups and the bottles, quick degradation occurred, changing the particle concentrations from 5.37 ± 0.11 ng/mL to 3,751 ± 0.19 ng/mL after 240 min. The polymer with the highest detection limit was PE, observing the stretching of C−C in 1,064–1,132 cm^{-1}, CH_2 twisting in 1,294 cm^{-1}, bending in 1,450 cm^{-1}, C−H stretching in 2,725 cm^{-1} and CH_2 stretching in the spectral

range of 2,855–2,895 cm^{-1}. Peng *et al.* [161] studied plastic samples of food packages concerning the photoinitiators, Irgacure 907, employing SERS and diverse solvents (acetone, acetonitrile, methanol, xylene, and vinyl acetate). To evaluate the penetration detection of Irgacure 907 in plastics, samples corresponding to the blank were prepared with the photoinitiators and plastics, and samples containing the plastics were prepared with solvents. These samples were deposited on blades and dried at room temperature. The results indicated that the use of photoinitiators combined with the tested solvents acted safely, obtaining a higher penetration capacity of Irgacure 907 for acetone, followed by acetonitrile, methanol, ethyl acetate, and xylene. In addition, the polymers PET and PP were more efficient concerning the penetration of photoinitiators compared to PE. Part of the tests were performed with a gold blade to aid in the detection of the SERS peak, identifying that at 1,083 cm^{-1}, there is the characteristic peak of food packages. The authors chose to use this technique to obtain real-time results and because of the quick monitoring.

The Raman tweezer (RT) is another Raman technique variation that combines optical tweezers with RS and operates in the detection range of micro- and nanoplastics with size ≥50 nm in water from different sources [6, 78, 159]. This technique presents the advantages of being able to differentiate plastic from other components and determining the size and shape of the particles [6], in addition to being a fast analysis [78]. The disadvantages of the Raman tweezer are that the counting and determination of the particle can be affected by diffraction, requiring high concentrations of micro- and nanoplastics [6]. Gillibert *et al.* [78] identified submicro- and nanoplastics in seawater by the RT technique. The authors reported that the analysis allowed them to overcome the limitation associated with the counting and identification of particles with approximately 500 nm, when they occur in an associated form, highlighting the presence of PE and PP at a concentration of ~10^4–10^5 particles per mL, in the spectral range of 300–1,800 cm^{-1}. Bernatova *et al.* [159] also evaluated micro- and nanoplastics in seawater using RT and identified 50 nm in 1,001 cm^{-1}, being that the analysis occurred in the variable absorption range of 500 to 6,000 cm^{-1}. The author mentioned that it was possible to detect nanoplastics in liquids with high sensitivity and indicated that this technique has the potential to evaluate polymers and aid in the monitoring of environmental pollution.

Both FTIR and Raman techniques and their variations can be coupled with other techniques as alternatives to better identify polymeric particles, such as micro- and nanoplastics [6]. Concerning the Raman and FTIR techniques, the combination of these technologies is fundamental to promote a better comprehension of the spectra and, consequently, of the samples' characteristics [147], because they are complementary techniques that act from vibrations, being that the inactive Raman vibrations are considered active IR and vice versa. Plastic polymers have highly specific IR and Raman spectra with distinct band patterns, which allow both techniques to be efficient in identifying microplastics [144]. As the light sources used are normally in the visible range, another possibility is to couple the Raman technique to a standard optical microscope, using m-RAMAN with spatial resolutions of 1 µm to 20 µm [143]. These studies are presented in Table 19.3.

19.4.4.3 Imaging Spectroscopy—Hyperspectral Images (HSI)

Imaging spectroscopy, or hyperspectral images (HSI), consists of the signature capture, called digital impressions, and processing of a 3D image formed with a great number of wavelengths of the electromagnetic spectrum (continuous spectral bands), involving tens or hundreds of colors to identify and detect characteristics of materials at the macro-, micro-, and nanoscales. The hyperspectral image provides information about the chemical constituents of the materials and is useful for the triage of micro- and nanoplastics, distinguishing several plastic materials, such as PET and PP [162, 163].

The advantages of the HSI technique are the ease, quickness, and flexibility to change the wavelength; higher objectivity for being a more automated technique; non-destructive; and the possibility to work at a camera resolution. In addition, the HSI provides high-resolution spectra, allowing better identification of the surface of the material. The disadvantages are the complexity, the cost, and the instrumentation (sensibility of the detector, data storage capacity, and fast computers). Studies point out that this technique identifies microplastics of different colors, kinds, and sizes; however, the decrease in the pixel size in the image and the narrowing of the field of view with proper lens selection would allow the characterization of microplastics smaller than 100 mm using HSI, which would enable additional and valuable data about the environmental effects of the microplastics [162, 163]. Zhang *et al.* [164] evaluated the detection of microplastics in the intestinal tract of fish using HSI and identified five kinds of microplastics with particles bigger than 0.2 mm. The technique was promising with a precision of 96.22% and was more efficient at identifying microplastics in marine fish. In addition, the technique does not need a digestion pre-treatment, thus saving reagents. Zhu *et al.* [163] characterized microplastics in filtering substrates using HSI-NIR with wavelengths of 900 to 1,700 nm. With this system, three types of polymers were identified: PE, PP, and PS, with a size of up to 100 µm.

19.4.4.4 X-Ray Photoelectron Spectroscopy (XPS)

X-ray photoelectron spectroscopy (XPS) corresponds to a spectroscopic technique obtained by a photoelectron spectrophotometer with a monochromatic source of X-rays. Using this technique, qualitative and quantitative compositions can be obtained from the percentage and atomic weight of the elements in a sample [30].

Hernandez *et al.* [62], after characterizing nanomaterials by FTIR, employed the XPS technique to evaluate the sensibility of polymer surfaces in samples from a water package. The authors identified typical peaks in the range 280–300 eV, indicating that polyolefins are present, and verified PE with repetition of its -CH_2 units, with intense bands in bounding energy of ~13 eV (molecular orbital antibonding) and ~19 eV (analogous bounding), confirming the presence of PE. Melo-Agustín *et al.* [30] used XPS to evaluate the qualitative and quantitative composition of surface waters from the Manzanillo and Santiago Bays concerning the presence of microplastics and metallic contaminants. In the Santiago samples, arsenic, chromium, and zinc were detected, while mercury was not detected in any of the bays. The authors concluded that at both investigation sites, metal contamination was detected. In addition, PE was the microplastic that presented the highest accumulation of these elemental compounds on its surface, proving the role of microplastics as a holder for heavy metals. These studies are better explained in Table 19.3.

19.4.5 Identification and Quantification Methods by Thermal Analysis, Gas Chromatography, and Mass Spectrometry

The methods that involve thermal analysis, gas chromatography, and mass spectrometry are called thermo-analytical methods. These methods have also been used to identify and quantify micro- and nanoplastics in different matrices based on the evaluation of their physical-chemical properties. Among these methods, those can be highlighted [30, 94, 97, 143, 156, 165–174]: thermogravimetric analysis (TGA), differential scanning calorimetry (DSC), automated thermal extraction–desorption gas chromatography mass spectrometry (TED–GC–MS), pyrolysis–gas chromatography–mass spectrometry (Py–GC–MS), secondary ion mass spectrometry (SIMS), time-of-flight secondary ion mass spectrometry (ToF-SIMS), matrix assisted laser desorption ionization time of flight mass spectrometry (MALDI-TOF-MS), inductively coupled plasma mass spectrometry (ICP-MS), and single particle inductively coupled plasma mass spectrometry (SP-ICP-MS).

The thermal analyses are employed to evaluate polymeric samples through TGA coupled or not to DSC, aiming to consider the thermal degradation, also known as thermal stability, phase transitions, and/or the materials' purity [156]. This thermal degradation of the polymers is caused by the change in the polymers' properties during processing at high temperatures. Samples of micro- and nanoplastics are completely degraded because the chemical bonds rupture, resulting in a volatilization process [6]. In general, thermal analyses present as their advantages the characterization of microplastics and low-solubility additives that cannot be easily dissolved, extracted, or hydrolyzed, but they are destructive techniques and require complex data acquisition [94].

Thermogravimetric analysis (TGA) is a quantitative thermal analysis technique [94] in which the sample mass variation is evaluated because of transformations that occur according to temperature or time in a precisely controlled atmosphere [156]. It is a robust method to evaluate samples that release a high quantity of compounds with a high molecular weight [165]. The polymer samples reach complete pyrolysis at around 600 °C [166]. However, TGA equipment can operate at controlled temperatures up to 1,000 °C [165]. Thus, the method evaluates all the degradation mechanisms by mass loss, and it is highlighted that polymeric material degradation is detected by enthalpy changes [94]. TGA presents the following advantages: the possibility to analyze a small quantity of sample (1–20 mg), and, depending on the atmosphere employed, up to 100 mg can be studied; quick measurements [29, 156, 165]; effective decomposition of the samples; and the regeneration possibility of the filtering materials when used [6]. Among the TGA disadvantages, it is highlighted that particles out of the range of 200 to 500 µm can present overlayed peaks due to lower reactivity during the reading [156]. Depending on the temperature employed, part of the matrix under evaluation gets stuck in the solid phase adsorbents, and in this case, an association of TGA to gas chromatography coupled to mass spectrometry is necessary to solve this problem [166], high energy requirement, and high cost [6].

Dümichen *et al.* [167] used TGA to identify PE in environmental samples. The operation conditions were 4 mg of samples in 150 µL aluminum oxide crucibles in a decomposition range of 25 to 600 °C with a heating rate of 10 °C/min and nitrogen flow of 90 mL/min. The authors observed that the PE decomposition in the evaluated matrices was in the range of 400 to 550 °C due to the split of the main carbonic chain split, releasing smaller aliphatic chains without producing other residues. Still, an overlay of PE with the environmental samples was verified during the thermal decomposition process. Dümichen *et al.* [29] employed TGA in a decomposition range of 375 and 500 °C in environmental samples, which were initially subjected to complete thermal decomposition. From the TGA results, the polymers PET, PA, PP, and PE initiated their degradation right after the beginning of the test; however, the highest mass loss occurred around 404, 394, 410, and 446 °C, respectively.

TGA can be used individually or coupled to Differential Scanning Calorimetry (DSC), composing the TGA-DSC. This combination aims to evaluate the thermal events that do not cause modifications in the mass of the samples but in the energy flow [6]. DSC is another thermal analysis technique that involves gradually heating the samples and evaluating the heat flow according to temperature or time compared to a reference sample [156]. DSC considers temperature and heat flow in an associated way, aiming to see the transitions that occur in a sample, such as fusion enthalpies (change from solid to liquid), glass transitions, and the kinetics of crystallization of materials, in this case, polymers. The main information about the samples is given by the heating and/or cooling of a reference sample [94]. DSC has the advantages of easy equipment operation, uses a low sample quantity, ranging from 1 to 20 mg, is an economic analysis, and can also identify coextracted impurities (non-plastic material) [156], as well as promote the total decomposition of the evaluated microplastics, in addition to the regeneration of

adsorbents and filtering materials (if used) [6]. The disadvantages of this technique are that the samples that are out of the compatibility range (200 to 500 µm) are not reactive to the technique and thus can exhibit overlayed peaks [156], its expensive equipment, and it requires too much energy [6].

Majewsky *et al.* [156] qualitatively and quantitatively determined mass concentrations (weight per volume) of microplastics in environmental samples of freshwater by the TGA-DSC method. The tests indicated significant peaks in temperatures lower than 450 °C, specifically 101, 164, and 216 °C for the polymers PE, PP, and PA, respectively. The samples that contained PVC and polyurethane presented large peaks in the temperature range of 250 to 291 °C, while the samples with PA filtered through a PES filtration membrane suffered decomposition at 291 °C. PP and PVC waste were noted at high temperatures (800 °C). Dumichen *et al.* [166] studied environmental samples, aiming to identify microplastics by TGA-DSC. The authors observed overlaying of the decomposition ranges of the different polymers, making it impossible to uniquely identify the polymers in the sample by TGA decomposition. However, they verified that the decomposition of the polymers happened in the range of 350 to 450 °C. Dumichen *et al.* [166] employed TGA as a solid-phase adsorbent for thermal extraction to evaluate PA and PA doped with 2% in weight of metal oxide particles. The PA samples were decomposed at 400 °C with maximum decomposition at 453 °C under nitrogen gas, but when subjected to pure air, the degradation process was even faster, beginning at 20 °C with intensified deterioration at 470 °C and with maximum decomposition between 480 and 650 °C, reaching 12% of the initial mass. The authors employed the main decomposition products, CO_2, CO, NH_3, and H_2O, and hydrocarbon fractions.

The combination of chromatography and mass spectrometry is another way to identify and quantify materials. Gas chromatography coupled to mass spectrometry (GC–MS) is a method that includes the capacity for the separation of elements from gas chromatography with the ability to identify elements from mass spectrometry, with the advantage of improving the dynamic range and analysis resolution [104]. This is a destructive method with the potential to evaluate monomeric particles, but it needs a specific pre-treatment for each kind of sample, which can be time-consuming. Plastic samples primarily need to vaporize so they can be analyzed by GC–MS [97]. Thus, to identify and quantify micro- and nanoplastics from different sources, the main techniques involving chromatography and mass spectrometry are automated thermal extraction-desorption–gas chromatography-mass spectrometry (TED–GC–MS) and pyrolysis-gas chromatography–mass spectrometry (Py–GC–MS) [152].

Automated Thermal Extraction–Desorption–Gas Chromatography Mass Spectrometry (TED–GC–MS) is a two-stage analytical method, in which, firstly, the sample is decomposed by TGA, and then the gaseous products from pyrolysis are adsorbed in a solid adsorbent, which is then analyzed by GC–MS [97]. This technique presents the advantages of quickness of analysis, economy, and robustness because it shows the impurities in the samples investigated [97, 160]. However, it is a destructive method by the method principle itself [152]. In this technique, the polymeric samples can be quantitatively and qualitatively studied after the TGA analysis [97].

Käpper *et al.* [152] analyzed environmental samples enriched with PE employing TED–GC–MS and thermogravimetric analysis coupled to solid phase extraction (TGA-SPE). The study identified fragility concerning the representation of the results in weight percentage due to the microplastic particle sizes tested (~100 µm), which did not enable the comparison with other scientific studies. Dümichen *et al.* [167] also used TED–GC–MS to quantify the polymer PE in environmental samples after TGA. They verified the composition of this material by the random split process of the macromolecule on the twister surface, with ions overlapping in the chromatogram at (m/z = 55) as detected for

aliphatic polymers. The most characteristic peaks of this plastic sample correspond to the double bonds in the extremity of dialkene compounds, followed by a double bond at one end identifying it as a monoalkene, and there is also the presence of a saturated alkane. The authors also mentioned that TED–GC–MS can be the first method to be able to evaluate PE in a single analysis; however, studies must be developed with other polymeric matrices because the potential of the method has not been fully explored.

Pyrolysis–gas chromatography–mass spectrometry (Py–GC–MS) is a standard thermoanalytical technique for polymeric samples, where the polymers are introduced in an oven and thermally degraded in a controlled manner (pyrolysis) into fragments or stable compounds, which are then separated and detected by GC–MS [29, 162]. Thus, Py–GC–MS is widely used to identify microplastics [48, 104, 168]. This thermochemical method allows the identification of the kind of polymer [94, 97, 144] from the comparison among the pyrograms of the combustion products and the reference samples of known polymers [144]. However, the characterization of the sample quantity is not possible because they are pyrolyzed [168], but information about the kinds of polymers and their degradation capacity when pyrolyzed can be obtained [162]. Py–GC–MS has the advantage of having the analytical capacity to identify plastics economically and rapidly [160] and operating sensitively [162]. But this technique has disadvantages, such as the limited sample quantity (>>0.5 mg), which is not adequate for environmental samples of microplastics because, to ensure microplastic homogeneity, at least 20 mg of sample is necessary [29], and still, it is a destructive method [152, 162, 168].

Fries *et al.* [116] evaluated samples from marine environments concerning the contamination of plastic materials by Py–GC–MS to know the polymer and verify the microplastic content. In this evaluation, the authors promoted the pre-treatment of the sample by heating it at 40 °C for 60 min, aiming to remove the contaminants from the sample. Then, the sample was desorbed at 350 °C for 10 min at a rate of 10 °C/min, and, to recover the compounds, the sample was then cooled through a cold injection system at 50 °C. Lastly, the sample was pyrolyzed for 60 s at 700 °C, and the compounds were analyzed. According to the authors, the temperature of 350 °C is equivalent to the sample data transfer line, and the polymers identified were PE, PP, PS, PA, chlorinated polyethylene, and chlorosulfonated polyethylene. Bouzid *et al.* [169] evaluated microplastics in environmental sediment samples by Py–GC–MS and identified the presence of PE, PP, PS, PET, and PVC. The authors mentioned that even after the purification, a residual mineral fraction remained in one sediment sample, including the matrix effects in the analysis results.

Other methods that can be employed in the identification and quantification of micro- and nanoplastics are surface identification methods. Those methods are employed to evaluate the sample surface with high spatial resolution [145], among them secondary-ion mass spectrometry (SIMS) and time-of-flight secondary ion mass spectrometry (ToF-SIMS), matrix-assisted laser desorption ionization time of flight mass spectrometry (MALDI-TOF-MS), inductively coupled plasma mass spectrometry (ICP-MS), and single particle inductively coupled plasma mass spectrometry (SP-ICP-MS).

Secondary-ion mass spectrometry (SIMS) and time-of-flight secondary ion mass spectrometry (ToF-SIMS) are similar techniques used to evaluate the surface composition using a beam of pulsed primary ions focused on the surface and the collection and analysis of secondary ions ejected from the surface [145, 170]. Jungnickel *et al.* [170] evaluated sand from freshwater in Ottawa to detect microplastics by ToF-SIMS for 14 days. The results showed slow disintegration in the surface of the PE samples, revealing m/z 113 ($C_8H_{17}^+$). In addition, the fragmented ions were at m/z = 29, indicating the presence of silicon or CH_3–CH_2^+, and still m/z between 46 and 109, showing the influence of the organic matter in the samples. Thus, the authors concluded that the technique was

satisfactory to identify PE in the investigated samples and indicated the presence of sodium, potassium, and calcium in ionic form, which were adsorbed on the surface of the samples tested. Du *et al.* [171] analyzed soils in the suburbs southeast of Baoding City in China by ToF-SIMS to verify the presence of microplastics. The results evidenced that the characteristic ions present on the surface of the sample were: Na^+, Al^+, Si^+, and Ca^+, and the ions $C_4H_9^+$, $C_5H_{11}^+$, $C_6H_9^+$, $C_6H_{11}^+$, and $C_7H_{11}^+$ pointed out the presence of PP. The presence of $C_5H_{11}^+$ helps to differentiate polymers. The polymeric materials PA, PET, and PVC are respectively in m/z = 30.034 (CH_4N^+), m/z = 149.027 ($C_8H_5O_3^+$), and m/z = 83.982 (C_4HCl^+). The particles of the polymers presented a size variation in the range of 0 to 35 µm.

Matrix-assisted laser desorption ionization time-of-flight mass spectrometry (MALDI-TOF-MS) was also cited by Kundu *et al.* [145] as an alternative technique to evaluate plastic particles from different sources to predict their removal and mitigate the environmental pollution of this material. It is a soft ionization technique that allows the analysis of fragile molecules that would disintegrate in conventional analysis. Lin *et al.* [160] used MALDI-TOF-MS, a simple, easy, and efficient method, to identify and quantify micro- and nanoplastics such as PS. The authors identified digital impression peaks in regions of low mass (m/z 90, 104, 128, 130, and 312–318) and high mass (repeated peaks with Δm/z 104 in the range of m/z 350–5,000), and quantification occurred at m/z 315.3. Wu *et al.* [168] also evaluate the presence of polymers by MALDI-TOF-MS in environmental samples. The authors tested several conditions concerning laser energy, matrix, analyte, cationization agent, and volume ratio to optimize the measurement. Under optimum conditions, through representative peaks, they identified a significant presence of PS and PET, identified as $[DIT]H^+$ and $[DIT]Ag^+$, with molecular weights of 227 and 334 Da, respectively. PS was also analyzed with a mass of 58 Da, indicating that the composition was C_4H_{10}, or even with a molecular weight of 224 Da for $C_{16}H_{16}O$ and 14 Da for CH_2 peaks.

Inductively coupled plasma mass spectrometry (ICP-MS) is based on the use of an inductive plasma source to atomize and ionize the elements from the sample, the separation by a magnetic field, and the identification of the ions generated. It is a robust method, able to differentiate polymeric particles on a nanometric scale. One advantage of the ICP-MS is the high detection of both pure particles as well as particles enriched by trace elements [48, 145]. Flores *et al.* [172] mentioned the evaluation of nanoparticles in the environment and biological systems with ICP-MS. The authors indicated, as an advantage of this technique, the quick results obtained from the samples analyzed. Bolea-Fernandez *et al.* [173] evaluated the detection of microplastics by ICP-MS. With the monitoring, the authors verified the presence of $^{13}C^+$ and $^{165}Ho^+$, and the characterization of the metallic nanoparticles in microplastics could be performed, obtaining microspheres of PS in the range of 1 and 2.5 mm. Correia and Loeschner [48] used hydrodynamic chromatography (HDC) and the AF4 coupled to the ICP-MS to evaluate the presence of polymers in fish. According to the authors, it was possible to detect, quantify, and characterize polymeric nanoparticles, and PE presented the highest concentration among the polymers present, being between 1 and 100 nm, while the other polymers were in different shapes and with sizes between 2 and 500 nm.

Single-particle-inductively coupled plasma-mass spectrometry (SP-ICP-MS) can detect ion pulses in a particle-to-particle basis in an individualized form from the velocity of the data acquisition through a combination of permanence times (length of the analysis) and accommodation (electronic stability of the sample time). These characteristics indicate the capacity to analyze the mass of elements and the total concentration of the number of particles in a nanomaterial [172]. Flores *et al.* [172] indicated that the SP-ICP-MS method

is reliable and safe and can be satisfactorily used to evaluate nanoparticles in environmental and biological matrices. Von Goetz *et al.* [174] studied four plastic storages with silver incorporation by SP-ICP-MS to evaluate the migration of silver from commercial plastic storages containing micro- and nanosilver in different food simulators (water, 10% ethanol, 3% acetic acid, oil). The results indicated that the highest migration was 0.5 ng/cm^2 in food that simulated acidity corresponding to 3% of acetic acid and that approximately 12% of total silver was released in the range of 100 and 350 nm. Bolea-Fernandez *et al.* [173] evaluated the detection of microplastics by SP-ICP MS in water and found microspheres of PS in the range of 1 and 2.5 µm. These studies are better explained in Table 19.4, which elucidates the techniques mentioned and shows that there is no standardization among the methods employed by the same experimental technique.

To characterize, identify, and quantify micro- and nanoplastics in several different matrices, usually several methods are used individually or combined, and the coupling of techniques always leads to better results. Since each technique is more adequate for a certain condition and the material and presents limitations, one technique complements the other.

19.4.6 Quantification Methods by Counting, Weighing, Absorbance, and Turbidity

When the qualitative and quantitative presence of micro- and nanoplastics is desired to be identified in the environment and advanced technical resources are not available, such as the methods previously mentioned, the analytical challenges are even bigger. In this sense, simpler techniques, which will provide relevant information, can be used to study these emerging contaminants. Among them, it can be cited: 1) counting performed in counting chambers and by fluorescence optical microscopy; b) weighing, which is determined by the density difference between the particles; c) absorbance; and d) turbidity. Those two latter techniques use the principle of visible light passage through the analyzed medium that contains the pollutant.

Regardless of the kind or association of the applied technique to identify and quantify the materials, it is fundamental that the sampling points specifically represent the real contamination condition. Still, as already mentioned, the micro- and nanoplastics in the ecosystems are mixed with other pollutants, so, before being submitted to any qualitative or quantitative technique, simple or advanced, their separation from the medium is necessary. For example, the identification of micro- and nanoplastics in soil samples is extremely complex due to the presence of organic material, chemical elements, vegetable and animal residues, in addition to the natural composition of the soil [175]. In the case of the aquatic ecosystem, the same complexity is present because residual water contains emerging contaminants, organic and inorganic materials that can present similar form, density, size, and color to micro- and nanoplastics [6, 148, 176, 177]. Thus, regardless of whether the medium is solid or liquid, before applying the determination technique, it is necessary to perform the separation of the organic matter from the micro-and nanoplastics, and for that, low-cost compounds without toxicity, such as NaCl and H_2O_2, can contribute [176, 178].

When macro- (2.5 cm–100 cm) and microplastics (5 mm–1 µm) are present, the semiquantitative identification using the counting technique is favored; however, for nanoplastics (<1 µm), the visual counting is extremely hard and inaccurate [6, 17, 28]. In addition, depending on the nature of the sample, the counting technique may not be able to identify its characteristics. On the other hand, counting may have a good semiquantitative result for concentrated samples. In turn, the weighing technique presents low cost, quickness, and determined precision; however, the quantification may be influenced by the presence of mixtures of micro- and nanoplastics. Thus, to have the

desired result, a sequence of stages is necessary, involving filtration and drying, before performing the weighting [6, 177].

Table 19.4 Identification and quantification methods of micro- and nanoplastics by thermal analysis, gas chromatography, and mass spectrometry.

Technique	Range	Origin or source of samples	With or without pre-treatment	Description or consideration about the technique and/or the research	Reference
TGA	Microplastics	Environmental samples from rivers	Fermentation	Presence of PET, PA, and PP at a degradation temperature of 446 °C	Dümichen *et al.* [29]
TGA-DSC	Microplastics	Residual water	Samples filtered and mechanically pulverized until 200–500 μm	Heating of 20–800 °C at 5 °C/min in N_2 atmosphere, detection of PE, PP, PVC, PA, PES, PET, and polyurethane	Majewsky *et al.* [156]
TGA-DSC	Microplastics	Environmental samples	Homogenization by cryomilling	PE detection	Dümichen *et al.* [167]
TED–GC–MS	Microplastics	Environmental samples Enriched with polyethylene	Extraction and purification	PE ~100 μm idenfication	Käpper *et al.* [152]
TED–GC–MS	Microplastics	Environmental samples	Homogenization	PE presence in a thermal sorption of 25 at 200 °C/5 min	Dümichen *et al.* [167]
Py–GC–MS	Microplastics	Marine water	Extraction	Presence of PE, PP, PS, PA, chlorinated polyethylene, and chlorosulfonated polyethylene	Fries *et al.* [116]
Py–GC–MS	Microplastics	Sediments	Purification	Presence of PE, PP, PS, PET, and polyvinyl chloride	Bouzid *et al.* [169]
ToF-SIMS	Microplastics	Ottawa sand	Without treatment	Identification of PE and sodium, potassium and calcium in ionic form on the polyethylene surface	Jungnickel *et al.* [170]
ToF-SIMS	Micro- and nanoplastics	China soil	Drying, sieving, and washing	Identification of PP, PVC, PET, and PA in the	Du *et al.* [171]

Technique	Range	Origin or source of samples	With or without pre-treatment	Description or consideration about the technique and/or the research	Reference
				range from 0 to 35 µm	
MALDI-TOF-MS	Micro- and nanoplastics	Packages of mineral water	Extraction and filtering	Contaminated water with PS packaging	Lin *et al.* [160]
MALDI-TOF-MS	Micro- and nanoplastics	Environmental water	Without treatment	Detection of significant amounts of PS and PET	Wu *et al.* [168]
ICP-MS	Microplastics	Water	Filtering	Characterization of metallic nanoparticles in microplastics, obtaining spherical PS microspheres in the range of 1 and 2.5 mm	Bolea-Fernandez *et al.* [173]
SP-ICP-MS	Microplastics	Plastic packages	Without treatment	Silver migration from packaging during acidity simulation with acetic acid	Von Goetz *et al.* [174]
SP-ICP-MS	Microplastics	Water	Filtering	Presence of spherical PS microparticles in the range of 1 and 2.5 mm	Bolea-Fernandez *et al.* [173]

Liebezeit and Liebezeit [179] analyzed, by the counting technique with a dissection microscope, honey samples collected from producers from Germany, France, Italy, Spain, and Mexico and refined sugar bought in regional markets. The samples were initially sieved and washed with deionized water, and the authors observed that both tap water and deionized water had a small number of fibers and granular particles and thus had to filter the water in a glass fiber filter. The same happened with the laboratory environment where the samples were exposed, where the air had to be directed to go through a filter of 0.8 µm because it presented identification for fibers and fragments. Then, the authors quantified only colored fibers and fragments and found in honey a mean value of 166 ± 147 fibers/kg and 9 ± 9 fragments/kg. In the sugar samples, the mean values of 217 ± 123 fibers/kg and 32 ± 7 fragments/kg were determined using the counting technique. Zhang *et al.* [175] developed a simple method for the extraction, distinction, and quantification of PE and PP microplastics in soil samples collected (collection depths of 0–10 cm and 10–30 cm) in agricultural and fruit fields and greenhouse fields. The authors used the principle of the weight difference technique from the fluctuation of particulate material present in the soil in water, followed by a quick heating at 130 °C. For identification, a chamber connected to a microscope was used. The authors observed that

not only the microplastics fluctuated but also the organic matter and other pollutants, thus enabling their mechanical removal. However, to identify the microplastics, H_2O_2 was necessary to separate the organic matter. For a depth of 0 to 10 cm, mean quantities of microplastics of 0.008 ± 0.025, 0.540 ± 0.603, and 0.130 ± 0.307 g/kg were found in the agricultural field, fruit field, and a greenhouse field, respectively. At 10 to 30 cm, the mean microplastic counting was 0.368 ± 0.740, 0.460 ± 0.735, and 0.024 ± 0.051 g/kg for the agricultural field, fruit field, and a greenhouse field, respectively. Still considering the agricultural and fruit fields, most of the microplastics found had sizes higher than 100 µm, and for the greenhouse field, samples lower than 50 µm and higher than 100 µm were found. Thus, this study demonstrated that even in deeper soil, microplastics remain and are found.

Concerning the absorbance and turbidity techniques, microplastics cannot be detected at low densities. The technique using absorbance is still not selective concerning compounds that have the same wavelength as microplastics. In the same way, turbidity does not differentiate when suspense particles from other sources are present with micro- and nanoplastics. Nonetheless, both techniques are easy to use, have a wide range of measurements, and provide qualitative and quantitative (through a standard curve) information for micro- and nanoplastics, thus making it possible to identify the nature of the pollutants [6, 177].

Yuan *et al*. [180] prepared a synthetic solution of PS microplastic and used graphene oxide as the adsorbent. To quantify the removal efficiency, the authors built a standard curve of microplastic solution with concentrations that ranged from 0.1 to 0.8 g/L and, at the maximum absorption wavelength of 720 nm, determined the absorbance. From this curve, the maximum adsorption capacity of 617.28 mg/g was determined in conditions of pH 6 and an initial micropollutant concentration of 600 mg/L with an equilibrium time of 120 min. Rajala *et al*. [176] performed the removal of PS using coagulation-flocculation with the coagulant's ferric chloride, poly aluminum, and polyamine chloride. Then, the authors sedimented the effluent prepared with PS with sizes lower than 10 µm, justifying that few studies use this size range for emerging pollutants. The concentration and microparticle size were determined by flow cytometry, and turbidity and chemical oxygen demand were also used to aid in the interpretation of the results obtained by flow cytometry. The results showed that the turbidity technique can be used to aid in the evaluation of the efficiency of microplastic remediation processes.

19.5 Conclusions

This review approaches the problem of pollution of terrestrial, aquatic, and atmospheric ecosystems by micro- and nanoplastics, presenting the pre-treatment methods that can be applied to the samples as a function of their characteristics and the kind of analysis to be posteriorly performed, as well as a vast review of countless techniques that can be used to characterize, identify, and quantify micro- and nanoplastics. From what was presented in this chapter, several techniques can be used depending on what is desired to be evaluated, for example, size, morphology, kind of polymer, concentration, or quantity, among others, and each technique presents its advantages and disadvantages. The techniques can be used individually, but for a more complete analysis, the coupling of techniques is the best option because it provides faster and more precise results. Still, based on the studies evaluated, a methodology standardization does not exist for these techniques, in a way that each research is performed in a way that hampers the comparison of the results because some use simple analyses such as visualization, and others use technological equipment that presents higher accuracy in the results. Among other factors, this is due to the great diversity of the polymeric materials and matrices in which these materials are found and the difficulty of accessing robust equipment. Thus,

from what has been exposed, this review contributes to the knowledge about pollution of micro- and nanoplastics, but mainly about the technological advances in this area up to now, giving an overview of the several possibilities of techniques that can be used.

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20
New Approaches for Micro(Nano)Plastics Analysis

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Abstract

In the 21st century, micro- and nanoplastics have become among the emerging global pollutants resulting from the excessive use of plastic products and the disposal of plastic waste extensively. These pollutants are persistent, tend to accumulate in the environment, and have a negative impact on both human health and ecology. Accurate identification and quantification using suitable analytical methods are important to better understand and mitigate the effects of these pollutants. Various conventional and advanced methods have been applied for the analysis of micro- and nanoplastics in the environment. Owing to the severity of this issue, new analytical approaches are emerging based on the combination of two or more analytical methods. Different analytical methods based on microscopic, spectroscopic, thermal, and electroanalytical techniques have been widely used for the analysis of micro- and nanoplastics in different environmental compartments. Each of these techniques has advantages and disadvantages. Recently, the development of analytical techniques for this application has seen an exponential growth. Therefore, this chapter presents developments in the analysis of micro- and nanoplastics using single

analytical methods, bi-analytical methods, and multi-analytical approaches by emphasizing new approaches developed or employed for the analysis of micro- and nanoplastics.

***Keywords*:** Microplastics, nanoplastics, plastic pollution, analytical methods, new approaches

20.1 Introduction

The generation of plastic waste has increased exponentially since its production began in the 1940s. Thus, it is practically impossible to cross plastic products in our daily lives multiple times. This is because of the wider applications of plastic materials in the food industry, agriculture, transportation, medicine, offices, electronics, construction, homes, military, and so on. These wider applications are owing to their advantages, including low cost, durability, corrosion resistance, malleability, and abundance, which make plastics extensively used in everyday life [1]. Commonly used plastics include polypropylene (PP), polyethylene (PE), polystyrene (PS), polyvinylchloride (PVC), high-density polyethylene (HDPE), low-density polyethylene (LDPE), polycarbonate (PC), polyamide (PA), acrylonitrile Butadiene Styrene (ABS), ethylene-propylene-diene-monomer rubber (EPDM), poly(methyl methacrylate) (PMMA), styrene-butadiene rubber (SBR), and polyethylene terephthalate (PET) [2, 3]. Most used plastics (approximately 50%) are discarded after a single use directly into the environment to end up in water systems, landfills, and biota. Plastic wastes in the environment undergo movement from one system to another, and different degradation processes are aided by biological or non-biological agents [4]. Thus, it is becoming difficult to find a place that is free of micro- and nanoplastic pollutants. The generation and discharge of

large quantities of plastic waste into the environment have raised concerns regarding environmental, human safety, and health issues. This concern is mainly related to their persistence in the environment and slow degradation through biotic and abiotic means [5]. Furthermore, the wide distribution of micro- and nanoplastics in different environmental segments causes the release of chemical additives during production. Plastic additives form secondary pollutants that may be toxic to the environment and biota [6]. Additionally, tiny plastic particles are suspected to be carriers of other chemical and biological contaminants, which may worsen the effect of plastic waste pollution [7, 8]. A better understanding of the environmental and human health effects of micro- and nanoplastics in the environment requires identification, quantification, and detailed ecotoxicological studies. Such investigations require a holistic approach that comprises representative sample collection, preservation, preparation, measurement, and interpretation of results [9]. Micro- and nanoplastics have been detected in different environments including atmospheric fallout, wastewater, treated water, freshwater, and human blood [10, 11]. Different analytical methods have been employed to detect plastic particles. These include microscopic, spectroscopic, thermal, chromatographic, and electroanalytical methods [12–15]. Many studies have revealed that micro-nano plastics are emerging contaminants in food [16], wastewater treatment plants [17], table salts [18], neustonic/planktonic samples, and fin whales [19]. The worst scenario is the bioaccumulation of these pollutants in organisms living in water bodies, including freshwater, marine species, and other food chains [19, 20].

20.2 Global Plastic Production and Its Waste Generation

It has been reported that approximately 390 million metric tons of plastics will be produced in 2021 for various applications. It is estimated that production will be 445 and 589 million metric tons in 2025 and 2050, respectively. Of these quantities of plastics, more than 70% are degraded into micro- and nanoplastics and found in various compartments of the environment, while the remaining portions go to energy recovery, landfills, and recycling, respectively [15] (Figure 20.1). The wide use of plastics and their products is presumably due to their low density, strength, durability, abundance, easy processing, and resistance to corrosion and flames, as well as the huge demand from people around the world for quality of life. In the 21st century, the increase in the production and use of plastic products and the poor management of their waste have become among the main environmental issues. Furthermore, in addition to the disposal of large amounts of plastic waste into the environment, the degradation of these wastes also generates micro- and nano-sized particles that are abundantly found in the environment, including in food and water bodies.

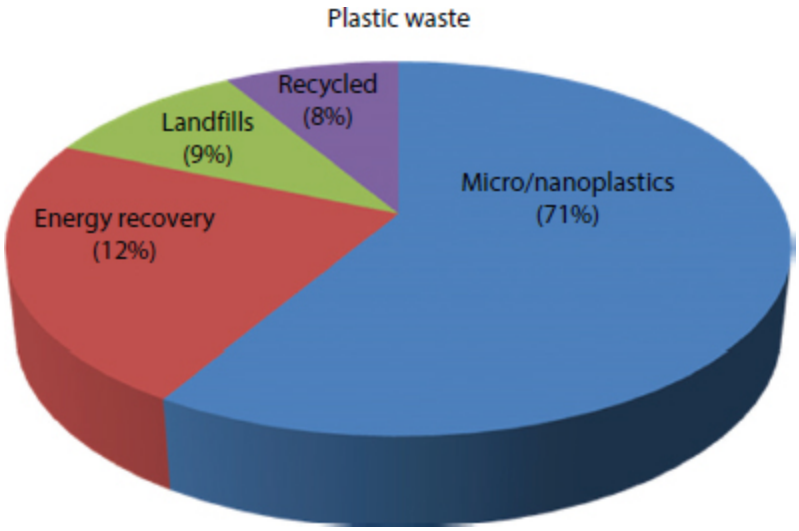


Figure 20.1 Annual plastic waste generation and management [15].

The widespread presence of these pollutants has raised issues related to the impact of these tiny particles on the environment and human health [21]. Different types of plastics, such as thermoplastics (PE, HDPE, LDPE, PP, PS, PVC, PET, PC, PA, ABS, etc.) and thermosetting plastics (epoxy, silicone, polyurethane, and phenolic), are commonly produced. The most commonly produced plastic materials are PP, PVC, PET, and PS, with approximate annual productions of 68, 38, 33, and millions of metric tons, respectively [22, 23].

20.3 Sources and Health Effects of Micro- and Nanoplastics

Plastic materials and their breakdown products are ubiquitous in all environmental segments at the present time. These products are released into the environment as primary and secondary plastics, respectively. The former is purposefully prepared for various industrial applications, including pre-production plastic pellets and consumer product ingredients. Secondary plastics are the remains of

large plastic debris such as plastic bottles, bags, and food packaging. The breakdown products of plastic materials are released into the environment [4], wastewater treatment plants [24], dumped plastic waste and trash [25], greenhouse materials, domestic waste, and irrigation [26]. Micro- and nanoplastics are small particles of manmade organic polymers such as EPDM, HDPE, LDPE, PE, PET, PS, SBR, PE, PP, PET, and PVC [27–29]. Frequent exposure to small plastic particles in air, food, water, indoor dust, and personal care products leads to oral ingestion, inhalation, and dermal contact (Figure 20.2). Owing to their abundance, micro- and nano-level plastic particles are the main threats to living organisms and human beings. Many studies have indicated that plastic particles are found in the human blood [11], feces [30], colectomy specimens [31], and placentas of pregnant women [32]. This confirms that humans can be exposed to microplastics and nanoplastics through different routes, including oral ingestion, body contact, and inhalation. Furthermore, the incidence of the COVID-19 pandemic has increased the use of single-use face masks, thus amplifying the magnitude of plastic waste generation and disposal in the environment [33]. COVID-19 face masks, including N95, surgical masks, and other single-use face masks, contain a significant portion of persistent plastic materials [34, 35]. The presence of nanoplastics in water has become an increasingly concerning issue, as they are not only present in oceans and rivers but also in tap water, bottled water, and even in some foods. These tiny plastic particles, measuring less than 100 nm, are not easily filtered by traditional water treatment methods. The persistence of these particles in the environment and their negative effects on human health have made it crucial to explore and implement effective removal techniques.

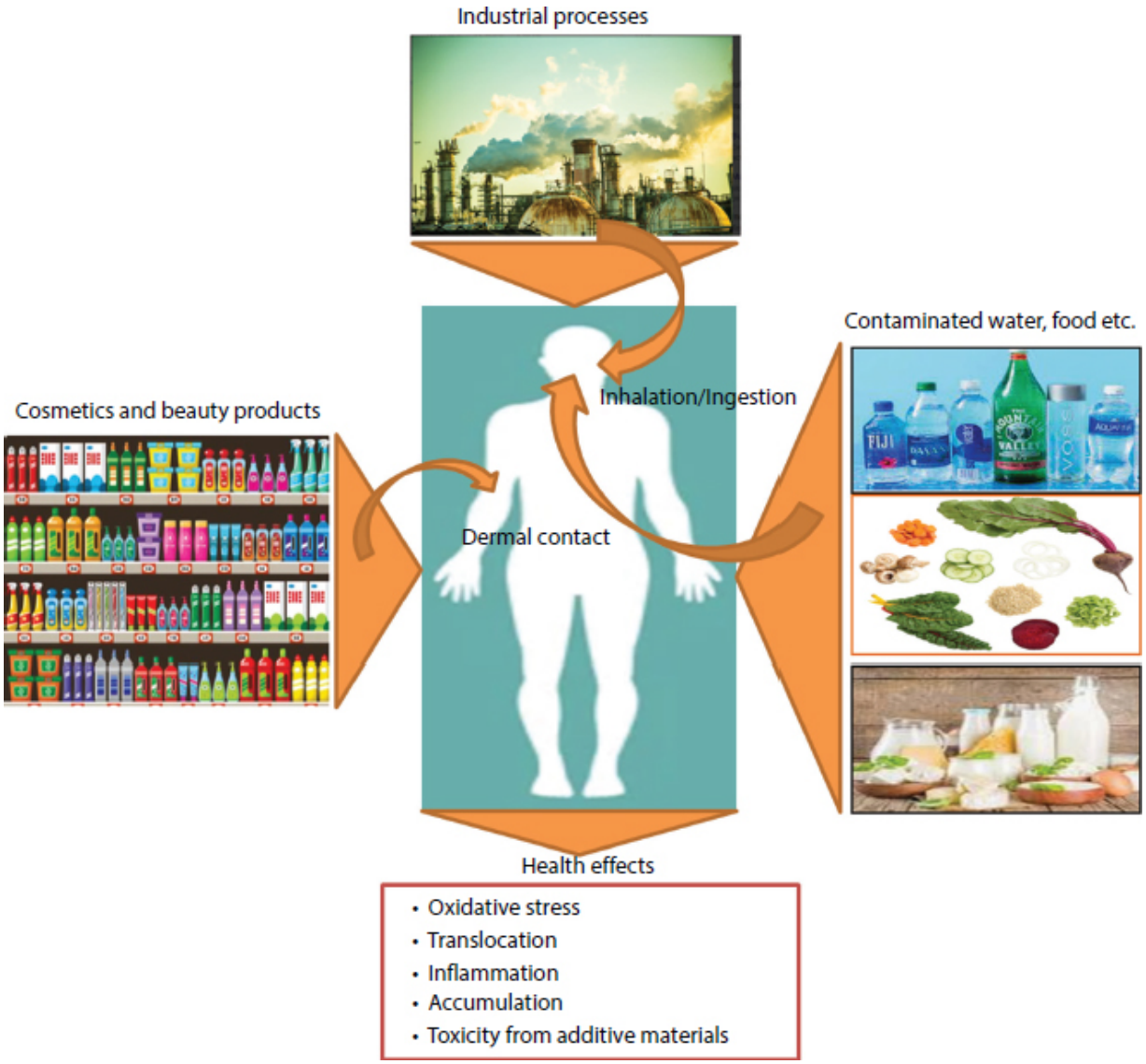


Figure 20.2 Possible exposure routes and adverse effects of micro- and nanoplastics on humans.

The implications of nanoplastics in water are numerous because they can be ingested by aquatic organisms, potentially harming their health [36]. These particles can also make their way up into the food chain, potentially affecting human health. Additionally, nanoplastics can interfere with water treatment processes, making it difficult to provide clean and safe water for human consumption. Therefore, it is important to note that more research is needed to fully understand the potential effects

of micro- and nanoplastics on human health and the environment and to improve the effectiveness of removal methods [37]. It is also important to mention that prevention is still the best way to tackle this problem, by reducing the use and disposal of single-use plastics and encouraging proper solid waste management. An *in vitro* study showed that micro-nanoplastics generated from PS caused multiple biological distresses in mammals, such as cytotoxicity, oxidative stress, inflammation, and carcinogenicity; however, *in vivo* models in rodents indicated that plastic particles bioaccumulate in different body parts [37].

20.4 Sample Collection Methods

The collection of representative samples from environmental components, such as water, soil, air, sludge, or biota is the most important step for the proper detection and quantification of micronanoplastics [38]. Sampling is the process of collecting or obtaining a representative small portion of the bulk material to make statistical inferences and estimate the characteristics of the entire material. Thus, it is regarded as a significant step in qualitative (identification) and quantitative (quantification) analysis through physical and chemical measurements. The effective analysis of a given target substance requires a representative sample that truly reflects the composition of the material to be analyzed within the context of a defined analytical problem. A sample is the portion of the material intended to be representative of the entire material under investigation. However, there is no standard protocol for the identification and quantification of microplastics and nanoplastics in the environment [39, 40]. A better understanding of the existing analytical approaches in real sample collection, preservation, preparation, and

determination of micro- and nanoplastics is very important (Figure 20.3). The analysis of plastic particles existing in soil, water, food, air, and living organisms is in its infancy and requires significant work in the future. Therefore, great emphasis must be placed on sampling methods because the final result is directly dependent on the sample collected [41].

Water bodies are homogenous (e.g., rivers, lakes, oceans, and groundwater) and contain fewer quantities of micro- and nanoplastics owing to natural dilution processes. Thus, nets with different mesh sizes (50–3,000 µm) are commonly used for sampling and concentrating them through filtration [42]. However, this sampling technique has the disadvantage of the loss of particles of very small sizes [43, 44]. In addition to net, acoustic Doppler current profilers [45], rotatating drum samplers [46], continuous-flow centrifuges [47], and cascades of different filter cartridges [48] were also used for sample water. Each material used for water sample collection required proper cleaning after each sample collection to avoid possible errors.

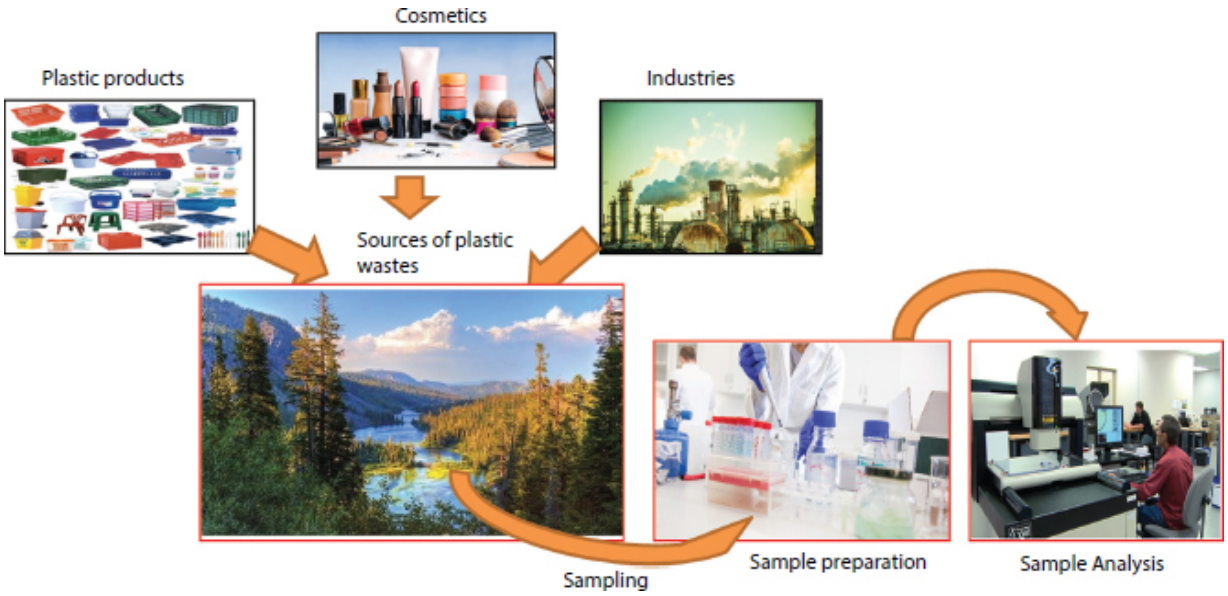


Figure 20.3 Schematic of sources, sample collection, preparation, and micro- and nanoplastic analyses.

Solid samples, such as soil, rocks, and sediments, are heterogeneous, and their sampling requires more attention than liquid samples. The sampling process must obtain a representative sample to provide meaningful insights into the occurrence and abundance of micro- and nanoplastics. Plastic-based apparatus must not be used for sampling; therefore, there is no sample contamination [49]. Grabbers or box corers are widely used for sediment sample collection at different depths during the analysis of micro- and nanoplastics [50–52]. The composite sampling method is recommended to obtain a representative sample that provides full insight into the quantity of plastic particles.

From the viewpoint of analytical chemistry, collection of representative samples plays a significant role in obtaining reliable results. Therefore, it is important to select a suitable sampling technique to obtain validated data regarding micro- and nanoplastics in the environment. Generally, it is possible to classify sampling techniques for the collection of samples from environmental segments as bulk, volume-reduced, and selective sampling [15, 53]. The

three sampling techniques are summarized in Table 20.1. The environmental samples collected for the analysis of micro- and nanoplastics require sample preparation using alkaline, enzymatic, and acidic digestion. These sample digestion methods remove unwanted matrices and produce available plastic particles [54]. However, acidic and alkaline digestion may degrade plastic particles and affect the final results. However, enzymebased sample digestions are more specific, with little or no impact [55, 59]. Therefore, using different digestion methods and comparing the results may minimize these limitations in the future.

Table 20.1 Common sampling methods employed for the analysis of microplastics and nanoplastics.

Method	Advantage	Disadvantage
Bulk sampling	Entire volume of the sample is taken without reducing, Suitable when microplastics cannot be easily identified visually	Needs handling large volume samples, Require further processing in the laboratory
Volume-reduced sampling	Suitable for obtain and preserving the portion of interest	Needs filtering large volumes of water with nets, Require further processing in the laboratory
Selective sampling	Direct extraction of plastic pollutants from the environment using naked eye visualization	Great risk of overlooking small size plastics and needs particular care in the field

20.5 Emerging Analytical Approaches

Monitoring the present status of our environment and future trends of environmental systems requires the availability or development of suitable analytical methods. For example, there is growing concern about the possible health and environmental impacts of plastic particles at the present time [15]. To assess the actual risks associated with micro- and nanoplastics, reliable data on the occurrence of these particles in environmental, water, air, and food samples are required. In the past decade, several assessments have partially or fully focused on methods for the detection, identification, and quantification of micro- and nanoplastics. In most cases, the identification and quantification of tiny plastic particles in different environmental samples can be performed using microscopy [15, 57], spectroscopy [50, 58], thermal degradation [59, 60], attenuated total reflectance (ATR) [61], and gas chromatography coupled with mass spectrometry (GC–MS) [62]. Following the increase in the production, consumption, and waste generation of plastics, many efforts have been made to develop new or apply existing analytical methods. This implies that the analysis of micro- and nanoplastics in environmental samples is very important to understand their current status, and possible environmental and human health risks, and to evaluate their removal efficiency [63].

20.6 Single Analytical Methods

20.6.1 Microscopic Techniques

In every analysis, cost, experimental operation, method availability, and demand for chemicals were the primary criteria. Analytical methods based on the visualization process are advantageous [64]. As a result, the use of the

naked eye or microscopy makes visual analysis among the most commonly applied methods for the physical analysis of microplastics [65]. Different forms of microscopy such as light and stereoscopic microscopy, optical microscopy, electron microscopy, scanning probe microscopy, and fluorescence microscopy, are widely used to analyze plastic waste particles [66]. Small particles of plastics were detected visually using light microscopy and stereoscopic microscopy. This involves the visual sorting of micro- and nanoplastics based on their nature, size/shape, breaking down of the stage, and color of the pollutants. Light microscopy is an old and important method for visually detecting microsized plastic wastes in the environment. To date, micro- and nanoplastics have been identified using various microscopic techniques because of their simplicity, accessibility, and fastness. However, this method does not provide chemical evidence required to identify the types and nature of plastic particles [67]. Physical characterization of plastic wastes in various samples was performed using visual sorting with the help of light microscopy, stereoscopic microscopy [53], and optical microscopy [68]. Microscopic methods are widely used; however, they provide no chemical information and suffer from false results and inability to identify very small plastic particles.

The morphological characteristics of plastic waste can be determined using scanning electron microscopy (SEM). This is one of the most powerful and important methods for characterizing different materials [69]. Its combination with energy-dispersive X-ray spectroscopy (EDX) provides another advantage in obtaining the type and elemental composition of the sample [70]. Hence, SEM-EDX has been applied to analyze the surface morphology and elemental composition of micro- and nanoplastics [52, 60]. It is obvious that the application of SEM is limited because of

the poor conductive nature of plastic waste, the requirement of gold plating, and the inability to analyze the color of micro- and nanoplastics [69].

20.6.2 Thermal Techniques

A thermal analytical method is a technique in which the physical properties of a substance under study are measured based on the change in temperature. Thermogravimetric analysis (TGA) and differential scanning calorimetry (DSC) are two common techniques that are based on the thermal stability of micro- and nanoplastics [71]. TGA is based on the determination of the product formed from the thermal degradation of the sample mass as a function of temperature. In contrast, DSC relies on the changing temperature of the sample and reference to measure their responses. The identification and quantification of plastic particles were possible based on the peak position and area under the peak, respectively. However, thermal analysis is a destructive method that offers chemical identification of micro- and nanoplastics [72]. DSC was employed for the detection and identification of microplastics in wastewater treatment plants. Accordingly, PE, PS, PP, and PET were found to exist at different wastewater treatment stages. The method was able to confirm that the microplastic removal efficiency of the plant was in the range of 90%–95% [73]. Similarly, microplastics generated from PE and PP in wastewater samples were detected using TGA-DSC [74]. The combination of TGA-DSC offers the complementary advantages of each method and alternative methods to other spectroscopic methods.

20.6.3 Spectroscopic Techniques

The interaction of light with matter is the basis of all the spectroscopic techniques. These techniques are based on

certain characteristics of the sample generated as signals during the interaction. This phenomenon is related to the type and quantity of sample under investigation. Various spectroscopic techniques are available, such as ultraviolet and visible light spectroscopy, fluorescence spectroscopy, infrared (IR) and Raman spectroscopy, nuclear magnetic resonance spectroscopy (NMR), and atomic spectroscopy. These techniques rely on the energy absorption, emission, and scattering.

Fourier transform infrared (FTIR) and Raman spectroscopy are vibrational spectroscopic techniques that provide full information on the chemical bonds and functional groups of the samples. Hence, both the FTIR and Raman methods were successfully used for the identification and detection of micro- and nanoplastics in different environmental segment samples. Chemical identification and quantification of micro- and nanoplastics are commonly performed using FTIR spectroscopy. Thus, this method is the most popular approach for routine analysis of plastic particles in different sample matrices [53, 75]. Various plastic pollutants have been detected in different sample matrices such as wastewater [76], air [77], and freshwater [78]. In another study, Langknecht *et al.* (2023) employed both Raman and FTIR techniques to identify and detect microplastics in sediments with greater than 96% recovery. The analysis was supported by complete sample extraction and method validation; thus, it is recommended for regulatory applications [79]. The sampling, preparation, identification, and detection of micro- and nanoplastics in different environmental components, such as air, water, plants, and soil, were performed using various spectroscopic techniques (Figure 20.4).

Many studies have indicated that Infrared and Raman spectroscopic methods are the most commonly used techniques for reliable identification of the chemical

composition of micro- and nanoplastics [53, 56, 80]. The proper application of FTIR fingerprinting will greatly improve the determination of plastic composition and its abundance by avoiding false-positive results [68]. Microsized plastic wastes were sampled using an active sampling technique and analyzed by FTIR. The study confirmed that higher concentrations of plastic particles were found indoors than outdoors, which was further correlated to the lifestyle of the occupants [81].

Surface-enhanced Raman spectroscopy (SERS) is among the most widely used low-cost methods for the chemical characterization of micro- and nanoplastics owing to its high sensitivity and fast characterization capability [82]. For example, SERS combined with a bifunctionalized silver nanowire membrane was developed for the detection of trace nanoplastics in water [83], PS of different sizes in different water samples [84, 85], and spiked real water [86]. Similarly, PS-, PE-, and PP-based nanoplastics were qualitatively detected in water using silver colloids as the active substrate [87]. It was confirmed that the method is promising for the identification and quantification of different plastics at the nanoscale.

Nuclear magnetic resonance (NMR) spectroscopy is an analytical technique used to study molecular structure by measuring the interaction of electromagnetic radiation at radio frequency with the nuclei of molecules placed in a strong magnetic field. Thus, NMR affords both qualitative and quantitative information regarding the composition of a sample by detecting hydrogen, carbon, and phosphorus atoms [88, 89]. Microplastics obtained from PE, PET, PS, PVC, ABS, and PA have been successfully identified and quantified in different environmental samples by using NMR techniques following proper sampling and calibration methods [90, 91, 88].

Mass spectrometry (MS) is a fast and reliable analytical method that offers better accuracy when applied to the characterization of plastic particles in the environment. Thus, their combination with other analytical tools leads to enhanced signal generation, which is directly related to the target analyte. Matrix-Assisted Laser Desorption/Ionization time-of-flight mass spectrometry (MALDI-TOF-MS) was first proposed as an effective method for the analysis of micro- and nanoplastics [92]. Thermal destruction and MALDI-TOF-MS have also been employed for the detection of PS and PET particles in different sample matrices [93]. Nanoplastics derived from PS were analyzed by MALDI-TOF-MS using silver cluster ions as cation-forming agents [94]. The tiresome sample preparation step was simplified and eliminated by a novel approach based on the application of metal oxide nanoparticles combined with MS to detect 1×10^{-12} g of micro- and nanoplastics in water and snow [95]. This method is promising for the analysis of plastic pollutants in water, air, soil, and other environmental samples; however, further work is required to validate and standardize this method.

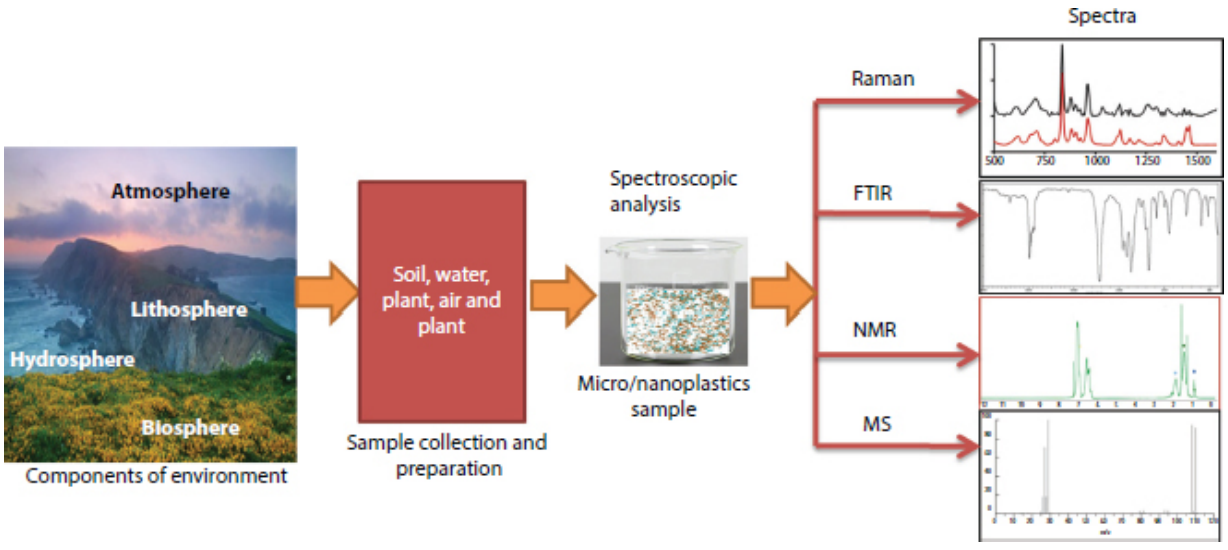


Figure 20.4 Spectroscopic analysis of micro- and nanoplastics in different environmental samples.

20.6.4 Electroanalytical Techniques

The electroanalytical method is an important technique in analytical chemistry that relies on the measurement of the potential or current in an analytical cell containing the analyte. These highly sensitive and selective methods are used for various analytical measurements to detect one or more analytes. The effectiveness of these methods depends on the proper preparation of the working electrode or the surface modification for a particular application. Generally, the electroanalytical methods used for the analysis of micro- and nanoplastics include potentiometry, voltammetry, chronoamperometry, and electrochemical impedance spectroscopy (Figure 20.5). Electrochemical methods are fast, selective, low-cost, portable, and relatively green for removing micro- and nanoplastic pollutants from environmental samples. The removal and detection of PS nanoplastics using electrosorption [96], electrocoagulation [97], electrooxidation [98], electrochemical sensing [99], serial Faradaic ion concentration polarization (fICP) [100], and electrochemical impedance spectroscopy (EIS) [101] have been successfully performed by different researchers. Most attention has been paid to the removal of plastic waste as a means of remediation using electroanalytical methods. Recently, the detection of plastic particles using electrochemical sensors has become a promising technology [102]. As a result, the design and development of electrochemical sensors for monitoring micro- and nanoplastics are advantageous because of their fast response, humble procedure, cost-effectiveness, possible miniaturization, and selectivity. Subsequently, voltammetry, amperometry, and EIS were mostly used as label-free methods. PE particles were successfully detected through the development and employment of an EIS-based sensor in combination with flow cytometry [101]. Similarly,

chronoamperometry and fICP technique-based sensor measurements have been used to detect PE and PS particles [100, 103]. Furthermore, voltammetry and impedance sensors have been developed based on exoelectrogenic biofilms for the detection of PE particles [104]. Electroanalytical methods are powerful techniques that are not well developed for monitoring and remediating of micro- and nanoplastics. Therefore, needs further attention and investment are required in the future.



Figure 20.5 Schematic of electroanalytical methods used for the analysis of micro- and nanoplastics in the environment.

The application of impedance to the study of plastic particles in the environment is of major significance because the measurement enables the detection of substances with poor conductivity.

20.7 Hyphenated Analytical Techniques

The coupling of two or more analytical techniques can enhance the performance of the analysis of plastic particles by combining the advantages of individual techniques. Some combined techniques with state-of-the-art proofs are presented in the following subsections.

20.7.1 Chromatographic-Based Hyphenated Techniques

New approaches based on the combination of two or more different analytical techniques have emerged for reliable and fast analyses of micro- and nanoplastics in different environmental samples. These include pyrolysis–gas chromatography/mass spectrometry (Py–GC/MS), pyrolysis–mass spectrometry (Py–MS), thermal extraction–desorption gas chromatography/mass spectrometry (TED-GC/MS), headspace solid-phase microextraction–gas chromatography/mass spectrometry (HS SPME–GC/MS), thermogravimetric analysis–mass spectrometry (TGA–MS), thermogravimetric analysis-Fourier-transform infrared spectroscopy (TGA-FTIR) [105], and atomic force microscopy-based infrared spectroscopy (AFM-IR) [106].

The Py–GC/MS combined technique is a destructive technique that is commonly used for the analysis of plastic pollutants because of its high sensitivity, simple sample treatment requirements, and good accuracy and reproducibility [107]. This approach involves the degradation of plastic waste at high temperatures, followed by the separation of the products formed using GC and their detection by MS. Pyrolysis involves breaking down the molecular bonds in the plastic chain to form smaller and simpler ones at high temperatures in the range of 600–800

°C. This method was successfully used for the detection of PS and PMMA in water [108], and cloud-point extraction combined with Py–GC–MS was used for the detection of plastic particles in water [109]. In another study, PE, PC, and PET microplastics were detected using automated fluorescence microscopy combined with Py–GC–MS [110], which provided a method to determine the chemical compositions and mass of plastic pollutants in the atmosphere. The same method has been used to measure the quantity of micro- and nanoplastics in wastewater treatment plants [111] and nanoplastics (PP and PS) in different organic matter suspensions [112].

Currently, the increase in the magnitude of plastic waste contamination in aquatic systems worldwide is attracting the attention of researchers from various countries. In this regard, three techniques, namely microscopy, Pyr–GC–MS, and FTIR, were used simultaneously to identify and quantify micro- and nanoplastics in water. Interestingly, three common plastic materials, namely PP, PVC, and PET, have been successsfully detected in different water systems [113].

Thermal methods, such as DSC, are combined with Pyr–GC/MS for the identification and quantification of microplastics in air. DSC provides the melting point for identification, whereas the Pyr–GC/MS generates fragmentation patterns to confirm the occurrence and chemical composition of microplastics in the samples [114].

Microscopic techniques, such as SEM, have been successfully combined with Raman spectroscopy to identify and detect nanoplastics in water, and fluids from humans and seawater. However, the detection limit of this method depends on the sampling environment used. Furthermore, this approach requires a suitable sample preparation method for the effective analysis of small plastic waste

particles in various environmental samples [115]. Another multi-analytical approach based on the thermal desorption TD–GC–MS was proposed to study the degradation products of PE, PP, PS, and PET at different temperatures and time intervals. The fragmentation products of each polymer were successfully identified, and this method seems quite interesting for the detection of micro/nanoplastics in complex matrices [12]. Further development, other multi-analytical approaches, such as TGA–TD–GC–MS [116] and TGA–FTIR–GC–MS [117] have also been used in the identification and quantification of plastic waste particles.

20.7.2 Spectroscopic-Based Hyphenated Techniques

Hyphenated techniques based on different spectroscopic methods provide the advantages of spectral and coupled methods for the identification of samples. The most innovative approach was based on the combination of conjugation of plastic particles with functionalized metal nanoparticles, enabling its detection by highly sensitive inductively coupled plasma mass spectrometry (ICP-MS). In particular, single-particle inductively coupled plasma mass spectrometry (SP-ICP-MS) has emerged as a novel analytical method for the detection and quantification of microplastics and nanoplastics at levels relevant to different environmental monitoring standards [118].

A new multi-analytical approach based on SP-ICP-MS was first applied to the analysis of microplastics by quantitatively determining the particle number density [119–121]. This approach is based on the detection of microplastics in water samples, dissolved organic carbon, and dissolved particles in natural water. A real water sample was digested using strong acids to detect nanoplastics labeled with gold nanoparticles to achieve a

good spiked recovery within the range of approximately 73%–93% [122]. In addition, the detection and quantification of PS from food packaging and personal care products have been reported using ICP-MS by monitoring the 13C isotope and confirmed by attenuated total reflectance FTIR [123].

Interestingly, PE and PVC-derived nanoplastics that abundantly exist in the food chain were detected successfully after doping with gold nanoparticles by employing a very interesting Single-Cell ICP-MS approach [124]. The standard obtained results were in good agreement with the standard environmental conditions.

Matrix-assisted laser desorption ionization-Fourier transform ion cyclotron resonance mass spectrometry (MALDI-FTICR-MS) has also emerged for the detection and quantification of various types of micro- and nanoplastics generated from various sources [125]. Moreover, SEM was combined with SERS for the identification and detection of polystyrene (PS) and PMMA [126]. The ultrahigh spatial resolution of SEM and the structural fingerprint of SERS were combined for better performance.

20.8 Current Trends and Future Perspectives

In the present century, increased plastic production usage, disposal of plastic waste, and plastic pollution have become burning issues. This may be due to the biotic and abiotic fragmentation of plastic pollutants to release micro- and nanoplastics into the environment and their transportation from place to place. Thus, plastic pollutants are available on Earth. The presence of plastic pollutants in all environmental compartments has a negative impact on the environment, global economy, and the social and health

sectors [127]. The ubiquitous presence of these emerging pollutants, their persistence, and the toxic additives used during their synthesis have increased their threat. In particular, the presence of small particles of plastic pollutants in fresh water, food, soil, air, biota, bioaccumulation, and ecotoxicological effects makes the issue more complex [128–131]. However, the correct identification and determination of plastic waste-based particles in environmental samples is labor-intensive. Practically, it is impossible to ban plastic products from the market because of their numerous advantages, but their waste disposal can be minimized significantly if strict monitoring is set. This can be achieved through the implementation of three R systems: reduce, recycle, and reuse.

The complexity of real samples obtained from various environmental components makes the analysis of microplastics and nanoplastics more difficult. Therefore, the collection of representative samples and the selection of suitable analytical methods based on their sensitivity, accuracy, and non-invasiveness are very important [132, 133]. The analysis can be performed using visual analysis with the naked eye and microscopy; chemical analysis using FTIR, Raman, NMR, and MS; separation analysis using chromatography; thermal analysis using TGA and DSC; and electroanalytical methods. Thus far, it is clear that the investigation of highly complex and diversified plastic analytes requires the application of multiple analytical methods rather than a single method to obtain more information about them. This is because single analytical methods are less accurate than the combination of the two methods, which is far less important than the application of multi-analytical approaches (Figure 20.6). Therefore, it is essential to implement it through exhaustive

optimization and to develop new analytical methods to address this urgent issue.

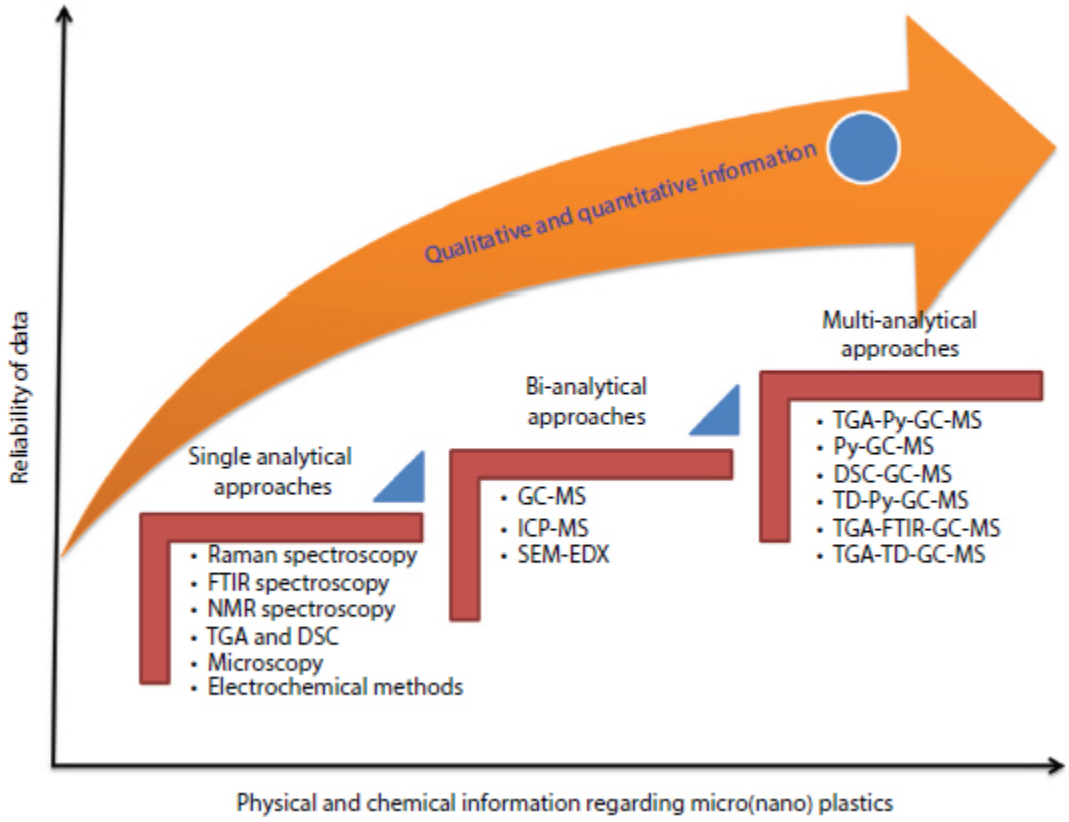


Figure 20.6 Conventional and new advanced analytical approaches are used for micro- and nanoplastic analyses.

20.9 Conclusions

The ubiquitous presence of micro- and nanoplastics in the air, water, soil, sediment, and biota has become a threat to human health and the environment in the 21st century. Thus, the identification and quantification of these emerging pollutants are very important for future mitigation programs. For this purpose, both conventional and modern analytical methods have been used. These methods include single analytical methods, such as spectroscopic, microscopic, thermal, and electroanalytical methods have been widely used. New approaches based on

multi-analytical methods have become promising for providing reliable results regarding the identity and quantity of plastic contaminants in the environment. These are bi-analytical methods approach (GC–MS, ICP-MS, SEM-EDX, SEM-EDX, etc.), and multi-analytical approaches (TGA–Py–GC–MS, Py–GC–MS, DSC–GC–MS, TD–Py–GC–MS, TGA–FTIR–GC–MS, TGA–TD–GC–MS, etc.). All of these methods provide physical, chemical, or both information regarding the type and quantity of plastic particles in the environment. Hence, it can be concluded that there is no single effective and accurate analytical method for obtaining full information about the pollution status of different environmental segments. However, the combination of two or more analytical methods has emerged as a new approach for generating reliable data on the nature and chemical composition of micro- and nanoplastics following careful collection of representative samples and analytical preparation. Considering the complexity of environmental samples and plastic waste diversity, it is strongly advised to limit the utilization of plastic products for single-use and indiscriminate release of their waste into the environment. The development of fast, cost-effective, real-time, and effective remedial and detection methods is urgently needed to pave the way for a better future that is less affected by microplastic and nanoplastic pollutants.

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Note

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21
Enzyme-Catalyzed Biodegradation of Micro- and Nanoplastics

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Abstract

This chapter details the various enzymes used for the biodegradation of micro- and nanoplastics. In this chapter, the advantages of enzymatic approaches over conventional methods are presented. The mechanism of enzyme-catalyzed degradation of plastics is discussed, and examples of biodegradation of synthetic polymers using various enzymes are detailed.

***Keywords*:** Enzyme, biodegradation, plastic, toxic effects

21.1 Introduction

Plastics have a relatively short history compared with many other materials. The first synthetic plastic, bakelite, was invented in 1907 by the Belgian chemist Leo Baekeland. Bakelite is a thermosetting phenol formaldehyde resin that has been used for several applications ranging from electrical insulation to jewelry [1]. In the 1920s and the 1930s, new types of plastics including polystyrene, polyvinyl chloride, and nylon were developed. These materials are used in a variety of applications including packaging, toys, and textiles [2].

During World War II, plastics played a critical role in war. Plastics have been used in various military applications, including aircraft windshields, radios, and insulation for electrical wiring. After the war, the use of plastics exploded [3]. Plastics have been used to replace traditional materials such as wood, metal, and glass in many applications. In the 1950s and the 1960s, plastic products became more affordable and more widely available, leading to a boom in consumer goods made from plastic. Nevertheless, the widespread use of synthetic polymers has also caused environmental concerns, as plastic waste has become a major problem. Currently, efforts are being made to reduce plastic waste and develop sustainable alternatives to traditional plastics [4].

Plastics have become a major environmental concern because they do not biodegrade quickly and remain in the ecosystem for several centuries. Plastic waste can be found in landfills, waterways, and oceans, where it can harm wildlife and ecosystems [5].

When plastic waste ends up in the ocean, it can be ingested by marine animals, causing harm or even death. Plastics can also entangle animals by causing injuries or drowning. Additionally, plastic waste can be broken down into smaller particles termed microplastics, which can be consumed by marine organisms and enter the nutrition chain [6]. The plastic waste in landfills poses environmental risks. As plastics break down over time, they can release harmful chemicals into the soil and groundwater. Efforts are being made to reduce plastic waste and develop more sustainable alternatives to traditional plastics [7]. These include initiatives such as recycling, reducing single-use plastics, and developing biodegradable and compostable plastics. Governments and organizations around the world are working to reduce plastic pollution through policy and education campaigns [8].

Several policies have been implemented worldwide to reduce plastic pollution around the world [9–12]. Here are some examples:

1. Plastic bag bans and fees: Many countries and cities have implemented bans or fees for single-use plastic bags. For example, in 2021, the European Union implemented a prohibition on disposable plastic bags, and many cities in the United States, such as San Francisco and Seattle, implemented fees or bans on plastic bags.
2. Extended producer responsibility (EPR) programs: EPR programs require manufacturers to take accountability for end-of-life management of crops, including plastics. This can include requirements for producers to collect and recycle their products or pay fees to support recycling and waste management programs.

3. Plastic bottle deposit programs: Many countries and states have implemented deposit programs for plastic bottles, where consumers pay a small deposit back when they purchase a bottle and can receive the deposit back when they return the bottle for recycling.
4. Plastic straw bans: Many cities and businesses have implemented bans on plastic straw, or switched to paper or biodegradable alternatives.
5. Microbead bans: Microbeads, which are small plastic particles used in personal care products, have been banned in several countries and states owing to their environmental impact.

These policies are just a few examples of many initiatives implemented to reduce plastic pollution. There is still much work to be done to address this issue, but these policies represent important steps towards a more sustainable future.

21.2 Degradation of Plastics

Plastic degradation refers to the process by which plastics break down over time into smaller pieces. There are two main types of plastic degradation: physical and chemical [13].

Physical degradation occurs when plastics are exposed to sunlight, wind, and other environmental factors, which cause them to break down into smaller pieces. This process is known as photodegradation, and it can lead to the formation of microplastics as small pieces of plastic that are less than five millimeters in size. Chemical degradation occurs when plastics are exposed to chemicals, such as acids and bases, which break down the chemical bonds in the plastic [14]. This process is known as chemical

degradation and can lead to the formation of microplastics. Although plastic degradation can occur naturally in the environment, it can take hundreds of years for plastics to fully degrade. Additionally, the degradation of plastics can release harmful chemicals into the environment, which can negatively impact wildlife and ecosystems [15]. Efforts are being made to develop more sustainable plastics that can biodegrade quickly and safely in the environment. This includes the development of bioplastics that are made from renewable resources and can be broken down by microorganisms in the environment. Additionally, initiatives such as recycling and reducing single-use plastics can aid in decreasing the quantity of polymeric waste that ends up in the ecosystem in the first place [16].

21.2.1 Chemical Degradation

The chemical degradation of plastics refers to the breakdown of plastic materials through chemical reactions. This process can occur because of exposure to a variety of factors such as heat, light, oxygen, and moisture. The chemical structure of plastics can be altered by these factors, which can cause the material to become brittle, discolored, or weakened [17]. As the chemical bonds within the plastic break, the material becomes more susceptible to further degradation. Chemical degradation can also result in the release of harmful chemicals into the environment. For example, when plastics containing additives, such as plasticizers, stabilizers, or flame retardants, degrade, these chemicals can be released into the environment, which can have negative impacts on wildlife and ecosystems.

Efforts are being made to develop more sustainable plastics that are less prone to chemical degradation and that release fewer harmful chemicals into the environment. This includes the development of bioplastics that are made from renewable resources and can be broken down by

microorganisms in the environment. Additionally, initiatives such as recycling and reducing single-use plastics reduce the amount of waste, which can help mitigate the negative effects of chemical degradation [18]. Chemical degradation of plastics can occur through a variety of mechanisms, depending on the type of plastic and environmental conditions. For example, some plastics may degrade owing to oxidation, which occurs when the plastic is exposed to oxygen in the environment. This can cause the plastic to become brittle and break down into smaller pieces.

Other plastics may degrade owing to hydrolysis, which occurs when the plastic is exposed to water or moisture. This can cause plastic to break down into its component parts, such as monomers or oligomers, which can be released into the environment. UV radiation is another factor contributing to the chemical degradation of plastics [19]. When plastics are exposed to sunlight, UV radiation can break down the chemical bonds within the plastic, causing it to become brittle and discolored. This can also result in the release of harmful substances into the ecosystem. The chemical degradation of plastics can have several negative environmental impacts. For example, when plastics break down into smaller fragments, they can cause harm or death in wildlife. Additionally, the release of harmful chemicals into the environment adversely affects the environment. To address these issues, efforts are underway to develop sustainable plastics that are less prone to chemical degradation and release fewer harmful chemicals into the environment. This includes the development of bioplastics and other environmentally friendly materials as well as initiatives to reduce plastic waste and promote recycling [20].

21.2.2 Photodegradation

Plastic photodegradation is a type of degradation that happens when plastics are exposed to sunlight or ultraviolet (UV) radiation. This process can cause plastics to break down into smaller pieces, eventually leading to the formation of microplastics. When plastics are exposed to sunlight or UV radiation, the energy from the radiation can break down the chemical bonds within the plastic, causing it to become brittle and discolored. Over time, this can cause the plastic to break down into smaller pieces that can be ingested by wildlife and enter the food chain [21].

Plastic photodegradation can occur with a variety of plastics including polyethylene, polypropylene, and polystyrene. However, some plastics are more prone to photodegradation than others, depending on factors such as their chemical composition and the amount of UV radiation to which they are exposed. Efforts have been made to address the issue of plastic photodegradation [22]. This includes the development of more sustainable plastics that are less prone to degradation as well as initiatives to reduce plastic waste and promote recycling. Additionally, some researchers have explored the use of additives that can make plastics more resistant to photodegradation, or that can break down plastics more quickly and safely in the environment [23].

One photocatalyst that has been extensively studied for plastic degradation is titanium dioxide (TiO_2). When exposed to UV radiation, TiO_2 can produce free radicals that break down the chemical bonds in plastics, causing them to degrade. This process can be enhanced by modifying the surface of TiO_2 with other compounds such as metal oxides or carbon materials [24]. Other photocatalysts studied for plastic degradation include zinc oxide (ZnO) and metal-organic frameworks (MOFs). These

materials have shown promise in laboratory studies, but more research is needed to determine their effectiveness and potential environmental impact. Although the use of photocatalysts for the degradation of plastic is in the experimental phase, it represents a promising area of research for addressing the problem of plastic pollution. However, it is important to note that photocatalysis is only one potential solution, and efforts are needed across a range of areas to reduce plastic waste and promote sustainability [25].

21.2.3 Biodegradation

Biodegradation is a procedure in which organisms such as bacteria and fungi break down organic materials, including plastics, into simpler compounds. Recently, there has been growing interest in the development of decomposable plastics as a possible remedy for the issue of plastic contamination [26–29]. Biodegradable plastics are designed to break down more quickly and safely in the environment than traditional plastics do. They are typically made from renewable resources such as plant-based materials and are designed to be broken down by microorganisms in the environment [30, 31].

Bioplastics and oxo-biodegradable plastics are the two main types of biodegradable plastics. Bioplastics are composed of renewable and natural resources such as corn starch, sugarcane, or cellulose. They can be broken down by microorganisms in the environment through a process called aerobic biodegradation. Oxo-biodegradable plastics, on the other hand, are made from traditional plastics modified with additives that help them break down more quickly [32]. When exposed to oxygen and UV radiation, these plastics may break down into minor parts, which might then be further broken down by microorganisms in the environment. While recyclable plastics represent a

potential solution to the challenge of plastic-based pollution, there are also some concerns about their use. For example, some biodegradable plastics may not break down as quickly or completely as advertised and may release harmful chemicals into the environment during the process. Additionally, some experts argue that recyclable plastics may not be a complete solution to the problem of plastic contamination and that efforts to reduce plastic waste through recycling and reducing single-use plastics may be more effective in the long run [33].

21.3 Enzyme-Based Degradation of Plastics

Enzymes play crucial roles in the degradation of plastics by microorganisms. These enzymes can be either extracellular, meaning they are secreted by the microorganism into the surrounding environment, or intracellular, meaning they are located within the microorganism cell. The degradation of plastics by microorganisms typically involves a process called biodegradation. During biodegradation, the polymer chains of the plastic are broken down into smaller molecules by the action of enzymes. These smaller molecules can then be assimilated by microorganisms for use as energy sources or building blocks for cellular processes [34]. The metabolic products produced during the degradation process, such as carbon dioxide, water, methane, and nitrogen, are released into the environment. This is the main procedure for reducing the influence of plastic pollution, as it helps to break down plastic into its component parts and prevent it from accumulating in the environment. Enzymes that can degrade various types of plastic materials have been isolated from various microorganisms, such as algae, actinomycetes, bacteria, and fungi [34].

For example, some bacteria, such as Pseudomonas, produce enzymes that can degrade polyethylene, one of the most commonly used plastics. Other bacteria, such as Bacillus and Streptomyces, can degrade polypropylene and polystyrene, respectively. Fungi, such as *Aspergillus* and *Trichoderma* are also known to produce enzymes that can degrade plastics. The discovery and characterization of these enzymes is an important step toward the development of more effective methods for biodegrading plastic waste. By studying the mechanisms by which these enzymes function, researchers can develop strategies to enhance their activity and make them more effective at breaking down plastic materials [35].

Various hydrolysates are used for the degradation of plastics, such as esterases, lipases, PETase, and proteases. These enzymes can target specific types of chemical bonds within the polymer chain depending on the type of plastic being degraded. These enzymes work in a similar way by breaking down the long carbon chains of the plastic polymer through hydrolysis and then taking up the resulting minor subunits into the microbe’s cell for additional enzymatic breakdown. This process ultimately leads to the release of metabolic products, as plastic waste is broken down by microorganisms [36].

21.3.1 Mechanism of Degradation

Hydrolases are involved in a wide range of biological processes, including food digestion, breakdown of cellular components, and metabolism of drugs and toxins. Hydrolases can target specific types of chemical bonds within a molecule depending on the type of enzyme and the substrate being acted upon. Examples of hydrolases include esterases, lipases, proteases, and nucleases [37].

The hydrolase enzyme facilitates the breaking of chemical bonds within the substrate using a molecule of water. This allows enzymes to break down large molecules into smaller subunits, which can then be further processed by the organism for energy or other cellular functions.

In the context of plastic biodegradation, hydrolases act on the polymer chains of plastic materials through hydrolysis, breaking the bonds between monomers that make up the polymer. This results in the breakdown of plastic into smaller subunits, which can then be further processed by microorganisms and ultimately released as metabolic products [38].

21.3.2 Esterase

Esterases catalyze the hydrolysis of ester bonds. Ester bonds are a type of chemical bond that link an alcohol molecule with a carboxylic acid molecule, creating an ester molecule. Esterases can break this bond using a molecule of water, resulting in the production of alcohol and carboxylic acids.

Esterases are found in a wide range of organisms including bacteria, fungi, plants, and animals. They play key roles in various biological processes such as lipid metabolism, toxin breakdown, and food digestion [39]. Esterases are important enzymes for the breakdown of polyethylene terephthalate (PET). PET is a common plastic used in packaging materials and can be difficult to degrade. However, esterases produced by certain bacteria, such as *Ideonella sakaiensis*, have been found to be capable of breaking down PET into its component parts, which can then be further processed by the microorganism. This discovery has led to the development of new strategies to manage plastic waste in the environment.

The structures of two genes from Roseateles depolymerans strain TB-87, which encode esterases capable of degrading aliphatic-aromatic copolyesters, were analyzed by Ahmad *et al.* [40]. Two open reading frames were predicted, one containing 1,083 bp nucleotides encoding esterase Est-H and another consisting of 870 bp nucleotides encoding esterase Est-L. A protein with 244 amino acids was identified in the intergenic regions. These genes were found to be polycistronically expressed in the promoter and terminator regions. Est-H and Est-L are homologous to plastic-degrading enzymes. However, Est-Ch was homologous.

The absence of specific esterase-encoding genes in the mutated strain led to the loss of its ability to create a distinct area on plates containing poly(butylenesuccinate-co-adipate) overlying nutrient agar. This research could play a role in the creation of new types of biodegradable plastics with distinctive features, such as monomers that can be recycled [40].

Noor *et al.* [41] showed that the *P. aeruginosa* strain S3 possesses considerable esterolytic activity for removal of polylactic acid) at 25 °C. A thorough investigation was carried out to optimize the physical and chemical factors for efficient production of esterase enzymes using response surface methodology. The ability of *Pseudomonas aeruginosa* strain S3 to degrade the polymer was confirmed in preliminary studies using SEM and FT-IR. The findings of this research highlight the considerable potential of the strain in the development of a highly effective system for the recycling of polylactic acid [41].

Various types of biodegradable polyester-based plastics have been tested for their ability to be degraded by microorganisms isolated from the soil samples. Strain TB-13 was found to be the most effective at breaking down

these plastics and was therefore chosen as the optimal strain for this purpose. Upon further analysis of its physical and genetic characteristics, it was determined that strain TB-13 was thoroughly associated to *Paenibacillus amylolyticus*. This strain can break down poly (ethylene succinate), polylactic acid, polybutylene succinate, and poly (caprolactone). However, these plastics cannot be used as sole sources of carbon. Both enzymes found in the culture medium can degrade plastics [42].

Microplastic pollution is a significant environmental issue and a promising approach to address this problem is the use of enzymes for degradation. MGS0156 is a hydrolase enzyme that has been identified in environmental metagenomes, and it has been found to be effective in breaking down various commercial plastics, such as poly lactide and poly caprolactone. To understand the enzymatic depolymerization mechanism, a combination of quantum mechanics, molecular dynamics, molecular mechanics, and molecular mechanism methods was used. Through analysis of the binding procedures of nine oligomers, it was observed that longer oligomers possess stronger binding energies [43].

21.3.3 Lipase

Lipases are a class of hydrolases that catalyze the hydrolysis of ester groups in lipids, and are a type of fat or oil. Lipases are found in a wide range of organisms including fungi, bacteria, animals, and plants. They are involved in several complicated biological processes, such as the digestion of food, metabolism of drugs and toxins in the body, and the synthesis and breakdown of cellular components. Lipases can also be used in various industrial applications, such as the production of biodiesel from vegetable oils and modification of fats and oils for use in food products. In the context of plastic biodegradation,

lipases produced by certain microorganisms can degrade various types of plastics [44]. The ability of lipases to break down these plastics is due to their ability to hydrolyze ester groups that hold the plastic chains together. This process results in the breakdown of plastic into smaller subunits, which can then be further processed by microorganisms and ultimately released as metabolic products.

Amin *et al.* [45] reported an *Aspergillus melleus* lipase for degradation of polymers. They found that under optimum operating conditions, lipase showed good catalytic capability for the biodegradation of the polymeric film. A lipase derived from *A. melleus* showed great potential as an applicant for enzymatic remediation of waste because of its ability to function effectively in extensive pH ranges, maintain stability in the presence of various solvent systems, and effectively degrade poly(ε-caprolactone) [45].

Sun *et al.* [46] investigated the conversion of phthalate esters to the related monomers by the action of carboxylesterase and lipase enzymes. The results showed that phthalate esters containing long alkyl chains were difficult to degrade. However, the short alkyl chains were comparatively easily biodegraded by the enzyme-based catalytic system. According to these findings, the length of the alkyl chain significantly affects the enzymatic breakdown of phthalate esters [46].

A lipase from *Amycolatopsis mediteranei* was cloned and investigated for potential use in process biotechnology. The three-dimensional configuration of the enzyme was determined through comparative modeling. Based on the specific characteristics of the substrate and modeling studies, it was determined that *A. mediteranei* lipase quickly broke down aliphatic plastics, such as poly(1,4-butylenesuccinate) and poly(ε-caprolactone). These results

suggest that the enzymes can break down a variety of microplastics than previously thought [47].

Lipases produced by *A. niger* were isolated by chromatography. The lipase enzyme demonstrated the highest level of activity at 37 °C and pH 7.0, and was able to maintain its stability in a pH range of 4.0–10.0 at 50 °C. When olive oil was used, the values for V_{max} and K_m were determined to be 32.21 µmol/min/mg and 3.83 mM, respectively. *Pichia pastoris* was used for lipase expression. Both poly (ε-caprolactone) and polylactic acid were successfully degraded using recombinant lipase. This study provides a foundation for further research on the exploitation of this lipase in polymer degradation [48].

Candida rugosa lipase was utilized to break down solid polyurethane, which is commercially available in an aqueous environment, with varying concentrations of substrate and enzyme, pH, and temperature. A theoretical study was conducted and used to describe the degradation kinetics of polyurethane. The optimum conditions for the reaction were identified a pH 7 and 35 °C. Moreover, diethylene glycol was generated as a by-product of degradation process with a rate of 0.12 mg l^{-1} min^{-1}. In addition, the activation energy of the reaction was estimated to be 9.121 kcal/gmol K [49].

The effectiveness of commercial lipases in degrading aliphatic polyester films was investigated. After immersing the polymers in a solution of lipase for 100 days at pH 7.0 and 37 °C, *Chromobacterium viscosum* lipase was able to degrade synthetic polymers. However, *Rhizopus niveus* lipase was able to degrade succinate-adipate and poly(ε-caprolactone) within two weeks. These outcomes may be attributed to the serial reactions of chemically assisted hydrolysis of synthetic polymers under harsh conditions, followed by an enzymatic process [50].

21.3.4 Cutinase

Cutinases are a class of hydrolase enzymes that can break down cutin, a waxy polymer that is a major component of the cuticle and the protective layer on the surface of plant leaves and stems. In addition to their natural role in degrading cutin, cutinases can break down various types of synthetic polymers, including polyethylene terephthalate, which is a common plastic used in packaging materials. This discovery has led to an increased interest in the potential of cutinases for biodegrading plastic waste [51].

Similar to other hydrolases, cutinases catalyze the hydrolysis reaction, in which a molecule of water is used to break the chemical bonds within a molecule. Cutinases can target ester bonds within the polymer chains of cutin and other polymers, breaking them down into smaller subunits that can be further processed by microorganisms. Research on the mechanisms of cutinase activity and the development of more effective cutinases are ongoing, with the ultimate goal of developing more efficient and sustainable methods for managing plastic waste in the environment [52].

Cutinases exhibit potential as effective agents in the bio-recycling of poly (ethylene terephthalate) because of their ability to produce monomers at a high rate and with high efficiency using gentle and mild reaction conditions. Furukawa *et al*. [53] used cationic surfactant to attract cutinases isolated from *Thermobifida fusca* near the low-crystallinity poly (ethylene terephthalate) film surface *via* electrostatic interactions.

Liu *et al*. [54] reported removal of poly(ε-caprolactone) using a dual-functional cutinase and lipase catalytic system.

The lipase obtained from yeast showed a distant similarity to proteins in the cutinase group, which was higher than

that of lipases. However, the enzyme was able to efficiently break down heavy compounds such as poly (ε-caprolactone), polylactic acid, and poly(3-hydroxybutyrate) [55].

Cutinases have been utilized in various reactions, including ester hydrolysis. To broaden the potential use of cutinases in the biotechnology industry for the breakdown of synthetic polyesters, a study was conducted to compare the performance and durability of five different cutinases. These cutinases were isolated from *Humicola insolens*, *Alternaria brassicicola*, *Fusarium solani*, *Aspergillus oryzae*, and *Aspergillus fumigatus*, which are well characterized. *H. insolens* cutinase demonstrated the highest level of hydrolysis of poly(ε-caprolactone). These findings provide useful insights for engineering enzymes with enhanced stability and function [56].

F. solani cutinase expressed in *P. pastoris*, resulting in the production of an engineered cutinase. The enzyme was found to be capable of degrading poly (butylene succinate) with an optimal pH of 8.0 and an optimal temperature of 50 °C. Its activity was enhanced in the presence of potassium and sodium ions, and inhibited by zinc, iron, manganese, and cobalt ions. The effects of diverse substances on enzyme activity were also studied, and mercaptoethanol and EDTA were found to have major inhibitory effects. The study further investigated the enzymatic biodegradation of polymeric films using recombinant enzymes, and it was observed that complete degradation occurred after approximately 6 h. The characteristics of the poly (butylene succinate) films after degradation were analyzed, revealing that the crystalline and amorphous areas of the polymer were biodegraded by the engineered enzymes. X-ray diffraction analysis showed that the crystallinity of the polymer decreased in the presence of the enzyme.

Additionally, gel permeation chromatography showed no significant variation in the molecular weight of poly (butylene succinate) before and after degradation [57].

21.4 Conclusion

Synthetic polymer degradation has been a promising research area for addressing polymer pollution in the ecosystem. The identification and characterization of microorganisms and enzymes capable of degrading these synthetic polymers may offer insights to the underlying mechanisms of biodegradation and lead to the development of novel approaches for plastic waste management. Genetic engineering methods to produce recombinant enzymes that enhance biodegradation of plastic waste could be a promising strategy. This approach could lead to the development of more sustainable and effective methods for plastic waste management, contributing to cleaner and healthier environments.

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22
Remediation Strategies for Micro(Nano)Plastics

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Abstract

Microplastics and nanoplastics are emerging pollutants that are dispersed in all terrestrial and aquatic environments, can be toxic to some species, and have an affinity with other hydrophobic pollutants and low biodegradation rates. Therefore, it is necessary to implement remediation and containment strategies to reduce its presence in the environment. This chapter reviews the physical, chemical, and biological methodologies currently being used, and discusses the advantages and disadvantages of each of them are exposed. Some of the most relevant findings were that most of the MP distributed in different matrices were impossible to remove because they are economically, technically, and energetically unfeasible; therefore, they would have to wait for natural attenuation processes. Another relevant point is the need to establish protocols for removing nanoplastics, because they are not considered because of their size.

***Keywords*:** Biodegradation, fungus, aluminum, biosolid, fenton, coagulants, flocculants

22.1 Introduction

Plastic pollution has become a focus of interest for researchers, governments, civil associations, and non-governmental organizations because it is present in different ecosystems worldwide. It has been estimated that if measures are not implemented to prevent or minimize plastic pollution, by 2040, plastic leakage into the ocean will triple by an average of 29 million metric tons per year (interval of 23 to 37 million metric tons per year), which is equivalent to 50 kg of plastic per meter of coastline worldwide [1]. The high presence of plastic waste in the environment is linked to different factors, including population growth, current consumption patterns that focus on the use of plastic items with a short useful life (known as single-use plastics), high production of plastic items to meet human needs, and inadequate management once plastic items reach the end of their useful life.

The production of plastics, year after year, is increasing; in 1950, worldwide, approximately 2 million tons of plastic were produced [2], but by 2021, this figure increased almost 196 times, reaching 390.7 million tons per year [3]. Moreover, it has been estimated that, from 1950 to 2019, the cumulative production of plastics reached 9,500 million tons, equivalent to more than one ton of plastic for each person alive today [2]. Of all the plastics produced in 2021, 44% corresponded to the packaging applications classified as single-use plastics because they were designed to be used and discarded immediately [3].

Within plastic waste, microplastics (MP) have stood out, and plastic particles smaller than or equal to 5 mm facilitate their transport, distribution, and ingestion by fauna, making their removal from the environment difficult. These particles are classified into primary ones, intentionally manufactured, of the size mentioned above for use in industrial applications, and secondary ones, which originate from the wear or fragmentation of larger plastic waste. Secondary MP have the most significant presence in the

environment; however, primary MP have received more public attention because they can be legally regulated [4]. Some estimates indicate that 3 million tons of MP leak into the environment annually, and that by 2050, approximately 22 million tons of microfibers from synthetic textiles will enter the environment [5].

Microplastics have been documented in various ecosystems, including aquatic [6], terrestrial, soil [4], coastal, and marine environments [7]. The presence of MP in the different ecosystems poses a faunal risk because they can ingest them by confusing them with their food; the flora; ecosystem services by causing accumulations in bodies of water or the soil and hindering oxygenation or the passage of sunlight; the economy of tourist areas and the population, since in the first instance, they cause contamination and damage to the scenic landscape and people are exposed to MP through the consumption of foods that contain them.

The smaller the MP, the more difficult it is to remove it from the environment because it is hardly visible to the human eye. The greatest challenge is to use effective techniques that allow them to be removed from the environment without affecting the environments themselves and what surrounds them. Some technologies that have become known worldwide for the removal of plastic waste are the placement of barriers for retention and subsequent removal, as well as trawling nets towed to ships, both for use in ocean water [8], while for river water, there is an interceptor with a conveyor belt that directs the waste towards the boat [9]. However, some disadvantages of these technologies are that their manufacture is expensive, so they are not accessible to all countries or locations, their operation can put biota at risk, and they only focus on removing larger plastic waste on the surface.

Therefore, it is essential to implement other reliable, robust, and replicable methods at an affordable cost that do not involve long periods and do not put the population and environment at risk. In this sense, this work aims to analyze and compare different methods (physical, chemical, and biological) to remove microplastics through relevant information such as their characteristics, conditions, type of MP that can be removed, costs, and efficiency, among others. This work will generate relevant information so that, when seeking to implement a method, there will be a baseline from which to start.

22.2 Methods for the Removal of Micro(Nano)Plastics from the Environment

The following sections explore the different physical, chemical, and biological removal methods of microplastics in different matrices. Physical methods include filtration and adsorption, while chemical methods include coagulation, flocculation, chemical degradation, and advanced oxidation processes. Each section mentions the most important findings of different investigations, including the most relevant characteristics and efficiencies of each method.

22.2.1 Physical Removal Methods

22.2.1.1 Microplastics Adsorption

Adsorption is a physical treatment process with very few records within the scope of this review, accounting for six articles that approach the use of different materials at a laboratory scale.

Table 22.1 shows the materials tested in the literature, with their main characteristics. A common way to present results is in terms of removal efficiency percentages and maximum absorption capacity in units of mass of adsorbed pollutant (mg) per mass of used adsorbent (g). Up to 100 % removal and adsorption capacities up to 29.9 mg/g were

reported. Granular activated carbon (GAC) was obtained commercially, whereas the rest of the adsorbent materials were prepared using different methods to give them the desired properties, where an oat protein-based sponge (OPS) was the most innovative. The most reported characteristics were superficial area, pore volume, and size.

22.2.1.2 Microplastics Used in Adsorption Removal Experiments

The treated MP solutions were prepared by diluting micro- and nanoplastics (mainly PS) in ultrapure water or by directly adding commercially acquired suspensions to the adsorbent suspensions.

Many studies have used PS microspheres for their experiments, especially fluorescent ones because they facilitate the visualization and quantification of the removed MP. However, they are not representative of MP found in nature, where they are exposed to weathering processes that modify their chemical properties, which compromises their perspective on removal mechanisms. MP can receive chemical treatment to simulate the usual degradation of plastics, for example, subjecting the MP to a potassium persulfate hot bath for several days according to the degradation sought. The reviewed study in this chapter was conducted for 0.5 and 1.0 days due to the rapid chemical degradation and slow weathering process within nature [14].

22.2.1.3 Influence of Morphology in Microplastic Removal

The morphology of the absorbent influences the physical adsorption of the contaminant; the roughness, porosity, and irregularity of the adsorbent material facilitate the retention of MP between the pores. Although spruce bark has a smaller surface area than pine bark biochar, probably because of its high ash content, it presented particularly spaced pores that favored the reaction, perhaps because of its longer activation time or the greater steam ratio it was prepared with [13]. The oat protein-based sponge (OPS) presented a highly porous and interconnected structure that provided fast adsorption of MP, adsorbing 38% within 10 s and maintaining this property for up to 20 cycles; the mechanical properties were also preserved up to 90% after 100 cycles, which is very close fatigue resistance to that of polyurethane sponges [15].

Table 22.1 Adsorbents reported in the reviewed literature: characteristics and removal efficiencies.

Material	MP types	Characteristics	Maximum adsorption capacity	Removal efficiency	Reference
Granular activated carbon	Amidine PSNP spheres (90 nm)	Granular coconut shell-based activated carbon: surface area 1,150 m^2/g, mean particle diameter 1.4 mm, iodine number min. 1,100, hardness 98%, and total ash máx. 4%	2.2 mg/g (Ultrapure water) 6.33 mg/g (Lake Geneva water)	26%–98% (Ultrapure water) 78%–90% (Lake Geneva water)	[10]
CuNi carbon material	PSNP sphere (100 nm)	Prepared by hydrothermal method, using $Cu(NO_3)_2$, $C_9H_6O_6$, CTAB $C_{19}H_{42}NBr$, $Ni(NO_3)_2$, C_6H_{14}: surface area 110.48 m^2/g, average pore width 8.25 nm and micropore volume 0.02083 cm^3/g	NR	32.72%–99.18%	[11]
Magnetic zeolite adsorbent	PSNP spheres	Prepared by co-precipitation, using Fe_2 and Fe_3: surface area 165.88 m^2/g, pore volume 0.165 cm^3/g, and average pore size 3.98 nm	29.9 mg/g	Up to 85%	[12]
Activated scots pine bark carbon	PE spheres (10 µm) and cylinders (2–3 mm), and fleece shirt fibers	Prepared by slow pyrolysis: surface area 556 m^2/g, pore volume 0.206 cm^3/g, 86.9% micropores (<2 nm), 13.1% mesopores (2–50 nm) Surface area 603 m^2/g, pore volume 0.240 cm^3/g, 79.6% micropores (<2 nm), and 20.4% mesopores (2–50 nm)	NR	Up to 100%	[13]
Activated spruce bark carbon	PE spheres (10 µm), cylinders and pieces (2–3 mm),	Prepared by slow pyrolysis: surface area 187 m^2/g, pore volume 0.071 cm^3/g, 85.4 % micropores (<2 nm), 10.4 mesopores (2–50	NR	Up to 100%	[13]

Material	MP types	Characteristics	Maximum adsorption capacity	Removal efficiency	Reference
	and fleece shirt fibers	nm), and 4.2% macropores (>50 nm)			
Magnetic corncob biochar	PA powder (27–307 µm)	Prepared with a mixture of $Fe(NO_3)_3$, $FeSO_4$: surface area 27.82 m^2/g	NR	Up to 25% (pristine PA) Up to 75% (aged PA)	[14]
Oat proteinbased sponge	PS (1 µm)	Prepared through chemical crosslink method with oat protein isolate, NaCl, and epichlorohydrin: strength 176 kPa, Youngs Modulus 60.1 kPa	NR	Up to 81.2%	[15]

PSNP, polystyrene nanoplastics; PE, polyethylene; PA, polyamide; NR, not reported.

22.2.1.4 Influence of Concentration of Adsorbent and Microplastics on Removal

It has been reported that PS adsorption increases when the adsorbent concentration decreases, in the case of GAC [10], while increasing the dose of CuNi carbon material to remove PS and MCCBC to remove PA also increases the adsorption of the plastic particles, probably because more adsorption sites are provided; however, at the beginning of the reaction, the efficiency increases rapidly, due to physical adsorption. After more than 11 h, the adsorption slows down because the adsorption sites are continuously occupied while the concentration of nanoplastics is constantly decreasing; at this point, the predominant adsorption mechanism is chemisorption [11, 14].

An innovative discovery is the affinity of MCCBC for aged PA, which has been attributed to the fact that, derived from the degradation process, MP decrease in size, decrease in crystallinity, and increase in functional groups that contain oxygen, which facilitates its adsorption and suggests that this method can be used to remove MP that have undergone weathering in the environment [14].

22.2.1.5 Influence of Electrostatic Interactions on Microplastic Removal

Electrostatic attraction is one of the main adsorption mechanisms between plastic particles and the adsorbent surface; magnetic zeolite (MZ) was reported to be negatively charged, while the treated NPs were positively charged [10]. However, different factors influence this property, such as the pH, on which the charge distribution in the adsorbent material depends. The removal efficiency was generally higher in acidic solutions of CuNi carbon [11] and MZ. Moreover, ZM and MCCBC perform well in neutral solutions and have many active sites, heterogeneous charges, and iron oxide components on their surface, which trap particles by electrostatic attraction [12, 14]. About MCCBC, the removal of pristine PA was not affected by pH; nonetheless, the best removal efficiencies for both this type of MP and aged PA were obtained with neutral pH; in the case of the second MP, both the charge of the adsorbent material and of the plastic were negative, but the predominant process was not electrostatic attractions but rather hydrophobic interactions. The OPS's adsorption mechanisms are also hydrophobic interactions; aside from intraparticular diffusion, its MP adsorption capacity is enhanced by the numerous active side chains of the amino acid residues [15].

22.2.1.6 Influence of Ions on Microplastic Removal

The ions present in the water can also interfere with the adsorption of MP. In the study of CuNi carbon, the content of OH^- ions is higher in alkaline conditions so that it could compete with PS for the adsorption sites in the material. In addition, in the presence of HCO_3 in the solution, a significant decrease in the removal efficiency was recorded, which suggests that it hydrolyzes, generating OH^- anions that change the zeta potential of the PS surface. The high concentration of electrolytes results in the agglomeration of the adsorbent and a decrease in the number of the active sites [11]. The same situation occurs for MCCBC when the concentration of CaCl2 increases, in addition to the fact that Ca^{2+} competes for available adsorption sites [14]. For MZ, the anions interfere with the reaction in the order $CO_3^{2-} > PO_4^{3-} > NO_3^- > Cl^-$ [12].

22.2.1.7 Influence of Dissolved Organic Matter on Microplastic Removal

Lake water contains dissolved organic matter (DOM), so its effect on the adsorption process was explored in Lake Geneva water, in which these elements improved the adsorption capacity because the negative charge of DOM changes the stability of positive PS particles upon neutralization, and the presence of divalent ions allows the binding of PS–DOM complexes and aggregation of PS particles [10]. On the other hand, DOM decreased MP removal with MZ because at a higher concentration, it also caused a negative charge, resulting in repulsion forces [12].

22.2.1.8 Reaction Kinetics in Microplastic Removal

The reaction was adjusted to a Langmuir isotherm for almost all cases, which means that monolayer coverage predominates in a uniform adsorbent structure, and that the reaction is was spontaneous and endothermic and is also responsible for the binding of PS to the adsorbents during complexation [11, 12]. In the case of Lake Geneva water, the Langmuir and Freundlich isotherms did not fit the reaction; the adsorption rate was controlled by chemisorption, aggregation, and surface charge of PS [10].

22.2.1.9 Adsorption as a Feasible Option for Microplastic Removal

Because of the common application of adsorption treatment for drinking water and wastewater, this method is presented as a convenient, practical, and easily applicable method for removing MP. Furthermore, its application is economically feasible, utilizing conventional materials such as high lignocellulosic organic waste, and some materials are biodegradable, such as the OPS adsorbent; these characteristics also make the process environmentally friendly.

22.2.2 Filtration Removal

Filtration is a typical operation during traditional water treatment, removing macro- and micropollutants, such as suspended solids and pathogens [16]. Thus, it has been explored in different scientific studies as a low-cost, accessible form of microplastic pollution treatment in wastewater of different kinds.

According to Martín-García *et al.* (2022), the main removal mechanism for filtration is size exclusion, which is achieved through mechanical retention or random contact. However, separation also depends on the characteristics of the filtering media, filtration speed, grain size of the filter, particle size, and characteristics of the wastewater being treated. Flow rates can also affect the surface area required for treatment. Microplastics can be transported to the surface of the filtering material and removed by mechanisms such as inertial impact, interception, adhesion, and sedimentation. Smaller particles, such as those in the micron range, may be retained by diffusion. Brownian motion also affects microplastics, which gives them random movement that eventually leads them to collide

with porous media. Other processes, such as chemical and physical adsorption, flocculation, and biological growth within the pore, can play a role in MP removal.

Li *et al.* (2023) also explore the governing mechanisms of nanoplastic (NP) removal by filter media. Their study indicates that pH could control NP retention; the higher this parameter, the more increase in mobility, and the effects of cation screening and bridging can also make colloid retention easier in the porous media.

22.2.2.1 Sand Filtration

Table 22.2 summarizes the sand filtration methods reported in the reviewed literature, describing the system characteristics and MP (size, type, and chemical composition) and removal efficiencies for different configurations and water types.

Sand filtration is a common process in wastewater treatment plants; hence, it is important to test its applicability for removing microplastics. As observed in Table 22.1, within the reviewed articles, siliceous sand is the most frequent material employed for MP removal in the literature, reporting efficiencies of 35.6%–95.3% for urban wastewater (UWW) treatment plants, 25.41%–96.79% for artificial wastewater (AWW), 47.39% for highway stormwater (HSW), and 49%–100% for plastic recycling industry wastewater (PIRWW). The lower removal efficiencies in UWW were attributed to the accumulation of suspended solids in the granular media [16], the small size and morphology of fibers that facilitated their longitudinal passage through the filtering material [19], water turbidity, water flowrates, and operating conditions of the plant [17]. In the case of AWW, increasing the flow velocity was not favorable [18], and for HSW, high peak flows probably contributed to the higher mobility of smaller MP, such as those in the size range of 20–100 µm [21].

Sand amendment materials, such as granular limestone, coke, and vegetation, have also been explored, with efficiencies of up to 99.65% [18], 94.0% [17], and 88.48% [21], respectively, noticeably enhancing MP removal.

Some materials have also been studied for use as filter media. In the case of coke, its efficiency was slightly lower than that when combined with sand [17]. In contrast, spent activated carbon, which was thought to remove MP by size exclusion, showed no significant removal [16].

Microplastic sizes in samples reported in the reviewed literature for sand filtration were in the range of 25–6,300 µm in wastewater, with the most common MP chemical composition being PE and PP. Microplastics were significantly smaller. In addition to the abundance of PP and PE in this water type, black EPDM and EVA -materials for the construction of vehicle infrastructure [21], were also present in highway stormwater (20–100 µm).

Only two of the seven reviewed articles on sand filtration reported MP types, and in both cases, fibers and fragments were most common in the inflow and outflow. Martín-García *et al.* (2022) indicated an increase in fragments as the water flow also increased, probably because of the progressive clogging of the porous media that, at the same time, could reduce adsorption by surface charge and polarity, leading particles to escape through larger pores and decrease the frequency of mechanical retention through inertial impact, interception, or sedimentation of smaller particles.

Table 22.2 Sand filtration in literature for MP removal,

Materials	Treated waters	Characteristics	Flow rate	MP size	MP types	MP chemical composition
Siliceous sand (SS) and spent activated carbon (SAC)	Urban wastewater (UWW)	Two 20 cm gravel layers for SS and SAC media Media filtering area 0.07 m^2 Bottom nozzles for backwashing activated when head loss reached 80% max. with air pulses at 95 Nm^3 $m^{-2}{\cdot}h^{-1}$ and water at 40–45 Nm^3 $m^{-2}{\cdot}h^{-1}$	14.3–42.9 $m^3{\cdot}m^{-2}{\cdot}h^{-1}$	Inflow: 25 to >500 µm	–	Inflow: PP and PE
Siliceous sand	Urban wastewater	Three RSF columns in parallel Effective grain size 0.9 mm Effective surface 29.96 m^2 Allowed filtered water for backwashing 1.5%	Hydraulic loading rate 6.68 $m{\cdot}s^{-1}$	Inflow: 210 µm–6.3 mm	–	Inflow: LDPE, HDPE, AC
Siliceous sand (SS)-granular limestone amended sand (GLAS)	Artificial secondary treated wastewater (AWW)	Quartz sand sizes 0.60–0.80 mm and 0.3–425 mm Granular limestone size 0.4–0.7 mm, thickness 2.5, 5, and 7 cm Acrylic columns 16.2 in length and 2.5 cm inner diameter Stainless steel screens pore size of 50 µm to seal the inlet and outlet. Peristaltic pump	3, 7, and 21 $ml{\cdot}min^{-1}$	100 nm	Spheres	PS
Siliceous sand (SS) and coke (C)	Urban wastewater	Stainless steel columns, with 2 m in height and 0.36 m diameter	300, 500, and 1,000 $L{\cdot}m^{-2}{\cdot}d^{-1}$	>100 µm	Inlet: Fibers, fragments, filaments,	PE, PES, PET

Materials	Treated waters	Characteristics	Flow rate	MP size	MP types	MP chemical composition
		Gravel base Silica sand particle size 0.25 mm Coke particle size 0.2–0.9 mm			flakes Outlets: Fibers, fragments, flakes filaments	
Sand	Plastic recycling industry wastewater (PRIW), both untreated (U) and treated (T) by an on-site sand column	Two sand-filter columns (inner diameter 2.4 cm); porous material was washed with water and burnt at 500 °C, size 0.7–1.4 mm, depth 16 cm	7.24 $cm^3 \cdot min^{-1}$	30–5,000 µm	Pretreatment: Fragments, fibers Sand treated: Fragments and fibers	Pretreatment PRIW-U: PE, unidentified polymer, PP Pretreatment PRIW-T: PP, PE, EPS, Sand-treated PRIW-U and PRIW-T: significant reduction of all polymers
Sand and biochar (slow and fast filtration)	Microplastic suspension	Plexiglass column (length 10 cm and 2 cm inner diameter) Sand 0.30-0.45 mm Biochar (lignin, cellulose, woodchip) layer in the middle of columns at 0.1 or 1.0 wt%	4 or 260 m/d (up-flow)			
Sand (S) filter and Vegetated sand filter (VS)	Highway stormwater	Gross pollutant trap (GPT) Sand according to recommendations for bioretention design according to the DWA (German Association for Water, Wastewater and Waste) Top gravel layer Top turf layer 30 mm deep for VS	Depending on rainfall. GPT-VS: 0.3–7.9 L/s GPT-S:1–10 L/s	20–100 µm	–	Inlet: PP, black EPDM, EVA GPT-V: PP, PE, PS GPT-S: PP, black EPDM, PE

PP, polypropylene; PE, polyethylene; LDPE, low density polyethylene; HDPLE, high density polyethylene; AC, acrylic; PS, polystyrene; PES, polyester; PET, polyethylene terephthalate; EPS, expanded polystyrene; EPDM, ethylene

propylene diene monomer.

22.2.2.2 Ultrafiltration

Ultrafiltration was also tested for MP removal, as shown in Table 22.3, in which the systems, MP characteristics, and removal efficiencies are described.

Ultrafiltration in drinking water (DW) and UWW treatment resulted in efficiencies of up to 99.9 100%. Yang *et al.* (2023) found the best removal rates for smaller pore sizes and lower MP concentrations, with more prominent MP preferably retained and smaller plastics separated from them under the forces of water flow and interception, and suggested that transport pipes in treatment processes and external instruments may cause secondary MP contamination of water. González-Camejo *et al.* (2023), despite complete MP removal in UWW and artificial plastic industry wastewater produced during PET and HDPE recycling, found ultrafiltration to be best for the latter, not only because it allows plastic recovery, but also because organic particles in UWW have been found to increase membrane fouling and consequently increase energy consumption and costs.

Slightly lower removal was reported by Yang *et al.* (2023) for treated shellfish seawater (69%–96%) with on-site ultrafiltration pre-existent equipment. For this reason, they proposed that membrane aging could reduce the MP removal capacity.

As shown in Table 22.4, the MP sizes in the inflow are larger than those in the outflow, where they decrease in size, which could be due to their breakage during treatment. PE and PP were the most common chemical components of MP removal-treated waters.

22.2.2.3 Membrane Filtration

Within the reviewed literature, the most widely studied method for MP removal was membrane treatment, especially those fabricated with polymer materials such as polyvinylidene fluoride (PVDF), polyethylene terephthalate (PET), polycarbonates (PC), polytetrafluoroethylene (PTFE), polyester (PEST), and cellulose acetate (CA), as shown in Table 22.3, mainly at the laboratory scale, testing removal primarily for plastics such as polystyrene spheres (PS) and polyamide fibers (PA), registering efficiencies of 92.7 to >99%.

According to Yahyanezhad *et al.* (2021), who studied domestic and commercial wastewater, the most critical factors for MP removal using membranes are membrane shape, pore size, polymer type, and fouling persistence. They also observed that MP concentration affected efficiency because with a higher amount of MP in the influent, the removal efficiency declined from 100% with 81 MP/L to 98.1% with 106 MP/L [23].

Electrostatic charges also play an essential role, as proposed by Pizzichetti *et al.* (2021), who reported that most of the commercially available membranes are negatively charged, while plastic is a hydrophobic material that enhances the adsorption of hydroxide ions that simultaneously affects the particle charge depending on its size and can induce electrostatic interactions between the charged particle and membrane [24]. Lee *et al.* (2023) used an electrokinetic-assisted membrane to enhance MP removal, creating an electric field with uniform direction and intensity throughout the channel width by using a CEM membrane with micro holes and suppressing electroconvection with a microstructure as a physical filter sieve. It was found that the larger the zeta potential, the lower the voltage required to reach peak efficiency. Removal of more than 99.9% was achieved with critical fluxes of 8,000, 10,000, and 14,000 L m^{-2} h^{-1} when zeta potentials were -12.5 ± 0.79, -25.7 ± 0.82, and -6.29 ± 0.98 mV, respectively. This study suggests that MP removal does not depend on MP type, but instead on their zeta potentials, and low zeta potentials can be resolved by adjusting the pH conditions of the feed [30].

Table 22.3 Ultrafiltration in literature for MP removal.

Materials	Treated waters	Characteristics	MP size	MP shapes	Common MP types	Removal efficiency
Ultrafiltration	Urban wastewater and drinking water	Unspecified	DWWTP Inflow: 250–>600 µm Outflow: 80–200 WWTP Inflow: 150–400 µm Outflow: 80–200 µM	–	Inflow and outflow UW: PE, PP, PS Inflow and outflow DW: PE, PP, PA	UWWTP: 87.8%–99.8% DW: 82.3%–99.9%
Hollow fiber bundle (KMS Puron® Koch Membrane Systems), pore size 0.03 µm	Urban wastewater (UWW) and artificial plastic industry wastewater produced during PET (PRIWPET) and HDPE recycling (PRIW-HDPE)	Two 2.5 m high membrane tanks Membrane area 3.4 m^2 50 L clean-in-place tank Blower with air demand 0.1–1.2 $Nm^3\ h^{-1}$ Transmembrane flux 20–50 LMH Recirculation factor 2	100–300 µm	Pretreatment: Fragments, fibers	Pretreatment PRIW-U: PE, unidentified polymer, PP	UWW: 100% PRIW-PE and PRIW-HDPE: 100%
Semi-industrial scale ultrafiltration and industrial scale ultrafiltration	Treated shellfish seawater	Two 220 kDa UF modules: a semiindustrial UF membrane area of 9 m^2 and an industrial scale UF membrane area of 240 m^2	Inflow: 500–600 µm SISUF: 150–600 µm ISUF: 80–400 µm	–	PP, PE, PA	69%–96%

PP, polypropylene; PE, polyethylene; PS, polystyrene; PA, polyamides.

Table 22.4 Membrane filtration in literature for MP removal.

Materials	Treated waters	Characteristics	MP size	MP shapes	Common MP types	Removal efficienc
Microfiltration membrane (PVDF + PET)	Wastewater from domestic and limited commercial sources	Pore size 0.1 µm Intermittent filtration/aeration system Depth 60 cm, length 30 cm, width 28 cm Backwash flush 50–85 LMH Working volume capacity 50 L	Influent: 1-5,000 µm (6%–14%) Effluent: <0.1 µm in the	Influent: fibers 34%, pellets 22%, fragments 38%	–	98.1%–100%
PC membrane pore size 5 µm	Suspensions of PA and PS	Zeta potential -11.5 Mv at pH 7.4 Hydrophobic Shore hardness 71.0 ± 2.7	Influent: 20–300 µm Effluent: 10–100 µm	Fibers and spheres	PA and PS*	96.8 ± 4.5–99.6 ± 0.4%
CA membrane pore size 5 µm	Suspensions of PA and PS	Zeta potential -35 Mv at pH 6 Hydrophilic Shore hardness 62.7 ± 0.6	Influent: 20–300 µm Effluent: 7–70 µm	Fibers and spheres	PA and PS*	94.3 ± 5.1–99.8 ± 0.1
PTFE membrane pore size 5 µm	Suspensions of PA and PS	Hydrophobic Shore hardness 76.3 ± 3.2	Influent: 20–300 µm Effluent: 8–220 µm	Fibers and spheres	PA and PS*	96.0 ± 4.6–99.6 ± 0.3
Nanofibrous membrane (made with PVDF-QA dope solution)	Synthetic lake waters spiked with PS	Glass sand core filtration apparatus Water flux 109 $Lm^{-2}h^{-1}$	50–1,900 µm	Spheres	PS*	92.7%
Biomimetic gill-inspired membranes (PVDF-$DMSO_2$) Average pore size 10.5 µm	Suspension of PS	One filtration cell Active area 2 × 10^{-4} m^2 Oblique pore angle 25° Flow rate 0.51 m/s in reversed direction	700 nm	Spheres	PS*	97.6%
Steel magnetic filter mesh size 3 mm	Suspension of PVC	Steel permanent magnet, remanence 1.39 T, coercive force	48–374 µm	–	PVC*	74.01%–100%

Materials	Treated waters	Characteristics	MP size	MP shapes	Common MP types	Removal efficienc
		10,500 Oe, and size 50 mm × 50 mm × 50 mm Magnetic matrices box with surrounding magnetic intensity of magnetic matrices 4–8 T Magnetic medium (Fe^3O^4) Reaction times: 5, 10, 15, and 20 min				
Ceramic membrane (Al_2O_3) pore size 0.1 µm	Synthetic wastewater	Hydrophilic Effective surface area 13.1 cm^2 Peristaltic pump pressure 1 bar	1 µm	Spheres and other nonspecified shapes	PS and PE*	99% (in terms of turbidity)
Holey Ti_3C_2 nanosheets based membranes pore size 25 nm	Suspension of PS	Surface area 51.32 m^2/g Water flux 196.7 L m^{-2} h^{-1} kPa^{-1}	80, 100, 200, 500, and 1,000 µm	Spheres	PS*	>99.3%
Electrokinetic-assisted membrane filtration (PEST/CEM + electric field) pore size 82.6 µm	Suspension of PE, PS, PP, PEST, acrylic, and nylon	A system with a main channel and buffer channels, with cation exchange membranes in between (CEMs with microholes) and a branch channel in the middle of the main channel for continuous MP flush repelled from the electric force barrier. Flux 8000 and 10,000 m^{-2} h^{-1} Voltage of 150 V and 200 V	10–85 µm	Fragments, microspheres, and fibers (no specific order)	PP, PE, PS, PEST, acrylic, nylon	>99.9%

PTFE, polytetrafluoroethylene; PC, Polycarbonate; CA, cellulose acetate; PVDF, polyvinylidene fluoride; QA, quaternary-ammonium; PEST, polyester; CEM, cation exchange membrane; PP, polypropylene; PE, polyethylene; PS, polystyrene; PA, polyamides; PAN, polyacrylonitrile; PEI, polyethylenimine; PAA, polyacrylic acid; –, not mentioned.

The PVDF membrane used by Wan *et al.* (2022) had the advantage of presenting a strong positive surface, allowing the exchange of anions and electrostatic interactions with negatively charged molecules in a wide pH range (4.1–11.9). However, the smallest tested particles (50–100 µm) exhibited ineffective electrostatic attraction under acidic conditions (2.7). In this study, synthetic lake water was used to test the treatment, and it was found that natural and synthetic organic matter did not change the membrane flux significantly. Competitive adsorption was observed between the anionic surfactant SDS and the smallest nanoplastics, and higher PS (50–100 nm) removal was observed using the surfactant CTAB, which was attributed to its enhancement of the electrostatic forces [25].

Pizzichetti *et al.* (2021) demonstrated the zeta potential for PA and PS particles to change from negative to positive, from basic to acidic conditions, while PA showed high aggregation and hydrophobic behavior, and PS had a slightly negative potential (20 mV), conferring instability and a slightly negative surface charge. On the PC membrane they tested, PA particles were smaller than the defined testing ranges, an occurrence that was attributed to particle breakdown when they collided with the membrane, which could lead to a greater release of MP in the effluent, while PS presented the same effect but also produced higher membrane fouling. However, MP removal was not enhanced because of the abrasion induced by these particles on the membrane, as the PS spheres were broken down to make the MP suspension for treatment and their irregular shape and brittleness produced this effect. The tested CA membranes wet easily and lose hardness owing to their hydrophobicity, making them more prone to abrasion, an effect seen with both PA and PS. However, fewer particles passed the membrane, and a higher water flux was achieved in comparison to PC, which is why this material was deemed best [24].

A biomimetic gill-inspired membrane (PVDF-$DMSO_2$) was created by Zhang *et al.* (2022) based on the morphology of manta ray and whale shark gills, resulting in a 97.6% efficiency. They tried different PS sizes, oblique pore angles, and flow directions, obtaining the best results for 65°, a flow rate of 0.43 m/s and 0.51 m/s in the reverse direction (opposite to the oblique membrane pores), being that the pores on the membrane reduce size under these conditions, inducing MP rejection. Turbulence also increases the possibility of contact between the particles and membrane pores. The rejection mechanism of MP is membrane retention due to the contact of particles with their pores, while MP smaller than the membrane pores can be rejected by ricocheting away.

Ceramic membranes were studied by Kook and Park (2022), who reported turbidity efficiencies of 99%. There was a linear correlation between the increase in the microplastic concentration in the feed solution and turbidity and total solids. Pore blocking with PS and PE was classified into four types: standard (when the particle deposits into the pore), intermediate (particles deposit on top of others, while others block the membrane pores), and cake filtration (accumulation of particles on the membrane surface when other particles cover the membrane pores). Because PE particles with an average sized of 0.8629 ± 0.0080 µm were found in the permeate, while the membrane pores were approximately 0.0273–0.0737 µm, it was deemed that different MP shapes and broad particle size distributions could be responsible for the slight ineffectiveness [28].

A magnetic filter mesh was designed and tested by Zhang *et al.* (2023), and Fe_3O_4 was chosen as the magnetic medium because of its prevalence in the literature for MP removal, presenting properties such as good stability, biocompatibility, superparamagnetism, and rich surface hydroxyl groups, as well as the characteristic of Fe acting as an adsorption site. The magnetic medium and the PVC solution were mixed at different reaction times before they were poured into the treatment system. It was found that the turbidity of the mixture decreased with the increase in stirring time, reaching

100% removal with the maximum tested time (20 min), being able to trap all the magnetic aggregates in the magnetic matrices under external magnetic forces (which must be greater than the fluid drag force). They were able to recuperate and recycle the magnetic medium, with 100% efficiency in up to three cycles and 94.1% in the fifth cycle.

Another innovative membrane type was Holey Ti_3C_2 nanosheet-based membranes by Yang *et al.* (2022), which were obtained by etching Co_3O_4 nanoparticles embedded on Ti_3C_2Tx nanosheets followed by simple vacuum filtration using polymeric membranes as a supporting matrix, leading to a planar porous structure with 25 µm nanoholes that promoted a greater water flux. The best removal efficiencies were achieved as the membrane thickness increased (although this data was not specified), with the thinnest membrane accounting for 45% of the efficiency. In comparison, the thickest sphere removed up to 98.5% of the PS spheres.

22.2.3 Chemical Removal Methods

22.2.3.1 Coagulation and Flocculation

The increasing frequency of microplastics in surface waters has attracted public attention because of the associated environmental problems [31]. Many microplastics enter the environment through the discharge of treated and untreated wastewater, especially those that come from processes where textiles [32] are involved so their removal from these environments is urgent [33].

On the other hand, coagulation is a process widely used in wastewater treatment plants to eliminate colloidal solids [34], such as clays, activated carbon, and even viruses [35]. Some authors mention that coagulation processes can be effective in eliminating microplastics [36]; however, there are discrepancies since others mention that current coagulation processes are not effective for their elimination [33].

Coagulation is affected by the characteristics of microplastics, such as surface area, density, particle size, morphology, adsorbed substances, functional groups, and the degree of weathering [37, 38]. In addition to the parameters of the process itself, such as the type of coagulant, dose, temperature, pH, sedimentation times, and mixing speed [34], the synergistic effect of these factors involved in coagulation processes should be studied further [39].

Most studies have focused on polyethylene and polystyrene because they are two of the most widely used plastics and because their density is less than or equal to that of water, which allows them to remain floating [38, 40]. The coagulation process has been combined with flotation, filtration, and electrochemical techniques to increase contaminant removal rates (Park *et al*. 2023; Tang *et al*. 2022; Wang *et al*. 2023) [39–41].

Studies have evaluated the coagulation of two different microplastics, polyethylene and polyethylene terephthalate (PE and PET), using the same coagulant (ferrate). Approximately >99% of PE was removed with 2.7 mg/L coagulant, while PET removal efficiency reached 75% with the highest dose (3.6 mg/L ferrate), demonstrating that the MP tends to exhibit different floc formation and, therefore, in an accurate sample, the possibility of removing all microplastics is complex [42].

In another study, three types of microplastics (polyethylene, polystyrene, and polyamide), five coagulants (polyaluminum chloride (PAC), ferric chloride ($FeCl_3$), aluminum chloride ($AlCl_3$), aluminum hydroxide ($Al(OH)_3$), and aluminum sulfate ($Al_2(SO_4)_3$ were used. In global terms, the removal order was 65% PA, 22% PS, and 12% PE, although the removal efficiency by coagulant type was not statistically significant [34]. In another study, a polyaluminum sulfate (PAS) coagulant was evaluated to assess MP removal from polystyrene. The optimal dose was 7.5 g/l with a removal percentage of 62% [37].

The removal mechanism of polystyrene (PS)-MP using Al- and Fe-based coagulants was investigated. The results showed that the Al-based salt (92.7%) was better than the Fe-based salt (91.2%) in the removal efficiency of coagulant PS of the hybrid PS-HA system (12 mg/l), which was higher than that of the individual PS system (9 mg/l). This study aimed to improve the traditional drinking water treatment process to remove MP [43].

In this study, a magnetic magnesium hydroxide coagulant (MMHC) was prepared by adding Fe_3O_4 magnetic particles at three different concentrations to remove polyethylene microplastics from wastewater samples. Microplastic removal efficiency after coagulation was compared with that of a traditional magnesium hydroxide (MHC) coagulant. The results showed that with 120 mg/L Fe_3O_4 and 200 mg/L $Mg(OH)_2$, the microplastic removal efficiency was 92% [44].

The improved performance of anionic polyacrylamide (PAM), sodium alginate (SA), and activated silicic acid (ASA) for the removal of polyethylene terephthalate (PET) microplastics using polyaluminum chloride (PAC) was evaluated. The highest elimination efficiency was obtained with ASA (54.7%) at doses of 20 mg/L PAC and 5 mg/L ASA, while PAM achieved removal of the order of 91.45% at high doses (PAC and PAM were 200 mg/L and 100 mg/L, respectively). Current coagulation treatments can be improved by adding coagulant aids [33].

Usually tests with coagulants are performed in volumes greater than 2 l, generating significant amounts of waste; however, it is possible to estimate PM removal based on the linearity between PM concentration and turbidity. These findings suggest that turbidity measurements can provide rational optimization of water treatment processes for the effective removal of nano- and microplastics [36].

In Table 22.5, studies related to the removal of microplastics using different types of coagulants are presented. The removal rates vary owing to the source water used; however, it is observed that in those where there is a greater complexity of contaminants the efficiency is lower due to the affinity of the coagulant with certain contaminants [45, 46]. Aluminum is the most commonly used element in most studies, owing to its high reactivity [34, 47].

The removal of PM by means of coagulants presents several challenges, one of which is how to dispose of the sludge generated in the process [39], and the other is how to evaluate the behavior of nanoplastics because its removal is not clear [53].

22.2.4 Oxidation Processes

Advanced oxidation processes (AOPs) are water treatment methods that generate sufficient amounts of hydroxyl radicals (OH•) to decompose organic or inorganic pollutants, thus giving rise to less toxic or non-toxic compounds [54]. In AOPs, H_2O_2, ozone, and ultraviolet light can be used as oxidants, along with a catalyst, to speed up the process. The classification of AOPs is based on the source used to generate OH•, within the types of AOPs are Fenton (H_2O_2/Fe^{2+}) and photo-Fenton ($H_2O_2/UV/Fe^{2+}$), which have proved to be the most effective, cost-effective and practical techniques when either used independently or in combination with conventional and biological methods (Ameta *et al.*, 2018) [55].

Table 22.5 Investigation of coagulants for microplastic removal from different water sources.

Location	Source water	Polymer	Coagulant	Removal efficiency (%)	References
Iran	Tap water	PE, PS, PA	PAC	18.6, 18.8, 38.8	[34]
			$AlCl_3$	14.4, 17.0, 75.7	
			$FeCl_3$	12.6, 28.9, 73.9	
			$Al_2(SO_4)_3$	10.1, 6.7, 63.5	
			$Al(OH)_3$	6.7, 14.4, 73.1	
China	Deionized water	PS, PE	PAC	77.8, 29.7	[48]
India	Simulated water	PE	$Al_2(SO_4)_3$	99	[47]
China	Deionized water	PET	PAC	35.5	[33]
			PAC-PAM	91.45	
			PAC-SA	69.9	
			PAC-ASA	77.6	
China	Deionized water	PE	$AlCl_3 \cdot 6H_2O$	61	[49]
			$FeCl_3 \cdot 6H_2O$		
China	Simulated water	PS	$AlCl_3$	98	[50]
Republic of Korea	WWTP	PE	PAC	15.8–42.3	[45]
Canada	Real surface waters	PS	$Al_2(SO_4)_3$	75.6–85.2	[51]
Czech Republic	WWTP	PE	Salt Al	62.0	[46]
Germany	DWTPs	PE, PA	PAC-PAM	88.0–93.6	[52]

PE, Polyethylene; PS, Polystyrene; PA, Polyamide; PAC, poly-aluminum chloride; PAM, anionic polyacrylamide; SA, sodium alginate; ASA, activated silicic acid; WWTS, wastewater treatment plants; DWTPs, drinking water treatment plants.

In the Fenton AOP, the OH• as an oxidizing agent originates from the decomposition of hydrogen peroxide by the action of the Fe^{2+} catalyst, under acid conditions, known as "Fenton's reagents." The hydroxyl radical is a potent, non-selective oxidant; therefore, in addition to being used to remove contaminants from water, it is also useful for killing pathogenic microorganisms [56]. There are also processes called the Fenton type, in which metals in a low oxidation state are used, such as cobalt and copper (Ameta *et al.*, 2018) [55]. The AOP Foto-Fenton process is an improvement to the conventional Fenton process in which ultraviolet light from the visible light region (UV–vis) is used to improve the production of OH•, which increases the rate of pollutant degradation. The addition of more iron is not required, as compared to the Fenton process, because it is continuously reduced. [56].

22.2.4.1 Degradation of Nano and/or Microplastics by Fenton and Photo-Fenton Processes

An alternative to address NP and MP contamination in bodies of water could be AOPs, specifically those mentioned above, because they are usually used to degrade organic and inorganic pollutants. Regarding the conventional Fenton method, Liu *et al.* [57]

investigated the degradation of nylon 6 (PA6) and polystyrene (PS) MP, individually, during four Fenton treatment cycles (5 h each), at pH = 3.0, and 150 rpm, with H2O2 30% and 0.2 M Fe^{2+} solution. The results showed mass loss of 25.6% and 22.1% for PA6 and PS, respectively. In addition, gas chromatography–mass spectrometry (GC–MS) revealed the formation of oxidation intermediates such as alcohols, aldehydes/ketones, and carboxylic acids.

Mackuľak *et al.* [58] evaluated the degradation of polyvinyl chloride (PVC) MP in two stages: the first corresponded to the Fenton process and the second to anaerobic digestion. The experimental conditions were pH = 3.0, H_2O_2 30%, 20 min of reaction at 300 rpm and 16 °C, and 15 min of sedimentation. The Fenton treatment resulted in a mass loss of 10 ± 2% and the presence of low-molecular-weight molecules (benzene, ethylbenzene, and o-xylene), which were removed in the second stage, such as PVC particles.

Regarding the photo-Fenton process, Feng *et al.* (2011) studied the decomposition of sulfonated PS MP using UV irradiation from a 250 W Hg (Xe) light source, initial pH = 2.0, volume of the reaction mixture of 25 mL, H_2O_2 = 0 to 176 mM, and Fe^{3+} = 0 to 0.63 mM. This process achieved a PS mineralization efficiency of >99% of PS after 300 min of treatment. For its part, the study by Liu *et al.* (2020) used Photo-Fenton to accelerate the aging of PS MP under the conditions: pH = 6.5, 365 nm UV light, 0.50 mL of 20 mM Fe^{3+}, 0.51 mL of 30 % H_2O_2, and reaction time of 108 h. This experiment resulted in low-molecular-weight species, such as benzoic acid, formic acid, and acetophenone, which affected the adsorption of pharmaceuticals on the MP [59].

In addition, MP degradation has also been reported in Fenton processes with variations in other conditions or in combination with other processes. For example, Hu *et al.* [60] implemented a Fenton process in which hydrothermal conditions were combined to degrade the MP of low-density polyethylene (LDPE), high-density polyethylene (HDPE), ultra-high molecular weight polyethylene (UHMWPE), PS, PVC, polypropylene (PP), and polyethylene terephthalate (PET). Of all the MP, the one with the highest mass loss was UHMWPE, with a proportion of 95.9% in 16 h at 140 °C and a mineralization efficiency based on the total organic carbon content (TOC) of 75.6% in 12 h.

Miao *et al.* [61] combined the Fenton process with electrochemistry based on the TiO_2/graphite (TiO_2/C) cathode to degrade PVC MP in a size range of 100–200 µm. The process conditions included a temperature of 100 °C, pH of 3.0, duration of 6 h, and stirring speed of 800 rpm. This process resulted in a mass loss of 56% and dichlorination efficiency of 75%. The intermediate products formed during PVC microplastic degradation are alkenes, alcohols, monocarboxylic acids, dicarboxylic acids, and esters. The surface morphology and molecular weight of PVC changed accordingly.

The degradation of MP by Fenton methods is relatively new; therefore, in addition to the removal efficiency and the necessary reaction time, there are still several aspects to consider, such as environmental impacts, process costs, and the toxicity of the intermediate products generated. This information makes it possible to determine the viability of each process and choose the most appropriate one.

22.2.5 Biological Processes for Remediation of Microplastics

Unlike physical and chemical microplastic remediation techniques, biological removal methods are used less frequently because specific conditions are required in experimental procedures to guarantee significant degradation percentages.

In recent years, there have been few published research related to this subject (6), and most of them have focused on the degradation of plastics through microorganisms, mainly

bacteria [62–66]. The removal through other types of microorganisms (such as protozoa or fungi), as well as the one that is carried out through the ingestion of microplastics by some organisms such as plankton, was not found reported in the investigations collected for this chapter, except for remediation through worms in a vermicomposting process [67].

22.2.5.1 Removal by Microorganisms

Remediation processes are mainly carried out at the laboratory scale. However, an *in situ* test (in mangrove soil) has also been conducted to confirm the results obtained in the laboratory [62]. The process generally consists of introducing an inoculum of degrading microorganisms into a sample of soil, leachate, or seawater (microcosm) together with samples of polymers (microplastics) to be studied. After incubation, microplastics were analyzed to determine whether degradation by microorganisms was carried out. Figure 22.1 shows the experimental process carried out for the biological removal of microplastics from microorganisms.

22.2.5.2 Preparation of the Microcosm

In this type of research, a microcosm is used as the test medium, an experimental unit that simulates the processes of an ecosystem [65]. The microcosms used tend to originate from different sources, such as seawater [66], mangrove soil [62], landfill [65], leachate from landfills (under aerobic and anaerobic conditions) [64], and rhizospheric soil [63]. Because of their nature, these microcosms already contain a large microbial load; however, the bacterial strains or strains of interest are usually added to them to show their polymer-degrading activity.

Microcosms usually do not add any additional elements, such as nutrients or other types of solutions before the degradation process. The only previous preparation was the determination of some parameters such as pH, humidity, C:N ratio, and analysis of the concentration of some other nutrients such as phosphorus, carbon, and nitrogen [62, 63, 65].

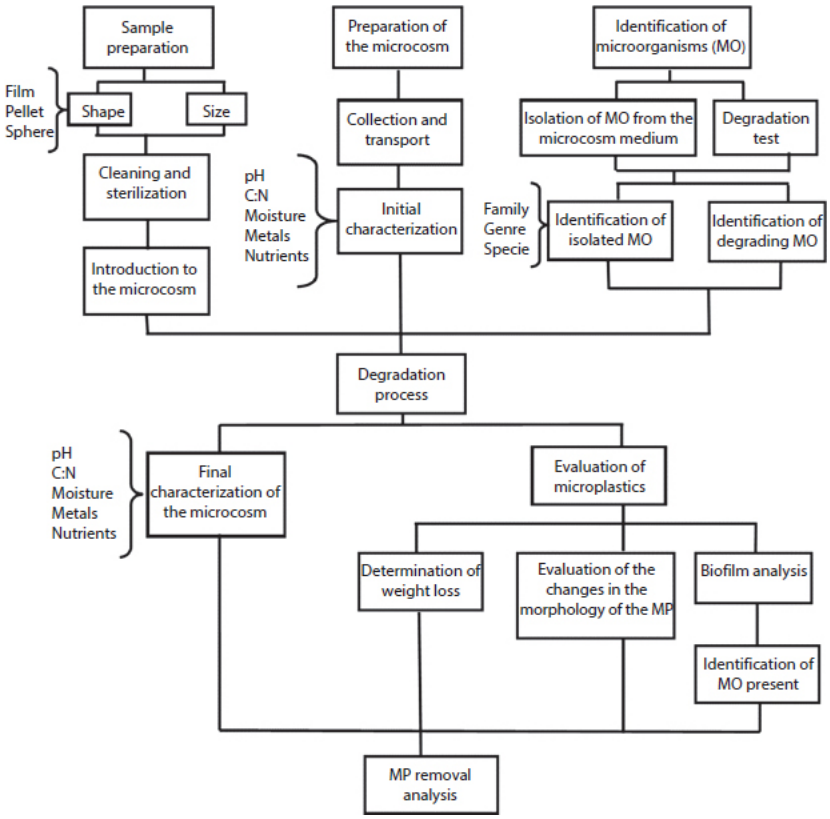


Figure 22.1 Stages of the microplastic removal process by microorganisms.

22.2.5.3 Identification of Degrader Microorganisms

A microplastic degradation test was carried out with previously isolated bacterial communities before carrying out the microplastic removal test in different microcosms. These communities are inoculated in solutions (saline solution or Bushnell–Haas broth) loaded with a high content of nutrients, except for the carbon source, because the main objective of biological removal is to use the polymers as the primary source of carbon [62–65].

The bacterial strains used in this process were isolated from the media used in the microcosms. Isolation occurs as follows: a portion of the soil, sludge, or leachate samples is mixed with saline solution, from which a suspension will then be obtained from which a series of dilutions will be made that will be inoculated into nutrient agar after incubation at approximately 37 °C for 24 h. From the colonies developed in the agar, pure individual cultures are obtained that are identified through the DNA sequence that is contrasted with different databases [62–65].

Once the bacterial communities (family, genus, and species) were identified, a degradation test was performed. Their potential as degraders was evaluated through their ability to use polymers as the main carbon source. This test consists of identifying which of all previously isolated bacterial strains do have the ability to degrade polymers and which do not [62–65].

For the test, as mentioned above, samples of the different polymers used in the microcosm were added to the solution loaded with high nutritional content, and the previously isolated bacterial communities are inoculated. This process lasts approximately 40 days of incubation [62]; after this time, the microplastics are extracted from the solution to analyze their weight loss and changes in their morphology to determine if the

inoculated bacteria would have or the degradation process has not been carried out [62–65].

After verifying that the biofilm had formed, to determine which bacterial strains were involved in this process (degraders), the bacteria present in the biofilm were isolated using a process similar to that described above. Once the most predominant strains in the biofilms of each polymer have been determined, these are used in the “real” degradation test (the one that involves the microcosm) to determine the behavior and interaction of the bacteria, the polymers and the medium used (soil, leachate, sludge, among others), thus simulating what would occur in natural conditions [62–65].

Although in most cases this degradation test is carried out to identify the degrading bacterial strains, there is another scenario in which the type of strains involved in the degradation is unknown until after the incubation process has finished, which means that the native microorganisms of the medium are left to act by themselves without adding any specialized inoculum [66].

22.2.5.4 Sample Preparation

The removal tests involved the use of primary microplastics (virgin resins) of different types of polymers, among which polypropylene (PP), polyethylene terephthalate (PET), and polystyrene (PS) stand out. Other types of polymers studied are polyethylene (PE) [64, 66], low-density polyethylene (LDPE) [65], and polyvinyl chloride (PVC)[64]. The microplastic samples studied are mainly films, pellets, spheres (in the case of expanded polystyrene), and even dust [63]. They are crushed into small fractions so that the microorganisms can degrade them. The size of the samples to be divided depends on the study’s objectives.

The sizes chosen for the samples can be between 1 and 5 mm [66], which allows changes that occur in the samples to be more easily observed throughout the process. However, microplastics smaller than 0.5 mm [64] or larger than 2 cm [63] can also be observed.

In addition to being crushed for the removal test, the microplastic samples were washed with distilled water and ethanol and dried in an oven for 24 h to avoid external contamination by microorganisms and to obtain erroneous results [62–66].

Inside the microcosm, the samples were inserted to be visible throughout the test and at the end, and to prevent fragments that form during degradation from spreading throughout the microcosm. Some ways to prevent them from dispersing are to place them inside nylon bags (0.15 mm pore diameter) [66] or to sew the microplastics in small bags of non-biodegradable material (1.8 mm mesh) inside the microcosms [62].

22.2.5.5 Microplastic Removal Process

To start the “real” removal process, microplastics of the different polymers to be studied were placed inside the microcosm, and inoculums of the degrading bacterial strains were added. This inoculum can be injected at different depths in the microcosm or placed only on the surface [62]. The degradation process began when the inoculum was added. The degradation time of microplastics varies from approximately 18 [63] to 90 days [62].

Some measures are usually taken to take care of the process and prevent some factors, such as exposure to sunlight from acting as degraders of microplastics [66] or external microorganisms from entering the microcosm. Quality control and contamination prevention include covering the microcosms with aluminum foil, placing them inside a laminar flow hood at room temperature, and avoiding the passage of light or air currents. In addition to these measures, the temperature, moisture content, and the amount of oxygen present are monitored and controlled throughout the process [62–66].

As in various laboratory tests, in the bioremediation tests carried out, a blank was also carried out, which is a test carried out under the same conditions as the study microcosms, but which does not have the specialized degradation inoculum but only contains native soil microorganisms or activated sludge used in the microcosms. The purpose of the blank was to determine whether the inoculated strain would have a positive effect on the degradation rate of microplastics or if the degradation time would be the same as that of native microorganisms [62–65].

As previously mentioned, most tests for the removal of microplastics by microorganisms are carried out in the laboratory; however, to show its efficiency in environments where conditions cannot be controlled, this process has also been tested *in situ* after its application in the laboratory. The process is similar to removal in the laboratory; the only difference is that the inoculum is injected into some sections of the chosen ecosystem [62].

During the incubation period, in addition to monitoring the properties of the different microcosms, the growth and characteristics of biofilms were also evaluated, such as the increase in their mass, to show how bacteria grow in contaminated environments, as well as the time in which the biofilm begins to form on the surface of the polymers [62, 66].

22.2.5.6 Determination of the Removal of Microplastics by Microorganisms

To determine whether the action of bacteria has carried out the microplastic removal process, the following were mainly evaluated: the loss of mass of the microplastics used in the test, the observation of the formation of the biofilm on the surface of the polymer (as well as changes in their density and morphology), and changes in the surface of microplastics. In general, the process could already be considered successful if biofilm formation occurs because it is considered that the bacteria would use microplastics as the primary source of carbon [62–66].

The evaluation of the microplastic weight loss allows for an approximation of the degradation percentage. In this type of research, removal efficiencies by microorganisms have been reported to range from 14% removal [65] to 69.23% [63]. Table 22.6 shows the different removal percentages reported, as well as the microorganisms involved in the process.

The ability of bacteria to remove microplastics can also be observed through enzymes that are potentially involved in polymer degradation. Some enzymes identified in the degradation process are alkane 1-monooxygenase, Cutinase, Feruloyl Esterase, Triacylglycerol Lipase, medium-chain dehydrogenated Acyl-CoA and Protocatechuate 4,5-dioxygenase [66].

In the analysis of the characteristics of biofilms that form in microplastics, different specialized techniques are used, such as staining with crystal violet and observation with a UV–Visible Spectrophotometer at 595 nm to quantify the total amount of biofilm biomass [66], Fourier Transform Infrared Spectroscopy (FTIR), High-Performance Liquid Chromatography (HLPC), and Scanning Electron Microscopy (SEM) to evaluate biofilm morphology, the presence of cavities, cracks, erosion surface, or fissure formation on the surface of microplastics, and changes produced by biofilm formation in the functional chemical groups present on the surface of microplastics [62–66].

Table 22.6 Microplastic removal efficiency.

Microorganism/organism	Specie	Microcosm medium	Duration time	MP types	MP size
Bacterium	*Phylum Proteobacteria*	Seawater	30 days	PE, PP and PS	PE (5 × 5 mm), PP y EPS (3–5 mm)
Earthworm	*Eudrilus eugeniae*	Anaerobically Digested Dewatered Sewage Sludge and Fresh Cow Manure	14 weeks	PP and HDPE	1,650–2,000 µm and 146–500 µm
Bacterium	*Pseudomonadaceae, Chitinophagaceae, Burkholderiaceae, Unclassified family belonging to Betaproteobacteriales*	Leachate from a landfill bioreactor (aerobic and anaerobic)	Anaerobic cultures (6 weeks), aerobic cultures (4 weeks)	PE, PVC and PET	PE and PVC <0.5 mm, PET 300 µm, and 50–300 nm
Bacterium	*B. cereus, A. faecalis, B. sonorensis, S. epidermidis,B. vietnamensis, R. ruber, B. flexus, S. globispora B. gottheilli*	Mangrove soils (laboratory and *in situ*)	90 days	PET and PS	NR
Bacterium	*Pseudomonas* sp.	Landfill soil	45 days	PUR and LDPE	NR
Bacterium	*Priestia aryabhattai (VT3.12), Bacillus pseudomycoides (VT3.15) Bacillus pumilus (VT3.16)*	Rhizospheric soil	28 days (films), 18 days (PET powder)	PET	Powder (300 µm), PET films (2 x 2 cm)

NR, Not Reported.

22.2.5.7 Main Findings

Although there have been few investigations related to the subject, it is possible to obtain valuable information on how the removal of microplastics works through microorganisms and the interaction of the different components involved (microcosms, microorganisms, and polymers) that can serve as a guide for future research on the subject. The main findings are as follows:

The incubation time and type of polymer present affected the biofilm formation. Some polymers are more susceptible to biological remediation than others; for example, PVC is not considered a carbon source by microorganisms, unlike PE and PET [64].

Media with a high load of microorganisms can contain many microbial species capable of degrading polymers, especially sanitary landfill media where, due to a large amount of plastic waste, microorganisms have adapted to use as a carbon source [64].

Control and blank tests carried out without the inoculum of degrading bacterial strains have also shown interesting results, such as the natural ability of soils to remediate microplastics by themselves. However, this more “natural” remediation process can take even longer and even not appear because the bacterial strains, when found in a contaminated environment, would not have the same metabolic strength to carry out the removal that ecosystems enriched with inoculums present [62].

Some microorganisms native to ecosystems would have a greater preference for one type of polymer to be used as a carbon source. For example, Auta *et al.* (2022) reported that in the blank test, there was greater degradation of PS than PET; however, when performing the test in the microcosm, the results were inverted, thus showing that the isolated bacterial communities had a better prediction when using PET as their primary carbon source [62].

The size of microplastics present in the soil also influences the removal time and efficiency. Dhaka *et al.* (2022) reported a much faster degradation of PET powder compared to larger sheets due to the much smaller surface area that bacteria have available to adhere to microplastics [63].

22.2.5.8 Removal by Larger Organisms

As previously mentioned, the removal of microplastics by larger organisms is reported to be lower proportion than that carried out by microorganisms; however, both processes share certain similarities, especially in the phase related to the evaluation of microplastic removal.

The removal process by larger organisms has been reported mainly for earthworms in the vermicomposting process [67]. For this process, different combinations of composts were used according to the media chosen for the study (anaerobically digested dehydrated sewage sludge (D-AD-SS) and fresh cow manure (PT-CD), whose mixture was named CDSSM), as well as different sizes of microplastics and polymers, as shown in Table 22.7.

To start the vermicomposting process, the samples of the polymers to be studied and the worms were added to the mixtures of the chosen media, which will be the organisms that oversee the degradation of microplastics. For this research, a proportion of three worms/100 g of CDSSM was maintained. As in any composting process, specific characteristics were maintained and controlled, such as a temperature of 16–20 °C and a humidity of 70%–75%.

Table 22.7 Composition of studied vermicomposts. Source: Author's elaboration using data from Ragoobur *et al*. (2022) [67].

Sample	Composition	Type of polymer	MP size
V1 (Blank)	100% PT-CD	NA.	NA.
V2 (Blank)	100% CDSSM	NA.	NA.
V3	100% CDSSM	PP	1,650–2,000 µm
V4	100% CDSSM	PP	146–500 µm
V5	100% CDSSM	HDPE	1,650–2,000 µm
V6	100% CDSSM	HDPE	146–500 µm

Note: NA, Not Applicable.

After 14 weeks of vermicomposting, different parameters of the worms were analyzed, such as their activity, the number of organisms present, the change in their biomass, the formation of cocoons (and their hatching), mortality, and the analysis of their epidermis in search of attached microplastics.

Unlike the removal of microplastics from microorganisms, where the formation of biofilm and the microorganisms present are observed to establish the removal efficiency, in the case of vermicompost, the level of remaining microplastics was determined through the filtration of a vermicompost solution with distilled water. The microplastics were determined by observing the filters in a Stereoscopic Microscope (NB-XT3B) and using a Fourier Transform Infrared Spectrophotometer (ATR-FTIR).

The main findings of this study are as follows.

There was an increase in earthworm biomass after 14 weeks, between 4% and 37% for the five samples.

As in the case of microorganisms, the relationship between the worms, the polymers, and their size did not remain the same for all the polymers studied; however, in the V6 sample that contained HDPE (146–500 µm), there was a reduction of 23% in the biomass of the worms present, unlike the other five samples that presented an increase. However, for the percentage of removal of HDPE of different sizes, the values were greater than 30% with 56 and 31% for samples V6 and V5, respectively.

Contrary to what was expected, the presence of microplastics in the vermicomposts did not affect the survival of the worms or their chemical properties (pH, electrical conductivity, and C:N ratio), and they also had a positive effect, with an increase in phosphorus content and reduction in cadmium.

22.3 Comparison of Different Removal Methods

This section lists the most important advantages and disadvantages of each microplastic removal method explored in the previous sections. This information is presented in Table 22.8 below.

Table 22.8 Comparison of different methods for the removal of microplastics.

Removal method	Advantages	Limitations
Adsorption	Adsorbents can be obtained from renewable sources The process has been proven to be especially efficient with aged MP removal Adsorbents can be recycled several times depending on the material and MP concentration Some adsorbents, like the OPS, can biodegrade Bio-based carbon adsorbents are low-cost	Studies have been conducted at lab-scale and mainly with one type of virgin polymer, therefore not being representative of the behavior of MP in the environment The elaboration of adsorbents could compete with the alimentary use of the source materials
Filtration	Conventional sand filtration is less energy consuming Sand filtration allows a lower operational and maintenance cost Granular limestone is an abundant, low-cost material Ultrafiltration can allow the recovery of plastic materials Filtration can be capable of removing acrylates Water can be reused Filtration membranes can be regenerated Filtration is a widely used method in water treatment. Therefore, it is generally accessible Gravity filtration implies a lower energy consumption The magnetic mediums used in filtration can be recovered Holey Ti3C2 nanosheets membrane has physicochemical stability, large water reflux, high MP removal efficiency and has been used in water purification	Sand filtration can have a significantly low efficiency as a tertiary treatment for MP Small leaks between seals, breakage of membrane or some contamination due to the pipelines can occur Ultrafiltration can be more energydemanding when water is richer in organic matter Coke is four times more costly than siliceous sand Membrane filters that have small pore sizes require higher energy consumption to provide a flux that allows separating tiny particles Membrane fouling can cause flux decline, affecting the produced waters quality Microfibers can longitudinally penetrate pores and gaps in filters Polymeric membranes have poor chemical resistance, can have low water permeability, and are prone to contamination during their use.
Coagulation	Coagulants accelerate the sedimentation time of PM High adaptability and versatility in already installed processes The use of coagulants is cheaper than other chemical processes	Generation of sludge with high aluminum content which could be toxic Low removal of PM when the wastewater brings other contaminants Coagulant dosage may require continual adjustment based on the variable composition of the wastewater source
Fenton and photo-Fenton	In addition to the oxidation of MP, it can oxidize other contaminants Short reaction times	Generation of substances with lower molecular weights, which could be toxic

Removal method	Advantages	Limitations
	Degradation rates for MP above 50%	Release of plasticizers, dyes and additives
Biological degradation	Degrading strains of microorganisms can be used to accelerate the degradation process carried out by native microorganisms The process can be adapted to the conditions of each type of plastic and medium where they are found. The processes are minimally invasive, so a large amount of waste is not generated It can be applied to different components, from seawater to sewage treatment plant sludge or leachate from landfills.	It has yet to be reported whether its industrial or *in situ* application would be profitable and viable. This type of process has only been tested on a laboratory scale under controlled conditions, so what would happen under natural conditions is unknown. The time in which plastics would be entirely degraded by the action of microorganisms is unknown. Under uncontrolled conditions, the percentages of removal efficiency could present considerable variations concerning the processes carried out in the laboratory.

22.4 Prevention and Reduction of Microplastic Pollution

As mentioned above, microplastic pollution has become a significant concern in recent years, as it has been detected in all types of environments, including oceans, freshwater, soil, air, and even food. Owing to the persistence of plastics in the environment, it is necessary to propose measures to reduce, avoid and mitigate microplastic pollution, including the generation of public policies and laws that complement and push to implement these measures [5, 68].

Interest has also arisen in generating policy responses to manage sources of unintentional microplastics. For example, the OECD (2021) generated a report focused on solutions to reduce microplastics in water derived from the release of synthetic textile fibers and tire wear. Likewise, the research carried out by [68] mentions the existence of a European regulation, “Strategy for plastics in the circular economy,” whose among its objectives are to reduce the addition of microplastics in certain products, control the loss of pellets and. It is also intended to allocate more resources to the eco-design and recycling of plastic products, in addition to improving tire designs regarding abrasion and durability and avoiding the generation of microplastics.

The OECD report (2021) mentions the importance of adopting best practices at all product life cycle stages. First, it mentions the adoption of interventions at the source, such as the sustainable design and manufacturing of textiles, tires, and products such as washing machines, detergents, and vehicles. Second are best practices in product use, such as improved washing processes, and eco-driving and mitigation measures, such as using microfiber filters in washing equipment. Finally, at the end-of-life, for example, in the treatment of wastewater, stormwater, and road runoff, to prevent microplastics from reaching water bodies. In the case of textiles and tires, preventive options early in the life cycle generated the most significant mitigation benefits.

Calero *et al*. (2021) described some alternatives for reducing microplastic pollution. The first alternative is to manage plastic waste according to the four R’s (reduce, reuse, recycle, and recover) to avoid fragmentation of larger plastics. Exemplifying the four R’s: reducing the production of plastic materials, changing their use to materials such as glass

or cardboard, reusing plastics that can no longer be used for their initial purpose, creating other products, or giving them a new function; recycling plastics, which has benefits in saving energy, resources and polluting emissions; and recovering stored energy from plastics when they can no longer be used.

Emphasis is placed on the industrial production and redesign of plastics for circularity. This measure involves changing how plastics are produced to reduce, prevent, and reuse plastics; produce plastics without toxic additives; or produce long-life plastics [68].

Operation Clean Sweep® (OCS) is a program that aims to minimize the loss of plastic pellets. It promotes good cleaning and control practices in processes in which pellets are used. Measures include preventing pellets from going down the drain, which can be prevented by using screens or similar physical barriers; avoiding loss of pellets during transportation due to accidents, which can be prevented by using resistant packaging to avoid breakage; the use of vacuum cleaners for powder cleaning; and the creation, publication, and dissemination of internal and external work manuals to specify activities and measures against pellet loss [68].

Finally, improving wastewater treatment technology is one of the main strategies for reducing microplastic pollution. Alternatives for effluent treatment include ozone, rapid sand filtration, reverse osmosis, dissolved air flotation, and membrane bioreactors, as illustrated in the previous sections. In the future, it will also be essential to develop technologies to reduce microplastics in effluents, and sludge, and to evaluate recovered microplastics. Although there is already an interest in taking specific measures to mitigate and prevent microplastic pollution, it is crucial to follow up on research related to this issue and generate laws, rules, and regulations governing this problem [68].

22.5 Conclusions

MP are omnipresent in all environments; however, one significant challenge is eliminating them because of the complexity of different matrices (water, soil, and sediments). Therefore, techniques have been adapted for their removal and degradation, which involve physical, chemical, and biological processes, each with advantages and disadvantages.

This work analyzed adsorption techniques, filtration, coagulation-flocculation, Fenton and photo-Fenton processes, and biodegradation with micro- and macroorganisms.

The advantages of physical treatments are their low cost compared to chemical and biological methods. MP can be removed by backwashing, energy consumption is low, beds can be regenerated and obtained from recyclable materials, and they are fast processes. However, the disadvantages are that more MP can be generated because of the abrasion they suffer when passing through the bed, and membranes can fragment depending on the material, generating new MP.

In the case of chemical processes, the advantages are that other types of pollutants can be eliminated in addition to MP, they are relatively fast processes, and they can be adapted to wastewater treatment plants. The disadvantages are the generation of waste sludge, which may contain metals such as aluminum in high concentrations or low-molecular-weight intermediates that are toxic, expensive and have high energy requirements.

The advantages of biological processes are that they are non-invasive, native microorganisms can be used, and PM may reach mineralization. The disadvantage is that there is no estimate of the time at which the complete degradation of plastic waste would occur because laboratory tests do not last longer than 40–90 days. None of the reviewed investigations in this chapter mentioned that after this time, the studied microplastics

were completely degraded, and it is also important to mention that to obtain high percentages of degradation in the polymers throughout the experimental procedure, the conditions of the microcosms such as the temperature, humidity, and, above all, the type of microorganisms present, *in situ* conditions, or on a larger scale, these factors could not be monitored and cared for in the same way as on a laboratory scale, so the percentages of degradation efficiency could present considerable variations.

In some cases, removal and degradation methods are not viable because they are costly, impractical, have low removal efficiencies, and can directly affect the biota. In these cases, the most viable option is for natural attenuation to occur, if measures are taken to prevent and minimize the entry of more PM and NP into the environment.

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Note

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23
Removal of Microplastics and Nanoplastics From Water

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Abstract

Over the last few decades, plastics produced from fossil raw materials have been used for different applications. However, traditional plastics have environmental disadvantages. These plastics can be transformed through several processes, such as oxidation, photodegradation, and mechanical degradation. Microplastics (MPs) and nanoplastics (NPs) can be obtained from these processes. The MPs and NPs are present in different locations, such as terrestrial, atmospheric and aquatic environments. There are various strategies for removing MPs and NPs from the environment. MPs and NPs can be removed using sponges/aerogels, materials with metals, biochar, and activated carbons.

In this work, we review the different materials used to remove MPs and NPs from the environment will be done.

***Keywords*:** Plastics, microplastics, nanoplastics, sponge/areogel, materials with metals, biochar

23.1 Introduction

Due to the mechanic and chemical properties and low price, plastics, like poly(ethylene) (PE), poly(styrene) (PS), poly(propylene) (PP), poly(urethane) (PU), poly(amide) (PA), poly(carbonate) (PC), poly(ester) (PES), poly(ethylene terephthalate) (PET), and poly(vinyl chloride) (PVC), have been used in different activities, namely on industry and agriculture [1]. According to Sharma *et al*. [2], by 2050, the amount of plastic produced will be 33 billion tons (estimated value). However, some of these materials were not recycled. Approximately 20% of plastics used were recycled. Plastics can be degraded into smaller particles. Plastic fragments are generally separated into microplastics (MPs) and nanoplastics (NPs). Normally, MPs vary between 1 µm and 5 mm. These MPs have various arrangements, such as fragments, grains, fibers, beads, and pellets. The NPs were in the range of 1 nm to 1 µm, similar to colloidal presentation [3].

According to their sources, MPs and NPs can be divided into primary and secondary materials. Primary plastics (MPs and NPs) are mass-produced based on their dimensions and granulometric properties. These materials are linked to the abrasives and washing products industry [4], cosmetics PE microspheres [5], poly(acrylamide) [6], tints, and microchip technology [7]. Secondary plastics (MPs and NPs) are obtained by the degradation of materials with large dimensions, such as mesoplastics and macroplastics. This process occurs before the materials are released into the environment. As example, the microfibers are obtained by washing the dress [8], and/or by different processes like oxidation, hydrolysis, and photodegradation [9, 10]. MPs and NPs can be found in water, lakes, rivers, sewage drains, sediments, oceans, [11], water for

consumption [12], air [13], soil [14], food [15], and the human body [16].

Owing to the presence of MPs and NPs in different environments, animals can consume these materials [17]. For example, because of the dimensions of MPs and NPs, these materials can be used as food by aquatic species, enter the food chain, and can be toxic to humans when they eat these animals. Consequently, the elimination of MPs and NPs is imperative to defend aquatic environments [17].

In this chapter, a review of the use of different adsorbents to remove MPs and NPs from aqueous media is presented.

23.2 Sponge/Aerogel Materials to Remove MPs and NPs

Because of their high surface area, textural properties, exceptional mechanical properties, low cost, and the existence of specific groups on their surfaces, sponge/aerogel materials have been used to remove different contaminants from aqueous media [18]. It is also important to indicate that these materials can be obtained from renewable raw materials and become biodegradable sponges/aerogels biodegradable after utilization [19]. These materials can be manufactured by crosslinking (a chemical reaction between two chains of the polymer), self-polymerization, and desilylation, and have been applied for the elimination of MPs and NP [18, 20–22]. Owing to their surface properties, these materials have hydrophilic groups that can be used to remove MPs and NPs through different interactions between the material and MPs and NPs, such as hydrogen bonding, van der Waals forces, ion exchange, and diffusion inside the material.

Adsorbents prepared from chitin can be used to remove MPs from PS, and PS-modified (PS-COOH, PS-NH_2) at a

concentration of 1 mg/L and with a size of 1 µm. After 48 h, the removal efficiency was between 43.4% and 70.4%, owing to the presence of carboxylic and hydroxyl groups, which can interact *via* electrostatic and hydrogen bonds [19]. In addition, an adsorbent prepared from chitin and graphene oxide was used to remove MPs of PS with a size of 1 µm and a concentration of 1 mg/L. After 24 h, the removal efficiency was 89.8%, which could be attributed to the presence of −OH, −COOH, and $-NH_2$ groups. The presence of these groups can promote different interactions between the pollutant and adsorbent, such as electrostatic interactions or hydrogen bonds [18].

A material prepared from protein (oat protein sponges) can be applied to remove PS MPs with a size of 1 µm and a concentration of 1 mg/L. After 12 h, owing to the presence of −OH and −COOH groups, the removal efficiency was 81.2%, which can be attributed to the hydrophobic interactions between the adsorbent and the PS MPs [21].

A composite material consisting of Fe_3O_4-PDMS/halloysitePDMS@PDMS-coated sponge can be used to remove the MPs of PP (150 µm in size), PVC (150 µm in size), and PE (250 µm in size). This material can remove approximately 100% of the MPs of PP, PVC, and PE owing to the presence of −OH, −COOH, and −Si−O groups on its surface. The presence of these groups can improve the electrostatic interactions and hydrogen bonds between MPs and the composite surface [20].

A polydopamine-enhanced magnetic chitosan aerogel (PDA-MCS) can be used to remove MPs from PET (with a size of 10–20 µm). After 18 h, approximately 91.6% of the PET MPs were removed, which could be due to the presence of −COOH and $-NH_2$ groups. The presence of these groups can improve the interactions between the MPs and PDA-

MCS, which can be attributed to electrostatic and hydrogen bond interactions [23].

A material consisting of a three-dimensional graphene-like carbon-assembled layered double oxide material from a layered double hydroxide was used to remove the PS NPs (80 nm in size) NPs. This material can remove approximately 80% of the NPs after 12 h, which can be attributed to the presence of −COOH and −OH groups on its surface and the interactions such as hydrogen and electrostatic bonds between the material and the NPs [24].

NPs in PVC (98 nm in size), PMMA (76 nm in size), and PVAc (71 nm in size) can be removed using a polyethylenimine/cellulose fiber material. After 30 min, the removal of NPs from PVA, PMMA, and PVAc was approximately 97%, 98%, and 99%, respectively. This behavior can be attributed to the presence of −OH and NH_2 groups, on the material, which can improve the interactions (electrostatic) between the NPs and the material [25].

A composite material consisting of zinc metal incorporated on MOF (an aerogel material) was used to remove PVDF NPs with a size of 60–110 nm and PS NPs with a size of 90–140 nm. This material removed 91.4% of the PVDF NPs and 85.8% of the PS NPs, which could be due to the van der Waals forces and hydrogen bonds [26].

Table 23.1 summarizes the sponge/aerogel materials used to remove MPs and NPs from the water.

Table 23.1 Summary of sponge/aerogel materials for the removal of MPs and NPs.

Material	Nature of MPs and NPs	Removal rate	Reference
chitin with graphene oxide	PS	89.8%	[18]
chitin	PS	43.4%–70.4%	[19]
Composite material of PDMS	PP, PVC and PE	~100%	[20]
Oat protein sponges	PS	81.2%	[21]
PDA-MCS		91.6%	[23]
Graphene	PS	80%	[24]
Polyethylenimine/cellulose	PVA, PMMA and PVAc	97%, 98%, and 99%	[25]
Zn-MOF	PVDF and PS	91.4% and 85.8%	[26]

23.3 Materials With Metals to Remove MPs and NPs

MPs and NPs can be removed by the application of metal-containg materials. An example is the preparation of materials with iron. This metal can be introduced into the material by calcination, chemical vapor deposition (CVD), hydrothermal methods, and co-precipitation. The use of Fe improves the separation of these materials through the application of a magnetic field [27]. In addition, Co, Mn,

and Pt, kaolin, and lauric acid have been used to prepare materials with different properties (highly hydrophobic surfaces) to remove MPs and NPs [28–30].

The MPs of PE, PP, PS, and PET (200–900 µm in size) can be removed using Fe_3O_4 nanoparticles. After 150 min, the removal efficiencies were 79–94% for PE, 81–93% for PS, 80–90% for PP, and 57–72% for PET. This removal efficiency can be explained by the number of −OH groups present on the surface of the material. The interactions between MPs and the material are electrostatic and hydrophobic [27].

CuFeCo alloy was used to remove MPs of high-density polyethylene (HDPE) with a size of 270–1,240 µm. The removal efficiency was 83%, owing to the presence of carboxylic acid groups on the surface. The interactions are hydrophobic and electrostatic between the material and MPs [29].

An Fe-carbon nanotube material was used to remove MPs from the PE (48 µm in size). The removal was approximately 100% after 300 min. This material was also used to remove the PET MPs. After 300 min, removal was 100%. These results can be explained by strong interactions between MPs and the material [31].

PS NPs with carboxylic acid groups (55 nm in size) were removed using Zn−Al-layered double hydroxide (LDH). After 2 h, the removal of NPs was 100%, which could be attributed to the presence of −OH groups. The interactions between the NPs and the material are electrostatic [32].

A composite material consisting of cellulose and Mg−Al LDH was used to remove the PS NPs (100 nm in size). The adsorption capacity was 6.08 mg/g. The contact between the NPs and the material was due to hydrogen bonding and

electrostatic interactions, owing to the presence of carboxylic acid groups [33].

Fly ash with Fe was used to remove PS NPs (80 nm in size). The adsorption capacity was 89.9 mg/g. The interaction between the NPs and the material (electrostatic relationship) was due to the presence of Fe−OH and carboxylic acid groups [34].

PS NPs (100 nm in size) were removed using CuNi@C. After 24 h, the removal of NPs was approximately 99.18%. The adsorption capacity was 38 mg/g. These results can be attributed to the presence of hydroxyl and carboxylic acid groups on the surface of the material. The interaction between the NPs and material is due to electrostatic forces [35].

PS NPs and carboxylic acid groups (50 nm in size) were removed using γ-Fe_2O_3/Pt/TiO_2. The removal of NPs was approximately 97%, which could be attributed to the presence of carboxylic acid and hydroxyl groups. The interaction between the NPs and material is due to electrostatic forces [30].

Table 23.2 shows a summary of the materials and metals used to remove MPs and NPs from water.

Table 23.2 Summary of materials and metals used to remove MPs and NPs.

Material	Nature of MPs and NPs	Removal rate	Reference
Nanoparticles of Fe_3O_4	PE, PS, PP, PET	79–94% to PE, 81–93% to PS, 80–90% to PP, 57–72% to PET	[27]
CuFeCo alloy	HDPE	83%	[29]
γ-Fe_2O_3/Pt/TiO_2	PS	97%	[30]
Fe-carbon nanotubes	PS	100%	[31]
Al layered double hydroxide	PS	100%	[32]
CuNi@C	PS	99.18%	[35]

23.4 Biochar as Material to Remove the MPs and NPs

Biochar is a material that can be obtained using different processes, such as carbonization and pyrolysis from diverse biomass, such as sugarcane bagasse and corncobs. This material has been used for the removal of MPs and NPs and is cheaper than sponges and materials with metals [36–38]. Biochar can be modified by immobilizing metals (Mg, Fe) and Fe_3O_4, as shown in Figure 23.1. Also, the biochar can be modified using HNO_3, and H_2SO_4, as oxidizing agents [36, 38, 39].

Magnetic biochar was used to remove PS MPs (1 µm in size). After 2 h, the removal was 94.81%, and the adsorption capacity was 100.6 mg/g. These results can be attributed to the presence of carboxylic acid and hydroxyl groups on the surface. The interactions between the MPs and the material were electrostatic and chemical bonding. In addition, magnetic biochar with Mg and magnetic biochar with Zn were applied to remove MPs from the PS. The removal efficiencies of biohar with Mg and Zn are 98.75% and 99.21%, respectively [38].

The biochar obtained by pyrolysis and modified with Fe was heated to 550 °C. This material was used to remove MPs from PS with carboxylic acid (1 µm in size) and PS with NH_2 groups (1 µm in size). The removal efficiency was 99.5%, and the adsorption capacity was 206 mg/g for MPs with COOH groups. When biochar was applied to the removal of MPs from PS with NH_2, it was 99.1% and the adsorption capacity was 206 mg/g, which could be due to the presence of −OH and −COOH groups. The relationship between the MPs and the material involves electrostatic interactions [39]. In addition, the biochar obtained by pyrolysis and modified with Fe was heated to 850 °C. This biochar was used to remove PS MPs with −COOH (1 µm in size) and PS MPs with $-NH_2$ groups (1 µm in size). The removal efficiency of PS MPs with carboxylic acid was 99.5% and the adsorption capacity was 225 mg/g. For PS MPs with amine groups, the removal was 99.1%, and the adsorption capacity of biochar was 168.79 mg/g [39]. In addition, the biochar obtained by pyrolysis and modified with Fe was subjected to 550 °C and used to remove PS NPs with carboxylic acid groups (30 nm in size). The removal efficiency was 49%, and the adsorption capacity was 159 mg/g, which can be attributed to the presence of carboxylic acid groups on the surface. The relationship between biochar and NPs was determined by electrostatic

interactions. The biochar obtained by pyrolysis, modified with Fe, and subjected to 850 °C was used to remove NPs of PS with carboxylic acid groups (30 nm in size). The removal rate was 51% and the adsorption capacity was 169 mg/g [39].

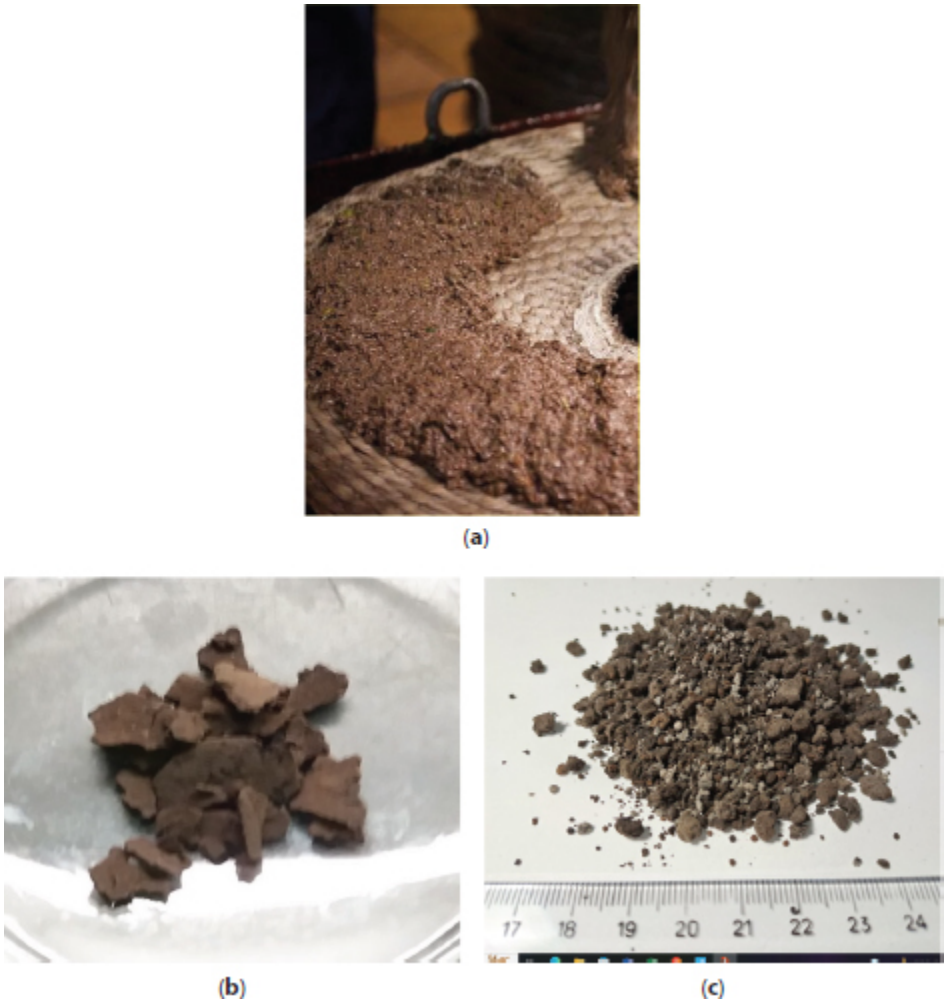


Figure 23.1 Magnetic biochar produced from olive pomace. (a) Olive pomace, (b) magnetic biochar film, and (c) granulated biochar.

Corncob biochar treated at 500 °C was used as to remove NPs of PS (50 nm in size). After 8 h, the removal efficiency was high (>90%), and the adsorption capacity was 12.3 mg/g. Corncob biochar treated at 700 °C was also used to remove PS NPs (50 nm in size). The removal efficiency

increased (>90%), and the adsorption capacity was 13.8 mg/g. A material obtained from corncob biochar treated at 900 °C was used to remove PS NPs (50 nm in size). The removal was higher (>90%), and the adsorption capacity was 15.5 mg/g. These results could be attributed to the presence of hydroxyl and carboxylic acid groups present on the surface. The relationship between the material and NPs is due to hydrogen bonding and hydrophobic interactions [36].

Mesoporous biochar was used to remove PS NPs (100 nm in size). After 24 h, the adsorption capacity was 56.02 mg/g, which could be attributed to the presence of carboxylic acid and hydroxyl groups. The nature of the interaction between NPs and biochar involves electrostatic interactions and chemical bonding [40].

Table 23.3 Summary of the biochar to remove MPs and NPs.

Material	Nature of MPs and NPs	Removal rate	Reference
Corncob biochar	PS	>90%	[36]
Biochar obtained by pyrolysis at 350 °C, 550 °C, and 750 °C	PS	24%, 39%, 99%	[37]
Magnetic biochar with Mg and magnetic biochar with Zn	PS	98.75% and 99.21%	[38]
Biochar obtained by pyrolysis and modified with Fe	PS	99.5%	[39]

PS NPs (less than 500 nm in size) were removed using biochar obtained by pyrolysis at 350 °C, 550 °C, and 750 °C. The material prepared at 750 °C removed 99% of PS NPs after 5 min. The adsorption capacity was 44.9 mg/g. The biochar prepared at 550 °C removed approximately 39% of the PS NPs and the material prepared at 350 °C removed 24%. These results can be attributed to the presence of hydroxyl and carboxylic acid groups [37].

Table 23.3 shows a summary of biochar as a material to remove MPs and NPs from water.

23.5 Additional Materials to Remove MPs and NPs

Powder and granulated activated carbons, such as those shown in Figure 23.2, are widely used in both research and industrial processes because of their excellent physical and chemical properties. Granular activated carbon was used to remove PS NPs (97 nm in size). The adsorption capacity was 2.20 mg/g, which could be attributed to the hydroxyl and carboxylic acid groups present on the surface. The interactions between the granular activated carbon and NPs were electrostatic [41].

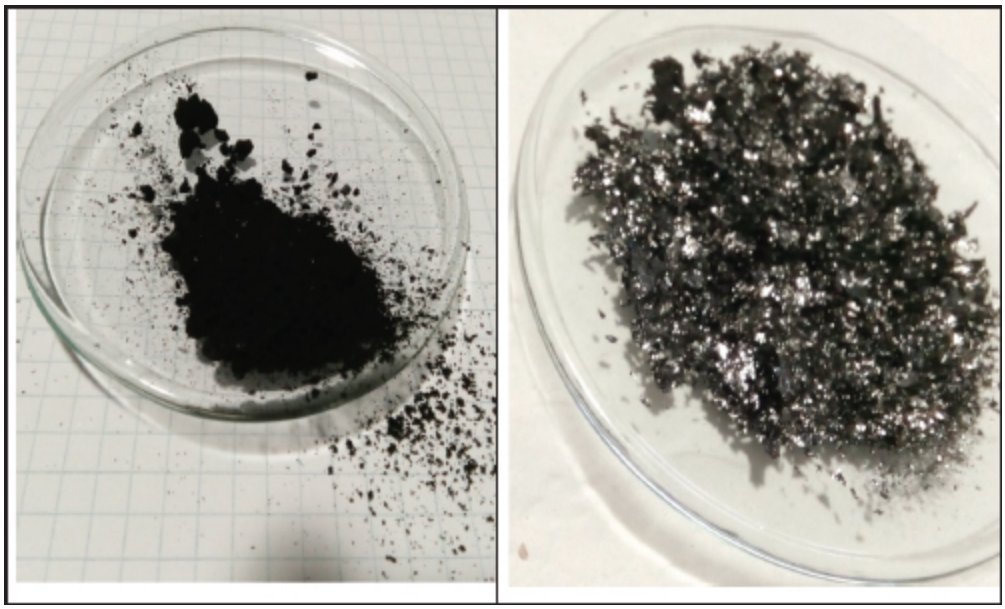


Figure 23.2 Different forms of activated carbon used as adsorbents.

Coffee ground material was used to remove the PS NPs with $-NH_2$ groups (100 nm in size). After 40 min, the removal efficiency was 74%, and the adsorption capacity was 4 mg/g, which can be attributed to the presence of −OH and −COOH groups on the surface. There are some electrostatic and hydrogen bonds between the NPs and the adsorbent material [42].

Magnetic material was used to remove the PS NPs (100 nm in size). After 3–5 min, the removal efficiency was 94.0% owing to the presence of hydroxyl and carboxylic acid groups. Different interactions, such as electrostatic and hydrogen bonding, have been established between the material and NPs [43].

Cigarette butt material was used to remove PS NPs of three different sizes: 50 nm, 180 nm, and 500 nm. After 2 h, the removal efficiencies of PS with sizes 50, 180, and 500 nm were 99%, 98.5%, and 90%, respectively. This behavior can be attributed to the presence of oxygenated groups, such

as −OH and −COOH. There are electrostatic interactions between the material and PS NPs [44].

23.6 Conclusion

The presence of MPs and NPs in aquatic, terrestrial, and atmospheric environments has increased in recent years. Microplastics and nanoplastics are particles obtained from traditional plastics, such as PE, PS, PET, and PP. Different materials, such as sponges/aerogels, materials with metal on their surface, biochar, and granular activated carbon have been developed or used to remove MPs and NPs, from the aqueous phase. When tested on sponge/aerogel materials, the removal of NP was superior to that of the MPs, which could be due to the diffusion capability of the NPs on these materials. MPs and NPs can also be removed using materials that contain metals on their surfaces. Different relationships between the material and MPs/NPs are presented, such as hydrogen bonding and electrostatic interactions. In addition, NPs can be removed using materials with metals, owing to their diffusion into the porous material system. MPs can be removed by adsorption onto the surface of materials containing metals.

Biochar can also be used to remove MPs and NPs from the aquatic environment. The removal efficiency of biochar was high. This material can be used to remove MPs and NPs with positive and negative charges, respectively, which can be attributed to the presence of different functional groups on its surface.

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Note

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24
Microplastics and Nanoplastics in Aquatic Systems

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Abstract

The predilection for having a sustainable and well-developed economy, with amelioration in population and living standards, focused only on successful development and implementation, not on the billions of waste generated. Plastics are one such material that has pioneered sophisticated living standards since their discovery and has been augmented with industrialization and commercialization. However, with the overall consumption, only approximately 9% of the million tons of waste plastics generated undergo recycling. The uncontrolled littering of these materials results in a critical system, outpacing the expertise to access these materials beyond planetary boundaries. The non-biodegradable and toxic nature of plastic materials leads to the generation of micro- and nanoplastics, polluting the ecosystem cognate with it. This chapter focuses on the presence of nano/microplastics in aquatic ecosystems, primarily focusing on their toxicity and subsequent effects on the marine environment and its flora and fauna.

Keywords: Microplastics, nanoplastics, toxicity, aquatic ecosystem, pollution

24.1 Introduction

The introduction of a material that can be crafted into various shapes and made to imitate natural substances and plastics in 1869, followed by an idea presented in the 1862 London Exhibition, revolutionized the human world [1]. Since manufacturing the very first synthetic polymer, bakelite, in 1907 [1], plastic polymeric materials that may contain other materials to improve performance and/or reduce cost, according to IUPAC, have found enormous applications [2]. The economic feasibility, ease in processing, and tailor-made properties of plastics have received pervasive attention and led to their rapid application augmentation. Plastics have played a crucial role in upholding the quality and comfort of modern lifestyles owing to their convenience and impressive cost-to-performance ratio. Owing to these characteristics, the use of plastics has covered various aspects ranging from daily life to commercial manufacturing in recent years. Global plastic production has soared since the 1950s with an annual increment of 4% and is an estimated value of has reached an estimated value of 445.25 million metric tons by 2025 from 390.7 million metric tons in 2021 [3]. Although the introduction of plastics has changed the global perspective, the scenario of the marine environment and dump yards covered with plastic bottles, bags, and other plastic wastes implies the unpreparedness of humankind. For decades, humankind has leaned on traditional methods, such as landfills and incineration, to dispose of such waste materials, creating an adverse environmental impact. This vital impact is ascribed to the harmful ecological impediments associated with their production and inadequate waste treatment technologies. Compared with the total production rate, only approximately 9% of the waste generated is recycled, and approximately 80% of the generated waste is reported to accrue in the natural environment or landfills.

Global scale data on plastic leakage into marine and terrestrial ecosystems are still under estimation. Reliable quantitative estimations of sources, input loads, and originating sectors signify a knowledge gap; however, approximately 8 million tons of plastics are

estimated to leak into the ocean. It is estimated that the Ocean may already account for 150 million tons of plastic, of which 2.5 lakh tons are fragmented into 5 trillion plastic pieces [4]. In addition to plastic pellets, microbeads, and scraps leaked by the polymer, cosmetic, food, and textile industries during production, machine runoff, packaging, and scrubbing agents, photodegradation and degradation of macroplastics under weathering processes lead to the formation of micro- and/or nanoplastics. Microplastics (MPs) and nanoplastics (NPs), which were first reported in the 1970s, are considered particles with sizes ranging between 1 µm and 5 mm and 1 nm and 1 µm, which can enter the environment directly or via numerous sources, including packaging, cosmetics, cleaning products, industrial feedstock, and many more [5–8], and are defined as primary MPs and NPs. The breakdown of larger plastics, attributed to microbial or human activities, leads to the generation of secondary NPs and MPs [9].

Although the polymeric materials that lead to the production of MPs and NPs found in the natural environment may be considered biochemically inert, ingesting these particles by various organisms may lead to numerous hazardous effects on both the environment and organisms [10–12]. This includes utilizing compounding agents, including pigments, plasticizers, and chemicals, that impart flame retardancy and UV stabilization to improve the material properties that are susceptible to direct leaching into the environment due to poor binding with plastics. Accordingly, the ingestion of these particles not only leads to energy disbursement and false satiety but also imparts other toxicologic effects owing to the leakage of such fatal toxins, highlighting the probable perils that MPs and NPs can cause to the environment and biota [6, 13–15].

This chapter focuses on MPs and NPs in aquatic ecosystems, encompassing the toxicity of these entities to aquatic flora and fauna, with a comprehensive overview of various studies. Among several factors such as shape, size, and incubation time, the type and nature of plastic play a crucial role, as plastic toxicity may vary based on the polymer type. The presence of MPs and NPs and their migration to freshwater and ocean environments were thoroughly examined and explored. A comparative analysis of the various NP and MP toxicities was performed. Furthermore, a propensity towards environmental implications, prospects, and associated challenges is presented.

24.2 A Theoretical Assessment of MP and NP Migration and Fate Aquatic Environment

The environmental providence of MPs and NPs in the aquatic environment is substantially influenced by the flow between fresh water bodies and long-distance travel with the ocean [16, 17]. Plastic manufacturing, consumption, and disposal mostly occur in land ecosystems, making land a widespread source of MPs and NPs [18, 19]. Although MP and NP accumulation is believed to occur in the deep ocean, recent reports have demonstrated the vast accumulation of MPs and NPs in freshwater bodies [16, 19–21]. Figure 24.1 depicts the migration paths of MPs in water.

24.2.1 Migration *Via* Land to Freshwater to Ocean

The high-end demand for plastic products from multifarious sources leads to the high input dominance of MPs and NPs in terrestrial ecosystems, followed by dumping in landfills and sewage treatment. The ineffective disposal management system led to the existence of 21%–42% of the total production remaining in landfills, which increased with the increase in production [22]. Even though wastewater treatment plants have gained attention and applications in removing these materials, the significant increase in MPs and NPs in effluents leads to the liberation of these MPs and NPs into freshwater bodies [23]. Several countries, including China, Europe, India, North America, and Australia, utilize sewage and industrial sludge as agricultural fertilizers, increasing the input impact

of MPs and NPs on land, which further enters water bodies [24, 25]. An obvious route for transferring MPs and NPs into the ecosystem is from land to ocean. Approximately 70%–80% of total plastic debris accounts for land sources demonstrating over 4.8–12.7 metric tons of plastic waste accumulation in the ocean from 192 coastal countries, as reported by Jambeck *et al.* [18, 19].

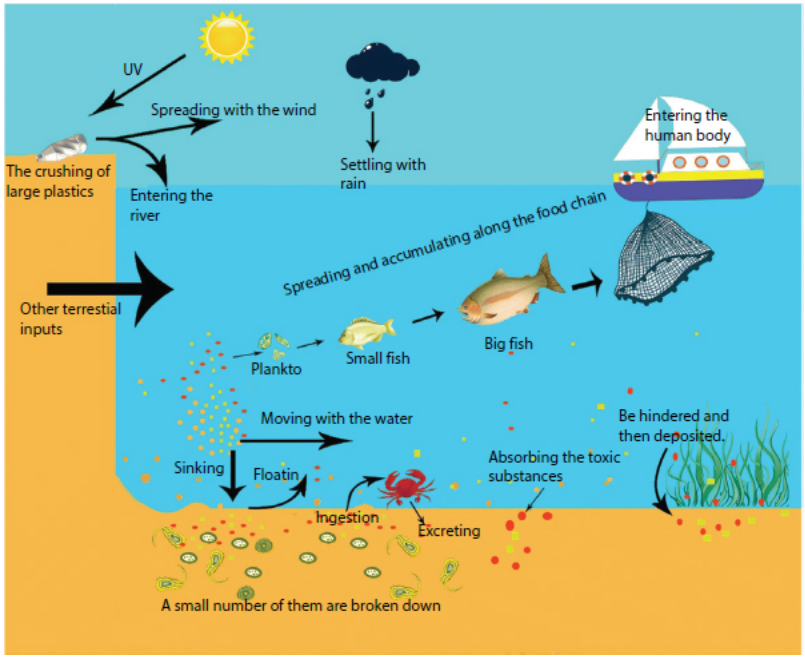


Figure 24.1 The exodus route of MPs in aquatic ecosystem.

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24.3 Pollution in Marine Environment

Research on MPs in marine environments was pioneered in the late 20th century and gained pervasive attention globally, because of its widespread source and distribution, small size, detection difficulties, and adverse effects [7, 26]. The physical and chemical features of MPs are usually identified by Scanning Electron and stereoscopic microscopy and Raman and vibrational spectroscopic techniques, respectively, which also help improve MP identification accuracy [27, 28]. Auta *et al.* [29] reported the active ingestion of small MPs by marine organisms, which has greater consequences for marine ecosystems. Several bacteria and fungi have been reported to actively degrade MPs in the ocean, thus playing a vital role in marine MP pollution control and management [30].

24.3.1 Toxicity Assessment of MPs in Aquatic Environment

Many studies have reported the fatal consequences of MP ingestion, including reproductive complications, reduced growth rate, digestive tract obstruction, and false satiation, which can lead to the death of several millions of marine animals, posing a severe threat to marine ecosystems [31, 32]. This is the first report of the presence of MPs in sea turtles. Furthermore, many studies have reported the detection of MPs in other marine organisms as they can be easily ingested by marine organisms, including seabirds, crustaceans, and fish because of their small size [33, 34]. Moreover, ingestion of these non-biodegradable MPs by marine organisms leads to bioaccumulation in the food chain, reaching higher trophic levels. Owing to their feeding pattern and filtration abilities, which make them susceptible to MP ingestion, bivalves such as oysters are

endorsed as biological MP contamination indicators [35]. Thomas *et al.* [36] reported the detection of MP microbeads in oysters' digestive tubules and intestines after being exposed to 6 mm polystyrene microbeads for 80 days. The study also demonstrated an increase in mortality among juvenile oysters with an increase in microbead dosage and that the continual ingestion of MPs can lead to death. Table 24.1 shows the occurrence of MPs in various aquatic organisms, as reported in various studies.

Bessa *et al.* [37] analyzed the MP toxicity in the Antarctic Ocean by determining the presence of MPs in gentoo penguins. Even though the Antarctic region displayed the presence of MPs, the studies lacked an in-depth exploration of their ingestion and bioaccumulation via the food chain. Seabirds, which are recognized as biological markers that can determine environmental changes, are also considered indicators of environmental MP contamination. The study demonstrated the presence of films (16%), fragments (26%), and microfibers (58%) within the gentoo penguin scats collected from different islands, which accumulate in their guts, preventing further food intake, resulting in the absorption of toxic matter from water, affecting their natural growth, and even resulting in death.

Table 24.1 Demonstration of the occurrence of MPs in various aquatic organisms as reported by various studies [34].

Aquatic organisms	Place	Plastic form	Plastic dimensions	Occurrence (%)
Atlantic horse Mackerel, blue whiting, Dragonet, John Dory, Poor cod, red band fish, red gurnard, Solenette, Thick back sole, and Whiting	English Channel	Macro & MP	0.13–14.3 mm	36.5
Atlantic mackerel, Cod, Gray gurnard, Haddock, Herring, Horse mackerel, and Whiting	North Sea	MP	0.04–4.8 mm	2.6
Cololabis saira, *Symbolophorus californiensis* *Hygophum reinhardtii*, *Myctophum aurolaternatum*, and *Loweina interrupta*	North Pacific	MP	1–2.79mm	35
Stolephorus commersonnii	Alappuzha, India	MP	1.14–2.5 mm	37.5
Caretta caretta	Mediterranean Sea	Macroplastics & MP	-	37
	Adriatic Sea			35.2
	Northern Pacific			35
Chelonia mydas	Southwest Atlantic		0.5–3.0 cm	90
	Ubatuba, Brazilian coast	Macroplastics & MP	(76%) 0–5 cm, (1%) N10 cm, (23%) 5–10 cm	45
	Brazilian coast	Macroplastics & MP		70
Euphausia pacifica	Pacific Ocean - Northeast	MP	816.1 ± 107.7 µm	5.8
Crangon crangon	Belgium	MP	300–1,000 µm	
Lampris sp.	North Pacific	Macroplastics & MP	49.1	29
Mesoplodon mirus	North and west coast – Ireland	Macroplastics & MP	0.3 mm–7.1 cm	85
Mytilus edulis	North Sea, Germany	MP	5 to 25 µm	
Neocalanus cristatus	Pacific Ocean— Northeast	MP	555.5 ± 148.7 µm	2.6

Aquatic organisms	Place	Plastic form	Plastic dimensions	Occurrence (%)
Pontoporia blainvillei	Argentina—Northern coast	Macro plastic	0.2–11.4 cm	28.1
Zooplankton	Western English Channel Portuguese coastal waters	MP	7.3–30.6 µm	87

Cole *et al.* [38] demonstrated the fatal impact of MP on the food habitat, functioning, and fertility of *Calanus helgolandicus* copepods, mostly found in the Atlantic region. Copepods, subjected to polystyrene 20 mm microbead exposure, reported a 40% carbon biomass reduction with energy deficiency, demonstrating rapid lipid consumption, adversely affecting their growth, and leading to death.

Feng *et al.* [39] reported that the presence of MPs in freshwater promotes *Microcystis aeruginosa*, an alga, which can cause the blooming of cyanobacteria, adversely affecting the aquatic ecosystem. Lei *et al.* [40] analyzed the effects of MPs on zebrafish by exposing them to different MP concentrations in freshwater. The results demonstrated the crucial and fatal impact of MPs on zebrafish, which led to death with excessive abdominal swelling and fatal damage, including enterocyte splitting and villus cracking in surviving zebrafish.

A large amount of MPs, with a concentration of 3,400 to 10,000 particles/m^3, have been detected in water and fish from major river tributaries of Lake Michigan [41], and MPs can easily absorb toxic matter from the environment, including drugs, cosmetics, personal care products, and toxic chemicals [42, 43]. However, the effects of MPs on freshwater organisms remain unclear. The study illustrated by Imhof *et al.* [44] reported that freshwater Daphnia had no significant aftereffects on morphology, mortality, reproductivity, or other processes even after consuming excess MPs. A study by Critchell and Hoogenboom [45] demonstrated that the evaluation of effects of MPs on planktivorous reef fish showed no potential toxic effects on the behavior and physical conditions, even though 2,102 MPs were detected in its gastrointestinal tract. These reports demonstrate that MPs with particle sizes less than 2 mm are more likely to be ingested by fish.

24.3.2 Toxicity Assessment of NPs in Aquatic Environment

Nanoplastics (NPs) are much smaller than MPs and have marginally discrete properties from MPs in one ecotoxicity study in a marine ecosystem. Table 24.2 shows the ecotoxicity of NPs in freshwater and marine ecosystems. Della Torre *et al.* [46] explored the effects of NP on *Paracentrotus lividus* using amine and carboxyl functionalized PS of 40 and 50 nm size, respectively. The sea urchins showed abnormal gene expression and embryotoxicity 48 h post-fertilization. On the other hand, Bergami *et al.* [47] employed carboxyl- and amino-functionalized PS NPs to examine the mortality and sub-lethal effects on *Artemia franciscana,* demonstrating the adverse effects of NPs on the motility, feeding, and causing multiple molting of *A. franciscana* affecting not only the organism and growth but also the ecosystem.

24.4 Toxicity Comparison of MPs and NPs

Lee *et al.* [48] demonstrated a toxicity assessment using plain and fluorescent NPs and MPs of 6 µm, 500 nm, and 50 nm dimensions and investigated the sex ratio, ingestion, mortality, and fecundity of *Tigriopus japonicus*. Both NPs and MPs caused physical

fertilization inhibition, and the 50 nm PS nanobeads did not affect the fecundity of *T. japonicus*. Cole and Galloway [49] employed fluorescently labeled PS MPs (1.84, 4.1, 7.3, 10.2, and 20.3 µm) and NPs (180, 160, and 70 nm), fluorescently labeled and carboxyl-functionalized PS (990 and 940 nm), and microplastic beads (1 and 10 µm) on *Crassostrea gigas* larvae. The consumption of NPs and MPs by *C. gigas* larvae was evaluated based on plastic size, and growth and ingestion were investigated using 10 and 1 µm MP beads, which varied within the 24 h exposure time based on the plastic properties, size, and age of the larvae; however, no significant effects of plastics were reported on the feeding and growth.

Table 24.2 Illustration of ecotoxicity investigates using NPs in marine and freshwater ecosystems [56].

Materials			Test species	Exposure conditions	
Plastic type	**Option and details**	**Size**		**Concentration**	**Media**
PS	Green carboxylated polystyrene (PS-COOH) NPs (ex. 505 nm, em. 515 nm)	40 nm	*Paracentrotus lividus*	2.5-50 µg/mL	Filtered natural seawater
					Filtered natural seawater
	Amino modified polystyrene (PS-NH2) NPs (ex. 358 nm, em. 410 nm)	50 nm	*Paracentrotus lividus*	1-50 µg/mL	Filtered natural seawater
					Filtered natural seawater
PS	Amino modified polystyrene (PS-NH2) NPs (ex. 358 nm, em. 410 nm)	50 nm	*Mytilus galloprovincialis* Lam. (*in vitro*)	1, 5, 50 mg/mL	Artificial seawater
PS	Green carboxylated PS NPs (PS-COOH) (ex. 505 nm, em. 515 nm)	40 nm	*Artemia franciscana*	0, 5, 10, 25, 50, 100 mg/mL	Natural seawater
	Unlabeled amino modified PS NPs (PS-NH2)	50 nm	*Artemia franciscana*	0, 5, 10, 25, 50, 100 mg/mL	Natural seawater
	Blue fluorescently labeled amino modified (PS-NH2) (ex. 358 nm, em. 410 nm)	50 nm	*Artemia franciscana*	0, 5, 10, 25, 50, 100 mg/mL	Natural seawater
PS	Amino modified polystyrene (PS-NH2) NPs (ex. 358 nm, em. 410 nm)	50 nm	*Mytilus galloprovincialis* Lam. (*in vitro*)	1, 5, 50 mg/mL	Artificial seawater

Materials			Test species	Exposure conditions	
Plastic type	**Option and details**	**Size**		**Concentration**	**Media**
PS	Fluorescent NPs	39.4 nm	*Oryzias latipes* (ST II strain)	1 mg/L	2.78% solids-latex (polystyrene) solution
PS	Positively (Amidine Latex) charged NPs	20 nm	*Chlorella sp.*	0.08–0.8 mg/mL	Alga-Gro freshwater medium
			Scenedesmus sp.	0.08–0.8 mg/mL	Alga-Gro freshwater medium
	Negatively (Carboxyl Latex) charged NPs	20 nm	*Chlorella sp.*	0.08–0.8 mg/mL	Alga-Gro freshwater medium
			Scenedesmus sp.	0.08–0.8 mg/mL	Alga-Gro freshwater medium
PS	Functionalized with carboxylic acid groups NPs	~70 nm	*Scenedesmus obliquus* (SAG 276/3A)	44-1,100 mg/L	WC-medium
			Daphnia magna	0.22-150 mg/L	RT medium
			Daphnia magna	0.88 and 1.8 mg/L	RT medium with fish kairomones
			Daphnia magna	32 mg/L	RT medium
			Daphnia magna	32 mg/L	Filtered RT medium
PS	Labeled with Bodipy fluorophores NPs	41 nm	*Pimephales promelas* neutrophil (*in vitro*)	0.025, 0.05, 0.1, 0.2 mg/mL	Standard stimulants
PC	–	158 nm	*Pimephales promelas* neutrophil (*in vitro*)	0.025, 0.05, 0.1, 0.2 mg/mL	Standard stimulants

Materials			Test species	Exposure conditions	
Plastic type	**Option and details**	**Size**		**Concentration**	**Media**
PS	Carboxylated NPs	<100 nm	*Daphnia magna*	0.0001–1 mg/mL	HH Combo medium
	Amino functionalized NPs	<100 nm	*Daphnia magna*	0.0001–1 mg/mL	HH Combo medium
	Fluorescently labeled carboxylic-acid functionalized NPs (ex. 441 nm, em. 486 nm)	<100 nm	*Daphnia magna*	10 µg/mL	HH Combo medium
Silica	Plain NPs	50 and 100 nm	*Vibrio fischeri*	3–1,000 µg/mL	Microtox Diluent
			Thamnocephalus platyurus	0.1–1,000 µg/mL	Thamnotox medium
			Daphnia magna	0.1, 1.0, 10, 100, 1,000 µg/mL	Elendt M4 Daphnia medium
	Green fluorescently labeled NPs (ex. 485 nm, em. 510 nm)	50 and 100 nm	*Pseudokirchneriella subcapitata*	~100 µg/mL	Algal medium
			Thamnocephalus platyurus	0.1–1,000 µg/mL	Thamnotox medium
			Daphnia magna	0.1, 1.0, 10, 100, 1,000 µg/mL	Elendt M4 Daphnia medium
			Rainbow trout gonadal cell line (RTG-2) (*in vitro*)	~1,000 µg/mL	Dulbecco's modified nutrient mix/F-12 Ham
			Rainbow trout gonadal cell line (RTG-2) (*in vitro*)	~1,000 µg/mL	Dulbecco's modified nutrient mix/F-12 Ham

Materials			Test species	Exposure conditions	
Plastic type	**Option and details**	**Size**		**Concentration**	**Media**
PS	Polyethyleneimine polystyrene NPs	55 nm	*Vibrio fischeri*	3 to 1,000 µg/mL	Microtox Diluent
			Pseudokirchneriella subcapitata	0.1–1.0 µg/mL	Algal medium
			Thamnocephalus platyurus	3–20 µg/mL	Thamnotox medium

Materials			Test species	Exposure conditions	
Plastic type	**Option and details**	**Size**		**Concentration**	**Media**
			Daphnia magna	0.33, 1.0, 1.5, 2.0, 3.3 µg/mL	Elendt M4 Daphnia medium
			Rainbow trout gonadal cell line (RTG-2) (*in vitro*)	~200 µg/mL	Dulbecco's modified nutrient mix/F-12 Ham
			Rainbow trout gonadal cell line (RTG-2) (*in vitro*)	~200 µg/mL	Dulbecco's modified nutrient mix/F-12 Ham
		110 nm	*Vibrio fischeri*	3 to 1,000 µg/mL	Microtox Diluent
			Pseudokirchneriella subcapitata	0.1–0.8 µg/mL	Algal medium
			Thamnocephalus platyurus	2–15 µg/mL	Thamnotox medium
			Daphnia magna	0.33, 1.0, 1.5, 2.0, 3.3 µg/mL	Elendt M4 Daphnia medium
			Rainbow trout gonadal cell line (RTG-2) (*in vitro*)	~200 µg/mL	Dulbecco's modified nutrient mix/F-12 Ham
			Rainbow trout gonadal cell line (RTG-2) (*in vitro*)	~200 µg/mL	Dulbecco's modified nutrient mix/F-12 Ham

Materials			Test species	Exposure conditions	
Plastic type	**Option and details**	**Size**		**Concentration**	**Media**
PMMA	Poly(methylmethacrylate) plastic nanoparticle, medium, PMMA PNP	86–125 nm	*Vibrio fischeri*	0.01, 0.1, 1, 1,000 mg/L	Microtox diluent Direct

Materials			Test species	Exposure conditions	
Plastic type	**Option and details**	**Size**		**Concentration**	**Media**
			Daphnia magna	0.01, 0.1, 1, 1,000 mg/L	M7 medium Direct
			Daphnia magna	0.01, 0.1, 1, 1,000 mg/L	M7 medium Direct
			Corophium volutator	0.01, 0.1, 1, 500 mg/L	Diluted seawater Direct
			Corophium volutator	0.01, 0.1, 1, 500 mg/L	Diluted seawater Direct
	Poly(methylmethacrylate) fluorescent analog of plastic nanoparticles, medium, PMMA FPNP		*Vibrio fischeri*	0.01, 0.1, 1, 1,000 mg/L	Microtox diluent Direct
			Daphnia magna	0.01, 0.1, 1, 1,000 mg/L	M7 medium Direct
			Daphnia magna	0.01, 0.1, 1, 1,000 mg/L	M7 medium Direct
			Corophium volutator	0.01, 0.1, 1, 500 mg/L	Diluted seawater Direct
			Corophium volutator	0.01, 0.1, 1, 500 mg/L	Diluted seawater Direct
	Poly(methylmethacrylate-co-stearylmethacrylate) copolymer plastic nanoparticle, hydrophobic, PMMA-PSMA PNP		*Vibrio fischeri*	0.01, 0.1, 1, 1,000 mg/L	Microtox diluent Direct
			Daphnia magna	0.01, 0.1, 1, 1,000 mg/L	M7 medium Direct
			Daphnia magna	0.01, 0.1, 1, 1,000 mg/L	M7 medium Direct
			Corophium volutator	0.01, 0.1, 1, 500 mg/L	Diluted seawater Direct
			Corophium volutator	0.01, 0.1, 1, 500 mg/L	Diluted seawater Direct

Materials			Test species	Exposure conditions	
Plastic type	**Option and details**	**Size**		**Concentration**	**Media**
	Poly(methylmethacrylate-co-stearylmethacrylate) copolymerfluorescent analog of plastic nanoparticles, hydrophobic, PMMA-PSMA FPNP		*Vibrio fischeri*	0.01, 0.1, 1, 1,000 mg/L	Microtox diluent Direct
			Daphnia magna	0.01, 0.1, 1,	M7 medium
			Daphnia magna	0.01, 0.1, 1, 1,000 mg/L	M7 medium Direct
			Corophium volutator	0.01, 0.1, 1, 500 mg/L	Diluted seawater Direct
			Corophium volutator	0.01, 0.1, 1, 500 mg/L	Diluted seawater Direct
PS	–	24 nm	*Scenedesmus sp.*	100 mg/L (0.01%, w/v)	Green algal laboratory culture
			Daphnia magna	–	–
			Carassius carassius	–	Water
PS	Encapsulated fluorophore	28 nm	*Scenedesmus sp.*	100 mg/L (0.01%, w/v)	Green algal laboratory culture
			Daphnia magna	–	–
			Carassius carassius	–	Water
PS	Sulfonated NPs	24 and 27 nm	*Scenedesmus sp.*	100 mg/L (0.01%, w/v)	Tap water
			Daphnia magna	–	Tap water
			Carassius carassius	–	Copper-free water

Materials			Test species	Exposure conditions	
Plastic type	**Option and details**	**Size**		**Concentration**	**Media**
PS	Plain PS	50 nm	*Tigriopus japonicus*	0, 6, 13, 31, 63, 187, 250, 313 µg/mL	Filtered seawater
		500 nm	*Tigriopus japonicus*	0, 6, 13, 31, 63, 187, 250, 313 µg/mL	Filtered seawater
		6 µm	*Tigriopus japonicus*	0, 6, 13, 31, 63, 187, 250, 313 µg/mL	Filtered seawater
		50 nm	*Tigriopus japonicus*	0, 6, 13, 31, 63, 187, 250, 313 µg/mL	Filtered seawater
		500 nm	*Tigriopus japonicus*	0, 6, 13, 31, 63, 187, 250, 313 µg/mL	Filtered seawater

The lifespan and fecundity of Brachionus koreanus were investigated by Jeong *et al*. [50] using PS micro- and nanobeads to analyze their effect on antioxidant enzymatic activity, egestion, ingestion, and ROS induction, demonstrating augmentation in antioxidant enzymes and oxidative stress and decrease in reproduction time, growth rate, body size, lifespan, and fecundity. This study reports the effect of size dependency on MP and NP toxicities, demonstrating that the smaller the size, the greater is the toxic effect.

24.5 Regulatory Policy

With the vast exploration of the potential direct or indirect harmful effects, various regulatory policies immediately regulated and banned the application of MPs and NPs in consumer products and their littering into the environment to develop stratagies to protect the oceans and prevent plastic pollution. A meta-analysis by the European Food and Safety Agency deduced the presence of both MPs and NPs in food, especially seafood [51]. Assessing the impact of NPs on the environment has emerged to be tough with existing gaps, although plastic wastes tend to cause an annual loss of USD 13.3 billion globally to marine ecosystems [52].

Numerous government organizations have decreed the legislature principally on MPs in single-use cosmetics to prohibit MPs precisely. In 1999, the Canadian government categorized microbeads as toxic contaminants, which were combined with forbidding the introduction, production, and consumption of these microbeads. Like other legislation, this prohibition covers only personal care products and does not include cleaning or other household products. In 2014, nine US states proclaimed regulations for the application of microbeads, with Illinois substantially banning microbeads in personal care products [53]. Figure 24.2 depicts the boundaries of MP/NP regulatory policies.

24.6 Environmental Implication and Conclusion

Studies on plastic accumulation and its toxic effects on aquatic ecosystems began in the early 1970s and became a matter of substantial concern. The introduction of several studies demonstrating ecotoxicities and problems associated with plastic pollution in the early 1990s gained pervasive attention. A more detailed exploration of the toxic effects of

NPs and MPs on aquatic organisms and their bioaccumulation *via* the food chain and the potential hazardous effects on both aquatic and terrestrial environments has gained more attention over the last 20 years. This chapter describes the toxic effects of various NPs and MPs on aquatic ecosystems. However, the provision and transportation of NPs and MPs in aquatic environments strongly depend on the physicochemical properties of both water and plastics. Despite the current research on plastic pollution, the literature and studies lack in-depth exploration of detection methodologies and the development of new resolution methodologies for detecting plastics at the nanoscale, especially their impact on groundwater sources. The interaction ability of MPs and NPs with other toxic contaminants, including pharmaceutical drugs and cosmetics is still unclear. More research is required in these fields to understand the transport and fate of this nano/microplastic in water, thereby avoiding its potentially hazardous effects. Recognizing the performance and predominance of MPs and NPs in the environment is the first measure used to moderate the influence of these contaminants.

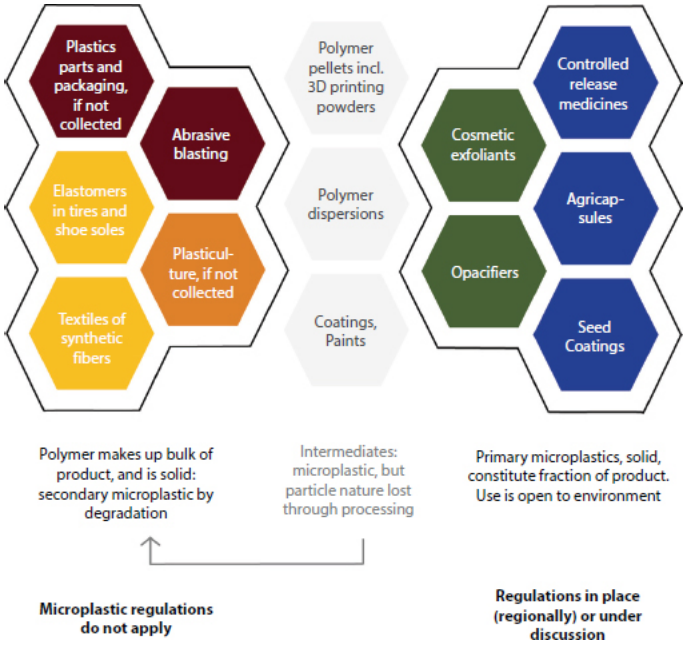


Figure 24.2 Boundaries within MP/NP regulatory policies.

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Note

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